ANL/RTR/TM-22/25

MITR & NBSR DDE Irradiations in BR2 – Fluence in LEU Cladding and Structural Materials

Nuclear Science & Engineering Division

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Abbreviations

ANL	Argonne National Laboratory
BNL	Brookhaven National Laboratory
BOC	beginning of cycle
BOL	beginning of life
BR2	Belgian Reactor 2 (at Belgian Nuclear Research Centre, SCK CEN)
DDE	design demonstration element
EOC	end of cycle
EOL	end of life
HEU	highly enriched uranium with ≥ 20 wt.% ^{235}U enrichment
INL	Idaho National Laboratory
LANL	Los Alamos National Laboratory
LEU	low-enriched uranium with < 20 wt.% ^{235}U enrichment
MIT	Massachusetts Institute of Technology
MITR	Massachusetts Institute of Technology Reactor
MURR®	University of Missouri Research Reactor
NBSR	National Bureau of Standards Reactor
NNSA	U.S. National Nuclear Security Administration
U-10Mo	uranium – A binary alloy comprised of 90 wt.% uranium and 10 wt.% molybdenum.
USHPRR	U.S. high-performance research reactors

1 Introduction

The BR2 nuclear reactor is a material testing reactor (MTR) located in Mol, Belgium, and operated by the Belgian Nuclear Research Centre (SCK CEN) since 1963. The reactor is highly versatile as the number and location of fuel elements and control rods can change significantly from cycle to cycle to accommodate different needs. Argonne National Laboratory (ANL or Argonne) Reactor Conversion (RC) team has collaborated with SCK CEN for over a decade on the conversion of domestic and international research reactors from highly enriched uranium (HEU, ≥20 wt.% of ^{235}U) to low-enriched uranium (LEU, <20 wt.% of ^{235}U) fuel. The U.S. High-Performance Research Reactor (USHPRR) project within the M3 Reactor Conversion Program aims at converting five U.S. high-performance research reactors (MITR, MURR, NBSR, HFIR, and ATR) and one critical facility (ATR-C) to LEU fuel [NNSA, 2019]. These USHPRRs still use and regularly refuel with HEU fuel. Each facility has a unique reactor design, operating conditions, and fuel element design to accomplish its mission. The goal of the USHPRR project is to convert the USHPRRs and the critical facility to LEU fuel while maintaining experimental performance and ensuring safe facility operation.

The current technical report focuses on two reactors requiring very high-density LEU fuel: the Massachusetts Institute of Technology Reactor (MITR) and the National Bureau of Standards Reactor (NBSR). To support the conversion of these reactors, so-called design demonstration elements (DDE) are planned to be irradiated in the BR2 reactor under conditions similar to the targeted reactors [Stillman, 2020] and using a prototypic geometry. In support of this experiment, SCK CEN studied and modeled the DDE irradiations [Kalcheva, 2022] using MCNP6.2 [Werner, 2017] to investigate the feasibility of irradiating the MITR DDE and NBSR DDE in BR2. Argonne reviewed and confirmed the conclusions of this study [Rykhlevskii, 2022].

Structural analysis is another step toward converting USHPRR to LEU fuel. The objective of the current report is to provide information useful to the structural analysis of the NBSR & MITR DDEs to support its irradiation in BR2. Specifically, the goal is to provide the fast neutron (E>0.1MeV) fluence in the cladding of the fuel plates in BR2 for the whole period of irradiation (8 cycles for MITR DDE and 10 cycles for NBSR DDE). Additionally, fast neutron fluences in the side plates and in the NBSR DDE's outside plates were calculated and reported. Neutronic calculations were performed using MCNP6.2 on the RTRHPC cluster.

2 Calculation methodology

The three-dimensional MCNP6.2 model of the BR2 reactor with the MITR and NBSR DDE in the core was provided by SCK CEN [Kalcheva, 2022] and independently reviewed by Argonne [Rykhlevskii, 2022]. For the purpose of this report, the models are modified to obtain the detailed spatial distribution of neutron fluxes in the fuel cladding, side plates, and NBSR DDE's outside plates because the original models have no spatial discretization of non-fuel materials.

2.1 Experiment Irradiation Cycle Plan

The rigorous analysis performed by SCK CEN suggested a basket and plug made of aluminum alloy (AlMgSi0.5) placed in the large 200mm irradiation channels H3 (NBSR DDE) and H5 (MITR DDE). To reduce power density in MITR DDE, SCK CEN suggested placing 12 absorbing rods made of hafnium evenly spaced in the outer ring of the irradiation basket. Table 2.1 describes the irradiation plan for

the MITR DDE and NBSR DDE irradiation experiments in BR2. Notably, the reactor power needs to be increased to achieve the target maximum fission density in NBSR DDE. Simultaneous irradiation of both DDEs in Cycles 5-8 presents significant challenges because of substantial differences in target test values in the two different DDEs and must be carefully evaluated.

Table 2.1. Irradiation cycle plan for simultaneous irradiation of MITR DDE & NBSR DDE in Al-basket & Al-plug in BR2. Cycle length is 31.5 days.

Cycle #	Reactor Power (MW)	Position in the BR2 core	
		MITR DDE	NBSR DDE
1	55	H5	not loaded
2	56	H5	not loaded
3	56	H5	not loaded
4	56	H5	not loaded
5	56	H5	not loaded
6	56	H5	H3
7	56	H5	H3
8	56	H5	H3
9	60	unloaded	H3
10	60	unloaded	H3
11	60	unloaded	H3
12	60	unloaded	H3
13	60	unloaded	H3
14	65	unloaded	H3
15	65	unloaded	H3

2.2 DDE and Basket Geometry

The MITR DDE is the full-size LEU MITR fuel assembly designed to mimic MITR irradiation conditions in the BR2 core. The MITR DDE geometry and fuel composition are described in the USHPRR technical specification [INL, 2020A], technical drawings [Housley, 2021A], and technical report [Jaluvka, 2018]. The MITR DDE is fitted in an aluminum basket surrounded by an aluminum plug [Rossaert, 2021A]. The basket and plug are separated by a 2.5mm water gap. The Al-plug has 12 holes that are filled with non-cladded, solid hafnium rods to reduce neutron flux in the basket [Kalcheva, 2022].

The NBSR DDE is the full-size LEU NBSR fuel assembly designed to mimic NBSR irradiation conditions on the BR2 core. The non-fueled aluminum hardware regions are shortened to meet dimensional constraints. The NSBR DDE geometry and composition are described in the USHPRR specification [INL, 2020B], technical drawing [Housley, 2021B], and technical report [Diamond, 2018]. The NBSR DDE is fitted inside an aluminum basket surrounded by an aluminum plug without any holes [Rossaert, 2021B]. The basket and plug are separated by a 2.5mm water gap. The key DDE dimensions from the documents above are reproduced in Table 2.2. The main basket dimensions from [Kalcheva, 2022] are summarized in Table 2.3.

Table 2.2. Summary of MITR DDE & NBSR DDE dimensions ([Sun, 2020], [Diamond, 2020], [Housley, 2021A]), [Housley, 2021B]).

Parameter	MITR DDE	NBSR DDE

Plate length, in (mm)	23 (584.2)	13.0 (330.2)
Fuel core length, in (mm)	22.375 (568.325)	11.0 (279.4)
Plate thickness, in (mm)	0.049 (1.2446)	0.0501 (1.2725)
Fuel core thickness, in (mm)	0.025 (0.6350*) 0.017 (0.4318†) 0.013 (0.3302‡)	0.0085 (0.2159)
Zirconium interlayer thickness, in (mm)	0.001 (0.0254)	
Plate width, in (mm)	2.526 (64.1604)	2.793 (70.942)
Fuel core width, in (mm)	2.082 (52.8828)	2.415 (61.341)

Table 2.3. Summary of basket dimensions for irradiation of MITR DDE in channel H5 and NBSR DDE in channel H3.

Parameter	MITR DDE	NBSR DDE
Vertical position of the DDE mid-plane relative to the reactor mid-plane (mm)	+141.13625	-50.00000
Number of absorbing rods (Hf-rods) in the Al-plug (mm)	12	0
Diameter of absorbing rods (mm)	10.5	—
Outer/inner radius of outer Al-plug (mm)	99.8/75.0	
Inner radius of aluminum basket (mm)	72.5	

2.3 Fuel and Cladding Material Composition

The fuel and cladding material compositions are described in the USHPRR documents [INL, 2020A], [INL, 2020B], [Jaluvka, 2018], [Diamond, 2018]. The cladding material for both DDEs is aluminum alloy 6061 per ASTM B209 T6 (UNS A96061) [ASTM B209] and is well-defined in the specifications (see Table 2.4).

Table 2.4. Cladding (AA6061) isotopic composition [Diamond, 2018].

Isotope	Weight fraction (-)
^{27}Al	0.9741
Si	0.0060
Mg	0.0100
Ti	0.0007
^{55}Mn	0.0007
^{50}Cr	8.70E-5
^{52}Cr	1.68E-3
^{53}Cr	1.90E-4
^{54}Cr	4.72E-5
^{54}Fe	2.05E-4
^{56}Fe	3.21E-3
^{57}Fe	7.42E-5
^{58}Fe	9.80E-6
^{63}Cu	2.08E-3
^{65}Cu	9.25E-4

* For fuel plates from 4th to 16th.
† For 2nd, 3rd, 17th, 18th fuel plates.
‡ For 1st and 19th fuel plates.

The isotopic composition of U-10Mo is also defined in the specifications ([INL, 2020A], [INL, 2020B]). It is uranium alloyed with molybdenum (natural abundance) to a nominal molybdenum content of 10.0±1.0 wt.%. The density of the U-10Mo alloy in both DDEs used for this study is 17.0 g/cm^3 [Jaluvka, 2018]. Other more recent sources ([Rabin, 2020], [Jamison, 2020]) reported very similar values (<1% difference). Table 2.5 shows U-10Mo requested composition that is defined in the specifications [INL, 2020A], [INL, 2020B]. Based on this information, the number densities for MCNP material card entries are calculated and presented in Table 2.6. Fuel meat composition at BOC for cycles 2-15 was taken from previous depletion calculations performed with VESTA-2.1.5 [Rykhlevskii, 2022].

Table 2.5. Fresh LEU isotopic composition for MITR and NBSR DDEs ([INL, 2020A], [INL, 2020B]).

Isotope	Units	Value
^{232}U	μg/gU	≤0.002
^{234}U	wt.%	≤0.260
^{235}U	wt.%	19.75±0.20%
^{236}U	μg/gU	≤4600.0
^{238}U	wt.%	Balance to yield a total of 100 wt.%.

Table 2.6. Fresh LEU fuel meat isotopic composition for MITR and NBSR DDE used in the current neutronics analysis.

Isotope	Number density, cm^{-3}
^{234}U	1.0248E+20
^{235}U	7.7513E+21
^{236}U	1.7977E+20
^{238}U	3.0819E+22
^{92}Mo	1.5775E+21
^{94}Mo	9.8567E+20
^{95}Mo	1.6986E+21
^{96}Mo	1.7812E+21
^{97}Mo	1.0213E+21
^{98}Mo	2.5844E+21
^{100}Mo	1.0326E+21
Total	4.9534E+22

2.4 Neutronics Model and Discretization

The current analysis relies on the MCNP input files provided by SCK CEN [Kalcheva, 2022A]. To accurately calculate the evolution of the compositions during depletion and to capture the gradient of power within a fuel plate, it is necessary to implement adequate fuel core discretization within the fuel plates.

2.4.1 Fuel Meat Discretization

The magnitude of the calculated peak values will typically depend on the size of the nodes used in the calculations. Designing the appropriate discretization will also largely depend on the power profile (i.e., if the profile is flat, the node size is irrelevant; if the profile exhibits steep gradients, node size is highly relevant). Table 2.7 summarizes fuel meat discretization used in MITR and NBSR DDE calculations to determine the target test values.

Table 2.7. Axial and lateral discretization of the fuel meat ([Stillman, 2020], [Sun, 2020], [Diamond, 2020]).

USHPRR	Node Size for fission density		Node Size for power density	
	Width, inch (cm)	Height, inch (cm)	Width, inch (cm)	Height, inch (cm)
MITR	0.52 (1.32)	1.40 (3.55)	0.52 (1.32)	1.40 (3.55)
NBSR	0.805 (2.0447)	0.79 (2.00)	0.805 (2.0447)	0.79 (2.00)

2.4.2 Fuel Cladding Discretization

The previous analysis of MITR & NBSR DDE irradiation in BR2 was done using a detailed geometric model [Kalcheva, 2022A] but was focusing only on the fuel meat [Rykhlevskii, 2022]. Hence, the model was edited to capture energy-dependent neutron flux in the cladding of each plate over the irradiation history of the experiment. A two-dimensional mesh (axial and lateral) was used for each test plate in the MITR DDE (Figure 2.1**Error! Reference source not found.**) and NBSR DDE (Figure 2.2). Table 2.8 summarizes cladding discretization (node sizes) used in MITR and NBSR DDEs calculations. For MITR DDE, the thickness in the table is of both the cladding and the Zr interlayer. For NBSR DDE, the thickness in the table is of just cladding. The number of nodes is limited by Cycles 5-8 when both the DDEs are loaded, and the memory footprint is maximized (>300GB), so a more detailed mesh is unfeasible.

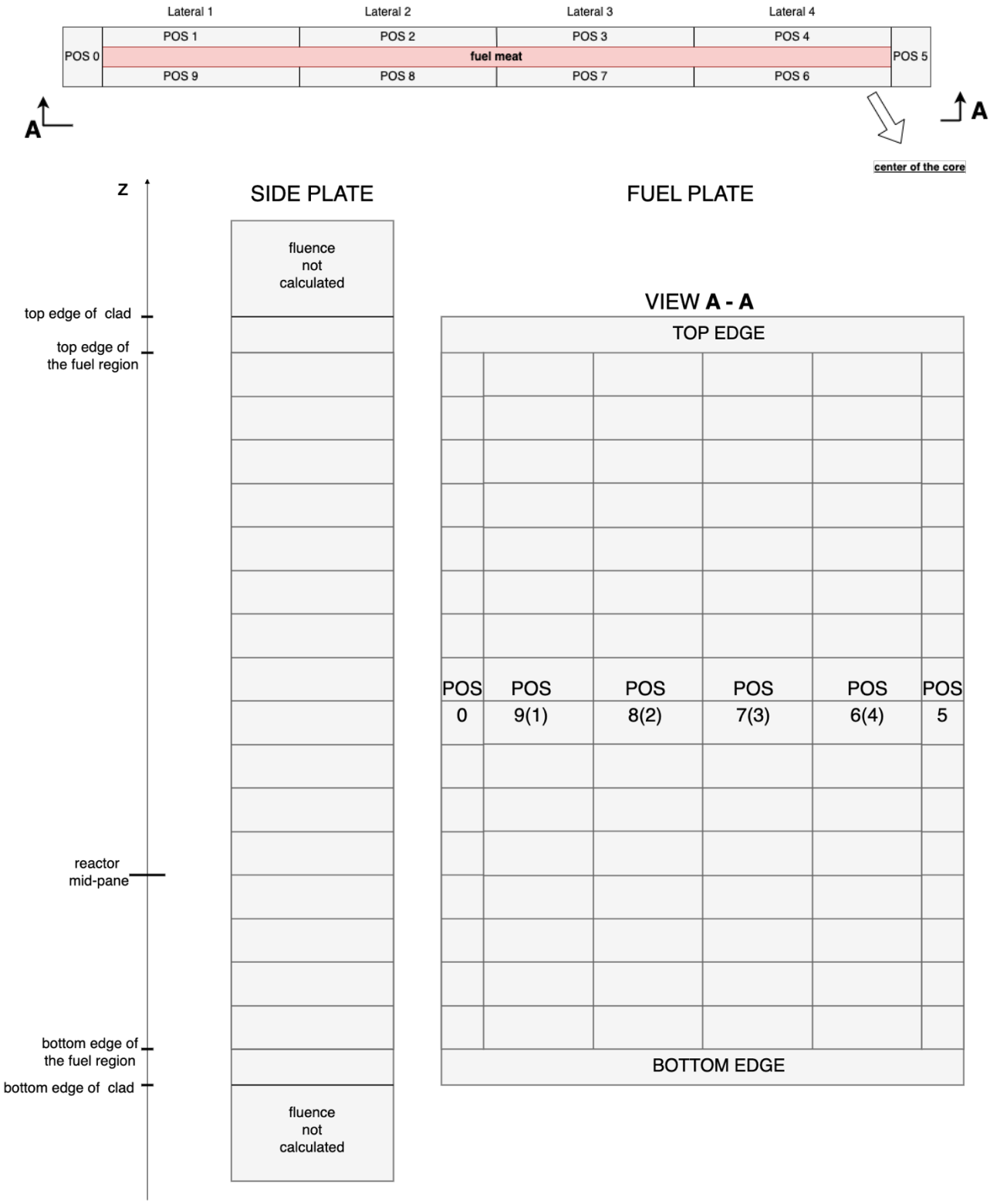

Figure 2.1. MITR DDE fuel plate cladding and side plate discretization.

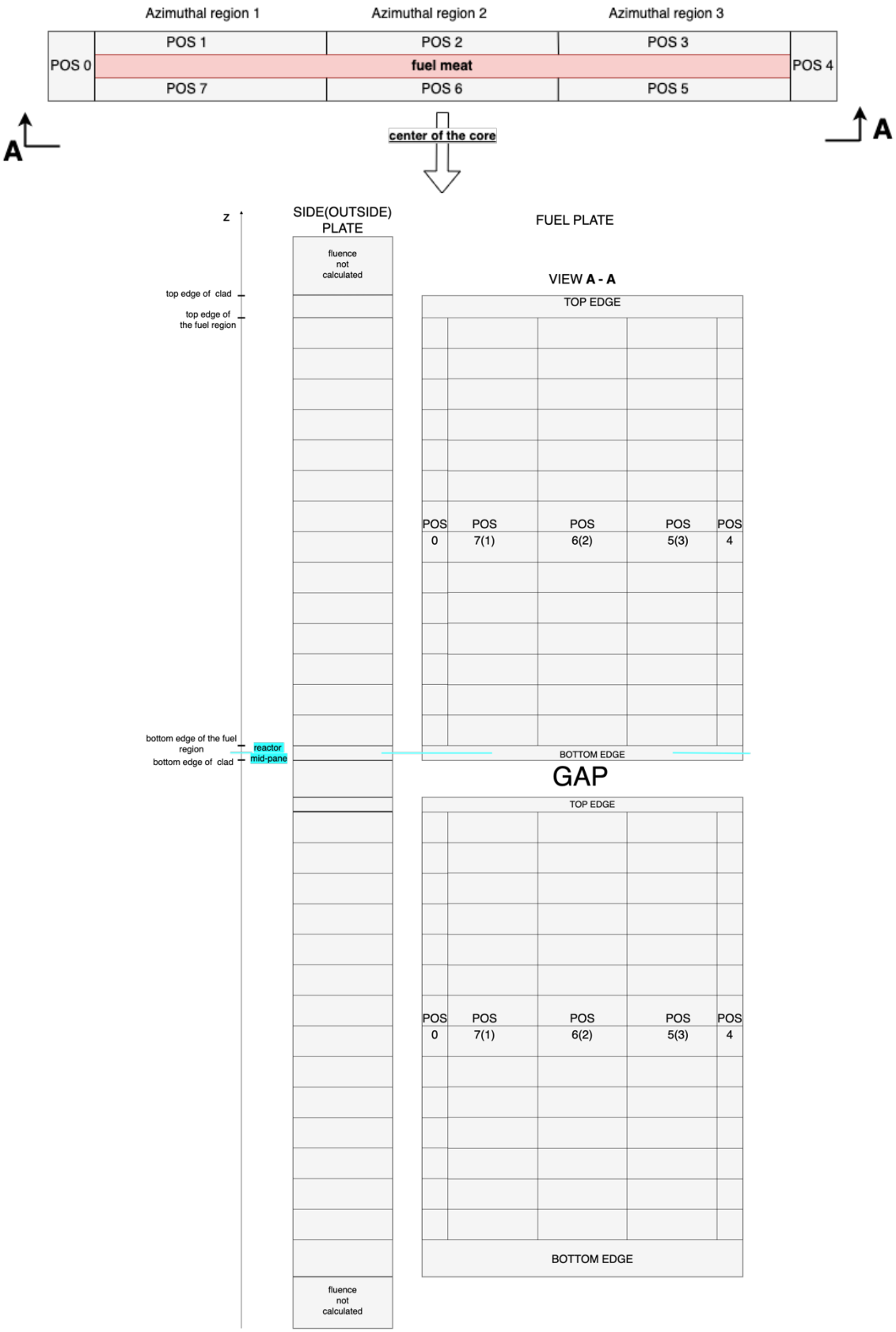

Figure 2.2. NBSR DDE structural elements discretization.

Table 2.8. Axial and lateral discretization of the cladding in the MITR & NBSR DDEs.

USHPRR	Node position	Node dimensions for fluence calculations		
		Width, cm	Height, cm	Thickness, cm
MITR	POS 0 and POS5 (unheated edges)	0.287	3.552	0.12446
	POS 1,2,3,4 and POS 6,7,8,9	1.322	3.552	0.03048§ 0.04064** 0.04572††
	Top edge and bottom edge	5.862	0.79375	0.12446
NBSR	POS 0 and POS 4 (unheated edges)	0.26295	1.99571	0.12700
	POS 1,2,3 and POS 5,6,7	2.04470	1.99571	0.05271
	Top edge (upper plate) & bottom edge (lower plate)	6.66000	3.49200	0.12700
	Bottom edge (upper plate) & top edge (lower plate)		1.58800	

2.4.3 Side and Outside Plates Discretization

MITR DDE side plates have axial discretization only: 18 axial nodes from z=-15.07375cm (the bottom edge of the fuel plate) to z=43.34625cm (the top edge of the fuel plate). The fast neutron fluence in side plates above and below those planes was not captured (see Figure 2.1). Similarly, NBSR DDE side plates have axial discretization only (see Figure 2.2). NBSR DDE also has two unfueled outside plates. Discretization of side and outside plates in NBSR DDE is similar, but the thickness and width are different. The thicknesses are 0.475cm and 0.165cm for the side (left & right) and outside plates, respectively. The widths are 8.471cm and 6.66cm for the side (left & right) and outside plates, respectively. Table 2.9 summarizes mesh dimensions for fast fluence calculations in those structural components.

Table 2.9. Axial and lateral discretization of the side and outside plates in the MITR & NBSR DDE.

USHPRR	Node position	Node dimensions for fluence calculations		
		Width, cm	Height, cm	Thickness, cm
MITR	Left/right side plate	7.0537	3.5520‡‡ 0.7937§§	0.4816
NBSR	Left/right side(outside) plate: - upper plate, top unheated edge - upper plate, fuel meat region - upper plate, bottom unheated - gap between upper and lower - lower plate, top unheated edge - lower plate, fuel meat region - lower plate, bottom unheated	8.4710(6.6600)	 3.4920 1.9957 1.5880 5.2400 1.5880 1.9957 3.4920	0.4750(0.1650)

§ For fuel plates from 4th to 16th.
** For 2nd, 3rd, 17th, 18th fuel plates.
†† For 1st and 19th fuel plates.
‡‡ In the region of the fuel meat (see Figure 2.1, left).
§§ In the region of unheated top and bottom edge of the cladding.

3 Methods and Calculations Results

To accurately determine how the experiment design meets the target test values (fission rate density and power density), Argonne carried out a series of independent calculations [Rykhlevskii, 2022] and compared the results with those of SCK CEN [Kalcheva, 2022A].

2.1 Scope

[Kalcheva, 2022] reports data for the three irradiation campaigns, i.e., a series of consecutive irradiation cycles. In the first campaign, from Cycle 1 to Cycle 5, only the MITR DDE was loaded in the large experimental channel H5. The second campaign consists of Cycles 6, 7, and 8, in which both the MITR DDE (H5) and the NBSR DDE (H3) are irradiated simultaneously. In the third campaign (Cycles 9-15), only NBSR DDE (H3) is loaded, and MITR DDE is substituted with so-called "DG" devices used for medical isotope production. In the current report, the following parameters are calculated to inform ongoing thermo-mechanical analysis at Argonne:

1) Fast neutron (E>0.1MeV) fluence distribution in the fuel plates cladding.
2) Fast neutron (E>0.1MeV) fluence distribution in the side plates.
3) Fast neutron (E>0.1MeV) fluence distribution in the NBSR DDE's outside plates.

Notably, fission and power density distributions for the MITR & NBSR DDEs irradiation in BR2 are reported in [Rykhlevskii, 2022]. The current analysis relies on the assumptions listed in Section 3 of [Rykhlevskii, 2022].

2.2 Tools and Calculation Methods

This section covers the various analytical tools and methods that were employed to perform the independent steady-state and depletion calculations.

3.2.1 Computation tools

3.2.1.1 Steady-State Calculations – MCNP 6.2

The Monte Carlo N-Particle (MCNP) Transport computer code has been selected to perform the steady-state calculations. This software package is developed by Los Alamos National Laboratory (LANL) [Werner, 2017]. It is primarily used for the simulation of nuclear processes and particle interactions such as neutrons, photons, and electrons. Its range of applications is very large and covers, among other things, medical physics, fission and fusion reactor design, criticality calculations, and radiation protection. The MCNP code has a large base of users worldwide and has been benchmarked against many experiments. Its ability to model complex geometries in three dimensions (3D) and calculate various parameters (eigenvalues, neutron flux, spectrum...) makes the code an excellent tool for studying complex systems such as BR2. Version MCNP6.2 of the code [Werner, 2017] has been used for the study (referred to as MCNP throughout the rest of the report). Both SCK CEN and Argonne used MCNP6.2 with ENDF/B-VII.1 Neutron Cross Section Data [Chadwick, 2011].

3.2.2 Methodology and Normalization

3.2.2.1 Fast neutron flux

The *F4:N* MCNP tallies are used to determine neutron flux distribution in the cladding and structures of interest. *E0* card is used to obtain the flux for neutrons with energy above 0.1MeV as follows:

E0 0.1 20

MCNP reports neutron flux normalized to the source neutron. One can obtain the absolute value by normalizing it according to the total reactor power during a cycle as follows:

$$\phi_i = \frac{Power[W]\times\bar{\nu}}{199\frac{MeV}{f}\times 1.6021773\times 10^{-13}\frac{J}{MeV}} \times \frac{1}{k_{eff}} \times \frac{1}{V_i} \times F4n \qquad \text{(Equation 1)}$$

where ϕ_i is a local neutron flux, V_i is cell volume, $\bar{\nu}$ is the number of neutrons created per fission, Power [W] is the total reactor power during the cycle and 199 MeV/f is the average amount of energy released per fission for BR2 with DDEs loaded. F4n tallies, for the second energy bin from 0.1MeV to 20MeV, are extracted from the MCNP output to specifically obtain fast neutron flux. BOC neutron flux calculations used 10^9 neutron histories with an uncertainty ~1% in the hot spots. Uncertainty quantification (UQ) of the Monte Carlo simulations in a complex reactor system like BR2 is a complex topic that would require a separate study and is outside of the scope of this report.

3.2.2.2 Fast neutron fluence

The neutron fluence can be obtained by explicitly integrating the neutron flux over time:

$$Fluence_i(t) = \int_0^{T_{cycle}} \phi_i \cdot dt \qquad \text{(Equation 2)}$$

Where ϕ_i is a local neutron flux. In practice, time cannot be treated as a continuous variable in depletion calculations. Instead, to obtain the fluence distribution in a reasonable time, it was assumed that the fast neutron flux during a given cycle does not change significantly. Therefore, the fast neutron fluence in cell *i* at the end of each cycle k is calculated as follows:

$$Fluence_{i,k} = \phi_{i,k} \cdot \Delta T_k \qquad \text{(Equation 3)}$$

where

$\phi_{i,k}$ = fast neutron flux in i^{th} cell at BOC of kth cycle [$1/cm^2s^{-1}$]
ΔT_k = duration of the kth cycle [s]

The fast neutron flux in the current report is only calculated for BOC and assumed to be constant over the duration of the cycle. The duration of the cycle is 31.5 days for all fifteen cycles.

2.3 Fast neutron fluence calculation results

This section presents fast neutron fluences calculated in the cladding and structural components of MITR & NBSR DDEs. Detailed distributions of fast neutron flux in cladding, side plates, and outside plates (NBSR DDE) are listed in the Appendices.

3.3.1 MITR DDE

Maximum fast neutron fluences in the fuel cladding of MITR DDE at the end of each of the eight irradiation cycles are presented in Table 3.1-Table 3.4. The maximum fluence at end of cycle number 1 (EOC1) is 2.522×10^{20} n/cm^2 and is observed in Plate #19, Position #4 (see Figure 2.1). But the maximum fluence per irradiation cycle (4.64×10^{20} n/cm^2, marked in red in Table 3.2) is observed in Cycle 6 when NBSR DDE is loaded. The presence of a second DDE with fresh fissile material led to a fission rate density increase and, consequently, a fluence increase. The axial distribution of fast

neutron fluence integrated over eight cycles in different locations in the cladding of a 19th plate is shown in Figure 3.1. The maximum cumulative fluence is ~2.6x10^{21} n/cm^2 (Position #3&4) and has a statistical error below 0.5%. Figure 3.2 shows cumulative fast neutron fluence in the left and right side plates. The maximum fluences are 1.635x10^{21} n/cm^2 and 1.815x10^{21} n/cm^2, respectively. Detailed distributions of fast neutron fluence in the cladding at the end of each cycle are reported in APPENDIX A: . The fast neutron fluence in the side plates is detailed in APPENDIX B: .

Table 3.1. Maximum fast neutron fluence (n/cm^2) at the end of cycle in the MITR DDE cladding, POS 3.

	Power	Plate number			
		16	17	18	19
EOC1	55MW	1.652E+20	2.264E+20	2.278E+20	2.495E+20
EOC2	56MW	1.665E+20	2.262E+20	2.324E+20	2.544E+20
EOC3	56MW	1.664E+20	2.250E+20	2.271E+20	2.517E+20
EOC4	56MW	1.678E+20	2.252E+20	2.285E+20	2.506E+20
EOC5	56MW	1.668E+20	2.219E+20	2.252E+20	2.457E+20
EOC6	56MW	3.155E+20	4.206E+20	4.233E+20	4.617E+20
EOC7	56MW	3.101E+20	4.117E+20	4.124E+20	4.524E+20
EOC8	56MW	3.094E+20	4.101E+20	4.095E+20	4.477E+20

Table 3.2. Maximum fast neutron fluence (n/cm^2) at the end of cycle in the MITR DDE cladding, POS 4.

	Power	Plate number			
		16	17	18	19
EOC1	55MW	1.706E+20	2.264E+20	2.284E+20	2.522E+20
EOC2	56MW	1.707E+20	2.276E+20	2.295E+20	2.464E+20
EOC3	56MW	1.713E+20	2.238E+20	2.274E+20	2.461E+20
EOC4	56MW	1.715E+20	2.246E+20	2.279E+20	2.484E+20
EOC5	56MW	1.701E+20	2.248E+20	2.247E+20	2.471E+20
EOC6	56MW	3.197E+20	4.229E+20	4.203E+20	4.640E+20
EOC7	56MW	3.163E+20	4.152E+20	4.146E+20	4.535E+20
EOC8	56MW	3.155E+20	4.166E+20	4.157E+20	4.477E+20

Table 3.3. Maximum fast neutron fluence (n/cm^2) at the end of cycle in the MITR DDE cladding, POS 6.

	Power	Plate number			
		16	17	18	19
EOC1	55MW	1.708E+20	2.282E+20	2.252E+20	2.454E+20
EOC2	56MW	1.707E+20	2.267E+20	2.254E+20	2.474E+20
EOC3	56MW	1.710E+20	2.245E+20	2.257E+20	2.432E+20
EOC4	56MW	1.702E+20	2.272E+20	2.254E+20	2.444E+20
EOC5	56MW	1.680E+20	2.221E+20	2.230E+20	2.407E+20
EOC6	56MW	3.211E+20	4.226E+20	4.201E+20	4.565E+20
EOC7	56MW	3.171E+20	4.176E+20	4.120E+20	4.513E+20
EOC8	56MW	3.160E+20	4.161E+20	4.150E+20	4.479E+20

Table 3.4. Maximum fast neutron fluence (n/cm²) at the end of cycle in the MITR DDE cladding, POS 7.

	Power	Plate number			
		16	17	18	19
EOC1	55MW	1.667E+20	2.248E+20	2.281E+20	2.471E+20
EOC2	56MW	1.700E+20	2.260E+20	2.297E+20	2.488E+20
EOC3	56MW	1.678E+20	2.265E+20	2.259E+20	2.468E+20
EOC4	56MW	1.695E+20	2.237E+20	2.295E+20	2.436E+20
EOC5	56MW	1.676E+20	2.222E+20	2.270E+20	2.432E+20
EOC6	56MW	3.174E+20	4.169E+20	4.199E+20	4.539E+20
EOC7	56MW	3.118E+20	4.113E+20	4.121E+20	4.456E+20
EOC8	56MW	3.097E+20	4.108E+20	4.050E+20	4.401E+20

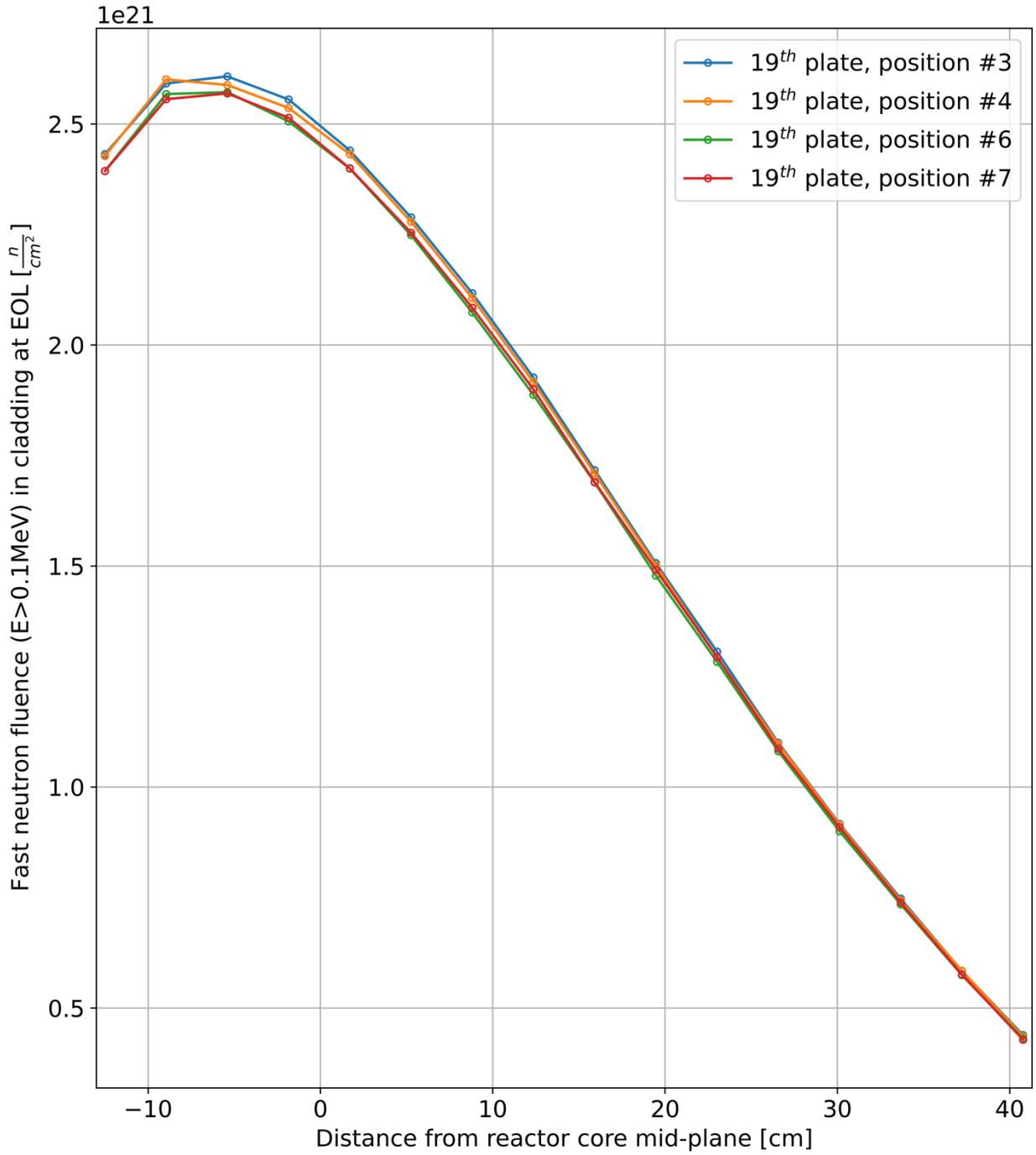

Figure 3.1. Cumulative fast neutron (E>0.1MeV) fluence axial distribution in MITR DDE fuel cladding after eight irradiation cycles (EOL).

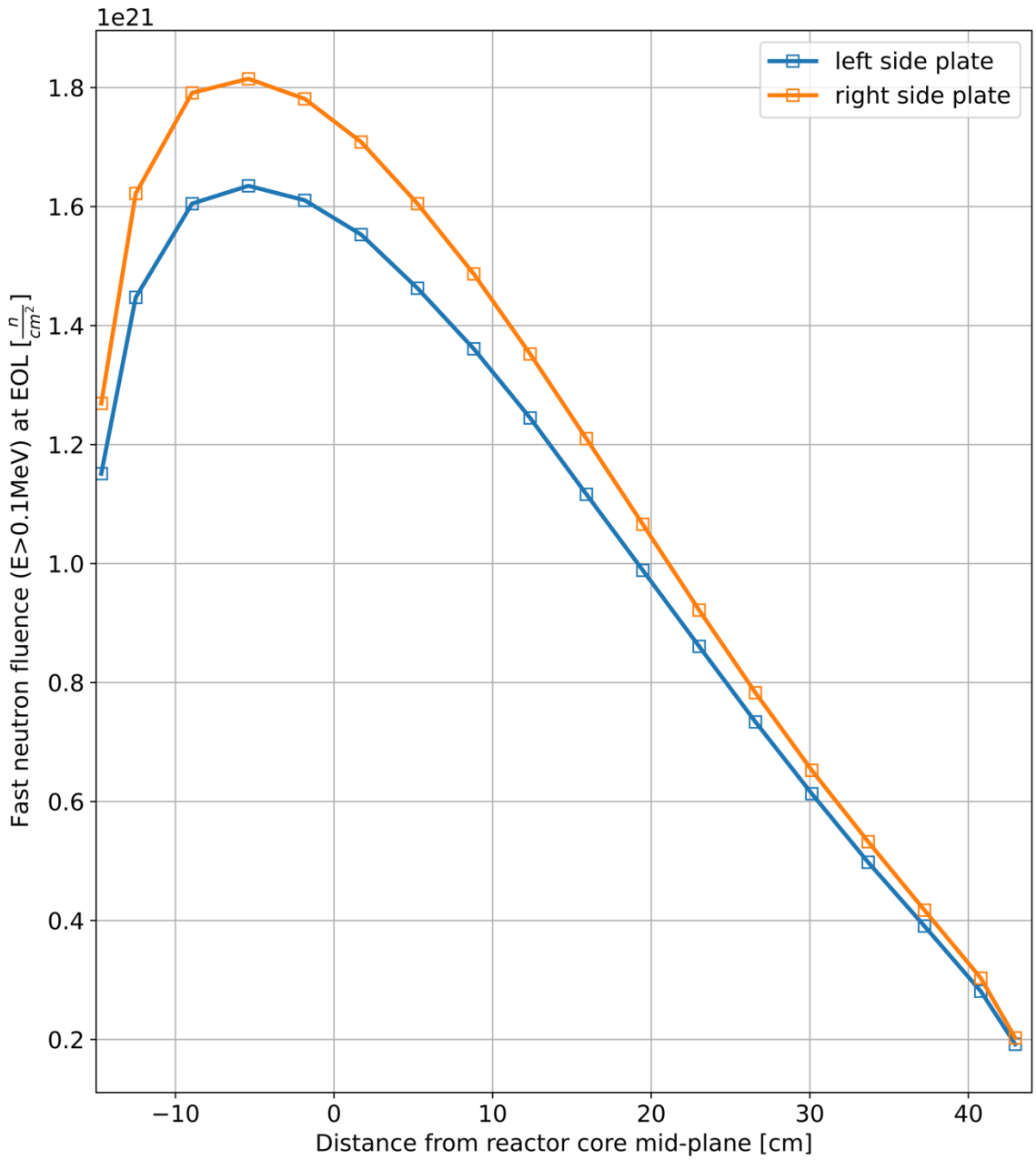

Figure 3.2. Cumulative fast neutron (E>0.1MeV) fluence axial distribution in MITR DDE side plates after eight irradiation cycles (EOL).

3.3.2 NBSR DDE

Maximum fast neutron fluences in cladding and structural components of NBSR DDE at the end of each of the ten irradiation cycles in Position #5 (maximum is always observed in this location, see Figure 2.2) are presented in Table 3.5. The maximum fluence at EOC6 is 7.52x10^{20} n/cm^2 and is observed in lower plate #16. But the maximum fluence per irradiation cycle (4.64x10^{20} n/cm^2, marked in red) is observed in Cycle 6 when NBSR DDE is loaded. The axial distribution of fast neutron fluence integrated over ten cycles in position #5 is shown in Figure 3.3. The maximum cumulative fluence is 7.5x10^{20} n/cm^2 (Lower plate #16, POS 5) and has a statistical error below 0.5%. Figure 3.4

shows cumulative fast neutron fluence in the outside and side plates. Detailed distributions of fast neutron fluence at the end of each cycle are reported in APPENDIX C: and APPENDIX D: .

Table 3.5. Maximum fast neutron fluence (x10^{20}, n/cm^2) in the NBSR DDE upper/lower fuel plate cladding, POS 5.

	Plate number					
	12	**13**	**14**	**15**	**16**	**17**
EOC6	6.434/7.100	6.538/7.279	6.528/7.359	6.621/7.421	6.664/7.520	6.383/7.193
EOC7	6.156/6.827	6.242/6.919	6.276/7.059	6.297/7.105	6.268/7.041	6.067/6.765
EOC8	5.921/6.566	6.057/6.692	6.178/6.775	6.083/6.873	6.010/6.799	5.736/6.511
EOC9	6.325/6.976	6.312/7.024	6.404/7.114	6.417/7.095	6.390/7.094	6.055/6.760
EOC10	6.042/6.707	6.144/6.874	6.206/6.855	6.186/6.872	6.056/6.752	5.777/6.446
EOC11	5.815/6.487	5.927/6.528	5.907/6.601	5.883/6.601	5.753/6.422	5.462/6.090
EOC12	5.627/6.163	5.599/6.255	5.611/6.277	5.615/6.250	5.507/6.148	5.189/5.754
EOC13	5.385/5.929	5.469/6.079	5.400/6.004	5.360/5.926	5.237/5.781	5.020/5.505
EOC14	5.557/6.157	5.585/6.216	5.631/6.171	5.567/6.103	5.428/5.975	5.183/5.616
EOC15	5.301/5.939	5.329/5.886	5.341/5.849	5.272/5.799	5.110/5.636	4.838/5.377

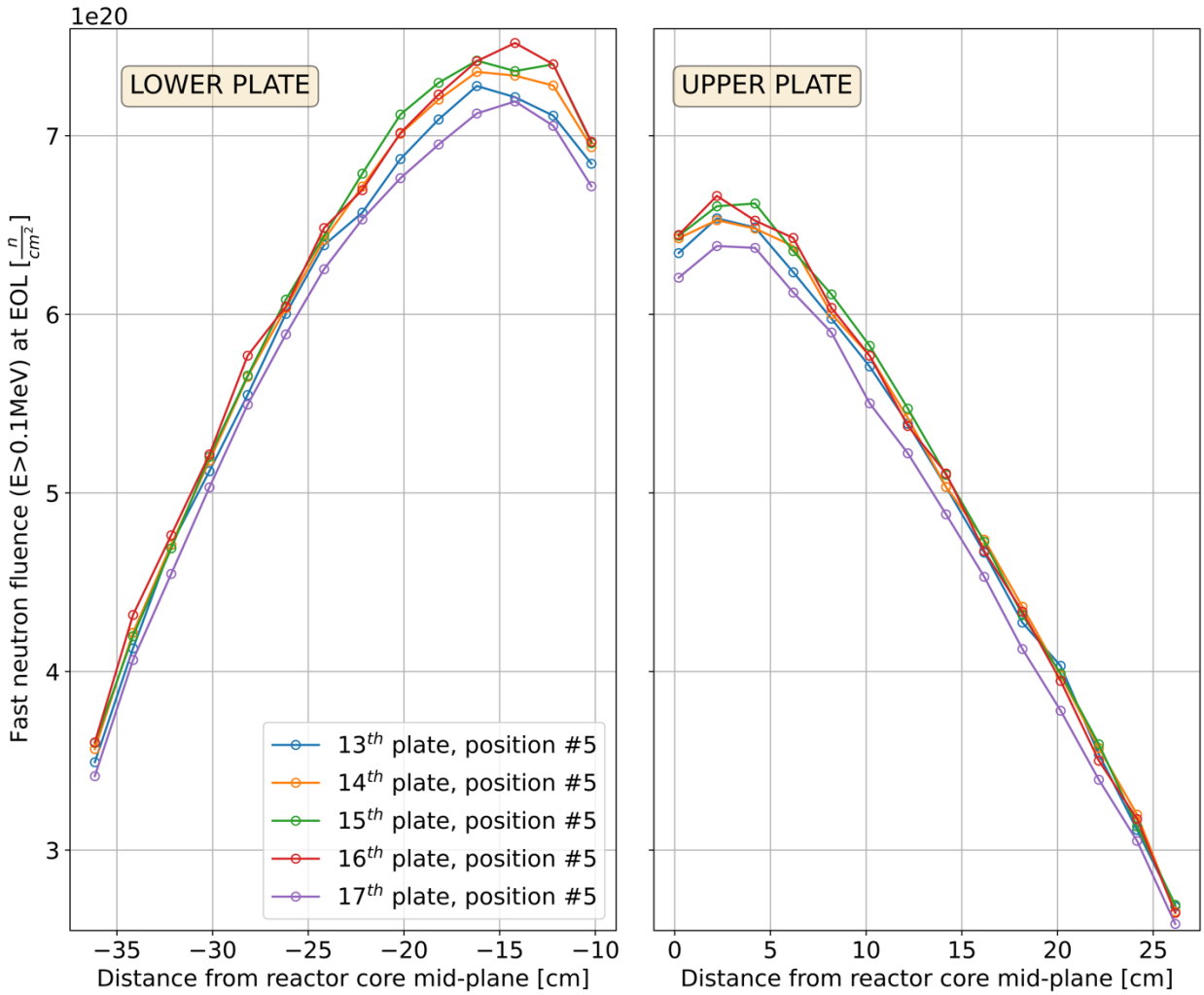

Figure 3.3. Cumulative fast neutron (E>0.1MeV) fluence axial distribution in NBSR DDE fuel cladding after ten irradiation cycles (EOL).

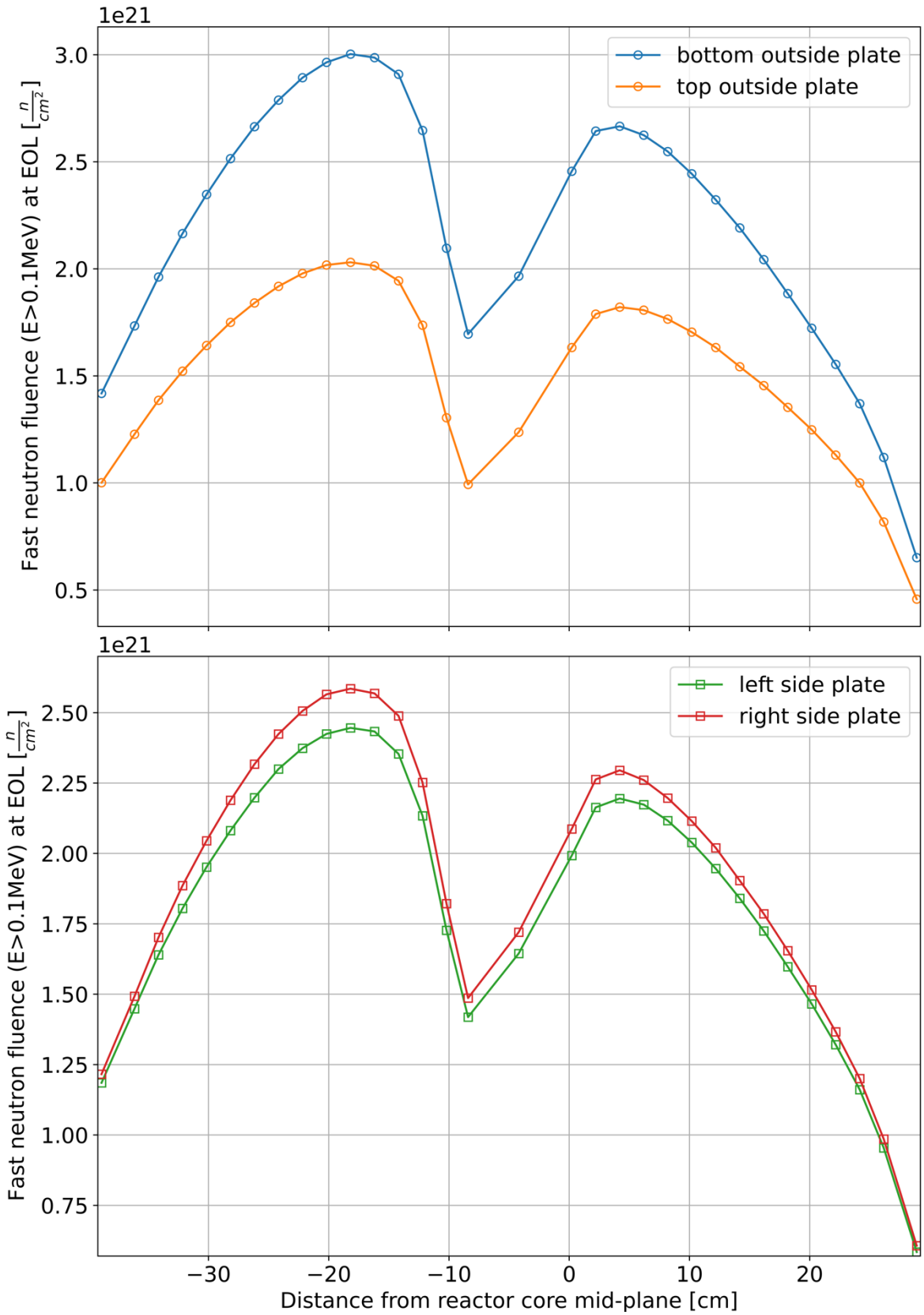

Figure 3.4. Cumulative fast neutron (E>0.1MeV) fluence axial distribution in NBSR DDE outside (top) and side (bottom) plates after ten irradiation cycles (EOL).

4 Conclusions

This technical report supports the conversion of the National Bureau of Standards Reactor (NBSR) at the National Institute of Standards and Technology (NIST) Center for Neutron Research (NCNR) and the Massachusetts Institute of Technology Reactor (MITR) from HEU to LEU. The objective is to provide a fast neutron (E>0.1 MeV) fluence distribution in the cladding of the fuel plates, side and outside (for NBSR DDE) plates throughout the irradiation experiment. It provides complimentary information regarding power and fission density previously supplied [Rykhlevskii, 2022]. The data from the current study is needed to inform the thermo-mechanical analysis of MITR & NBSR DDEs at ANL.

For the MITR DDE, the local peak fluence (E>0.1 MeV) in the cladding of the 19th plate at EOL is 2.608×10^{21} n/cm^2. The maximum local peak fluence in the side plates is observed in the right side plate and is 1.815×10^{21} n/cm^2. For NBSR DDE, the local peak fluences (E>0.1 MeV) in the lower 16th and upper 16th plates are 7.520×10^{21} n/cm^2 and 6.664×10^{21} n/cm^2, respectively. The maximum local peak fluences in the bottom and top outside plates are 3×10^{21} n/cm^2 and 2×10^{21} n/cm^2, respectively. Finally, the maximum local peak fluence in the side plates of NBSR DDE is observed in the right side plate and is 2.585×10^{21} n/cm^2.

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APPENDIX A: Fast neutron fluence in MITR-DDE Cladding

Appendices A.1-A.80 report the fast neutron fluence (E>0.1 MeV) for each cell position in the cladding (see Figure 2.1) during eight irradiation cycles. Fast neutron fluence in the top and bottom edges of the cladding (unfueled sections of the cladding, see Figure 2.1) is reported in A.81-A.88 for eight irradiation cycles.

The text under A.1-A.88 is generated in the comma-separated csv file format, which could be parsed using any spreadsheet software, text editor, or Python. Once the text is imported (copied) to spreadsheet software, the first line represents the plate number as a column title. The fast neutron fluence data listed under the plate number corresponds to axial coordinate (z-coordinate) given in the first column of the spreadsheet.

A.1 "MITR_Cycle1_cladding_fluence_position_0.csv" File


```
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61575e+20,1.666680540388325e+20,1.5901647278156392e+20
−1.847890625,1.2983978763129004e+20,1.4015071138588197e+20,1.4176347964024671e+20,1.5161879415128526e+20,1.5284141934466544e+20,
1.574930570196242e+20,1.589793162995654e+20,1.6083537104320623e+20,1.6712897135124287e+20,1.6614167054385336e+20,1.7639862893173
316e+20,1.770090568502804e+20,1.7996565120359196e+20,1.8028855872572205e+20,1.8362291064601864e+20,1.8308060294446868e+20,1.78641
28802378727e+20,1.6963437985171512e+20,1.6360264427395383e+20
−5.399921875,1.3274949404303201e+20,1.3919437431349125e+20,1.4632222611020915e+20,1.5195851055812896e+20,1.5589709764997307e+20,
1.586670249151492e+20,1.5913767368713054e+20,1.679658768743526e+20,1.6536138442188423e+20,1.7091273976809308e+20,1.7722580299527
183e+20,1.7745404995611994e+20,1.8100072463069384e+20,1.8596288433378322e+20,1.838679664915804e+20,1.8733148142072902e+20,1.81747
```

39298323572e+20,1.766056436171535e+20,1.7005017857884153e+20
−8.951953125,1.2994860304285714e+20,1.3436668568811089e+20,1.4126540584583784e+20,1.5124280594058586e+20,1.5402246466533297e+20,
1.5415782042118478e+20,1.582795358885931e+20,1.6306299060683234e+20,1.632921222458233e+20,1.684418337151909e+20,1.70090873773411
34e+20,1.7488848334193574e+20,1.760491810653184e+20,1.799028390554516e+20,1.812325103041133e+20,1.821764618825044 8e+20,1.83190303
03417863e+20,1.7322705778971578e+20,1.6401048089779483e+20
−12.503984375,1.1108461100346086e+20,1.1968987529869173e+20,1.2390094325852509e+20,1.296504665087261e+20,1.372940856627093e+20,1.
4000827820488761e+20,1.4456260128413594e+20,1.4646465929120353e+20,1.493823278062309e+20,1.5377475478534278e+20,1.5717103417563
696e+20,1.61075118819911e+20,1.6366280238766575e+20,1.6709535358181563e+20,1.680180728847791e+20,1.6774470733864708e+20,1.6437496
829263754e+20,1.631258027549727e+20,1.4790933869843205e+20

A.2 "MITR_Cycle1_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.90055488862467e+19,2.734182534786914e+19,2.786529098585148 2e+19,2.250991125823528 6e+19,2.25895395209847e+19,2.2713
03391140762e+19,2.380618408709028e+19,2.405515965321812e+19,2.424677986528417e+19,2.452085166655299e+19,2.4381515277697327e+19,2.
41799611392213e+19,2.496552558295728e+19,2.572175880187224 5e+19,2.5456313831433142e+19,2.5561247464616374e+19,3.397381918681322
3e+19,3.3491448633874235e+19,3.5869996603184136e+19
37.224453125,3.742951848190339 5e+19,3.498692282916078 4e+19,3.716616486439272e+19,2.827106573778725 3e+19,2.8627170857673163e+19,2.
98436585723091 15e+19,3.095997048167397e+19,3.134598194843622e+19,3.246966587687374 4e+19,3.155030712954239 4e+19,3.22242978945474
27e+19,3.352663564689482 5e+19,3.230774287639119 5e+19,3.352004788517032e+19,3.453984385688881 6e+19,3.576234421119396e+19,4.6073812
10856866e+19,4.554517037208849e+19,4.815083926942428e+19
33.672421875,4.691129438875255e+19,4.480179897368328e+19,4.592109106097113 5e+19,3.515339452226264 3e+19,3.672305946268472e+19,3.71
1623381322679e+19,3.896995676515088e+19,3.940548101249336e+19,3.982500640993346e+19,4.112661218875592e+19,4.035537351257958e+19,
4.072575211620192e+19,4.299753651851622e+19,4.349904295075 02e+19,4.348795878022959e+19,4.415049938795155 5e+19,5.7962369249261306
e+19,5.792106502892510 4e+19,6.205305557724157 5e+19
30.120390625,5.882714368516102e+19,5.336818970376507e+19,5.587363051200655e+19,4.485000465868326e+19,4.512428559524491e+19,4.5178
45163609087e+19,4.779562297453295e+19,4.839547527822565e+19,4.953531490803519 5e+19,5.033938782518765e+19,5.019743724517147e+19,5.
205272871179186e+19,5.154092236638543e+19,5.355797998201876e+19,5.349628507063 05e+19,5.496493766461012e+19,7.183797309106453e+1
9,7.1449504284611396e+19,7.921888041367411e+19
26.568359375,6.754233417609956e+19,6.458202410592774 6e+19,6.813209570191267e+19,5.371169442225727e+19,5.384815520083637e+19,5.510
714966374236e+19,5.593532542339481e+19,5.729888753272158e+19,5.769153904503153 5e+19,5.935050475549694e+19,6.0453693425236e+19,6.1
03038399524651e+19,6.172419032924828e+19,6.290946460142763e+19,6.547555464459317e+19,6.535321049828087e+19,8.608792910705407e+1
9,8.590702707874614e+19,9.500232096442417e+19
23.016328125,7.950800995602751e+19,7.578958444930518e+19,7.712177626470 57e+19,6.053421051297999e+19,6.277718652870536e+19,6.36168
647294641e+19,6.526903354291215 6e+19,6.726888978070926e+19,6.802752805549 19e+19,6.987210133835418e+19,7.031860518857084e+19,7.196
136291384102e+19,7.302220168677967 5e+19,7.432877442880 71e+19,7.534046640792 8e+19,7.713327870581198e+19,1.0274189531425697e+20,1.0
013502388898734e+20,1.0849102451447053e+20
19.464296875000002,9.050852635949185e+19,8.548248243427785e+19,8.788283275786648e+19,6.937958772371238e+19,7.1614721165956194e+1
9,7.259713420407927e+19,7.460640153005423e+19,7.592290819849165e+19,7.809425537641 88e+19,8.009777148184063e+19,8.067237069892272e
+19,8.451680021958 21e+19,8.425590394176 23e+19,8.721935104132673e+19,8.784257421399448e+19,8.877636329653 19e+19,1.19371288124551e+
20,1.185645487324148e+20,1.290845768005757e+20
15.912265625,1.006474053564490 7e+20,9.701315680509544e+19,1.019194707751576 5e+20,8.013384731985 58e+19,8.045434715613544e+19,8.324
473480087352e+19,8.455653592522198e+19,8.692185608725966e+19,8.937605874875498e+19,8.894105733964461e+19,9.171837402858008e+19,9.
289580572727789e+19,9.422642902798 21e+19,9.700845126100 65e+19,9.868623914782768e+19,1.0025632235883543e+20,1.347537117512774e+2
0,1.339098508446618 4e+20,1.457798472281148 1e+20
12.360234375000001,1.138359997612678 7e+20,1.076523919901760 4e+20,1.125775281365939 8e+20,8.676918732348534e+19,9.033337555173707e
+19,9.270706112548 84e+19,9.429753502754823e+19,9.712033864267673e+19,9.814928428345703e+19,1.005857104450608 3e+20,1.019743687895
2856e+20,1.071504672873563 6e+20,1.070464224791676 8e+20,1.101808376806300 9e+20,1.114576086434276e+20,1.132812683970057 3e+20,1.4980
936148293894e+20,1.505873447723094 2e+20,1.635396162962172 8e+20
8.808203125,1.241892424333876 8e+20,1.175774301502027 9e+20,1.213497079948318 5e+20,9.671775320395678e+19,9.909143877770 81e+19,1.001
7737378578774e+20,1.038372414105144 7e+20,1.062083127931052 9e+20,1.085694502492861 2e+20,1.116123687601303 2e+20,1.136064737773742
8e+20,1.150212740334472 4e+20,1.174801822390314 8e+20,1.211363899961334 7e+20,1.240344823166849 5e+20,1.250174181930401 1e+20,1.653183
119618345e+20,1.673537211994146 4e+20,1.806850504415977 8e+20
5.256171875,1.310938441265506 7e+20,1.281162803947194 4e+20,1.297506727082759 8e+20,1.036134666471740 4e+20,1.074746269912607 1e+20,1.
086949314249910 4e+20,1.133042732792182 9e+20,1.148252096964082 9e+20,1.179496909714607 1e+20,1.189132818332280 6e+20,1.209784928500
3811e+20,1.238943616704811e+20,1.277021925148932 2e+20,1.297548554141328e+20,1.336306552287184e+20,1.334398192740005 2e+20,1.763590
8690917078e+20,1.790527494809696 7e+20,1.913744780969603 2e+20
1.704140625,1.40341806776003e+20,1.309364698186874 4e+20,1.380313846283362 5e+20,1.076675542989070 5e+20,1.129382865167456e+20,1.141
810729944564e+20,1.191041177879459 4e+20,1.208927473799731 1e+20,1.236177802456982e+20,1.254241863376168 7e+20,1.298693569869635 4e
+20,1.300528732064319 7e+20,1.367117409305062e+20,1.364884890053978 5e+20,1.413064433142345 6e+20,1.441987844142328 7e+20,1.90572967
08714516e+20,1.894169717559636e+20,2.049829116021584 7e+20
−1.847890625,1.447148257493193 4e+20,1.373663343971001 8e+20,1.403663801729118 9e+20,1.128222164292185 8e+20,1.153574590166899 9e+20,
1.190120982590956 8e+20,1.231001703959154e+20,1.263694778612605 9e+20,1.279139419988952 8e+20,1.317458234019841 5e+20,1.348483454712
882e+20,1.365857369165691 8e+20,1.391110455776306 1e+20,1.424112004986699 3e+20,1.467559862074526 4e+20,1.489315160912366 5e+20,1.9825
60749079108e+20,1.978482610868698e+20,2.129096620390844 7e+20
−5.399921875,1.456433864495357 1e+20,1.373799281911348 8e+20,1.435530791975846 7e+20,1.106153162515083 6e+20,1.162264161584465 2e+20,
1.173881627101812 2e+20,1.247131263419557e+20,1.249829108697212 5e+20,1.283290755551857e+20,1.333713274655949 1e+20,1.346339817961 2
562e+20,1.393520740026304 8e+20,1.409080405813714 5e+20,1.464705165327239 6e+20,1.487443400041434 6e+20,1.523383300116251e+20,1.99864
84814809422e+20,2.014626417854034 7e+20,2.194963780871283 7e+20
−8.951953125,1.402560613059379 9e+20,1.355546999228604 6e+20,1.425141996303943 8e+20,1.105797632517252 9e+20,1.148680824314408 3e+20,
1.182879673076318 7e+20,1.206187801463507e+20,1.242786477710774 1e+20,1.253630142644607 5e+20,1.304235655129935 9e+20,1.331402329670
0501e+20,1.372821574417314 6e+20,1.401750213799619e+20,1.443247884281699e+20,1.481587611841872e+20,1.512957905768101e+20,2.0009541
980845202e+20,2.004284677623021 6e+20,2.167776192801885e+20
−12.503984375,1.296811352234061 8e+20,1.237897940240603 5e+20,1.281199402623441 8e+20,1.033112661490180 3e+20,1.068100995982567 7e+20
,1.094906912142530 5e+20,1.132938165145762 2e+20,1.157072377939674 6e+20,1.179104781040529 2e+20,1.201837787372403 3e+20,1.2324604222
6724e+20,1.297145968702608 2e+20,1.325306035883719 7e+20,1.348352745154855 9e+20,1.382907123914597 3e+20,1.400521543954175e+20,1.881
2085577857827e+20,1.891346391106275 6e+20,2.037003894188078e+20

A.3 "MITR_Cycle1_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.935569365028662e+19,2.737382304767389e+19,2.8579697146198135e+19,2.2375541832584606e+19,2.2766101991966155e+19,2.3
216265709807542e+19,2.3575194156146815e+19,2.397997551544159e+19,2.430063220319086e+19,2.41895290788688e+19,2.5101463523304276e
+19,2.5574684407181447e+19,2.6946768779691495e+19,2.6368823397923934e+19,2.6914091390185005e+19,2.7623582871149883e+19,3.6901399
58365534e+19,3.73489491103362e+19,3.9321513192417485e+19
37.224453125,3.919190159467895e+19,3.701051592269540 6e+19,3.8345426496902865e+19,2.8270333764262306e+19,2.983194699590999e+19,3.1
17930112004153e+19,3.2133480893630992e+19,3.113904257616953e+19,3.267681438443328e+19,3.272010539005148e+19,3.2196587468245926e
+19,3.3822666653912007e+19,3.4483481895468024e+19,3.4854331053499257e+19,3.576527210529374e+19,3.695567019214771e+19,4.916545914
264496e+19,4.892782916615377e+19,5.40708842877126e+19
33.672421875,4.73753656035679e+19,4.5613923599610135e+19,4.680871352761378e+19,3.7465908022857834e+19,3.830924609124128e+19,3.858
78143013062e+19,3.946728049152803e+19,4.035856282579541e+19,4.077855877764441e+19,4.171851735132065e+19,4.196231681895066e+19,4.2
77282064635799e+19,4.380166171949189e+19,4.516736746557027e+19,4.519816263744119e+19,4.580836713812955e+19,6.090176579014898e+1
9,6.268464416162299e+19,6.863140621357181e+19
30.120390625,6.06183874683487e+19,5.536700026509796e+19,5.783897942648481e+19,4.523036947253878e+19,4.624457107517377e+19,4.57375
22557679485e+19,4.704137654089998e+19,4.799534717919661e+19,5.046858115234051e+19,5.076722635051821e+19,5.132880689562092e+19,5.2
499232562008515e+19,5.375404431905769e+19,5.6169034113145225e+19,5.570475376303704e+19,5.626366783315601e+19,7.572684386145272e
+19,7.717719711730872e+19,8.376966438590574e+19
26.568359375,6.98355026621069e+19,6.641561778591584e+19,6.966401172197686e+19,5.339590013006657e+19,5.466901122523936e+19,5.60623
75113796035e+19,5.806641405744997e+19,5.976563831178737e+19,5.944775266666825e+19,6.073811742350048e+19,6.1417284287003345e+19,6.
320329968786998e+19,6.607368158211994e+19,6.674239168098072e+19,6.635862841861652 5e+19,6.811745623141377e+19,9.296168334452297e
+19,9.299043944728868e+19,1.027727427699511e+20
23.016328125,8.271823670114497e+19,7.884348256302357e+19,8.162341344311956e+19,6.332564383418228e+19,6.495899047127 4594e+19,6.661
2204961186865e+19,6.654841869687021 6e+19,6.8789303359667175e+19,7.097424433162902e+19,7.355445100706135e+19,7.522596483509726e+
19,7.411127372425193e+19,7.652678635657155e+19,7.881890916611 47e+19,8.008260917310962e+19,8.22027182042906e+19,1.089192290265635
7e+20,1.0919371909841807e+20,1.2165452268414837e+20
19.464296875000002,9.247805797982691e+19,8.720471157082782e+19,9.00635910239715e+19,7.1444275902290354e+19,7.322976846492488e+19
,7.493474393981544e+19,7.636366082815517e+19,8.053852411150415e+19,8.071262924279472e+19,8.333048027093854e+19,8.5146820289267 2e+
19,8.840567098997031e+19,8.938651551339705e+19,9.031821324300606e+19,9.249897150911108e+19,9.415480019018388e+19,1.2641130491982
889e+20,1.2751083372194323e+20,1.3887733688788022e+20
15.912265625,1.0466803136132743e+20,9.962839364207875e+19,1.0350367061843223e+20,7.992523486524637e+19,8.445771949935437e+19,8.69
0930796968916e+19,8.70107385867173e+19,8.966361977641207e+19,9.259203671442557e+19,9.473149076019438e+19,9.450719315862184e+19,9.
840076947309899e+19,1.0159949377711012e+20,1.0195502377494073e+20,1.0515322524071979e+20,1.0762415872564242e+20,1.4451301019172
726e+20,1.469959689559883e+20,1.6164275919014465e+20
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+19,9.54190230354109e+19,9.886452698497506e+19,1.0073158231181779e+20,1.0240780168394264e+20,1.05375954327596e+20,1.083514267064
9881e+20,1.0932861136230085e+20,1.1264288291560697e+20,1.1486912810790502e+20,1.1743364963637423e+20,1.2189555110914821e+20,1.598
5203824518906e+20,1.6386325316188956e+20,1.7921744352408237e+20
8.808203125,1.2758193972150934e+20,1.2163204064016792e+20,1.2676526640296316e+20,9.922057982103776e+19,1.0178196432011436e+20,1.0
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21753319388e+20,1.8217932210895048e+20,1.9746815769213033e+20
5.256171875,1.3757651536640593e+20,1.3193770503316626e+20,1.3585010352399912e+20,1.0657377671734583e+20,1.083770457798719e+20,1.1
330950166153934e+20,1.1609361524749217e+20,1.1775467231088597e+20,1.2101143165866068e+20,1.2425668856532907e+20,1.2680813913799
569e+20,1.2966231304705044e+20,1.3334518555398971e+20,1.3379325791890268e+20,1.395868283688451e+20,1.4190927579585026e+20,1.91231
743259596e+20,1.9139539162624446e+20,2.130947467732492e+20
1.704140625,1.44723191161033e+20,1.357826573920577e+20,1.419040474135292e+20,1.108966032203802e+20,1.1545052422200446e+20,1.19274
56305161178e+20,1.2128592173051517e+20,1.2521400536831115e+20,1.2698328994575047e+20,1.3061335579124728e+20,1.3367561931667933e
+20,1.3630863265355417e+20,1.4238296723413631e+20,1.4702158802936137e+20,1.4756272559958883e+20,1.5241466439351226e+20,2.0310958
221653053e+20,2.0461744767791792e+20,2.2612596687020474e+20
−1.847890625,1.511614211511594e+20,1.4216232950018848e+20,1.4923789929524947e+20,1.1629752215801266e+20,1.1898229647986577e+20,1.
2363921661321444e+20,1.254670590726494e+20,1.3020292777904577e+20,1.3215677425241773e+20,1.3680114626819596e+20,1.4062413942133
907e+20,1.4230872420517757e+20,1.4753449233505522e+20,1.5134284601769945e+20,1.5578226544649296e+20,1.5959061912913714e+20,2.1170
608842878147e+20,2.1644457132633833e+20,2.3867513011716057e+20
−5.399921875,1.5184843058814383e+20,1.4477129227838654e+20,1.4881596884194168e+20,1.1718843850551756e+20,1.2046349719141586e+20,
1.230306329110456e+20,1.2616557095074008e+20,1.3017208032335166e+20,1.3179601587226611e+20,1.3695329219373816e+20,1.420750155154
2714e+20,1.4445445230973164e+20,1.4792034695034782e+20,1.525699473484471e+20,1.6001098106774864e+20,1.6227434777452606e+20,2.1781
649884737877e+20,2.181871911539404e+20,2.411680228078316e+20
−8.951953125,1.4806465030240852e+20,1.4015358501244564e+20,1.4568103080224719e+20,1.1393743037829602e+20,1.1684075108116 85e+20,1.
2175699897764071e+20,1.2326486443902812e+20,1.2826267509970852e+20,1.314802215800754e+20,1.3581611903891238e+20,1.3864624238929
034e+20,1.4180732234059e+20,1.4796687955300506e+20,1.499285685998586e+20,1.5408565538331607e+20,1.6043238868282432e+20,2.17123215
35160913e+20,2.2018547887704116e+20,2.408768019125498e+20
−12.503984375,1.3336923611266651e+20,1.273430026494379e+20,1.3433282697443384e+20,1.0512394629972197e+20,1.0960519278708377e+20,
1.125367467544899e+20,1.1528426166419544e+20,1.2048911626478893e+20,1.2465247710703164e+20,1.2778846082319034e+20,1.317008593140
2324e+20,1.3513538366071318e+20,1.3619831378658024e+20,1.4295442942182579e+20,1.484238401678638e+20,1.5135591697350204e+20,2.0271
274799836368e+20,2.0804099242173373e+20,2.2745136178858797e+20

A.4 "MITR_Cycle1_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,3.0226689861148373e+19,2.8342642291762258e+19,2.9461777527580484e+19,2.2776820175724286e+19,2.412704991013239e+19,2.
3982066868370006e+19,2.46649458833208e+19,2.5073021123477828e+19,2.4839626136666685e+19,2.5255073395896377e+19,2.63950175933523
3e+19,2.701489460133461 6e+19,2.6889152006513648e+19,2.736488251390492e+19,2.72952927452119e+19,2.761364894473991e+19,3.7168256217
32113e+19,3.7137827032212685e+19,4.07956555878342e+19
37.224453125,3.816834118768931e+19,3.767598442451714e+19,3.8568207867602305e+19,3.087239507779658e+19,3.0668017612867203e+19,3.13
78345635003445e+19,3.2300998263197057e+19,3.2730719006163186e+19,3.3323774412838543e+19,3.444839945009386e+19,3.470349222353730
4e+19,3.550008855397068e+19,3.5105397972555506e+19,3.5975662209892315e+19,3.630881473138887e+19,3.6404284992571027e+19,4.99971902
022757e+19,5.130109646931941e+19,5.499055673798321e+19
33.672421875,4.966121435432579e+19,4.508109915727314e+19,4.8186078566268084e+19,3.6777486922646737e+19,3.906046006312805e+19,3.92
4319202524834e+19,4.02613149146241e+19,4.112880810933075e+19,4.193251503972074e+19,4.266804386464439 5e+19,4.2258818380376736e+1
9,4.342291770415588e+19,4.555452917644315e+19,4.608547140114457e+19,4.7705694798611096e+19,4.749629808665352e+19,6.3450079333422
97e+19,6.471116514925739e+19,7.035938657067493e+19
30.120390625,5.877172283255802e+19,5.59337569086985e+19,5.756187516346979e+19,4.538094688338468e+19,4.715185225934352e+19,4.80281
8142017273e+19,4.831124603903373e+19,4.997324421124535e+19,5.078981296214509e+19,5.246890794454651e+19,5.4095457684621476e+19,5.4
790832533319565e+19,5.53188991477444e+19,5.681107946216871e+19,5.836286333505284e+19,5.8977721096006935e+19,7.95179438824375e+1
9,7.986510846855443e+19,8.781120392006826e+19
26.568359375,7.0451928937757295e+19,6.702001878222785e+19,6.950193187002468e+19,5.4277928227625705e+19,5.565090142513033e+19,5.79
9217102849123e+19,5.83163307323956e+19,6.088503496672166e+19,6.133310733163462e+19,6.240858557507218e+19,6.383907097810823e+19,6.
632359825706557e+19,6.598898178851912e+19,6.9861121735479984e+19,6.932625822403779e+19,7.081164164144472e+19,9.775924696230758e
+19,9.64448316467986e+19,1.0460424509701077e+20
23.016328125,8.207200864626465e+19,7.81334682438266e+19,8.180902101551643e+19,6.446490834193649e+19,6.644332821221734e+19,6.79156
406738217e+19,7.134807366758325e+19,7.043990365841891e+19,7.254014483677996e+19,7.349641596329783e+19,7.573050372907745e+19,7.745
691557148425e+19,8.011241095233954e+19,8.139702448861862e+19,8.304344208151355e+19,8.60236200045053e+19,1.1272130865042255e+20,1.
1426054440573618e+20,1.2640764505220415e+20
19.464296875000002,9.639620769121295e+19,8.990935374550088e+19,9.468338964284083e+19,7.3363092214111355e+19,7.459699044187637e+1
9,7.787936886302414e+19,8.002614264404242e+19,8.1736869339486 1e+19,8.312500484572175e+19,8.771552452359325e+19,8.65846254275527e+
19,8.94623270570521e+19,9.155681701486e+19,9.231179542201791e+19,9.462221756968468e+19,9.625451853031281e+19,1.31931430974381e+20
,1.3365000024330623e+20,1.4402049657708546e+20
15.912265625,1.0475063980199984e+20,1.0186614127548308e+20,1.0565410426707524e+20,8.24338127028805e+19,8.591905235808454e+19,8.93
7815010168339e+19,9.026226955217096e+19,9.175235851366683e+19,9.464469961366515e+19,9.776708953578915e+19,1.0063119737125385e+2
0,1.0142068310173062e+20,1.0294632506300955e+20,1.0481494890454861e+20,1.0823012823665076e+20,1.1069165063339552e+20,1.476740901
4302694e+20,1.513370947971463e+20,1.6344236838504 6e+20
12.360234375000001,1.1723183407878216e+20,1.1241649396110603e+20,1.1892007873024539e+20,9.375848881024918e+19,9.70335474961475e+
19,9.841697745829418e+19,1.0053238094538626e+20,1.0421682196702185e+20,1.0580886438377796e+20,1.0836345198583721e+20,1.118784934
2027117e+20,1.145789528890874e+20,1.154154940604535e+20,1.1897654525931261e+20,1.2124566318664316e+20,1.247978261355565e+20,1.637
9685270641236e+20,1.669134914079832e+20,1.8366941107044637e+20
8.808203125,1.295363090331134e+20,1.2331035136522114e+20,1.282323504822465e+20,1.0293273126897485e+20,1.0628621568968874e+20,1.07
60219951989406e+20,1.1050447454630234e+20,1.1260105585703867e+20,1.1712988062352192e+20,1.2007136851733799e+20,1.20968036085396
05e+20,1.2413173022785623e+20,1.2657547612470949e+20,1.2961787179732 16e+20,1.3256563749922 94e+20,1.3619988230127655e+20,1.840066
417301533e+20,1.843036138459883e+20,2.0455470709006546e+20
5.256171875,1.3772029588023447e+20,1.3229218935453265e+20,1.3980903461748921e+20,1.0896262459982817e+20,1.118518286704339e+20,1.1
87271514225991e+20,1.186319948643562e+20,1.2275980270681584e+20,1.2315140854266159e+20,1.2766193397102123e+20,1.309422210392405
6e+20,1.3470351928099543e+20,1.3804131855474621e+20,1.4370313377019847e+20,1.4463274014687904e+20,1.4753606084975154e+20,2.011290
7099332126e+20,2.0046454360031733e+20,2.2309873350632364e+20
1.704140625,1.4675128066336372e+20,1.4152446685702185e+20,1.4637954268033789e+20,1.1473789571164696e+20,1.188406073189656e+20,1.2
3362635188431 54e+20,1.2578233052660802e+20,1.2841586670171493e+20,1.3158688057 94246e+20,1.3690362256168 83e+20,1.401561992036061
5e+20,1.4284881609894083e+20,1.4448948247128257e+20,1.5174072591233044e+20,1.5347027478412988e+20,1.553890910959509e+20,2.1343929
716820564e+20,2.1349419518257653e+20,2.3544921822508e+20
−1.847890625,1.5351942157794764e+20,1.4548496646520824e+20,1.5216317920387033e+20,1.1979269573962668e+20,1.2253341375231487e+20,
1.268787222993297e+20,1.3140493287465247e+20,1.3506270914645077e+20,1.3813542943652487e+20,1.4252831626266116e+20,1.452465522313
6892e+20,1.4793446358261465e+20,1.5330505790278507e+20,1.5701564083602584e+20,1.6004235136167484e+20,1.6334616615033887e+20,2.210
157459896221e+20,2.242139474554011e+20,2.471153076980125e+20
−5.399921875,1.5375835865001 91e+20,1.4880655775376382e+20,1.5433086651417274e+20,1.2201162119667525e+20,1.2804412871868912e+20,1.
2919750985871013e+20,1.3063217796760302e+20,1.3637137324140662e+20,1.3719902616282697e+20,1.4340616165436349e+20,1.464422832681
9036e+20,1.4981354418879575e+20,1.5500585067181877e+20,1.5863696219379 98e+20,1.6204795882002 51e+20,1.6520747025662847e+20,2.26400
45694205926e+20,2.2778702393359863e+20,2.4949108462469225e+20
−8.951953125,1.5036461568543 32e+20,1.4477495214601126e+20,1.5057061394888 21e+20,1.1850337665925865e+20,1.222516039452109e+20,1.26
722291650616277e+20,1.3133539538978265e+20,1.3215416006125722e+20,1.3601950311120075e+20,1.4005999696889904e+20,1.43442237492378
65e+20,1.4708955699953487e+20,1.5058943612523784e+20,1.5696283417458337e+20,1.600193464794623e+20,1.6514943521286495e+20,2.254457
543302377e+20,2.255853521382094e+20,2.4909163621536494e+20
−12.503984375,1.3448863276760075e+20,1.3135421756613838e+20,1.3622341002172124e+20,1.1037794769413323e+20,1.1343655135194055e+20
,1.170028309331207e+20,1.2058165863187133e+20,1.2573475224748655e+20,1.2764102044173708e+20,1.3299436110024807e+20,1.36856567120
79897e+20,1.3913143626868267e+20,1.4476815524899393e+20,1.4766311054015277e+20,1.510469195783287e+20,1.5582984372561438e+20,2.101
4175643832684e+20,2.132683290663077e+20,2.3213233248061345e+20

A.5 "MITR_Cycle1_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.806423093316698e+19,2.904591199776511e+19,2.9717131720139993e+19,2.3029141906537587e+19,2.4372888446867608e+19,2.4
731503190267613e+19,2.556867176751225e+19,2.6026050652956668e+19,2.714016664174669e+19,2.64381517475009e+19,2.758562481549915e+
19,2.771853029409994e+19,2.895284679245063e+19,2.8305468493459685e+19,2.8751658640737083e+19,2.8892302125173006e+19,3.7890138964
386865e+19,3.866827910522697e+19,3.995179468121864e+19
37.224453125,3.697506749055876e+19,3.667793852325416 6e+19,3.7423558126057415e+19,3.1077399948604494e+19,3.2597552108446343e+19,3.
2851285502486323e+19,3.304128491603285e+19,3.3959702554546414e+19,3.493751461622698e+19,3.583731921367765e+19,3.662602068680625
8e+19,3.6308501028449608e+19,3.7497016897668006e+19,3.5735784029003084e+19,3.775137769758651e+19,3.745811773319949e+19,5.13193435
2361984e+19,5.305553244096699e+19,5.405781333191e+19
33.672421875,4.711107087723942e+19,4.622553976352519e+19,4.828505184360533e+19,3.858373616309579e+19,3.993229281516116e+19,4.0674
82767239501e+19,4.161259032549 6414e+19,4.296501598047936 5e+19,4.395150715681284e+19,4.467171682153585e+19,4.4850788916031414e+1
9,4.608202066881268e+19,4.7164923215146115e+19,4.757733801262961e+19,4.72499889955094e+19,4.8161034614950306e+19,6.4573135855981
98e+19,6.452921744448526e+19,6.980099533878805e+19
30.120390625,5.563835330755985e+19,5.499212525267953e+19,5.826509258564943e+19,4.662514502432213e+19,4.745781719277 0675e+19,4.863
033421208669e+19,5.137554863357101e+19,5.292691423586945 4e+19,5.507421085511982e+19,5.451059124091190 5e+19,5.491840506195289e+1
9,5.65094018022448e+19,5.714621876894725e+19,5.85625875397165e+19,5.9368804093620584e+19,5.9656887959509795e+19,7.81434021702365
7e+19,7.776852715781813e+19,8.580193659409328e+19
26.568359375,6.692956776807389e+19,6.744419423553563e+19,6.998712574941701e+19,5.469567597507666e+19,5.703537706374125e+19,5.8675
520597850915e+19,6.169020584416154e+19,6.244309289839102e+19,6.436504623960465e+19,6.52047244403634e+19,6.641457210945162e+19,6.9
12548834290991e+19,6.926926885673847e+19,7.031076261508927 5e+19,7.23817248524525e+19,7.216526982436152e+19,9.559940222548673e+1
9,9.680611286518235e+19,1.0332381426658853e+20
23.016328125,7.854964747658124e+19,7.451333632473976e+19,8.21608911457223e+19,6.592048998011352e+19,6.85759853609688e+19,6.965146
360440636e+19,7.217834078016412e+19,7.413114157707187e+19,7.513080827685437e+19,7.74631896302695e+19,7.842678049203683e+19,8.0032
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38597980562938e+20,1.2333492476213004e+20
19.464296875000002,9.00374491123663e+19,8.854892866556676e+19,9.453751777608386e+19,7.508009296834029e+19,7.91472515758759e+19,8.
005908145266496e+19,8.273339900987597e+19,8.566129310965735e+19,8.682303966139205e+19,8.919045117635812e+19,9.095659872440482e+
19,9.289685140374209e+19,9.6401436073534e+19,9.755011166946607e+19,9.87343402651812e+19,9.892203919050647e+19,1.3132859849276529e
+20,1.3037337304271163e+20,1.430804334357628e+20
15.912265625,1.030200452537362e+20,1.024287152132268e+20,1.0626216513101198e+20,8.515100299512403e+19,8.83805747548293e+19,9.1799
41395455617e+19,9.468338964284083e+19,9.600407901713508e+19,9.840704353188422e+19,1.0172602062927926e+20,1.0314604926767322e+20,
1.0538536541577388e+20,1.0694080915628273e+20,1.0898824367320126e+20,1.1032566387092283e+20,1.131840204858344e+20,1.488473391358
679e+20,1.4916417910452283e+20,1.636776455894927e+20
12.360234375000001,1.1270666917992363e+20,1.1194437103751625e+20,1.186325177025883e+20,9.480573378915313e+19,1.0007228330113488e
+20,1.0255262787423538e+20,1.0637562102737848e+20,1.1006738178426354e+20,1.1161289159836246e+20,1.1350452032211404e+20,1.1710792
141777356e+20,1.1907902155280495e+20,1.205058470082163e+20,1.2298305463192415e+20,1.2465195426879957e+20,1.2613420065681387e+20,
1.6603826020744144e+20,1.6991406002202703e+20,1.8544862957429565e+20
8.808203125,1.2359634387818193e+20,1.2342223874689137e+20,1.306008076736768e+20,1.0401605208589397e+20,1.0963813159570632e+20,1.1
212109035996735e+20,1.1510754234174436e+20,1.183193376015581e+20,1.2087810790947417e+20,1.257148843946666e+20,1.2696080790177e+
20,1.2983380398718047e+20,1.3219807847275395e+20,1.351766878810494e+20,1.3784525421770729e+20,1.3956539200132884e+20,1.867724559
7798253e+20,1.8690943959479373e+20,2.0406689901951258e+20
5.256171875,1.3430982209222127e+20,1.328599916745974e+20,1.4082490930246694e+20,1.135385048072008e+20,1.1589545955752482e+20,1.22
255263812835 66e+20,1.2462581235719437e+20,1.2856696695079294e+20,1.3252642088251517e+20,1.3470665631038808e+20,1.37883421408650
87e+20,1.3925116622383443e+20,1.4315676781764996e+20,1.4764742539318965e+20,1.477316023485584e+20,1.507353079919948e+20,2.0049748
240893985e+20,2.029888065849145e+20,2.189531491639725e+20
1.704140625,1.4428923542838682e+20,1.4069995096499413e+20,1.5031546889161543e+20,1.1815882626430222e+20,1.2428073912400583e+20,1.
2917032227064072e+20,1.3339537802427168e+20,1.3844860953755507e+20,1.4004117479254334e+20,1.444382443245364e+20,1.4704041020571
71e+20,1.494888616466593e+20,1.5207325102794844e+20,1.5557574434481193e+20,1.5827881800478867e+20,1.6252949283179273e+20,2.142261
687075219e+20,2.123266974102887e+20,2.3448510452508063e+20
−1.847890625,1.4538405868641223e+20,1.4433106248695515e+20,1.5181444610305707e+20,1.2430688103561103e+20,1.2676840343235582e+20,
1.3357732572904382e+20,1.3614707563983407e+20,1.4132369697589399e+20,1.4452869533869038e+20,1.4779329725994664e+20,1.51295267738
57796e+20,1.5479410118781675e+20,1.594468386153086e+20,1.6180849890972156e+20,1.6445667455532738e+20,1.674190759784276e+20,2.2427
616520502143e+20,2.2272594984683363e+20,2.425849441683294e+20
−5.399921875,1.482403239483954e+20,1.4563554387605417e+20,1.552421735527297e+20,1.2554182493984026e+20,1.3267386126396842e+20,1.3
550084758495378e+20,1.3979544082345453e+20,1.4550849418565293e+20,1.4661220569362409e+20,1.5054604055197322e+20,1.5433557205826
17e+20,1.568723831604294e+20,1.6146865405885406e+20,1.6590807348764757e+20,1.6856984292728812e+20,1.7056917632685313e+20,2.264255
5317720026e+20,2.2843586617963948e+20,2.4739136628456332e+20
−8.951953125,1.4661952542887354e+20,1.4401788238592495e+20,1.5388906820804503e+20,1.2401304594916866e+20,1.3044604755697407e+20,
1.3559025292264351e+20,1.372466044419484e+20,1.4010182402746735e+20,1.463784970038737e+20,1.4931423367713663e+20,1.5286273675842
522e+20,1.5542568977219813e+20,1.5919221639627404e+20,1.632421213421502e+20,1.6694747589306997e+20,1.672230116413887e+20,2.242149
931318653e+20,2.2685166633636486e+20,2.522087977551679e+20
−12.503984375,1.3115031065561791e+20,1.3239205145686447e+20,1.392882877383138e+20,1.1267320753306 9e+20,1.1865343123187245e+20,1.2
474397379764981e+20,1.2918809877053227e+20,1.309887536418978e+20,1.3548725379091907e+20,1.3863630846288036e+20,1.41022542154202
19e+20,1.436942455202527e+20,1.4930377691249456e+20,1.529260001845098e+20,1.5696649404220808e+20,1.6002666621471172e+20,2.1129043
203425894e+20,2.0975276479364163e+20,2.302532518744323e+20

A.6 "MITR_Cycle1_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.1962754170259374e+19,2.6001044488799556e+19,2.701391249451661e+19,2.6734973476084797e+19,3.0183537344619745e+19,3.
063428085838762e+19,3.1454642900226585e+19,3.2418057397759885e+19,3.262976087733722e+19,3.494478664147417e+19,3.436603020043993
5e+19,3.6241105524151435e+19,3.6601346463908614e+19,3.601772429308885e+19,3.6475722167627866e+19,3.603780648693091e+19,3.47611274
59024294e+19,3.4759888909624345e+19,3.37449861441788e+19
37.224453125,2.892579042896952e+19,3.3646256063439847e+19,3.466398979894242e+19,3.7346245460164846e+19,3.862911723507121e+19,4.10
6587471166049e+19,4.310346541020843e+19,4.3892863717048615e+19,4.455380675755102e+19,4.5653815560336065e+19,4.641685045852015e+
19,4.5455382252901065e+19,4.676700606744914e+19,4.817797923743617e+19,4.7737940329196446e+19,4.770945369299757e+19,4.75770173750
1709e+19,4.748969964232055e+19,4.580810342844424e+19
33.672421875,3.889540535606068e+19,4.2269567792784286e+19,4.346821820849399e+19,4.8225840324962845e+19,5.156903902668767e+19,5.24
8724647112276e+19,5.344464515728488e+19,5.383045329536962e+19,5.640716685415335e+19,5.779301516488421e+19,6.022897643114495e+19,
5.803833641388879e+19,6.102810619755619e+19,6.096609025974434e+19,6.159182311016244e+19,6.150140900396603e+19,6.185386477606635e
+19,6.129041326690295e+19,5.926529653016881e+19
30.120390625,4.718094697047563e+19,5.022008179451249e+19,5.48476562239865e+19,5.783459503759686e+19,6.116700066597926e+19,5.91140
1656774622e+19,6.620170397678e+19,6.793549620108281 4e+19,6.957321237907514e+19,6.940875071232450 6e+19,7.040454442298851e+19,7.1478
27829182833e+19,7.467249719430 19e+19,7.359885180017297e+19,7.702388323010884e+19,7.3711648263382794e+19,7.616990341884264e+19,7.4
12320053542365e+19,7.255758562607133e+19
26.568359375,5.476670817391827e+19,6.145664428993923e+19,6.364931906692388e+19,6.847629994978999e+19,7.253237229900089e+19,7.6516
43184738607e+19,7.8207936456462e+19,7.91723240999524e+19,8.053570158585558e+19,8.612713285193428e+19,8.585031706104526e+19,8.5005
09556339297e+19,8.762577762587499e+19,9.057334826212976e+19,9.422176092312822e+19,9.363787334886562e+19,9.05193828954176e+19,9.17
3935405436933e+19,8.791789834863483e+19
23.016328125,6.293458759533787 5e+19,6.885449985584645e+19,7.644326896497467e+19,8.176832364224964e+19,8.478012191167332e+19,8.765
992620218792e+19,9.13907908660974e+19,9.227900772149084e+19,9.885216632265925e+19,9.769412263370511e+19,1.0224225296595368e+20,1.
0316585694705998e+20,1.0363119765018442e+20,1.0508030044812706e+20,1.1044056531548635e+20,1.0859955010027351e+20,1.096408162743
7515e+20,1.0608087142765889e+20,1.0576504133067139e+20
19.464296875000002,7.062801412876197e+19,8.167083211091065e+19,8.865802008291985e+19,9.354852085644059e+19,1.0187511153668248e+2
0,1.0157609032440861e+20,1.0437167325572653e+20,1.1017781590706817e+20,1.1276019140596597e+20,1.1658377033924323e+20,1.177807398
6648156e+20,1.2012248291053176e+20,1.2218909105216425e+20,1.2468742212749396e+20,1.252005354503308e+20,1.2899668936118054e+20,1.2
755554866652332e+20,1.2390713600552483e+20,1.2137695651705355e+20
15.912265625,8.603415317912368e+19,9.309291161288719e+19,9.839124901024893e+19,1.0698147377705183e+20,1.1191443910142802e+20,1.19
23338137699552e+20,1.1893347548657882e+20,1.25577408339173e+20,1.2693008121954805e+20,1.2915681610503135e+20,1.3429414208039946
e+20,1.3993838863160467e+20,1.4133441074097799e+20,1.392085291637764e+20,1.3974110540575531e+20,1.4491116447240787e+20,1.41882026
51138487e+20,1.4080448853342754e+20,1.4057447221629377e+20
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20,1.289949200048949e+20,1.3328295496315376e+20,1.4161839242482388e+20,1.4357353112046081e+20,1.464115786026342e+20,1.4923900994
709373e+20,1.5260609495867474e+20,1.5744705375619747e+20,1.622296237962939e+20,1.5806986716874424e+20,1.6073274837863884e+20,1.62
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3782181141278e+20,1.4899130006710354e+20,1.525078956848215e+20,1.585555554691536e+20,1.646447951261983e+20,1.669502663663928e+2
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3105e+20,1.7687104706000018e+20,1.694167490285809e+20
5.256171875,1.0516611422798081e+20,1.1890251175158004e+20,1.3024850893327388e+20,1.4225978407836993e+20,1.4790403062957513e+20,1.
5376944671648586e+20,1.5822291648745246e+20,1.644669748194911e+20,1.6879924368489107e+20,1.7640216764430444e+20,1.7489644544522
12e+20,1.8036817975857604e+20,1.8524894907252572e+20,1.887956237470997e+20,1.9271209388537322e+20,1.9018899182204453e+20,1.949034
4164514365e+20,1.9236972344410105e+20,1.8512420945438782e+20
1.704140625,1.1585391087141498e+20,1.3008749751128025e+20,1.399083095747487e+20,1.4843218348083993e+20,1.5598821949868376e+20,1.6
152011192575056e+20,1.740736947723966e+20,1.7501587699450216e+20,1.7862094042650234e+20,1.8640433872705143e+20,1.88147154668411
03e+20,1.92678476115946e+20,1.950078336659967e+20,1.984837340891448e+20,2.0201094584457665e+20,2.0077505047905416e+20,2.021976129
3271214e+20,1.9727260971162134e+20,1.971540628404 8315e+20
−1.847890625,1.1796917631090269e+20,1.331396371040166e+20,1.440494879612991e+20,1.5466562567516468e+20,1.597312927209642e+20,1.69
2663537443011e+20,1.7683919864685855e+20,1.8177127929309194e+20,1.86938684325316e+20,1.905632106764583e+20,1.9513699667484872e+
20,1.9963823906552768e+20,2.0147217685559797e+20,2.124837656993050 8e+20,2.093077711665736e+20,2.101526387929687e+20,2.0820103880
990314e+20,2.0608400401412974e+20,2.0312121691381842e+20
−5.399921875,1.223633726463002e+20,1.324531268651866 2e+20,1.4773594178243894e+20,1.561775406212477e+20,1.6429446258164076e+20,1.7
25387781946002e+20,1.7606510527188923e+20,1.8380603902158288e+20,1.887611212995296e+20,1.9189553595954843e+20,2.017729674241575
e+20,2.0452166241390585e+20,2.0431287837219983e+20,2.132419348677036e+20,2.1517938000048407e+20,2.1488566685706712e+20,2.17810412
79723707e+20,2.1062417224309293e+20,2.111868275419278e+20
−8.951953125,1.1738705809292573e+20,1.316330302267905e+20,1.4326123973604458e+20,1.5274056603638373e+20,1.6141218119232625e+20,1.
6726432710709464e+20,1.7319786341100267e+20,1.819181358648004e+20,1.8951752111164254e+20,1.9038804440417947e+20,1.9216447811496
637e+20,1.9913220316783342e+20,1.9852442928371463e+20,2.0708988306251848e+20,2.133728672328413e+20,2.1323928083327513e+20,2.12144
93397060413e+20,2.103224969963906e+20,2.084399019084651e+20
−12.503984375,1.0437167325572653e+20,1.1441807824561465e+20,1.2734234123410317e+20,1.3513016292536638e+20,1.4501644117140369e+20
,1.540268880560471e+20,1.5597141061397014e+20,1.6247025625114152e+20,1.697776977108523e+20,1.7519989004820916e+20,1.765676024570
1214e+20,1.838679664915804e+20,1.894175524815036e+20,1.9175664149112534e+20,1.899191649884838e+20,1.9436024926545088e+20,1.930181
925227897e+20,1.880294924754157e+20,1.903836210134654e+20

A.7 "MITR_Cycle1_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.890715073096476e+19,2.888885139284112e+19,3.0185228789342536e+19,2.2975655555393364e+19,2.459922511754535e+19,2.47
3877064169386e+19,2.5695303187327795e+19,2.6690368910667776e+19,2.64291066460855e+19,2.6741031935358644e+19,2.7379835687343084e
+19,2.83881292179553e+19,2.8495049636420526e+19,2.849248772908322e+19,2.8234780764479242e+19,2.841327773691949e+19,3.80507548692
8916e+19,3.735511860147502e+19,4.056471794071395e+19
37.224453125,3.91795626124013e+19,3.6850266004555583e+19,3.82599947297771e+19,3.123806813733e+19,3.264497353609816e+19,3.33579680
33218134e+19,3.361938714927004e+19,3.361577956546853e+19,3.4791224478884327e+19,3.6349125559084077e+19,3.6456307396665352e+19,3.6
834319438476415e+19,3.722660496402391e+19,3.6199070986470277e+19,3.760911341463106e+19,3.808829465435421e+19,5.1977858276954595
e+19,5.002422093887548e+19,5.383194721564115e+19
33.672421875,4.789318458864352e+19,4.609169317610661e+19,4.917947120726534e+19,3.952029628826336e+19,3.935544539368102e+19,4.1451
29473089239e+19,4.172860812920025e+19,4.303638339916153e+19,4.480577254424727e+19,4.421611558608058e+19,4.606816545566193e+19,4.5
88015282739741e+19,4.6670475099045536e+19,4.744333457374139e+19,4.921146890707009e+19,4.853355685532427e+19,6.514668939659987e+
19,6.308670676211083e+19,6.948938375245418e+19
30.120390625,5.9354687461353775e+19,5.5814549791778824e+19,5.9049872772037255e+19,4.720052849875239e+19,4.877479441561698e+19,5.0
29003189607705e+19,5.178932281045796e+19,5.236120326873312e+19,5.375404431905769e+19,5.4687310563363e+19,5.593741677632323e+19,5.
718961434221188e+19,5.7880283646821016e+19,5.901170558109367e+19,5.868127181840406e+19,5.9442001446115115e+19,7.881472646025788
e+19,7.814444784670078e+19,8.662331545672838e+19
26.568359375,6.904706260809435e+19,6.820215602501457e+19,7.187038906145497e+19,5.67702980800646e+19,5.8004719146061726e+19,6.1276
640802567406e+19,6.203214204795743e+19,6.293456083656862e+19,6.4588298164713e+19,6.5858795068725264e+19,6.673611762219548e+19,6.9
0664076226822e+19,6.930534469475363e+19,7.161001562186727e+19,7.132663730006699e+19,7.105057871351618e+19,9.508231521393603e+19,
9.548856052028072e+19,1.020444291126305e+20
23.016328125,8.034455112739363e+19,7.819568599344695e+19,8.285835734734879e+19,6.748534480880025e+19,6.815457774589313e+19,6.9303
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0,1.1136454343811313e+20,1.2206233650518934e+20
19.464296875000002,9.230813555439318e+19,9.098326347424211e+19,9.51994309779273e+19,7.609387630038961e+19,7.96826379255502e+19,8.
158576909040081e+19,8.359085371052623e+19,8.71205346154591e+19,8.814320619745418e+19,9.044108022755046e+19,9.15291065885585e+19,9.
367222050195205e+19,9.514819283118113e+19,9.792237249072398e+19,9.910032702765389e+19,9.931835057044118e+19,1.3086588665735343e
+20,1.3185300523956542e+20,1.4051329771613305e+20
15.912265625,1.0627105338095773e+20,1.0231839634625289e+20,1.0809523597276796e+20,8.678382679398425e+19,8.93656019841129e+19,9.29
4076981523882e+19,9.47884801274937e+19,9.80750412544983e+19,9.954787655433475e+19,1.0396010839505887e+20,1.04333414892781e+20,1.0
548993306219463e+20,1.0839952782385237e+20,1.098964136823656e+20,1.1340622673447852e+20,1.1407493683333931e+20,1.47119358778764
79e+20,1.4869676172502204e+20,1.6230258103905963e+20
12.360234375000001,1.1684807081641796e+20,1.1459568371251472e+20,1.2063707948447433e+20,9.692375146740567e+19,1.0193149605449605
e+20,1.0214376837673022e+20,1.0669873505481867e+20,1.1003235162271259e+20,1.123490478291646e+20,1.1355889549825286e+20,1.16622204
70014912e+20,1.2035892954499511e+20,1.2285025372096979e+20,1.2385148893544861e+20,1.2673389610903696e+20,1.2770219251489322e+20,
1.6686277609946916e+20,1.656199896217584e+20,1.7998392437234654e+20
8.808203125,1.2790139388132477e+20,1.2504617429580584e+20,1.3317526312855598e+20,1.0605250699993833e+20,1.1256602569548769e+20,1.
1396670931929383e+20,1.1886884058349922e+20,1.1953964203528842e+20,1.2384103217080653e+20,1.2607825696597875e+20,1.264395381843
6251e+20,1.3104992571505397e+20,1.3198528331228768e+20,1.3548202540859803e+20,1.3818666758327106e+20,1.4037369990816134e+20,1.864
1640314191983e+20,1.857952713221805e+20,2.0187463831230126e+20
5.256171875,1.3757965239579854e+20,1.3607178693441113e+20,1.4291207952502537e+20,1.1475305802037797e+20,1.1839567198344528e+20,1.
2149035147926775e+20,1.2622569734743204e+20,1.2966701859113936e+20,1.3204540970897963e+20,1.3617792309552821e+20,1.382028755684
663e+20,1.3999045948402924e+20,1.4370417944666264e+20,1.480928835669421e+20,1.4685061992746343e+20,1.5314088669790444e+20,2.00073
4606027036e+20,2.0308814584901426e+20,2.173496043061101e+20
1.704140625,1.474335845562592e+20,1.403282129819683e+20,1.510918836662896e+20,1.2098267555589495e+20,1.249897077667386e+20,1.2959
016137102012e+20,1.3510558188148326e+20,1.3749390692573353e+20,1.413247426523582e+20,1.4431590017822414e+20,1.4837992175636713e
+20,1.5062028358093196e+20,1.541170256772423e+20,1.565424722359719e+20,1.5880165623689252e+20,1.5936057030701148e+20,2.130612851
2639458e+20,2.1242289964499583e+20,2.2943291868826144e+20
−1.847890625,1.5015809458375216e+20,1.487098326808246e+20,1.5546594831607015e+20,1.2488304876738943e+20,1.2871859003810302e+20,1.
3480233570686306e+20,1.392307755327824e+20,1.4324931018473233e+20,1.4689140130956752e+20,1.4981302135056368e+20,1.5360516704801
266e+20,1.567725210580976e+20,1.6177869713049164e+20,1.6272660284529584e+20,1.639254709115099e+20,1.6857454847137707e+20,2.185798
4266625034e+20,2.206037494627242e+20,2.3920005970219283e+20
−5.399921875,1.5437321641097318e+20,1.4682186382469772e+20,1.5947402620337802e+20,1.259747349960222e+20,1.353816404680341e+20,1.3
759063199867272e+20,1.4078987914091597e+20,1.4519740543755115e+20,1.4900262209080276e+20,1.5229336592366417e+20,1.5543666937507
234e+20,1.5941808251254 29e+20,1.6314121356335 42e+20,1.6635666869079268e+20,1.6994333896302484e+20,1.707856313549441e+20,2.2815510
204899967e+20,2.2515191924379537e+20,2.4540562667903304e+20
−8.951953125,1.504195136998041e+20,1.4795903697952352e+20,1.5655815738293502e+20,1.2547228745497043e+20,1.3094744942156163e+20,1.
3569011502497535e+20,1.3830692037665495e+20,1.4317924986163043e+20,1.4633405575414486e+20,1.4912392056065085e+20,1.539340322960
0596e+20,1.543512572052248e+20,1.5973701383412623e+20,1.6286201794741073e+20,1.6535595631454595e+20,1.69087975615303e+20,2.234955
6772449046e+20,2.2336904087232132e+20,2.452576634593477e+20
−12.503984375,1.370113272375017e+20,1.354381069971013e+20,1.4180836801705424e+20,1.148001134612673e+20,1.2039395970654606e+20,1.2
558103780724803e+20,1.3017469451451218e+20,1.3172177284330738e+20,1.353121029831643e+20,1.3905196485740288e+20,1.42131482044494
38e+20,1.443545902073998e+20,1.504357216849993e+20,1.531618002271886e+20,1.572801969814704e+20,1.6076752798960284e+20,2.118608485
4548418e+20,2.1243910763019102e+20,2.2782205409514953e+20

A.8 "MITR_Cycle1_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9890348026435985e+19,2.8687663241127576e+19,2.9870271038323196e+19,2.355412377539303e+19,2.386191864263255e+19,2.4
34151815294138e+19,2.445105276256713e+19,2.589481825669861e+19,2.5591833501194445e+19,2.5790459745570693e+19,2.61465648654566e+
19,2.7093215768503775e+19,2.744421028843307e+19,2.759404251103602e+19,2.8332080959473766e+19,2.8089222600661545e+19,3.7897354131
989905e+19,3.810005851457656e+19,4.081484375095241e+19
37.224453125,3.977324542495519e+19,3.734779886622558e+19,3.923582000617567e+19,3.1052721984049197e+19,3.114625774377256e+19,3.184
4194499807953e+19,3.280234784396140 5e+19,3.3330571309855887e+19,3.4038860262886937e+19,3.445446437358626e+19,3.559336289457801e
+19,3.5366398818021736e+19,3.494687342058164e+19,3.589002130747371e+19,3.6396076432327e+19,3.6965185847972e+19,5.024114652137534e
+19,5.029400546664104e+19,5.357209661428555e+19
33.672421875,5.012073687652183e+19,4.674226078831338e+19,4.808156320367053e+19,3.757361269867122e+19,3.915922420517246e+19,3.9987
66138394096e+19,4.04743714942064e+19,4.091156882389161 6e+19,4.3304913115170054e+19,4.257916136518674e+19,4.292867872348144e+19,
4.349559221841831e+19,4.66219557111063e+19,4.593824015498414e+19,4.710516280521665e+19,4.768828428548204e+19,6.288645971921507e+
19,6.420453490234878e+19,7.245230801378651e+19
30.120390625,6.108580484784952e+19,5.757180908987976e+19,5.870218534768822e+19,4.590425566989739e+19,4.799116447333978e+19,4.8336
76054476041e+19,4.878148674498791e+19,4.995912757897854e+19,5.170634838302309e+19,5.272353016358106e+19,5.397834192063022e+19,5.4
3997495357059e+19,5.761154479551966e+19,5.560123179308047e+19,5.849514140777 5105e+19,5.880623015587687e+19,7.858206344697166e+1
9,8.026664823081017e+19,8.684918157299723e+19
26.568359375,7.137055571156371e+19,6.767618076351814e+19,7.049950721687876e+19,5.460783915208322e+19,5.5353406471063265e+19,5.693
551496140942e+19,5.906921778662511e+19,6.007829557458547e+19,6.189777262230675e+19,6.389501466894333e+19,6.5259099616502194e+19,
6.653848477046023e+19,6.689506044475503 4e+19,6.870042086020951e+19,7.00394095726274e+19,7.095333080234486e+19,9.617347860433671e
+19,9.764631390417315e+19,1.018692783048757e+20
23.016328125,8.331897782983225e+19,7.970511996953068e+19,8.261942027527735e+19,6.58211507160138e+19,6.762598829323617e+19,6.84970
3678792113e+19,7.019050982170540 4e+19,7.211873722170428e+19,7.230329911763692e+19,7.397481294567283e+19,7.572527534675642e+19,7.6
39137125445668e+19,8.016940031963885e+19,8.074190818379253e+19,8.228898651258774e+19,8.5639333903909e+19,1.1394056740768863e+20,
1.1444824333106143e+20,1.2297939476429944e+20
19.464296875000002,9.703929871670062e+19,9.075007762272382e+19,9.413545517559603e+19,7.480717141118211e+19,7.620733219675613e+19
,7.954879133813162e+19,8.097979957939978e+19,8.15036834879678e+19,8.31741516395395e+19,8.87209424439289e+19,8.733228409946117e+19
,8.907960947115211e+19,9.278130415444713e+19,9.326179248975054e+19,9.489775331800341e+19,9.754697464007342e+19,1.310922756118543
6e+20,1.3298129014444548e+20,1.4311337224438537e+20
15.912265625,1.0918587652493653e+20,1.0404323967396338e+20,1.0846069989700852e+20,8.357987410765205e+19,8.90356910596554e+19,9.09
2052288638966e+19,9.131160588400332e+19,9.345733398855737e+19,9.581533441534561e+19,9.808497518090828e+19,9.911444365992069e+19,
1.0058675612152504e+20,1.056206426202206e+20,1.0552810025313822e+20,1.0966584202200783e+20,1.1058394595758213e+20,1.511671723717
1254e+20,1.4899007397323227e+20,1.626727505073891 6e+20
12.360234375000001,1.1982615738648132e+20,1.1574488214667893e+20,1.1872924277552749e+20,9.425100242489098e+19,9.624197041274231e
+19,9.963885040672082e+19,1.0140133808714277e+20,1.0460633644993918e+20,1.0637666670384272e+20,1.0825104176593489e+20,1.11690794
4949459e+20,1.1391599401077975e+20,1.1650038339206894e+20,1.1976603098978936e+20,1.2135023083306395e+20,1.2476331881223766e+20,1.
6857559414784125e+20,1.6741855314019552e+20,1.830472335742428e+20
8.808203125,1.3385390715382674e+20,1.2580742676174899e+20,1.2868042284715945e+20,1.0407460996788961e+20,1.0661508093768206e+20,1.
0924129737753952e+20,1.1051911401680124e+20,1.141418601270486e+20,1.1745874587151522e+20,1.2052676061750043e+20,1.2149819405274
933e+20,1.2553189101343028e+20,1.285350738186346e+20,1.3118586365540096e+20,1.3496023285295843e+20,1.3648639765246945e+20,1.83625
49265894962e+20,1.8299181272163982e+20,1.9873708608144628e+20
5.256171875,1.399141251021421e+20,1.3488023860344653e+20,1.4076112303815026e+20,1.1140793901137774e+20,1.1450627837482497e+20,1.1
7800159237079e+20,1.2094241701202295e+20,1.2351739530513426e+20,1.261524999949375e+20,1.2918600741760387e+20,1.329734475709639e
+20,1.3675827353316344e+20,1.3925221190029864e+20,1.4282162851087142e+20,1.4521622761390688e+20,1.4875531960701764e+20,1.9855932
1082531e+20,2.0220454923675877e+20,2.1870218681256267e+20
1.704140625,1.492833862214425e+20,1.4138905175490696e+20,1.4793916912670356e+20,1.1653175368599518e+20,1.210135230115891e+20,1.24
37850987340925e+20,1.2668945485930812e+20,1.2991432107492447e+20,1.3458169797291524e+20,1.3817987068625371e+20,1.40993786051436
46e+20,1.4359752044731348e+20,1.4812529953733254e+20,1.502422715391209e+20,1.5336988984356592e+20,1.5740881518656792e+20,2.119737
8160361862e+20,2.1329028827205606e+20,2.315012667344641e+20
−1.847890625,1.5812144369692544e+20,1.489268105471477e+20,1.5383992141422726e+20,1.2107835495236996e+20,1.2475077069466717e+20,1.
270972686803491e+20,1.3107031640610601e+20,1.3646966682904212e+20,1.3993190160203361e+20,1.4212939069156596e+20,1.4579866940447
054e+20,1.5106992446054125e+20,1.5595218787192668e+20,1.5679500310207804e+20,1.5963035483477705e+20,1.654343820493615e+20,2.20359
06117009968e+20,2.2454385837985858e+20,2.4500983813733043e+20
−5.399921875,1.56542995074204e+20,1.5021978949514042e+20,1.571897459673164e+20,1.2352837490800846e+20,1.2826999483495798e+20,1.30
37755574856843e+20,1.3237532063343713e+20,1.3752370870496343e+20,1.3918110590073253e+20,1.4331780199313795e+20,1.47639582819708
08e+20,1.5554123702149307e+20,1.5570279403521317e+20,1.6130552853043767e+20,1.6361647351633658e+20,1.6653652504263636e+20,2.24779
65842253742e+20,2.2809445281407566e+20,2.470954398451926e+20
−8.951953125,1.527529407296834e+20,1.4538667287757275e+20,1.516952389861374e+20,1.2138630667107911e+20,1.2256739823740163e+20,1.2
70862890774749e+20,1.300157516919526e+20,1.3287515398332839e+20,1.370139414286622e+20,1.4025396995300957e+20,1.4458568470598969
e+20,1.494010248236659e+20,1.5321408405039897e+20,1.57243075466991e+20,1.6112514934036185e+20,1.6667037163005497e+20,2.2278346205
236504e+20,2.2641248222139767e+20,2.4523309006243876e+20
−12.503984375,1.3759690605745798e+20,1.3323382101055162e+20,1.3836286406749005e+20,1.0931083486240932e+20,1.1368437667395776e+20
,1.1930122780144909e+20,1.225213884729765e+20,1.2608871373062083e+20,1.2799289057194295e+20,1.3388632312421715e+20,1.36647954666
18954e+20,1.4063668753890956e+20,1.4375437191694464e+20,1.4977276280669166e+20,1.5237597436433 66e+20,1.581036671970339e+20,2.1146
349148908533e+20,2.0951225920687387e+20,2.278183942275248e+20

A.9 "MITR_Cycle1_cladding_fluence_position_8.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.960979303108908e+19,2.8700315926344487e+19,2.8174392948671255e+19,2.2531504477221175e+19,2.3032383503576633e+19,2.
3155668758706713e+19,2.4344655182334005e+19,2.4280502931254866e+19,2.4241185496200655e+19,2.5181719191932215e+19,2.549406275179
1034e+19,2.603561859260417e+19,2.6037500810239746e+19,2.6968989404555907e+19,2.704255274381291e+19,2.747112324266841e+19,3.644396
8414387708e+19,3.6750508469870174e+19,3.855016994859725e+19
37.224453125,4.0125115555161055e+19,3.764790801145316e+19,3.80640872442078e+19,2.9480808839229067e+19,3.024148618311691e+19,3.061
238762497136e+19,3.1536451916391645e+19,3.2374561602454065e+19,3.1872427764341555e+19,3.2938599487247663e+19,3.264225477729122e
+19,3.4142329949020283e+19,3.412899757410163e+19,3.585023331801061e+19,3.640208907199619e+19,3.6095758151806562e+19,4.8425852179
510895e+19,4.951806124637577e+19,5.2634124825891316e+19
33.672421875,4.928226120369695e+19,4.564681012440947e+19,4.773466003666965e+19,3.7638340071805665e+19,3.815349258189755e+19,3.978
6943786636304e+19,3.968237614021554e+19,4.069626403991126e+19,4.048331202797538e+19,4.148825939390213e+19,4.181550384337591e+19,
4.257084823729629e+19,4.386398403675866e+19,4.5324218935201415e+19,4.502405750615061e+19,4.672803958840017e+19,6.10288154805502e
+19,6.2962794101102215e+19,6.670840719589398e+19
30.120390625,6.08249085700297e+19,5.673840494790628e+19,5.799269386672334e+19,4.521525944763098e+19,4.632184656587871e+19,4.57287
9115920335e+19,4.729145006731524e+19,4.878723796554105e+19,4.984002502970529e+19,5.118711773472079e+19,5.2494527017919594e+19,5.4
1519242136887e+19,5.487344097399197e+19,5.569900254248388e+19,5.6897347770465845e+19,5.636928115604098e+19,7.57503715818974e+19,
7.692362057473835e+19,8.349935701990808e+19
26.568359375,7.028409786525199e+19,6.736927472127319e+19,7.009901313108723e+19,5.27690170897741e+19,5.523315367767939e+19,5.71132
7996032471e+19,5.753102770775664e+19,5.994392614893479e+19,6.050650008667848e+19,6.138800534600553e+19,6.2009137165744865e+19,6.
244988979540838e+19,6.516028319063456e+19,6.579762299556911 5e+19,6.788217902696704e+19,6.8536249655328915e+19,9.153904051496849
e+19,9.327224925439263e+19,9.879132963248053e+19
23.016328125,8.214259180759866e+19,8.078687227175346e+19,8.208507960206724e+19,6.292410407192655e+19,6.521465836677337e+19,6.6900
81166530818e+19,6.7265229913084535e+19,6.933200944459092e+19,7.208213854545701e+19,7.359470955093335e+19,7.378345415272283e+19,7.
52029599528847e+19,7.697381304502033e+19,7.992052932115744e+19,8.160093139913912e+19,8.206730310217572e+19,1.1033036941501176e+
20,1.0927319050969786e+20,1.1738032013669966e+20
19.464296875000002,9.528517644799233e+19,8.7981126345502e+19,9.060995697651999e+19,7.2233761632767115e+19,7.3683069212158886e+19
,7.508584418889345e+19,7.757716836486812e+19,7.865578363769831e+19,8.23261080270671e+19,8.344184481437665e+19,8.566390730081788e+
19,8.803341016871238e+19,9.020580302310374e+19,9.107894287071712e+19,9.348243022369836e+19,9.634601522093097e+19,1.2795681473392
778e+20,1.2697544737226891e+20,1.3702648954623271e+20
15.912265625,1.0674474481924378e+20,1.0168105654131833e+20,1.0500996756512334e+20,8.25179896582492e+19,8.51201555394299e+19,8.710
118960087127e+19,8.874447016437357e+19,9.0695702446585e+19,9.226630849582488e+19,9.562345278416349e+19,9.534739419761268e+19,9.83
5580538513804e+19,1.0163452393866109e+20,1.0199109961295589e+20,1.0605198416170625e+20,1.0734287175677057e+20,1.445234669563693
3e+20,1.4552522500908025e+20,1.5783545118396465e+20
12.360234375000001,1.1742423854819638e+20,1.120745577573101e+20,1.1533549981094162e+20,9.25010628620395e+19,9.286757246274427e+1
9,9.663828179267702e+19,9.916620464489896e+19,1.0179294392298855e+20,1.0278319953459318e+20,1.0581984398665214e+20,1.09447295640
98843e+20,1.10467353031823e+20,1.1323473579434849e+20,1.1455071962455378e+20,1.1864192879076616e+20,1.2088229061533103e+20,1.5896
94873093978e+20,1.6314853329860366e+20,1.7812000607489645e+20
8.808203125,1.3059557929135576e+20,1.218803888004172e+20,1.2847965296603159e+20,9.809438626908614e+19,1.0203188099505999e+20,1.05
26040707830106e+20,1.0760115384342987e+20,1.0965015687504472e+20,1.1373561482070393e+20,1.171978495936954e+20,1.196170220936397
8e+20,1.217559533011765e+20,1.2369097759819273e+20,1.2725673434114077e+20,1.2995091975117174e+20,1.3360555899357743e+20,1.7763324
368080778e+20,1.8054649831009026e+20,1.9426577352049443e+20
5.256171875,1.3942161148750029e+20,1.3399036793240584e+20,1.3640326637356496e+20,1.0620622144017688e+20,1.1021063945986e+20,1.156
6331938247072e+20,1.165380277447804e+20,1.1872401439320647e+20,1.226714430455903e+20,1.2520302576543696e+20,1.2774820227931836e
+20,1.3244642663300325e+20,1.3365888849325202e+20,1.3559182143733981e+20,1.3851082728717543e+20,1.4434413344275774e+20,1.9202384
318123327e+20,1.930569715278704e+20,2.0938834654586523e+20
1.704140625,1.4898902829676803e+20,1.395768944424351e+20,1.426443863501882e+20,1.1169654571549906e+20,1.1650874880378257e+20,1.19
63793562292391e+20,1.2325545335085025e+20,1.273973778255767e+20,1.2811941742411206e+20,1.3118795500832937e+20,1.384005084202015
5e+20,1.3997372866060191e+20,1.4278502983462416e+20,1.4702733924991454e+20,1.4976962577729903e+20,1.535701368864617e+20,2.0352628
428751726e+20,2.062607282414202e+20,2.2154956382460006e+20
−1.847890625,1.5345040693130992e+20,1.4432635694286622e+20,1.5168373654503113e+20,1.1719262121137439e+20,1.2038925416245712e+20,
1.2364496783376758e+20,1.2668265796229076e+20,1.299043871485145e+20,1.3419793471055105e+20,1.3845697494926875e+20,1.397802785147
2352e+20,1.4286502408413602e+20,1.4940363901482634e+20,1.527351642297919e+20,1.5575298650549512e+20,1.6213893267241114e+20,2.1274
026245188285e+20,2.1512545046674044e+20,2.3429322289389847e+20
−5.399921875,1.5517943296487725e+20,1.4578350709573953e+20,1.5010215089291706e+20,1.1918358919922573e+20,1.2194103803534125e+20,
1.249719312668471e+20,1.2672500785909 12e+20,1.2953787754780973e+20,1.3298233582090967e+20,1.3864467387459402e+20,1.4150407616596
979e+20,1.4532184093679188e+20,1.4922482833944687e+20,1.5488925774605962e+20,1.5631503760500677e+20,1.614446035001773e+20,2.18745
05954759515e+20,2.198702074230826e+20,2.3768330599085962e+20
−8.951953125,1.5233571582046457e+20,1.4200286383939682e+20,1.479166870827231e+20,1.1579559745519298e+20,1.1752462348876032e+20,1.
2352942058447266e+20,1.2577971633544749e+20,1.2827051767319006e+20,1.3319669949607225e+20,1.3645607303500744e+20,1.389254380052
3378e+20,1.4272647195262851e+20,1.4928495473613878e+20,1.4984909718857882e+20,1.5665122258824947e+20,1.6009724937604576e+20,2.176
4553074548084e+20,2.1634157219461392e+20,2.3584186973738998e+20
−12.503984375,1.35738738980561e+20,1.2723843500301715e+20,1.359719248320793e+20,1.0584860008941784e+20,1.1016358401897066e+20,1.1
468561188843658e+20,1.1694270453642876e+20,1.2081223029222913e+20,1.2416257768355037e+20,1.2834946624623775e+20,1.3162138790274
345e+20,1.3479815300100624e+20,1.3817411946570057e+20,1.4505937614427575e+20,1.4824136962485957e+20,1.5159276269264506e+20,2.0225
160467764815e+20,2.054142531436441e+20,2.2387723963392626e+20

A.10 "MITR_Cycle1_cladding_fluence_position_9.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9235754559842005e+19,2.758808215519004e+19,2.8760337755390005e+19,2.3099725067871605e+19,2.245982335559974e+19,2.3
618642013234643e+19,2.4238309885924086e+19,2.38375020971933e+19,2.3784695435750814e+19,2.4521635923901145e+19,2.452963534885233
e+19,2.4721778399150486e+19,2.577430404419868e+19,2.5705080262268133e+19,2.5588853323271455e+19,2.6669141678444364e+19,3.28175101
5269242e+19,3.311871725820743e+19,3.6079341031318508e+19
37.224453125,3.801943685918614e+19,3.6415526014561264e+19,3.798189707412109e+19,2.846733921011903e+19,2.918754887484203e+19,2.959
567639882227e+19,3.144328214343074e+19,3.1692571412497842e+19,3.1009064991668523e+19,3.1862180134992327e+19,3.229456735294218e+
19,3.31392125169059e+19,3.2598597784910545e+19,3.387406165212781e+19,3.4448033463331385e+19,3.469554508240933e+19,4.533870155423
069e+19,4.545905891526099e+19,4.7196868631127654e+19
33.672421875,4.793684158102419e+19,4.573422867681723e+19,4.645815049298818e+19,3.6564325775418008e+19,3.6176954929252286e+19,3.71
7421657316711e+19,3.898898807679946e+19,3.9259452294266765e+19,3.952196937060609e+19,4.190961472515459 5e+19,4.024981247351782e+
19,4.1263648089390326e+19,4.258888615630388e+19,4.389728883214368e+19,4.332744744297373e+19,4.323255230384688e+19,5.806118567512
893e+19,5.77134982507799e+19,6.064296086525759e+19
30.120390625,5.937507815240583e+19,5.467842231341724e+19,5.615910018673525e+19,4.365782892184012e+19,4.479599546930693e+19,4.6118
41020976712e+19,4.659952595094905e+19,4.874122820111592e+19,4.907438072261246e+19,4.916342007353975e+19,4.972630771422271e+19,5.0
68681383042064e+19,5.250027823847274e+19,5.323382027811439e+19,5.470665557795085e+19,5.444471362366683e+19,7.171196907712751e+1
9,7.056643051058804e+19,7.530909611400177e+19
26.568359375,6.852840708184736e+19,6.553672671774932e+19,6.6997536738247385e+19,5.300847700007764e+19,5.490324275322187e+19,5.482
6908371334726e+19,5.5588160837277884e+19,5.676454685951147e+19,5.782852266184274e+19,5.963545159199353e+19,6.070151874725321e+1
9,6.251995011851028e+19,6.2350027693076546e+19,6.295076882176383e+19,6.515348629361721e+19,6.42615242696481e+19,8.67979434262510
6e+19,8.491572579067732e+19,9.231963799549947e+19
23.016328125,8.114135659311985e+19,7.588003546345914e+19,7.813869662614762e+19,6.21513491648771e+19,6.144917741916167e+19,6.44711
8240072174e+19,6.507087785294481e+19,6.781839276265038e+19,6.828424162745487e+19,7.1016071390197326e+19,7.09988177285379e+19,7.22
2330486812504e+19,7.388697612267938e+19,7.510884907110602e+19,7.674951544344778e+19,7.818418355234067e+19,1.0178300999657857e+2
0,1.0037866650514771e+20,1.0621876955774735e+20
19.464296875000002,9.151237576513117e+19,8.58040279470217e+19,9.078144791665004e+19,7.034004155608709e+19,7.2817771938027086e+19
,7.2167361177289925e+19,7.449346847191982e+19,7.717719711730872e+19,8.028756176009432e+19,8.099862175575551e+19,8.24207417470778
9e+19,8.473220957120887e+19,8.563724255098059e+19,8.620713622397375e+19,9.059531750602108e+19,8.854788298910253e+19,1.15903302131
00636e+20,1.1732437644586456e+20,1.2380338781809505e+20
15.912265625,1.0441184062759654e+20,9.962002823036507e+19,1.0197593730422489e+20,8.000732046768669e+19,8.124069585721958e+19,8.32
3427803623144e+19,8.62986329145919e+19,8.728784284973233e+19,9.000921584783272e+19,9.073700666692121e+19,9.381390966285219e+19,9.
367169766371995e+19,9.558946829907675e+19,9.808706653383669e+19,9.94454002608424e+19,1.0170824412938774e+20,1.3524831671884759e
+20,1.3296821918864287e+20,1.422590545731277e+20
12.360234375000001,1.154405902955945e+20,1.0752220527038218e+20,1.120604411250433e+20,8.851964972456893e+19,9.103084175336356e+1
9,9.310389534365519e+19,9.52226185659967e+19,9.854298147223121e+19,1.0094385463405194e+20,1.0227134090536354e+20,1.0383828708697
866e+20,1.0623445470471047e+20,1.0851298372021892e+20,1.0780767494511085e+20,1.1096038948469688e+20,1.1370110749738508e+20,1.4860
631071086805e+20,1.4915372233988076e+20,1.602117509488765e+20
8.808203125,1.25742071982736e+20,1.179502138096928e+20,1.2265628073685929e+20,9.712086148090883e+19,9.987517328763173e+19,1.01636
6152915895e+20,1.0425603483442962e+20,1.0510564696159833e+20,1.089621017615961e+20,1.1155903926045575e+20,1.1284940401728797e+2
0,1.1689930896316413e+20,1.1792668608924815e+20,1.2105064452606845e+20,1.245510464900035e+20,1.2536144574976444e+20,1.6628242566
18339e+20,1.657344911945891e+20,1.7539654172386764e+20
5.256171875,1.3496284704411894e+20,1.2684787484363558e+20,1.3373626855160336e+20,1.0422361886403918e+20,1.0669612086365815e+20,1.
10376902017669e+20,1.1292730691387145e+20,1.1483984916690721e+20,1.1709328194727466e+20,1.1787806213366248e+20,1.2178470940394 2
19e+20,1.2469587268029627e+20,1.2871127030285361e+20,1.3082981081933827e+20,1.3306389858511788e+20,1.3270000317557 36e+20,1.790407
242016313e+20,1.7605113519046164e+20,1.8992726187049694e+20
1.704140625,1.4393318259232412e+20,1.3324532345165788e+20,1.3964120354498388e+20,1.0836763469169403e+20,1.128389472526459e+20,1.1
44550402280788e+20,1.1832247463095072e+20,1.2214860481348647e+20,1.2478736937091442e+20,1.2600296826055582e+20,1.29233585696725
29e+20,1.3337707868614805e+20,1.3601793459650442e+20,1.3956800619248937e+20,1.4206455875078508e+20,1.4397762384205295e+20,1.92577
0060307991e+20,1.8821078395450012e+20,2.0248269917623806e+20
−1.847890625,1.4580076075739896e+20,1.3962447272155655e+20,1.4391854312182522e+20,1.1366869152699466e+20,1.1633830354011675e+20,
1.2015188560508199e+20,1.2290985727942964e+20,1.2567201165963408e+20,1.2874734614086877e+20,1.3275019564585558e+20,1.36122502242
92522e+20,1.3681317154753434e+20,1.4079040197914809e+20,1.4343125788950446e+20,1.4859010272567283e+20,1.482978361539268e+20,1.966
0599744739115e+20,1.9643607502195737e+20,2.1174320994326086e+20
−5.399921875,1.4904915469345997e+20,1.402173712767623e+20,1.4491036724812616e+20,1.1211011075709318e+20,1.1793191447156918e+20,1.
209340516003093e+20,1.2246178491451666e+20,1.2671036838859227e+20,1.3052238193886121e+20,1.3273137346949983e+20,1.3569743476022
482e+20,1.3883184996168717e+20,1.4221618183809519e+20,1.4715700313147626e+20,1.4856082378467502e+20,1.503102405092944e+20,2.01061
10202314775e+20,2.0025854533686837e+20,2.140614746644092e+20
−8.951953125,1.434767448156975e+20,1.3775846307117803e+20,1.4119612444726064e+20,1.1111462676316751e+20,1.1489161015188547e+20,1.
1983975118051602e+20,1.2069093182238104e+20,1.2514551355990557e+20,1.2754586388349418e+20,1.3060917308539044e+20,1.340186011969
3943e+20,1.3926528285610125e+20,1.4215448692670693e+20,1.4566482281705197e+20,1.4735620449790783e+20,1.5168269086856692e+20,2.000
1856258833275e+20,1.9728830134028658e+20,2.1303305186186094e+20
−12.503984375,1.3301684314422852e+20,1.2569083383598983e+20,1.3248668517687525e+20,1.042905421577485e+20,1.0820660051620608e+20,
1.1046264748773404e+20,1.1308311270703838e+20,1.1586095223420597e+20,1.1895929159765316e+20,1.229997854553515e+20,1.251601530304
0446e+20,1.2840279574591234e+20,1.3357732572904382e+20,1.3646652979964951e+20,1.3827607292096083e+20,1.4149884778364876e+20,1.891
2052247836076e+20,1.8667991361090014e+20,2.0111495436105443e+20

A.11 "MITR_Cycle2_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.5310275125859095e+19,2.4882320573755535e+19,2.7653146485789053e+19,2.9056136146624356e+19,2.9357772180722553e+19,
2.8877132467372634e+19,3.04399757403172e+19,3.1081188317124465e+19,2.966821904383465e+19,3.134892963144471e+19,3.089103624197760
6e+19,3.118251285078428e+19,3.22261465568385e+19,3.2338799299230466e+19,3.1847460722077954e+19,3.1416539258163593e+19,3.151849313
6752976e+19,3.026402887929479e+19,2.603716851950558e+19
37.224453125,3.347980165653096e+19,3.577798952646101e+19,3.8283052065381474e+19,3.7669440759056515e+19,3.90113839617241e+19,3.93
0384954113437e+19,4.096882650550533e+19,3.9968437786754015e+19,4.19333425632715e+19,4.2168268034835915e+19,4.348090174506952e+1
9,4.2930584557375e+19,4.311956784908123e+19,4.501065945600272e+19,4.199852471669064e+19,4.353268784213078e+19,4.203295887497964e+
19,4.036897088121228e+19,3.719833112606857e+19
33.672421875,4.087972924476277e+19,4.692584598308464e+19,4.574258760908571e+19,4.893534434318267e+19,4.92038948152695e+19,5.11515
3755943324e+19,5.201931431088356 6e+19,5.02709041901361e+19,5.382508472663985e+19,5.453102992477372e+19,5.141208636027276e+19,5.52
81388893651075e+19,5.545293034016653e+19,5.437036715489785e+19,5.584618101472556e+19,5.42834276481995e+19,5.204448810806613e+19,
5.116781062118339e+19,4.8640721009728184e+19
30.120390625,5.063718293914238e+19,5.4132025239430095e+19,5.6826610508567855e+19,5.857322250094512e+19,5.817385818992747e+19,6.19
2888966538066e+19,6.251471988838629e+19,6.4489334558102766e+19,6.438216610723986e+19,6.686771895334384 4e+19,6.6263727633799365e
+19,6.727022998900931e+19,6.848324738753477e+19,6.673816380427216e+19,6.924124840198542e+19,6.628018050838654e+19,6.578848230555
9994e+19,6.529768316691854e+19,5.912335987580262e+19
26.568359375,6.1794030037616935e+19,6.373825633787733e+19,6.566953611387239e+19,6.829030821341413e+19,7.055639949193727e+19,7.294
152686856653e+19,7.3565297600183026e+19,7.698560747952665e+19,7.817515930282123e+19,7.945057175579204e+19,8.194053001640296e+19,
8.0103741886261e+19,8.080168541314754e+19,8.290954139509459e+19,8.013565866483176e+19,8.13680958497552e+19,8.081409249890181e+19,
7.915415029396737e+19,7.218595332741563e+19
23.016328125,7.2173456335242854e+19,7.348968630221682e+19,7.812930702938156e+19,8.047658380381846e+19,8.285110222306363e+19,8.514
326636295443e+19,8.843411099964488e+19,8.955632291547611e+19,9.10850916558057e+19,9.125051946586253e+19,9.611535577139269e+19,9.6
18098745690438e+19,9.552646873015776e+19,9.519021872493352e+19,9.749182303876778e+19,9.541318664283621e+19,9.506434973902071e+1
9,9.093854419363578e+19,8.873574703374308e+19
19.464296875000002,8.2962586182015e+19,8.649842580914286e+19,8.903117952496414e+19,9.279870799259012e+19,9.659545604623155e+19,9.
839718067315494e+19,1.0274775226481274e+20,1.0491809320762365e+20,1.0485605777885233e+20,1.0871574032545015e+20,1.0996364141435
716e+20,1.0606170285106002e+20,1.1034844088557634e+20,1.088272242844015e+20,1.1033225773024469e+20,1.0716395554198223e+20,1.06980
54644822355e+20,1.063197342721813e+20,9.877928295181882e+19
15.912265625,9.385420934588754e+19,1.0014945676989828e+20,1.0301297619941473e+20,1.0674499163172957e+20,1.109553092105131e+20,1.0
97289856620483e+20,1.1324702381831135e+20,1.1707523911843098e+20,1.1627237480114428e+20,1.1914848112925201e+20,1.20866592786961
87e+20,1.2221608812878425e+20,1.2255413626237862e+20,1.2325001194163947e+20,1.2299108145633311e+20,1.247073949856728e+20,1.243711
4498044858e+20,1.2006642566223043e+20,1.10109289812342e+20
12.360234375000001,1.0652561997056724e+20,1.1164848769721867e+20,1.1484645900358915e+20,1.1852093432805813e+20,1.222799216859257
4e+20,1.2412929671465897e+20,1.2927374198175257e+20,1.3053692716180613e+20,1.2956144252098186e+20,1.3495313043897562e+20,1.371630
3020593055e+20,1.3367825742451587e+20,1.3710549009808471e+20,1.4100113521208618e+20,1.4009847477025432e+20,1.4061903293342228e+
20,1.4003374214892772e+20,1.3308937038328093e+20,1.264641664033416e+20
8.808203125,1.1731618832003552e+20,1.2065621176765045e+20,1.2275732476820932e+20,1.268040126653062e+20,1.3165446394387489e+20,1.3
650761241499882e+20,1.3922008906141993e+20,1.3977571072780647e+20,1.4215823081829895e+20,1.4702396618801419e+20,1.4782053705600
526e+20,1.5254152309192078e+20,1.522843907349846e+20,1.5473434063935898e+20,1.5665474173871443e+20,1.5355027310759343e+20,1.49295
00231955533e+20,1.4637304371800795e+20,1.3847386578779397e+20
5.256171875,1.2509758884200251e+20,1.269083041107768e+20,1.3192957758451288e+20,1.3771325748720653e+20,1.399753029768968e+20,1.46
8432542868108e+20,1.480183311767254e+20,1.5135206117504473e+20,1.5360601508706913e+20,1.5779655325378065e+20,1.5827935072117477
e+20,1.5989586812596932e+20,1.638661355673334e+20,1.661866202290546e+20,1.6451885616570984e+20,1.6678989229725096e+20,1.627288193
7319264e+20,1.5946162012457012e+20,1.507397984649974e+20
1.704140625,1.2481168643114341e+20,1.3435075743496431e+20,1.389782407956303e+20,1.4206562720723454e+20,1.5023632252134615e+20,1.5
55354068282755e+20,1.5402677712569195e+20,1.5746389950529677e+20,1.6373487219631004e+20,1.659951195576301e+20,1.703807546525064
6e+20,1.7117822458468264e+20,1.7247827139632493e+20,1.763271651727017e+20,1.766984786811445e+20,1.7953862244184857e+20,1.73217302
1564702e+20,1.6892247254428805e+20,1.584223019266043e+20
−1.847890625,1.3115098800022364e+20,1.3719449745240875e+20,1.4409661320135624e+20,1.5003223495133037e+20,1.5427671696914737e+20,
1.555848553584555e+20,1.5920628589600414e+20,1.705228067937509e+20,1.7200806082752207e+20,1.773233282897831e+20,1.72546600274391
9e+20,1.7453712838018453e+20,1.7997466857161795e+20,1.82362583047221e+20,1.836293644840149e+20,1.842955710451677e+20,1.8349360579
20661e+20,1.74355517414796e+20,1.6709737224855228e+20
−5.399921875,1.3430580422570974e+20,1.3848375549382997e+20,1.4650340802484622e+20,1.5285259929996244e+20,1.6003702120302867e+20,
1.6245999918185028e+20,1.6462494573955062e+20,1.6606254937151193e+20,1.6898630610142953e+20,1.726967439933022e+20,1.787258684185
2582e+20,1.792581144161e+20,1.8384244269588162e+20,1.861638264217879e+20,1.8654772682882194e+20,1.8534927427009498e+20,1.84392669
9771576e+20,1.792922788551335e+20,1.708006176269442e+20
−8.951953125,1.2831893581718538e+20,1.3817088115741812e+20,1.3981706768032067e+20,1.4398243204984963e+20,1.4928421354933427e+20,
1.535691534554 8037e+20,1.587019108881678e+20,1.6518686085523282e+20,1.6840011625274986e+20,1.664788160892093e+20,1.7336744587538
045e+20,1.7717318457087282e+20,1.8126752286977953e+20,1.8191754627560066e+20,1.8351967865343377e+20,1.8102028021887936e+20,1.7867
012643905018e+20,1.757724425705002e+20,1.632592672423966e+20
−12.503984375,1.1758141225463749e+20,1.2227722449337044e+20,1.2654867843674022e+20,1.3289607158348625e+20,1.3410980823335979e+20
,1.4117105834306848e+20,1.4613209451640342e+20,1.5263772295972556e+20,1.5339113874683226e+20,1.5643267288499683e+20,1.5618093491
31712e+20,1.6191426722149974e+20,1.6320712151966132e+20,1.688946015545502e+20,1.6898720516561463e+20,1.7023690438289185e+20,1.663
5744424242196e+20,1.5949848175615885e+20,1.4915384924249598e+20

A.12 "MITR_Cycle2_cladding_fluence_position_1.csv" File


```
,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.917940143219455e+19,2.7067855173111853e+19,2.775843813056776e+19,2.1767045171982823e+19,2.180519540384043e+19,2.24
1820268091736e+19,2.3711431780057117e+19,2.3475676099343733e+19,2.4220933971258524e+19,2.4045910484156908e+19,2.426657610101575
e+19,2.487469504754825e+19,2.5261245098751738e+19,2.5807197232925737e+19,2.6583538511019782e+19,2.705398719189342e+19,3.565038212
441011e+19,3.545692112971157e+19,3.659515827017633e+19
37.224453125,3.761522533313229e+19,3.5972321197982888e+19,3.6866726285530423e+19,2.9691454056723497e+19,2.875353220052501e+19,3.0
139639580008915e+19,3.041604279187978e+19,3.1158537694356423e+19,3.052688037357195e+19,3.104870965918974e+19,3.2085620208912912
e+19,3.3594785996450054e+19,3.4268472333111095e+19,3.495570784682462e+19,3.455242057690923e+19,3.616339116143687e+19,4.6742057263
74718e+19,4.685533901071156e+19,4.856492635057029e+19
33.672421875,4.790579873748722e+19,4.5394259518203855e+19,4.544638399933521e+19,3.5799369937959875e+19,3.6596858559061344e+19,3.6
981336638186622e+19,3.793349815879674e+19,3.885276997128767e+19,4.000540643324745e+19,4.063323810404063e+19,4.11144729925286e+1
9,4.28164621664324e+19,4.241391877290421e+19,4.357165610151903e+19,4.3159282912874316e+19,4.463375218035154e+19,5.80494565553710
1e+19,5.739325131380907e+19,6.25212163229704e+19
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70192352623e+19,4.692637920568875e+19,4.774076444758508e+19,4.912187722846926e+19,4.985879305804192e+19,5.088954006055691e+19,5.1
35010581228637e+19,5.302813153971694e+19,5.433060595966899e+19,5.425303027929e+19,5.43369820429878e+19,7.2407864849078075e+19,7.3
17086948623025e+19,7.902570799373713e+19
26.568359375,6.730912355512799e+19,6.362321605657321e+19,6.653389809161474e+19,5.204058250168697e+19,5.5556407977712206e+19,5.548
627106120518e+19,5.530720938800166e+19,5.740759750127642e+19,5.885603109520179e+19,5.871469458163461e+19,6.006589290494797e+19,6.
148935350587464e+19,6.274916130161824e+19,6.44165070894898e+19,6.476878569285461e+19,6.675546698694375e+19,8.760260273810376e+1
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23.016328125,7.919998760445156e+19,7.501568292647564e+19,7.691416173465443e+19,6.113813758306295e+19,6.3747018341013635e+19,6.475
762754704669e+19,6.610244978704123e+19,6.7940887143967785e+19,6.9151811634267095e+19,6.941269971006217e+19,7.200032685695014e+1
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+20,1.0111617999206713e+20,1.1159792962793133e+20
19.464296875000002,9.14739479931809e+19,8.479712607782275e+19,8.899258890160662e+19,6.857265073280757e+19,7.253591785573105e+19,
7.325960331241717e+19,7.69412600887594e+19,7.692478854018579e+19,7.950763362458463e+19,8.04172881780697e+19,8.197677188979786e+1
9,8.323657968554146e+19,8.53491886251773e+19,8.772481100171449e+19,8.94649504074759e+19,9.069553448800825e+19,1.1782258096792982e
+20,1.1969661812338645e+20,1.2806575881961483e+20
15.912265625,1.0172881533095045e+20,9.699457346672625e+19,1.0063531704177272e+20,7.963887467289703e+19,8.109634105152407e+19,8.30
6814481786929e+19,8.468288791836056e+19,8.620198976906953e+19,8.834701046557604e+19,9.122049868125779e+19,9.240007409523958e+19,
9.705036419576594e+19,9.721454834122557e+19,9.730381350768905e+19,1.0040843474367806e+20,1.007198001457471e+20,1.346862586656567
5e+20,1.371325492989775e+20,1.4449586285166199e+20
12.360234375000001,1.1407291263618682e+20,1.0851190530162226e+20,1.1221959775151636e+20,8.656702053907199e+19,9.02747129889661e+
19,9.304565253127014e+19,9.593826899690835e+19,9.871664730308436e+19,9.919432221171932e+19,1.0280690475210767e+20,1.036612999168
296e+20,1.05451916648865e+20,1.0751936166499253e+20,1.1016968696451557e+20,1.1303786177743163e+20,1.1321851747146488e+20,1.535525
5786576986e+20,1.5316255410276866e+20,1.617819560692608e+20
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56071253811e+20,1.0465915695622501e+20,1.0808205101787845e+20,1.0825845598969915e+20,1.1138433083675084e+20,1.1264573265332417e
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9182552e+20,1.6457096118096816e+20,1.7770144209552572e+20
5.256171875,1.340417429101793e+20,1.2699298280122332e+20,1.3103382560452575e+20,1.0388286881215862e+20,1.067170378473743e+20,1.08
03104235132789e+20,1.1303467373577221e+20,1.137392309425019e+20,1.1893573884734053e+20,1.1981882638699717e+20,1.227438546095060
6e+20,1.2543509311032485e+20,1.2605197917142072e+20,1.3058590575137862e+20,1.3229735278220535e+20,1.3606083596113912e+20,1.774065
4824203026e+20,1.7810207266405827e+20,1.9603161895658134e+20
1.704140625,1.4230567823164716e+20,1.3332602755764172e+20,1.3968723334871823e+20,1.0844336240594495e+20,1.1165690839863044e+20,1.
1458725002390499e+20,1.1942616592261317e+20,1.2295213999792087e+20,1.2453872206375399e+20,1.2648449015654739e+20,1.310327629239
7262e+20,1.3191106840114015e+20,1.3441102440239432e+20,1.3853316226801174e+20,1.4274031457788009e+20,1.4538373245380775e+20,1.904
2757305961474e+20,1.9003438125495416e+20,2.0622006875977987e+20
−1.847890625,1.4499266601025341e+20,1.368658164801402e+20,1.4455165358070165e+20,1.1223341259870713e+20,1.1537629033460903e+20,1.
198937453659933e+20,1.221636310274934e+20,1.2595208719942587e+20,1.2988878730852095e+20,1.3079525382034656e+20,1.34284034076294
48e+20,1.3829884120604508e+20,1.400326045284877e+20,1.4401978196385672e+20,1.4661006581212753e+20,1.487423343419964e+20,1.9985620
626732047e+20,1.9996513102401703e+20,2.169499543048015e+20
−5.399921875,1.4543846050229427e+20,1.389874582044777e+20,1.4255115743942176e+20,1.1160377437097361e+20,1.1885603780585528e+20,1.
2073485702380103e+20,1.2587185481766406e+20,1.2636068787210694e+20,1.3139460565231567e+20,1.3311561680812061e+20,1.371070449657
0222e+20,1.3960487560585014e+20,1.411595772550892e+20,1.4561858485605096e+20,1.4822108953068283e+20,1.524829698890377e+20,2.00842
37382063135e+20,2.0321640217633885e+20,2.1785110741386147e+20
−8.951953125,1.4278600984166497e+20,1.351607455326323e+20,1.4057404027031085e+20,1.10196253978344e+20,1.1466376302373087e+20,1.17
816736224488755e+20,1.2152655403588792e+20,1.2464711548017395e+20,1.2719276674521301e+20,1.3130427780529906e+20,1.33626766154179
4e+20,1.3521387956028914e+20,1.4093747701948362e+20,1.462312201949343e+20,1.4791556887165603e+20,1.5019023659564522e+20,1.9879299
437390715e+20,2.0115320788242386e+20,2.173718384843968e+20
−12.503984375,1.3121501263883557e+20,1.2529747597869366e+20,1.2951259839271051e+20,1.0252157502359044e+20,1.0690406962472637e+20
,1.1048636576935032e+20,1.1180090161358045e+20,1.1563186500763843e+20,1.197800385468077e+20,1.2474275672995625e+20,1.27332509237
95048e+20,1.3007953846780899e+20,1.3145889782578047e+20,1.3828130697691834e+20,1.3913995286385282e+20,1.4405006835962113e+20,1.89
49666489506696e+20,1.8697704930357975e+20,2.04873121158679e+20
```

A.13 "MITR_Cycle2_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9369780653289e+19,2.785200715327145 6e+19,2.8898906900194116e+19,2.1812899837850673e+19,2.253212203621362e+19,2.361
5577994164183e+19,2.4134166104094913e+19,2.395212892534259e+19,2.545656578441827e+19,2.5194561894042407e+19,2.6220579968095932e
+19,2.6284606471422423e+19,2.687827296243226e+19,2.677769024807787e+19,2.6779603073073512e+19,2.7018015455169737e+19,3.724445608
814287e+19,3.756230384158607e+19,4.013866657461079e+19
37.224453125,4.031135216449551e+19,3.69182131583299e+19,3.809869185078184e+19,2.9760262622539096e+19,3.0459240756364788e+19,3.149
3335202622153e+19,3.129296678432822e+19,3.243550778103319e+19,3.2746926317129912e+19,3.366263814976784e+19,3.4335208671848075e+
19,3.4776593039593423e+19,3.4960064837092483e+19,3.6844994468218774e+19,3.6353982918641947e+19,3.732787651156411e+19,4.987335178
16199e+19,5.0436997547003625e+19,5.424134079320551e+19
33.672421875,4.805972801560908e+19,4.495675393447745e+19,4.5761043711119e+19,3.581568208445053e+19,3.737590967256589e+19,3.738425
171490802e+19,3.833083441761456e+19,3.940647967349574e+19,4.160314664488853e+19,4.245849822210829 5e+19,4.182359972563675e+19,4.4
20533251535447e+19,4.345550511706118e+19,4.403705704976526 5e+19,4.522083005193194e+19,4.727616050975370 4e+19,6.26625528365376e+
19,6.157543063067872e+19,6.829475976816232e+19
30.120390625,5.974283801679441e+19,5.537575228367897e+19,5.757496968839545e+19,4.535318691482511e+19,4.651033976913571e+19,4.6462
8379484105e+19,4.894648180317403e+19,4.930795259332 35e+19,5.102997329565392e+19,5.151859381398619e+19,5.297292528498148e+19,5.330
458788561547e+19,5.387046528016078e+19,5.545492198488765e+19,5.650006830889764e+19,5.851012857515579e+19,7.848905431440307e+19,7.
55034533003654e+19,8.45596169741967e+19
26.568359375,6.993660122275858e+19,6.822834223359127 6e+19,7.000248741705306e+19,5.347993017688303e+19,5.486672829872646 6e+19,5.55
9094509568917e+19,5.709251271727137 6e+19,5.825933596461552e+19,5.955527489916577e+19,6.157064856818959e+19,6.236128289972334e+1
9,6.360727584827616e+19,6.623422217563018e+19,6.738191717301786e+19,6.766618422098194e+19,7.019323857634111 5e+19,9.1751838957826
16e+19,9.333151360006393e+19,1.0331433471623045e+20
23.016328125,8.012611370651022e+19,7.598962965342546e+19,7.880307641785498e+19,6.2401133420465955e+19,6.392182929200463e+19,6.727
03357149385e+19,6.570128787823211e+19,6.884682231551686e+19,6.991375359086614e+19,7.244718402954413e+19,7.43610717057434e+19,7.52
117474852935e+19,7.699439411641624e+19,7.894600695225187e+19,7.969041467972416e+19,8.208675932704752e+19,1.103301517280392e+20,
1.1106818337219268e+20,1.211689620297574e+20
19.464296875000002,9.347019341224827e+19,8.658827415013473e+19,9.000532346874593e+19,7.181967116291688e+19,7.439720284455005e+19
,7.682649058902064e+19,7.789767258658247e+19,8.126690128030251e+19,8.160536503647656e+19,8.194117209126778e+19,8.308249100533663e
+19,8.56786195966497e+19,9.012593771152694e+19,9.070509861298646e+19,9.275182135832782e+19,9.646854659292358e+19,1.2970813161448
767e+20,1.2816140006939717e+20,1.4134182696995216e+20
15.912265625,1.0556509212777408e+20,9.882822876116371e+19,1.0348276958390265e+20,8.139442294667892e+19,8.308674172754918e+19,8.54
1029275698266e+19,8.871416659668479e+19,9.010096471852823e+19,9.230177614407442e+19,9.302121087854798e+19,9.542127490780732e+19,
9.769541129151994e+19,1.0049026114626958e+20,1.0378244549988717e+20,1.0662671000035767e+20,1.0954164275761175e+20,1.461908383339
1509e+20,1.4580189725146705e+20,1.617001296666925e+20
12.360234375000001,1.176190776420041e+20,1.115867714821234e+20,1.145086116629 7288e+20,8.875348577715084e+19,9.158127872904772e+1
9,9.623316285040378e+19,9.846957407448008e+19,1.021097863092 4996e+20,1.0143764085939097e+20,1.0461558705354639e+20,1.08251017225
8272e+20,1.1098529428904801e+20,1.133561346030961 1e+20,1.145829993016924 7e+20,1.2008183982389851e+20,1.2181932252827709e+20,1.637
3463158564952e+20,1.628871438445229 8e+20,1.815685366283903e+20
8.808203125,1.2757267504295941e+20,1.2250953354753938e+20,1.267294380240454e+20,9.994138664057446e+19,1.0278618248132151e+20,1.05
3966572601019e+20,1.0766069817855972e+20,1.1110643987210561e+20,1.1466695106539025e+20,1.1559732788966146e+20,1.178263003498657
9e+20,1.2142719340416963e+20,1.2411683788415874e+20,1.2709712549543071e+20,1.3245250414296331e+20,1.3526329420600998e+20,1.800839
7189565828e+20,1.8113655698354025e+20,2.0013409723196575e+20
5.256171875,1.3646624859216077e+20,1.3093659033391374e+20,1.3654966901558202e+20,1.0721543502679543e+20,1.0846727271839053e+20,1.
1220206352238959e+20,1.1759251062817569e+20,1.1801545748832412e+20,1.2237935517978015e+20,1.2303609176161867e+20,1.268240165932
7457e+20,1.2970813161448767e+20,1.3350136984890927e+20,1.3654063623088036e+20,1.4064470852709445e+20,1.4345709261097083e+20,1.928
4942204021336e+20,1.9456724515435892e+20,2.1295002470279483e+20
1.704140625,1.4619668307695732e+20,1.3640461312007886e+20,1.4342999425686584e+20,1.1323711438114478e+20,1.159294155625167e+20,1.1
923807146470795e+20,1.2278901853301437e+20,1.2405148303014081e+20,1.2760189875817066e+20,1.3177079456812 61e+20,1.352919865809 44
68e+20,1.371240478545524e+20,1.3934823825226762e+20,1.4515844417654274e+20,1.489304287999016e+20,1.541099338158901e+20,2.05545797
9488146e+20,2.073380087016797e+20,2.2712777730246866e+20
−1.847890625,1.504675962200139e+20,1.4356814272877363e+20,1.4708827206103907e+20,1.1693949342827318e+20,1.197715371023826e+20,1.2
43787886405069e+20,1.2589045172734393e+20,1.3043234841145036e+20,1.3429094149988988e+20,1.3611450132907252e+20,1.40832802984999
65e+20,1.4511381159331103e+20,1.4663131942319027e+20,1.5278636518695826e+20,1.5509078796643 53e+20,1.6010557749668758e+20,2.133235
5691722238e+20,2.1959337218072914e+20,2.380090948263123e+20
−5.399921875,1.5150796048153477e+20,1.4476578371215873e+20,1.5008290585977838e+20,1.1701388106699275e+20,1.216306967300953e+20,1.
23944152294274e+20,1.284244135062985e+20,1.3302847700276342e+20,1.3411187982668633e+20,1.3906025182236756e+20,1.416388461845538
4e+20,1.4529340460679109e+20,1.5029119124819322e+20,1.540923995867633e+20,1.586433290555714e+20,1.627845951711453e+20,2.184276116
1393816e+20,2.2482760524520417e+20,2.4397020138913284e+20
−8.951953125,1.5154621698144767e+20,1.3738706129145378e+20,1.4550859741880128e+20,1.1483591727333899e+20,1.1818282967544316e+20,
1.2248881127675324e+20,1.241912255228783e+20,1.2911568720611395e+20,1.3312199289143946e+20,1.3483184590143647e+20,1.395474908559
8076e+20,1.4138167749069477e+20,1.4731940508134631e+20,1.5235332286155506e+20,1.5793611514745894e+20,1.6005722553151986e+20,2.177
9425400426866e+20,2.205226863244472e+20,2.4488782604676638e+20
−12.503984375,1.3393069279237652e+20,1.2901101317162998e+20,1.3734402272905173e+20,1.064062037855818e+20,1.1023185377687408e+20,
1.1508245916166673e+20,1.1687626393536156e+20,1.221949801038109e+20,1.2630117776113128e+20,1.2674962895455499e+20,1.317288186862
7719e+20,1.3427287593048655e+20,1.4185138229518123e+20,1.455155048423967e+20,1.4756700965022717e+20,1.5277786374253317e+20,2.0419
93816879903e+20,2.0589913923273255e+20,2.2754541075985138e+20

A.14 "MITR_Cycle2_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.997019516581126e+19,2.806927219236026e+19,2.8619049976525554e+19,2.1940793442420683e+19,2.2999170139317215e+19,2.3
87279982205093e+19,2.5827388163435336e+19,2.4663274751501693e+19,2.5057794906853708e+19,2.6089392053811204e+19,2.62658501596595
57e+19,2.665123126225459e+19,2.8239513616972767e+19,2.8245039555849077e+19,2.8402953886045196e+19,2.785269789563099e+19,3.8246085
64350191e+19,3.7553483592995045e+19,4.146919576116565e+19
37.224453125,4.0750558037106926e+19,3.715248108626889e+19,3.931089155774493e+19,3.023692798464858e+19,3.0611416611573973e+19,3.15
26703371990647e+19,3.2026003829881946e+19,3.2715205302618776e+19,3.268656606171174e+19,3.386205015556394e+19,3.4206199252697276
e+19,3.491017198512271e+19,3.5783854801884082e+19,3.650419281482782e+19,3.7260661966578205e+19,3.833593528426962e+19,5.0350335947
895325e+19,5.176099124815669e+19,5.659252151702053e+19
33.672421875,4.855982548391522e+19,4.574350948199224e+19,4.84515383355506e+19,3.802313526345382e+19,3.846988616799251e+19,3.97451
5596578425e+19,4.212720755966792e+19,4.111171002309045e+19,4.1606015882382e+19,4.25215151789093e+19,4.39914680540357e+19,4.326161
9050141385e+19,4.600014683557476e+19,4.629100250296829e+19,4.714268783227973e+19,4.786095361814485e+19,6.440906832561783e+19,6.48
968838695076e+19,7.0391428499501105e+19
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99632567212e+19,4.858665816788194e+19,5.134006348105923e+19,5.310995794230847e+19,5.328599097593557e+19,5.34214827464605e+19,5.44
99572167617724e+19,5.666318977380413e+19,5.7094106738101084e+19,5.910257298352951e+19,5.828855967982677e+19,8.050708468480973e+
19,7.966862972838484e+19,8.547883565265997e+19
26.568359375,7.114008694918595e+19,6.722782849281304e+19,7.2209143585641505e+19,5.477693179198642e+19,5.6707291016759304e+19,5.83
0609390895353e+19,5.895911110885605e+19,6.002285434254594e+19,6.1988282025572336e+19,6.331450735588698e+19,6.518110574747167e+1
9,6.6305953112966906e+19,6.721082560396285e+19,6.760242338779374e+19,6.936222238378818e+19,6.947539786269725e+19,9.4248075577144
36e+19,9.697650789732295e+19,1.050943246427345e+20
23.016328125,8.35787628236515e+19,7.851615266850806e+19,8.059103644850754e+19,6.395955445164099e+19,6.580489923216295e+19,6.83664
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66474289158171e+19,7.802785095434173e+19,8.093640762827699e+19,8.188963208444065e+19,8.492358506364604e+19,1.1238378189697596e+
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19.464296875000002,9.527834437341043e+19,9.175502699948558e+19,9.316148471156205e+19,7.400241701905975e+19,7.769310658010366e+19
,7.791467547543265e+19,8.031261414358571e+19,8.33954504282354e+19,8.463400461291628e+19,8.750749282859803e+19,8.786774153611136e+
19,8.980979024696877e+19,9.20839266306814e+19,9.407538998725963e+19,9.605728921885966e+19,9.82623513666184e+19,1.328196602740720
6e+20,1.3333559168261995e+20,1.4642197135422233e+20
15.912265625,1.0928713076513551e+20,1.0287969836999755e+20,1.0770373674096176e+20,8.261066083974391e+19,8.607446810269313e+19,8.8
01757949410366e+19,9.269762465011786e+19,9.36997324117258e+19,9.586547537901847e+19,9.816724145711266e+19,1.0021662090383686e+2
0,1.0081703541635912e+20,1.0353643495183602e+20,1.0610493384876755e+20,1.0844336240594495e+20,1.1240131612610273e+20,1.492858954
4492584e+20,1.5139372232207257e+20,1.6663574949571284e+20
12.360234375000001,1.1750058876032937e+20,1.135628259706812e+20,1.1863181220914343e+20,9.34292802109525e+19,9.711943843171983e+1
9,9.914968962848758e+19,1.0191425308747281e+20,1.0559006512077278e+20,1.079768456431179e+20,1.1015268407566539e+20,1.12785475146
06168e+20,1.1557766829942845e+20,1.1752981247554065e+20,1.188725093544289e+20,1.2169392622300694e+20,1.2621988269881634e+20,1.683
6260539456002e+20,1.7086574943747362e+20,1.885402523972439e+20
8.808203125,1.300455326901086e+20,1.252544374162916e+20,1.2989994545432887e+20,1.0252370038469673e+20,1.0488657059459626e+20,1.08
80361111345829e+20,1.1132110134383922e+20,1.13794490331265e+20,1.1685554166457537e+20,1.1845859527898217e+20,1.2280974080380053
e+20,1.25455815381111e+20,1.2938029466384497e+20,1.310481717919931e+20,1.3660333438351542e+20,1.3626061990512884e+20,1.8459026878
123462e+20,1.8595953267395133e+20,2.0523018182453294e+20
5.256171875,1.3937746196747887e+20,1.3283294378098626e+20,1.3847418349731263e+20,1.0872019069003706e+20,1.1222969321677113e+20,1.
1728858398997 86e+20,1.1944741953367589e+20,1.2140700247366004e+20,1.2561096674186899e+20,1.2985318750999085e+20,1.3148333947850
262e+20,1.3483184590143647e+20,1.3788917785281089e+20,1.428141708763231e+20,1.4610954327160013e+20,1.4894530632764552e+20,1.99046
4436858303e+20,2.0074513855001936e+20,2.2052693704665976e+20
1.704140625,1.4751387562257034e+20,1.4370947924234081e+20,1.469804099848957e+20,1.1578807904894951e+20,1.2248349787398753e+20,1.2
5754428616542 42e+20,1.2920017031008831e+20,1.3041747088370645e+20,1.3332602755764172e+20,1.3757674977018869e+20,1.4073025431162
197e+20,1.4138858491429015e+20,1.4653142745119542e+20,1.492821760629899e+20,1.5548769915303186e+20,1.5911940994337666e+20,2.12384
14730824948e+20,2.1620342121622297e+20,2.3601125538641527e+20
−1.847890625,1.5333789639403625e+20,1.4628754226425053e+20,1.52461716277975e+20,1.2088894570400586e+20,1.231583000252294e+20,1.29
3553216708463e+20,1.3096368868801874e+20,1.3565010992735176e+20,1.3784401392930256e+20,1.4313563174364696e+20,1.454182695717846
7e+20,1.5117055940591383e+20,1.5533148511172077e+20,1.561508118181892e+20,1.5917838871407575e+20,1.657744469073955e+20,2.21234150
95477223e+20,2.2621015264483508e+20,2.4850997271213297e+20
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2026201234193e+20,1.313356268816166e+20,1.3587539820461675e+20,1.4166753855948854e+20,1.4245392216880972e+20,1.484362823426 93e+
20,1.5313864379032 31e+20,1.5718320597556154e+20,1.598717877749975e+20,1.6289670796950123e+20,1.665443589681431e+20,2.26248409144
74798e+20,2.323917654224315e+20,2.5436056049742732e+20
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2719967416880839e+20,1.3172828734600063e+20,1.3456458174232258e+20,1.3804911127605797e+20,1.4156286452500457e+20,1.471801939288
8539e+20,1.4901119252194e+20,1.5253610391669457e+20,1.5800040732092367e+20,1.6257259040079453e+20,1.6631800801032493e+20,2.259954
9117310147e+20,2.2698856615000772e+20,2.498356667021711e+20
−12.503984375,1.3596200666969738e+20,1.3205240491470733e+20,1.4084342979053103e+20,1.1020103604083312e+20,1.174984633992231e+20,
1.1937462591578602e+20,1.2237138507563164e+20,1.2767044165384798e+20,1.3146952463131184e+20,1.3381060988987205e+20,1.39085224815
36629e+20,1.4141355790728888e+20,1.4498682126721116e+20,1.493709098891768e+20,1.5371514799039978e+20,1.5694729089276517e+20,2.081
4723994289334e+20,2.1303822718870517e+20,2.3371745941246958e+20

A.15 "MITR_Cycle2_cladding_fluence_position_4.csv" File


```
,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.7084114185574842e+19,2.6698148608675586e+19,2.8714903762418487e+19,2.27286116704886e+19,2.3920089106665513e+19,2.4
497868523405963e+19,2.552574628842747e+19,2.540215654009767e+19,2.6196988459816292e+19,2.6116649809999155e+19,2.735759502592458
3e+19,2.823435961629005e+19,2.8455025233148895e+19,2.921303527170133e+19,2.914252641700071e+19,2.8838334108665324e+19,3.802472928
428352e+19,3.7385473797544124e+19,4.083413786261113e+19
37.224453125,3.706656336354779e+19,3.6645688730477978e+19,3.9376193277735174e+19,3.0864175181137547e+19,3.175374507216831e+19,3.2
399695446392476e+19,3.488062946574551e+19,3.4718570681392157e+19,3.4923827430230516e+19,3.457664969352075e+19,3.551845033373818
5e+19,3.626004195774466e+19,3.708728563433395e+19,3.8723016675749675e+19,3.757255870892384e+19,3.878826526171228e+19,5.0929868787
54845e+19,5.160429900059668e+19,5.431891647358448e+19
33.672421875,4.5560834694908035e+19,4.597262340924853e+19,4.780909480715178e+19,3.801112697320338e+19,4.012469232533705e+19,4.122
956129643331e+19,4.187625554704467e+19,4.323473323214703e+19,4.446457343629217e+19,4.587421919002806e+19,4.601656525012073e+19,
4.595965870650025e+19,4.687313890997659e+19,4.769512231782787e+19,4.779862740370338e+19,4.7944905381842665e+19,6.361418327187154
5e+19,6.43171464577715e+19,7.116134056024867e+19
30.120390625,5.797188087499202e+19,5.602930082385806e+19,5.9851231433214345e+19,4.6453858297736495e+19,4.804065289968027e+19,5.02
0033858782007e+19,5.1047401256725365e+19,5.289317110946857e+19,5.5072356985758425e+19,5.489488933338459e+19,5.625990250388873e+
19,5.704734879376307e+19,5.752236700101518e+19,5.829546710342217e+19,5.923540805267162e+19,5.874923169961155e+19,7.7871636913030
63e+19,7.759427728866194e+19,8.579498311721815e+19
26.568359375,6.86528831145694e+19,6.744833470758891e+19,7.142435399715002e+19,5.484706870849344e+19,5.7984101701353095e+19,6.0977
14147926273e+19,6.187351252583357e+19,6.297551225943637e+19,6.5420740212204e+19,6.526930823338201e+19,6.662741398029078e+19,6.846
957071915331e+19,7.008484515992115e+19,6.960876427211589e+19,7.111670797701694e+19,7.118100015048171e+19,9.45530648958946e+19,9.4
63276593737987e+19,1.0515436609398671e+20
23.016328125,8.026532485897113e+19,7.773348844112285e+19,8.269886332565425e+19,6.74701196589282e+19,6.8765527253201895e+19,7.0425
96561747806e+19,7.218151389125995e+19,7.473141587851156e+19,7.478508124644496e+19,7.632437402766352e+19,7.805229260706388e+19,7.9
78605592950645e+19,8.15325714185867e+19,8.329024505347487e+19,8.472539514048602e+19,8.575141321453955e+19,1.1316963416602059e+2
0,1.1403093675433792e+20,1.24266675842151e+20
19.464296875000002,9.159668759706819e+19,8.974071601101488e+19,9.55392324492055e+19,7.629568165272884e+19,7.89050937509561e+19,8.
169728690432288e+19,8.480031411948218e+19,8.782470297370934e+19,8.789005782772725e+19,8.75144002521934e+19,9.028533979449745e+1
9,9.284268054562102e+19,9.455731561810716e+19,9.55828023518841e+19,9.92920888226079e+19,9.973416393271278e+19,1.3150565577011847e
+20,1.31762293123701e+20,1.4217603120416449e+20
15.912265625,1.0594340640469076e+20,1.0250191543335741e+20,1.0859585706532007e+20,8.627744008834225e+19,8.953561866425947e+19,9.1
797002881334448e+19,9.608226221185837e+19,9.794514122150707e+19,9.974479073824414e+19,1.033876596743969e+20,1.0507679041360773e+
20,1.0630737449414009e+20,1.0834825249643921e+20,1.1027223563789325e+20,1.1165850241946015e+20,1.128566747431218e+20,1.491631558
4103855e+20,1.5192346857781123e+20,1.6228938603338357e+20
12.360234375000001,1.1577373286148217e+20,1.1376473527577716e+20,1.206987258849944e+20,9.461629438880624e+19,1.0158694747710669e
+20,1.0360444650723679e+20,1.0748801258867502e+20,1.0879139028709723e+20,1.1181631048160094e+20,1.142849174065376e+20,1.15652587
27842459e+20,1.176498953780451e+20,1.2112007872431312e+20,1.2377412340577213e+20,1.2489843943099079e+20,1.2788829116724103e+20,1.
6940456367691062e+20,1.6777494304867543e+20,1.861242481596875e+20
8.808203125,1.245456294873494e+20,1.2521033617333641e+20,1.3258002580933971e+20,1.052021867188779e+20,1.1067871094946808e+20,1.11
783336738445373e+20,1.1649635563761517e+20,1.2038523512181906e+20,1.2223748732593639e+20,1.253219176314158e+20,1.274881919389850
3e+20,1.3190734901920419e+20,1.3185102694988792e+20,1.3569580519113662e+20,1.3745932356906705e+20,1.3940827970351982e+20,1.845955
8218400032e+20,1.834468245060595e+20,2.0151505061076694e+20
5.256171875,1.3311402278729094e+20,1.3295090132238444e+20,1.4155701978196235e+20,1.1268823987544965e+20,1.1757550773932551e+20,1.
2142506804306338e+20,1.2589789049121589e+20,1.2945521364284113e+20,1.3446468977032775e+20,1.3535043401136721e+20,1.389566404684
3673e+20,1.4240291350225918e+20,1.4570413064057846e+20,1.4645278909026329e+20,1.4981351633955825e+20,1.514606711969202e+20,1.9750
5556883782e+20,2.0044068057154565e+20,2.2108643835788627e+20
1.704140625,1.4198315468377016e+20,1.3912985739859801e+20,1.4964720683299235e+20,1.2283099441486325e+20,1.270354900233488e+20,1.3
176069910287129e+20,1.3462249783246855e+20,1.3849809380975824e+20,1.4063726976322249e+20,1.4298207440371871e+20,1.4757019769188
658e+20,1.5058236571975267e+20,1.537071778862513e+20,1.5710191091324656e+20,1.6094403245311243e+20,1.63357379989286e+20,2.1612425
151501428e+20,2.1413810156120172e+20,2.3527641178392117e+20
−1.847890625,1.4636777464601235e+20,1.4634014495163079e+20,1.5579215713150553e+20,1.2449462082079883e+20,1.2926977588631878e+20,
1.3530527008785888e+20,1.384598373098453e+20,1.4147253667798796e+20,1.4496981837836096e+20,1.5009831472779887e+20,1.514325101622
6207e+20,1.5663220610876015e+20,1.5896585260344843e+20,1.6272136567823365e+20,1.6767877045861656e+20,1.6856770274131545e+20,2.235
2210218567565e+20,2.2210926839028036e+20,2.4545317210103516e+20
−5.399921875,1.505154168490505e+20,1.48039902496373e+20,1.595338553591e+20,1.2757373772351252e+20,1.3230213484469445e+20,1.35669
23817730821e+20,1.4115904591481263e+20,1.4305486802160856e+20,1.48987813549771e+20,1.5231453502136556e+20,1.547395720436236e+20,
1.585413117224703e+20,1.632580193575677e+20,1.660799675664223e+20,1.659997351846605e+20,1.6959531483619865e+20,2.23940266983335e
+20,2.2951243246370747e+20,2.4586708617648194e+20
−8.951953125,1.4721154300520292e+20,1.4559945660609449e+20,1.5395690781623838e+20,1.2291919690077363e+20,1.3011726362744532e+20,
1.3409328291700644e+20,1.3958096529340459e+20,1.413843341920776e+20,1.4458672203895518e+20,1.494070410279834e+20,1.5441970519712
943e+20,1.5657694671999703e+20,1.6284782466405692e+20,1.6481218966653016e+20,1.7043430113290001e+20,1.7072972632667213e+20,2.27585
7926208706e+20,2.2569581525711684e+20,2.4643615161268666e+20
−12.503984375,1.3407787404898596e+20,1.3356141130016152e+20,1.4337845425003869e+20,1.1710102087234997e+20,1.2106960139803913e+20
,1.2319761920569544e+20,1.2966403037153249e+20,1.3332177683542915e+20,1.372319099306958e+20,1.402456719793916e+20,1.443513382964
354e+20,1.4742620447693654e+20,1.5201326508455127e+20,1.564749293868959e+20,1.587751014441604e+20,1.59782522608534e+20,2.11355472
53281312e+20,2.1284428798775768e+20,2.314241947788005e+20
```

A.16 "MITR_Cycle2_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.2647426822455075e+19,2.428588139336583e+19,2.615206892236027e+19,2.9023230397450002e+19,2.9560511154460692e+19,3.1
91534006805237e+19,3.0464160566896165e+19,3.144494968641249e+19,3.3673909614091354e+19,3.368137184682761e+19,3.498753229492855e
+19,3.4980789313540366e+19,3.6770466480383525e+19,3.811555640769893e+19,3.700035719251142e+19,3.6578785996122018e+19,3.550845008
3770573e+19,3.5173548674823987e+19,3.365242198006767e+19
37.224453125,3.042613015186679e+19,3.342109276524362e+19,3.474757206392763e+19,3.8784190442151485e+19,4.195896589254661e+19,4.172
053407066034e+19,4.420069253165377e+19,4.341455080820976e+19,4.407203644676718e+19,4.543645625406205e+19,4.997061675231554e+19,4.
474453645721562e+19,4.819325676480815e+19,4.82076417917696e+19,4.8490667197236404e+19,5.044010806977032e+19,4.746276711442129e+
19,4.753010702188465e+19,4.674441483053317e+19
33.672421875,3.643178900185955e+19,4.228856282280115e+19,4.606939744036648e+19,4.707383194795071e+19,4.995614181893557e+19,5.2657
82969513554e+19,5.179203088489243e+19,5.459827992581855e+19,5.7173469471176155e+19,6.010513796592255e+19,5.8024433722365256e+19,
6.1041333501858324e+19,6.248891674627416e+19,6.2779314478058725e+19,6.178593845995112e+19,5.98217529347817e+19,6.079328169319158
e+19,6.141507448360087e+19,5.87871997769969e+19
30.120390625,4.596906187731026e+19,5.039227785512346e+19,5.571338923458754e+19,5.826799021010656e+19,5.984980373735656e+19,6.4091
6784690368e+19,6.580466546089164e+19,6.568805683608528e+19,6.926732126335307e+19,7.15617330637066e+19,7.260222004511301e+19,7.285
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7.34864496711505e+19,7.234562712668789e+19
26.568359375,5.566942499593657e+19,6.083895415379423e+19,6.604534494324063e+19,6.935219292242571e+19,7.266578388299897e+19,7.6057
77324051222e+19,7.742264257989963e+19,8.019346849193316e+19,8.177725996039034e+19,8.618159559031662e+19,8.591187633478917e+19,8.9
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23.016328125,6.315665171654162e+19,7.476240156263237e+19,7.722071276392807e+19,8.218858182506971e+19,8.465786160942354e+19,8.9380
55586729071e+19,9.153372468416637e+19,9.656938318486389e+19,9.900584712646189e+19,9.939694004697668e+19,1.0080667268920017e+20,1.
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6011e+20,1.0869236465663779e+20,1.0507453037582955e+20
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1.0475805978267735e+20,1.0658765539933856e+20,1.0854312000191259e+20,1.1252597434186798e+20,1.1711389887838991e+20,1.20096094780
33845e+20,1.2329226795833876e+20,1.2399353802271015e+20,1.2360514229475061e+20,1.2562084419772578e+20,1.2558937695124759e+20,1.23
34711087362936e+20,1.1904598781215157e+20,1.1926266228075864e+20
15.912265625,8.548410159552263e+19,9.107969727069515e+19,1.001377689354921e+20,1.0668205713877316e+20,1.1299618491067082e+20,1.16
07368161623905e+20,1.2155797314529726e+20,1.2569726465345854e+20,1.3087227810284526e+20,1.3196194389517617e+20,1.34356151820074
86e+20,1.3644737511459771e+20,1.3647794329689081e+20,1.4407413659672897e+20,1.4335488524865577e+20,1.440903197520606e+20,1.406747
7491289796e+20,1.3949250550950265e+20,1.372942935769539e+20
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+20,1.3124898599639861e+20,1.3712257231760143e+20,1.4195594137665338e+20,1.4146505233159342e+20,1.4736561057834897e+20,1.5002863
869459002e+20,1.552818707280797e+20,1.57112365408926e+20,1.5754301715358482e+20,1.6254900653617434e+20,1.6115905330602285e+20,1.6
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78270543e+20,1.7616623268357033e+20,1.7216359893154293e+20
5.256171875,1.1000859462361175e+20,1.1813613485683897e+20,1.2985453744532169e+20,1.4126186382576273e+20,1.494990898895711e+20,1.5
225921693780204e+20,1.5990126251107985e+20,1.6316576456714712e+20,1.6986648993863415e+20,1.7510263975260707e+20,1.8307194468925
82e+20,1.8309172410133017e+20,1.8481163388741023e+20,1.914844882691594e+20,1.896521954599429e+20,1.9456198497472764e+20,1.8836923
086781733e+20,1.912336493615189e+20,1.8738565464932722e+20
1.704140625,1.1645668295908803e+20,1.3022495188957938e+20,1.3963905297167255e+20,1.5136914339456146e+20,1.6077695102735896e+20,1.
6596095511859662e+20,1.6976489568571877e+20,1.7379180417074364e+20,1.8340280030937186e+20,1.8795116602174977e+20,1.950259020942
349e+20,1.9470943150108262e+20,1.992586962776457e+20,2.0154501650033338e+20,2.0355802121075327e+20,2.0329279727615125e+20,2.05233
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043559756779704e+20,1.7688098871071808e+20,1.771525060946157e+20,1.854949226680798e+20,1.9280701168542902e+20,1.951230010262247
5e+20,2.033323561002953e+20,2.057796088121144e+20,2.121413869858219e+20,2.1171433149790342e+20,2.1567381016904643e+20,2.110993715
9530083e+20,2.1284355611437836e+20,2.1037832211885746e+20
−5.399921875,1.2050696711292528e+20,1.3389852814986327e+20,1.44683702114221e+20,1.5534120896429572e+20,1.6285738555166074e+20,1.7
02279137410409e+20,1.8150847207138404e+20,1.8679946480064754e+20,1.9321608588964567e+20,1.96196483663224e+20,1.9818161738390608
e+20,2.052779309968333e+20,2.0745186819638457e+20,2.1360416441496578e+20,2.1175209219367725e+20,2.1459313501856642e+20,2.15600985
97005404e+20,2.1461021723808317e+20,2.1100766704842152e+20
−8.951953125,1.1535263213979566e+20,1.3133080083724191e+20,1.4220857841266408e+20,1.531510886094128e+20,1.623754871484517e+20,1.7
112517979776224e+20,1.7713722200346916e+20,1.8144463851424255e+20,1.8721123619741948e+20,1.8896620948671814e+20,1.9872015683077
584e+20,2.0311208537494785e+20,2.0590098247710173e+20,2.121144150602691e+20,2.1433869985418553e+20,2.1508941844873698e+20,2.14909
60561171867e+20,2.1208384687797604e+20,2.1163431478543024e+20
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,1.5644166352684774e+20,1.6018087147264334e+20,1.636018106969165e+20,1.7141737565791696e+20,1.768027701266151e+20,1.833839199614
8492e+20,1.8225829160175036e+20,1.8724000625134243e+20,1.9148358920497434e+20,1.9210574162105768e+20,1.9804136337103176e+20,1.940
270417845982e+20,1.9254088868664194e+20,1.895128405112537e+20

A.17 "MITR_Cycle2_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.7865981402545205e+19,2.7484532217996767e+19,2.9103579174728245e+19,2.286187181185195e+19,2.4249679480220873e+19,2.
490965723774645e+19,2.5804168593349296e+19,2.6177488271666233e+19,2.6772376845312184e+19,2.617695693138967e+19,2.79164587288191
96e+19,2.8540571017676403e+19,2.9000020954825073e+19,2.8523143056604963e+19,2.932089734784471e+19,2.977599029472552e+19,3.6833995
72449381e+19,3.701587350116316e+19,4.096697293175322e+19
37.224453125,3.836404318490009e+19,3.73845173850463e+19,3.843279861668803e+19,3.1455981981179396e+19,3.1800503016506323e+19,3.376
221131759675e+19,3.437989438910748e+19,3.420354255131444e+19,3.5230782708004065e+19,3.539613580207214e+19,3.5180730453951324e+1
9,3.726730372003532e+19,3.749445168826829e+19,3.825517156223123e+19,3.745231640433642e+19,3.852121363870901e+19,5.03208465625457
9e+19,4.974721159996257e+19,5.3060596430615265e+19
33.672421875,4.956985021564405e+19,4.6634513991769735e+19,4.822672826453452e+19,3.839831463273874e+19,4.110299604255472e+19,4.075
252399613023e+19,4.211780283677265e+19,4.412217776207151e+19,4.477466362169748e+19,4.544930637085634e+19,4.570519984805166e+19,4.
495643513031152e+19,4.676113237967598e+19,4.879032089589059 6e+19,4.815568806955733e+19,4.857645643457182e+19,6.278104171821234e
+19,6.468749063053965e+19,7.071023266544213e+19
30.120390625,5.903509276840533e+19,5.836666670048232e+19,5.9764622968133706e+19,4.768927757478561e+19,4.950199806232626e+19,5.180
408294458638e+19,5.381148650946168 4e+19,5.27849370951316e+19,5.437417586234759 6e+19,5.480296746553826 5e+19,5.641611654519984e+1
9,5.670197761399362e+19,5.772746434777057e+19,5.811268604828264e+19,5.806274006228523e+19,5.971414564185971e+19,7.81633427248666
5e+19,7.703318195660574e+19,8.514781066035788e+19
26.568359375,7.102372342861747e+19,6.8384024934625804e+19,7.2629433744407085e+19,5.665150028771963e+19,5.8473997436349145e+19,5.9
988831722456899e+19,6.041870284858937e+19,6.379271360479852e+19,6.480226013027842e+19,6.557429755213226e+19,6.838030555268982e+1
9,7.056092604772642e+19,6.938081929346806 6e+19,6.917253390505327e+19,7.103594425497855e+19,7.204389675962873 4e+19,9.500682949208
4e+19,9.669755425212454e+19,1.0305025859877598e+20
23.016328125,8.042685230304792e+19,7.904589892424671e+19,8.435664498854758e+19,6.639203023777099e+19,6.778467310265668e+19,7.0933
92692187741e+19,7.281593418148258e+19,7.438232531680613e+19,7.562353620286985e+19,7.765166203853133e+19,7.894600695225187e+19,8.0
17287165084823e+19,8.206709973681447e+19,8.3291307734028e+19,8.516534488948464e+19,8.718496928072101e+19,1.1396186251838405e+20,
1.1125893453148073e+20,1.2208871204849726e+20
19.464296875000002,9.358496291198702e+19,9.170986307597725e+19,9.534263654687521e+19,7.643967486767887e+19,7.970157282553209e+19
,8.033864981713758e+19,8.390553709374104e+19,8.639699165057013e+19,8.915730438734281e+19,8.96881133236346e+19,9.113920361894281e+
19,9.338783566938017e+19,9.48607109160277e+19,9.652911938445237e+19,9.86805161642777e+19,9.9770826411796e+19,1.3021077951612137e+
20,1.3077346886900725e+20,1.4162024927487397e+20
15.912265625,1.0771276952566343e+20,1.0361082259055562e+20,1.098636349652122e+20,8.686244573284401e+19,9.200263156836644e+19,9.35
3714228709587e+19,9.74573708476173e+19,9.877190669184747e+19,9.996582829329659e+19,1.051899658925168e+20,1.0492217039312634e+20,
1.072680377141757e+20,1.0863039418329699e+20,1.101404632493043e+20,1.1344965049177214e+20,1.131977952006787e+20,1.50180672470666
97e+20,1.4972584519392444e+20,1.600519121287542e+20
12.360234375000001,1.1980979360229551e+20,1.1612707414540014e+20,1.2207914792351901e+20,9.780327336766331e+19,1.0319372047344945
e+20,1.0535574405880614e+20,1.0689716220113099e+20,1.1024566862406484e+20,1.122759198208326e+20,1.1532900104999446e+20,1.17189223
35826031e+20,1.216328220912016e+20,1.221413147358775e+20,1.2496645098639157e+20,1.259430544147242e+20,1.2763005979282878e+20,1.66
1830475800766e+20,1.6763573189621452e+20,1.8262802913986763e+20
8.808203125,1.2885533047059544e+20,1.2819221780543812e+20,1.3324260713422048e+20,1.0687325188868542e+20,1.1012346036045411e+20,1.
1398099076834048e+20,1.1700537962256766e+20,1.192959875548539e+20,1.2330335592073257e+20,1.2562106220712378e+20,1.2918741814345
068e+20,1.308871756881929e+20,1.3438339470801276e+20,1.3581004335059884e+20,1.3794656260268024e+20,1.3943165867568885e+20,1.85322
45568234583e+20,1.847602976697365e+20,1.971155531207808e+20
5.256171875,1.3791521352636272e+20,1.3489188735268872e+20,1.4373073285340352e+20,1.1444697619089095e+20,1.1820620864761217e+20,1.
2355999327431506e+20,1.2575602263737213e+20,1.3122670212492008e+20,1.3337916158529854e+20,1.3558262971222757e+20,1.389210406699
0665e+20,1.4440287830326254e+20,1.4648998290962309e+20,1.497641016938373 8e+20,1.508119047192302e+20,1.5056589417117906e+20,1.9860
383723544884e+20,2.01974128609722e+20,2.1591862282798232e+20
1.704140625,1.4621368596580753e+20,1.443816246921998e+20,1.5303609511694542e+20,1.2297179958815387e+20,1.2845363722150974e+20,1.3
300084730838185e+20,1.3556828352476024e+20,1.3797791167899779e+20,1.4257984981435644e+20,1.4486301898277074e+20,1.4806328146854
203e+20,1.5216044634116073e+20,1.5477570318243027e+20,1.5674910096960517e+20,1.6113372093184737e+20,1.6479465543740346e+20,2.1830
008994756172e+20,2.1373587697183946e+20,2.3205117630515115e+20
−1.847890625,1.514112565511993e+20,1.5074920656659513e+20,1.5876181793724616e+20,1.2754717070968413e+20,1.3064807256373712e+20,1.
3488232322771046e+20,1.3841733008771985e+20,1.4310534534788255e+20,1.4648520084713395e+20,1.5087991627463095e+20,1.5314183183 19
825e+20,1.5892653342298237e+20,1.6035265072529185e+20,1.6392697676576725e+20,1.6645031173919048e+20,1.696723591763011e+20,2.21781
43143963774e+20,2.2316132213788577e+20,2.4239902819132018e+20
−5.399921875,1.5363013354614886e+20,1.523543855421082e+20,1.596241832061166e+20,1.2772251300095168e+20,1.3225325153925017e+20,1.3
714211342395576e+20,1.4108997167885876e+20,1.4625725586848612e+20,1.4864297371027813e+20,1.5193622074444 89e+20,1.54922353098763
1e+20,1.5999718408026 76e+20,1.6364855446084544e+20,1.6625212181603048e+20,1.6873560626871104e+20,1.7028765121656724e+20,2.2481113
369663055e+20,2.2533556654960352e+20,2.474239131868272e+20
−8.951953125,1.5068703975423662e+20,1.4680453635335157e+20,1.5625707987350287e+20,1.2650202438567413e+20,1.3146846195075868e+20,
1.3566605013564883e+20,1.3802945168582494e+20,1.4409895166506541e+20,1.4686404646432722e+20,1.518299526891352e+20,1.542932462113
0616e+20,1.584520465560068e+20,1.6311455748289426e+20,1.6560122997723424e+20,1.7260801420434132e+20,1.7073344570860 81e+20,2.26735
64817836117e+20,2.2540729748694024e+20,2.434388611125645e+20
−12.503984375,1.3853263092773518e+20,1.3497477643583337e+20,1.455665135089 4726e+20,1.188374408961754e+20,1.2135068040434378e+20,
1.2558014900582801e+20,1.2868264488071073e+20,1.3498699726219443e+20,1.3807833499126922e+20,1.418917641562004e+20,1.443954395393
9058e+20,1.479963325936944e+20,1.5172049659216213e+20,1.5633306153305214e+20,1.599015428304853e+20,1.622086223113452e+20,2.133804
103268152e+20,2.129675589319216e+20,2.2525480282756514e+20

A.18 "MITR_Cycle2_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9502350052292817e+19,2.807862378122786e+19,2.8636584205652312e+19,2.291341181867908e+19,2.328604075463648e+19,2.40
22797182126182e+19,2.541767167617347e+19,2.60800404649436e+19,2.4835110196943905e+19,2.620979376048159e+19,2.6344222850453385e+
19,2.688688067491267e+19,2.7588356108038226e+19,2.793909382460101e+19,2.8503642868454904e+19,2.8946196184808698e+19,3.7982594000
35166e+19,3.742516491620378e+19,3.875983855691586e+19
37.224453125,4.007628722614166e+19,3.721151299099564e+19,3.929367613278411e+19,3.0826768825667133e+19,3.0260360090845245e+19,3.30
06539176261214e+19,3.2743844543525814e+19,3.2551499363408056e+19,3.361704915403827e+19,3.434413518849443e+19,3.42122033978225e+
19,3.5419567908268806e+19,3.6467370933661635e+19,3.670822748103008e+19,3.743823588700736e+19,3.8888688573983695e+19,5.0896553752
207606e+19,5.092811536463577e+19,5.663874812108199e+19
33.672421875,4.960045541557438e+19,4.665029479798382e+19,4.878963015353106e+19,3.788126740961007e+19,3.889798702882365e+19,4.0394
98512402737e+19,4.166749195238096e+19,4.0907675356888195e+19,4.162726949344473e+19,4.370555385121426e+19,4.359312224869239e+19,4.
488140988326006e+19,4.450394575078589e+19,4.60260762410713e+19,4.724497083551914e+19,4.879754712365192e+19,6.455943762388668e+1
9,6.389526227817621e+19,6.980907955638217e+19
30.120390625,5.994049659967784e+19,5.719027932815996e+19,5.891022780341176e+19,4.613893291581442 5e+19,4.728875327430838e+19,4.820
366809653146e+19,4.931448807872529e+19,5.215115441324085e+19,5.25996587406922e+19,5.300523077379684e+19,5.4353984931837985e+19,5.
507926440935382e+19,5.688794671079255e+19,5.721578366143522e+19,5.931776579553 9706e+19,5.883690284524533e+19,8.03365244560313e+
19,7.869733970281787e+19,8.457608852277032e+19
26.568359375,7.251997764743399e+19,6.796373477586022e+19,7.210712625254038e+19,5.543366837382492e+19,5.678486669713829e+19,5.8950
60966443096e+19,6.040701336250486e+19,6.060520328566487e+19,6.253184312850178e+19,6.360302512606361e+19,6.463010588067028e+19,6.6
78947276464412e+19,6.835639524024425e+19,6.778307908182697 6e+19,6.946052033495332 5e+19,7.128301748358284e+19,9.564709452534889e
+19,9.455412757644773e+19,1.0392750139539035e+20
23.016328125,8.259046990923432e+19,7.853687493929422e+19,8.153310275886326e+19,6.63532423975815e+19,6.7236861277514695e+19,6.7485
5285269487e+19,7.029631858999538e+19,7.105772920631785e+19,7.360816253384602 4e+19,7.567507620969698e+19,7.499974271817859e+19,7.8
33443429392168e+19,7.878554218872822e+19,8.033280507409531e+19,8.345974260170018e+19,8.485876154990469e+19,1.127599708127864e+2
0,1.1290130732635356e+20,1.247321299244249e+20
19.464296875000002,9.924479953799332e+19,9.1832071339588e+19,9.506049486001739e+19,7.537008689094674e+19,7.68844066791666e+19,7.8
46886338389348e+19,8.091993607970336e+19,8.27169288950576e+19,8.486194959156409e+19,8.821311271588082e+19,8.961585104602132e+19,
8.852500945822645e+19,9.224757943586444e+19,9.379696768233782e+19,9.553870110892894e+19,9.8333019623402e+19,1.3331752611321663e+
20,1.3388499752859165e+20,1.4498150786444549e+20
15.912265625,1.1119304833718626e+20,1.0565595131506727e+20,1.0724519008228327e+20,8.482581845275745e+19,8.800217062608316e+19,9.0
6004245785025e+19,9.201432105445093e+19,9.361843734941083e+19,9.703229862636262e+19,9.909761828138389e+19,9.982077239779343e+19,
1.0045253598663321e+20,1.025327331693984e+20,1.0630099841082127e+20,1.0921114910558623e+20,1.1159899230848449e+20,1.498762144921
933e+20,1.5082253152476155e+20,1.6276971764340136e+20
12.360234375000001,1.196333886304748e+20,1.146903300375592 6e+20,1.2086078466934774e+20,9.628364017667778e+19,9.664867094668026e+
19,1.0047538361852566e+20,1.032585439871908e+20,1.0517987042726201e+20,1.0763200580362504e+20,1.1116648132335785e+20,1.136260554
6359282e+20,1.1560954871602253e+20,1.157763895628650 1e+20,1.2017004230980887e+20,1.2286712555366993e+20,1.2792601632687736e+20,1.
6740353619535413e+20,1.71082004930037e+20,1.844733739203896e+20
8.808203125,1.3388340350776194e+20,1.2648874087875992e+20,1.3243125053190057e+20,1.0404439625623539e+20,1.0612140539734116e+20,1.
0990667352761424e+20,1.1078179096312232e+20,1.159517318541326e+20,1.155920144868958e+20,1.2121625131437199e+20,1.24052014370417
36e+20,1.2727884387001708e+20,1.3058856245276146e+20,1.3199130078290197e+20,1.3877598477440347e+20,1.3724466209733344e+20,1.84852
21953758285e+20,1.8550683075831507e+20,2.0070156864734075e+20
5.256171875,1.4018616186841594e+20,1.3531749091421995e+20,1.3958149663368113e+20,1.0981315763893821e+20,1.1616214260365366e+20,1.
1664459957477774e+20,1.2018066911534024e+20,1.2288306576196695e+20,1.282363190483933e+20,1.3171766054046925e+20,1.3251520229559
838e+20,1.3625955722457568e+20,1.3911710523196038e+20,1.4279079190415409e+20,1.460410003759228e+20,1.5017589040817786e+20,2.01577
748763402e+20,1.9872445147822988e+20,2.177783137959715 8e+20
1.704140625,1.4894796302902837e+20,1.4551497350212013e+20,1.4761961233760744e+20,1.1818814307820883e+20,1.2352651883689127e+20,1.
2717523251608627e+20,1.2943077199011899e+20,1.3205559295636675e+20,1.3385258577172097e+20,1.3824783253949451e+20,1.408694654640
8286e+20,1.4246720567572395e+20,1.4580880467506243e+20,1.5234269605602368e+20,1.5519758736202554e+20,1.590444909643805e+20,2.1213
06979963264e+20,2.139834815407203e+20,2.3105756998796827e+20
−1.847890625,1.5520874550783348e+20,1.4859196504372753e+20,1.5538568181993072e+20,1.2204142276388268e+20,1.2750094410562268e+20,
1.2994617205839032e+20,1.3288129574615399e+20,1.3722978456958953e+20,1.3907619203066462e+20,1.4351022663862768e+20,1.46951717609
961e+20,1.505148855046285e+20,1.546370233702459e+20,1.576072155162631e+20,1.619094777356372e+20,1.674662343479892e+20,2.229753530
4108684e+20,2.218791980505263e+20,2.450844219490967 2e+20
−5.399921875,1.5545794409754404e+20,1.5100530517412e+20,1.586884929790797e+20,1.2300208598391828e+20,1.2780115136188378e+20,1.30
79472248006998e+20,1.3478349393626874e+20,1.3841201668495414e+20,1.4103736899147848e+20,1.435059759164151e+20,1.480515919824575
2e+20,1.5431024910015634e+20,1.5687715397625817e+20,1.592315227417326e+20,1.643010403204714e+20,1.699858499394764e+20,2.260332163
3273777e+20,2.2971593578963318e+20,2.4876926676709835e+20
−8.951953125,1.5287403633259207e+20,1.4886135456394772e+20,1.5322472091512717e+20,1.2093889169000033e+20,1.2607323278248346e+20,1.
2962364851051332e+20,1.325619602399364e+20,1.3537540700436591e+20,1.3854644577492597e+20,1.4299110718842038e+20,1.4867379144631
91e+20,1.4991447099210623e+20,1.5368592427518853e+20,1.5843610634770976e+20,1.6472930058338553e+20,1.6505501217292196e+20,2.24280
32476033876e+20,2.2777760646071176e+20,2.4856629478144924e+20
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,1.2006802497670773e+20,1.2419919562702681e+20,1.2812154954865451e+20,1.3128196151368319e+20,1.3597369615578189e+20,1.3933070402
314086e+20,1.4252459042559335e+20,1.4594482778586392e+20,1.5003827327654668e+20,1.5336977681063035e+20,1.5896372724234217e+20,2.1
08204128743088e+20,2.1330230330615965e+20,2.3166967398657507e+20

A.19 "MITR_Cycle2_cladding_fluence_position_8.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.8848801512113713e+19,2.7493299332560146e+19,2.928875126111233e+19,2.2760438953055048e+19,2.2291956231204717e+19,2.
426524775032433e+19,2.4703709746548548e+19,2.418506850259016e+19,2.5248386664058778e+19,2.605506747194488e+19,2.558950712161567
e+19,2.7424915838965797e+19,2.7161530463870857e+19,2.6640710724778545e+19,2.7078481978643223e+19,2.749096143534325e+19,3.72054025
778151e+19,3.604660256864714e+19,3.904362739863104e+19
37.224453125,3.903937667641849e+19,3.696528990683386e+19,3.9251912787045835e+19,3.0554935140174754e+19,3.1251628510811197e+19,3.1
195518977605583e+19,3.1884348512148816e+19,3.2360429399954076e+19,3.217860475731238e+19,3.3054200399069397e+19,3.40985497126645
27e+19,3.462308883369282e+19,3.4398969505036284e+19,3.5952289669556257e+19,3.6197556341220217e+19,3.707740270518978e+19,4.9205244
517862834e+19,5.017047726427693e+19,5.3639863600130105e+19
33.672421875,4.8933038894176846e+19,4.649912848930011 5e+19,4.803533949691459e+19,3.711704068982178e+19,3.766607459759988e+19,3.82
2180339286273e+19,3.8315213013483454e+19,4.00948310017939e+19,4.089370110761444e+19,4.264228882377329e+19,4.273718619716 84e+19,4.
420193193758443e+19,4.453582616737999e+19,4.438078107467733e+19,4.665422671603043e+19,4.721510951197599e+19,6.277307161406382e+
19,6.135651843673255e+19,6.720816890258001e+19
30.120390625,6.087087342394905e+19,5.573972037312 83e+19,5.799897922909702e+19,4.54435678958694e+19,4.641182928185994e+19,4.853357
7274252755e+19,4.854802972977541e+19,4.835504694132578e+19,5.140106134480928e+19,5.269041165993009e+19,5.294306396143835e+19,5.44
4059339691864e+19,5.4221681202972475e+19,5.548839642231147e+19,5.7338523265322525e+19,5.861214590825691e+19,7.650928044390932e+
19,7.504065591947434e+19,8.395442039918533e+19
26.568359375,7.123200881703227e+19,6.903969883591117e+19,6.937656857125553e+19,5.416589047393279e+19,5.590018513665195e+19,5.6286
46951771715e+19,5.689326011355823e+19,5.906697318499944e+19,5.9794909363898106e+19,6.167797930405641e+19,6.34319335570086e+19,6.4
399504200639594e+19,6.6309141154626314e+19,6.732878314536102e+19,6.8981782745765224e+19,7.035370333986476e+19,9.337667752357223
e+19,9.148138675705286e+19,1.0006890830695085e+20
23.016328125,8.258834454812803e+19,7.7307353539315e+19,8.130303241910916e+19,6.3372954786309505e+19,6.480704219276755e+19,6.74292
0645763245e+19,6.709605610422408e+19,6.88882668570892e+19,7.049716521453822e+19,7.252422836964655e+19,7.41426908520738e+19,7.6540
0981799503e+19,7.62239507153921e+19,8.019571928274067e+19,8.010911081766003e+19,8.09459717532552e+19,1.08579916857023e+20,1.10856
1786018419e+20,1.1962116780411373e+20
19.464296875000002,9.707480584848808e+19,8.896974126971417e+19,9.243567389376966e+19,7.3193185777846125e+19,7.498114580849869e+1
9,7.713626197026e+19,7.796515280170667e+19,8.106339795437683e+19,8.203203127856097e+19,8.205115952851742e+19,8.325836463688076e+1
9,8.699581214226267e+19,9.006164553806217e+19,9.25589448379335e+19,9.497494907548988e+19,9.581924877495704e+19,1.290248280188207
4e+20,1.2793132972964307e+20,1.3696623979241162e+20
15.912265625,1.0812774628166332e+20,1.0067091684030282e+20,1.0522662837160003e+20,8.121482993319882e+19,8.47679023626115e+19,8.81
009999917524 9e+19,8.79681648483828e+19,8.989852407315569e+19,9.281664487206917e+19,9.35488317731804e+19,9.588194692759211e+19,9.98
1545899502774e+19,1.0145039302602862e+20,1.0381060653454533e+20,1.0662086525731542e+20,1.074481620679324e+20,1.4494272002425599
e+20,1.4226157698869197e+20,1.6027082432270035e+20
12.360234375000001,1.1933371271449025e+20,1.1176636449560353e+20,1.1631038654081624e+20,9.199094208228193e+19,9.33740208221894e+
19,9.678310003665206e+19,9.865873121293841e+19,1.017064990393346e+20,1.030263482863304e+20,1.0702680922861365e+20,1.088960643215
812e+20,1.1114044564980598e+20,1.1400171303912666e+20,1.1718869201798375e+20,1.2085919064851805e+20,1.2268806388046637e+20,1.6429
732093853545e+20,1.6570165328950565e+20,1.7955794502185556e+20
8.808203125,1.2940473631656713e+20,1.2393299414846605e+20,1.2970653759365797e+20,1.0035476937574466e+20,1.022755644755393e+20,1.0
5727682252404e+20,1.0812030751779137e+20,1.1091887675447696e+20,1.1530721609865514e+20,1.1771259353068015e+20,1.196004455333275
9e+20,1.2027471634429284e+20,1.2516623493038126e+20,1.2796427282679028e+20,1.3316078073162896e+20,1.3527976575458363e+20,1.802008
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5.256171875,1.408838116515502e+20,1.3371549998036633e+20,1.385724814484778e+20,1.0956980379226989e+20,1.1111759801791354e+20,1.13
94804767119324e+20,1.1797348160647524e+20,1.1851279198719212e+20,1.2123591090460502e+20,1.2502064769460152e+20,1.27402646154457
51e+20,1.3089780249372425e+20,1.3463206195744676e+20,1.3833390966429858e+20,1.4452136718493727e+20,1.4644906970832732e+20,1.93295
2165322542e+20,1.9397745744736803e+20,2.1335331197271022e+20
1.704140625,1.4813129302394277e+20,1.3921859122478495e+20,1.4606544202864494e+20,1.1374135630360815e+20,1.1679231217166375e+20,1.
1965464224153754e+20,1.2239635806863032e+20,1.2554614322812764e+20,1.2785587941037035e+20,1.3326067270362379e+20,1.375475260549
774e+20,1.3744816542325914e+20,1.4398365082505008e+20,1.4796392083682374e+20,1.5199838755680738e+20,1.5310198131123988e+20,2.0557
39589834727e+20,2.0827157356761032e+20,2.252553341678417e+20
−1.847890625,1.516684252450584e+20,1.4324189979896064e+20,1.490446669593638e+20,1.1790440737052133e+20,1.2131561194609028e+20,1.2
523105844412259e+20,1.2831708277043169e+20,1.3130959120806476e+20,1.3436001573584375e+20,1.3769683267269313e+20,1.4253574857140
127e+20,1.4400862381804878e+20,1.4962648456220616e+20,1.5441439179436376e+20,1.573617363084885e+20,1.613919523062596e+20,2.154738
9101649455e+20,2.1556740690517066e+20,2.3441298383449758e+20
−5.399921875,1.5427199260024344e+20,1.4671633386744121e+20,1.53452665893775e+20,1.1890173306964017e+20,1.2258285850570582e+20,1.2
372949082254038e+20,1.2926339980299995e+20,1.336591779110501e+20,1.3443440337456334e+20,1.3954483415459791e+20,1.42066575107191
42e+20,1.4756435294884433e+20,1.5144048026641 06e+20,1.5434266085702704e+20,1.594344947273817e+20,1.6193391938835933e+20,2.1984947
819403513e+20,2.2192808135597054e+20,2.3964084081565375e+20
−8.951953125,1.510605719686642e+20,1.436908823326609e+20,1.4787624969118997e+20,1.1587149947237075e+20,1.1911586320109722e+20,1.2
232728383267645e+20,1.2535751742994586e+20,1.2887073933861593e+20,1.3336959746032032e+20,1.3618463824557954e+20,1.4152248266398
54e+20,1.4369566439515002e+20,1.4970937364535083e+20,1.5084006575388833e+20,1.5881548330517955e+20,1.6058272106504598e+20,2.16379
82618804368e+20,2.2124158971864423e+20,2.4076621952142557e+20
−12.503984375,1.3558528641361043e+20,1.3318894176628707e+20,1.3789821063751256e+20,1.1035831276269733e+20,1.1217762186966743e+20
,1.1679337485221686e+20,1.1953562201958621e+20,1.229059133938594e+20,1.2650733778843982e+20,1.2849455042280553e+20,1.35195282650
60922e+20,1.3857779485124347e+20,1.4190239096173176e+20,1.438555978183971e+20,1.4954040743740208e+20,1.5354246240051508e+20,2.027
6370026070262e+20,2.0730984766702158e+20,2.228207330206054e+20

A.20 "MITR_Cycle2_cladding_fluence_position_9.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.955022381121162e+19,2.7036612364849635e+19,2.793920009265632e+19,2.1997965656179438e+19,2.250226071267048e+19,2.32
8264017686644e+19,2.349018168894046e+19,2.311792469113024 5e+19,2.399607076621479e+19,2.57789299302123e+19,2.471949055276263e+1
9,2.5704914229686325e+19,2.55185731946938e+19,2.5858949775863493e+19,2.655484613608509e+19,2.7598770377745897e+19,3.5400120854146
404e+19,3.4154712379897803e+19,3.6070140942899126e+19
37.224453125,3.862280589958889e+19,3.517717047409832e+19,3.7416769739834e+19,2.9724344019843076e+19,2.9413244287912296e+19,3.0452
917807073624e+19,2.9920939922173374e+19,3.129982107389595e+19,3.0843452910351376e+19,3.279081502397445 3e+19,3.2871950684206445e
+19,3.2916955205631783e+19,3.523088897605938e+19,3.4387598823117718e+19,3.463371563922419e+19,3.5554528338517176e+19,4.690794169
809183e+19,4.607363119582417e+19,4.8296493242847945e+19
33.672421875,4.7910208861782745e+19,4.485096408541269e+19,4.743997271701973e+19,3.733781257473594e+19,3.6823050114796507e+19,3.72
8005588667295e+19,3.840235281884067e+19,3.93762995457905e+19,4.062558680405805e+19,4.058897745900248e+19,4.105873539751658e+19,4.
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19,5.614194496249056e+19,6.032093623770078e+19
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4636815978671e+19,4.675268406927854e+19,4.79655213845735e+19,5.003758906110718e+19,5.00699476839502e+19,5.111291551282626e+19,5.2
68642660785581e+19,5.265412111904046e+19,5.448788268153322e+19,5.392838137030672e+19,5.451126165370222e+19,7.2970022861687415e+
19,7.162679464252258e+19,7.532970502992755e+19
26.568359375,6.8687420232546345e+19,6.55434798160913e+19,6.734472335365808e+19,5.39406021966678e+19,5.447513051489558e+19,5.49033
9077780968e+19,5.5939504317118005e+19,5.752236700101518e+19,5.87247900468894e+19,6.0413389445823685e+19,6.033847046682754e+19,6.2
54300127430971e+19,6.395211568776902e+19,6.334745045303422e+19,6.5519038163369165e+19,6.533466308739993e+19,8.768655450180156e+
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23.016328125,8.023928918541928e+19,7.483396455188924e+19,8.003047245672792e+19,6.273747181553374e+19,6.491384158835779e+19,6.6311
266515732595e+19,6.632561270319994 5e+19,6.793929312313808e+19,6.863694290627236e+19,7.010291072932449e+19,7.081278133881983e+19,
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+20,1.0086273068014399e+20,1.073488014362141e+20
19.464296875000002,9.223695263033308e+19,8.485504216796871e+19,8.998619521878947e+19,6.923470071741177e+19,7.305556864621492e+19
,7.430421829615059e+19,7.627442804166 61e+19,7.809798787084876e+19,7.97887126308893e+19,8.00841378246613e+19,8.18364980567838e+19,
8.446822644662693e+19,8.43008542595079e+19,8.83071599448334e+19,9.035441403045134e+19,9.053878910642056e+19,1.1896868194448779e+
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9480226020452e+19,8.575141321453955e+19,8.731886703041624e+19,8.921203243582934e+19,9.107172340381865e+19,9.459132139580752e+19,
9.738563991028058e+19,9.80960418600525e+19,9.850783057439298e+19,1.0109226967962152e+20,1.0201627042057395e+20,1.343143204720588
9e+20,1.3377819813300139e+20,1.4447460924059927e+20
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19,9.475019213850147e+19,9.624644635731801e+19,9.884044958752478e+19,1.0042118691031568e+20,1.0457254849114435e+20,1.04788803983
7077e+20,1.075517734218632e+20,1.0996990302052586e+20,1.112440570037368e+20,1.1197943194650744e+20,1.1305433332600526e+20,1.53462
23001875323e+20,1.5327519824140114e+20,1.588957156869414e+20
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3579558355432e+20,1.0589027237703392e+20,1.065284120491925 5e+20,1.086032958291920 3e+20,1.1368662825512162e+20,1.147854399470649
8e+20,1.1787837169696948e+20,1.1849100703585283e+20,1.20796492495883e+20,1.2495582418086019e+20,1.2519280194420967e+20,1.65808452
68509586e+20,1.654051654151805e+20,1.7689167951403552e+20
5.256171875,1.3637910878680357e+20,1.2827563822885934e+20,1.3309861391927045e+20,1.0465596891456561e+20,1.0903846351570151e+20,1.
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620852155e+20,1.79514375119177e+20,1.920040596601931e+20
1.704140625,1.4324455650034349e+20,1.3425587304163636e+20,1.4179984228835408e+20,1.0923346539720211e+20,1.1248154850786453e+20,1.
1776253951667757e+20,1.1938790942270023e+20,1.224994380822846e+20,1.2624113630987906e+20,1.274281504877328e+20,1.32348892789032
47e+20,1.3387012000084774e+20,1.359301262531033e+20,1.4066649347843375e+20,1.42681335807181e+20,1.444905494488963e+20,1.901178016
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1.2202282585420276e+20,1.2544146919364367e+20,1.2914278556021894e+20,1.3229735278220535e+20,1.3397691939643797e+20,1.37282918597
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808e+20,1.3939605887715877e+20,1.4257506775186733e+20,1.4521742294724184e+20,1.4751706366422976e+20,1.4935337566005002e+20,1.9832
541493052696e+20,1.9937906269896206e+20,2.1314821462595482e+20
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,1.1222012909179291e+20,1.1423444008026361e+20,1.1773809786395543e+20,1.20147726018193e+20,1.2370770587120109e+20,1.285567172351
6404e+20,1.305646521403159e+20,1.3444131079815873e+20,1.3909372625979136e+20,1.3889447365607824e+20,1.4215955965559087e+20,1.8880
114047303896e+20,1.8832080886302114e+20,2.0137477677775287e+20

A.21 "MITR_Cycle3_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.4636808559677985e+19,2.4723087314478825e+19,2.72928461020002e+19,2.9570155860227887e+19,2.930646641586781e+19,2.89
7977550576253e+19,2.7017203455152095e+19,2.9444692170954994e+19,2.9464464385596854e+19,3.074777098955649e+19,3.0842857367243248
e+19,3.00737182176749e+19,3.1080573311460155e+19,3.127937394231377e+19,2.9401822414663328e+19,3.0365897625859838e+19,2.9816949048
439464e+19,2.8403595196358152e+19,2.5344024727936147e+19
37.224453125,3.3419176935077605e+19,3.5098197452983194e+19,3.6065418243781816e+19,3.945114038009159e+19,3.985287583213302e+19,4.0
723302644889444e+19,4.059783895561655e+19,4.067719743529274 6e+19,4.303647201058123e+19,4.01262716364082e+19,4.147896073902017e+
19,4.237167623010015e+19,4.13987035223148e+19,4.277925347283122e+19,4.217530218922531e+19,4.255519833145777e+19,4.15385470040544
95e+19,3.892942353465523e+19,3.769123352956021e+19
33.672421875,4.169348926788435e+19,4.199861048928942e+19,4.659259468762267e+19,4.759037253741326e+19,4.767808927146078e+19,4.9227
60178346218e+19,4.964164993280331e+19,5.105221770009419e+19,5.216871871143886e+19,5.2201612486706676e+19,5.374905790354097e+19,5.
3072668415383 52e+19,5.503829617189318e+19,5.546223042855523e+19,5.402290307633073e+19,5.394947626104711e+19,5.138699724346204e+
19,5.240014349645154e+19,4.879935358906008e+19
30.120390625,5.213879076836733e+19,5.310879764395639e+19,5.6676154533862515e+19,5.781359611798697e+19,5.773657435458664e+19,5.737
312509998808e+19,6.108410016715646e+19,6.443495130673421e+19,6.375245040677837e+19,6.604746528448082e+19,6.632427628946686e+19,6.
7642903258673095e+19,6.769485025895 9426e+19,6.775147069179748e+19,6.857004037797048e+19,6.619225181988099e+19,6.314670164911411
e+19,6.1401084717346636e+19,6.0297435645519176e+19
26.568359375,5.976798966163191e+19,6.4997650560700965e+19,6.591885601560582e+19,7.021940257391328e+19,7.136196695910403e+19,7.161
6938654281105e+19,7.362300957710364e+19,7.347876228392099e+19,7.468181167117525e+19,7.57562517895545e+19,7.788949400200536e+19,7.
856049106798776e+19,8.108468388812993e+19,8.196373857636645e+19,8.181895204096629e+19,8.103875842593907e+19,7.836681323820043e+
19,7.491224784545584e+19,7.240899559809344e+19
23.016328125,7.208769711016322e+19,7.273388903414037e+19,8.016329868581926e+19,8.442735652074219e+19,8.467316109822168e+19,8.4848
68444001966e+19,8.731257200550417e+19,8.931720494908001e+19,8.913287398439613e+19,8.980297231334936e+19,9.228771057791073e+19,9.2
21671035260587e+19,9.489314922549038e+19,9.544497376140411e+19,9.45372493619369e+19,9.696473807773978e+19,9.089107323457208e+19,
8.935998483166876e+19,8.371142260330104e+19
19.464296875000002,7.907978132344534e+19,8.38643876456667e+19,8.89216707825399e+19,9.487517448490687e+19,9.424066614230899e+19,9.
969420243534566e+19,1.0056238240552914e+20,1.0205608334801876e+20,1.0317590968637135e+20,1.018125256131122e+20,1.04807117394324
82e+20,1.0845958468089386e+20,1.086123699758537e+20,1.0947066383871625e+20,1.0918846041155517e+20,1.0919205535967186e+20,1.038032
2813273584e+20,1.0378165844403564e+20,9.429459036405952e+19
15.912265625,9.41858431835293e+19,9.725862508628017e+19,1.0010312778362048e+20,1.0558542366159076e+20,1.0680860475829854e+20,1.08
761560322269682e+20,1.1089965571510523e+20,1.1571958240257319e+20,1.1892357991158368e+20,1.180140580380581e+20,1.197674939819794
4e+20,1.2525518228212475e+20,1.2347208801624064e+20,1.2289509884351e+20,1.2800891253951835e+20,1.2514284015347782e+20,1.203930149
5428553e+20,1.1667404112755751e+20,1.1070732599086168e+20
12.360234375000001,1.0392725384276206e+20,1.088244719147391e+20,1.1330108105706204e+20,1.1721238460803362e+20,1.1971806344537478
e+20,1.230928209899286e+20,1.2321055554075061e+20,1.3123088478911236e+20,1.3045707220699228e+20,1.3485549122777697e+20,1.34399831
553985e+20,1.3551516420719175e+20,1.4278504803619198e+20,1.393177205776331e+20,1.4098308029269521e+20,1.3869938950156037e+20,1.35
05321337419558e+20,1.3443488229812285e+20,1.2639118588700251e+20
8.808203125,1.1247783793833732e+20,1.1545445497896664e+20,1.2499275106960553e+20,1.2961675308471322e+20,1.2849512927230227e+20,1.3
23273439647064e+20,1.3611552054268096e+20,1.4218738791179031e+20,1.4485483941438307e+20,1.478593173029166e+20,1.482754325474248
6e+20,1.517050130507584e+20,1.5115408725187382e+20,1.4842911657941385e+20,1.5087188382471274e+20,1.5075235179983238e+20,1.5131316
37060379e+20,1.4672061748695132e+20,1.3578208910485686e+20
5.256171875,1.2146700570415024e+20,1.2782467144853738e+20,1.325655092774379e+20,1.3730814458039678e+20,1.398488741618758e+20,1.41
77307014134043e+20,1.4861245893336564e+20,1.5118823925898248e+20,1.5165108882900786e+20,1.570066627858644e+20,1.550465173252327
3e+20,1.5853451573546267e+20,1.6528942324674537e+20,1.6471602902213145e+20,1.6644789527735255e+20,1.618679313766744e+20,1.6286552
947905918e+20,1.5440481908640147e+20,1.4980508297108147e+20
1.704140625,1.2275130091884195e+20,1.324450785155284e+20,1.3615596370899383e+20,1.4582907035400924e+20,1.4788448193973354e+20,1.5
7835298326764 16e+20,1.5822624893445548e+20,1.5994193792315143e+20,1.6133318284431504e+20,1.6798563433427173e+20,1.7012642593776
766e+20,1.7460123760603224e+20,1.7300058695707076e+20,1.748996182997185e+20,1.7588014039854894e+20,1.7621626804746056e+20,1.68254
3567059952e+20,1.6565700669167813e+20,1.561250017602433e+20
−1.847890625,1.3485459249074779e+20,1.3696392829822258e+20,1.4692552952960333e+20,1.502948946519821e+20,1.5320051146730635e+20,1.
6007135605535267e+20,1.6327355609030482e+20,1.6719923943374324e+20,1.6839276220848823e+20,1.738525896607291e+20,1.7714016971345
294e+20,1.7434869250083396e+20,1.7853680705679157e+20,1.8071624435254208e+20,1.812527903589598e+20,1.8276985846420796e+20,1.78251
90741854296e+20,1.761641412997684e+20,1.655419683519437e+20
−5.399921875,1.2981088028301512e+20,1.4005917862670284e+20,1.433925942679146e+20,1.4821971085161598e+20,1.51287999069221e+20,1.57
19539756199123e+20,1.619308429687167e+20,1.6743740474647472e+20,1.694550693769736e+20,1.724622434765947e+20,1.7985435554156282e
+20,1.825990984286646e+20,1.8442623080897833e+20,1.8345469608043967e+20,1.833881895402807e+20,1.8077286478538015e+20,1.798372795
3800846e+20,1.7485827639637646e+20,1.6598324823326884e+20
−8.951953125,1.2921681510673016e+20,1.3385609565133385e+20,1.4321015065099198e+20,1.4550372754944773e+20,1.5086020024333346e+20,
1.5401566595276846e+20,1.597999374725417 2e+20,1.6368607638669636e+20,1.6580889324960878e+20,1.6876034565342098e+20,1.74126704454
6276e+20,1.783570596509565e+20,1.7927916384289053e+20,1.82857035956038e+20,1.823222874236786e+20,1.8139389207254034e+20,1.7420579
331319507e+20,1.7792296966586473e+20,1.638199882040435e+20
−12.503984375,1.1336758759722102e+20,1.1745324613185264e+20,1.2355117687480812e+20,1.2961675308471322e+20,1.3580725374167378e+20
,1.4002772283068 17e+20,1.4332698646478478e+20,1.4814331820413606e+20,1.536480825078357e+20,1.5532692327833544e+20,1.590485933161
51e+20,1.5824512241206818e+20,1.6483196609889508e+20,1.668487319923648e+20,1.652121318622363e+20,1.6675166839321385e+20,1.6534694
24166126e+20,1.5621038177801493e+20,1.4807860913803543e+20

A.22 "MITR_Cycle3_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.921663982586644e+19,2.674680660209045e+19,2.8048329038980923e+19,2.2170763342709866e+19,2.1993838300378735e+19,2.2
4370804133338e+19,2.236872180346284e+19,2.3406264216220692e+19,2.3064842969716777e+19,2.384504469496507e+19,2.5159420887557775e
+19,2.4837864536230035e+19,2.5432908441734603e+19,2.4457563334461518e+19,2.5341551169801384e+19,2.5643720658189808e+19,3.3710142
852349215e+19,3.405108606661628e+19,3.6156552496519074e+19
37.224453125,3.7521653220911915e+19,3.571081399299374e+19,3.6507693731839144e+19,2.9072858351957586e+19,2.9702851725445915e+19,3.
0055373942783574e+19,2.9572986301098983e+19,3.030469431165249e+19,3.1312492496339067e+19,3.176614508911908e+19,3.21999327867229
2e+19,3.243278760076666e+19,3.379544504928221e+19,3.3587713485020107e+19,3.4345447695135986e+19,3.466865060194624e+19,4.636477157
058264e+19,4.564777632998582e+19,4.909927530950005e+19
33.672421875,4.717243359210389e+19,4.412519054089415e+19,4.616336065478569e+19,3.615830528138757e+19,3.697223483668097e+19,3.8806
65698828422e+19,3.87787717744674e+19,3.9505911921424245e+19,3.8976783349913354e+19,4.0050921786137485e+19,4.145304345154046e+19,
4.130634066951747e+19,4.3391889089551114e+19,4.3447500173105816e+19,4.381797515667222e+19,4.4369624358008504e+19,5.6672846267456
94e+19,5.765706152845842e+19,6.276935072820982e+19
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179462827581e+19,4.605357258438687e+19,4.732949373925885e+19,4.833059947262938e+19,4.940356938560785e+19,5.101549408829933e+19,5.
056627444530005e+19,5.265721613375508e+19,5.383917739673759e+19,5.388220029805499e+19,5.541190345600657e+19,7.191357627240927e+
19,6.932264154862877e+19,7.511426767165102e+19
26.568359375,6.752842721961474 5e+19,6.390866089272584e+19,6.690167384239846e+19,5.197787921048422e+19,5.439475708535474e+19,5.484
410738800302e+19,5.488659914239056e+19,5.717743585080902e+19,5.80787921907548e+19,5.94395906250159e+19,5.9508639725895655e+19,6.0
88590371498193e+19,6.235186924135218e+19,6.328403210322892e+19,6.493430561425517e+19,6.6162848462985036e+19,8.751707963044526e+
19,8.486931218517144e+19,9.060835476213909e+19
23.016328125,8.045813693281452e+19,7.444077336460843e+19,7.766855325727226e+19,6.021081596714983e+19,6.187914847379077e+19,6.3904
41171728708e+19,6.583725539499048e+19,6.6739142881866105e+19,6.7669712302953325e+19,6.9532444585917284e+19,7.1389334252652945e+
19,7.107861329869403e+19,7.401426237994348e+19,7.483594668041262e+19,7.472971729444374e+19,7.797608732965211e+19,1.0223834793800
989e+20,1.0155157495772119e+20,1.0801350850620695e+20
19.464296875000002,9.07990365099532e+19,8.654454960190035e+19,8.696043764796845e+19,6.9164359763535176e+19,7.155398980090467e+19
,7.312777815403335e+19,7.481894997865759e+19,7.854813257309443e+19,7.770945157087027e+19,8.080550702493268e+19,8.152043079250312e
+19,8.23330855951649e+19,8.338369422239693e+19,8.646859559093263e+19,8.818632476204909e+19,8.926402188270317e+19,1.1563971612491
14e+20,1.1605560417097948e+20,1.2641456274373288e+20
15.912265625,1.047804171442447e+20,9.546888031714429e+19,1.0136567353227569e+20,7.864533246125593e+19,8.10439919964328e+19,8.2721
88514781092e+19,8.404125412154417e+19,8.789844312607346e+19,9.016272248799971e+19,9.146243902532873e+19,9.296877171836715e+19,9.3
69272498374494e+19,9.500837592896927e+19,9.860424064401518e+19,1.0004949144012151e+20,9.99915964247685e+19,1.3382671814971007e+
20,1.3302096825713626e+20,1.436885231961292e+20
12.360234375000001,1.1359692503273023e+20,1.084458621071002e+20,1.1378229531124589e+20,8.826121647915713e+19,9.186079922271194e+
19,9.283651613283592e+19,9.50519299772165e+19,9.64631873698128e+19,9.96718459730022e+19,1.0111550332831903e+20,1.040532769972878
5e+20,1.0686835572546263e+20,1.0671963458510625e+20,1.0979391301504505e+20,1.1071863981990396e+20,1.1576347335956513e+20,1.500383
847424178e+20,1.5161854685870457e+20,1.642577192012795e+20
8.808203125,1.2551533099150646e+20,1.1932746925882037e+20,1.237078379892463e+20,9.637129895094973e+19,1.0035755665943121e+20,1.01
471902912 91824455e+20,1.028199538261894e+20,1.0791046600181714e+20,1.0854359314219154e+20,1.1087214128262896e+20,1.132012205699962
3e+20,1.168868491161858e+20,1.1948681333777365e+20,1.1871824373028897e+20,1.2420499151558058e+20,1.2452474196734686e+20,1.6366283
46398539e+20,1.6422053891619045e+20,1.780542607039852e+20
5.256171875,1.343185602067459e+20,1.254223802787837e+20,1.310668787022391e+20,1.028693504906649e+20,1.0639351037018184e+20,1.1086
842325412007e+20,1.1172888128046783e+20,1.1699360964908451e+20,1.1634189236616557e+20,1.1946344287286052e+20,1.2176702710759524
e+20,1.247855351099004e+20,1.277243710727289e+20,1.2988932595877428e+20,1.3181367128560019e+20,1.3477481541948216e+20,1.754723554
780121e+20,1.767625113706039e+20,1.899848830421479e+20
1.704140625,1.368701900577179e+20,1.3185350730533849e+20,1.379335462112662e+20,1.0824880659612798e+20,1.1117064585720147e+20,1.13
91880007221586e+20,1.1757521553726403e+20,1.2196248917777795e+20,1.2406583101996138e+20,1.281965606933605e+20,1.285476488139875
6e+20,1.3123418998514006e+20,1.3499152336685862e+20,1.377694218099443e+20,1.406588611082973e+20,1.4297519286934828e+20,1.89094149
640799e+20,1.8851201260568966e+20,2.025300423781404e+20
−1.847890625,1.4499355120275662e+20,1.3658337071560201e+20,1.4395515895491104e+20,1.1224674953706602e+20,1.1640191196923796e+20,
1.202580386799076e+20,1.2203206942558755e+20,1.2545265565378485e+20,1.2729467320648488e+20,1.3127721288645743e+20,1.346090975773
7071e+20,1.357659355905716e+20,1.4078899210610916e+20,1.4118629000963267e+20,1.4565004880804418e+20,1.4781287910637017e+20,1.9643
1944476598e+20,1.9715908462355484e+20,2.1431300586980632e+20
−5.399921875,1.4486076447029554e+20,1.3805624115206023e+20,1.4258851790442163e+20,1.1446269452837644e+20,1.1788328075657373e+20,
1.2145258812512743e+20,1.2314588453747106e+20,1.2645652335119057e+20,1.2842070469775478e+20,1.3416665218481 04e+20,1.372520847002
7597e+20,1.3987913741528585e+20,1.4219759376405624e+20,1.4499992496591477e+20,1.4841254399016 44e+20,1.5078677076656842e+20,2.0344
14905097532e+20,2.0062534948771876e+20,2.2057310358495127e+20
−8.951953125,1.4084157565216373e+20,1.3572716186469297e+20,1.4031255331003882e+20,1.0949169041196365e+20,1.1353106281342953e+20,
1.1789177910745122e+20,1.2159546664925554e+20,1.228574717545656e+20,1.2558438009238625e+20,1.2982452603333326e+20,1.348597989282
5724e+20,1.3866334209287225e+20,1.4140459139779867e+20,1.4242333120924005e+20,1.460075106918294e+20,1.4953273286520603e+20,1.9962
466867189206e+20,1.9866913534510 22e+20,2.1644768538085058e+20
−12.503984375,1.2938526752235202e+20,1.2389851973706043e+20,1.2842495387319355e+20,1.0261918028670822e+20,1.0709621775836584e+20
,1.096255394382844e+20,1.120560677892519e+20,1.163626070964295e+20,1.1817434927412838e+20,1.208555789759824e+20,1.23905955794078
24e+20,1.2617714006609247e+20,1.2994190950482885e+20,1.3515352318046113e+20,1.3803765100951568e+20,1.3996412092406096e+20,1.86325
28069552066e+20,1.864086707635062e+20,1.9983818973768948e+20

A.23 "MITR_Cycle3_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
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A.24 "MITR_Cycle3_cladding_fluence_position_3.csv" File

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826523827327e+20,2.1217567062411284e+20,2.2948203103922972e+20

A.25 "MITR_Cycle3_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.847680526728632e+19,2.807616113810476e+19,2.926476173771034e+19,2.322636475108243e+19,2.4564111408588284e+19,2.491
1056583162577e+19,2.571824057244696e+19,2.620774558299146e+19,2.655601862489037e+19,2.8177238398854128e+19,2.702369349661828
e+19,2.779178507186612e+19,2.901698169493797e+19,2.921499327038392e+19,2.930226071095734e+19,2.966296259101461e+19,3.781128764
175601e+19,3.702758034677074e+19,3.967317008963222e+19
37.224453125,3.702858952593745e+19,3.710656189523859e+19,3.83888568132687e+19,3.0273197298712723e+19,3.171032154679244e+19,3.302
0182990481474e+19,3.389370723130340e+19,3.4927212927394447e+19,3.56198816386044e+19,3.5536172882460934e+19,3.6552788106182922e
+19,3.695699091979443e+19,3.8519678302089355e+19,3.8616931304943845e+19,3.84318265998931e+19,3.907037143895192e+19,5.07861979586
8555e+19,5.0450353754944995e+19,5.46220879713281e+19
33.672421875,4.616410426048746e+19,4.453603269112871e+19,4.845568457460772e+19,3.899627644223865e+19,3.9686608106957275e+19,4.090
84585043711e+19,4.223882221955212e+19,4.385127806917345e+19,4.4510165835645305e+19,4.5193858163740885e+19,4.57276608282344e+19,4.
759527966295293e+19,4.749239650264209e+19,4.898258232901326e+19,4.890105127528216e+19,4.88381103640956e+19,6.39947066953606e+19,
6.464589283134972e+19,7.138083590177544e+19
30.120390625,5.837517217760791e+19,5.547989026302665e+19,5.897483706140213e+19,4.689963652893585e+19,4.858868376584072e+19,4.9982
89134198904e+19,5.120527288633269 5e+19,5.233587224119928 4e+19,5.340363691426527e+19,5.431136701736917e+19,5.492537286826919e+19,
5.656130541218963e+19,5.847768353506786 5e+19,5.751365185740046e+19,5.768521231574017e+19,5.935726285089003e+19,7.686174107083878
e+19,7.88811616981068e+19,8.681809027077017e+19
26.568359375,6.972950009688951e+19,6.63386580967635e+19,6.94463987832825e+19,5.629785653498686e+19,5.7324563550375895e+19,5.93668
2349562723e+19,6.09862904847225e+19,6.276085237733229e+19,6.3891133044040974e+19,6.480258117565379e+19,6.5494134478311055e+19,6.9
023074680196596e+19,6.909955983809418e+19,7.022506018243425e+19,7.054534178113036e+19,7.159754384915191e+19,9.658110198823825e+
19,9.718554719440106e+19,1.0366819547315074e+20
23.016328125,7.877546345906777e+19,7.748424527261629e+19,8.281908503597243e+19,6.615913043447613e+19,6.714334569547762e+19,7.0196
9093951525e+19,7.356172519571614e+19,7.364086608826294e+19,7.449441920452272e+19,7.7348802805506e+19,7.826662470027695e+19,8.1344
62115872467e+19,8.175360429470476e+19,8.482682043078389e+19,8.445395528603319e+19,8.696309338261766e+19,1.1148296025194992e+20,1.
122802117936462e+20,1.2193752527207527e+20
19.464296875000002,9.200633348148929e+19,8.977817211079244e+19,9.65911937799053e+19,7.544464106201417e+19,7.82336935906266e+19,8.
074176939335138e+19,8.268895403816059e+19,8.532025592860924e+19,8.73795125756156e+19,9.017334542659661e+19,9.108904273364818e+1
9,9.386587888287418e+19,9.455318301009 27e+19,9.626560071191072e+19,9.793871354092028e+19,9.900047625367903e+19,1.304879285487088e
+20,1.2942297895437097e+20,1.4364178226630287e+20
15.912265625,1.0195418433054317e+20,1.0043457296425864e+20,1.0629896621666956e+20,8.506264966763476e+19,8.906696637173093e+19,9.3
792049459625 82e+19,9.416385231051684e+19,9.800192002557172e+19,9.917150556508889e+19,1.0323318613760826e+20,1.0378876582622539e
+20,1.0427582756089261e+20,1.0775112192286387e+20,1.0894460907422 4e+20,1.1232907731119188e+20,1.1347210550421683e+20,1.477990692
8619422e+20,1.4584497973129704e+20,1.6534657040746034e+20
12.360234375000001,1.1568380132008849e+20,1.122355954515393e+20,1.1724377985304119e+20,9.642282020314463e+19,1.0059604163093132e
+20,1.0492170222758329e+20,1.0613696640306705e+20,1.0909439250844012e+20,1.1146861928484412e+20,1.1430653733100221e+20,1.1684701
309644751e+20,1.1782697918201025e+20,1.1946875434215897e+20,1.2202835139707865e+20,1.2334718922388203e+20,1.2666738868233876e+2
0,1.6885851390759086e+20,1.691575496290932e+20,1.8395902112306433e+20
8.808203125,1.232266188708074e+20,1.1955161326321466e+20,1.3028556156843812e+20,1.0403043767930456e+20,1.0916184816853033e+20,1.1
383753459194968e+20,1.17936926596488e+20,1.190002827500363e+20,1.2047368433342438e+20,1.24476407596731e+20,1.2776845626790597e+
20,1.3104988200048409e+20,1.3305175477906724e+20,1.3487413989536303e+20,1.3710814388228815e+20,1.3925822665429788e+20,1.84972980
61213706e+20,1.846511055726514e+20,2.013673617487112e+20
5.256171875,1.3170903534042086e+20,1.3201710055973054e+20,1.4103916231006582e+20,1.1405530483318586e+20,1.1755715654164932e+20,1.
2180208280496497e+20,1.2295520278965695e+20,1.294697198841973e+20,1.3262898182291118e+20,1.3539307044582089e+20,1.3860597822444
91e+20,1.4146620444166062e+20,1.4230222970923557e+20,1.4616951050543191e+20,1.4869033383447295e+20,1.507995182928847e+20,1.987971
4175519464e+20,2.018416759570622e+20,2.183985880541687e+20
1.704140625,1.426347276873181e+20,1.3888376806875765e+20,1.4815228199454068e+20,1.2015605846937749e+20,1.2497727915157419e+20,1.2
843982598722919e+20,1.3223593309482638e+20,1.3644049219147388e+20,1.4156977809298029e+20,1.4382024763473057e+20,1.4454048287159
943e+20,1.4834296374235477e+20,1.5168175334335604e+20,1.5462430733469345e+20,1.5571422083473398e+20,1.5910453168793013e+20,2.1047
812503633052e+20,2.120322609530549e+20,2.3240121456565397e+20
−1.847890625,1.4745807295723418e+20,1.4331087772900988e+20,1.561980956878221e+20,1.2469364669103735e+20,1.292131759170825e+20,1.3
481518258615 03e+20,1.3680539013227689e+20,1.4095949027058917e+20,1.4222255766975893e+20,1.47990813327868e+20,1.5176620570520132
e+20,1.5721630435233366e+20,1.596362097647043e+20,1.6359856586134274e+20,1.6423434873636638e+20,1.6534763270132004e+20,2.23727585
20129654e+20,2.213337059884883e+20,2.3856145665798816e+20
−5.399921875,1.4765565961513625e+20,1.4669906399448669e+20,1.555601882250791e+20,1.2555994733361339e+20,1.2889767464075497e+20,1.
3591625017171753e+20,1.3868989943936449e+20,1.4425207008869399e+20,1.4786811838707397e+20,1.514512355758036e+20,1.5536950647726
503e+20,1.5565154549701234e+20,1.605094153173683e+20,1.6486428899516174e+20,1.668295326355856e+20,1.7126354720592583e+20,2.215498
8278893488e+20,2.273680662584494e+20,2.4553807158149307e+20
−8.951953125,1.4615145150981721e+20,1.4452401731677428e+20,1.523297525977661e+20,1.2290952415369034e+20,1.2792673805299958e+20,1.
3356380041943715e+20,1.3564642753135664e+20,1.4186350234518415e+20,1.473900861502141e+20,1.485182422292034e+20,1.53927973709667
6e+20,1.555803718084132e+20,1.6181338103013605e+20,1.6251183924288134e+20,1.6700852915094317e+20,1.6931848714883603e+20,2.2379344
742059724e+20,2.2523126215968578e+20,2.4609365127011018e+20
−12.503984375,1.303381451144927e+20,1.321530741737707e+20,1.4270590137591724e+20,1.1563121777403388e+20,1.178901856666617e+20,1.2
451996164497826e+20,1.2662755266260042e+20,1.3338267931636023e+20,1.3812157222443106e+20,1.411730113363866e+20,1.42615075250913
85e+20,1.4711707662827414e+20,1.4934683143976052e+20,1.5348234143552822e+20,1.5639462005186452e+20,1.6083713297308225e+20,2.11836
26773594237e+20,2.0919752978847587e+20,2.3026547276075005e+20

A.26 "MITR_Cycle3_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.1671515605616493e+19,2.420757175454379e+19,2.6201599601175368e+19,2.7153811483586765e+19,2.8608417365307236e+19,3.083611683952443e+19,3.1897255649871897e+19,3.2372687538305712e+19,3.256519700995509e+19,3.5681657532323893e+19,3.50779758698267 5e+19,3.475227357045356e+19,3.533411592314175e+19,3.64894423741468e+19,3.595945714804203e+19,3.76230193890458e+19,3.4979743912537 862e+19,3.5359460307364495e+19,3.2872924568744767e+19
37.224453125,2.9148108951327097e+19,3.271465697790697e+19,3.436294068941476e+19,3.837004960769644e+19,3.985943661244595e+19,4.20 94505730302435e+19,4.327751328180609e+19,4.462570869927218e+19,4.638229022279561e+19,4.483942836481011e+19,4.723627014791813e+1 9,4.804729044304606e+19,4.801844098440953e+19,4.892778311052925e+19,4.960965489456467e+19,5.032280272721539e+19,4.90145112338446 9e+19,4.813168185008564e+19,4.6887380433192215e+19
33.672421875,3.7146419142474056e+19,4.050517916790856e+19,4.528061837243234e+19,4.827754686992081e+19,4.927532471971 14e+19,5.2384 864966955565e+19,5.445465634514663e+19,5.445537533476996e+19,5.469992168040862e+19,5.826970516029352e+19,5.985184182645399e+19,6. 1420587310879736e+19,6.445544251099942e+19,6.431209395484593e+19,6.503279117854173e+19,6.362662722269381e+19,6.3273243822822015 e+19,6.219943282036319e+19,6.123194240845582e+19
30.120390625,4.601281943010158e+19,5.156090285860749e+19,5.432164326482865e+19,6.005531588985931e+19,6.008110964259665e+19,6.4364 04095513227e+19,6.299175938528427e+19,6.888639581224024e+19,6.753550418368662e+19,6.91756093882289e+19,7.09955518723092e+19,7.527 273126785812e+19,7.520020318960365e+19,7.445407170798218e+19,7.472243458489398e+19,7.697916326515353e+19,7.479091834651714e+19,7. 451940989000324e+19,7.397989805138921e+19
26.568359375,5.365540950510089e+19,5.833729018488751e+19,6.407545649506403e+19,6.53727833966788e+19,7.228479014066138e+19,7.45847 480720243e+19,7.801289059611116e+19,7.66388115522048e+19,8.20178425455228e+19,8.567678073870192e+19,8.349644470592227e+19,8.62247 407053902e+19,8.820906219210668e+19,9.159388559138728e+19,9.193450692544478e+19,9.286829469875808e+19,9.003098189765116e+19,9.073 019930634967e+19,8.945417247232636e+19
23.016328125,6.6016728478083015e+19,7.006517929970677e+19,7.64518742501363e+19,8.27712537970806e+19,8.578867349883426e+19,8.78197 2931106787e+19,9.303186483806801e+19,9.640662238262185e+19,9.590512712034193e+19,9.803693135354611e+19,9.875052855471143e+19,1.04 37033119814553e+20,1.0746827773771334e+20,1.0731908739087021e+20,1.084182427775518e+20,1.0940954972073 23e+20,1.0865550935325411 e+20,1.043029259209574e+20,1.034158724731612e+20
19.464296875000002,7.397998792509214e+19,8.374584423151845e+19,8.64501439523074e+19,9.50639092610337e+19,1.0004560861375327e+20, 1.015779552484974e+20,1.0487901635665886e+20,1.0772082284291162e+20,1.1252457226385444e+20,1.1783610810628137e+20,1.187114779726 9827e+20,1.1949967034728514e+20,1.2059523078585002e+20,1.2510149825013575e+20,1.2287263041778062e+20,1.2507633361331885e+20,1.220 9162793942716e+20,1.2336693578382713e+20,1.2084328220590244e+20
15.912265625,8.41051592957828e+19,9.285840859143714e+19,9.90776688333313e+19,1.0458872429623517e+20,1.1079630095675005e+20,1.1821 088144744754e+20,1.216539430062187e+20,1.2698974474843339e+20,1.2930129638747267e+20,1.3701695378294394e+20,1.3583960827472408e +20,1.370690805306361e+20,1.3913617569773966e+20,1.430591628300905e+20,1.4381230446053954e+20,1.447074465415983e+20,1.4109362494 72838e+20,1.405184332486115e+20,1.3874342761598996e+20
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8.808203125,1.0023883707502597e+20,1.0999552626375472e+20,1.2156946172547621e+20,1.267911238649856e+20,1.3746092987535661e+20,1.4 433447067449049e+20,1.475995823014849e+20,1.551867203017841e+20,1.5779395642342208e+20,1.6165133575264315e+20,1.640060267690828 4e+20,1.7172168416455413e+20,1.699215138951157e+20,1.7214858425341247e+20,1.7948946830771755e+20,1.7647510431186308e+20,1.7349399 35860881e+20,1.7348500621579636e+20,1.692834106044011e+20
5.256171875,1.1113422607972e+20,1.232249353332174e+20,1.308668962922963e+20,1.4421763486069768e+20,1.4956781639537915e+20,1.59013 5425720132e+20,1.638460515778896e+20,1.6354407593608665e+20,1.6827143270954952e+20,1.7624053394724828e+20,1.8167250055158474e+2 0,1.8361197506054537e+20,1.8355445589067812e+20,1.8522970171306118e+20,1.8962003210058328e+20,1.906445923138433e+20,1.9205201450 15321e+20,1.937416401163819e+20,1.846976493917893e+20
1.704140625,1.1496464329806581e+20,1.2576206996657971e+20,1.3644535703238836e+20,1.4488539647337503e+20,1.5813997017965465e+20,1. 6006955858129433e+20,1.7331873986539887e+20,1.7525461942624282e+20,1.7906706190400514e+20,1.8565300685380282e+20,1.885918769392 0656e+20,1.9766462724873273e+20,1.992841513753069 5e+20,1.977050704150457e+20,2.001073944940317e+20,1.974219682508554e+20,2.005873 200676114e+20,1.9494504899844787e+20,1.9145255690307207e+20
−1.847890625,1.1934329010420862e+20,1.2912424519272509e+20,1.424704900759806e+20,1.5096984616089285e+20,1.6532717020197074e+20,1. 708319345056704e+20,1.7959641801418945e+20,1.8367668412664604e+20,1.894447783798941e+20,1.9115147999829826e+20,2.01427639189890 37e+20,1.9802142584931546e+20,2.0346237982394365e+20,2.076091524765592e+20,2.110477203501845e+20,2.1086168178514516e+20,2.1084999 820376585e+20,2.0941920885331853e+20,2.0475566240892715e+20
−5.399921875,1.1937204968914221e+20,1.3509275780347929e+20,1.4485124446626637e+20,1.5273137073807675e+20,1.6452459803491706e+20, 1.681123562553855e+20,1.784001990283569e+20,1.8394091281322358e+20,1.8971260201458834e+20,1.935079684887963e+20,1.98710757150693 e+20,1.987035672544596e+20,2.0461905438049242e+20,2.1237874989039334e+20,2.1242188926779376e+20,2.1286496662317724e+20,2.11568089 0900771e+20,2.1581372281590193e+20,2.0831555978149108e+20
−8.951953125,1.136722594501115e+20,1.2835582503278007e+20,1.394534298690386e+20,1.477173168523069e+20,1.58228945145543e+20,1.6422 441986717247e+20,1.7457247802109864e+20,1.811656128671298e+20,1.8926592971108817e+20,1.939474508960631e+20,1.966679278833772e+2 0,2.0263913670521894e+20,2.043566231679732e+20,2.0611095784892367e+20,2.1213788836657427e+20,2.1452763012715184e+20,2.1067923816 822253e+20,2.0886918179146318e+20,2.030013277279766e+20
−12.503984375,1.0252432534021915e+20,1.1569801271387297e+20,1.2806463423532723e+20,1.3818261570978449e+20,1.4408192556929217e+20 ,1.5129069528030852e+20,1.5860461722373836e+20,1.6214474238166047e+20,1.6877202923480026e+20,1.7662069971058953e+20,1.8028934426 36837e+20,1.8136872743572344e+20,1.8634143941815124e+20,1.89930995112678e+20,1.9138964531102974e+20,1.9322037263946016e+20,1.9286 716898699425e+20,1.9083602330105768e+20,1.8779379845729878e+20

A.27 "MITR_Cycle3_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.960204003816147e+19,2.816773086880992e+19,3.0269904187747684e+19,2.4409069619766735e+19,2.5535897831431414e+19,2.4
909303079829409 4e+19,2.599794254570297e+19,2.6350092960189743e+19,2.729112597579489e+19,2.7360281306060616e+19,2.75745459775598e
+19,2.8713378109838963e+19,2.912894746774915e+19,2.880192030304401e+19,2.917356380985607e+19,2.911843075853823e+19,3.77543486908
767e+19,3.777888767903551e+19,4.020033341750269e+19
37.224453125,3.80927423998805e+19,3.7754454920262664e+19,3.8647153565251985e+19,3.0095634880065774e+19,3.2253684856023163e+19,3.3
281719738736804e+19,3.387166463371487e+19,3.5789954885540545e+19,3.5137387767533834e+19,3.5870795448262844e+19,3.63642309460882
e+19,3.7091530437124e+19,3.797185335864794e+19,3.793122061851485e+19,3.7830780734081294e+19,3.775588901697325e+19,5.0814348745967
305e+19,5.120224534883259e+19,5.350402368400585e+19
33.672421875,4.784720265177809e+19,4.5680548095557214e+19,4.849350223601263e+19,3.9522536820328374e+19,3.955116563984698e+19,4.13
7045010394968e+19,4.298115316870251e+19,4.397253891325691e+19,4.604624275675501e+19,4.475008490385596 4e+19,4.65460520177385e+19,
4.717870112587605e+19,4.7084316316442714e+19,4.9338557001394905e+19,4.9469272260829585e+19,4.89582026849334e+19,6.38858215747425
3e+19,6.473512551556356e+19,6.934388742582255e+19
30.120390625,5.923138102851694e+19,5.695754102185348e+19,6.059111716891832e+19,4.657218444668685e+19,4.8614391277245194e+19,5.048
795895757798e+19,5.262046076620984e+19,5.2265176584837e+19,5.387157735945811e+19,5.411802953490586e+19,5.7142911300369146e+19,5.6
67922003061507e+19,5.8690673453935436e+19,5.8035769289437405e+19,5.921332203290223e+19,6.013486195618208e+19,7.927633501391095e
+19,7.859274891520134e+19,8.341503189125774e+19
26.568359375,7.027180111226054e+19,6.670939865379482e+19,7.106639691930762e+19,5.618365994507033e+19,5.842881801752219e+19,6.1418
112938685915e+19,6.160082748255235e+19,6.151584397377726e+19,6.392565759448085e+19,6.4938023642764075e+19,6.794962673498126e+19,
6.911708768677904e+19,6.964239200039505e+19,6.971197224820466e+19,7.189392383600503e+19,7.35564137264177e+19,9.651789550358677e+
19,9.458930100132212e+19,1.0091632322962742e+20
23.016328125,8.16664961982103e+19,8.026214371570198e+19,8.357862514564977e+19,6.672533306169016e+19,6.829115421087115e+19,7.07227
4485569835e+19,7.313468306412132e+19,7.361059071326181e+19,7.551793933833268e+19,7.728187829234562e+19,7.921153508846995e+19,8.10
9498210169784e+19,8.155229960829377e+19,8.413579827505645e+19,8.486984333210128e+19,8.62465761742577e+19,1.1113293442518252e+20,
1.1013119131549616e+20,1.2136760461635232e+20
19.464296875000002,9.531431656055959e+19,9.058232856257672e+19,9.578385044654195e+19,7.624879751379845e+19,8.002525218499142e+19
,8.207972850962917e+19,8.403966068075464e+19,8.607714030363736e+19,8.821925587169942e+19,9.088933148802672e+19,9.019459130379038e
+19,9.325877794206215e+19,9.60111813325153e+19,9.788772343565522e+19,9.902597130631155e+19,9.913751216157884e+19,1.2935339870656
14e+20,1.2987710957938785e+20,1.395652295797479e+20
15.912265625,1.0520161665961122e+20,1.0260696390732182e+20,1.080283806202426e+20,8.573083250537888e+19,9.043838774458892e+19,9.31
3183382582936e+19,9.595912893339057e+19,9.938555777781614e+19,1.0011588480635206e+20,1.0127962772964091e+20,1.0436612253896616e
+20,1.0620442206315726e+20,1.0848516697990868e+20,1.1061081699314559e+20,1.1392092465993525e+20,1.126063360085706e+20,1.48714235
44631594e+20,1.4804020999234355e+20,1.5959902947961517e+20
12.360234375000001,1.1972848519085282e+20,1.148063465919857e+20,1.2086035929835102e+20,9.732736342466948e+19,1.0134867683052066e
+20,1.0409205072316649e+20,1.0740640756539492e+20,1.1110212790325155e+20,1.1330213848666664e+20,1.1394110824326934e+20,1.1601842
388589039e+20,1.2038657623692993e+20,1.2102342140581321e+20,1.2231410844533488e+20,1.2435052577435792e+20,1.2755440405517872e+2
0,1.6482285953463386e+20,1.6709935527594652e+20,1.7909690212726958e+20
8.808203125,1.2846585218679156e+20,1.2481952851341043e+20,1.3443063220894302e+20,1.0412923100825557e+20,1.0954480510494807e+20,1.
1267644740331006e+20,1.1801978551754372e+20,1.1891529924126119e+20,1.2346988416467606e+20,1.2676565086435995e+20,1.284100817591
5791e+20,1.313547603382147e+20,1.3250894261676635e+20,1.3579143064320411e+20,1.3693498998315891e+20,1.3922688898543706e+20,1.8510
258046301907e+20,1.8315167778970095e+20,1.9944089183416595e+20
5.256171875,1.3862191263234441e+20,1.3425004225279599e+20,1.4380059519832633e+20,1.1451740266215039e+20,1.2095649689265283e+20,1.
2144674550889913e+20,1.2646608399592777e+20,1.3079227573950959e+20,1.3224124456412484e+20,1.3438123554446752e+20,1.397792817924
7515e+20,1.42638445715827e+20,1.4287161921802864e+20,1.4620403505587179e+20,1.4930540197923265e+20,1.5002032574680308e+20,2.01212
7979921265e+20,1.9944354756881514e+20,2.1749032680413495e+20
1.704140625,1.4498558399880897e+20,1.4124524731884539e+20,1.4874822884982597e+20,1.1977257038602989e+20,1.2698926372182442e+20,1.
2847913086003765e+20,1.3359726267601733e+20,1.3600973203137017e+20,1.4104341148550457e+20,1.4229691823993712e+20,1.458497600536
6564e+20,1.490276121349241e+20,1.510289737665774e+20,1.538562688741386e+20,1.579083888019208e+20,1.598704455607656e+20,2.09917233
8784149e+20,2.1307224664169005e+20,2.2939173606115616e+20
−1.847890625,1.5171734018765563e+20,1.479488527204103e+20,1.5882302381511265e+20,1.2501286599587375e+20,1.3192999246323609e+20,1.
3430528153349979e+20,1.3987913741528585e+20,1.4256674088029802e+20,1.4473116461941354e+20,1.498275194112696e+20,1.5367302318334
234e+20,1.58371548924745e+20,1.591135611857375e+20,1.640654440126759e+20,1.6461624337892442e+20,1.6624580215968675e+20,2.17997572
12213628e+20,2.2001061898624618e+20,2.37367969506628e+20
−5.399921875,1.5322367288069403e+20,1.499990798696093e+20,1.582706310080746e+20,1.25694327506864e+20,1.3204737593473168e+20,1.371
6338316299197e+20,1.4049686129469478e+20,1.443110273979067e+20,1.4826966546603626e+20,1.5123452762842717e+20,1.565045674663423e
+20,1.5988319308708187e+20,1.5988637996866096e+20,1.656466684228224e+20,1.6779675119483208e+20,1.710000983287231e+20,2.241859650
0175215e+20,2.2572841568602004e+20,2.4315587760114136e+20
−8.951953125,1.512329341876376e+20,1.4493300045275439e+20,1.5736024517032146e+20,1.2636675952004689e+20,1.2833997036441846e+20,1.
3302203055099596e+20,1.380068444875847 1e+20,1.430389305009296e+20,1.451805149220618 2e+20,1.4972500805380964e+20,1.5261763423374
17e+20,1.588368336352886e+20,1.611042998787939 4e+20,1.6416105046004787e+20,1.6690654894041304e+20,1.6795290839220634e+20,2.245174
00685975e+20,2.24336810729828e+20,2.4263429131603424e+20
−12.503984375,1.3489113659711804e+20,1.365679674546365e+20,1.4459997132774198e+20,1.1543150652841243e+20,1.203339926908753e+20,1.
2467186966691371e+20,1.2844832433810668e+20,1.3303902725275097e+20,1.3880675176393022e+20,1.4203081362808511e+20,1.434808447465
6006e+20,1.492039529156324e+20,1.5182834989599307e+20,1.534855283171073e+20,1.5622943335668295e+20,1.598704455607656e+20,2.110177
703170523e+20,2.101084467731589e+20,2.2404786679999267e+20

A.28 "MITR_Cycle3_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,3.0786391462328283e+19,2.7738245461337817e+19,3.0797279974390096e+19,2.3580639753288577e+19,2.3593812197148717e+19,
2.3921742311634588e+19,2.491249067987316e+19,2.5054147566062633e+19,2.5597032843056497e+19,2.585575451258365e+19,2.5962993077719
22e+19,2.6446602357342454e+19,2.6984016820958913e+19,2.7842297144894317e+19,2.8455240701934637e+19,2.8806328822562172e+19,3.949953
815826611e+19,3.774085755885866e+19,4.1250676471269786e+19
37.224453125,4.150403355680552e+19,3.791762325711083e+19,3.874998361086984e+19,3.0886937576147812e+19,3.1403477965421388e+19,3.12
85085314759094e+19,3.1832697799428575e+19,3.258565168717585e+19,3.5246219773458936e+19,3.4972732219282104e+19,3.471725054602699
6e+19,3.461500476203197e+19,3.490628573835858e+19,3.6956141084706685e+19,3.622793864389015e+19,3.712552384063403e+19,5.0391768248
58317e+19,4.9969347095278e+19,5.360016127830766e+19
33.672421875,5.012199872291525e+19,4.776620274497683e+19,4.975147062465587e+19,3.7854310543073395e+19,3.8883673293111165e+19,4.046
951868154778e+19,4.106079144385045e+19,4.167793106163654e+19,4.361996358122625 4e+19,4.3023698037783044e+19,4.450581043082057e+1
9,4.630459262343129e+19,4.520495913457463e+19,4.673636196270172e+19,4.710125990350475e+19,4.8214915671309296e+19,6.3169835513312
41e+19,6.400320504623811e+19,6.847386875473759e+19
30.120390625,5.985335408336461e+19,5.704305567755841e+19,5.909646970833648e+19,4.796097432415074e+19,4.802927981932872e+19,4.8346
95879806859e+19,5.023056515537542e+19,5.042979836876002e+19,5.170486968854425e+19,5.2129149856103875e+19,5.3663367762959 14e+19,5.
4704415745453965e+19,5.548785746697431e+19,5.72815406490585e+19,5.9249440024131625e+19,5.951660692984332e+19,7.93379480577729e+
19,8.008155375955491e+19,8.651586766768877e+19
26.568359375,7.214250059917217e+19,6.697869014722589e+19,7.073443008815493e+19,5.569075559417483e+19,5.7001626217030566e+19,5.744
725849116992e+19,6.0453550114088665e+19,6.052950412505639e+19,6.178566661413817e+19,6.241348228521412e+19,6.532098057918182e+19,
6.5603550745859e+19,6.76851155639188e+19,6.773610566918386e+19,6.849033430956276e+19,7.0645197403941085e+19,9.532865752766538e+1
9,9.690032129307466e+19,1.0323159269681873e+20
23.016328125,8.503874805579176e+19,7.976339674857818e+19,8.288441610834328e+19,6.515844961864947e+19,6.792997429857703e+19,6.9079
90740168993e+19,6.78290563819066e+19,7.062926299604576e+19,7.421769165407384e+19,7.55312180115788e+19,7.651330868486092e+19,7.878
874213231388e+19,8.020637328806832e+19,8.1312752342934e+19,8.316433054037123e+19,8.401204104040274e+19,1.1233545107435e+20,1.1189
99105918777e+20,1.2410194901119081e+20
19.464296875000002,9.709100304088878e+19,9.275684409335928e+19,9.428336036973181e+19,7.486622205541373e+19,7.756657304674216e+19
,7.977986230340336e+19,8.005074723762394e+19,8.245737397674847e+19,8.27479113473733e+19,8.604845836942578e+19,8.829733447038655e+
19,8.996407353623796e+19,9.275737524028912e+19,9.281527025564217e+19,9.41707572206048e+19,9.584068316803529e+19,1.29599850882009
14e+20,1.3156775025708225e+20,1.4301077971364789e+20
15.912265625,1.1004673895365093e+20,1.0374839865955721e+20,1.0822809186586406e+20,8.416766709084712e+19,8.620727130144922e+19,8.9
417523345428 17e+19,9.211574974903722e+19,9.435240947061155e+19,9.616840082374921e+19,9.844914574050063e+19,1.0063853338531886e+
20,1.0126900479104401e+20,1.0388330997973767e+20,1.0755672214654085e+20,1.0996547347338476e+20,1.083210425785868e+20,1.477443611
5242025e+20,1.4847203244630694e+20,1.602911139292023e+20
12.360234375000001,1.2071323159878417e+20,1.161432434144038e+20,1.2084017571501693e+20,9.431629147938215e+19,9.709312762860814e+
19,1.002794780607441e+20,1.0326771068804814e+20,1.0575772749515822e+20,1.0853297020359465e+20,1.1053645642296736e+20,1.134598891
248304e+20,1.1465443857005024e+20,1.1610393854159533e+20,1.2056663504614713e+20,1.2198320390804188e+20,1.253034033664986e+20,1.68
03948534177097e+20,1.6605883844038155e+20,1.7894286951761474e+20
8.808203125,1.3249088362115165e+20,1.248418366844639e+20,1.3004229627456945e+20,1.031848517669924e+20,1.0531528210259791e+20,1.09
12094985493232e+20,1.1097359034622924e+20,1.148063465919857e+20,1.1817594271491795e+20,1.1959357387067236e+20,1.214382471580216
e+20,1.2576709463625266e+20,1.2838511785345522e+20,1.3333222035802502e+20,1.3536810654011823e+20,1.3649042000287924e+20,1.8333864
150900618e+20,1.838777556427981e+20,2.031435170821106e+20
5.256171875,1.4248653769389154e+20,1.3549929983178976e+20,1.4331937607988742e+20,1.1133211452387413e+20,1.1349388252834043e+20,1.
1729317651751672e+20,1.2075997252861046e+20,1.2206287594751853e+20,1.266020576099679e+20,1.3077421674389488e+20,1.3273733579659
94e+20,1.3326210896328558e+20,1.3861288313453706e+20,1.4349040539129727e+20,1.4656680840895545e+20,1.496814540055624e+20,2.024105
3431892545e+20,2.0053664795043476e+20,2.175285693830837e+20
1.704140625,1.5001873230601355e+20,1.4407626045491552e+20,1.496272770187183e+20,1.1663880349994854e+20,1.2039295000008804e+20,1.2
338171377432191e+20,1.2653672653759706e+20,1.2799419371308 98e+20,1.3285896844353377e+20,1.3483589731641424e+20,1.40285464816616
75e+20,1.4371348710183186e+20,1.4690727359098564e+20,1.4991250292004468e+20,1.5291985683682312e+20,1.579939034576257e+20,2.101541
254091255e+20,2.1174119243550025e+20,2.3085345241208766e+20
−1.847890625,1.5453932382591838e+20,1.4765884649671534e+20,1.5395346876230012e+20,1.2205331530278134e+20,1.2761548595211084e+20,
1.2914784484471164e+20,1.3203144152683633e+20,1.343929207769241e+20,1.3872548628366405e+20,1.436895854899889e+20,1.4573609461067
896e+20,1.5101994426877005e+20,1.5232072309995874e+20,1.5776869715937172e+20,1.6115635227791868e+20,1.655340652736954e+20,2.22264
27540957554e+20,2.2345670026707593e+20,2.4100207680062272e+20
−5.399921875,1.5708086188522334e+20,1.482999408410374e+20,1.549206873215466e+20,1.2276080301333393e+20,1.2705725052884446e+20,1.2
941872977893222e+20,1.328510012395861e+20,1.3703697019368902e+20,1.4226345598335694e+20,1.438685820053464e+20,1.479212330800584
e+20,1.5142786511089048e+20,1.552027263412939e+20,1.5931061669670974e+20,1.6238755086129775e+20,1.677940954601829e+20,2.264534312
4525754e+20,2.2594193675181746e+20,2.4683778811882203e+20
−8.951953125,1.5149903879948963e+20,1.4441194531457712e+20,1.5055094152971754e+20,1.1993244561191304e+20,1.2403130646952149e+20,
1.2718950611437571e+20,1.3072588237327904e+20,1.3637091194366427e+20,1.3987276365212774e+20,1.4140140451621962e+20,1.45606494759
79693e+20,1.4983283088056805e+20,1.532773187206083e+20,1.582217654905289e+20,1.6451426316839433e+20,1.641180275587305e+20,2.25394
324267148e+20,2.2396128985042806e+20,2.4643677218678958e+20
−12.503984375,1.4002785855564227e+20,1.3581108307960837e+20,1.4267828173556533e+20,1.1294839463139033e+20,1.168517934188161e+20,
1.2244317714928704e+20,1.2534483282702646e+20,1.276547908249193e+20,1.3162723871322482e+20,1.3620731868927222e+20,1.386904305862
9432e+20,1.4294279290662778e+20,1.4512899366986691e+20,1.4850815043753637e+20,1.544198157667034e+20,1.580873853172783e+20,2.07524
9481063962e+20,2.1050308894203317e+20,2.239124243328824e+20

A.29 "MITR_Cycle3_cladding_fluence_position_8.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,3.0148855802436174e+19,2.8344762140527022e+19,2.892875818989083e+19,2.3020173512916873e+19,2.2910597901289992e+19,2.
378210378377852e+19,2.3594768261622436e+19,2.4267943880507097e+19,2.4449224327662965e+19,2.4846097313642623e+19,2.5526390301387
207e+19,2.6429658770280415e+19,2.6803798667662746e+19,2.6630272965682606e+19,2.7310034806497346e+19,2.7131781896841597e+19,3.7102
950096115655e+19,3.6409165976353026e+19,3.941476710826297e+19
37.224453125,3.9184355570096505e+19,3.668950532592484 8e+19,3.83857761610756e+19,2.9535699786623914e+19,2.920054607389216e+19,3.12
4689585050329e+19,3.134159934809453e+19,3.1582208907313996e+19,3.2171835114134155e+19,3.376474475673721e+19,3.3780732279325524e
+19,3.466413585304257e+19,3.467385584185871e+19,3.5684097302422573e+19,3.4733716100852167e+19,3.6588056262324584e+19,4.846152719
0836e+19,4.904329242309446e+19,5.314762409408032e+19
33.672421875,4.8617949961675145e+19,4.695062663420 09e+19,4.8447611141274075e+19,3.721831520927783e+19,3.7357263246125105e+19,3.81
4495414308419e+19,3.868444007972705e+19,4.032106311465629e+19,4.137384944430068e+19,4.189591376164464e+19,4.258119953052976e+19,
4.345594540929034e+19,4.443065314024761e+19,4.469845742227511e+19,4.5153172308914815e+19,4.668393776072608e+19,6.172936503957468
e+19,6.269817703961067e+19,6.806063644331872e+19
30.120390625,5.9649393662304395e+19,5.5777863690669285e+19,5.852336217103448e+19,4.453571400297081e+19,4.704426783793246e+19,4.84
0073491 05302e+19,4.863345945202 66e+19,4.925925676476915e+19,5.1836328553680716e+19,5.158413999139064e+19,5.186357639118173e+19,
5.395337398665412e+19,5.462846173448622e+19,5.54458968595166e+19,5.694001317316862e+19,5.901520422807029e+19,7.568418832737396e+
19,7.6025184656334e+19,8.507167916544211e+19
26.568359375,7.0478417267969974e+19,6.64417006011533e+19,6.917126467362315e+19,5.430499325421105e+19,5.613054525208591e+19,5.7240
11118853066e+19,5.7043586824488255e+19,5.889410272806579e+19,6.0489668105318064e+19,6.137562118429837e+19,6.338654346068889e+19,
6.3524110515518554e+19,6.533532154628761e+19,6.534381989716513e+19,6.656705127659654e+19,7.043273863200337e+19,9.277968341134259
e+19,9.247480507361195e+19,1.0136514238534584e+20
23.016328125,8.419050640883042e+19,7.969275420690887e+19,8.098875271572898e+19,6.216331208125746e+19,6.466288953310 4734e+19,6.653
677590159543e+19,6.6719490445461864e+19,6.822210510999138e+19,7.09415773907942e+19,7.318248628780732e+19,7.488109416944938e+19,7.
500272681638371e+19,7.832186398098075e+19,7.836010655992953e+19,7.897198782311018e+19,8.030994693938797e+19,1.0840177691192312e
+20,1.0822968530665359e+20,1.1952240018207321e+20
19.464296875000002,9.572542428425907e+19,9.011969958668234e+19,9.252260829729794e+19,7.40312590816985e+19,7.458843221110517e+19,
7.578563739097422e+19,7.70779178712854e+19,8.01654749744703e+19,8.10094674459929e+19,8.258166235833203e+19,8.522571177509696e+19,
8.655304795277785e+19,8.773538101861127e+19,9.103592804066374e+19,9.4623825551762e+19,9.35556890758451e+19,1.2565874066256442e+2
0,1.2404724087741682e+20,1.3766319242397545e+20
15.912265625,1.0769482034830036e+20,1.0184583035685493e+20,1.0629524818816064e+20,8.257103941973515e+19,8.496916780798217e+19,8.6
78037883875122e+19,8.873606183443794e+19,9.066731207135181e+19,9.434178653201467e+19,9.53876148368781e+19,9.76848253084547e+19,9.
886450263963887e+19,1.0088286097304722e+20,1.028475734665413e+20,1.048871776771434e+20,1.0878685843606025e+20,1.4266765879696844
e+20,1.433183137860277e+20,1.5371179690922095e+20
12.360234375000001,1.205002416799166e+20,1.116996681993264e+20,1.174551763311 1923e+20,9.200686462841913e+19,9.502484148379445e+1
9,9.644353493340858e+19,9.811717890934795e+19,9.912423348833273e+19,1.0293096353452684e+20,1.0497163003898865e+20,1.087916387584
2884e+20,1.1148774057431853e+20,1.1336003350201968e+20,1.1611509262712203e+20,1.1681301969293746e+20,1.2033877301324392e+20,1.589
6430889845126e+20,1.6045895635903313e+20,1.7620958741663597e+20
8.808203125,1.3325307946547823e+20,1.2340242850458585e+20,1.299052603666696e+20,9.952365597957566e+19,1.0007657993354358e+20,1.05
46240980216476e+20,1.0765126630005313e+20,1.1105113779798648e+20,1.1467780903496337e+20,1.1504270697576641e+20,1.17031852228033
31e+20,1.2034780251105128e+20,1.2367596917345565e+20,1.2888174023285963e+20,1.3061699725266097e+20,1.3262367035361273e+20,1.78853
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5.256171875,1.3793620194591542e+20,1.3008478802895698e+20,1.3703272101825026e+20,1.058692683504255e+20,1.101604043966376e+20,1.111
76499927169722e+20,1.1527747391875758e+20,1.1769047442104023e+20,1.2126243752424315e+20,1.2486892517788597e+20,1.26253625223990
03e+20,1.294686575903376e+20,1.3523213292607809e+20,1.3911481698323992e+20,1.430426485294385e+20,1.452894000426799e+20,1.91077452
2768376e+20,1.917477597023011e+20,2.0851500598362597e+20
1.704140625,1.4714947659099465e+20,1.3856826679243014e+20,1.4451180093738785e+20,1.123965329712821e+20,1.1840646048247038e+20,1.1
99494423136 6805e+20,1.2048058924351234e+20,1.2332169417124951e+20,1.2692552609024 31e+20,1.2997006029211061e+20,1.36067627046723
17e+20,1.3880993864550928e+20,1.4385264759745107e+20,1.445208304351952e+20,1.5142361593545173e+20,1.5370223626448375e+20,2.048909
904812983e+20,2.0347229703168416e+20,2.2219469516176592e+20
−1.847890625,1.5277857175348452e+20,1.4479702683871424e+20,1.4912693661080496e+20,1.1924567263162437e+20,1.2030053043429512e+20,
1.2433724710111181e+20,1.2743967631833237e+20,1.3248291641720398e+20,1.3439026504227488e+20,1.3679370489982034e+20,1.40393287643
37514e+20,1.4127074237147795e+20,1.4587206822471908e+20,1.5074268557139136e+20,1.5669365577336685e+20,1.6043399245333044e+20,2.11
24563234995552e+20,2.1382753757592866e+20,2.3045243648005525e+20
−5.399921875,1.553742867996336e+20,1.4562402260848181e+20,1.4959328361520828e+20,1.183841523114169e+20,1.2149348643872544e+20,1.2
521257724149522e+20,1.2853383899381162e+20,1.3303159119573315e+20,1.3565014555986554e+20,1.3834571622882537e+20,1.4146779788245
017e+20,1.476046695098712e+20,1.4953538859985522e+20,1.525082179661938e+20,1.5851124056729407e+20,1.630164288262334e+20,2.1833113
239407847e+20,2.205183954511773e+20,2.3774720841453686e+20
−8.951953125,1.529421650078766e+20,1.4356848398998436e+20,1.479594756590072e+20,1.1613633850431581e+20,1.1993828822814132e+20,1.2
22578068707714e+20,1.2576072087309455e+20,1.2974007367148803e+20,1.3349315787776786e+20,1.3580364702259054e+20,1.38946443406479
28e+20,1.436895854899889e+20,1.472833256173154e+20,1.5081757728849936e+20,1.578956412756045e+20,1.5867802070326516e+20,2.15860768
02337263e+20,2.1786478538967515e+20,2.356210272543701e+20
−12.503984375,1.3790805115863368e+20,1.317010681364732e+20,1.348279301124666e+20,1.0847135715973272e+20,1.1248098533312735e+20,1.
147452646950536e+20,1.185079095460706e+20,1.2159918467776445e+20,1.260480713621403e+20,1.2949946411226856e+20,1.325381556979078
1e+20,1.3911428583631009e+20,1.3954239026176459e+20,1.4444487642422747e+20,1.4816131149234802e+20,1.540586358544093e+20,2.0364598
207774327e+20,2.0284713709525742e+20,2.1639935101023473e+20

A.30 "MITR_Cycle3_cladding_fluence_position_9.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9053790177176183e+19,2.6737670874897125e+19,2.8220739332408386e+19,2.2876392039008023e+19,2.1979284874501e+19,2.29
6631521423066e+19,2.304152561949661e+19,2.3650432459870118e+19,2.417127513927544e+19,2.433205331493931e+19,2.510959930553838e+1
9,2.4449383671741915e+19,2.473577809631396e+19,2.5052288551808176e+19,2.559251809415282e+19,2.5821707994380632e+19,3.36002485525
64433e+19,3.3422633019224494e+19,3.5496071289257693e+19
37.224453125,3.772062086083159e+19,3.5788839476987867e+19,3.749036866674408e+19,2.9711296961630437e+19,3.0196924599587082e+19,3.0
276809097835667e+19,3.036827259915485e+19,3.154301026389149e+19,3.1932181619388408e+19,3.1943548163687076e+19,3.21692856088709e
+19,3.3626062293354865e+19,3.2455520689363997e+19,3.3480846722735436e+19,3.3990694660692976e+19,3.5886251823921316e+19,4.6294022
79952739e+19,4.54019084161609e+19,4.8795937297865966e+19
33.672421875,4.781188138094344e+19,4.443421182467757e+19,4.675102161796542e+19,3.7118831389318e+19,3.7340744576606945e+19,3.83812
6141217192e+19,3.928293644027561 6e+19,3.95548836683559e+19,4.022396945588076e+19,4.070099251357392e+19,4.206375619147544e+19,4.29
8418070620262e+19,4.272476854566667e+19,4.331513835818861e+19,4.2892026713874645e+19,4.427810774199633e+19,5.673180357666966e+1
9,5.677482647798704e+19,6.250112152863844e+19
30.120390625,5.7405297883712225e+19,5.348065321909269e+19,5.588674881128737e+19,4.497842496899604e+19,4.562111275410764e+19,4.534
396028611488e+19,4.694728040854289e+19,4.737867794962425e+19,4.899463936432071 5e+19,5.015280524484622e+19,5.1067652716810035e+
19,5.160251767516325e+19,5.231441390523357e+19,5.464970761168001e+19,5.423753759412083e+19,5.4644396142381556e+19,7.158957664520
425e+19,6.9339107103453946e+19,7.405728528126086e+19
26.568359375,6.922809739511648e+19,6.621012053974 1176e+19,6.7718046673569145e+19,5.245670816773885e+19,5.4195045839733285e+19,5.5
2190971204731e+19,5.539278216653217e+19,5.624845987051133e+19,5.854513919515809e+19,5.876025370174503e+19,5.989372125003276e+19,
6.0475858285142114e+19,6.2536708372938015e+19,6.365264807254088e+19,6.380243150675697e+19,6.582450786867423e+19,8.51131086259699
5e+19,8.409011963908984e+19,9.137533092883425e+19
23.016328125,7.915948268934521e+19,7.461233382294815e+19,7.661794463004023e+19,6.218030878301247e+19,6.233168565801811e+19,6.4708
03702214151e+19,6.610017312526341e+19,6.629297946079689e+19,6.8058511855599346e+19,7.008324395216582e+19,7.1976782757060755e+19,
7.232893317154751e+19,7.327065667816145e+19,7.486515976155405e+19,7.616912547432181e+19,7.611972880984628e+19,1.0159990932833703
e+20,9.776077931942244e+19,1.080873379294553e+20
19.464296875000002,9.217364476439023e+19,8.690466722033479e+19,8.989183755377911e+19,6.981289016487507e+19,7.089802334254697e+19
,7.321382395666812e+19,7.538887063438054e+19,7.806372657307643e+19,7.970868861480423e+19,8.13642735951289e+19,8.155017502057441e+
19,8.256838368508594e+19,8.498722680359687e+19,8.687279840454414e+19,8.778902685852554e+19,8.760153199229051e+19,1.1514946750866
51e+20,1.1505226762050362e+20,1.2406529987303154e+20
15.912265625,1.0673981816844032e+20,9.85994603216466e+19,1.0259899670337416e+20,7.959342973102801e+19,8.168296175303547e+19,8.485
762695271485e+19,8.451450603603544e+19,8.783948581686075e+19,9.03974894309909e+19,8.98578441502691e+19,9.281739484336153e+19,9.50
19530014496e+19,9.499722184344255e+19,9.862814225585819e+19,9.916566294886061e+19,1.0067677596426763e+20,1.3433396346771137e+20,
1.312766817395276e+20,1.4347128410182286e+20
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19,9.385206906269822e+19,9.645522016586514e+19,9.662306259569594e+19,9.850172928655524e+19,1.0063109732830103e+20,1.031349239555
8704e+20,1.0682798855879449e+20,1.0641953656974421e+20,1.0973070653039359e+20,1.1213892671030764e+20,1.1506235941217065e+20,1.493
707330516035e+20,1.5130623246395615e+20,1.5909125301468404e+20
8.808203125,1.2553870145641964e+20,1.1897425655047392e+20,1.2472498435989815e+20,9.776980881722979e+19,9.950241010238189e+19,1.01
39010629104855e+20,1.0527119690742083e+20,1.0570408165524393e+20,1.0804697076278713e+20,1.1167045511818494e+20,1.13090742008588
62e+20,1.158686404516743e+20,1.1791302498464501e+20,1.2093631330931874e+20,1.2466868278533464e+20,1.2487689238183363e+20,1.632453
531529963e+20,1.6226857394901264e+20,1.7442280914463975e+20
5.256171875,1.3456235664754444e+20,1.2659302811216056e+20,1.3454642223964909e+20,1.0364376271437788e+20,1.0627825148640562e+20,1.
1121048187693978e+20,1.117299435743275e+20,1.1491948088804255e+20,1.1851640789694813e+20,1.1991385546936848e+20,1.2262004907692
52e+20,1.2476428923270663e+20,1.2765532197184915e+20,1.2962481478771183e+20,1.337900690115508e+20,1.3372208220453075e+20,1.764061
1178067837e+20,1.7457684175429458e+20,1.8606183181831786e+20
1.704140625,1.4106412621576849e+20,1.3374917069795282e+20,1.3970970154466551e+20,1.1009772905891598e+20,1.1268388346032788e+20,1.
1684541965565796e+20,1.1758477618200124e+20,1.2257649502867795e+20,1.2396385080943126e+20,1.2829535402231154e+20,1.301713649785
216e+20,1.3297688306195918e+20,1.3557047352038892e+20,1.3638259717612084e+20,1.3740824189765018e+20,1.4293110767417 12e+20,1.90113
95174610004e+20,1.857489862766396e+20,1.992172789767015e+20
−1.847890625,1.4869033383447295e+20,1.3984939318721459e+20,1.4339214320927606e+20,1.1254153608312961e+20,1.1844523420834898e+20,
1.204821826843019e+20,1.2290102580281282e+20,1.2655956585558036e+20,1.2996421767588233e+20,1.3152525850269472e+20,1.359295288449
6366e+20,1.3738912060817578e+20,1.4081289371795215e+20,1.4337036618515245e+20,1.4504347901416199e+20,1.4868289777745514e+20,1.992
1568553591197e+20,1.961201612287794e+20,2.0673300803399834e+20
−5.399921875,1.4922041847045757e+20,1.4001298644160663e+20,1.4278451112153417e+20,1.1487433339900577e+20,1.1844523420834898e+20,
1.2103882466677871e+20,1.2576390775467359e+20,1.2677999183146574e+20,1.3005079462544695e+20,1.3434246181858887e+20,1.37612202318
71039e+20,1.4167600747894913e+20,1.4260817034082589e+20,1.4631185788826302e+20,1.4826222940901845e+20,1.5201000214599985e+20,2.010
4973588466434e+20,1.9861761409290728e+20,2.1502314931500817e+20
−8.951953125,1.4310054354479153e+20,1.3647076756647499e+20,1.423362231127456e+20,1.114479045545802e+20,1.1553879820824101e+20,1.1
975451139041518e+20,1.2091134940361608e+20,1.260863139410891e+20,1.2776686282711643e+20,1.3371995761681136e+20,1.36880281849384
94e+20,1.3904629902929001e+20,1.4211048566756175e+20,1.4596289434972247e+20,1.4712451268529196e+20,1.5013983380601805e+20,1.98131
0835051699e+20,1.9853953549422015e+20,2.1050521352975257e+20
−12.503984375,1.3361160364312312e+20,1.2630089730074617e+20,1.3135210460356547e+20,1.0396829348851278e+20,1.0914060229133656e+20
,1.1134273746247103e+20,1.1265573267304612e+20,1.1670147883767017e+20,1.2021129775008129e+20,1.221791971251544e+20,1.24800938370
86587e+20,1.301724272723813e+20,1.3070888567152402e+20,1.3589712888224314e+20,1.384466341454958e+20,1.4040391058197203e+20,1.8673
532612536045e+20,1.8671726712974574e+20,1.9419262902037443e+20

A.31 "MITR_Cycle4_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.501904009654935e+19,2.464473765685148e+19,2.7528888484516246e+19,2.876017725433629e+19,2.9336863630216536e+19,2.71
70043432484168e+19,2.961950016631493e+19,3.0162216236743602e+19,3.059241690393057e+19,3.10512856331256e+19,3.1104218199027655e+
19,3.2366960955376935e+19,3.1889309702797308e+19,3.1858394927465718e+19,3.1894611946240807e+19,3.190584551285839e+19,3.060751481
7464603e+19,2.9479395123460436e+19,2.6570800054835814e+19
37.224453125,3.382489816527334e+19,3.593869592857073e+19,3.6693861210871132e+19,3.7419369877301125e+19,3.8149371970378146e+19,4.0
1411282659421e+19,3.823151180948591e+19,4.096585179273858e+19,4.206665145272878e+19,4.187622003142752e+19,4.329767061695001e+19,
4.1885476490320404e+19,4.3577071885862535e+19,4.295608032324255e+19,4.130573458432017e+19,4.174114762641769e+19,4.17027737628520
4e+19,3.9846269609363775e+19,3.786943149026797e+19
33.672421875,4.15774071593998e+19,4.424731140453401e+19,4.51557924040312e+19,4.841622277911853e+19,4.860647446335391e+19,4.878297
626204938e+19,4.843311806331364e+19,5.1297947156393886e+19,5.021610975685417e+19,5.210676395285987e+19,5.349595173505667e+19,5.3
477888159935594e+19,5.381067133741488e+19,5.606709046248909e+19,5.520336399239635e+19,5.586021309965968e+19,5.144919589733302e+
19,5.045183491875753e+19,4.673810766351749e+19
30.120390625,5.007097207615499e+19,5.168815632642226e+19,5.445637674659356e+19,5.870688874908819e+19,6.2456922891636244e+19,6.203
651789453981e+19,6.100150200066218e+19,6.246051763295388e+19,6.394388763767249e+19,6.614980064723406e+19,6.479961580833351e+19,6.
7713962463066325e+19,6.714024174877312e+19,6.762481287838919e+19,6.671669135302377e+19,6.895540637710868e+19,6.361964197082258e
+19,6.206446700828436e+19,5.788306390762111e+19
26.568359375,6.119148407929874e+19,6.305284113356580 5e+19,6.599459769084552e+19,6.905093662762459e+19,7.1037570416810934e+19,7.05
5102217947018e+19,7.44406221536751e+19,7.424237217000799e+19,7.469441089069954e+19,7.677109294989242e+19,7.905824711323233e+19,7.
9109651914074 39e+19,7.924085997216778e+19,8.147103748562331e+19,7.929100661354866e+19,8.286229224407774e+19,7.649888116361514e+
19,7.483550448741638e+19,7.391273439118162e+19
23.016328125,7.138392374276425e+19,7.446839153035375e+19,7.664437831844608e+19,8.057280149888136e+19,8.251153536001032e+19,8.3444
99981166502e+19,8.689307568353244e+19,8.66008232144094e+19,8.848707385230102e+19,8.946457388509662e+19,9.125969783058642e+19,9.27
5061679207206e+19,9.473311662874311e+19,9.688097456602502e+19,9.660148342857954e+19,9.373827196908995e+19,9.191214337973564e+19,
9.203975669651138e+19,8.621286075770456e+19
19.464296875000002,8.140040081873196e+19,8.382900805292047e+19,9.104401335152881e+19,9.160389431174916e+19,9.560394271293817e+19
,9.549250573209174e+19,9.896412715958962e+19,1.0190642292806699e+20,1.0089809798847272e+20,1.0150740664181044e+20,1.067799934694
3925e+20,1.0683301590387425e+20,1.0813700831684328e+20,1.0897278567319146e+20,1.0843177710488866e+20,1.0954434954269411e+20,1.059
2534372117353e+20,1.0414774413960718e+20,9.748489110738628e+19
15.912265625,9.420109491373438e+19,9.854444111075672e+19,1.0452609066328737e+20,1.045845052096988e+20,1.0541129571275291e+20,1.09
47155603101218e+20,1.114495624410362e+20,1.1682010596957033e+20,1.1719665512259171e+20,1.2062424096894868e+20,1.202440970746096
7e+20,1.2355125908682613e+20,1.23480262945803e+20,1.2356743542275547e+20,1.2519854929562852e+20,1.22936558321512e+20,1.2054246060
39727e+20,1.1901289817332256e+20,1.110298763922033e+20
12.360234375000001,1.0217512984155958e+20,1.0930799530106015e+20,1.1183240239136344e+20,1.161955196656327e+20,1.1620270914826795
e+20,1.1948380928593163e+20,1.238693936934361e+20,1.2699591995444183e+20,1.2639290209840996e+20,1.3212561581469498e+20,1.32616298
00455101e+20,1.3256417425544543e+20,1.394534959906768e+20,1.3839933809928279e+20,1.372912590881244e+20,1.3621643143415403e+20,1.3
390860750823778e+20,1.3184073256527305e+20,1.2423965204915164e+20
8.808203125,1.092882242238132e+20,1.1693963111838142e+20,1.2423875336382222e+20,1.2310281510745222e+20,1.2821273989045843e+20,1.3
31033854530894e+20,1.3435705148761167e+20,1.392908339460542e+20,1.3866175421546953e+20,1.4432886390270786e+20,1.478427235406878
5e+20,1.4640932044028425e+20,1.4887800904016432e+20,1.4763512722959493e+20,1.482965596320382e+20,1.491952449614449e+20,1.45434246
85787806e+20,1.4076108314496347e+20,1.36317982876377e+20
5.256171875,1.229680123080412e+20,1.2568473805883752e+20,1.2841853883089255e+20,1.3974017661075753e+20,1.397806174505808e+20,1.41
52496567495911e+20,1.45968964628875e+20,1.4966256133273633e+20,1.527234835646954e+20,1.550924180930113e+20,1.5844990648367455e+
20,1.565671607185676e+20,1.5894687947083642e+20,1.6373417622058566e+20,1.6475058932814456e+20,1.604557721389102e+20,1.6070650534
581463e+20,1.597278370220908e+20,1.4736821768676116e+20
1.704140625,1.255346576088266e+20,1.2782001440150772e+20,1.3776935968336873e+20,1.4528686246385536e+20,1.504821623531552e+20,1.48
97596574106966e+20,1.5366440710458416e+20,1.5826387862048735e+20,1.6456186540896915e+20,1.6657402186151064e+20,1.68050561857725
8e+20,1.7118068286004914e+20,1.7453727256538297e+20,1.761360337663974e+20,1.7133256068071886e+20,1.7212160639993787e+20,1.6608513
704231345e+20,1.687928759398157e+20,1.575350448183386e+20
−1.847890625,1.3103730788078353e+20,1.3631618550571819e+20,1.426815736939055e+20,1.4884206162698807e+20,1.5507084964510555e+20,1.
569832520260829e+20,1.6048902349609825e+20,1.6568072864408044e+20,1.6281212507261442e+20,1.679858565140085e+20,1.72357061956242
47e+20,1.7421823927344362e+20,1.800731741945279e+20,1.751088364348856e+20,1.812324782694625e+20,1.803796258918556e+20,1.773043246
9462607e+20,1.705812597453349e+20,1.6435516778320563e+20
−5.399921875,1.2943495193845146e+20,1.3856110145857598e+20,1.4420484532724974e+20,1.5373809930159548e+20,1.5098183139630527e+20,
1.5584551639905406e+20,1.630269108663426e+20,1.6416284912271262e+20,1.7007350253422015e+20,1.7415083787373812e+20,1.830882634746
8725e+20,1.8087749756434688e+20,1.8261555499141936e+20,1.837425063944953e+20,1.8335337564686218e+20,1.8522803324400448e+20,1.8133
76244530031e+20,1.7550425797982454e+20,1.662019661351363e+20
−8.951953125,1.2633898097864557e+20,1.343022316825179e+20,1.4290894108224537e+20,1.4530843091176112e+20,1.5373360587494847e+20,1.
5479405456364827e+20,1.6067595004461482e+20,1.653230518829766e+20,1.686356060071695e+20,1.668247550684151e+20,1.755150422037774
e+20,1.7536046832711945e+20,1.7693945845088695e+20,1.8146074434313178e+20,1.85728600972484e+20,1.810680188541811e+20,1.7743643143
804885e+20,1.728019111942987e+20,1.6554143241802242e+20
−12.503984375,1.1318582249744984e+20,1.185186212421489e+20,1.2337242070627423e+20,1.3224514096350608e+20,1.36534566040764e+20,1.4
10360808557619e+20,1.4374651580925236e+20,1.4578473413634662e+20,1.4790473282841693e+20,1.546035332738141e+20,1.544911976076382
4e+20,1.617166276560677e+20,1.6433000459398224e+20,1.6704403428879034e+20,1.6843609786404123e+20,1.6591798157104382e+20,1.6415296
358408913e+20,1.5471766631064873e+20,1.5023502388756834e+20

A.32 "MITR_Cycle4_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.7472313201031803e+19,2.6402644820445053e+19,2.7489149590140346e+19,2.203994868746805e+19,2.2776925770339037e+19,2.
2891487572569706e+19,2.3379689745079804e+19,2.340406798672277e+19,2.386693590811371e+19,2.361842655594065e+19,2.390273315183344
4e+19,2.583960835060393e+19,2.5286078863887196e+19,2.6119772238766555e+19,2.592474630562283e+19,2.581177785251915e+19,3.526756135
828593e+19,3.534032430175186e+19,3.71286462502492e+19
37.224453125,3.764005820837581e+19,3.409480328918285e+19,3.677141737597778e+19,2.8909195443752456e+19,2.8989022235014713e+19,2.97
4012701347836e+19,3.06555060869348e+19,3.037794466901294e+19,3.1885146719871468e+19,3.1577789673274868e+19,3.2150014456414966e+
19,3.2709439334901576e+19,3.3217823929686897e+19,3.3372272571729908e+19,3.4113551697243783e+19,3.5189859032526328e+19,4.54108750
12349395e+19,4.453756035584553e+19,4.8989112258471076e+19
33.672421875,4.65571834859442e+19,4.462142363156283e+19,4.541501772007957e+19,3.618389644121856e+19,3.763671217520913e+19,3.7775
22732598223e+19,3.753792452933567e+19,3.894081532401385e+19,4.010799667117247e+19,4.05527004125157e+19,4.089840406143522e+19,4.21
086058350226e+19,4.2402844307140305e+19,4.2999712890111205e+19,4.4407118173982515e+19,4.431544748754296e+19,5.917433099457442e+
19,5.886522126393813e+19,6.264836320779088e+19
30.120390625,5.3985324004339786e+19,5.2934563366726664e+19,5.4320989553759265e+19,4.280022557941191e+19,4.548815244500847e+19,4.6
04747110021995e+19,4.7206951259931435e+19,4.738253833372587e+19,4.875600528119227e+19,4.9254351776477544e+19,5.0747532355018555
e+19,5.2528259339343905e+19,5.230720870379266e+19,5.4021439917884916e+19,5.375003944992074e+19,5.337082235769685e+19,7.1454803832
671175e+19,7.019818248785079e+19,7.673834953777396e+19
26.568359375,6.797014929194144 6e+19,6.420612753761259e+19,6.735883434355249e+19,5.199751474514314e+19,5.286519956806497e+19,5.405
755583143005e+19,5.522760520701725e+19,5.658620089596506e+19,5.796604123993944e+19,5.861240986912219e+19,6.0417143313627546e+19,
6.197756322532758e+19,6.210874897011653e+19,6.432297314025862e+19,6.441698073875109e+19,6.632475076013521e+19,8.702182480337509e
+19,8.612476924487904e+19,9.285401372453849e+19
23.016328125,7.853458512026868e+19,7.480667927948496e+19,7.74904103257138e+19,6.1163361821435085e+19,6.246512805818688e+19,6.3135
39692427452e+19,6.48020401111073e+19,6.680912889474054e+19,6.740344811910825e+19,6.804397446815871e+19,7.075319910041939e+19,7.26
00952971353506e+19,7.465053106503983e+19,7.380871160961281e+19,7.560919612311287e+19,7.632407876475624e+19,1.0113889806851701e+
20,1.01922294722621e+20,1.1020133678646723e+20
19.464296875000002,9.090906555685791e+19,8.617575641694277e+19,8.779725471184413e+19,6.920605709811094e+19,7.1429841363014975e+1
9,7.194343089828 18e+19,7.437753724794867e+19,7.670223362422882e+19,7.901418320749303e+19,7.745907445954963e+19,8.064949052816166e
+19,8.200011947147452e+19,8.44676861528082e+19,8.52011578675998e+19,8.637704952331931e+19,8.8422909802376e+19,1.1788499739319439e
+20,1.1472857277262486e+20,1.2514217155026359e+20
15.912265625,1.0243747760701483e+20,9.619314237833637e+19,1.0070656933872673e+20,7.86838288218302e+19,8.180307529610328e+19,8.165
542494366876e+19,8.370606527010642e+19,8.701067135948615e+19,8.819346752808927e+19,8.94554000366663e+19,9.261182465723584e+19,9.2
5140992441137e+19,9.473204122888544e+19,9.650968773823193e+19,9.851996322011918e+19,1.001855441742006e+20,1.314507718603974e+20,
1.336719005434231e+20,1.4248683903031401e+20
12.360234375000001,1.1318674193407305e+20,1.0474092932826843e+20,1.0904987648402825e+20,8.578485476445426e+19,8.936192355454947e
+19,9.178859427495705e+19,9.36841486197009e+19,9.644967158778194e+19,9.9390462959832e+19,1.008579375057909e+20,1.038523716317830
2e+20,1.0388689419620114e+20,1.0674854922827733e+20,1.1040900328935177e+20,1.0987948026281499e+20,1.1197685883031108e+20,1.471011
781020805e+20,1.4744003034975396e+20,1.582572775728971e+20
8.808203125,1.2227892316906044e+20,1.179476691255227e+20,1.1988411943119267e+20,9.410532390560224e+19,9.750871764085542e+19,1.003
114187552329e+20,1.0351936166424183e+20,1.0603897775039057e+20,1.0866269264616941e+20,1.0856868504767693e+20,1.1081424508398315
e+20,1.1297907543118855e+20,1.1507698511506026e+20,1.185042790872183e+20,1.2104620206114492e+20,1.237803891630618e+20,1.617934504
0206627e+20,1.607248442542308e+20,1.7588981012855728e+20
5.256171875,1.3274032242050892e+20,1.2366991695692374e+20,1.2990947213822122e+20,1.0374986617127992e+20,1.0553760389176404e+20,1.
0748626987407424e+20,1.119688920846761e+20,1.1361535284923369e+20,1.1585029055802665e+20,1.1738521688369485e+20,1.2062449565886
792e+20,1.235270466518702e+20,1.2716891163979638e+20,1.2723530118675436e+20,1.3094355572163851e+20,1.3368411622006335e+20,1.75942
390649748e+20,1.7755432884988743e+20,1.8922561120509788e+20
1.704140625,1.4121640865972637e+20,1.3130046592608454e+20,1.381630206160356e+20,1.077868817426999e+20,1.1019018334257829e+20,1.15
20179746334125e+20,1.1826740118367229e+20,1.1952402252849268e+20,1.2087996263556217e+20,1.2424192929351357e+20,1.28205650805091
99e+20,1.3047298661280047e+20,1.3564074894800855e+20,1.3696163537428424e+20,1.397282205751166e+20,1.3993694931075246e+20,1.886328
8532985715e+20,1.8590613385719955e+20,2.0271702937970793e+20
−1.847890625,1.452428019036331e+20,1.3677680687555327e+20,1.4038043148443166e+20,1.1093215291938053e+20,1.1609035515982668e+20,1.
1872203680124043e+20,1.2037062203130066e+20,1.2181791415498433e+20,1.2557556251280522e+20,1.2935976668940927e+20,1.3303774759 08
8061e+20,1.3650593752396476e+20,1.3992632698323917e+20,1.4189729985332729e+20,1.4544250166088267e+20,1.4768328164980795e+20,1.967
8233499804143e+20,1.9814039957061357e+20,2.114804495781595e+20
−5.399921875,1.4560236768995746e+20,1.3896659969241481e+20,1.4418003803593e+20,1.1260676285184827e+20,1.1610203972009125e+20,1.19
92820209037278e+20,1.2178657828882016e+20,1.2437311503830252e+20,1.2885626836528007e+20,1.3156814857941906e+20,1.36083168888936
45e+20,1.3866439447466213e+20,1.4283843807100 34e+20,1.48092241259069e+20,1.51048966122389e+20,1.5136869818053856e+20,2.0136374485
451676e+20,1.977282532630985e+20,2.162849283124136e+20
−8.951953125,1.3993270037974716e+20,1.347819337685603e+20,1.3912859018699225e+20,1.0955709262278712e+20,1.1240706309459871e+20,1.
1852180592761522e+20,1.2072115883923872e+20,1.2419306658695252e+20,1.2728894394069639e+20,1.2920786740596944e+20,1.326962397613
2882e+20,1.3591799169610501e+20,1.3947275359842234e+20,1.4458740429606407e+20,1.4826910301216506e+20,1.5038294618730666e+20,1.974
5101051500206e+20,1.978063273703211e+20,2.127864647459166e+20
−12.503984375,1.279932042548265e+20,1.2185084337027545e+20,1.2949467024882788e+20,1.018845854599489e+20,1.0494859583115293e+20,1.
0949548312321014e+20,1.1267315239880622e+20,1.1442742978762352e+20,1.1759129003745236e+20,1.212443084692675e+20,1.2611039670309
857e+20,1.2787370307030216e+20,1.3190009631420888e+20,1.341477808160178e+20,1.3690002587470723e+20,1.4200724094308966e+20,1.85748
9234100031e+20,1.847483001582526e+20,2.034308497886e+20

A.33 "MITR_Cycle4_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.8759579960727986e+19,2.8133234418907746e+19,2.8906592973511705e+19,2.2276189251363275e+19,2.283285232469643e+19,2.
3603980190522597e+19,2.3742070448195166e+19,2.4575285818336424e+19,2.5124460150772724e+19,2.4924813505160724e+19,2.569541025461
1235e+19,2.504224333581998e+19,2.6804115688809267e+19,2.6880012218891616e+19,2.7181792543543742e+19,2.7921265873380356e+19,3.7544
13859093094e+19,3.692087352408955e+19,4.036866858834822e+19
37.224453125,3.8132987316629316e+19,3.57858247176586e+19,3.76356499424578e+19,2.900415905172113e+19,3.0029319880027263e+19,3.0215
104388234437e+19,3.0992233869105607e+19,3.1296457329085792e+19,3.280923610188879e+19,3.3686640354485273e+19,3.4003769942394085e
+19,3.4138036162161873e+19,3.3764926908258107e+19,3.5929332362362933e+19,3.5605245149932925e+19,3.6419765223650824e+19,4.8385870
27899222e+19,5.09562610906544e+19,5.336816677581853e+19
33.672421875,4.886132565848638e+19,4.454175617521327e+19,4.718251990665091e+19,3.6520677335026938e+19,3.757733336440992e+19,3.855
453438399362e+19,3.899966301843739e+19,3.977010043297519e+19,4.12643963559051e+19,4.11441516084548 3e+19,4.27770157937954e+19,4.36
0927515446047e+19,4.449565527380566e+19,4.489441744865399e+19,4.583916725768463e+19,4.664683593015647e+19,6.378335890258426e+19,
6.168120028770722e+19,6.930803144223838e+19
30.120390625,5.78699091759443e+19,5.5269032284319015e+19,5.8232130544146964e+19,4.3793838095003615e+19,4.556426142164109e+19,4.63
45268052054606e+19,4.760242051325065e+19,4.810384748351477e+19,5.023617350852952e+19,5.074955059724608e+19,5.1215711440166126e+
19,5.29207543409594e+19,5.409260951222386e+19,5.45254693583898e+19,5.619848594173054e+19,5.739349778697394e+19,7.773844167314876e
+19,7.535479137916995e+19,8.344953606066084e+19
26.568359375,6.868981198096579e+19,6.549780256322678e+19,6.685799160130159 4e+19,5.296616479107865e+19,5.54156204040022e+19,5.6502
28450861019e+19,5.710775717686685e+19,5.898525356483812 6e+19,6.041395661537357e+19,6.106191859368331e+19,6.240564302411254e+19,6.
364579976128736e+19,6.429641732147543e+19,6.616541584743609e+19,6.7919693236253385e+19,6.847258538331933e+19,9.307124032218495e
+19,9.113744559839332e+19,1.0176614650817588e+20
23.016328125,8.058203874845236e+19,7.663159514626554e+19,7.98873385290842e+19,6.159144162022006e+19,6.466235650430774e+19,6.53607
7453830553e+19,6.843381388789588e+19,6.902866422863925e+19,7.056996395081539e+19,7.222067364637827e+19,7.28218973836296e+19,7.273
585653077208e+19,7.618598850708369e+19,7.81447457005315e+19,7.953786395389747e+19,8.144563397528158e+19,1.0709855491983973e+20,1.
097536056817827 2e+20,1.1977577169055728e+20
19.464296875000002,9.211523084599026e+19,8.711689463461888e+19,8.991481570161541e+19,6.970052644385387e+19,7.283783087489952e+19
,7.450341182898097e+19,7.684457281290671e+19,7.863337276614215e+19,8.111581070599442e+19,8.07233157043789e+19,8.309209473983916e+
19,8.623471033464144e+19,8.811804900274502e+19,8.967315775068842e+19,9.312063414512168e+19,9.430289919734915e+19,1.2577844896830
875e+20,1.2568869030082159e+20,1.3802280589286038e+20
15.912265625,1.0725842094891449e+20,9.85911328144581e+19,1.0491619773223744e+20,8.261409000174179e+19,8.374961681291085e+19,8.452
292225587724e+19,8.605625523241841e+19,8.903103805251097e+19,9.19930740795876e+19,9.34456773670279e+19,9.554411816727529e+19,9.85
8316606882315e+19,1.0068904249832984e+20,1.0181766479661528e+20,1.039331013208839e+20,1.0619459484846005e+20,1.4295900148827908
e+20,1.4418056915230564e+20,1.5858497637668163e+20
12.360234375000001,1.1585294613990497e+20,1.1052850447387609e+20,1.1383098609775315e+20,8.93746703475654e+19,9.42524431416611e+1
9,9.367458852493895e+19,9.573319559701158e+19,9.959441164808688e+19,1.0209490754471171e+20,1.0494062908551799e+20,1.057957264503
3661e+20,1.0702632309274947e+20,1.1071705078723672e+20,1.1416877611267529e+20,1.1648603685969614e+20,1.193275094694971e+20,1.6176
530123415604e+20,1.6055860482864808e+20,1.7845032217563215e+20
8.808203125,1.2554953781039768e+20,1.2060749993484668e+20,1.2373152645650073e+20,9.862724872800323e+19,1.0111659118073913e+20,1.0
360221581884539e+20,1.0586317823004588e+20,1.0847308410005745e+20,1.109879201388252e+20,1.1506105162379036e+20,1.17152056794778
47e+20,1.2033822393238518e+20,1.2057085290492587e+20,1.2548633496169369e+20,1.285806189663106e+20,1.2913297999700088e+20,1.742943
3653606344e+20,1.7665408659313738e+20,1.944103692643272e+20
5.256171875,1.3637581401192715e+20,1.2938950920644644e+20,1.3496251333628595e+20,1.0726479434542247e+20,1.0880609406759862e+20,1.
1203740609713673e+20,1.1579558557133328e+20,1.1781489003160678e+20,1.2173134218575113e+20,1.2377667134843214e+20,1.264115396880
9994e+20,1.2880953012422166e+20,1.3161276235497482e+20,1.3568376937443731e+20,1.4047868801392946e+20,1.4329041810669324e+20,1.915
8058121479084e+20,1.939031531255683e+20,2.1216452747001433e+20
1.704140625,1.4482640666511273e+20,1.3577671474017847e+20,1.42191538325445e+20,1.1362172624574164e+20,1.1499306872770539e+20,1.16
612973673 47977e+20,1.2307825331443435e+20,1.2398274450218967e+20,1.277940356139526e+20,1.293932270210761e+20,1.3524878506276872
e+20,1.3707476316230061e+20,1.3919126191932057e+20,1.4361121239759415e+20,1.4861592200547346e+20,1.5311022877633993e+20,2.0332143
98152133e+20,2.0271012486682427e+20,2.2340879225919114e+20
−1.847890625,1.4908755334706282e+20,1.4149046470956886e+20,1.4576010925352958e+20,1.153924682422045e+20,1.190709802600515e+20,1.2
068504292569358e+20,1.2526273496753927e+20,1.287165847584805e+20,1.3215874998915713e+20,1.3494179979763507e+20,1.38679265733180
75e+20,1.4003148802562061e+20,1.4428626131106272e+20,1.5066709344828677e+20,1.5257433235329522e+20,1.5799331273419227e+20,2.10065
0244370156e+20,2.128273607068427e+20,2.3607007553863883e+20
−5.399921875,1.5229390290694478e+20,1.4410939955796671e+20,1.481809376938049e+20,1.1690083874908951e+20,1.1989314840957896e+20,1.
2303045284062462e+20,1.2753378858987736e+20,1.2857424556980262e+20,1.3305633666402882e+20,1.376430576842608e+20,1.3936281250865
996e+20,1.4406000573502998e+20,1.4903815952412613e+20,1.5392336794748 11e+20,1.573267616827343e+20,1.623803339971747e+20,2.1934362
751986097e+20,2.1739814823580474e+20,2.3967051344926327e+20
−8.951953125,1.467129320314703e+20,1.3917161061342102e+20,1.440509767566437e+20,1.1450975282585141e+20,1.1777558741980766e+20,1.2
122306381424094e+20,1.2549642617283132e+20,1.2806702943104377e+20,1.300783671456823e+20,1.3333411052850096e+20,1.37554361249524
96e+20,1.4118985284094319e+20,1.4737842085017697e+20,1.5038188395455536e+20,1.5612856313923692e+20,1.5998977919031 22e+20,2.147824
000856609e+20,2.1750118481268346e+20,2.3731235674131625e+20
−12.503984375,1.3139819133920667e+20,1.2547677486693174e+20,1.3380680410284166e+20,1.0687707939118794e+20,1.1018168548056769e+20
,1.1277300227743102e+20,1.1771397792023067e+20,1.1941620590423292e+20,1.2505400623190344e+20,1.256005249824614e+20,1.31465112002
5403e+20,1.3361507109122708e+20,1.3787674888955286e+20,1.4367175966441978e+20,1.476312322449929e+20,1.5218927298093905e+20,2.0294
00982574867e+20,2.0552026161046113e+20,2.2318784784691505e+20

A.34 "MITR_Cycle4_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9358413674288837e+19,2.684894191091528e+19,2.753132023036805e+19,2.304195284179524e+19,2.34288180098287e+19,2.4235
95556592487e+19,2.4474851711698416e+19,2.4338726584615805e+19,2.6172618318145094e+19,2.594455694643509e+19,2.5745282082286055e+
19,2.6186055562449388e+19,2.636695380000045e+19,2.70092328330906e+19,2.7614174384971588e+19,2.8766643808524317e+19,3.78009333585
6435e+19,3.930829474433558e+19,4.126413079771727e+19
37.224453125,3.904826016681061 6e+19,3.635661548658441e+19,3.774410390636833e+19,2.9803701643645313e+19,2.94622469257311e+19,3.154
1036420078936e+19,3.2151820252092223e+19,3.201665113448581e+19,3.274555524844671e+19,3.3337272002573672e+19,3.398847379077497e+
19,3.5327789955286196e+19,3.5925348989545456e+19,3.679383048703079e+19,3.710299332930465e+19,3.835478151510649e+19,5.02155661931
5376e+19,5.0503059487300534e+19,5.604764889104204e+19
33.672421875,4.676447820736598e+19,4.553536869080498e+19,4.814261897893823e+19,3.818376204214277e+19,3.818912631753697e+19,3.9250
66861757607e+19,4.015882450832348e+19,4.1718713303647855e+19,4.199207890220197 5e+19,4.17590781481983e+19,4.352923591664794e+19,4.
422319257309017e+19,4.4938022103095984e+19,4.534984974078564e+19,4.651134814272 4645e+19,4.877799349914474e+19,6.550311372698341
e+19,6.317045060506832e+19,7.032511930163442e+19
30.120390625,5.721451156837525e+19,5.716299327993588e+19,5.792195858075935e+19,4.5252389885851345e+19,4.705234328297573e+19,4.710
524247399183e+19,4.945325485916361e+19,5.146974440264609e+19,5.220895217429488e+19,5.271154760058546e+19,5.2319211933882655e+19,
5.304907205731976e+19,5.503905889365663e+19,5.707004791419472e+19,5.70323386515226e+19,5.936765735531602e+19,7.96706430478134e+1
9,7.914112002127667e+19,8.592506948762948e+19
26.568359375,6.987420149869592e+19,6.6778324144952025e+19,6.956509176805963e+19,5.298597543189091e+19,5.722354054676153e+19,5.760
8068802742075e+19,5.909944358560583e+19,5.811528494150094e+19,6.124568485966297e+19,6.293569716702494e+19,6.536608570206217e+19,
6.473405721502235e+19,6.6447438642913526e+19,6.804769228278835e+19,6.755162958791842e+19,7.049454542547114e+19,9.457323743256199
e+19,9.577727825619167e+19,1.0383856260601576e+20
23.016328125,8.069516653646874e+19,7.762318941962972e+19,8.02915180909643e+19,6.484134272290641e+19,6.674911274429053e+19,6.75866
8326871223e+19,7.0361235215179555e+19,7.1976360113572954e+19,7.3412498793367675e+19,7.43435457999062e+19,7.534576240078366e+19,7.
822281980775408e+19,7.962921597051163e+19,8.061603019649484e+19,8.275855365592233e+19,8.42085013614843e+19,1.1151160088522965e+
20,1.1194924077877656e+20,1.2399071124782462e+20
19.464296875000002,9.335751204866772e+19,9.082674251863004e+19,9.332192725149824e+19,7.2321585757754384e+19,7.598628874983413e+1
9,7.712818895751112e+19,8.001268199374085e+19,8.300711611973294e+19,8.36784472185719e+19,8.539713981021974e+19,8.700217349747553e
+19,8.93528945761632e+19,9.10280356250066e+19,9.333042511350886e+19,9.691970958024434e+19,9.489296949071156e+19,1.28947620381894
21e+20,1.3050007354795929e+20,1.4341257487309 59e+20
15.912265625,1.056448893996481e+20,1.0027264725980946e+20,1.0748786322320122e+20,8.321053369161217e+19,8.579600820834319e+19,8.79
1941147824679e+19,9.004706367915568e+19,9.201962989837078e+19,9.449941225634474e+19,9.693617418788993e+19,9.720332572484878e+19,
1.006162795548639e+20,1.0067523347256256e+20,1.06085715991449e+20,1.0761639338611186e+20,1.1050407312059556e+20,1.48573963811796
e+20,1.4793662416099954e+20,1.65158072641896e+20
12.360234375000001,1.191766724188086e+20,1.128563875484102e+20,1.167786819826868 6e+20,9.107530498244067e+19,9.523022838925802e+1
9,9.657660840156558e+19,1.0128654842095154e+20,1.0163283629788427e+20,1.0573358583438393e+20,1.0642138154086847e+20,1.1115628402
991063e+20,1.1246336143041905e+20,1.1492508583162044e+20,1.1644673424789702e+20,1.191750790696816e+20,1.2234637494876973e+20,1.61
30588556920694e+20,1.676968089175686e+20,1.8155575962413797e+20
8.808203125,1.3045333530690091e+20,1.2257794168855912e+20,1.2634515014114194e+20,1.0336055786791838e+20,1.0504525901152376e+20,1.
0773961238526583e+20,1.1203793721351242e+20,1.1243149444787924e+20,1.1651684160948463e+20,1.1740805488784838e+20,1.184511674496
5192e+20,1.228695245787985e+20,1.2630000524921055e+20,1.2952600611499205e+20,1.3421788817760541e+20,1.3523816273525544e+20,1.8152
442375797378e+20,1.830981215790654e+20,2.030877486099212e+20
5.256171875,1.384556657390263e+20,1.3308236136643635e+20,1.397723032342967e+20,1.1030543559609732e+20,1.1383098609775315e+20,1.16
17374043080586e+20,1.178775617639351e+20,1.2153004907937458e+20,1.260572850655322e+20,1.2673923849188442e+20,1.2942615623636725
e+20,1.3491152616422226e+20,1.379404828546325e+20,1.397006025235821e+20,1.434258527824875e+20,1.4711870494247738e+20,1.9707232453
91538e+20,1.9784881668037417e+20,2.178278213837167e+20
1.704140625,1.4714738522676324e+20,1.4070281912445955e+20,1.469960170596991e+20,1.1720304396684216e+20,1.1862431138811832e+20,1.2
349995971671135e+20,1.2601320240635 21e+20,1.2864753962964422e+20,1.3265268821852442e+20,1.3700412468433733e+20,1.38457259088153
28e+20,1.4113939678525514e+20,1.4625564083202382e+20,1.4872851867711416e+20,1.5189928343982663e+20,1.5477315414854307e+20,2.09977
92135140678e+20,2.1350347185306262e+20,2.321052917943091e+20
−1.847890625,1.5226947155366422e+20,1.4520243705908268e+20,1.5157955138167708e+20,1.2140629896384494e+20,1.2463336206237775e+20,
1.285891168283212e+20,1.3275731814453015e+20,1.3559985298708244e+20,1.3926508709553783e+20,1.4065395641789848e+20,1.432203107451
0563e+20,1.4904559515338542e+20,1.5169692810069875e+20,1.5371729479372358e+20,1.610132404462162e+20,1.653790170541721e+20,2.19259
71113250613e+20,2.2334452717773586e+20,2.4605931233212223e+20
−5.399921875,1.5151953523122707e+20,1.4972329964873233e+20,1.5576793512016124e+20,1.2078223722244006e+20,1.2600895347534683e+20,
1.2813766790900702e+20,1.303407386352602e+20,1.3478671381594127e+20,1.4177195638867065e+20,1.4306310029790916e+20,1.463363705211
2472e+20,1.5068940033606464e+20,1.5488362635468112e+20,1.591421174547529e+20,1.620457306805065e+20,1.6755978089264733e+20,2.25184
8454194107e+20,2.284512111297426e+20,2.5058920390015818e+20
−8.951953125,1.4743047025499202e+20,1.4221065851496885e+20,1.49302124362831e+20,1.1825199880877803e+20,1.2462486420036711e+20,1.2
564991880539811e+20,1.3021805075248185e+20,1.3399057036882133e+20,1.3693348620637404e+20,1.41255711271 5255e+20,1.43826314529737
97e+20,1.4936798279341 33e+20,1.5334338886525 63e+20,1.5654230279587896e+20,1.5997756351367194e+20,1.6780409442545268e+20,2.2285802
457762785e+20,2.2414598178861246e+20,2.4504966010198544e+20
−12.503984375,1.3577512139105147e+20,1.3118893148719517e+20,1.3740830424621744e+20,1.0868287506844464e+20,1.1377840557656246e+20
,1.1899237503645327e+20,1.2255882149903524e+20,1.2572746179624503e+20,1.2943412298200218e+20,1.318926606849496e+20,1.35916398346
978e+20,1.4008300631405997e+20,1.4346037534690563e+20,1.4595768454527648e+20,1.507690677924142e+20,1.573517241523905e+20,2.109530
510171254e+20,2.1235519824887762e+20,2.3178662196891084e+20

A.35 "MITR_Cycle4_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.7631648113730925e+19,2.73109069344676e+19,2.929834441220127e+19,2.2708624204428685e+19,2.3731554343957025e+19,2.54
4445776711012e+19,2.4911270037581304e+19,2.581199029906942e+19,2.6189242260703367e+19,2.6553056978033025e+19,2.772454036783451e
+19,2.685786466602644e+19,2.7692832720207385e+19,2.765034341015429e+19,2.889134993353016e+19,2.8648682861489398e+19,3.7315917884
308234e+19,3.69977260636481e+19,4.168435007414241e+19
37.224453125,3.6596202083646317e+19,3.6185914683446084e+19,3.852266740145379e+19,3.0341244527454577e+19,3.1502264924655485e+19,3.
1747481355299426e+19,3.336244691878013e+19,3.34894899558389e+19,3.4171390270553563e+19,3.4812022842879152e+19,3.552876439183735
e+19,3.694594221702088e+19,3.6678896903337157e+19,3.933246053942828e+19,3.895781104803508e+19,3.9153580544104735e+19,4.9602551672
36269e+19,5.035801160510677e+19,5.601418855937523e+19
33.672421875,4.5835449443054985e+19,4.495661117624216e+19,4.705622043251807e+19,3.8534192626805694e+19,4.049342782499163e+19,4.0
3791315809488e+19,3.991636988283299e+19,4.204104783203817e+19,4.256435679697964e+19,4.3715073536492675e+19,4.518971815352302e+1
9,4.609994539813552e+19,4.5828757376721625e+19,4.740245519781325e+19,4.751930080045928e+19,4.8807470457994084e+19,6.324480689766
124e+19,6.473830614602765e+19,7.0100457074728665e+19
30.120390625,5.653202702564736e+19,5.448722897934201e+19,5.871703979512796e+19,4.657773768968261e+19,4.837832842645779e+19,4.9398
44364919511e+19,5.01910286165981e+19,5.252167349628567e+19,5.180142657924809 5e+19,5.37888109453442e+19,5.479368312809999e+19,5.73
5844410618013e+19,5.823531724240095e+19,5.839943220248103e+19,5.9618344284629295e+19,6.017867206095453e+19,7.966533188405676e+1
9,8.022619077675767e+19,8.676317112842684e+19
26.568359375,6.7636077091648954e+19,6.550205149423209e+19,6.966175494843041e+19,5.5552117312547774e+19,5.856938944269343e+19,6.02
5727728455275e+19,6.09317950816457e+19,6.28485940814161e+19,6.377804773882763e+19,6.424596126578738e+19,6.655791084905158e+19,6.6
31519066537326e+19,6.7631828160643645e+19,6.908814926271359e+19,6.950985566499059e+19,7.235610832217252e+19,9.553402695613768e+
19,9.530352244909962e+19,1.0237586810743785e+20
23.016328125,7.774109725502708e+19,7.750156376960274e+19,8.146687863030813e+19,6.643469184989761e+19,6.8554908421547205e+19,7.167
2561546693304e+19,7.250535202373403e+19,7.519651869922214e+19,7.609304314134253e+19,7.67813699642027e+19,7.772781934563548e+19,7.
881235898474081e+19,7.878739651508463e+19,8.303845198589711e+19,8.382981538563606e+19,8.394028759177413e+19,1.1357286353918058e
+20,1.1283408066063235e+20,1.2308197112906401e+20
19.464296875000002,8.91170789053685e+19,8.835492690629104e+19,9.230590162485353e+19,7.372160852400397e+19,7.613234575314162e+19,
7.996275705442845e+19,8.383300208389005e+19,8.503173174376309e+19,8.70860898848304e+19,8.821418106674015e+19,9.01846228204526e+1
9,9.378452961470135e+19,9.280621325072877e+19,9.48642892064257e+19,9.875896558916783e+19,9.919129431895812e+19,1.302138018214765
5e+20,1.306041723575894e+20,1.4416622901016271e+20
15.912265625,1.0154573321227541e+20,9.972241069462186e+19,1.053240951087472e+20,8.510821250185866e+19,8.698517777345428e+19,9.193
465127826458e+19,9.534282506089873e+19,9.668389390944964e+19,9.862193756424659e+19,1.0133009996375595e+20,1.0272215598437058e+2
0,1.0407278492768344e+20,1.0454441626927283e+20,1.084284703245017e+20,1.0879706508921235e+20,1.1210857569147568e+20,1.4907427543
767125e+20,1.4730565790671104e+20,1.6343884893387 25e+20
12.360234375000001,1.1480292906521777e+20,1.1059064508982877e+20,1.1878205295169043e+20,9.514312530364917e+19,9.903302163901034e
+19,1.0520140722596887e+20,1.0609421385345963e+20,1.0724248745764458e+20,1.1001757052048758e+20,1.114568958985363e+20,1.15305365
15659568e+20,1.1716692805329702e+20,1.1814949334827495e+20,1.2181419634035466e+20,1.2326414404591665e+20,1.2435771266340826e+20,
1.655643766692787e+20,1.6635361560351503e+20,1.7898143855129585e+20
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37688454818005e+20,1.1645841880816162e+20,1.1900618406222052e+20,1.2243878919813521e+20,1.2424511599176756e+20,1.28334180968002
62e+20,1.3122982744812128e+20,1.3179652862095445e+20,1.3487328578517446e+20,1.3685275651727316e+20,1.3832501111061302e+20,1.82270
11114940567e+20,1.8324417858237294e+20,1.9906666652977113e+20
5.256171875,1.3400172381271027e+20,1.3192240320198677e+20,1.3823578355950151e+20,1.1207033531242791e+20,1.1939283678370374e+20,1.
201093127744741e+20,1.2327370414067 86e+20,1.2777438430805303e+20,1.3104393671663896e+20,1.3353168582024788e+20,1.37979254350055
95e+20,1.3883169613299622e+20,1.433599943519052e+20,1.4582065652035523e+20,1.4907002650666595e+20,1.517250772686089e+20,1.9881226
17858282e+20,1.9714508748261976e+20,2.1634494446286353e+20
1.704140625,1.4073627945612637e+20,1.3900962011884357e+20,1.485537813895208e+20,1.1815852232666123e+20,1.2370390840496623e+20,1.2
8054282630278 3e+20,1.3386522690416471e+20,1.3485735229390455e+20,1.3983763054850333e+20,1.4490076295770567e+20,1.4531662707985
036e+20,1.4704965981364111e+20,1.5187272762104345e+20,1.5609297834206744e+20,1.5948150081880205e+20,1.5970510081295647e+20,2.0927
89722010333e+20,2.1115965528725858e+20,2.313760690105228e+20
−1.847890625,1.4274974163626757e+20,1.4316135682740696e+20,1.541894572516886e+20,1.2224333837189099e+20,1.2893699805438095e+20,1.
3354390149688815e+20,1.3876265100415995e+20,1.4264298724475915e+20,1.4595874677802782e+20,1.4810073912107965e+20,1.5202515802 08
5894e+20,1.5536375555828115e+20,1.5948574974980735e+20,1.6050177537645206e+20,1.6542522417885482e+20,1.653025362960765e+20,2.2068
310301928486e+20,2.2127211107989596e+20,2.407056592654318e+20
−5.399921875,1.4881721511185003e+20,1.4735770731152607e+20,1.549351446431205e+20,1.2638657721844372e+20,1.3208864262756952e+20,1.
3702749380486655e+20,1.3937715265080289e+20,1.4297652832867597e+20,1.4687492252604775e+20,1.520841119385576e+20,1.5555655080264
71e+20,1.5658850992056 17e+20,1.6150186751182687e+20,1.618221306863521e+20,1.686034245708266e+20,1.7150225774919922e+20,2.24624517
6430854e+20,2.279020367973063e+20,2.4835904623874618e+20
−8.951953125,1.4373549362949944e+20,1.4173955828975514e+20,1.5262531952535896e+20,1.2375170887877596e+20,1.2925354341427652e+20,
1.3363790909538063e+20,1.3583726200700412e+20,1.4115001911276842e+20,1.4634805508138933e+20,1.5056883691878898e+20,1.50513069699
3443e+20,1.5566118072865282e+20,1.5888080819792635e+20,1.6153426561074233e+20,1.6550807833345838e+20,1.688610160130235e+20,2.2440
357323080932e+20,2.2534152475023147e+20,2.4616022444349833e+20
−12.503984375,1.3070402223621418e+20,1.3028231583393718e+20,1.4038627376456396e+20,1.1353249869463015e+20,1.1944382395576744e+20
,1.2402523381224278e+20,1.2737657814268 09e+20,1.305276915994938e+20,1.346061342482156e+20,1.3977708328167766e+20,1.4183144142274
498e+20,1.4606975010054155e+20,1.493722317244186e+20,1.5082164831360493e+20,1.5419051948443995e+20,1.5758169754305285e+20,2.08310
21593182267e+20,2.1110548141694088e+20,2.2899029425104133e+20

A.36 "MITR_Cycle4_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.035288612920415e+19,2.3200640201027936e+19,2.5107920075627667e+19,2.760698423964168e+19,3.0212542615190376e+19,3.1
619883841041195e+19,3.085240656972791e+19,3.1910428808038363e+19,3.4468087255529685e+19,3.371435986975633e+19,3.45790748937114e
+19,3.5998458502976266e+19,3.516178246129867 6e+19,3.531788410301661e+19,3.502374439470181e+19,3.692689031678628e+19,3.4185540587
964236e+19,3.3725413699308028e+19,3.2484239390864503e+19
37.224453125,2.8899114006262563e+19,3.3479533393182368e+19,3.463092903721817e+19,3.780247943322717e+19,3.92169202731803e+19,4.139
820930471612e+19,4.425602865222926e+19,4.49747971786887e+19,4.491323723362434e+19,4.524422304044482e+19,4.637503879043721e+19,4.6
88827798206135e+19,4.58147983560851e+19,4.88203615717527e+19,4.916941095369423e+19,4.975382602340739e+19,4.779595016476205e+19,4.
888111270002058e+19,4.488429956601745e+19
33.672421875,3.8123130358759465e+19,4.140836444893841e+19,4.458926117237324e+19,4.770293623316846e+19,4.927015357912073e+19,5.092
2027083103085e+19,5.290497626243885e+19,5.482924128976438e+19,5.772040186299851e+19,5.810773523997277e+19,5.699030990138855e+19,
5.997574256567743e+19,6.3108110281324315e+19,5.981272104692307e+19,6.2030316965766906e+19,6.323823991702239e+19,5.97856706185079
2e+19,6.040630270699616e+19,5.555789535484728e+19
30.120390625,4.5281967824279904e+19,4.965236444971737e+19,5.349487331266138e+19,5.797616770774764e+19,6.183503264368685e+19,6.463
147178320151e+19,6.701900909783617e+19,6.898137838312852e+19,6.908769285759733e+19,7.065544941474723e+19,7.094743227827146e+19,7.
2791264968615076e+19,7.40830352611042e+19,7.578703251419213e+19,7.760570201531238e+19,7.644451070118604e+19,7.535898869179576e+
19,7.306850939273702e+19,7.184090523276754e+19
26.568359375,5.4336761729249526e+19,5.952864661429763e+19,6.376990215789937e+19,7.09349405521927e+19,7.333838459715784e+19,7.7507
83518294e+19,7.765000720205213e+19,8.024882543763027e+19,8.3323857029261e+19,8.629104638136294e+19,8.621429865423161e+19,8.769335
496936905e+19,9.049132187394372e+19,9.118600563357506e+19,9.01776806939808e+19,8.998266597749957e+19,9.06764510518015e+19,8.83537
9881795e+19,8.71882039457096e+19
23.016328125,6.352536987976781e+19,6.754177435395 2014e+19,7.428622801408303e+19,7.857906809559271e+19,8.564399294419014e+19,8.939
51055091335e+19,9.182856564410083e+19,9.4503053184415e+19,9.429186213200445e+19,1.0198191249573716e+20,1.0112816143280082e+20,1.0
18677794589025e+20,1.0552003663761113e+20,1.0889819479085072e+20,1.0712418995060199e+20,1.0786021323538604e+20,1.08383248097100
73e+20,1.0418548892344225e+20,1.059487095397381e+20
19.464296875000002,7.222500334255593e+19,7.931095742786149e+19,8.398582864290193e+19,9.098470011978798e+19,9.520312905602279e+19
,1.0249685918948716e+20,1.064780351987586e+20,1.0954704559868232e+20,1.099217973810449e+20,1.109615763071684e+20,1.1369268102323
52e+20,1.1740065669236703e+20,1.2310011905146402e+20,1.2128387600073317e+20,1.2277928838886584e+20,1.2589233436993044e+20,1.23712
12376078993e+20,1.2334635883172141e+20,1.1962759893863671e+20
15.912265625,8.008571405034296e+19,8.869404108366338e+19,9.533164105812795e+19,1.0308639676557792e+20,1.1071084310026394e+20,1.15
600589977565 51e+20,1.2197586370437628e+20,1.2342005102873277e+20,1.2264987770143128e+20,1.3021231474838821e+20,1.34112609078013
07e+20,1.360609588721667e+20,1.3734158546657116e+20,1.4318034405172616e+20,1.4343107725863063e+20,1.4445288247816597e+20,1.417864
8310581646e+20,1.3987138466885088e+20,1.4046990909823572e+20
12.360234375000001,9.20415540671702e+19,1.0488466610972063e+20,1.099235947517037e+20,1.17261360466309e+20,1.2456497613839684e+20
,1.3074613383405578e+20,1.3679428610096251e+20,1.4030365231229549e+20,1.3840562889658864e+20,1.4450051280062453e+20,1.5013077638
93572e+20,1.530910458644227e+20,1.5097913534031708e+20,1.5741641835485692e+20,1.6039915496315758e+20,1.591859297684586e+20,1.5542
133692357414e+20,1.5629665643441622e+20,1.5039678724686158e+20
8.808203125,1.0306662568833096e+20,1.1130217804701352e+20,1.268296631685016e+20,1.2990676173638997e+20,1.37155557603384e+20,1.436
8810126284092e+20,1.490613408473633e+20,1.519461207547586e+20,1.5475990452113084e+20,1.6143983257461047e+20,1.638051723616088e+
20,1.7078885605642784e+20,1.702424553761486e+20,1.7323777357906097e+20,1.7600662307896286e+20,1.7653415136732457e+20,1.754638171
400012e+20,1.7344896463147152e+20,1.6988298124438595e+20
5.256171875,1.086249944507111e+20,1.2052628426804335e+20,1.2868724574438513e+20,1.3872016876188097e+20,1.489319301599287e+20,1.54
590052993873e+20,1.5778847408123124e+20,1.6698022763040247e+20,1.7161295050349373e+20,1.7984670549151747e+20,1.8252209171716106
e+20,1.8668390347764325e+20,1.8441652039155024e+20,1.90893345560584e+20,1.90296618501858e+20,1.8785129572054247e+20,1.88604394026
58523e+20,1.8627500165276323e+20,1.8829524627326933e+20
1.704140625,1.1311842109774435e+20,1.2670474590771406e+20,1.3960357644068769e+20,1.488303787177058e+20,1.5411644582527566e+20,1.6
176156192253805e+20,1.6982906012462154e+20,1.7620523253676173e+20,1.7967056316695373e+20,1.8574837204 97309e+20,1.86665929771055
12e+20,1.925514199933393e+20,1.954757420552285e+20,2.0267061680245817e+20,1.989383766294323e+20,1.9919719800430143e+20,2.02572660
10155283e+20,2.0119677286223125e+20,1.938086807691792e+20
−1.847890625,1.1889067696852325e+20,1.2950864413546281e+20,1.4208125189386184e+20,1.5460712801513 17e+20,1.6384830925742028e+20,1.
7107194193519093e+20,1.7888151744773472e+20,1.8322216758876884e+20,1.862318647569517e+20,1.9236269607416385e+20,1.9660808557028
085e+20,2.0125608609397205e+20,2.0391529598368632e+20,2.0868102428552983e+20,2.150302361377878e+20,2.1225509584058e+20,2.09873579
7176524e+20,2.0697172478899834e+20,2.0196604750420335e+20
−5.399921875,1.1780326771994121e+20,1.319404866368372e+20,1.4426775330030823e+20,1.5547346067267974e+20,1.6366318007956252e+20,1.
7442313952854832e+20,1.8182111716022387e+20,1.8550123358414412e+20,1.8989580484494262e+20,1.9119889857258222e+20,1.986400131000
6932e+20,2.0212870954882594e+20,2.0328352019711345e+20,2.0761248742886526e+20,2.1206187849475765e+20,2.1167364643245395e+20,2.182
4483356107532e+20,2.101854235269565e+20,2.064486899272837e+20
−8.951953125,1.1214155014467931e+20,1.2846167572670408e+20,1.4088060829377454e+20,1.4906942901532793e+20,1.5972693833676138e+20,
1.6527092813387104e+20,1.7209015241340866e+20,1.786748198219712e+20,1.8309095953067547e+20,1.9105241286388898e+20,1.965613539331
517e+20,1.969747491846788e+20,2.033527189674778e+20,2.0971451241434746e+20,2.101944103802506e+20,2.0983942967513498e+20,2.1421872
328533357e+20,2.0198581858145026e+20,2.0205052392516754e+20
−12.503984375,9.994459285397229e+19,1.1318761986810867e+20,1.2479683695338376e+20,1.3471742430470377e+20,1.452518137360085e+20,1.
5050912291303742e+20,1.5553816601639702e+20,1.5985095691221952e+20,1.6674836681541553e+20,1.7307151679312072e+20,1.790621531989
4547e+20,1.8213475834018677e+20,1.807993119406885e+20,1.9035952647491643e+20,1.9248311990830436e+20,1.9230877495439947e+20,1.9119
710120192344e+20,1.8713414482767597e+20,1.8774435216634308e+20

A.37 "MITR_Cycle4_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9114100141483524e+19,2.821316743344514 3e+19,2.9707357133099913e+19,2.3497066464101487e+19,2.4561901685669704e+19,
2.507958081702914e+19,2.567182868753176e+19,2.580195219956938e+19,2.647429241952209e+19,2.673013117767931e+19,2.734436726613441e
+19,2.8134031093471244e+19,2.836899697806488e+19,2.7837455709300617e+19,2.8539644669565633e+19,2.8455887617123463e+19,3.75790860
4844961e+19,3.742081336850182e+19,4.098598515178217e+19
37.224453125,3.780608518740829e+19,3.626542280488295e+19,3.869464288389371e+19,2.9721644163605266e+19,3.2142578827155677e+19,3.23
87476587974218e+19,3.3386931383698227e+19,3.3850967761115628e+19,3.4427388363623477e+19,3.5267720693198627e+19,3.63799846071136
1e+19,3.675633367090893e+19,3.6497361326135296e+19,3.739728491305992e+19,3.747068519617664e+19,3.809156023932754e+19,4.9808943495
94561e+19,5.021370728583894e+19,5.337400905595082e+19
33.672421875,4.803416501502769e+19,4.519141772592515e+19,4.774518459502906e+19,3.884909152593672e+19,4.0857826770334515e+19,4.030
610307929503e+19,4.1539992643237e+19,4.239997627871171e+19,4.4556202540631335e+19,4.448609517904372e+19,4.463539199224279e+19,4.5
141386563337626e+19,4.664301189225169e+19,4.649323707431451e+19,4.79311815497865e+19,4.720737615303197e+19,6.468997455584225e+1
9,6.412539784851172e+19,6.692438114825956e+19
30.120390625,5.781148637462129e+19,5.557389308395e+19,5.944679369528991e+19,4.646503479476677e+19,4.859481146117832e+19,4.9668303
87966986e+19,5.015953341552124e+19,5.263772242436819e+19,5.249851682230674e+19,5.5538839403156185e+19,5.583467122440089e+19,5.643
323937977391e+19,5.738234434308499e+19,5.891780178512884e+19,6.063118321302004e+19,6.091745493950278e+19,7.884741266553461e+19,7.
808738513195983e+19,8.452079779037458e+19
26.568359375,6.921455496012156e+19,6.721012175836665e+19,7.2367261766061466e+19,5.689477951022569e+19,5.776527924993855e+19,6.049
681076997711e+19,6.105820077905367e+19,6.26786368412037e+19,6.390551566898692e+19,6.530341396973385e+19,6.607671941270024e+19,6.7
76513837093524e+19,6.950454450123396e+19,6.92273017531375e+19,7.161520097812161e+19,7.206664989743579e+19,9.398104267369693e+19,
9.466405833280048e+19,1.0185856075754137e+20
23.016328125,8.056876083906077e+19,7.794823264153592e+19,8.159965772422406e+19,6.6062379270557336e+19,6.874504808403482e+19,7.160
245418510569e+19,7.308745557146147e+19,7.560654054123454e+19,7.566868115718719e+19,7.652908968576244e+19,7.870241789497842e+19,7.
876402739455541e+19,8.068295085982846e+19,8.419575456846838e+19,8.231560259861878e+19,8.576732792405736e+19,1.1146167594591727e
+20,1.1082221182961813e+20,1.1980445197484312e+20
19.464296875000002,9.193624462739158e+19,8.942034635587248e+19,9.353649826726638e+19,7.595548400004561e+19,7.790680556423415e+19
,8.028036464707536e+19,8.398808806558386e+19,8.41368006507697e+19,8.581619063061842e+19,9.044274537902516e+19,9.064032067077208e+
19,9.440221795959826e+19,9.418552247832746e+19,9.442452484737614e+19,9.768823497582977e+19,9.723200600913461e+19,1.2845793108353
226e+20,1.3104871676401993e+20,1.3990083339720732e+20
15.912265625,1.0452688942887595e+20,1.0215333034603474e+20,1.0602357537549633e+20,8.576307899305204e+19,8.976185418542427e+19,9.1
61120140548538e+19,9.54532972670368e+19,9.763140552363373e+19,1.0038683728057718e+20,1.0101143013835771e+20,1.0312899112812901e
+20,1.0478448087107286e+20,1.0683777677938883e+20,1.0750326559809547e+20,1.1141812440311282e+20,1.1180318377546906e+20,1.4680959
521184113e+20,1.4674586124676147e+20,1.603020756192025e+20
12.360234375000001,1.1804805012052317e+20,1.146871456953231e+20,1.204524139531529e+20,9.685810008066736e+19,1.0032150996637052e+
20,1.0512386423512197e+20,1.0761480003698488e+20,1.0820115251571761e+20,1.0921399144410835e+20,1.1167784031081238e+20,1.14620225
03198946e+20,1.1738256130181652e+20,1.2008541253756925e+20,1.2179507615083078e+20,1.2424724045727021e+20,1.2510552452034 28e+20,1.
6506937620716015e+20,1.661889695270593e+20,1.794769701297901e+20
8.808203125,1.2774835960564552e+20,1.2403373167425341e+20,1.3448503971456426e+20,1.0609846278446491e+20,1.1087319900168184e+20,1.
1446142123566601e+20,1.1691146107660278e+20,1.1916127004391434e+20,1.204386049273856e+20,1.242971653965826e+20,1.28134481210753
06e+20,1.3072314242573808e+20,1.3345786064403061e+20,1.3552921450911916e+20,1.3644167244250946e+20,1.3966183102815864e+20,1.82089
00046530435e+20,1.8323302513848402e+20,1.9475400155938164e+20
5.256171875,1.3688993466356964e+20,1.3420885919921912e+20,1.4322402855973528e+20,1.1337050820005269e+20,1.1828280355856653e+20,1.
2086084244603827e+20,1.2452979436912332e+20,1.2856362324228936e+20,1.3055849634928232e+20,1.3511706820160409e+20,1.383632514896
6083e+20,1.4043726093662768e+20,1.4475523707077381e+20,1.4776507357166017e+20,1.5032186780410534e+20,1.5217546395517177e+20,1.995
276755438472e+20,1.980081515930733e+20,2.141875497449175e+20
1.704140625,1.4406478578241095e+20,1.4234025091063082e+20,1.5106543073003458e+20,1.2141798352410953e+20,1.2524255254526404e+20,1.
299015053925863e+20,1.3398632143781601e+20,1.3708591660618954e+20,1.4108044286755645e+20,1.429993663328295e+20,1.45616176715724
7e+20,1.4864832010438892e+20,1.5196620410316027e+20,1.5693745337937278e+20,1.578228243776042e+20,1.607455577928817e+20,2.09192931
3481758e+20,2.1137528853577803e+20,2.261583817360023e+20
−1.847890625,1.4870568067296064e+20,1.4630078572395527e+20,1.573984623934489e+20,1.2365663904753213e+20,1.3103809443650666e+20,1.
3458382736043771e+20,1.3970803815284138e+20,1.4373868032775343e+20,1.4562998574149196e+20,1.509671742005368e+20,1.5252493853035
853e+20,1.5683760350074798e+20,1.6097818676542243e+20,1.6175467890664276e+20,1.6652941512385973e+20,1.6546558902340524e+20,2.2034
371965523578e+20,2.196553928323756e+20,2.4065838990799775e+20
−5.399921875,1.524149974405961e+20,1.4940356759058275e+20,1.5950486993933124e+20,1.275582199431579e+20,1.3191231199084914e+20,1.3
661481638097582e+20,1.3979407900569892e+20,1.4429635252220033e+20,1.48761979008781e+20,1.5285051286864036e+20,1.54300991690578e
+20,1.5942307801747902e+20,1.6274308648175298e+20,1.6491694580734463e+20,1.6927316232053857e+20,1.702212050510983e+20,2.27156880
52225013e+20,2.2466541360401157e+20,2.443565532317443e+20
−8.951953125,1.486137975399708e+20,1.4452898149474106e+20,1.5511094416346517e+20,1.2514323378301492e+20,1.3068649539581728e+20,1.
3328471670556423e+20,1.3846469471741257e+20,1.4228129699293215e+20,1.460076094845889e+20,1.5066603121553542e+20,1.5330939741721
382e+20,1.5738996453143826e+20,1.608331919948662e+20,1.6379469690556717e+20,1.6835911103802127e+20,1.6954403167212703e+20,2.23180
41221765572e+20,2.2542756560308902e+20,2.415028649453031e+20
−12.503984375,1.3538687532044127e+20,1.3266331054603769e+20,1.4196103381840693e+20,1.1531333190223063e+20,1.1831307719197937e+20
,1.2493078723274945e+20,1.2802241565548801e+20,1.3331445922260142e+20,1.3589621592470279e+20,1.4170450460896133e+20,1.4214586231
713792e+20,1.4680322181533313e+20,1.517033014972067e+20,1.54604259141082e+20,1.558534448566431e+20,1.5948946756443698e+20,2.09882
32040378732e+20,2.0892577981121692e+20,2.2411730150432657e+20

A.38 "MITR_Cycle4_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9947209288349655e+19,2.733374493862114e+19,2.8779603048090505e+19,2.3478105609490293e+19,2.3448628650640957e+19,2.
358735624796432e+19,2.4577038502376116e+19,2.472660087376302e+19,2.5419176627628528e+19,2.588315989340836e+19,2.587715827836336
e+19,2.634953318287868e+19,2.7258963752927687e+19,2.7728789298839814e+19,2.859365920497064e+19,2.8654100248521167e+19,3.767495255
425692e+19,3.872210160051552e+19,4.10881188308223e+19
37.224453125,4.048408017677994e+19,3.825657809724627e+19,3.805804679602316e+19,2.97583974168012e+19,2.9699974615478186e+19,3.1179
771061352473e+19,3.181365845570713e+19,3.2563010550131085e+19,3.305556787692163e+19,3.3599377933963715e+19,3.4502913112242856e+
19,3.507173875057871e+19,3.491091671202773e+19,3.647107106553994e+19,3.761387417105559e+19,3.777963559190024e+19,4.9745103307590
84e+19,5.111044417450958e+19,5.521273394849866e+19
33.672421875,4.887980850835948e+19,4.6505187192766956e+19,4.893440727177771e+19,3.889965380489991e+19,3.802740138114737e+19,3.991
20678401901e+19,4.064946981616163e+19,4.138697801540828e+19,4.199871785689777e+19,4.275800182754664e+19,4.394743695084556e+19,4.4
199611006010696e+19,4.631419774407827e+19,4.579317257955216e+19,4.627978140293526e+19,4.8245708667454554e+19,6.3575692399699755
e+19,6.359109477459399 5e+19,6.921827277475121e+19
30.120390625,6.038952526209303e+19,5.7368004200942084e+19,5.918973336946866e+19,4.702265387757612e+19,4.652829075510832e+19,4.738
583125525498e+19,4.961062464127278e+19,5.162833575241929e+19,5.174342867102561e+19,5.302708383936728e+19,5.354184183066056e+19,5.
548678999834115e+19,5.584476243553849e+19,5.546448311056327e+19,5.814927638954343e+19,5.968101601695762e+19,7.773472385851911e+
19,7.799337753346736e+19,8.466419921180379e+19
26.568359375,7.0696900764599026e+19,6.7230304180641866e+19,7.003619199327334e+19,5.471985795188272e+19,5.725965646030669e+19,5.80
98289217479705e+19,5.890717945761557e+19,5.969641839185187e+19,6.150115183635721e+19,6.452479736301083e+19,6.525508237954846e+1
9,6.540060826648033e+19,6.613142439939361e+19,6.827023004419146e+19,6.836689322456224e+19,7.133636488089816e+19,9.36820241541982
5e+19,9.635938180391911e+19,1.0291442011236087e+20
23.016328125,8.297471802081747e+19,7.936047108442577e+19,8.2154143220417e+19,6.405157267229445e+19,6.641504054399804e+19,6.878488
181220961e+19,7.052959910626495e+19,7.290103372360351e+19,7.3103920179107045e+19,7.496654530855975e+19,7.643986213465093e+19,7.86
1425257661825e+19,7.879908107534923e+19,8.177014608081212e+19,8.080776320810944e+19,8.421912368899758e+19,1.1097251776393096e+2
0,1.1260941843372658e+20,1.2196768897292149e+20
19.464296875000002,9.554996044740759e+19,9.036520238817826e+19,9.371601560224073e+19,7.294192968452962e+19,7.768745450108504e+19
,7.781439131486867e+19,8.066011285567493e+19,8.269110187621304e+19,8.513901725164716e+19,8.741909985237156e+19,8.885736299766894e
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1635e+20,1.2985158045327388e+20,1.4114205236713346e+20
15.912265625,1.0887407696368357e+20,1.0347846570331575e+20,1.0780918863047778e+20,8.5047665235033e+19,8.611574026649276e+19,8.839
316728533883e+19,8.869749696859415e+19,9.273345030726282e+19,9.415259326303632e+19,9.832716797575322e+19,9.806479648617534e+19,1.
0147934366531746e+20,1.0446846662755294e+20,1.0572190127411932e+20,1.089909225663296e+20,1.1010308025696942e+20,1.4651588785609
905e+20,1.469312208618681e+20,1.6238086511355032e+20
12.360234375000001,1.196716728809272e+20,1.1185363983115711e+20,1.2080082629558827e+20,9.413931535364473e+19,9.468105405682172e+
19,1.0087015318243115e+20,1.0074640306690151e+20,1.0375570845141221e+20,1.0390335880384671e+20,1.0698436489907202e+20,1.10627823
2361252e+20,1.1273051296737791e+20,1.160356501731333e+20,1.172598734190382e+20,1.1971681777285859e+20,1.2405072739827465e+20,1.63
51532969196805e+20,1.6604025694187343e+20,1.8060134349707025e+20
8.808203125,1.3052131820298584e+20,1.2272187422636399e+20,1.3065728399515576e+20,1.0378651320120071e+20,1.046299260057547e+20,1.0
853310025050746e+20,1.1136766834742477e+20,1.131171656888611e+20,1.1681320454710498e+20,1.1764280832589174e+20,1.20916078549107
31e+20,1.2308197112906401e+20,1.2730381519921498e+20,1.2797727076355659e+20,1.3400862832559 39e+20,1.3458807629144305e+20,1.853951
9990381104e+20,1.8021150407733307e+20,1.9812127938108968e+20
5.256171875,1.4267219864542065e+20,1.3382486205961424e+20,1.4034272222175956e+20,1.1049398190945793e+20,1.1367483788330802e+20,1.
190518600705276e+20,1.1977630280693295e+20,1.2452979436912332e+20,1.2614704373301938e+20,1.2660008600146053e+20,1.2992381228036
414e+20,1.350076582282174e+20,1.3950355834821083e+20,1.4073043717599407e+20,1.4552163800085656e+20,1.4858405502293364e+20,1.97910
42617995117e+20,1.9710525375444497e+20,2.1657332450439895e+20
1.704140625,1.5180846253958814e+20,1.3982753933736572e+20,1.480821500479314e+20,1.1987190375455241e+20,1.1988411943119267e+20,1.2
293910082401046e+20,1.2737286032805128e+20,1.3147732767918057e+20,1.3224957088939563e+20,1.3678636697031521e+20,1.3991145572472
059e+20,1.4255694639190162e+20,1.4737523415192297e+20,1.5097354759704474e+20,1.519242459094828e+20,1.5522141636960325e+20,2.09867
9802616444e+20,2.1221179682744838e+20,2.3107917495652673e+20
−1.847890625,1.5459947909370102e+20,1.482770697578e+20,1.54482633491055e+20,1.2282225522136442e+20,1.2683908837050922e+20,1.29526
00611499205e+20,1.3465552807115234e+20,1.361203470352329e+20,1.3791498926860062e+20,1.4181444569872374e+20,1.4593219095924462e+
20,1.5078765686556246e+20,1.5272516940398372e+20,1.563133916379679e+20,1.6120550457420648e+20,1.6402891922723488e+20,2.229376920
3397742e+20,2.221250839792119e+20,2.39470282575638e+20
−5.399921875,1.585270846917343e+20,1.5126672383641112e+20,1.573511930360148e+20,1.219310419430007e+20,1.265113895667247e+20,1.287
5748071940661e+20,1.3360816657834344e+20,1.3733766576825414e+20,1.4181816351335337e+20,1.4419331594532158e+20,1.475632493489079
4e+20,1.4928899086861907e+20,1.5508385722830635e+20,1.603695273989117 7e+20,1.62919948234849e+20,1.694686131467828e+20,2.2374923785
599163e+20,2.2945236549786875e+20,2.435545675044921e+20
−8.951953125,1.5137932050805185e+20,1.4495865464265302e+20,1.4902275714923187e+20,1.1920322823759179e+20,1.237352442711304e+20,1.
278731719539265e+20,1.2973898378163318e+20,1.326378169600058e+20,1.3846681918291526e+20,1.4075168183102063e+20,1.45608741086465
41e+20,1.4916031629052877e+20,1.5308420407393242e+20,1.5521398074034396e+20,1.6146044043452508e+20,1.6644443650375352e+20,2.21727
80893021546e+20,2.2179048066254376e+20,2.399992744857991e+20
−12.503984375,1.376956382054515e+20,1.3213325640312527e+20,1.3981266807884713e+20,1.1081424508398315e+20,1.1504618036527176e+20,
1.1931794937473516e+20,1.241495150441481e+20,1.2504710171901978e+20,1.2709667981270612e+20,1.3460666536459126e+20,1.3622869477758
683e+20,1.4211240198547109e+20,1.4535539857527382e+20,1.482436094261332e+20,1.5458195225330414e+20,1.5612378309185592e+20,2.10024
65959246515e+20,2.1265368565200064e+20,2.2528894422904075e+20

A.39 "MITR_Cycle4_cladding_fluence_position_8.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,3.0135543155160007e+19,2.8290869759204745e+19,2.894802005081348e+19,2.2869127573154263e+19,2.2669534039179833e+19,2.
4486908053425983e+19,2.4305053806398722e+19,2.4944305476147585e+19,2.44410727102062e+19,2.5768544979540128e+19,2.54046240389353
4e+19,2.6458730709715145e+19,2.6409336886778417e+19,2.6832795973095113e+19,2.759675376784982e+19,2.7514218283071676e+19,3.7408491
46858643e+19,3.725298059379208e+19,4.070560881706929e+19
37.224453125,4.023360569401693e+19,3.5817319918735454e+19,3.812502057099436e+19,2.9353368068720038e+19,3.0514547800833647e+19,3.0
124071041445675e+19,3.080825515657569e+19,3.158140126462938e+19,3.2712201140055028e+19,3.349198620280452e+19,3.3970203387452133
e+19,3.41876955432864 4e+19,3.503078967801504e+19,3.563116362906532e+19,3.5781150893552755e+19,3.777219996264094e+19,4.94259554774
54496e+19,4.917213496152479e+19,5.1796858978417385e+19
33.672421875,4.996424192418968e+19,4.552055054392396e+19,4.735709785933157e+19,3.773125089007727e+19,3.794385677525546e+19,3.9430
87640383877e+19,3.951314633042908e+19,3.9865807603869794e+19,4.038189338610224e+19,4.232747889343362e+19,4.385013643082396 5e+19,
4.376712294130772e+19,4.420428483011654e+19,4.600827471169597e+19,4.531697363713205e+19,4.636428201830336e+19,6.116920410156739e
+19,6.237483827432405e+19,6.632581299288654e+19
30.120390625,6.036296944330985e+19,5.504755675566725e+19,5.893798420740406e+19,4.486775540659567e+19,4.61524196960511e+19,4.60566
59413518926e+19,4.841736548006908e+19,4.8977852591307014e+19,5.028631089439217e+19,5.089720094968059e+19,5.252124860318514e+19,5.
3252383405923836e+19,5.475756721455486e+19,5.522760520701725e+19,5.73340127528996e+19,5.760700656999075e+19,7.47578165729239e+1
9,7.485235528779204e+19,8.190664298935771e+19
26.568359375,6.976426040893352e+19,6.543991087827 9434e+19,6.846568087043571e+19,5.4613634676749976e+19,5.552503037738893e+19,5.62
6168879043452e+19,5.872872435539255e+19,5.864693243354034e+19,5.937243740269699e+19,6.1668984611066946e+19,6.396393847030993e+1
9,6.409937314610418e+19,6.531669187912545e+19,6.653029279751707e+19,6.803706995527507e+19,6.941797253200078e+19,9.20084764544818
4e+19,9.055321758516322e+19,9.874674991252757e+19
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53649907858e+19,6.893306328101979e+19,6.913276303826935e+19,7.154350026740702e+19,7.212507269875881e+19,7.393883512165043e+19,7.6
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0,1.095416902478929e+20,1.1568298889969258e+20
19.464296875000002,9.371442225311375e+19,8.803413261539015e+19,9.215028452678405e+19,7.169433731809552e+19,7.252819002788757e+19
,7.594911060353765e+19,7.686369300243058e+19,7.878208535132799e+19,7.98448492190311e+19,8.279254510396482e+19,8.501420490336618e+
19,8.535411938379098e+19,8.883186941163707e+19,9.080656009635483e+19,9.318702369207966e+19,9.554624263277794e+19,1.2423183808237
599e+20,1.2611836344873353e+20,1.3563862448250588e+20
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7717687844494e+19,8.7639513148272e+19,9.034767554778137e+19,9.228678143532964e+19,9.306433580930133e+19,9.60035338322244e+19,9.95
8485155332494e+19,1.0123556124888782e+20,1.016004381989688e+20,1.0389167424358213e+20,1.0741191358148133e+20,1.4371212450897025
e+20,1.4433671736675079e+20,1.5620876171196213e+20
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19,9.463112911750934e+19,9.719535897921382e+19,9.848490953932535e+19,1.038465293516507e+20,1.052874480788264e+20,1.0580156873046
89e+20,1.0758027747256672e+20,1.1229871535396332e+20,1.1486188298291644e+20,1.1821269619697893e+20,1.180273365818723e+20,1.587331
5784549184e+20,1.6057135162166398e+20,1.7498478782442632e+20
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247899337061e+20,1.07164413350422e+20,1.0961232872585614e+20,1.1271829729073765e+20,1.1545195327627887e+20,1.1657420217805632e+
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22700942e+20,1.770232124742237e+20,1.917585052006382e+20
5.256171875,1.3625047054727052e+20,1.306450683185155e+20,1.3574484775763863e+20,1.0630241147271979e+20,1.1131668117536106e+20,1.1
41746183928076e+20,1.1636494232604482e+20,1.197226600529909e+20,1.2148915311844847e+20,1.2433699912475738e+20,1.277781021226827
1e+20,1.292864726295677e+20,1.3382486205961424e+20,1.3731695222960328e+20,1.4085737398977769e+20,1.432192485123543e+20,1.93451704
20625418e+20,1.909443037967457e+20,2.0878290950616338e+20
1.704140625,1.4677932157842827e+20,1.3684532088801388e+20,1.4604797432913931e+20,1.1432333097799344e+20,1.1577168533442843e+20,1.
1813143539150237e+20,1.2164052128551264e+20,1.2566001001653574e+20,1.2828850495969554e+20,1.3127391010730138e+20,1.352705608341
7091e+20,1.3676937124629398e+20,1.413683079431662e+20,1.4475205037251982e+20,1.5044774238513765e+20,1.518180226343501e+20,2.03647
5452698708e+20,2.0083634629348265e+20,2.205747552786495e+20
−1.847890625,1.54184146087932e+20,1.4332228508923306e+20,1.4883686641774956e+20,1.1696032378316385e+20,1.1855526625928205e+20,1.2
032972607037455e+20,1.2654856771302113e+20,1.3065887734428274e+20,1.3186079370240978e+20,1.3474263115676117e+20,1.3846841253204
224e+20,1.4185959059065515e+20,1.484395913687531e+20,1.5180368249220717e+20,1.551284710038621e+20,1.5792586095448297e+20,2.094255
603207165e+20,2.1312903480821966e+20,2.3237244333126792e+20
−5.399921875,1.5460638360658467e+20,1.447876351696893e+20,1.5023688918399916e+20,1.1989580399145728e+20,1.2050127665971393e+20,1.
2456909698092245e+20,1.2835861232128315e+20,1.3107049253542216e+20,1.3581389288647492e+20,1.3659410284232493e+20,1.4106769607 45
4053e+20,1.4616216434990701e+20,1.4853094338536725e+20,1.5318777176718682e+20,1.5762843578411126e+20,1.600981269309476e+20,2.1631
57330622021e+20,2.1850605699543928e+20,2.358767491778972e+20
−8.951953125,1.484194089464779e+20,1.4129182718507064e+20,1.4713676289924997e+20,1.1582532808837046e+20,1.1976514936304399e+20,1.
2319509891708038e+20,1.2581509599822956e+20,1.2813395009437739e+20,1.31760943823785e+20,1.3491258839697356e+20,1.39153021540272
8e+20,1.4161740152335247e+20,1.474495904445159e+20,1.524633290307815e+20,1.5865083480726395e+20,1.5983309985949142e+20,2.16025212
40471395e+20,2.1616330266238658e+20,2.3558304182215518e+20
−12.503984375,1.3498747580594215e+20,1.284797068549345e+20,1.3509104349919658e+20,1.062126528052326e+20,1.1118549543057213e+20,1.
1376406543441953e+20,1.1697678839080942e+20,1.2113489849588076e+20,1.2559521381870476e+20,1.2860398808683979e+20,1.338344221543
7618e+20,1.3496941784916961e+20,1.401440846972613e+20,1.4289898533782908e+20,1.485436901783832e+20,1.5333223542136737e+20,2.01869
36764414863e+20,2.0575820174675843e+20,2.1921084842594507e+20

A.40 "MITR_Cycle4_cladding_fluence_position_9.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.8228198026876424e+19,2.6376513894762394e+19,2.846943108470289e+19,2.2114358091698536e+19,2.2658805488391426e+19,2.
3100481866393383e+19,2.3371510552894583e+19,2.2666506675838546e+19,2.4207753286377124e+19,2.3557932400752554e+19,2.488094329253
09e+19,2.5297444754326397e+19,2.5775661938974024e+19,2.6436317598662136e+19,2.5831854051519242e+19,2.645480044853523e+19,3.398634
932527232e+19,3.4117003953685594e+19,3.618039107313918e+19
37.224453125,3.7744847469294264e+19,3.5420841544302477e+19,3.663784160749835e+19,2.793927071851535e+19,2.924241785784388e+19,2.97
62327677981102e+19,3.060053554205361e+19,3.1796822066598593e+19,3.1306176758760436e+19,3.2105347569221652e+19,3.281656550787295
2e+19,3.279133748002892e+19,3.2732436673967817e+19,3.4313729459231437e+19,3.4198636540625105e+19,3.5573537502305804e+19,4.4855274
17176757e+19,4.458222724303885e+19,4.660615241578063e+19
33.672421875,4.917112584041103e+19,4.417581699238097e+19,4.57907294442241e+19,3.701849271393654e+19,3.727162277857788e+19,3.78693
41147749835e+19,3.908118938210177e+19,3.986527648749413e+19,4.012605462794503e+19,4.067480406728079e+19,3.962414965294281e+19,4.2
352813144552776e+19,4.275662092496991e+19,4.351914470551033e+19,4.543477524925427e+19,4.523056100281156e+19,5.8535929111026614e
+19,5.858957186496865e+19,6.062534093288772e+19
30.120390625,5.602746646876683e+19,5.421901520963182e+19,5.568967645384469e+19,4.329485426006753e+19,4.539010836206095e+19,4.5461
86218441312e+19,4.7359169213196665e+19,4.875292480621342e+19,4.903770940684431e+19,4.962475233686543 6e+19,5.155615703696658e+19,
5.231554723089059e+19,5.459876341823138e+19,5.4781467451459715e+19,5.367037199357119e+19,5.410163849061014e+19,6.981312311549459
e+19,7.0265634267560075e+19,7.480721039586063e+19
26.568359375,6.717400584482151e+19,6.481000685674225e+19,6.798767613233834e+19,5.077626575094197e+19,5.381164894949774e+19,5.4919
55770913228e+19,5.552237479551061e+19,5.679014958421994e+19,5.828949111271865e+19,6.099606016310102e+19,6.059081836846959e+19,6.1
79804589035324e+19,6.178476798096164 5e+19,6.446425009618516e+19,6.441910520425375e+19,6.503413796727234e+19,8.80320081498875e+1
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23.016328125,8.011571857061962e+19,7.564743650216064e+19,7.8898930953974e+19,6.192976275151784e+19,6.253098648876919e+19,6.425286
5778671e+19,6.554400968790952e+19,6.721012175836665e+19,6.882365330763305e+19,6.9002108409856066e+19,7.000432501073353e+19,7.1455
33494904683e+19,7.47843723917071e+19,7.459051491458982e+19,7.533885788790002e+19,7.712075332825183e+19,1.0176508427542454e+20,1.0
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19.464296875000002,9.242327834387522e+19,8.50694410064352e+19,8.97746009784402e+19,7.073036109626585e+19,7.203372068214464e+19,7.
3194209962969874e+19,7.512110017387789e+19,7.725671912042177e+19,7.847032003881338e+19,7.951237036786 56e+19,8.130701260123336e+
19,8.213183633263913e+19,8.505350751516531e+19,8.529728993159496e+19,8.650398633710295e+19,8.933164992113664e+19,1.1650090811821
474e+20,1.1430633525397219e+20,1.2168460394469271e+20
15.912265625,1.0496293597329587e+20,9.909834895321697e+19,1.0087387099706081e+20,7.86955133820948e+19,8.058841214496032e+19,8.209
99693500993e+19,8.428604435233122e+19,8.70786542555711e+19,8.833793118226981e+19,8.946920906243354e+19,9.273345030726282e+19,9.38
774749804425e+19,9.533220273338545e+19,9.753155564500897e+19,9.922847246525458e+19,9.98419118791462e+19,1.3184910914214519e+20,1.
314151870632279e+20,1.3944247996500951e+20
12.360234375000001,1.1450444166209477e+20,1.0686433259817202e+20,1.1170386501321993e+20,8.68911701749618e+19,8.976928981468355e+
19,9.251781705874335e+19,9.276850398805664e+19,9.644754712227928e+19,9.920988339210635e+19,1.0189414555471084e+20,1.033085084631
0334e+20,1.0551529700398614e+20,1.0891869073923932e+20,1.0969146506583004e+20,1.0985717337503715e+20,1.111987733399637e+20,1.4851
71343596e+20,1.4285915160965426e+20,1.554975968849484e+20
8.808203125,1.2468594258356845e+20,1.1964246148026566e+20,1.2322696589962019e+20,9.634663501090318e+19,1.0001558693398821e+20,1.0
214801918227808e+20,1.0381200678723255e+20,1.065600029149167e+20,1.0787451594468444e+20,1.098120284831057e+20,1.118361129907601
9e+20,1.131878041668244e+20,1.1648338127781782e+20,1.1930892039634887e+20,1.2156722722567102e+20,1.2200646046834495e+20,1.6383931
068112293e+20,1.6133031692248747e+20,1.739815089907975e+20
5.256171875,1.3586063112753334e+20,1.2637701712368178e+20,1.3178537517706553e+20,1.0455450748041046e+20,1.062275240637512e+20,1.1
083070969162875e+20,1.1064110114551682e+20,1.1498032193468947e+20,1.1552949626712575e+20,1.1837946673893732e+20,1.2166389040604
185e+20,1.2291838728535956e+20,1.262506114262738e+20,1.274355320603796e+20,1.322676288461682e+20,1.3293736659588014e+20,1.7576234
2198398e+20,1.755472400662 5418e+20,1.859438431198717e+20
1.704140625,1.4380347652558442e+20,1.3197073479217218e+20,1.3894907285201792e+20,1.0959267741995658e+20,1.118961291412102e+20,1.1
524800458802397e+20,1.1778727198007227e+20,1.2091873413098562e+20,1.2062662012437058e+20,1.2636958149442249e+20,1.3047617331105
446e+20,1.3250397563333855e+20,1.347479423205178e+20,1.3763190424037186e+20,1.4026677258003962e+20,1.403915849283206e+20,1.852995
989561916e+20,1.8710273905156994e+20,2.0018201091866496e+20
−1.847890625,1.500706497584164e+20,1.3897244197254711e+20,1.4235405993639808e+20,1.1056090257279158e+20,1.1706760929104791e+20,1.
1932644723674576e+20,1.2213658398038257e+20,1.237920737233264e+20,1.2692088029236144e+20,1.2919671396208052e+20,1.3331074140797
175e+20,1.3669607718645237e+20,1.4141292171872194e+20,1.4205769699877772e+20,1.460161073465995e+20,1.4796742891078805e+20,1.98860
0622596379e+20,1.9803311406272948e+20,2.093071213689435e+20
−5.399921875,1.4802744506123803e+20,1.3911478116122501e+20,1.42850122631268e+20,1.1374494524489564e+20,1.1762368813636785e+20,1.2
137336974855376e+20,1.2334168703676357e+20,1.2709189976532515e+20,1.3113688208238012e+20,1.3237810105230628e+20,1.3806210850465
948e+20,1.4087596306292592e+20,1.4306256918153 35e+20,1.449028874232083e+20,1.491072046529624e+20,1.542643446606572e+20,1.99743839
9087424e+20,1.9781216965045338e+20,2.1253099776922236e+20
−8.951953125,1.4213258440774631e+20,1.3660366293708687e+20,1.4057216449604626e+20,1.1179681037896106e+20,1.1547744686231072e+20,
1.1885959594253735e+20,1.2263264667525248e+20,1.244947406883 2952e+20,1.2649758054095744e+20,1.3108642602669207e+20,1.35812299537
34792e+20,1.3637262731367316e+20,1.3962093506723253e+20,1.4537504988117338e+20,1.4814853959488936e+20,1.50810494869716e+20,1.9724
493736124452e+20,1.9839639766368348e+20,2.1094136645686077e+20
−12.503984375,1.3111457519460223e+20,1.2394344189039056e+20,1.3125478991777746e+20,1.0220166193622013e+20,1.0703800765301404e+20
,1.1074891776977651e+20,1.114829206009438e+20,1.1541318178085542e+20,1.182966125843338e+20,1.2078170610606437e+20,1.261688195044
2158e+20,1.290065742995929e+20,1.321316630539983e+20,1.3538846866956825e+20,1.3869519922445065e+20,1.3934794125014137e+20,1.85932
68967598277e+20,1.834104180079557e+20,1.9785200337862817e+20

A.41 "MITR_Cycle5_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.45073394381351e+19,2.633500299240993e+19,2.63545992276466e+19,3.023546282339257e+19,2.974007359224912e+19,2.797398
5364291342e+19,2.9355340166501716e+19,2.859414328584072 4e+19,3.012426767207258e+19,3.060698227400332e+19,3.1126642071724294e+19,
2.9830414034509906e+19,3.2047575236860314e+19,3.1440091944523637e+19,2.998547598764408e+19,3.0752246109519163e+19,2.862596469535
348e+19,2.8763587796946768e+19,2.6656902618017747e+19
37.224453125,3.2832323556200235e+19,3.499824689577587e+19,3.5822187685589127e+19,3.6313531822302986e+19,3.804294452743251e+19,3.8
910931901034594e+19,4.030972555478585e+19,4.143066616671813e+19,4.15841100820804 7e+19,4.133034782486437e+19,4.05137780960117e+1
9,4.1276593014444515e+19,4.237227425893871e+19,4.194187621163063e+19,4.3449527851031446e+19,4.30631763875122e+19,4.1238119671869
776e+19,3.932371131482714e+19,3.6607475350852993e+19
33.672421875,4.340278453762289e+19,4.479582516818539e+19,4.612180712220408e+19,4.691185900979613e+19,4.877808579763302e+19,4.7732
383942091055e+19,5.089879397058461e+19,5.234540962959231e+19,5.153648063466219e+19,5.3236858550886015e+19,5.472572297393608e+19,
5.504915074632841e+19,5.613566311011553e+19,5.152677240803119e+19,5.611669611179014e+19,5.322804923412824e+19,5.034443625175651e
+19,4.966153442105486e+19,4.501560863219297e+19
30.120390625,5.1245952963630514e+19,5.404318070718372e+19,5.466675448625144e+19,5.702828061424444e+19,6.281932769062882e+19,6.129
71137112815e+19,6.334554953042446e+19,6.385370328176793e+19,6.522930506078955e+19,6.563048853721729e+19,6.758220165399945e+19,6.7
13265682639132e+19,6.452033484376391e+19,6.49757225855481e+19,6.674783350965659e+19,6.594357884606734e+19,6.400993381773733e+19,
6.291694930286 2864e+19,5.766452902252484e+19
26.568359375,6.081367998145216e+19,6.404598010365431e+19,6.444374772256372e+19,6.680239733896236e+19,7.169003999273892e+19,7.3202
72552742989e+19,7.429858655389875e+19,7.416815473129138e+19,7.851285582163001e+19,7.791157500741871e+19,7.872715593541081e+19,8.0
29818072087516e+19,7.964305520525668e+19,8.077774913824955e+19,7.752810005549384e+19,7.937410137118 28e+19,7.937086529563912e+19,
7.524073399204325e+19,7.359069502872097e+19
23.016328125,7.296298626423636e+19,7.452259489431054e+19,7.910667568389341e+19,8.310547625500869e+19,8.244765400977048e+19,8.5796
54274253208e+19,8.876654096372233e+19,8.656565003007758e+19,9.011661370234388e+19,8.935955180709981e+19,9.252479325275814e+19,9.7
46250518647426e+19,9.499050303505985e+19,9.470105405587605e+19,9.583098376654073e+19,9.239355241126486e+19,9.275581309018123e+1
9,9.06505661670494e+19,8.776075070655227e+19
19.464296875000002,8.163845534187838e+19,8.817290088343359e+19,8.986455937388695e+19,9.173824711367174e+19,9.762520787353099e+19
,9.763239915251692e+19,1.003156451241431e+20,1.014006293411457 4e+20,1.0074622295342583e+20,1.0407848185353244e+20,1.058286593767
3386e+20,1.0832043754535957e+20,1.0855954757164184e+20,1.1122930989516941e+20,1.1014342676829354e+20,1.0940811849198192e+20,1.058
3135610635357e+20,1.0295124887248749e+20,9.457071212425603e+19
15.912265625,9.381652674060634e+19,9.324212333160496e+19,9.760093730695348e+19,1.0196784147116117e+20,1.0823594001727486e+20,1.10
20814827916693e+20,1.1142437333766285e+20,1.1244733277341177e+20,1.1815271373887622e+20,1.1810686933534091e+20,1.21008550406164
8e+20,1.2331784987052247e+20,1.221321877477168e+20,1.2465273103228625e+20,1.224090519886752e+20,1.2695843485715096e+20,1.22446806
20335137e+20,1.1944264940697793e+20,1.085577497518953 8e+20
12.360234375000001,1.0295664233172692e+20,1.0869797969212105e+20,1.1020275481992749e+20,1.1627399210380127e+20,1.202579606620080
5e+20,1.2300862487412736e+20,1.2621323857223369e+20,1.2868524072364807e+20,1.27010571629799e+20,1.3092712194751264e+20,1.33388337
18044815e+20,1.3272763842361557e+20,1.3723117688855598e+20,1.3792873095019148e+20,1.3550706775167859e+20,1.3605899841384894e+20,
1.329964124757148e+20,1.2991494943024259e+20,1.2342032559607202e+20
8.808203125,1.0980184101646175e+20,1.156995886947999e+20,1.208341618907559e+20,1.2699349234220738e+20,1.3159591089320439e+20,1.33
2966483733775e+20,1.3819121263317801e+20,1.3569853555467906e+20,1.3768782310416274e+20,1.4214102261620163e+20,1.471731200866081
2e+20,1.4634881973284556e+20,1.4996333633315006e+20,1.5322817699276354e+20,1.5107528784634993e+20,1.5101236415522298e+20,1.471290
7350281929e+20,1.4397030420824827e+20,1.3966272809567453e+20
5.256171875,1.1984625994006341e+20,1.258231116872468e+20,1.309262230376394e+20,1.3533987051525567e+20,1.4042590257805666e+20,1.43
54242310858529e+20,1.4639286631663439e+20,1.48631151901006e+20,1.4970624810940295e+20,1.5359493222104613e+20,1.5879692365749528
e+20,1.5923289494601748e+20,1.6321686350422422e+20,1.6610775765656922e+20,1.656852700161457e+20,1.634065334874782e+20,1.572373150
274211e+20,1.5331357343072148e+20,1.435783795035149 4e+20
1.704140625,1.2373224732208685e+20,1.3192311408706435e+20,1.4303543794007702e+20,1.4600363834152079e+20,1.4725581979494636e+20,1.
4794887930721563e+20,1.5427450808521677e+20,1.6056507937816153e+20,1.6053271862272482e+20,1.6340293784798524e+20,1.680233345964
4708e+20,1.6689879834502185e+20,1.7000003740770537e+20,1.7160728826106135e+20,1.7032274605219912e+20,1.763373520140587e+20,1.6945
1702385028e+20,1.6807547136909512e+20,1.543940630983579e+20
−1.847890625,1.3106735188773832e+20,1.3423960483040793e+20,1.398523980789285e+20,1.4491326066527873e+20,1.490913937561057e+20,1.5
774609801567584e+20,1.638209309390426e+20,1.608023915846973e+20,1.6638911644689387e+20,1.7332959957819228e+20,1.749449406204074
e+20,1.7280193948259942e+20,1.7629240652039655e+20,1.781711281554715e+20,1.823843187313549e+20,1.76033520476903e+20,1.76037116116
39598e+20,1.6841166366168747e+20,1.6411217773797288e+20
−5.399921875,1.2926234086226918e+20,1.382640243329106e+20,1.4177606520766552e+20,1.4856373366051288e+20,1.5474463794892213e+20,1.
6311978123791414e+20,1.6373193886159 17e+20,1.6817614927489817e+20,1.6976272520116958e+20,1.697285666259864e+20,1.76704107242341
24e+20,1.7695759982659538e+20,1.7840304690276788e+20,1.830252414709762e+20,1.8741102274252197e+20,1.7883362473205062e+20,1.753251
7949678862e+20,1.7698996058203206e+20,1.5647773618453194e+20
−8.951953125,1.2587614736976809e+20,1.33303571050898e+20,1.381534584185019e+20,1.438885034097833e+20,1.480738277795962e+20,1.4974
40023240791e+20,1.5784947265109862e+20,1.612563410706843e+20,1.681689579959 1223e+20,1.710122099249754e+20,1.7213045380728796e+2
0,1.7405951439526442e+20,1.8089572398126685e+20,1.781747237949645e+20,1.817020461375 6453e+20,1.7971365749795412e+20,1.7913116390
009356e+20,1.7367837660901002e+20,1.6428297061388878e+20
−12.503984375,1.1237901562304541e+20,1.1697604071480295e+20,1.2673190956909409e+20,1.3088577209334351e+20,1.3483108752700093e+20
,1.3949642976912482e+20,1.4545350549909694e+20,1.4400446278343148e+20,1.4892329760980953e+20,1.5173239096368952e+20,1.5673122876
878606e+20,1.5675280260574387e+20,1.6056148373866856e+20,1.6296786546933632e+20,1.633481043457175e+20,1.6290763850782913e+20,1.62
63886445572987e+20,1.5609030602916482e+20,1.4630387423918349e+20

A.42 "MITR_Cycle5_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.8229566706470416e+19,2.6346235595062436e+19,2.759605217927683e+19,2.118398204064265e+19,2.263025453442197e+19,2.29
50916478752416e+19,2.2814066715897012e+19,2.2935404005633712e+19,2.293657275360841e+19,2.3814727481869218e+19,2.412827068853559
7e+19,2.498655670122656e+19,2.5003716046491423e+19,2.489257873907935e+19,2.532278424358365e+19,2.565019305122257e+19,3.4073625329
40196e+19,3.296113663221445e+19,3.5629388258447192e+19
37.224453125,3.7085807609643205e+19,3.3996381713256088e+19,3.579285360018092e+19,2.8151473091797496e+19,2.893517173373963e+19,2.9
724926615187e+19,3.0416984790931325e+19,3.0927727655873815e+19,3.1980450831627964e+19,3.2416446951097807e+19,3.2108428734857265
e+19,3.3690647868057e+19,3.3175973759926886e+19,3.3994469216570225e+19,3.397614112333066e+19,3.527403574923143e+19,4.591585793320
1195e+19,4.554339920362851e+19,4.765150180053385e+19
33.672421875,4.571191141161661e+19,4.40536174102457e+19,4.588828610597994e+19,3.6251215305893822e+19,3.6724823860185723e+19,3.777
8875158638395e+19,3.817954321432763e+19,3.858371751394094e+19,3.9726699908286456e+19,3.979788728492708e+19,4.145947503059031e+1
9,4.128511908272873e+19,4.319825326749147e+19,4.2705732245972664e+19,4.406047052337005e+19,4.371797424187596e+19,5.7672931309571
38e+19,5.654933950662426e+19,6.149898717946511e+19
30.120390625,5.635012110184638e+19,5.345428236999542e+19,5.440256197673803e+19,4.300163798320271e+19,4.4403710553575326e+19,4.629
219478105356e+19,4.66234285820642e+19,4.753887699569943e+19,4.825122888627712e+19,4.750785204946203e+19,5.018306303863501e+19,5.1
92566626890791e+19,5.140281092495493e+19,5.092410237950072e+19,5.462409284285101e+19,5.499862344383339e+19,7.291818614133205e+1
9,7.092547084447405e+19,7.778761520318236e+19
26.568359375,6.708878999299215e+19,6.359635854496643e+19,6.655913466082276e+19,5.095730544696368e+19,5.274586172260535e+19,5.4799
40503905553e+19,5.5335435360177816e+19,5.739668178827941e+19,5.762352514518648e+19,5.814893048472062e+19,5.932086595389384e+19,6.
095817561662814e+19,6.238776688931408e+19,6.346407752419393e+19,6.4328419776390136e+19,6.705213380651304e+19,8.680131833349156e
+19,8.717903642895039e+19,9.15814975500014e+19
23.016328125,7.986479910366618e+19,7.593143091973212e+19,7.649986743469828e+19,6.098526931967794e+19,6.286429731354271e+19,6.4998
856114602566e+19,6.686779037595859e+19,6.683804042751177e+19,6.859275613679515e+19,7.055625273428574e+19,7.0546159001777e+19,7.28
6452998431187e+19,7.3475997674710065e+19,7.549527542553844e+19,7.625071161645608e+19,7.727017859983933e+19,1.0433785044473733e+
20,1.0085816897461738e+20,1.0874456155842357e+20
19.464296875000002,9.080481139590744e+19,8.481922801822171e+19,8.757428574402965e+19,6.887909939059587e+19,7.2155312463302705e+1
9,7.397112181671797e+19,7.691742921111267e+19,7.83401142457663e+19,7.783011512953497e+19,7.957154961183316e+19,8.396285450221661e
+19,8.30565435727472e+19,8.370626119686272e+19,8.562194537720663e+19,8.628494422830735e+19,8.922859537730501e+19,1.1790382693649
048e+20,1.1884148156164494e+20,1.2884118298333472e+20
15.912265625,1.0275154069362614e+20,9.821732980088206e+19,9.965648355699733e+19,7.884480087120354e+19,8.054479792530794e+19,8.071
798512519484e+19,8.28753876366692e+19,8.45891971668382e+19,8.684063076536774e+19,8.913403304117045e+19,9.182215338297305e+19,9.38
2761865773684e+19,9.506808525815364e+19,9.65141452523012e+19,9.87156414373664e+19,9.966763978766488e+19,1.3220877089769967e+20,1.
3163874063549527e+20,1.435812199405787e+20
12.360234375000001,1.1508236307575659e+20,1.0726662661951159e+20,1.1237193027261887e+20,8.777616039420456e+19,9.064543667208516e
+19,9.258662080824061e+19,9.383771239024558e+19,9.599617739987876e+19,9.847498560439476e+19,1.016741675605875e+20,1.030362902001
8865e+20,1.0458860001021772e+20,1.0672794005299225e+20,1.0649365920897348e+20,1.1106718253359374e+20,1.126449922994344e+20,1.4653
337107484693e+20,1.4727871353325581e+20,1.6000691022678314e+20
8.808203125,1.2298203688654799e+20,1.1733857791600078e+20,1.2113435258836827e+20,9.598395867105239e+19,9.681005098953125e+19,9.99
1360811143062e+19,1.0367007035192194e+20,1.0387725749289091e+20,1.0507044292524045e+20,1.1000415312569908e+20,1.118582124128317
e+20,1.1251058628234427e+20,1.1502711317149819e+20,1.1915863726205131e+20,1.1958363652557739e+20,1.2283169339707562e+20,1.6390096
597884106e+20,1.603655033553833e+20,1.732950434500061e+20
5.256171875,1.3182414656420854e+20,1.2280991218481991e+20,1.2934108836705727e+20,1.0334388341716566e+20,1.052840050551623e+20,1.0
70594394785426e+20,1.0855278064075745e+20,1.119835871955719e+20,1.1550576859204446e+20,1.1775667094149458e+20,1.189721688351792
3e+20,1.2399778512637539e+20,1.2511181444589317e+20,1.2752049777192734e+20,1.2923643229841398e+20,1.3240692680431872e+20,1.768336
935679403e+20,1.7682359983543157e+20,1.9080076311464624e+20
1.704140625,1.3839091643476622e+20,1.2965399407482837e+20,1.3807747947791573e+20,1.0686075232284416e+20,1.0935390425250415e+20,1.
1601683020643459e+20,1.158632992224858e+20,1.1864545065134352e+20,1.2297406815035689e+20,1.2479784623996325e+20,1.2853465226451
652e+20,1.3374514323534651e+20,1.3228633326329319e+20,1.3637854492197011e+20,1.3882388443448343e+20,1.4011003845572929e+20,1.8905
029739799813e+20,1.8573158439893867e+20,2.010777765557868e+20
−1.847890625,1.4275459637302046e+20,1.3418448622401662e+20,1.4070556867374524e+20,1.094139353984772e+20,1.1389608388143935e+20,1.
1929516827545906e+20,1.1956557405687756e+20,1.2423578471395e+20,1.259416255079279e+20,1.2893999531210452e+20,1.3029255546827635
e+20,1.3390770545364525e+20,1.3760838654079877e+20,1.4098606818767246e+20,1.439828442446109e+20,1.478147438543781e+20,1.927685097
047721e+20,1.944536317846531e+20,2.120225700897427e+20
−5.399921875,1.4355253249029072e+20,1.3496595361982523e+20,1.425553779682426e+20,1.1215571189729994e+20,1.1629254847864716e+20,1.
1728704675529823e+20,1.229336932203219e+20,1.2567653221730347e+20,1.294393694467477e+20,1.3079777334279299e+20,1.33408331319002
09e+20,1.3665320069602386e+20,1.4014881963852607e+20,1.4555640401781627e+20,1.4679952686363012e+20,1.5038439565147282e+20,2.00446
65264945054e+20,2.0046365261999158e+20,2.1196200769469022e+20
−8.951953125,1.4049625653645863e+20,1.3399854904622395e+20,1.3731938704160103e+20,1.0932999804393079e+20,1.12785242056398e+20,1.1
592811161017352e+20,1.193621056594642e+20,1.2311963039811456e+20,1.2587893811655778e+20,1.2849852732711679e+20,1.3214820850264
718e+20,1.3774757379960359e+20,1.4024444447281942e+20,1.4373634467176576e+20,1.4601274697702742e+20,1.4886342953712878e+20,1.9799
865689154018e+20,1.9890709281732723e+20,2.1496303374426397e+20
−12.503984375,1.2754971647129477e+20,1.2122891492450283e+20,1.3011511827575421e+20,1.0373222649421264e+20,1.049424118971032e+20,
1.0613294108405572e+20,1.1157505665350743e+20,1.1423236454870439e+20,1.1778482714270317e+20,1.1957407404214808e+20,1.23591379580
62855e+20,1.2544862636223762e+20,1.3126633503083053e+20,1.3346783121589574e+20,1.379165110068552e+20,1.3885044688845378e+20,1.846
8714870882327e+20,1.8084409286838852e+20,1.983769062360784e+20

A.43 "MITR_Cycle5_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.939629593466545e+19,2.6939268942405173e+19,2.7893020414665683e+19,2.159803757313295e+19,2.2372279981461627e+19,2.3
244112845677486e+19,2.259795459039399e+19,2.4162748753789153e+19,2.377955879281243e+19,2.4337157826558677e+19,2.395364911613431
4e+19,2.5767758472495473e+19,2.6645169452045107e+19,2.6152807805250126e+19,2.606196421267142e+19,2.7803876819141087e+19,3.6228371
5954793e+19,3.71243762928082e+19,3.923056639302768e+19
37.224453125,3.733028843598659e+19,3.591822838292112e+19,3.657155850077662e+19,2.8531741182837477e+19,2.9771251534911353e+19,2.98
58004509578613e+19,3.122873338426618e+19,3.1662498257602503e+19,3.2640952812055466e+19,3.249592181337719e+19,3.2896271119618777
e+19,3.4409427872495518e+19,3.5158542199368212e+19,3.5991647005695255e+19,3.6284046499001217e+19,3.673024260079568e+19,4.97767637
4270406e+19,4.875155926926322e+19,5.280073975250816e+19
33.672421875,4.7547376980969955e+19,4.5252380957929005e+19,4.723224002706535e+19,3.6229274718914286e+19,3.748775066312302e+19,3.9
35976616913961e+19,4.008162741823869e+19,3.97241499127053e+19,4.0336945725802e+19,4.102873827700661e+19,4.189642739840309e+19,4.2
31090793015692e+19,4.34436903421778e+19,4.406307364385915e+19,4.567849584452186e+19,4.568811145285914e+19,6.148836219787695e+19,
6.079826964372645 5e+19,6.797491345744408e+19
30.120390625,5.9180616196930224e+19,5.576309086910096e+19,5.631399616444667e+19,4.463852264667349e+19,4.499530952840365e+19,4.636
959777192324e+19,4.8034479261878804e+19,4.877705922507479e+19,4.942417372870122e+19,5.023050358142612e+19,5.207053789286236e+19,
5.271956489317467e+19,5.406150006775833e+19,5.428781217558599e+19,5.689093266468335e+19,5.685321398004541e+19,7.547455671144155e
+19,7.769039662165076e+19,8.593060109234246e+19
26.568359375,6.8470037599452e+19,6.60475417973532e+19,6.854813121412491e+19,5.250849963392602e+19,5.457096793491025e+19,5.6532339
5360832e+19,5.702161993821763e+19,5.697327627199154e+19,5.901964772586972e+19,5.970974028002022e+19,6.04742077052878e+19,6.377538
94847268e+19,6.506260600413146e+19,6.542491787628748e+19,6.865119353553e+19,6.869581845820023e+19,9.200330931905105e+19,9.2412371
11019492e+19,1.0078220035626208e+20
23.016328125,8.009482995504968e+19,7.54580879899799e+19,7.904189427966378e+19,6.229161080594129e+19,6.481185643865107e+19,6.55061
9898543685e+19,6.661438456508116e+19,6.921219256338445e+19,7.092334584815 6406e+19,7.189287541807 5324e+19,7.42234651294366e+19,7.6
09983687790433e+19,7.689671049701576e+19,7.838367667027771e+19,8.023295471569566e+19,8.101654710782191e+19,1.075965322978237e+2
0,1.0968062243633984e+20,1.1712342203884069e+20
19.464296875000002,9.243415232245062e+19,8.783937903465406e+19,9.112037334907557e+19,7.068056501886712e+19,7.441418354894393e+19
,7.451405837587256e+19,7.645364876478982e+19,7.73721784230856e+19,8.101760960598073e+19,8.258320064299501e+19,8.423538527995272e+
19,8.681672455679438e+19,8.939168884468315e+19,9.143062281144962e+19,9.410864942074348e+19,9.448689876528 17e+19,1.24847783653427
6e+20,1.293756195572188e+20,1.4102219312507219e+20
15.912265625,1.0282060307394914e+20,9.895204727770279e+19,1.0439947533794861e+20,8.126198418250824e+19,8.221451378188612e+19,8.33
7210552591532e+19,8.678591211018874e+19,9.006584392645144e+19,9.121706068152775e+19,9.348602549967774e+19,9.469089841177423e+19,
9.787042415202887e+19,9.977867084526107e+19,1.0297200906158032e+20,1.0398350730877246e+20,1.068007211768711e+20,1.42364128299655
86e+20,1.4514946722299005e+20,1.564241664352581e+20
12.360234375000001,1.1719567191364013e+20,1.0799337536014121e+20,1.139826774813828e+20,8.905009568662405e+19,9.257280833217602e+
19,9.309874492078958e+19,9.69603944790036e+19,9.910239076717516e+19,1.0216663545719836e+20,1.0432138172327569e+20,1.052175989202
3637e+20,1.0760928227572947e+20,1.1125949470033928e+20,1.164418294699607e+20,1.172073593933 8709e+20,1.2109769640188915e+20,1.6030
547220941023e+20,1.6104709592426326e+20,1.7740106758474767e+20
8.808203125,1.2724956074142946e+20,1.18133857787874e+20,1.2491843978098878e+20,9.674842609631997e+19,1.0063345061402794e+20,1.044
0903782137795e+20,1.0811715639564317e+20,1.0964874749157537e+20,1.1341370971733723e+20,1.1296214799984073e+20,1.171988594081165
6e+20,1.2033216647846276e+20,1.2397706641227848e+20,1.2600590664653621e+20,1.2918755738310848e+20,1.3143686598532037e+20,1.774950
986718028e+20,1.804674372710885e+20,1.925942600067264e+20
5.256171875,1.3893225924668257e+20,1.3093217935988313e+20,1.334380812674489e+20,1.0549703593600477e+20,1.1012049667408934e+20,1.1
220405556352605e+20,1.1450170783196404e+20,1.1668726654464703e+20,1.1830279499512563e+20,1.2192963246024149e+20,1.2543428263709
358e+20,1.2842680870139676e+20,1.3212855228670909e+20,1.353734216637309e+20,1.403480380433039e+20,1.4188122288647432e+20,1.901903
579224069e+20,1.9337785239885264e+20,2.112756338840955 8e+20
1.704140625,1.43223158061058e+20,1.3458770427528702e+20,1.4099881816557827e+20,1.115469004522988e+20,1.1290902309189999e+20,1.175
4045256617568e+20,1.198843235045221e+20,1.2366681694990442e+20,1.2731649812543481e+20,1.2984258749801808e+20,1.33040706956052e+
20,1.3763760524016619e+20,1.4104344308824849e+20,1.4390368813177917e+20,1.4779296264212238e+20,1.5001092754864924e+20,2.02177993
39923997e+20,2.034269599849273e+20,2.2476670425565227e+20
−1.847890625,1.4682183932496526e+20,1.402497569636135e+20,1.4674958945016581e+20,1.1446133290192907e+20,1.1733910916508017e+20,1.
208442905910117e+20,1.2341712988258284e+20,1.2808043430162298e+20,1.3203186495425692e+20,1.323718643650778e+20,1.36334982497458
7e+20,1.4053716271557303e+20,1.4508784232977877e+20,1.4809099337567008e+20,1.5458126337879307e+20,1.567152909307735e+20,2.1195563
270573733e+20,2.129958184032175e+20,2.320958165551599e+20
−5.399921875,1.484060240797588e+20,1.4294265854713078e+20,1.4811436833516 4e+20,1.1676482891024055e+20,1.2071625956287447e+20,1.22
17878827848366e+20,1.2538647021994692e+20,1.3020224312477704e+20,1.3168708430172137e+20,1.3726732463181 91e+20,1.395623206548600
4e+20,1.41078505527489 4e+20,1.4700512025736089e+20,1.5139058140787084e+20,1.5661010361305078e+20,1.5923500531440386e+20,2.1481375
27529504e+20,2.170774050803063e+20,2.313520678439892e+20
−8.951953125,1.4469259301469949e+20,1.375462303985081e+20,1.4536568559830894e+20,1.1296374174707897e+20,1.1705648465483532e+20,1.
1838673234967201e+20,1.2509215822995508e+20,1.2723362326904724e+20,1.2973686893121597e+20,1.3317405047498331e+20,1.371228248822
202e+20,1.4274662763682932e+20,1.4560156018956594e+20,1.507063325935938e+20,1.5231601730419894e+20,1.583047881763611e+20,2.133150
9909994144e+20,2.138420981867138e+20,2.3616730949973993e+20
−12.503984375,1.3524698438283185e+20,1.2895805778080439e+20,1.3538298414716022e+20,1.0714603307848606e+20,1.0908615471648268e+20
,1.1334624108425247e+20,1.1618842365908325e+20,1.1986041729594878e+20,1.2217135079137196e+20,1.2559684485539232e+20,1.2892087034
524587e+20,1.3272195750840741e+20,1.3809660444477438e+20,1.4028163190837797e+20,1.448461239986483e+20,1.5057936406361542e+20,1.97
54550142680547e+20,2.0390242791099718e+20,2.1761131040511096e+20

A.44 "MITR_Cycle5_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.887551246212215e+19,2.7282030848438977e+19,2.850278810800976e+19,2.2849926028757025e+19,2.3371506374919434e+19,2.3980211570104693e+19,2.4016602132044116e+19,2.45699511731551e+19,2.566203990569336e+19,2.5412140338740007e+19,2.6181760880077836e+19,2.67211380704004e+19,2.67671442406771e+19,2.734716198557435e+19,2.71918778796635e+19,2.7939876583469445e+19,3.74097632982659 7e+19,3.81503245149602e+19,4.0531861013036655e+19
37.224453125,3.846041460261044e+19,3.5741056814938673e+19,3.826714618752194e+19,2.933923978353707e+19,2.9566083140444127e+19,3.0976390071547556e+19,3.2126756828096823e+19,3.2484978082341396e+19,3.3084758292992602e+19,3.3925300586431345e+19,3.4513446442243535e+19,3.5052717382750216e+19,3.5557935257266864e+19,3.6103209312370856e+19,3.683617366722956e+19,3.759267235630602e+19,5.165286986663209e+19,5.127238927596035e+19,5.567968476363396e+19
33.672421875,4.851196593445038e+19,4.564338028037302e+19,4.803538238531379e+19,3.797118732538395e+19,3.890847007618282e+19,3.9614765727255265e+19,4.000098380798461 5e+19,4.116649116329700 6e+19,4.174289641445428 4e+19,4.273696969184183e+19,4.293740996950233e+19,4.438729495702163e+19,4.504657506456649e+19,4.519266856140359e+19,4.574484885453988e+19,4.776981097051792e+19,6.3751483276153455e+19,6.293548469018334e+19,6.940556722828883e+19
30.120390625,5.791146214622541e+19,5.654774575938602e+19,5.887514797627085e+19,4.608075764744932e+19,4.8151354059348484e+19,4.6713794050471444e+19,4.978744184920016e+19,4.967545454326102 4e+19,5.010241942838093e+19,5.289089272128363e+19,5.327312643391743e+19,5.364287579318512e+19,5.580930953900942e+19,5.591502810581155e+19,5.823339908834643e+19,5.938886583605802e+19,7.78503025945524 6e+19,7.928998759974714e+19,8.705313039713078e+19
26.568359375,6.976362910780957e+19,6.559544883077732e+19,6.938431726511252e+19,5.478506131391153e+19,5.632568364419364e+19,5.76532750936333e+19,5.89771477995171e+19,6.0002989771853234e+19,6.169554933884594e+19,6.205467371652549e+19,6.409785767592722e+19,6.559066758906266e+19,6.586532336311641e+19,6.728960214500825 5e+19,7.008131605729532e+19,7.053287777479181e+19,9.409218069928183e+19,9.706930054028218e+19,1.0437291288397822e+20
23.016328125,8.249448204673394e+19,7.78168339025498e+19,8.094057848946663e+19,6.3626639742492656e+19,6.6949602734187364e+19,6.8958786752507e+19,6.8578412411651146e+19,7.100781445178223e+19,7.277156139541554e+19,7.2836905032182686e+19,7.540177558756272e+19,7.830823930100184e+19,7.895423818156152e+19,8.004489254158538e+19,8.195260798573815e+19,8.385394844093804e+19,1.1149536929159628e+20,1.0939640417885676e+20,1.25989969174154e+20
19.464296875000002,9.356252536711243e+19,8.92695015564194e+19,9.31911822606065e+19,7.360668494824433e+19,7.711558511773172e+19,7.887773831412 68e+19,7.962467451977394e+19,8.145960884004787e+19,8.33136681271805e+19,8.531116466575316e+19,8.818840967982486e+19,8.924718909508428e+19,9.191405947371058e+19,9.264771445237249e+19,9.342758810094289e+19,9.643445789039007e+19,1.2954136926999395e+20,1.3015389945855097e+20,1.4222919103348631e+20
15.912265625,1.0601978503014185e+20,1.0148982413003304e+20,1.0501625551914087e+20,8.311763721687908e+19,8.676625589425065e+19,8.691181814200834e+19,8.983740682230615e+19,9.325121340657957e+19,9.395989967850935e+19,9.795117401209884e+19,9.780454926618234e+19,1.0162848013975844e+20,1.0184416726599793e+20,1.054710047311138e+20,1.0765921968919382e+20,1.0937356046844222e+20,1.4752255686070392e+20,1.4956786581642327e+20,1.619263131506829e+20
12.360234375000001,1.1812748279892109e+20,1.1087380786868935e+20,1.1562529963491118e+20,9.225671512992848e+19,9.55658656455586e+19,9.804998634086865e+19,1.0108554358060383e+20,1.0218735417129524e+20,1.0715293931651834e+20,1.077144695934522e+20,1.1102627635447936e+20,1.119904934336042e+20,1.1629467347496478e+20,1.1953104286671605e+20,1.2262397500702725e+20,1.2359403582602558e+20,1.6545221329071132e+20,1.6763830325247368e+20,1.8354283819177925e+20
8.808203125,1.2875937062510595e+20,1.2254428764511 61e+20,1.2839334000939408e+20,1.0198122952848509e+20,1.0486378703335088e+20,1.0741697010898392e+20,1.103122775917555e+20,1.124983675535179e+20,1.1573526819434856e+20,1.1669682902807636e+20,1.2201941355466138e+20,1.2269569363274727e+20,1.245428466818476e+20,1.3018843064871246e+20,1.3245686421778303e+20,1.3459726675871638e+20,1.8310349520310914e+20,1.8324480745823155e+20,2.0343652246835665e+20
5.256171875,1.373379807593803e+20,1.3234848940558387e+20,1.398460076632637e+20,1.0897671740612474e+20,1.1402730240405302e+20,1.1643067323929315e+20,1.1966438638564737e+20,1.2118057125827672e+20,1.2453115920210064e+20,1.2790352835818024e+20,1.3059483619445929e+20,1.3557901505746164e+20,1.3901513410307018e+20,1.4067422497806018e+20,1.4152794224866827e+20,1.4681918307956821e+20,1.973170643226602e+20,1.9823878147543243e+20,2.2156805354853892e+20
1.704140625,1.46317152699528e+20,1.381290106386182 6e+20,1.471103075750836e+20,1.140049899427179e+20,1.1924151211843889e+20,1.2129638355758758e+20,1.2557400114497782e+20,1.2738927924931366e+20,1.3043280522523995e+20,1.3389867421929534e+20,1.3922285249311855e+20,1.4078950602829164e+20,1.4317747064022894e+20,1.484490552551908e+20,1.5264007924263757e+20,1.586309751111174e+20,2.110955404461764e+20,2.1007129222107847e+20,2.3157891120089625e+20
−1.847890625,1.5065852017644713e+20,1.4299418970783331e+20,1.5279945396645986e+20,1.1654276679504814e+20,1.2103447776143963e+20,1.2492481476994168e+20,1.3037861781914037e+20,1.3296898833033198e+20,1.3543876530049802e+20,1.3804454203499245e+20,1.4237740952664105e+20,1.4566106008645 96e+20,1.5038811439502865e+20,1.5519326231827064e+20,1.5738094602727126e+20,1.63702810072222e+20,2.205597427958233e+20,2.1945155721617896e+20,2.4046245830675058e+20
−5.399921875,1.5099373834555333e+20,1.4481584280112207e+20,1.5194308045045478e+20,1.2071307206839802e+20,1.2392500400249654e+20,1.2914505745675588e+20,1.3198299003895143e+20,1.3232405194793109e+20,1.3729016834223361e+20,1.3984919515774014e+20,1.44888092675 9215e+20,1.4697005781812e+20,1.5162379975373 08e+20,1.558100424994629e+20,1.6071878399318937e+20,1.652636198675216e+20,2.2191230295199508e+20,2.2523579719276918e+20,2.456830430100894e+20
−8.951953125,1.4749599440673351e+20,1.4307387706974446e+20,1.4927992881538433e+20,1.1800476426157793e+20,1.231153804054793e+20,1.2620884379486993e+20,1.308046679580825 3e+20,1.3334564392763197e+20,1.3575538975182497e+20,1.406949436921571e+20,1.4360246990375503e+20,1.4713049504010109e+20,1.5022873967120636e+20,1.548362629369087e+20,1.604834406510118e+20,1.6683239839901234e+20,2.1975171294604427e+20,2.2024418084265512e+20,2.399811466408073e+20
−12.503984375,1.351301095853622e+20,1.3405592394679999e+20,1.3789472979459949e+20,1.0840243715128507e+20,1.1335155357504653e+20,1.1854238832993847e+20,1.216948203671433e+20,1.227047248670972e+20,1.2716615463596247e+20,1.3102939794141472e+20,1.3434014220428304e+20,1.3889666555836228e+20,1.4268447149453866e+20,1.4552612282029004e+20,1.5056236409307434e+20,1.542216077520341e+20,2.0603326796850112e+20,2.0712976606839846e+20,2.2542810935951475e+20

A.45 "MITR_Cycle5_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.710756865076151e+19,2.741441811902735e+19,2.887641558555714e+19,2.3255428451068867e+19,2.4077855150899814e+19,2.52
32206275544646e+19,2.58472864596828e+19,2.5572099436549644e+19,2.6267716981125997e+19,2.6644956952413348e+19,2.747009302254928e
+19,2.8253260415412003e+19,2.8090910696745026e+19,2.8246247927563817e+19,2.8463953800305066e+19,2.8050482641802105e+19,3.7811812
60156166e+19,3.770195029194017e+19,4.086474168619347e+19
37.224453125,3.68095049634433e+19,3.5184307779719483e+19,3.761625981543172e+19,3.0577847012175954e+19,3.1086464880800817e+19,3.20
26085127549088e+19,3.22597815975805e+19,3.4360393582466195e+19,3.484600836595271e+19,3.4718614836710756e+19,3.620260601512802e+
19,3.605199690111596e+19,3.6264177783431365e+19,3.720528552760198e+19,3.722765111384504e+19,3.841695842791489e+19,5.043131573344
22e+19,5.133752041309572e+19,5.517393564003791e+19
33.672421875,4.61363794260633e+19,4.4760763059845186e+19,4.733992421546127e+19,3.7582631748705214e+19,3.990780271945652e+19,4.032
4248872804155e+19,4.024865212880445e+19,4.213809260462562e+19,4.396883005717224e+19,4.3510415226551394e+19,4.489952531938646e+1
9,4.550297114868558e+19,4.780285466325708e+19,4.659628175410648e+19,4.672282528482138e+19,4.8198847727047524e+19,6.4309294809531
466e+19,6.384126437057334e+19,6.846685010497554e+19
30.120390625,5.7143807226481385e+19,5.513621695539996e+19,5.747583790111115e+19,4.643366641089981e+19,4.908523681603916e+19,4.986
240109430456e+19,5.146209832221681e+19,5.094444921924025e+19,5.278299603325593e+19,5.377090682132236e+19,5.4062031316837745e+1
9,5.668374552371438e+19,5.743227547659973e+19,5.7848243505775895e+19,5.903452270009314e+19,5.931980345573503e+19,7.8026146039836
4e+19,7.812814586308266e+19,8.487925916419477e+19
26.568359375,6.757063290801487e+19,6.446548203887729e+19,6.960797312754313e+19,5.524352926944031e+19,5.8156367971832324e+19,5.931
980345573503e+19,6.13449249464369e+19,6.368613963938632e+19,6.408351395078322e+19,6.560394881604785e+19,6.7907976073438716e+19,6.
708878999299215e+19,6.859806862758923e+19,6.864216230118007e+19,6.973759790291859e+19,7.094831455488856e+19,9.405764950912034e+
19,9.409961818639353e+19,1.0258579098085097e+20
23.016328125,7.846123903587122e+19,7.825405189490226e+19,8.09246410170844e+19,6.558163635471273e+19,6.775550758764873e+19,6.96605
6678640448e+19,7.294687359162007e+19,7.442958977224675e+19,7.60419307282489e+19,7.81589583096883e+19,7.74986157039846e+19,7.99312
0523859213e+19,8.252423199518076e+19,8.227985741865325e+19,8.494832154451776e+19,8.464604081833483e+19,1.0987824709387949e+20,1.1
419783335854288e+20,1.1968563634882367e+20
19.464296875000002,8.948625118081771e+19,9.027409356557923e+19,9.414902435077846e+19,7.464580814756564e+19,7.87651135092924e+19,
8.204345157831687e+19,8.297473121451842e+19,8.634178787980398e+19,8.671472473354812e+19,8.795891007752079e+19,9.11368420705372e+
19,9.304987000548408e+19,9.487258559693164e+19,9.718139409603718e+19,9.686583214286905e+19,1.002966386976835e+20,1.2941599448725
375e+20,1.2916683866901158e+20,1.414886298167921e+20
15.912265625,1.023222914374648e+20,9.892229732925597e+19,1.0558150453963058e+20,8.405051060031888e+19,8.803594119403489e+19,9.110
76233711698e+19,9.41033369299494e+19,9.663527004240614e+19,9.787467414466414e+19,9.966338979502963e+19,1.02643165881427e+20,1.056
39410689286e+20,1.077787507320605e+20,1.094272166254624e+20,1.1005780928271925e+20,1.1130411962300953e+20,1.4748111943251013e+2
0,1.4717618246093015e+20,1.617005322919346 5e+20
12.360234375000001,1.1267846099143706e+20,1.1103583883790867e+20,1.194561367465196e+20,9.39014622797745e+19,9.97728271053876e+19
,1.0206038564131683e+20,1.051214428368636e+20,1.072119079643326e+20,1.116143690853836e+20,1.1404323987643526e+20,1.1714785949649
345e+20,1.1897376258241746e+20,1.2112425885585952e+20,1.2337250495991259e+20,1.2430484709427298e+20,1.2627365618255766e+20,1.6428
240281785572e+20,1.657204940758122e+20,1.7998400060882754e+20
8.808203125,1.2401797259139287e+20,1.1962454270469179e+20,1.2789449712383032e+20,1.0315263374857891e+20,1.0768206339960835e+20,1.
108143079717957e+20,1.1434711434985644e+20,1.1748148391836141e+20,1.1995019839036868e+20,1.2220056949073938e+20,1.2520584553294
832e+20,1.2904783887522428e+20,1.3022189934071515e+20,1.3583135837018025e+20,1.3631532628152061e+20,1.3505626596332452e+20,1.8093
54677100466e+20,1.8418830582325953e+20,1.9674331531689994e+20
5.256171875,1.357846084511924e+20,1.3072977346062882e+20,1.4036078802120968e+20,1.1321821005611522e+20,1.1779438962613253e+20,1.2
223563192998029e+20,1.2399034763926367e+20,1.2643728089901521e+20,1.3072924221154943e+20,1.3366651837159417e+20,1.3682585664683
133e+20,1.4075338109089192e+20,1.418663479122509e+20,1.4525093579715689e+20,1.4673949571765708e+20,1.489999605505365e+20,1.957307
54571549e+20,1.9889912408113607e+20,2.146867842229 7195e+20
1.704140625,1.4134891130890787e+20,1.3837338521514577e+20,1.47251619830206e+20,1.1883457532361264e+20,1.2300806809143897e+20,1.26
519624506323 39e+20,1.302877742265617e+20,1.345579543268401 8e+20,1.393997584365613e+20,1.4175053561294003e+20,1.4844480526255555
e+20,1.4854202384408714e+20,1.507961136880137e+20,1.535878276003008e+20,1.5793078882445813e+20,1.587080062276315e+20,2.1158641459
554904e+20,2.1019188576210403e+20,2.2928604017417296e+20
−1.847890625,1.4058231888732268e+20,1.4203687886674076e+20,1.5092626971246857e+20,1.2191156999154162e+20,1.2793009081215063e+20,
1.3398951781187402e+20,1.3640457612686108e+20,1.4018919456856103e+20,1.4031722559669826e+20,1.4842355529937922e+20,1.50677645143
3058e+20,1.5247220453354478e+20,1.5780063280000326e+20,1.6146997019147174e+20,1.625994057342924e+20,1.6403643449409e+20,2.1492212
756514954e+20,2.1670074948300626e+20,2.3896752339729754e+20
−5.399921875,1.4618646542599373e+20,1.4712890129286285e+20,1.5656654118853935e+20,1.2400628511164588e+20,1.2990580613846758e+20,
1.3333289394972 62e+20,1.3731460579988636e+20,1.4322422055921679e+20,1.4526687326953913e+20,1.4760596296617093e+20,1.517268620751
359e+20,1.5715197767404657e+20,1.6026562852845466e+20,1.628953114715224e+20,1.6576936899111767e+20,1.666565549537284e+20,2.247555
480249847e+20,2.2284730133175252e+20,2.3936702270501208e+20
−8.951953125,1.4281462751899353e+20,1.4261009662342159e+20,1.5298379739701 43e+20,1.2210228841104897e+20,1.2839971499834697e+20,1.
3312198806520137e+20,1.3764504272727789e+20,1.415948796326736e+20,1.4537471683265888e+20,1.4867642986117726e+20,1.5294182871974
11e+20,1.5618297935320705e+20,1.5876006863741344e+20,1.63032904983089e+20,1.658256813935349e+20,1.7014686140543 65e+20,2.223447397
026329e+20,2.2471729809126733e+20,2.471466342238574e+20
−12.503984375,1.2837049629897954e+20,1.3082858578939863e+20,1.3875482205416043e+20,1.1330267865974102e+20,1.1877720042303663e+20
,1.2157316432795895e+20,1.2745674788239845e+20,1.2975918139255108e+20,1.3377489318379335e+20,1.3786976108786734e+20,1.4082828721
10884e+20,1.449523738145298e+20,1.493829911367894e+20,1.513815501735209e+20,1.5398360816445948e+20,1.5727735245678677e+20,2.08325
0764970656e+20,2.0551105012344344e+20,2.225248331405521e+20

A.46 "MITR_Cycle5_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.098271382515478e+19,2.5576323059394023e+19,2.6163760661557404e+19,2.775528059213166e+19,2.9459883384759714e+19,3.2
884010873911628e+19,3.119226249247093e+19,3.0919353454954775e+19,3.260849499776307e+19,3.386202481599849e+19,3.60200377486896e+
19,3.621024707786753e+19,3.5900572626535793e+19,3.4931008437257404e+19,3.6319644409441026e+19,3.55277947021025e+19,3.71798112671
45925e+19,3.277146735778178e+19,3.2950979659468124e+19
37.224453125,3.081615860150665e+19,3.0954051376061907e+19,3.4888849564202373e+19,3.807872114038752e+19,4.033777154283099e+19,4.12
37130871009214e+19,4.189612169908263e+19,4.3527822900990796e+19,4.668227742817022e+19,4.456795151531865e+19,4.738396647522261e+
19,4.7488060238543995e+19,4.797976393920715e+19,4.962926355660549e+19,4.990747616237376e+19,5.086625343317326e+19,4.914358255209
306e+19,4.859129232597341e+19,4.514882707540736e+19
33.672421875,3.741532565393522e+19,4.016275379051085e+19,4.544097278421089e+19,4.747700364710312e+19,4.9327319730183635e+19,5.014
2541094226444e+19,5.163455170183285e+19,5.498676640112545e+19,5.529823867220365e+19,5.756600850041758e+19,5.817160408202044e+19,
5.940526799205722e+19,6.193623863115627e+19,6.124030260729262e+19,6.0215455460809875e+19,6.145729945069314e+19,5.990569111849081
e+19,6.1519773686883434e+19,5.909100910037194e+19
30.120390625,4.513012975004395e+19,4.854643672329866e+19,5.148749004657054e+19,5.7873166004104225e+19,6.022830987199722e+19,6.275
9100729121645e+19,6.58169224449276e+19,6.856228308879478e+19,6.967396492903267e+19,7.153560727651602e+19,7.24418882107182e+19,7.3
85407562158075e+19,7.403331825030513e+19,7.474192890338148e+19,7.564524343500197e+19,7.834404054743525e+19,7.54678885170114e+19,
7.429822698994944e+19,7.0445858837185225e+19
26.568359375,5.386834273683824e+19,5.979377683927223 5e+19,6.513743647174427e+19,6.6406697212761416e+19,7.294509795775885e+19,7.81
0564964905157e+19,7.973195739172028e+19,8.449770786668528e+19,8.28072179590671e+19,8.513988908012908e+19,8.791320582105406e+19,8.
468558003019278e+19,8.674606124163718e+19,9.083843833055689e+19,9.213376745789804e+19,9.025504582282309e+19,8.979309603896423e+
19,8.960863973297508e+19,8.621058063014715e+19
23.016328125,6.429668606730139e+19,7.0239559021276275e+19,7.443665911042864e+19,8.214481127347539e+19,8.675585935925551e+19,8.922
606369092343e+19,9.362685675735235e+19,9.653752692690865e+19,9.895559448592856e+19,9.948415349139464e+19,1.0261146094040216e+20,
1.0503761868828123e+20,1.0868449604402243e+20,1.076965940933299e+20,1.1094255764560532e+20,1.0796536814542915e+20,1.064857624940
7345e+20,1.050735750832109e+20,1.0370902989563016e+20
19.464296875000002,7.558834244002578e+19,7.946147541086187e+19,8.638353088975883e+19,9.325740479945004e+19,9.868322479433635e+19
,1.03031251851206e+20,1.04527037880 28e+20,1.0937575773654522e+20,1.1120503932859189e+20,1.1573554508972956e+20,1.169940189122678
1e+20,1.2059055731510744e+20,1.2306435728626834e+20,1.2287648512276085e+20,1.2314975372422627e+20,1.2606222171352908e+20,1.241457
45863778e+20,1.2210072590215333e+20,1.2043324808729017e+20
15.912265625,8.19633213700679e+19,8.65716727262283e+19,9.703192735719152e+19,1.0563269702436717e+20,1.1007241288830745e+20,1.1651
849558932301e+20,1.1868486838383526e+20,1.2205218476899831e+20,1.285710791697464e+20,1.2716518412799653e+20,1.3378565534442095e
+20,1.3599607472272204e+20,1.40843895669114e+20,1.3868651197333416e+20,1.4451594250130594e+20,1.4312263219778145e+20,1.393939540
4357529e+20,1.4184348344815865e+20,1.3452635707997202e+20
12.360234375000001,9.483768835660879e+19,1.0107792069665199e+20,1.0819369125323252e+20,1.19621532471753e+20,1.2469318197658213e+
20,1.3269887330767186e+20,1.2987180175632699e+20,1.3523829369957935e+20,1.420214676030605e+20,1.4692232423197372e+20,1.506501034
7630664e+20,1.565118947597151e+20,1.5494779158027475e+20,1.5974886921325812e+20,1.6069991585914775e+20,1.607700308292606e+20,1.59
25896333234147e+20,1.562763803729258e+20,1.5147620164981568e+20
8.808203125,1.0124961248244113e+20,1.0975599661292642e+20,1.177886552402134e+20,1.2961920808194612e+20,1.3125522405124583e+20,1.4
061557256131061e+20,1.5043885965609484e+20,1.524793850683533e+20,1.568768521682512e+20,1.5844005643781836e+20,1.643899408888045
2e+20,1.6945709584426746e+20,1.682354773265321e+20,1.7090254292043992e+20,1.7551035493067637e+20,1.7515528553074596e+20,1.7290980
866738844e+20,1.7242799297533093e+20,1.6845930588496927e+20
5.256171875,1.1264868858501787e+20,1.1871902695901841e+20,1.300794499370458e+20,1.391629342061522e+20,1.4493033995287033e+20,1.55
33881737513484e+20,1.614675848908961e+20,1.6484748601428453e+20,1.699290235277193e+20,1.7204505736933002e+20,1.8124809665157752
e+20,1.8354750810732957e+20,1.8674223379683023e+20,1.8895085535538482e+20,1.9030910817385287e+20,1.9039720134143056e+20,1.8799801
088974874e+20,1.881957710618619e+20,1.823708350832563e+20
1.704140625,1.1368962621823168e+20,1.2641369547396655e+20,1.3230874642268496e+20,1.4907071882902113e+20,1.5731372236664665e+20,1.
5997269777169 53e+20,1.6696801440526146e+20,1.7471661751260404e+20,1.7767223317582243e+20,1.8081122645318212e+20,1.8772743901790
3e+20,1.9530704706907616e+20,1.936063095889031e+20,2.0137648653320353e+20,2.0047128429084924e+20,2.0418288315746376e+20,1.9779253
28685892e+20,1.9903302849366265e+20,1.9156578417664465e+20
−1.847890625,1.1771404572073432e+20,1.3231953334116 39e+20,1.3850942672830557e+20,1.5299266260597427e+20,1.5828993848898703e+20,1.
7048634764912905e+20,1.7601284554981845e+20,1.781702292455983e+20,1.8660649840597074e+20,1.9115498236457324e+20,1.9034596347865
58e+20,1.9825277472368896e+20,2.0059443494348 33e+20,2.0515460473043796e+20,2.0924644247343373e+20,2.0859203608571383e+20,2.027311
4371217857e+20,2.0041015841946878e+20,1.974221820008137e+20
−5.399921875,1.198786206955001e+20,1.3191682171795164e+20,1.4228215146630057e+20,1.5700269955050504e+20,1.6285190609568812e+20,1.
6600438302114647e+20,1.7653960673553803e+20,1.8244634351260857e+20,1.9271908554401363e+20,1.8960616065297796e+20,1.980541156417
0258e+20,2.019059444485428e+20,2.0034004344935593e+20,2.0776593791220482e+20,2.099179281487452e+20,2.130137737521893e+20,2.102154
6731678817e+20,2.128627568934847e+20,2.102792899177883e+20
−8.951953125,1.0913305207076998e+20,1.2780700577749103e+20,1.3648508169376547e+20,1.489916147601759e+20,1.6085992181658475e+20,1.
6955417811057754e+20,1.732675747969386e+20,1.787922748778815e+20,1.835996448799 7755e+20,1.863871643968998e+20,1.934651807388041
5e+20,2.0370286528515277e+20,2.0497032820642343e+20,2.0595013996825677e+20,2.0829269909792437e+20,2.095260034440119e+20,2.1164563
29251156e+20,2.0795470898558553e+20,2.0142502766635853e+20
−12.503984375,1.0031834185376283e+20,1.1444291269200812e+20,1.2326930873736744e+20,1.341955582466191e+20,1.4354242310858529e+20,
1.4966040370586763e+20,1.5706652215150517e+20,1.6335080107533723e+20,1.689096597314633e+20,1.7399029833502487e+20,1.744496412802
513e+20,1.8290568645783506e+20,1.8529498890091125e+20,1.8873871262529982e+20,1.8751170064832505e+20,1.9119453439899586e+20,1.8969
515273042887e+20,1.916808446404196e+20,1.8445630598917682e+20

A.47 "MITR_Cycle5_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.719373725144143e+19,2.7779970610567746e+19,2.9676901698408554e+19,2.3743380730504774e+19,2.4223417398657503e+19,2.
5604399380577632e+19,2.58961613749883e+19,2.5947320661335253e+19,2.5830977112944984e+19,2.6931247081306116e+19,2.76788207858485
33e+19,2.8581466096670028e+19,2.9593442468033618e+19,2.867937530200485e+19,2.8554638018159948e+19,2.810397942409845e+19,3.7860900
01649893e+19,3.708527636056379e+19,4.053324226064312e+19
37.224453125,3.721569800955837e+19,3.678963624787345e+19,3.823080875049046e+19,3.0545653317963854e+19,3.1254445839709503e+19,3.22
4432224936974e+19,3.387026318180472e+19,3.431789365611359e+19,3.5253795159306007e+19,3.525358265967424e+19,3.710976694312449e+1
9,3.6958732829848895e+19,3.6286437119858545e+19,3.708517011074791e+19,3.740657580378953e+19,3.922440390370655e+19,5.106228026505
463e+19,5.118967379429659e+19,5.24180279157029e+19
33.672421875,4.759853626731691e+19,4.556225854594748e+19,4.765888616273762e+19,3.848846455400316e+19,3.952705650424508e+19,4.0277
55207872423e+19,4.167925277474125e+19,4.332628429562872e+19,4.457631337947486e+19,4.4018236221557146e+19,4.4729419364160135e+19,
4.551168363358787e+19,4.738863975604295e+19,4.727033058605888e+19,4.774101727041404e+19,4.780662653172088e+19,6.266189141428841e
+19,6.206529869811365e+19,6.835528779829994e+19
30.120390625,5.879864810883614e+19,5.5912903109493916e+19,5.938249084710513e+19,4.720318070242175e+19,4.886370594992618e+19,4.915
536169452096e+19,5.1027536575261385e+19,5.335387629398737e+19,5.290438644790058e+19,5.476859259244989e+19,5.664018309920295e+19,
5.664602683907644e+19,5.654137077043313e+19,5.7720212477638664e+19,5.966883410090584e+19,6.073026976156228e+19,7.887880081228562
e+19,7.825192689858462e+19,8.352191776630828e+19
26.568359375,6.9447004656482615e+19,6.741975816946311e+19,7.0463815394468815e+19,5.6473370888268956e+19,5.7641587613886325e+19,6.
0709019798385975e+19,6.199358007239361e+19,6.21598610342482e+19,6.420623248812639e+19,6.673338435886846e+19,6.747500807372151e+
19,6.8527943749107425e+19,6.913728644318797e+19,7.012009724009207e+19,7.130797018164753e+19,7.224934355035785e+19,9.354340040025
376e+19,9.267055816278702e+19,1.0067382554406291e+20
23.016328125,8.100485962807494e+19,7.633252397468487e+19,8.33566993026125e+19,6.60671980132913e+19,6.881322450474932e+19,7.029222
194182015e+19,7.289056118920285e+19,7.431112122753886e+19,7.57295562695572e+19,7.748692822423765e+19,7.898983186988181e+19,8.1339
01529902234e+19,8.263420055461814e+19,8.419713534623538e+19,8.438519752034568e+19,8.564425783854175e+19,1.112600259494187e+20,1.1
243514891306838e+20,1.2097391536638719e+20
19.464296875000002,9.309449492815431e+19,9.188218452894612e+19,9.524074120896112e+19,7.621299293181816e+19,8.065476648474532e+19
,8.256354442705692e+19,8.379710478944143e+19,8.581372629487278e+19,8.679813083901513e+19,8.863997139732136e+19,9.119156072571619e
+19,9.507286649986831e+19,9.575658406506594e+19,9.664483252583549e+19,9.780614301342055e+19,1.0011229526712907e+20,1.29957868548
24952e+20,1.2848312110381397e+20,1.4004841356251801e+20
15.912265625,1.04190163200662e+20,9.74533936246939e+19,1.0696222089701099e+20,8.403085438438079e+19,8.93869076029685e+19,9.137962
289982649e+19,9.36363689891501e+19,9.864870405336107e+19,9.857751667672043e+19,1.0141332426259834e+20,1.044058503269015e+20,1.048
3456833398347e+20,1.0730062656059366e+20,1.0881946767862009e+20,1.102639339255294e+20,1.1228427417451661e+20,1.484559614932231e
+20,1.483518366736592e+20,1.6052594057736438e+20
12.360234375000001,1.1594404908255576e+20,1.1280117952878025e+20,1.1805470167504224e+20,9.682758220915171e+19,9.965754605515614e
+19,1.0468263109727288e+20,1.0483084959042762e+20,1.0892412374726337e+20,1.1172752513929744e+20,1.150706755960096e+20,1.15816549
30349793e+20,1.1910604360318994e+20,1.2026097910182212e+20,1.2341181739178877e+20,1.2478934625469275e+20,1.2550122002109895e+20,
1.641904967271182e+20,1.6476583948011667e+20,1.765802877570629e+20
8.808203125,1.2409394120974816e+20,1.2134791471829013e+20,1.2837846503517066e+20,1.0468688108990815e+20,1.097613722964098e+20,1.1
241124270449503e+20,1.1438748927989139e+20,1.1855035706612957e+20,1.2171872657571666e+20,1.2442012814450444e+20,1.2556709490694
55e+20,1.288672141882257e+20,1.3156542826253702e+20,1.3275755119672774e+20,1.360783891921048e+20,1.3780972994189425e+20,1.8114371
734917438e+20,1.813535607355404e+20,1.957472232930106e+20
5.256171875,1.3950228950888697e+20,1.346976728347244e+20,1.4417621890951527e+20,1.1418561462971651e+20,1.1920591843011858e+20,1.2
212991336317816e+20,1.2586884438404902e+20,1.2772396616934048e+20,1.3106711662605268e+20,1.3503767224554527e+20,1.3843979135007
172e+20,1.4055363143703467e+20,1.429660335066247e+20,1.4408218782246013e+20,1.4507987359358765e+20,1.4880817963287036e+20,1.96160
53507678975e+20,1.9871849939413752e+20,2.1217025733381803e+20
1.704140625,1.4474040543184617e+20,1.4360725114546969e+20,1.4968899060652822e+20,1.1790807692912574e+20,1.2273659981186168e+20,1.
2879071432079098e+20,1.3330845649207345e+20,1.3772101134563332e+20,1.4024710071821646e+20,1.4296390851030707e+20,1.4716555747934
2e+20,1.476181816949973e+20,1.5184692436708196e+20,1.5597047972144398e+20,1.5552741798921804e+20,1.6099875225803715e+20,2.1077466
60022142e+20,2.1121135274548724e+20,2.283977917134034e+20
−1.847890625,1.4793002490460956e+20,1.4297559599005401e+20,1.5410845169812026e+20,1.2354941090335534e+20,1.2806396558016135e+20,
1.3401926776032087e+20,1.3737941818757408e+20,1.397211641296029e+20,1.4324600177147252e+20,1.4916305401791463e+20,1.507583950033
7578e+20,1.5450210726596133e+20,1.5798816372503416e+20,1.590772243378198e+20,1.627895929047203e+20,1.630716861658857e+20,2.192151
5137584254e+20,2.126903501825581e+20,2.3285390899147457e+20
−5.399921875,1.5253436067583548e+20,1.500688336983046 6e+20,1.5747922710696165e+20,1.2629331239849573e+20,1.3108942908738778e+20,
1.3482251636838518e+20,1.3998306992575087e+20,1.4543952922034663e+20,1.4570887250360628e+20,1.4865624239615977e+20,1.55328730833
51958e+20,1.567843533110965e+20,1.6183600080718358e+20,1.6259781198705417e+20,1.647318395390346e+20,1.6796820893078585e+20,2.2209
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1.3481826637574993e+20,1.4024763196729585e+20,1.4240025323705555e+20,1.4583743478082291e+20,1.484389615226821e+20,1.539682019411
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152135677293e+20,2.230125197954483e+20,2.407233016047397e+20
−12.503984375,1.3362986218511506e+20,1.316185531704778e+20,1.4262337785040678e+20,1.1631964218169696e+20,1.1949544917839575e+20,
1.2430803458874943e+20,1.274609978750337e+20,1.322879270105314e+20,1.372614808919456e+20,1.4039850670584763e+20,1.42015097654485
06e+20,1.4570409126189161e+20,1.5053898913358 04e+20,1.5347360904822817e+20,1.5559435537322338e+20,1.554535743671804e+20,2.0858007
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A.48 "MITR_Cycle5_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9198140028046406e+19,2.810865441599724e+19,2.8238916690267992e+19,2.2212692758007575e+19,2.303421633440353e+19,2.3
876246125264617e+19,2.4248917354469073e+19,2.5419205951496135e+19,2.4904691218089845e+19,2.5673249261268886e+19,2.62532670061661
1e+19,2.614483906905901e+19,2.7198784117695795e+19,2.686526594564369e+19,2.8319825925061775e+19,2.7872089200937026e+19,3.75343412
0738706e+19,3.745417572130445e+19,3.9991049450199695e+19
37.224453125,3.9375331767166255e+19,3.7433669506839314e+19,3.823713061453541e+19,3.0147482132947837e+19,3.0039213570564555e+19,3.
128536453613103e+19,3.2413578206069006e+19,3.2651684043459502e+19,3.3402976491557765e+19,3.3057558340126933e+19,3.4141572086658
195e+19,3.526080764715418e+19,3.5676350677706683e+19,3.5727138089058198e+19,3.725447919235512e+19,3.855688943543086e+19,5.19697068
1759081e+19,4.980072307618534e+19,5.416828113271928e+19
33.672421875,4.9152492949492154e+19,4.6475263213817 42e+19,4.740893347087633e+19,3.936582240864485e+19,3.922318203082391e+19,4.097
263837422116e+19,4.138451578548589e+19,4.144768130102746e+19,4.148120311793808e+19,4.24737888979033e+19,4.379978660010473e+19,4.4
230363978964615e+19,4.5273790295829135e+19,4.575547383612803e+19,4.700778729101563e+19,4.813568221150596e+19,6.381257692028532e
+19,6.360485853023694e+19,6.813535067942518e+19
30.120390625,6.097464433808978e+19,5.756774399184868e+19,5.895271034186435e+19,4.630239476337818e+19,4.729630866604191e+19,4.7740
0610220711e+19,4.963922335604543e+19,5.039683766818864e+19,5.1444248353148715e+19,5.2588611995100684e+19,5.357965715273561e+19,5.
384793793783647e+19,5.576255962002156e+19,5.556174746800548e+19,5.870089827822514e+19,5.9801115121678344e+19,7.798789610611904e
+19,7.993970522386267e+19,8.589606990218097e+19
26.568359375,6.967331676431027e+19,6.577607351777592e+19,7.077831484947812e+19,5.45459992281781e+19,5.808730559150934e+19,5.81542
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9.618530207214789e+19,1.0097238752669e+20
23.016328125,8.561132039561848e+19,7.816745829495882e+19,8.198767042497905e+19,6.650707225104081e+19,6.667760320553066e+19,6.8542
28747425143e+19,7.044256543129251e+19,7.183656301565813e+19,7.173987568320594e+19,7.400405925964123e+19,7.66061172505798e+19,7.93
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1150811926950206e+20,1.2223935067353612e+20
19.464296875000002,9.723292525673973e+19,9.133606047531506e+19,9.6414801674452e+19,7.476852668490881e+19,7.682552312037515e+19,7.
849577022603272e+19,8.176242081531023e+19,8.09597034563253e+19,8.530160218232383e+19,8.618135065782287e+19,8.781122283344547e+1
9,8.901662699462137e+19,9.078834267444581e+19,9.396946216193868e+19,9.49384604827782e+19,9.64928952891249e+19,1.307563359145992e+
20,1.2969543150302216e+20,1.3990391381291914e+20
15.912265625,1.0782284440565134e+20,1.0140588677548663e+20,1.060978786448148e+20,8.426247898300252e+19,8.643210022330327e+19,8.67
5988090529777e+19,8.889072096280176e+19,9.207927793740633e+19,9.491614802144307e+19,9.829861091003143e+19,9.836607954311619e+19,
1.0285088427147538e+20,1.0309791509339993e+20,1.0593531642651607e+20,1.0770543835910228e+20,1.1132430708802703e+20,1.49592834523
15543e+20,1.4666990208825467e+20,1.585597877344768e+20
12.360234375000001,1.1996613586275089e+20,1.1233739908245738e+20,1.1718345318481374e+20,9.372508758541117e+19,9.577252153744815e
+19,9.898073472799084e+19,1.0024457628790156e+20,1.0401697600077513e+20,1.0656165909113768e+20,1.0806615648402006e+20,1.11698837
68900942e+20,1.1321449131255936e+20,1.1569808075879003e+20,1.1957779278570393e+20,1.224109441261848e+20,1.2406153501590431e+20,1.
6846864556358782e+20,1.67406678653852e+20,1.7758806726069915e+20
8.808203125,1.3142783475097046e+20,1.2725699822854118e+20,1.3111280404688173e+20,1.0404141345842787e+20,1.0644903428630325e+20,1.
0724059541462062e+20,1.096992161541191e+20,1.140124274298296e+20,1.1547708114175646e+20,1.1911773108293693e+20,1.21163040038656
28e+20,1.230734117282061e+20,1.2544756386407878e+20,1.3190277142796085e+20,1.31870365234117e+20,1.3657138833779509e+20,1.82169028
07243112e+20,1.830652452693918e+20,1.9963331030887743e+20
5.256171875,1.4015041338576429e+20,1.3762432401318101e+20,1.4260903412526277e+20,1.1030484010464379e+20,1.130561790868959e+20,1.1
638073582582882e+20,1.1914323103874849e+20,1.2411625367108326e+20,1.2691274882508502e+20,1.2914877620031172e+20,1.3177420915074
422e+20,1.3542760906983047e+20,1.372614808919456e+20,1.3923135247838908e+20,1.4379637581773881e+20,1.4715971373946849e+20,1.97018
50234003312e+20,1.9581894191873068e+20,2.1368803595368563e+20
1.704140625,1.4735043215897584e+20,1.3997722618587739e+20,1.4657321475580248e+20,1.1595573656230271e+20,1.179394206248108e+20,1.2
267338117141217e+20,1.275507789694536e+20,1.300317121702872e+20,1.3331005023931166e+20,1.3401767401308263e+20,1.411380054243830
5e+20,1.436837510129044e+20,1.4783918131203083e+20,1.4977664670463046e+20,1.547873880216032e+20,1.5890828713056816e+20,2.08891919
2647936e+20,2.1056429136676877e+20,2.2834679180178027e+20
−1.847890625,1.5408507673862636e+20,1.4477440537292828e+20,1.5148833123848185e+20,1.1863641941699361e+20,1.2225369439868015e+20,
1.277409661398815e+20,1.3175986542560022e+20,1.3587173330021525e+20,1.3672279432542626e+20,1.4166181701667897e+20,1.436555948116
958e+20,1.4696527657640532e+20,1.502733645938766e+20,1.5572398014859887e+20,1.5943634871549936e+20,1.6331606074241325e+20,2.18612
7149197943e+20,2.1760281041984042e+20,2.3609346587770228e+20
−5.399921875,1.552809184163729e+20,1.4669487079498682e+20,1.5093848844129495e+20,1.2111416512335079e+20,1.2646703084746201e+20,1.
2907812007275051e+20,1.3153142832145493e+20,1.3459939175503397e+20,1.3841960388505423e+20,1.4144612989043948e+20,1.453242481701
1515e+20,1.510659882203528e+20,1.5342632788016087e+20,1.5919144288989246e+20,1.620368129591997e+20,1.675522409016097e+20,2.221508
337886491e+20,2.2700645037443485e+20,2.432047660546527 7e+20
−8.951953125,1.5303001606692277e+20,1.4490668639370075e+20,1.5277554775788654e+20,1.188866377333946e+20,1.2313344287417916e+20,1.
274673728639866e+20,1.306649610729411e+20,1.344548920054351e+20,1.3755101164022278e+20,1.4042241291442097e+20,1.446777180404760
6e+20,1.4848411769443172e+20,1.5237232970661613e+20,1.5554813670331492e+20,1.6206178166593182e+20,1.6529390106504787e+20,2.206054
3021665232e+20,2.2043436801308308e+20,2.391603668131225e+20
−12.503984375,1.3921807125140388e+20,1.3465092291573652e+20,1.3719932474965492e+20,1.1042649614382814e+20,1.1585267424089765e+20
,1.1838460735335437e+20,1.2153385189608283e+20,1.248084712215514e+20,1.2810593425743456e+20,1.3129555373019793e+20,1.35453640274
72144e+20,1.39921976281619e+20,1.4263240908475 67e+20,1.4949508469254445e+20,1.5313201589016907e+20,1.5521823102500277e+20,2.05987
04929859266e+20,2.0666970436563144e+20,2.213077414996292e+20

A.49 "MITR_Cycle5_cladding_fluence_position_8.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,2.9145333869553287e+19,2.774219880102186e+19,2.9132849516187206e+19,2.157689385977253e+19,2.2162117845647974e+19,2.3
206287791122366e+19,2.343329064285454e+19,2.407806765053158e+19,2.417347998519318 5e+19,2.419106432972158e+19,2.5090575270974575e
+19,2.5921661330799866e+19,2.6792431696856908e+19,2.619153586313894e+19,2.674382240609111e+19,2.7678395786585006e+19,3.555012589
5799575e+19,3.6404215040763216e+19,3.849457391841635e+19
37.224453125,3.829158364517469e+19,3.701488585754228e+19,3.746708507393406e+19,2.9020649710616318e+19,3.092682453243882e+19,2.994
953872596055e+19,3.0692756188051825e+19,3.133779882026856e+19,3.2357690802915324e+19,3.201243202620831e+19,3.3082367672135266e+
19,3.3867660061315625e+19,3.5199714003022303e+19,3.62317715895875e+19,3.6440286853254996e+19,3.7951477984537936e+19,4.8553775236
99976e+19,4.846356914331635e+19,5.0947955463166116e+19
33.672421875,4.769341735289911e+19,4.602864211275943e+19,4.802119803489362e+19,3.7644362891732386e+19,3.751261312003929e+19,3.930
212564402388e+19,3.881959210519794e+19,4.059013903704768e+19,4.071025445390174e+19,4.273989156177858e+19,4.353713705524 56e+19,4.2
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113880030362674e+19,6.731085210818455e+19
30.120390625,6.0056114679794e+19,5.6438308449028055e+19,5.846555493604757e+19,4.495705959468631e+19,4.519134043870507e+19,4.77403
26646610805e+19,4.731325551167501e+19,4.811517599704083e+19,4.976236689265212e+19,5.120449564361205e+19,5.093786173065738e+19,5.2
388437341979886e+19,5.31472204020978e+19,5.603774664315471e+19,5.671402672124061e+19,5.7470525410317074e+19,7.521530716069062e+
19,7.842192660399507e+19,8.244241963695199e+19
26.568359375,7.119853287128956e+19,6.650016601300852e+19,6.952350452391731e+19,5.377250056856058e+19,5.6032965401440035e+19,5.691
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1218482353567e+19,9.705814430961461e+19
23.016328125,8.21157014531163e+19,7.69338979325743e+19,7.91901127728185e+19,6.214498606002478e+19,6.522251197703317e+19,6.5686292
42335603e+19,6.7377789492189905e+19,6.976097286241253e+19,7.25590617636525e+19,7.315565447982726e+19,7.45454020715576e+19,7.74911
782168729e+19,7.788430253563455e+19,7.808830218212708e+19,7.988658031592189e+19,8.160676483504379e+19,1.0888640506262543e+20,1.09
61209130509625e+20,1.1614326748733361e+20
19.464296875000002,9.333143201757012e+19,9.0718217795964e+19,9.14991539426932e+19,7.1457251172961075e+19,7.289215493644109e+19,7.
631021151334973e+19,7.667677337814101e+19,7.946051855423696e+19,8.031529832300385e+19,8.41918228554413e+19,8.548063312208421e+1
9,8.914040803012334e+19,8.959887598565214e+19,9.231727752498094e+19,9.338880691814615e+19,9.32671508789618e+19,1.237619105351183
9e+20,1.2493278350613281e+20,1.3559548377892328e+20
15.912265625,1.0688093978786164e+20,1.006440755956161e+20,1.052064426895688e+20,7.927245638012667e+19,8.148776504125648e+19,8.447
763486016261e+19,8.939965758087427e+19,8.894756461429839e+19,9.446139880947014e+19,9.334630699179352e+19,9.529811610953715e+19,9.
843461067435978e+19,9.975476463668773e+19,1.0316485247740528e+20,1.0482075585791885e+20,1.0594647265718362e+20,1.44102375287477
6e+20,1.4322793930277267e+20,1.5175714327266 21e+20
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19,9.540489717449808e+19,9.855361046814709e+19,9.977601459986403e+19,1.032493210810311e+20,1.0544869226977867e+20,1.064293780703
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426506932085847e+20,1.0763796972601752e+20,1.0986443461781488e+20,1.1104540132133803e+20,1.1299083545012876e+20,1.1766317110351
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647763771474e+20,1.759879450335234e+20,1.909999815194241e+20
5.256171875,1.3923825871642137e+20,1.31577115742284e+20,1.3818479179195605e+20,1.0810918765945207e+20,1.0962590378116083e+20,1.12
41336770081266e+20,1.148263010194821e+20,1.1830651373868147e+20,1.2012232309209673e+20,1.2275891227319681e+20,1.256212823130450
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8652018024e+20,1.9333747746881765e+20,2.065576108098 7645e+20
1.704140625,1.4625074656460205e+20,1.3793138598107862e+20,1.4432231240635236e+20,1.1253449249091761e+20,1.1605189264567548e+20,1.
1936688690117909e+20,1.2053297863047884e+20,1.2603565659498304e+20,1.2870677696624456e+20,1.3060386742880923e+20,1.323426456657
1039e+20,1.3861669729351444e+20,1.4136484878129011e+20,1.4537418558357946e+20,1.4756877553061238e+20,1.5043539556309592e+20,2.030
439293986744e+20,2.039603340606526e+20,2.1974161921353554e+20
−1.847890625,1.5066755141079705e+20,1.4131384886966696e+20,1.4854786758396065e+20,1.1631432969090286e+20,1.1913419980439855e+20,
1.2165391418802892e+20,1.259023130760517e+20,1.2970021274473685e+20,1.314113660295088e+20,1.347040478236773e+20,1.37722605092871
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1.2206510097549042e+20,1.2438081571262828e+20,1.2841405872349099e+20,1.3028139923760878e+20,1.3496701611798407e+20,1.37631761500
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16e+20,1.3334564392763197e+20,1.3901354035583194e+20,1.414126611984368e+20,1.4508677983161993e+20,1.501049586357044e+20,2.0041212
145928903e+20,2.0027399669864307e+20,2.1553359525554776e+20

A.50 "MITR_Cycle5_cladding_fluence_position_9.csv" File

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A.51 "MITR_Cycle6_cladding_fluence_position_0.csv" File

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205e+20,3.099174064073342e+20,3.118445122086502e+20,3.150602465397369e+20,3.1888392617720336e+20,3.115840925057696e+20,3.09904834
42167795e+20,3.059015549891215e+20,2.9025122884497732e+20

A.52 "MITR_Cycle6_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.276379303405037e+19,4.997023781158596e+19,5.196666549422334e+19,4.142791765109597e+19,4.079764556967215e+19,4.2674
13262381358e+19,4.347014567107495e+19,4.495937309418323e+19,4.681006760609057e+19,4.607524550779197e+19,4.611542031123065e+19,4.7
67210107009927e+19,4.854071534735287e+19,4.905428413582865e+19,4.7911716971718386e+19,4.960542724508988e+19,6.426641773325887e+
19,6.459068199879436e+19,6.820959857274947e+19
37.224453125,6.966003104035751e+19,6.633035182273113e+19,6.899027407946422e+19,5.462711846170198e+19,5.602607198699097e+19,5.6553
598468074906e+19,5.721380263554817e+19,5.850183761501569e+19,6.029776277981534e+19,6.200558995718486e+19,6.214888185848531e+19,
6.2621745132776866e+19,6.293380305116453e+19,6.376542678945279e+19,6.619395916112291e+19,6.584315935830955e+19,8.716234071809106
e+19,8.61407225328933e+19,9.29423114479554e+19
33.672421875,9.076586668042499e+19,8.457247227977155e+19,8.69272358578092e+19,6.937132439477434e+19,6.928428783250296e+19,7.07097
7689506981e+19,7.232791395827395e+19,7.4941133669398e+19,7.55089941671443e+19,7.815193368001955e+19,7.81153146385761e+19,8.014475
252958714e+19,8.068607749005556e+19,8.059214168809193e+19,8.139988344283011e+19,8.28933034808283e+19,1.0693290812233643e+20,1.089
3315692234182e+20,1.175747199925102e+20
30.120390625,1.0753048842183392e+20,1.0153664126151e+20,1.0526382282755878e+20,8.182073706405703e+19,8.590243340813808e+19,8.7431
4110660886e+19,8.726636002422029e+19,9.138520612049034e+19,9.357332652442301e+19,9.514794530649147e+19,9.629746478136852e+19,9.70
3143774247423e+19,9.863524561554833e+19,9.962449044526712e+19,9.969560568517178e+19,1.011762886652766e+20,1.308663706147331e+20,
1.3224462642094534e+20,1.4261842936435352e+20
26.568359375,1.2816452220910097e+20,1.2276772463752901e+20,1.2548761720851005e+20,1.0009045447986641e+20,1.0245158658685075e+20,
1.0487056616510084e+20,1.0624351486385752e+20,1.0683578805589944e+20,1.089060906743184e+20,1.1293896162980816e+20,1.144435265934
6306e+20,1.1564611714289585e+20,1.1891104964956425e+20,1.1925866518790427e+20,1.1970180866044458e+20,1.2138363101311561e+20,1.615
154468814116e+20,1.589797109391389e+20,1.7317516196130505e+20
23.016328125,1.5191754304800817e+20,1.4088512806936345e+20,1.4682749628736831e+20,1.1517856097605988e+20,1.2043896588602365e+20,
1.2354362374753373e+20,1.2449041171760792e+20,1.2721454997455346e+20,1.3119328843399638e+20,1.3099745616888573e+20,1.33865947748
62284e+20,1.3672170227046659e+20,1.3775764964579 44e+20,1.3858927338408262e+20,1.4133782433532379e+20,1.4106981540881737e+20,1.894
0748082492e+20,1.8590850487946088e+20,2.0254840959566384e+20
19.464296875000002,1.7127309464922774e+20,1.631861243084259e+20,1.710119849624136e+20,1.3407982417908244e+20,1.3511895581888358e
+20,1.4038572925779404e+20,1.4170029977453943e+20,1.4760976392631973e+20,1.4923161596474278e+20,1.5188357756029252e+20,1.5348313
974740215e+20,1.5664086868346798e+20,1.579565006217045e+20,1.5992862175219532e+20,1.6377096755003005e+20,1.6388347822808816e+20,
2.1481738061145706e+20,2.1821392938302367e+20,2.3049722958894688e+20
15.912265625,1.9247339680200445e+20,1.822195344870922e+20,1.8966222198278753e+20,1.4627874137646208e+20,1.556887736059382e+20,1.5
900943074089012e+20,1.6012445401730597e+20,1.6456119585016483e+20,1.6857336908657784e+20,1.6939756287442833e+20,1.7304513782864
354e+20,1.7510641836438798e+20,1.8061572661401813e+20,1.8127646149223694e+20,1.833642775652593e+20,1.8534648219991572e+20,2.44411
9346267132e+20,2.4371139644257767e+20,2.6194980192439914e+20
12.360234375000001,2.1711005103226447e+20,2.0332643154865078e+20,2.1077283401956498e+20,1.6483291975189014e+20,1.705911313411866
6e+20,1.7504804018237 67e+20,1.7874815550040205e+20,1.8115015233479433e+20,1.871853949332717e+20,1.8884068174866484e+20,1.92187874
4208946e+20,1.9545970616725524e+20,2.016891888986066e+20,2.0202937448650883e+20,2.0304515485350543e+20,2.0631645588912053e+20,2.7
552432137426325e+20,2.6943016988302898e+20,2.942164845435362e+20
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9045563454739566e+20,1.9423482576650905e+20,1.9751939457076304e+20,2.0195136000691192e+20,2.104262799027161e+20,2.1149778489799
626e+20,2.148985793555273e+20,2.1947755166819552e+20,2.2083776330905882e+20,2.242879138659267e+20,2.264049190299547e+20,2.9500565
14221799e+20,2.9432528024637548e+20,3.156603829285237e+20
5.256171875,2.5301104083697785e+20,2.3673998008893722e+20,2.4572703585420417e+20,1.9506432666181507e+20,2.0129911650062202e+20,2.
0570401568874742e+20,2.093340771883592e+20,2.1448090999877373e+20,2.1966542327212284e+20,2.2829212644115638e+20,2.3021701431529
262e+20,2.339001468894602e+20,2.37294042107299e+20,2.3836448568108797e+20,2.4387273250922706e+20,2.433377760777053e+20,3.24435685
10631327e+20,3.1582808752411974e+20,3.441515893037663e+20
1.704140625,2.6790703004327962e+20,2.5084096459839413e+20,2.6205700549500168e+20,2.0458209317263933e+20,2.127815741914993e+20,2.1
72528122228194e+20,2.2130691160813145e+20,2.265784614437 5194e+20,2.3163082774145725e+20,2.3778017315022946e+20,2.42084768007444
56e+20,2.4680066369246655e+20,2.5304978272140352e+20,2.5522994246415286e+20,2.587894194346055e+20,2.6046858823354876e+20,3.430731
8506879386e+20,3.388901229723114e+20,3.67538419728239e+20
−1.847890625,2.804509092252715e+20,2.6235367280176824e+20,2.7270571660460854e+20,2.1617918438455702e+20,2.2169220760940606e+20,2.
2774443295174125e+20,2.349907574715804e+20,2.3784226630745968e+20,2.4272374374509553e+20,2.449007192233715e+20,2.50191905356577
57e+20,2.5604670630156553e+20,2.6135381375713834e+20,2.6444042745329946e+20,2.6828118111889746e+20,2.7430315594873592e+20,3.57415
11225673384e+20,3.573434663060836e+20,3.870722901399658e+20
−5.399921875,2.810193004337633e+20,2.6687638977540738e+20,2.7952216542054623e+20,2.180531240271198e+20,2.2520710487723193e+20,2.3
231172962800724e+20,2.370132961228992e+20,2.411698226821038e+20,2.460693442850886e+20,2.511291405332317e+20,2.5910147735299323e
+20,2.6450039776754742e+20,2.7074792466424775e+20,2.744066445441196e+20,2.7837423807790586e+20,2.7947227861046385e+20,3.67204071
9585379e+20,3.650955581664388e+20,3.98137609184835e+20
−8.951953125,2.7558959879596678e+20,2.6234677356207594e+20,2.730724377297886e+20,2.1626834378981068e+20,2.243505377339025e+20,2.2
83542195983866e+20,2.350953074884551 6e+20,2.4112365084724032e+20,2.44071749038811e+20,2.5011176803399837e+20,2.566320802539152e
+20,2.6251076318245316e+20,2.7045391091120898e+20,2.7200730126345513e+20,2.77365887661347e+20,2.7791782683672658e+20,3.648052593
88619e+20,3.603552497871212e+20,3.9168257438662184e+20
−12.503984375,2.5494070510782415e+20,2.4594197370615487e+20,2.562711969469362e+20,1.992505730227709e+20,2.096339287595991e+20,2.1
359462305369306e+20,2.2036171577029398e+20,2.26625163989361e+20,2.3262909465385052e+20,2.371162540075373e+20,2.4366310176473193
e+20,2.4961449206541124e+20,2.5814619801099007e+20,2.601809430094567e+20,2.619747453294403e+20,2.6447014725505067e+20,3.481696003
583805e+20,3.4340434857402165e+20,3.707322369950026e+20

A.53 "MITR_Cycle6_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.561954755589411e+19,5.105161402673347e+19,5.220458312145667e+19,4.282798566895063e+19,4.268150950317683e+19,4.4392
096378257056e+19,4.496521091238436e+19,4.5526702881183965e+19,4.716697058247785e+19,4.766164606841179e+19,4.827111428860977e+19,
4.982806040285117e+19,5.00539839672349e+19,5.017923170318641e+19,5.047022040497547e+19,5.269124487513269e+19,6.726387202416638e+
19,6.736895275178673e+19,7.3767732210970845e+19
37.224453125,7.306666331608968e+19,6.759078984342966e+19,7.019445676113369e+19,5.550809829932706e+19,5.627444461591178 4e+19,5.872
261328516752e+19,5.923156489015696e+19,6.050951636545888e+19,6.1194133227227775e+19,6.08757067798934e+19,6.427490910518778e+19,6.
4949973173536645e+19,6.50338254713347e+19,6.696295903143541e+19,6.783916247235048e+19,7.115928889655683e+19,9.174555871672374e+
19,9.046866866291291e+19,1.0058189263025244e+20
33.672421875,9.188195137833196e+19,8.710767751129865e+19,8.995334852897679e+19,7.071667613476206e+19,7.180197960942669e+19,7.4451
28765124864e+19,7.416576527013881e+19,7.661021896417567e+19,7.744768052066504e+19,8.00248119010912e+19,8.110374684680915e+19,8.41
6700927016575e+19,8.423600166708822e+19,8.618689436775678e+19,8.62277590951647e+19,8.859472902035015e+19,1.1697873182499019e+20,
1.1835168052374687e+20,1.2888098171560331e+20
30.120390625,1.1201605564328405e+20,1.043578995848925e+20,1.094490077670235e+20,8.695217926285037e+19,8.954788551937103e+19,8.977
9275404434e+19,9.115169339244513e+19,9.238028876841024e+19,9.505029452930892e+19,9.832053414343287e+19,1.0064186294450042e+20,1.0
335220272206145e+20,1.024399109504485e+20,1.0745353536372811e+20,1.0743442977688805e+20,1.074673338431126e+20,1.437743173681773
e+20,1.4393618414557225e+20,1.5794641711753888e+20
26.568359375,1.296229153378924e+20,1.2422240279110151e+20,1.2888894237678669e+20,1.0343658573060506e+20,1.0512902229818722e+20,1.
0721843050344622e+20,1.0937948465935547e+20,1.1180164850207891e+20,1.1291614106774921e+20,1.1735182147911695e+20,1.1984350842 95
0838e+20,1.2090492992062294e+20,1.230022987870653e+20,1.2511983466183885e+20,1.2958364274272115e+20,1.2986332730562984e+20,1.7371
807905401016e+20,1.7150183098056294e+20,1.9059998787018723e+20
23.016328125,1.525793393477181e+20,1.4882615295513703e+20,1.5246046014071328e+20,1.2147650539358814e+20,1.2352292602845702e+20,1.
258511540692168e+20,1.2832904254022374e+20,1.3054529061367094e+20,1.3263576024042108e+20,1.3626847529376065e+20,1.3800761440695
188e+20,1.4221668132996666e+20,1.4326908073840673e+20,1.4598154335895001e+20,1.5200776387475292e+20,1.513295155419307e+20,2.02899
20939847723e+20,2.0127788807079972e+20,2.2250578718234542e+20
19.464296875000002,1.7538398008431444e+20,1.6652004921201675e+20,1.7406834814607796e+20,1.3695043860180176e+20,1.409737567638715
e+20,1.4237642526437938e+20,1.462262010126519e+20,1.518220151138079e+20,1.5463372064377035e+20,1.5789440746447428e+20,1.604853373
2428494e+20,1.641254823280623e+20,1.67583062835368e+20,1.7175975640290378e+20,1.7430345300635982e+20,1.7665503231992414e+20,2.340
556451379085e+20,2.3358755826032692e+20,2.5501925029816658e+20
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12707446029897e+20,1.667471934111153e+20,1.7107460883038932e+20,1.74731205867279e+20,1.7818984779607576e+20,1.8187722605620776e
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1915623e+20,2.63727682922016e+20,2.9067505173843244e+20
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809595e+20,2.078114680593554e+20,2.142378444773085e+20,2.15645820085272e+20,2.2054799524198456e+20,2.2154413931139557e+20,2.94073
19264223573e+20,2.9311366761426816e+20,3.245460729413892e+20
8.808203125,2.4200887637082987e+20,2.314174820217432e+20,2.3871687761613806e+20,1.8778934376171586e+20,1.925174457938857e+20,1.99
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89780835720734e+20,2.3870414055824468e+20,2.3914940687376 72e+20,2.4500420781875518e+20,2.5318989035825 06e+20,2.5530424196853085
e+20,2.619991580237359e+20,2.6953259705692157e+20,2.7231776704960614e+20,2.7684897539517422e+20,2.8423116186587595e+20,3.75912504
5823873e+20,3.713950947162037e+20,4.0584459063181784e+20
−1.847890625,2.8846411077244086e+20,2.7318123343262784e+20,2.865599206173813e+20,2.2501180332286686e+20,2.3191210443660263e+20,2.
393473619818601e+20,2.451666053068957e+20,2.4852600432627325e+20,2.563269215752198e+20,2.6082628727605438e+20,2.670456865032401
e+20,2.7275401128245425e+20,2.7749219681878963e+20,2.831983987550215e+20,2.9045374535753512e+20,2.9344058543353148e+20,3.92054071
908512e+20,3.9244308288500544e+20,4.267832523870348e+20
−5.399921875,2.9142547673265046e+20,2.768680809820143e+20,2.9069521874676367e+20,2.270555704040079e+20,2.354604364813986e+20,2.43
94756272435062e+20,2.49319947601627e+20,2.5474381142122237e+20,2.616642795432893e+20,2.646696944953802e+20,2.729779712170794e+2
0,2.8069079048226338e+20,2.8273614969564116e+20,2.9371071720302012e+20,2.982562547387182e+20,2.98833137319139e+20,4.013356721375
6324e+20,3.979163028039377e+20,4.3559782715999565e+20
−8.951953125,2.875120156949111e+20,2.7269510238969738e+20,2.8674726151056304e+20,2.25323330530509e+20,2.3081671745777235e+20,2.38
98541725339006e+20,2.4612984531008215e+20,2.5134142483145464e+20,2.5679872342802016e+20,2.649127600168454e+20,2.714383793442177
4e+20,2.7854937262393973e+20,2.8586415882995574e+20,2.902462374560222e+20,2.985985631696027e+20,3.043620818663548e+20,4.011546997
733282e+20,3.9988470895920965e+20,4.3725099113240664e+20
−12.503984375,2.6962971712335856e+20,2.5768713321608305e+20,2.6973373642948775e+20,2.1435937723804113e+20,2.2129099028576476e+20
,2.2773222660459345e+20,2.3308444447353864e+20,2.398775420166718e+20,2.4646950018723883e+20,2.5295584691943994e+20,2.60041896794
12068e+20,2.659763043509422e+20,2.7117992321113132e+20,2.7643077532767506e+20,2.8472206930551643e+20,2.86427242930992e+20,3.79287
294213386e+20,3.783267077639274e+20,4.118251700235029e+20

A.54 "MITR_Cycle6_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.535790715833436 6e+19,5.3349166986400055e+19,5.465949181718098e+19,4.279253419114741e+19,4.328370698616067e+19,4.41
6489910808398e+19,4.554564925480036e+19,4.689142556338451e+19,4.828385134650314e+19,4.950581283814878 4e+19,5.0055894525918904e+
19,5.011119458560598e+19,5.200779557700403e+19,5.2336995452473205e+19,5.262442839226702e+19,5.258738478222713e+19,7.062114820056
175e+19,7.100962846630967e+19,7.730491933011013e+19
37.224453125,7.3436037994997555e+19,7.010529735588007e+19,7.229554060279496e+19,5.778537810851336e+19,5.898319226123615e+19,6.038
479934025291e+19,6.088950525927789e+19,6.213508337910083e+19,6.264668853781805e+19,6.411834943524839e+19,6.472919750338482e+19,6.
556135195241863e+19,6.676818818781589e+19,7.012493365346569e+19,6.9104907600504594e+19,7.142358284784435e+19,9.741673374374884e
+19,9.626721426887175e+19,1.0320095015957761e+20
33.672421875,9.385194966584058e+19,8.794938475375525e+19,8.963120710642352e+19,7.1579611807038185e+19,7.39932842778327e+19,7.39720
5584801042e+19,7.656033215409327e+19,7.752144931429 75e+19,7.864814822711561e+19,8.25589557111272e+19,8.391120669080717e+19,8.5688
02626693292e+19,8.618264868179232e+19,8.769252075290278e+19,9.000854244651475e+19,9.071385702736037e+19,1.214096358396479e+20,1.1
868019047524683e+20,1.3290270774543637e+20
30.120390625,1.0922398641090719e+20,1.0564805740734225e+20,1.1083203996994576e+20,8.713952015603209e+19,8.959883375094455e+19,9.1
75351937790709e+19,9.321987316788186e+19,9.573013499436779e+19,9.879658168219777e+19,9.870476872321635e+19,1.0386274645928758e+
20,1.0584282825096177e+20,1.0572766401917585e+20,1.0637566183950128e+20,1.0993195454548061e+20,1.1031565841451853e+20,1.49366416
49411434e+20,1.4794782667123973e+20,1.6039883147275908e+20
26.568359375,1.3027515884418228e+20,1.2579542944093328e+20,1.312017798059253e+20,1.0302953058876264e+20,1.0719667136287839e+20,1.
0973134588365996e+20,1.1229627091693829e+20,1.1477044441272636e+20,1.1615931443384975e+20,1.1884736436009737e+20,1.208009106144
9372e+20,1.2462786580070728e+20,1.2769643533151946e+20,1.3064294139085347e+20,1.3181209716331618e+20,1.3275251660444369e+20,1.781
1820184542554e+20,1.767150026341721e+20,1.9479737915679975e+20
23.016328125,1.5497337552092702e+20,1.4523854831516982e+20,1.552403230259423e+20,1.2067990856450666e+20,1.2602045079704956e+20,1.
3130792195503676e+20,1.3270050695137904e+20,1.3496027330596196e+20,1.3832604085428625e+20,1.41571337063369e+20,1.43435193201766
15e+20,1.4691877853560416e+20,1.4891690449262733e+20,1.530007236796906e+20,1.5419853783241336e+20,1.5706331443693155e+20,2.090984
4161733178e+20,2.093685733868205e+20,2.286795452854132e+20
19.464296875000002,1.772016643878481e+20,1.7055610443197986e+20,1.7858416588002486e+20,1.4248681309945532e+20,1.4451890454419413
e+20,1.488229686906637e+20,1.5182095369231676e+20,1.5382597888903217e+20,1.6000292125657334e+20,1.6362502209500178e+20,1.66068414
3675475e+20,1.6756608009151018e+20,1.7113139488016397e+20,1.7553257909307048e+20,1.798716701487468e+20,1.8190323088274008e+20,2.4
094957772269758e+20,2.4010521692651592e+20,2.6720171546243396e+20
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e+20,1.9273026080285413e+20,1.973500978429303e+20,1.9851819219390185e+20,2.005975168949953e+20,2.0763367995959368e+20,2.754346312
5826404e+20,2.7476911998333524e+20,3.0174302433702956e+20
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e+20,1.8806319050642345e+20,1.9174154668388096e+20,1.9485681876030217e+20,2.016085208652819e+20,2.0676596789060754e+20,2.10733561
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85e+20,2.548032510247248e+20,2.6091969236727244e+20,2.656435487134778e+20,2.7072510410218874e+20,2.755237906635177e+20,3.66241362
666097e+20,3.6309955505239785e+20,4.0170292397348893e+20
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5035650071588e+20,2.4257992113304948e+20,2.508213283008085e+20,2.572763630990217e+20,2.6359553594637222e+20,2.6696820273438877e
+20,2.7315682073833223e+20,2.7649286848490527e+20,2.8271916695178327e+20,2.87761449745323e+20,2.9347720447497493e+20,3.870011749
000611e+20,3.8814061087077256e+20,4.259001497064275e+20
−1.847890625,2.9509852580265243e+20,2.764721707658286e+20,2.918877257920309e+20,2.3075780856501554e+20,2.3841649533415267e+20,2.4
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8e+20,4.0925493788276897e+20,4.468441185691 0005e+20
−5.399921875,2.960320460040877e+20,2.835725498306 3942e+20,3.011226234754731e+20,2.3703505526346704e+20,2.4499147076086183e+20,2.5
259602503395203e+20,2.5997821150465383e+20,2.6710565681748813e+20,2.710016044006241e+20,2.7964847457798888e+20,2.82625761860565
24e+20,2.9293747164674317e+20,2.9743365308310 44e+20,3.0823680101966846e+20,3.072815216776653e+20,3.154932090436731e+20,4.15932340
4833707e+20,4.232534952183333e+20,4.617464763043485e+20
−8.951953125,2.9184526893238626e+20,2.8346163128481792e+20,2.948262711901815e+20,2.3027220823283063e+20,2.406741388457533e+20,2.4
907051355121503e+20,2.5732253493388516e+20,2.6388318117046433e+20,2.6783060769591935e+20,2.7633471668272923e+20,2.8139557435236
342e+20,2.8939869239536727e+20,2.960500901694366e+20,3.0040297970449744e+20,3.0890974224503505e+20,3.1443125684181303e+20,4.20554
83107717454e+20,4.173695051823398e+20,4.573436999592 0527e+20
−12.503984375,2.7051016625023808e+20,2.631210805398441e+20,2.797068527600002e+20,2.208213112759 4657e+20,2.2933072737021198e+20,2.
392820845601566e+20,2.4615744226885114e+20,2.5434790120503663e+20,2.5929837103959494e+20,2.6489312371925978e+20,2.7003093444699
98e+20,2.7878182393049383e+20,2.8703013033794506e+20,2.900610194058227e+20,2.962836028974818e+20,3.005091218536089e+20,3.97812283
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A.55 "MITR_Cycle6_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
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5267759e+19,4.749988543316594e+19,4.823874093313077e+19,4.846307236527784e+19,5.005186112425267e+19,5.0851376862434705e+19,5.1119
06736249381e+19,5.221424205702581e+19,5.282700068384625e+19,5.423598464222627e+19,5.378488050850258e+19,7.239902919817862e+19,7.2
033900205235225e+19,7.80059882249913e+19
37.224453125,6.962394270965961 5e+19,6.8588526045077365e+19,7.20179788828685e+19,5.7185144255288074e+19,5.910366360047764e+19,6.11
2514083030534e+19,6.16112718732358e+19,6.333077468884138e+19,6.549129813400506e+19,6.485922163604635e+19,6.87084666735733e+19,6.7
830671100421554e+19,7.021090879424596e+19,6.992963209910061e+19,7.044442152229116e+19,7.212093676750661e+19,9.679367932846458e+
19,9.520313922402941e+19,1.054432030595571 4e+20
33.672421875,8.714854223870657e+19,8.566096001890951e+19,9.149453253407513e+19,7.23671865534452e+19,7.3057641233415225e+19,7.6721
13750999712e+19,7.883071272358732e+19,8.137122506257002e+19,8.172255557612893e+19,8.396003207939842e+19,8.58934113254636e+19,8.57
6232577131096e+19,8.887335216176774e+19,8.940565503956168e+19,9.04612387124751e+19,9.213085471799832e+19,1.2105883603683456e+20,
1.217216937580356e+20,1.3137850648419589e+20
30.120390625,1.0387972920314541e+20,1.0134080899639937e+20,1.0676042713003031e+20,8.639970937672524e+19,9.051430978703083e+19,9.2
69871521574461e+19,9.534218543936543e+19,9.691839635367055e+19,9.998643517373717e+19,1.0334583419311474e+20,1.0682729668397053e
+20,1.0722851400761183e+20,1.0743549119837916e+20,1.1015273021563245e+20,1.1085486053200472e+20,1.1102893365654752e+20,1.4926398
93202218e+20,1.490272923277032e+20,1.628730049685 4712e+20
26.568359375,1.2597480967293164e+20,1.2171957091505337e+20,1.2965741153635362e+20,1.0250094268618759e+20,1.0761646356261421e+20,
1.1030292135662515e+20,1.1417021055950106e+20,1.1654938683183435e+20,1.2047452350597598e+20,1.2206187934593783e+20,1.26110140913
04873e+20,1.2689877708094688e+20,1.2790128967930456e+20,1.2980229556989075e+20,1.3271218258778132e+20,1.3420719475801617e+20,1.76
56162722870605e+20,1.7603728501209547e+20,1.962876149303246 2e+20
23.016328125,1.4733167149564772e+20,1.3998185838042494e+20,1.5104240102858424e+20,1.1898906412916118e+20,1.2525038950524595e+20,
1.2992542046286006e+20,1.3349338880524165e+20,1.3793490703481052e+20,1.4065108463057268e+20,1.4535742752217463e+20,1.46693757179
48788e+20,1.5263400255451 05e+20,1.5138046377350 42e+20,1.5580818352368 86e+20,1.5732654696672798e+20,1.5982407173532054e+20,2.08745
5189715362e+20,2.103668402992137e+20,2.283690794992622e+20
19.464296875000002,1.7018354548859866e+20,1.652484662656615e+20,1.7587435681320937e+20,1.3932908416338949e+20,1.434649130035174e
+20,1.5059607329157054e+20,1.5457481175101348e+20,1.5916970538604842e+20,1.6439030699009 54e+20,1.6572451380442638e+20,1.69615684
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24663478483953e+20,2.411825597399972e+20,2.6621671631867963e+20
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e+20,1.987612577153671e+20,1.9914071589844055e+20,2.0179267749399026e+20,2.0549279281201565e+20,2.089084471704223e+20,2.742553919
8163576e+20,2.735495466900446e+20,3.000877375216364e+20
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6e+20,2.7763655014158123e+20,2.8050398029982718e+20,2.8893962760046015e+20,2.9386674616221395e+20,2.955549370438317e+20,3.9128984
84349095e+20,3.879516778453542e+20,4.3212591746256e+20
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5335e+20,4.091859454858465e+20,4.492641595688412e+20
−5.399921875,2.8343881072275895e+20,2.7959062710672314e+20,2.967845938412879e+20,2.388394717983618e+20,2.5230625696687777e+20,2.5
923733930385588e+20,2.6614666250026608e+20,2.7355113882228127e+20,2.805045110105728e+20,2.8490834877720707e+20,2.91447235873218
33e+20,2.9861926088867945e+20,3.034800406072385e+20,3.127605794147987e+20,3.145697723464035e+20,3.193068964612 4774e+20,4.22873506
32451434e+20,4.199997076373217e+20,4.608506365658478e+20
−8.951953125,2.8434632609766192e+20,2.7819061215994302e+20,2.950470468603334e+20,2.3679888898169405e+20,2.485132672683799e+20,2.5
538809426632894e+20,2.619312270483046e+20,2.7286705267125794e+20,2.783025921272556e+20,2.82554115909915e+20,2.9194345042031437e
+20,2.9889788403009697e+20,3.028139986215641 4e+20,3.100958807613556e+20,3.151928267616877e+20,3.196768018509012e+20,4.2269094182
804265e+20,4.203080505804904e+20,4.639728078819613e+20
−12.503984375,2.6111977031834753e+20,2.602276455550657 5e+20,2.760624620702583e+20,2.2455486137094203e+20,2.3538454484478388e+20,
2.42292806619703e+20,2.5000721801712363e+20,2.597123254211296e+20,2.658048847801272e+20,2.730894204736464e+20,2.79167650642514e+
20,2.857038841847974e+20,2.892002065765288e+20,2.963165069637064e+20,2.994063049243408 5e+20,3.044846760485785e+20,3.990254882621
524e+20,3.9789719721709765e+20,4.329532955148838e+20

A.56 "MITR_Cycle6_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,4.2873075285015994e+19,4.677200723662518e+19,4.976503782180032e+19,5.097670783937513e+19,5.5563327406384e+19,5.53609
1843731754e+19,6.081653161286946e+19,6.013988938486782e+19,6.206816258485179e+19,6.558328977435362e+19,6.4909521143073415e+19,6.4
36093356897096e+19,6.41910321628158e+19,6.964296354256838e+19,6.656758645134445e+19,6.818452340653989e+19,6.878429692224301e+19,
6.762515984473203e+19,6.414891601086719e+19
37.224453125,5.696258940993033e+19,6.229814012246456e+19,6.6422200417219084e+19,7.339103166630195e+19,7.637598026069805e+19,7.953
40630575609e+19,8.00909122222375e+19,8.323390863631281e+19,8.530640047175406e+19,8.686694309129124e+19,8.670108268052559e+19,8.87
547165374824e+19,8.801772877833052e+19,9.443716425423303e+19,9.202962900105134e+19,9.43078524017682e+19,9.341075142529357e+19,8.9
57099760616653e+19,9.142078569569618e+19
33.672421875,7.05402441188381e+19,7.798627202347514e+19,8.533845903517765e+19,8.94575803355672e+19,9.562072690387624e+19,9.948930
649011521e+19,1.0274814477202357e+20,1.0594322512667499e+20,1.0712768577529364e+20,1.0926851533276665e+20,1.1169670456238372e+2
0,1.1274287336878306e+20,1.1726609420812514e+20,1.1716282432594839e+20,1.1776448363949995e+20,1.2129361961301878e+20,1.189471482
9016772e+20,1.1370014027661284e+20,1.1230105787286161e+20
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4706035069472e+20,1.274898125436244e+20,1.296037021318339e+20,1.3178673764115307e+20,1.3589777695076357e+20,1.388512955810189e+
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2646687e+20,1.4299735685067194e+20,1.4380824992550339e+20
26.568359375,1.0416249315835747e+20,1.1401084792211859e+20,1.1953084762421027e+20,1.310845024423511e+20,1.384947899877652e+20,1.4
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e+20,1.7033513966029898e+20,1.7307762853132354e+20,1.7028934171255102e+20,1.7542499785315005e+20,1.7682408025690132e+20,1.7538099
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676649173186032e+20,1.7362001991249537e+20,1.8002993659925814e+20,1.8594325985259694e+20,1.907412683784268e+20,1.93448735289408
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736096702e+20,2.0928404922249563e+20,2.0207221945065597e+20
19.464296875000002,1.4280877706582745e+20,1.5437679986757547e+20,1.6583257279739222e+20,1.7310097650468523e+20,1.860914296835462
3e+20,1.9451645607121892e+20,2.000526197548687e+20,2.0869316589664947e+20,2.1365012024113394e+20,2.185558886440178e+20,2.22833955
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6e+20,4.025468987239863e+20,3.9885522493591087e+20
−5.399921875,2.3774702474836165e+20,2.602984730188397e+20,2.7983713472668264e+20,3.0198178946128183e+20,3.181493630152852e+20,3.3
54026173306077e+20,3.467694883618549e+20,3.645615420624475e+20,3.726013268896522e+20,3.796425368561563e+20,3.8828128699998614e+
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75801034790725e+20,3.471583219182249e+20,3.523936559450989e+20,3.696145822973052e+20,3.794036691286866e+20,3.851813945367325e+2
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3.098599344729054e+20,3.221149264908728e+20,3.2598979206993995e+20,3.4247795125817895e+20,3.486732461898091e+20,3.62489858426084
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A.57 "MITR_Cycle6_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.428427932007197e+19,5.242058239489848e+19,5.550597545634484e+19,4.4787475883697226e+19,4.446167256999614e+19,4.567790237259323e+19,4.866893506347951e+19,4.824940821911648e+19,4.935721382939275e+19,5.033802035825716e+19,5.096680644959343e+19,5.240094609731286e+19,5.261041762858432e+19,5.413674173280706e+19,5.407730212930464e+19,5.513660077743698e+19,7.09533731272806e+19,7.084298529220468e+19,7.705283172597042e+19
37.224453125,7.047095705956903e+19,6.925403732000618e+19,7.292761710075368e+19,5.844558227598661e+19,6.040602777007521e+19,6.094629130905253e+19,6.296140000993351e+19,6.323471604389552e+19,6.598804339184669e+19,6.603686878043795e+19,6.822711202735285e+19,6.844735698675913 5e+19,6.926305940268066e+19,6.978687090854569 6e+19,7.165921841887178e+19,7.270418787687407e+19,9.560966365512629e+19,9.389706007921294e+19,1.0432977191537795e+20
33.672421875,9.087837735848314e+19,8.782307559630987e+19,9.29959132332567e+19,7.209970833768433e+19,7.541611978667177e+19,7.732508633844131e+19,7.971965322239576e+19,8.141951974041574e+19,8.176076674980905e+19,8.436018798154862e+19,8.535367849723185e+19,8.677067618786979e+19,8.825666627543015e+19,8.862391811135 58e+19,9.01295444965018e+19,8.98365921649542e+19,1.218978897255606e+20,1.2113685051643149e+20,1.2967439428021145e+20
30.120390625,1.0787120472048171e+20,1.0519642256287302e+20,1.101033741162956e+20,8.776575883578968e+19,9.120529517774643e+19,9.394906973227755e+19,9.606289063188 22e+19,9.908953401374538e+19,1.003154758359827e+20,1.026697087032748e+20,1.057096198538268 8e+20,1.0776877754658914e+20,1.1021004697615262e+20,1.1089147957344816e+20,1.1146730073237783e+20,1.153722703981883e+20,1.4707109251957909e+20,1.486313821115175e+20,1.5927531682441432e+20
26.568359375,1.2827915573014135e+20,1.2489852828094146e+20,1.3224303428870868e+20,1.0449270011426405e+20,1.0877978151687576e+20,1.1251439303336234e+20,1.1391546940163357e+20,1.1647349519521964e+20,1.1983289421459723e+20,1.2303095716732541e+20,1.2680855625420213e+20,1.2611650944199544e+20,1.3086318635025977e+20,1.300910022154739e+20,1.3368444467364225e+20,1.3489499588425841e+20,1.750989884139502e+20,1.775832454139038e+20,1.9276900268727982e+20
23.016328125,1.4916846138602147e+20,1.433762843090093e+20,1.5395547231094813e+20,1.2380897912031237e+20,1.2664775089829827e+20,1.320912510154793e+20,1.3393759369927308e+20,1.3882809321958341e+20,1.4174594089865735e+20,1.4395210546793895e+20,1.4780294263770258e+20,1.5065232863059 96e+20,1.5579013935833966e+20,1.5645777347625 07e+20,1.5704580098232818e+20,1.5894203047620434e+20,2.087964672031097e+20,2.099422717027679e+20,2.273665669009045e+20
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1.704140625,2.739714617327626e+20,2.6789641582836842e+20,2.8331303227606188e+20,2.2626905707909205e+20,2.3440538351923075e+20,2.4107482545864907e+20,2.5053368307671646e+20,2.5854688462388586e+20,2.6289818202670996e+20,2.6748564571130 71e+20,2.727731168692943 3e+20,2.775537592652743e+20,2.8530001330742837e+20,2.9246407766170606e+20,2.9422019951875513e+20,2.952901123817986e+20,3.878269608201482e+20,3.8841339619398903e+20,4.2334530817731474e+20
−1.847890625,2.845193378007136e+20,2.7823996825927985e+20,2.947785072230814e+20,2.3402698675764835e+20,2.4633629179010394e+20,2.530450063246935e+20,2.61906814354009e+20,2.685067331857593e+20,2.7399162874109385e+20,2.8041853586979245e+20,2.8613959770689995e+20,2.8933182284142703e+20,2.9791235417559715e+20,3.03849945996892e+20,3.0675240306434474e+20,3.0945849715594134e+20,4.125915163400877e+20,4.084201298800073e+20,4.4325545250764174e+20
−5.399921875,2.9377970959994257e+20,2.8673080947745076e+20,3.011528739879699e+20,2.400123425460434e+20,2.5476079416508023e+20,2.603566082662361 7e+20,2.664810102699672e+20,2.750323525131316e+20,2.813748766332867e+20,2.881589520937454e+20,2.9506349889344563e+20,3.033478936315948e+20,3.0792474310128075e+20,3.120244836107108e+20,3.1818125896992077e+20,3.1920393857660964e+20,4.196154730575382e+20,4.193219900152451e+20,4.5650411555973366e+20
−8.951953125,2.941581063615249e+20,2.8125705884777297e+20,2.9809863364728776e+20,2.409219807639286e+20,2.519092853292009e+20,2.588626575174924e+20,2.676469817779565e+20,2.7097984526005626e+20,2.7872981427742923e+20,2.860037357560373e+20,2.9397394973281655e+20,2.987705134511632e+20,3.091968567583816e+20,3.111466880375 59e+20,3.1850923821067515e+20,3.211298878722 37e+20,4.225996595798068e+20,4.2012124039805436e+20,4.5650358484898814e+20
−12.503984375,2.699497357029296e+20,2.6542489588630818e+20,2.8149853223700156e+20,2.2713411559435043e+20,2.3740867562833 94e+20,2.4436204781663083e+20,2.5275948394358368e+20,2.6044842122521 76e+20,2.6736411295057455e+20,2.7727991252056673e+20,2.823264410000709e+20,2.8869656207899496e+20,2.9359661439272534e+20,2.9623424679814496e+20,2.996615767929539e+20,3.017589456593963e+20,3.9825277341662105e+20,3.949347698353969e+20,4.2804315969698785e+20

A.58 "MITR_Cycle6_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.570870696114772e+19,5.311830781208263e+19,5.434637247730219e+19,4.294999606935425e+19,4.347693876861808e+19,4.5261
82514807632e+19,4.618191836764898e+19,4.641245911551907e+19,4.764009921214217e+19,4.9825088422676046e+19,4.971109175453034e+19,5.
011862453604378e+19,5.128709038453723e+19,5.2808266594528076e+19,5.287832041294164e+19,5.4251375253847425e+19,7.013448644688572
4e+19,7.105049319371759e+19,7.5969651094288e+19
37.224453125,7.597814246621692e+19,7.080530482927012e+19,7.358357558226249e+19,5.724139959431715e+19,5.9075005220217545e+19,6.043
4155439589745e+19,6.061671993606145e+19,6.1969501626486964e+19,6.32734579283212e+19,6.4313650989613466e+19,6.4755733040662675e+
19,6.650601707951059e+19,6.757911420702741e+19,6.8855473550092665e+19,6.929914773337855e+19,7.126277749194049e+19,9.532095700954
314e+19,9.497758715716757e+19,1.0189221746103339e+20
33.672421875,9.457265485830737e+19,8.884416307076207e+19,9.331911607730107e+19,7.200789537870291e+19,7.509450907486406e+19,7.5352
4344972049e+19,7.752728713249864e+19,7.733198557813355e+19,8.051200436551279e+19,8.280042910035578e+19,8.408687194758662e+19,8.52
310843150081e+19,8.75481674301112e+19,8.939981722136055e+19,8.995865563643237e+19,8.9334539799657e+19,1.2174504503084012e+20,1.21
08749441709466e+20,1.3050814086148194e+20
30.120390625,1.1279991541447218e+20,1.0842792029257127e+20,1.1303395885326292e+20,8.859685186333239e+19,8.96487205610269e+19,9.27
1092156289244e+19,9.419001241076058e+19,9.664773387343633e+19,1.0018439028183006e+20,1.006546000023938e+20,1.017812989152119e+2
0,1.0450119148619294e+20,1.0711865688328148e+20,1.069780185357088e+20,1.0943680141987566e+20,1.1175547666721541e+20,1.4723508213
99563e+20,1.4585629562299849e+20,1.603850329933746e+20
26.568359375,1.3383782007910831e+20,1.2862836340071804e+20,1.3367701472320445e+20,1.0413765462548623e+20,1.077592247531691e+20,1.
1242947931407319e+20,1.130169761094051e+20,1.1504004547146945e+20,1.174866220084885e+20,1.1943910684139373e+20,1.23784035715271
2e+20,1.2485766355353356e+20,1.2848931718538204e+20,1.2976779937142953e+20,1.3217669544551403e+20,1.3497990960354756e+20,1.777488
271665177e+20,1.754370511588702e+20,1.9381025717006326e+20
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3012019130647955e+20,1.3408460057579246e+20,1.3482175780137148e+20,1.3609121790474451e+20,1.4188710995697558e+20,1.455044343986
94e+20,1.4690020365950965e+20,1.5034080142295748e+20,1.5210063825522544e+20,1.5338389683798295e+20,1.6000504409955556e+20,2.10743
11421781384e+20,2.088845651868722e+20,2.275252494138261e+20
19.464296875000002,1.814399204018686e+20,1.7285673551397067e+20,1.791525570885167e+20,1.4250114228958536e+20,1.4355460311951655e
+20,1.5012851712473458e+20,1.525984449345582e+20,1.5753299344674977e+20,1.6358787234281275e+20,1.6466415373480293e+20,1.65090845
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2.416474623531054e+20,2.3935107695707898e+20,2.645417932057009e+20
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993835712415118e+20,1.7178469980794497e+20,1.7847961586315005e+20,1.7914778069180668e+20,1.8245145508290075e+20,1.8822930296978
286e+20,1.9368660156634838e+20,1.9625418015335452e+20,1.9986991246283624e+20,2.0342620516881557e+20,2.0726271314844916e+20,2.7602
63737395604e+20,2.7308888976290087e+20,2.979357054484016e+20
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+20,1.8804196207660116e+20,1.933978949207652e+20,1.9739308541332046e+20,2.0170617164246447e+20,2.0655262217089352e+20,2.13371724
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568454395e+20,3.335182688057806e+20,3.644926707594858e+20
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+20,2.583982856151298e+20,2.6073447431707296e+20,2.6871264895503555e+20,2.6923752188239167e+20,2.7657406722897553e+20,3.66187760
88079565e+20,3.643552166763864e+20,3.968145472961608e+20
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518563015e+20,2.4429093257672616e+20,2.5246812374427278e+20,2.6045319762192758e+20,2.6268377488550484e+20,2.6567751420119345e+2
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5120648915187e+20,2.5672070894842328e+20,2.5988692925641795e+20,2.663764602530924e+20,2.7439125393249842e+20,2.8032619220006548
e+20,2.8412130474154564e+20,2.9086239263161424e+20,2.974930926866069e+20,3.026776059599559e+20,3.073176100083633e+20,4.0481288894
24546e+20,4.0680676921351333e+20,4.371076992311062e+20
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91965143698e+20,2.6208460245377067e+20,2.6626607241801646e+20,2.7484554233069548e+20,2.7931147325456e+20,2.838612564762226e+20,2.
9205808394135477e+20,2.98569374078597e+20,3.0467467049548795e+20,3.13383633830083e+20,3.174127898103538e+20,4.153941997873756e+
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444208759682e+20,2.5672654676662436e+20,2.6457575869341657e+20,2.6924229827910173e+20,2.7795285374593335e+20,2.84819720082699e+
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05e+20,4.1640838802213555e+20,4.530900533335637e+20
−12.503984375,2.7833443477198907e+20,2.6883948882322375e+20,2.818546391472705e+20,2.2622288524422853e+20,2.3434010609752716e+20,
2.4054570684532844e+20,2.488645977819388e+20,2.5618150683093706e+20,2.6027487881142036e+20,2.6495362474425333e+20,2.715630963694
237e+20,2.804832825807505e+20,2.8697334428817048e+20,2.9216422609046625e+20,2.972691327519817e+20,3.0190170684995115e+20,3.954835
247463031e+20,3.9494856831478143e+20,4.324708794471722e+20

A.59 "MITR_Cycle6_cladding_fluence_position_8.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.559778841532626e+19,5.3219673564484076e+19,5.545078153880688e+19,4.3272880486951305e+19,4.253455569773202e+19,4.45
2031609438369e+19,4.524059671825403e+19,4.57809133283059e+19,4.651042831914893e+19,4.808685151775229e+19,4.859267192934293e+19,5.
0088586307845235e+19,5.027704169359262 5e+19,5.004941985482310 5e+19,5.187676309392594e+19,5.136743999141462e+19,6.74766870331348
6e+19,6.634680385584341e+19,7.287454602619796e+19
37.224453125,7.249827210759785e+19,6.8573135433456206e+19,7.1328054913644044e+19,5.6074366664836686e+19,5.675579926213224e+19,5.8
13777004356339e+19,5.887598869063357e+19,6.089905805269792e+19,6.1203686020647805e+19,6.178640641926969e+19,6.4615625403835556e
+19,6.5075751620233716e+19,6.688547526258404 6e+19,6.723733648688853e+19,6.874030931830674e+19,6.928906422921297e+19,9.3054822126
01355e+19,9.110339871459942e+19,9.94084911718253e+19
33.672421875,9.34889435158794e+19,8.887388287251328e+19,9.186974503118414e+19,7.001030013242531e+19,7.181365524582896e+19,7.34280
773338142e+19,7.448737598194654e+19,7.783881434014076e+19,7.877658022754048e+19,7.92706719316543e+19,8.209086883354568e+19,8.2514
3760085004e+19,8.459370070959381e+19,8.671760511331407e+19,8.661199367494815e+19,8.799237232414265e+19,1.1624847383910336e+20,1.1
775516164574049e+20,1.2355742222691823e+20
30.120390625,1.1366391250823941e+20,1.0737021377667562e+20,1.1102468797058307e+20,8.629515935985046e+19,8.988754039652768e+19,9.0
80620069708734e+19,9.124881345888212e+19,9.481306682604 48e+19,9.582884719304144e+19,9.715986974289 91e+19,1.008297345484277e+20,1.
0235765078488711e+20,1.0442317700659605e+20,1.0547133072907169e+20,1.0859403275593071e+20,1.0920647295630382e+20,1.447428644788
1932e+20,1.4505386097571588e+20,1.557705030607 5404e+20
26.568359375,1.3175637253503266e+20,1.2416508603058132e+20,1.2992807401658781e+20,1.0300193362999365e+20,1.043701059320403e+20,1.
0734261681790665e+20,1.1002801319042648e+20,1.1285351719977345e+20,1.1477787436316415e+20,1.1854698207811194e+20,1.210025806978
0549e+20,1.2109917005349691e+20,1.2467616047855298e+20,1.2590740940824587e+20,1.2944141226291179e+20,1.3117312142565 52e+20,1.7362
14896983187e+20,1.7116642178937074e+20,1.8643125496383478e+20
23.016328125,1.5692532964308668e+20,1.4888081616192943e+20,1.5492985723979 13e+20,1.1993797494221758e+20,1.2408176444352882e+20,1.
2657716636913916e+20,1.3019183725712979e+20,1.3140451131072818e+20,1.3422576963411067e+20,1.3809358954773214e+20,1.391502346421
3666e+20,1.4241994354551 51e+20,1.4422382939696643e+20,1.4768034845547885e+20,1.5129714218645173e+20,1.503768897536554e+20,2.027007
2357963878e+20,2.004706770268071e+20,2.1794432832428055e+20
19.464296875000002,1.7743358498365666e+20,1.6976269186737173e+20,1.7547101664658584e+20,1.3883286961629343e+20,1.419428345852591
e+20,1.4549541231601952e+20,1.479308439273819e+20,1.5450316580036323e+20,1.5723632613998323e+20,1.5834232733372462e+20,1.61868900
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677213681615648e+20,1.6854577212780888e+20,1.713633154759725e+20,1.7700795496571974e+20,1.7754662637246035e+20,1.84699015090335
8e+20,1.860560424667258e+20,1.8878867209560025e+20,1.937004000457329e+20,1.9808619364701825e+20,2.0029448105928206e+20,2.65947115
25993657e+20,2.6140741554243954e+20,2.8851505900401436e+20
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e+20,1.8322045495321323e+20,1.875377868683217e+20,1.92501524471519e+20,1.959352229952746e+20,1.99751563966577e+20,2.0341665237539
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8.808203125,2.4697155255253598e+20,2.319471313458094e+20,2.4436735492408643e+20,1.9008307560401443e+20,1.9566615264727705e+20,2.0
211004251983354e+20,2.059953758880584e+20,2.1046342965490513e+20,2.1500949790134878e+20,2.1747836428968123e+20,2.24537347916338
63e+20,2.2911419738602465e+20,2.3234251085124957e+20,2.3717197863582083e+20,2.4019968343922514e+20,2.4632780041817503e+20,3.25107
5649101888e+20,3.205397375231773e+20,3.5072868757345763e+20
5.256171875,2.654233037540715e+20,2.515223971956897e+20,2.623080316776503e+20,2.0450620153602464e+20,2.0966895566880588e+20,2.159
2444322668954e+20,2.2213747392492863e+20,2.2622288524422853e+20,2.3218488975981904e+20,2.355586179693267e+20,2.4145906003843256
e+20,2.500677190421172e+20,2.5095612883018003e+20,2.536516087068655e+20,2.607721547800075e+20,2.6476999882629048e+20,3.4912806396
48569e+20,3.4666609681621674e+20,3.7690440296583384e+20
1.704140625,2.8131172205456533e+20,2.6588767565643412e+20,2.7595048210294578e+20,2.174566051491134e+20,2.236101962438501e+20,2.30
7583392757611e+20,2.369856991641302e+20,2.3924334267573 09e+20,2.4527752385271713e+20,2.5459574312321194e+20,2.5716066815649022e
+20,2.6444838811448282e+20,2.7147765193938898e+20,2.7232731984302617e+20,2.8007835028189027e+20,2.8339476173087772e+20,3.7343090
11361614e+20,3.6813228505251755e+20,4.001298973236571e+20
−1.847890625,2.92338299215009e+20,2.7694131906490122e+20,2.8903409411316938e+20,2.2722062144587625e+20,2.3640722445147275e+20,2.4
091189725976294e+20,2.460698749958342e+20,2.5101503772293 69e+20,2.5491629241352847e+20,2.6093614440038472e+20,2.703626286629731
e+20,2.744400793210897e+20,2.8015477262925 05e+20,2.8781452081987876e+20,2.9060818218449227e+20,2.926646863235267e+20,3.9396303846
02815e+20,3.8773620928265794e+20,4.2162155967574475e+20
−5.399921875,2.9387842179861624e+20,2.8095349230131423e+20,2.957237030609189e+20,2.3050519025013028e+20,2.3871687761613806e+20,2.
4432277522145958e+20,2.5203824804037132e+20,2.6001854552131617e+20,2.6502686282714025e+20,2.6713113093327487e+20,2.724392998103
388e+20,2.7857962313643652e+20,2.8545551155587662e+20,2.936104128721098e+20,2.978507917291124e+20,3.01515880137931e+20,3.97577709
34827214e+20,4.0191043187500 18e+20,4.351345166791242e+20
−8.951953125,2.9616047800451256e+20,2.7593986788803458e+20,2.9185376030431517e+20,2.3109905557440886e+20,2.326970256292818e+20,2.
4036155021662008e+20,2.453024672577584e+20,2.5339792897048914e+20,2.601740437697 6448e+20,2.6571466395338247e+20,2.7401657214613
5e+20,2.8010117084394924e+20,2.87585253777798e+20,2.914238846004138e+20,2.980752823744832e+20,3.03414763185535e+20,4.024544103891
9795e+20,4.0122422288099626e+20,4.348028224631509e+20
−12.503984375,2.758390328463787e+20,2.605789760686246e+20,2.7467996057808165e+20,2.17712407728472e+20,2.2432400219662464e+20,2.29
56848578422165e+20,2.350597498685028e+20,2.4369494440946532e+20,2.4806693953136624e+20,2.56602360452164e+20,2.615676901875979e+
20,2.6763742898453647e+20,2.741243064274831e+20,2.7781911463805292e+20,2.8512487876139444e+20,2.8829534475535365e+20,3.795027627
7608225e+20,3.802701705141581e+20,4.076909333156117e+20

A.60 "MITR_Cycle6_cladding_fluence_position_9.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.259205503678804e+19,5.220182342557976e+19,5.283713725908639e+19,4.268830260071996e+19,4.231611515486065e+19,4.2991
92221825328e+19,4.41586367212864e+19,4.443741907592765e+19,4.58336659764143e+19,4.712875940879773e+19,4.6263594751390245e+19,4.76
7401162878327e+19,4.844099479826266e+19,4.8349394123579466e+19,4.843738596519286e+19,4.820530615616067e+19,6.368741230985587e+1
9,6.32957477796346e+19,6.669813436940233e+19
37.224453125,7.129302800443726e+19,6.808912723350796e+19,6.843621206110242e+19,5.3763652078680285e+19,5.565775872957423e+19,5.637
0503260857655e+19,5.785065553021691e+19,5.919972224542351e+19,6.005045157055 1824e+19,6.081308291191764e+19,6.159959623683354e+1
9,6.3136534555967414e+19,6.405041845981705e+19,6.400849231091802e+19,6.625605231835311e+19,6.587871697826189e+19,8.6800926700366
54e+19,8.332901700293081e+19,9.12875553433078e+19
33.672421875,9.068679077933695e+19,8.497262818192173e+19,8.881921966572089e+19,6.787737364603061e+19,7.015306132298022e+19,7.1203
86859918364e+19,7.345461287109207e+19,7.57860251763252e+19,7.664100018741797e+19,7.826019867211324e+19,7.905148839373914e+19,7.96
3686234608882e+19,8.127994281433416e+19,8.249367828942368e+19,8.159624641868631e+19,8.21317335609536e+19,1.089729602282 5863e+20,
1.0685542435348506e+20,1.1530062444753809e+20
30.120390625,1.1046107315880123e+20,1.0322270930014549e+20,1.0721896121419 18e+20,8.369255386363755e+19,8.566838996934731e+19,8.76
0442276914027e+19,8.823437642411676e+19,9.113789491306063e+19,9.35908399790264e+19,9.42329999811507e+19,9.656653512936607e+19,9.7
41036521480215e+19,9.845055827609441e+19,9.994557044632927e+19,1.0078037844909087e+20,1.0196863980839364e+20,1.3088759904455539
e+20,1.327111211662902e+20,1.4149969111271876e+20
26.568359375,1.323311322724712e+20,1.2496698996711837e+20,1.2727770455327477e+20,1.000867395046475e+20,1.0082177388724434e+20,1.0
505366137231809e+20,1.0485835981795302e+20,1.0813815222549702e+20,1.0889176148418834e+20,1.1250059455397785e+20,1.1437612632877
728e+20,1.1712467728001842e+20,1.1870141890506913e+20,1.1915411517102947e+20,1.196137106766821e+20,1.2323209653989163e+20,1.59335
2871386623e+20,1.5729417361 1249e+20,1.6979241166912294e+20
23.016328125,1.5257403224026255e+20,1.4304512080378156e+20,1.4768247129846107e+20,1.1758002709970659e+20,1.2018634757113838e+20,
1.2351443465652809e+20,1.2510232120723548e+20,1.2738703096685956e+20,1.3007614231459832e+20,1.3128775494670559e+20,1.34082477732
81021e+20,1.3710275258577671e+20,1.3956843470963583e+20,1.4201713408963712e+20,1.4304352867154489e+20,1.4214981177602644e+20,1.86
73269866731133e+20,1.846921158506436e+20,1.961485687149886e+20
19.464296875000002,1.7516691938938153e+20,1.6656728246837137e+20,1.7263277557934552e+20,1.3352310860699284e+20,1.37670613083523e
+20,1.4062242625031258e+20,1.4334762592874922e+20,1.4792341397694408e+20,1.4874654634330345e+20,1.5233892737998065e+20,1.5483379
85948454e+20,1.568367009485786e+20,1.592344520970064e+20,1.591537840636817e+20,1.6353373984676594e+20,1.6412866659253564e+20,2.12
85215872065847e+20,2.1321994126732965e+20,2.258768618381252e+20
15.912265625,1.955164922170299e+20,1.850885567775749e+20,1.9171872612182196e+20,1.4996983461181296e+20,1.5393159032739804e+20,1.5
886507741809852e+20,1.6167625223731544e+20,1.6458666996595157e+20,1.674424244877953e+20,1.716944789812002e+20,1.758419834577304
e+20,1.7603197790463987e+20,1.809797941854704e+20,1.835877067891389e+20,1.8536505707601022e+20,1.865777311296086e+20,2.4248651604
18314e+20,2.4141341891431462e+20,2.5897569890629612e+20
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7e+20,1.7718786590846367e+20,1.803063222493582e+20,1.828160533650986e+20,1.878747881917506e+20,1.8885076525283043e+20,1.945628050
0726343e+20,1.986779361283146e+20,2.0153422136090388e+20,2.051133346289422e+20,2.0588445734223695e+20,2.049090109919026 5e+20,2.70
79144294538343e+20,2.6587016220183072e+20,2.8933235355217258e+20
8.808203125,2.37627328455509e+20,2.2357782288837106e+20,2.3307011528340862e+20,1.8166228820425707e+20,1.8677621694844702e+20,1.92
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591758946e+20,2.8854743235949334e+20,3.156237638870802e+20
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e+20,2.3475352976831627e+20,2.3734339820663577e+20,2.388028527569183e+20,2.4170318698138888e+20,2.4337704867287653e+20,3.19989390
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7230756503493e+20,2.2433992351899 13e+20,2.2649195559222608e+20,2.3516164633164985e+20,2.4150629329478713e+20,2.4292116814244282
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−1.847890625,2.849131251739171e+20,2.65817091127275e+20,2.7938152707297355e+20,2.158703107306427e+20,2.2384636252562306e+20,2.283
1494700321535e+20,2.340874877826419e+20,2.3957397547021307e+20,2.4380586295528682e+20,2.4653053192297795e+20,2.5322969366414744
e+20,2.5923362432863692e+20,2.606978552756295e+20,2.686792141780654e+20,2.6880976902147254e+20,2.7393112771610028e+20,3.582318760
941465e+20,3.536120390540704e+20,3.7775725513394435e+20
−5.399921875,2.841255504275101e+20,2.6936276961834323e+20,2.861088164836576e+20,2.2045512086151204e+20,2.2385379247606088e+20,2.3
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774523359912e+20,2.3540365043162395e+20,2.4335263597858092e+20,2.462030833929 6905e+20,2.5262362199272102e+20,2.5805173149828088
e+20,2.6408166698930 27e+20,2.6922690766748058e+20,2.736026177646003e+20,2.7742638868634055e+20,2.7736217268612812e+20,3.675447882
571856e+20,3.588575840631585e+20,3.8765129556336876e+20
−12.503984375,2.6148543002203652e+20,2.4928492069242018e+20,2.5960777540425486e+20,2.0182080516350483e+20,2.110174916732669e+20,
2.1557895053133175e+20,2.241849559812886e+20,2.273310092809522e+20,2.3306533888669857e+20,2.4035305884469117e+20,2.4593295162348
044e+20,2.5318246040779286e+20,2.5618999820286598e+20,2.587140585087364e+20,2.645041127427663e+20,2.6555757357269752e+20,3.442582
621636233e+20,3.420881859250396e+20,3.623953018930433e+20

A.61 "MITR_Cycle7_cladding_fluence_position_0.csv" File

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5932993521148e+19,5.459699544209634e+19,5.607361458130137e+19,5.647574084481329e+19,5.719769943587265e+19,5.825656005735419e+19,
6.0873615029673665e+19,6.000252113981409e+19,5.885283173093073e+19,6.090371161105805e+19,5.816671951590828e+19,5.789549092128306
e+19,5.4573816582403285e+19,4.993741576000402e+19
37.224453125,6.329275128918779e+19,6.574369110037376e+19,6.964106362883897e+19,7.046157729386449e+19,7.1979972284841886e+19,7.636
419070740249e+19,7.568427748973981e+19,7.810673784928743e+19,7.764837140683037e+19,7.657899944199966e+19,7.79157368581734e+19,7.9
6442688755928e+19,8.204408941869605e+19,7.9432963922112e+19,8.087418588798737e+19,7.855674912139e+19,7.64810732518236e+19,7.24325
88932646396e+19,7.0259525916152635e+19
33.672421875,8.13898705958869e+19,8.507405151950094e+19,8.31765294436218e+19,8.847918772138395e+19,9.191945157497235e+19,9.314667
337112353e+19,9.453021770939061e+19,9.723801162857046e+19,9.775908676895675e+19,1.0401198845359237e+20,1.0234275119352729e+20,1.0
368856250438707e+20,1.025116514114456e+20,1.0387812604683793e+20,1.0588516374273968e+20,1.0254129879012275e+20,1.00586368608259
67e+20,9.541784225887624e+19,9.398219040657052e+19
30.120390625,9.821817193574538e+19,1.0139313667044396e+20,1.0608730496099297e+20,1.1013192613688802e+20,1.1560770713801646e+20,1.
1512795864669528e+20,1.1884016981924045e+20,1.2008715653450973e+20,1.2103497424676412e+20,1.2470406195941525e+20,1.2377511076086
451e+20,1.2606963818939313e+20,1.2431325560412552e+20,1.2364214675952455e+20,1.2443993076756426e+20,1.2490260955601071e+20,1.222
9813225949369e+20,1.2251195274813494e+20,1.1318560614063464e+20
26.568359375,1.1797231018887293e+20,1.2120567127551137e+20,1.2473550614892133e+20,1.299444607419554e+20,1.3354347283227872e+20,1.
3907675177993255e+20,1.3974696221911908e+20,1.4129671155906111e+20,1.4380416107081654e+20,1.4648410442214816e+20,1.4873011795829
6e+20,1.5096535062947034e+20,1.5330569673413364e+20,1.5442151625889464e+20,1.5054310008467453e+20,1.5199492323444405e+20,1.4822251
88991266e+20,1.464966820979506e+20,1.3490006500811204e+20
23.016328125,1.3573648044897347e+20,1.4248709873321945e+20,1.4562702565675414e+20,1.5104171508969937e+20,1.5467666339660104e+20,
1.5975534920453856e+20,1.6524101266522604e+20,1.6334088521364498e+20,1.682192266141581e+20,1.7002322468639208e+20,1.726178195233
5004e+20,1.7393308505011182e+20,1.8292702165426867e+20,1.766229108610089e+20,1.7879345834234216e+20,1.772392169753279e+20,1.720347
5440936606e+20,1.6893256051323866e+20,1.6265989390948496e+20
19.464296875000002,1.532526908146833e+20,1.6152430946560862e+20,1.6690126587114655e+20,1.725261821710752e+20,1.7981943732565452e
+20,1.7866229115183114e+20,1.883794441146212e+20,1.9056167086634248e+20,1.9204134458395666e+20,1.9520732526451065e+20,1.96270138
8698158e+20,1.998359099598041e+20,2.0051061242606295e+20,2.021717640373979e+20,2.0450761811499165e+20,2.0019437372017335e+20,1.96
8523055783853e+20,1.9575086054025842e+20,1.849053103769077e+20
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7966e+20,2.2632359679430292e+20,2.2964410320614392e+20,2.278266290526931e+20,2.315810652797178e+20,2.323743572606852e+20,2.225125
6102616724e+20,2.15086341870248e+20,2.085872771020506e+20
12.360234375000001,1.9517947469666242e+20,1.9929506990029973e+20,2.1629829077435338e+20,2.187850769615763e+20,2.241296907721937e
+20,2.306368411891213e+20,2.3532382223635458e+20,2.3871530267593782e+20,2.4269883228364967e+20,2.464218243211683e+20,2.523090750
0211906e+20,2.518904180789811e+20,2.5979548732080708e+20,2.5895188463662993e+20,2.56300690258561e+20,2.562099513117006 2e+20,2.504
942960649116e+20,2.4548927950095974e+20,2.3798669588481147e+20
8.808203125,2.090724160258585e+20,2.2142908409632955e+20,2.3224049485393 08e+20,2.3785193507264263e+20,2.426305534721508e+20,2.509
9470788076536e+20,2.5411576829059636e+20,2.6087896425064477e+20,2.6662247006528206e+20,2.6772661031965236e+20,2.768562061413861
6e+20,2.7664238565274485e+20,2.858025272585702e+20,2.8648711218438812e+20,2.8241194522440142e+20,2.8159978672973038e+20,2.7354109
01620319e+20,2.6691714704122465e+20,2.613317605795322e+20
5.256171875,2.3046165213330175e+20,2.4208162776391624e+20,2.4604718926333886e+20,2.587138072017983e+20,2.6371702695492125e+20,2.7
014781291161974e+20,2.7594701986195348e+20,2.7783996007021887e+20,2.8861184098958393e+20,2.8475408813989642e+20,2.9568768203386
415e+20,3.009963596279032e+20,3.0497180358688486e+20,3.035460341941383e+20,3.0235834223622324e+20,2.932170671441015e+20,2.9456467
52657902e+20,2.8844204236625117e+20,2.772856439294976e+20
1.704140625,2.4124790753929816e+20,2.5123637893725484e+20,2.6339180419488704e+20,2.6873372278926102e+20,2.7464792563264556e+20,2.
8533715325388037e+20,2.9226206218853145e+20,2.9883210098447116e+20,3.008714812752934e+20,3.0998221058332354e+20,3.1707781654672
18e+20,3.1437541306002866e+20,3.159790667248383e+20,3.2303693966082926e+20,3.2329118839312115e+20,3.2022043868649986e+20,3.136360
254039288e+20,3.057246673242016e+20,2.951935590559116e+20
−1.847890625,2.443860376520039e+20,2.596939675089732e+20,2.6857740024714517e+20,2.745706627670021e+20,2.8652125159013754e+20,2.94
41643837240446e+20,3.0512722772358634e+20,3.134392746181623e+20,3.1430354062687194e+20,3.1965084965373274e+20,3.234870407734733
e+20,3.3032660119375066e+20,3.291038714246718e+20,3.355885617062379e+20,3.3883360206326425e+20,3.379756248924558e+20,3.2539076184
67122e+20,3.216462080792465e+20,3.092796575492165e+20
−5.399921875,2.544203277260 98e+20,2.6809585494499505e+20,2.8067263234200856e+20,2.8875109382882512e+20,2.9840266319635966e+20,2.9
864882627992145e+20,3.055979921607629e+20,3.137815670810712e+20,3.236613314238784e+20,3.290194213157126e+20,3.362848259024437e+
20,3.3508006424165404e+20,3.377213761601639e+20,3.4755622023224807e+20,3.436059316248712e+20,3.376369260512047e+20,3.30864746037
01174e+20,3.237529687761532e+20,3.137618021619531e+20
−8.951953125,2.4816922285229136e+20,2.5586496363254835e+20,2.6864298384240067e+20,2.7915522559698705e+20,2.8782214263027437e+20,
2.9751324183604514e+20,3.033429945704705e+20,3.094817987674698e+20,3.2216279119256053e+20,3.219956877854711e+20,3.31090245796040
93e+20,3.3603866281888819e+20,3.3864942895330014e+20,3.3718233291148834e+20,3.432169220804104e+20,3.3556520316546194e+20,3.2983966
54591138e+20,3.2636373491057145e+20,3.10576056562281e+20
−12.503984375,2.2226729634801992e+20,2.33500957650417e+20,2.4424498800193382e+20,2.5201889005324875e+20,2.6758017023709548e+20,2.
7566761577805667e+20,2.8002039001111118e+20,2.8893526693878923e+20,2.9407504431490997e+20,3.0307706656779056e+20,3.034382255444
0316e+20,3.0734000025939916e+20,3.1732487803569806e+20,3.186302611029072e+20,3.1694575095079626e+20,3.146736836576292e+20,3.11196
85470367226e+20,3.003773582973409e+20,2.8357627864154046e+20

A.62 "MITR_Cycle7_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.351082939251457e+19,5.0507186784570835e+19,5.202990100834658e+19,3.997800472492705e+19,4.144279220027577e+19,4.266
7058896179446e+19,4.25403739999861e+19,4.463614358194073e+19,4.463667453288872e+19,4.567266602263079e+19,4.585706528687199e+19,4.
6803219876212064e+19,4.6783415405851574e+19,4.81341546175694e+19,4.8373932065687224e+19,4.910531699656038e+19,6.397321782289964
e+19,6.237558642036077e+19,6.672619848829151e+19
37.224453125,7.193164158250593 5e+19,6.663646777807913e+19,6.946590537998332e+19,5.2064465915061535e+19,5.477449264876001e+19,5.66
3016621202801e+19,5.7016167551225676e+19,5.821080718423083e+19,5.919041168329503 5e+19,6.090963085292645 6e+19,6.204745873449535e
+19,6.367694719391438e+19,6.464911837970656e+19,6.338545512346111e+19,6.3097679709643874e+19,6.5122726625324605e+19,8.2197047211
18216e+19,8.209351177632173e+19,9.052819853628606e+19
33.672421875,8.917214981508823e+19,8.486560667584168e+19,8.738868558074854e+19,6.905601124812555e+19,7.07380638513968e+19,7.21934
0039987107e+19,7.284116055643385e+19,7.251674952720446e+19,7.458214871493335e+19,7.679993082473891e+19,7.677656898302681e+19,7.92
8371935949362e+19,8.007536722296503e+19,8.178343642268839e+19,8.198201207724124e+19,8.267012450585222e+19,1.0907166039526597e+2
0,1.0610311364498517e+20,1.1510963457594797e+20
30.120390625,1.071772274127938e+20,1.000523966415511e+20,1.044152205812859e+20,8.293506902890535e+19,8.695224390149066e+19,8.8868
4458728309e+19,8.880207700433063e+19,9.053297709481809e+19,9.241466725453 82e+19,9.166602641785497e+19,9.452625917474 33e+19,9.8023
10211828636e+19,9.876749534738557e+19,9.84526414352202e+19,1.020450555494037e+20,1.0117854360226395e+20,1.3587300234852545e+20,1.
3077534229675547e+20,1.4226671366436898e+20
26.568359375,1.277059148663 5426e+20,1.2330220770362329e+20,1.2506231009625088e+20,9.741941089040776e+19,1.0017345345769564e+20,1.
0261317306376615e+20,1.0634575822822221e+20,1.0750535509865923e+20,1.0709121335921744e+20,1.1123528550837 53e+20,1.1272301006467
772e+20,1.1491212082329113e+20,1.1728600251180935e+20,1.1960944386026737e+20,1.1742404975828997e+20,1.2158829804347189e+20,1.6046
93050147314e+20,1.559105601751838e+20,1.7219217099567396e+20
23.016328125,1.4708296971369775e+20,1.4096482193986737e+20,1.4693961295773714e+20,1.1625595767268491e+20,1.176226254128428e+20,1.
2123096805546638e+20,1.2338875270814766e+20,1.2519133117661543e+20,1.2686860522135465e+20,1.3057304998556662e+20,1.314321286194
3431e+20,1.3376512708495637e+20,1.3687119013076974e+20,1.3898649870761086e+20,1.4055121115137362e+20,1.4133542570157297e+20,1.881
8653735329487e+20,1.8629210437082268e+20,2.0100103848333007e+20
19.464296875000002,1.694710473871622e+20,1.6047886213179 55e+20,1.6822171980651282e+20,1.312229339459214e+20,1.361406016263186e+2
0,1.3774566634212953e+20,1.4245838695659782e+20,1.4280350507279932e+20,1.4808381225068206e+20,1.5097802586824252e+20,1.525634453
9897738e+20,1.5598914091548813e+20,1.5720395668451738e+20,1.604353241540593e+20,1.623244476270514e+20,1.6584890001989062e+20,2.14
80417027854552e+20,2.1448082115121214e+20,2.298815843489665e+20
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671285739884e+20,1.6084946589350108e+20,1.6396880271301453e+20,1.668927495836631e+20,1.698607653829959e+20,1.7249056542845125e+
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5506104e+20,2.4175948800672566e+20,2.5902229617912404e+20
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e+20,1.7114513572621343e+20,1.7646207851950834e+20,1.8037518700628522e+20,1.8449749016657494e+20,1.8867235747071692e+20,1.9226211
683016042e+20,1.9537667509114185e+20,1.9739428869355 05e+20,2.0392020679544665e+20,2.0450584569109316e+20,2.067607943672589e+20,2.
742818264247098e+20,2.6894576939728683e+20,2.9325164189493556e+20
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848e+20,2.9166038190377 27e+20,3.159030712386092e+20
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8456951003696e+20,2.0912087133112903e+20,2.1554803255669673e+20,2.191388538180362e+20,2.23963074131585e+20,2.2642296987367958e+
20,2.304130662479168e+20,2.336624860496908e+20,2.3747683766013926e+20,2.4048467478057216e+20,2.4299607276462303e+20,3.1692939942
10976e+20,3.1352918955009094e+20,3.40234429381762e+20
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2808370860645e+20,2.2119097923206506e+20,2.250270998313816e+20,2.311808213187281e+20,2.350482680239768e+20,2.405292746602044e+2
0,2.464483158285339e+20,2.51112719906734e+20,2.5438231584453208e+20,2.5817171276042437e+20,2.596520040034548e+20,3.3998010387766
896e+20,3.3706518317313635e+20,3.6489975567120826e+20
−1.847890625,2.7589326255189675e+20,2.6340529625488297e+20,2.6809624788048318e+20,2.126570046448243e+20,2.1979670204261106e+20,2.
2492356439652113e+20,2.3164009388875013e+20,2.4037636078717973e+20,2.4003389742571825e+20,2.4659061068259856e+20,2.508005207493
0864e+20,2.5762430233303404e+20,2.5954103525532228e+20,2.6497744201191775e+20,2.66656308909501e+20,2.7091612836532332e+20,3.53043
6210023172e+20,3.511709570087131e+20,3.7916269098739374e+20
−5.399921875,2.7878906902230126e+20,2.6482187338415306e+20,2.7812644223919437e+20,2.1638428029980036e+20,2.2283002480854814e+20,
2.2968247774346566e+20,2.3422476310362525e+20,2.4059139592112062e+20,2.4595028383930778e+20,2.4977737827250825e+20,2.58242329236
5087e+20,2.6484364237302114e+20,2.666191423431408e+20,2.7257004056835044e+20,2.7423032418275357e+20,2.7633129708399863e+20,3.6315
451990512475e+20,3.618499734258831e+20,3.902590348496935e+20
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3638149535397e+20,2.324683773676337e+20,2.3692093201758087e+20,2.4398682723359533e+20,2.5124333383994255e+20,2.5340058754167587
e+20,2.5903503900187617e+20,2.6229348496976616e+20,2.7099789481131563e+20,2.7527204994273408e+20,2.756155752060916e+20,3.63490611
8552102e+20,3.616466192127983e+20,3.845391002868649e+20
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1252161215308372e+20,2.192248678716126e+20,2.2518744701767827e+20,2.3040510198369678e+20,2.3543798601981044e+20,2.4020964218950
7e+20,2.4550959455246585e+20,2.5219320508591863e+20,2.5373402473702126e+20,2.5800605606464766e+20,2.6245117740132286e+20,3.452508
539384876e+20,3.429528982355337e+20,3.65064881416037e+20

A.63 "MITR_Cycle7_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.490085897438456e+19,5.229086339928971e+19,5.295465827448217e+19,4.201080352444862e+19,4.207892453107731e+19,4.3660
94597574492e+19,4.427971621054679e+19,4.588977186526893e+19,4.700423790512573e+19,4.719851285699977e+19,4.829561680085689e+19,4.9
37058009018233e+19,5.037874975024907e+19,5.024776415137690 4e+19,5.119014898898617e+19,5.165786367908138e+19,6.893389253008503e+
19,6.8016940242885075e+19,7.406978105011115e+19
37.224453125,7.215782668635491 5e+19,6.834028937021846e+19,7.108902242802631e+19,5.56314474788357e+19,5.656804495111174e+19,5.7682
510990968545e+19,5.804939809603812e+19,6.069459571898552e+19,6.076574314601783e+19,6.337005754596905e+19,6.318156995942824e+19,6.
399180110607973 6e+19,6.67713293188717e+19,6.770845774209574 5e+19,6.936555565081088e+19,6.933476049582675 6e+19,9.140904615902187
e+19,9.170000727852712e+19,9.982196393011811e+19
33.672421875,9.207485864781673e+19,8.625616720865966e+19,8.866190595405801e+19,6.935759138659085e+19,7.097911558178984e+19,7.3419
89708975635e+19,7.646543172749748e+19,7.647976740309352e+19,7.814482957602 87e+19,7.855950226641 85e+19,7.980776794517186e+19,8.236
323485790688e+19,8.350743415085 18e+19,8.396299006423777e+19,8.582450408793381e+19,8.672712069953769e+19,1.149264564988872e+20,1.1
44836434082533e+20,1.2824270627478456e+20
30.120390625,1.109427315360260 3e+20,1.047837005391995e+20,1.0925802417801477e+20,8.578733752157364e+19,8.8657658346474e+19,8.8965
07894536733e+19,9.176584519607 94e+19,9.293393728168444e+19,9.569966076982836e+19,9.759727945798853e+19,1.000316895545790 1e+20,1.0
220911939233 64e+20,1.032688974845 4896e+20,1.040355906534 6426e+20,1.061009898411 9317e+20,1.074108458299 1482e+20,1.42911288115243
8e+20,1.4424291309283351e+20,1.5509661237189627e+20
26.568359375,1.3124788864047751e+20,1.230197417992 8606e+20,1.301817391368 8891e+20,1.006582116732 2171e+20,1.039586027660 0393e+20,
1.053215538495 2583e+20,1.100045412109 0599e+20,1.133898844553 6856e+20,1.145473575220 1355e+20,1.187424009621 7965e+20,1.18822043604
38e+20,1.203485275798 8655e+20,1.231758413779 9873e+20,1.249078033703 8221e+20,1.274898178305 1733e+20,1.280515639335 0375e+20,1.71818
38152828033e+20,1.7228243265683433e+20,1.8470456103629586e+20
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97757730948219e+20,1.282671300183 9265e+20,1.305788904459 9466e+20,1.315351331033 4676e+20,1.380127346689 7467e+20,1.39266840808156
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1963417e+20,2.029103380923463e+20,2.2380856740571637e+20
19.464296875000002,1.794906227269134e+20,1.678086399689 6707e+20,1.746218025337324e+20,1.375826644010928e+20,1.425667009499903e+2
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5716e+20,1.6226126446423917e+20,1.655627174589174e+20,1.704076448594 3824e+20,1.717456412484 04e+20,1.740467826570 4592e+20,2.335876
219660225e+20,2.323935132839 6532e+20,2.513282859916 2293e+20
15.912265625,1.978270137161 7244e+20,1.896535548226 2517e+20,1.942675185607 6505e+20,1.554820827601 4593e+20,1.597811925861 2046e+20,
1.654958176394691e+20,1.669676136673314e+20,1.695103377573144e+20,1.723514562800 7462e+20,1.794221300546211e+20,1.803035086283049
e+20,1.850863147679 0952e+20,1.881148589753 1454e+20,1.913345455240 0042e+20,1.936638273328 8646e+20,1.976979926358 0783e+20,2.6355927
202980364e+20,2.631116803806377e+20,2.888022729506764e+20
12.360234375000001,2.191144300744281e+20,2.097723481443 2785e+20,2.203425196171 5737e+20,1.712661925423 5795e+20,1.759523656094 2616
e+20,1.799828142557 1152e+20,1.865666060109 3987e+20,1.889144711030 0598e+20,1.954802105260 0228e+20,1.979454157775 7688e+20,2.0151075
13934123e+20,2.035782743849 3316e+20,2.108029239344003e+20,2.150930075942 5883e+20,2.184395914195172e+20,2.201030607396084e+20,2.88
58723781673543e+20,2.938924996891743e+20,3.196712301165814e+20
8.808203125,2.393011851174 7508e+20,2.281591794736471e+20,2.373048095529865e+20,1.846052732090 1943e+20,1.945314011819 2216e+20,1.98
560787926311 57e+20,2.051249344964638e+20,2.096682817585194e+20,2.113567057731667e+20,2.167532912086 6196e+20,2.234342469873 7472e
+20,2.253069109809 7877e+20,2.284474858384 1232e+20,2.345093528117 5446e+20,2.366294399471 2755e+20,2.426536094031 6154e+20,3.2317338
25696045e+20,3.247864115496354e+20,3.5233745624147414e+20
5.256171875,2.586543471721585e+20,2.461562928071326e+20,2.544253228713 2028e+20,1.998759534245132e+20,2.083817876115 0985e+20,2.119
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+20,2.431304033544676e+20,2.499648039571 5304e+20,2.521857717726466e+20,2.608551388516 2793e+20,2.615151108799 9484e+20,3.467295523
28674e+20,3.430734241007303e+20,3.7567009565143466e+20
1.704140625,2.770332142372577e+20,2.625881627459 0748e+20,2.726263213688 3864e+20,2.128380589180 9306e+20,2.201683677062 1266e+20,2.2
574972407161268e+20,2.340176922339 0432e+20,2.408658975612 378e+20,2.411791586205 5918e+20,2.500582513240 0145e+20,2.52660441920160
66e+20,2.593931070879 8564e+20,2.664991474288 9228e+20,2.703692488888 8096e+20,2.739664415615965e+20,2.809696845657466e+20,3.6893923
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342704248851 5343e+20,2.410209352380545e+20,2.501989533252 2207e+20,2.520769268283 0617e+20,2.603374616773 2575e+20,2.660244772813 7
827e+20,2.728450731594 1563e+20,2.780690995368 1015e+20,2.828115534043 6655e+20,2.889424440009 4898e+20,2.939838232522 3073e+20,3.8692
57247981352e+20,3.874880018520 6964e+20,4.2162018354439664e+20
−5.399921875,2.902268143441665e+20,2.743258955335 94e+20,2.875014431280 7077e+20,2.248566645770729e+20,2.336449646684067e+20,2.4301
624890064712e+20,2.457384344110 5483e+20,2.536527892419 7698e+20,2.602535714275 414e+20,2.666531232038 1295e+20,2.696083961803937e+
20,2.767226079326 7637e+20,2.863216701216 0972e+20,2.885267094086 632e+20,2.926336149914 609e+20,2.989434360575 201e+20,3.97014323761
1267e+20,3.962354187204073e+20,4.340216048368861e+20
−8.951953125,2.868807614698561e+20,2.722148343841 3693e+20,2.846189104313 6638e+20,2.210858509443 606e+20,2.316650485833 0618e+20,2.3
709455297757 76e+20,2.431638532641 9175e+20,2.514435023473 394e+20,2.571076870606 278e+20,2.632757442235 704e+20,2.713031916064 17e+2
0,2.752274500631019e+20,2.809548179392026e+20,2.875040978828 1078e+20,2.940804563247 6715e+20,2.974519948445 8166e+20,3.95820215079
06955e+20,3.957485367010892e+20,4.3113535548354567e+20
−12.503984375,2.630861947351336e+20,2.536841153479 0908e+20,2.660945628065 1455e+20,2.111607848733 5384e+20,2.192519463699 6074e+20,
2.259339640505 6953e+20,2.316862866212 2627e+20,2.381097311901 58e+20,2.436523281363 5384e+20,2.510235201474 6957e+20,2.5970297529446
305e+20,2.621671186441 4165e+20,2.693620349405 2063e+20,2.754642541859 1093e+20,2.817464658026 7398e+20,2.878364731762 602e+20,3.76779
25218181146e+20,3.732478974266 4824e+20,4.081780983938 227e+20

A.64 "MITR_Cycle7_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
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392949219e+19,4.635424775458133e+19,4.725532460843601e+19,4.868225528119216e+19,4.85508449215616e+19,4.92935391076272e+19,5.03321
8535210927e+19,5.183244035078454e+19,5.320075403888123e+19,5.376621679850367e+19,5.390638784877627e+19,7.137308118520754e+19,7.08
315112182452e+19,7.717690599782053e+19
37.224453125,7.284965577160189e+19,6.971545232554439e+19,7.177501105284527e+19,5.650910939588348e+19,5.7833301060201185e+19,5.851
344922459213e+19,6.1823397434438386e+19,6.133598446417229e+19,6.244195528886105e+19,6.393976791317551e+19,6.5929772066288075e+1
9,6.607312882224869e+19,6.877673104947634e+19,6.9312991506958655e+19,7.049010975867972e+19,7.026923416431078e+19,9.3897613252308
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8719320381e+19,7.80720892961524e+19,7.85812712552866e+19,7.994209853501645e+19,8.15747727001235e+19,8.376069775304892e+19,8.62014
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e+20,2.7142743412824952e+20,2.7380237771866374e+20,2.8306694081035567e+20,2.8545409627257396e+20,2.9138322550891548e+20,3.8619619
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24e+20,2.8704960387132083e+20,2.918244457467054e+20,2.9796648631319586e+20,3.069958381349228e+20,3.080094234946592e+20,4.09556978
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3727605e+20,3.9064875284552725e+20,4.216217763972 4076e+20

A.65 "MITR_Cycle7_cladding_fluence_position_4.csv" File

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7129053381e+19,4.780416860338598e+19,4.746839522386933e+19,4.90149491452104e+19,5.033314106381568e+19,5.082766877678501e+19,5.171
9135418480845e+19,5.341844392756217e+19,5.235463060814478e+19,5.333773938346581e+19,5.433167955812611e+19,7.09929203064379e+19,6.
986783524762104e+19,7.734893410497328e+19
37.224453125,6.971704517838839e+19,6.770261728166771e+19,7.2060131711922495e+19,5.783170820735718e+19,6.060751976351315e+19,6.143
261753670871e+19,6.255080023320152e+19,6.353836899648578e+19,6.585809368830778e+19,6.532395703461746e+19,6.7053264272260915e+19,
6.772597912337982e+19,6.913565389032589e+19,6.904963983674952e+19,7.144582146508385e+19,7.303017909392267e+19,9.627945920504685e
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33.672421875,8.471959516514102e+19,8.464791678716073e+19,9.015865667647649e+19,7.212172202189076e+19,7.520229942220002e+19,7.4302
86851628415e+19,7.744822193224971e+19,8.098116954025694e+19,8.198413588103325e+19,8.289206200211717e+19,8.55908856708128e+19,8.60
9582002236296e+19,8.763770157536161e+19,8.829979740752046e+19,9.06030626199544e+19,9.272421165722354e+19,1.2015419953291772e+20,
1.1784827956574378e+20,1.3081516361785567e+20
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6e+20,1.272057590733361e+20,1.3105037488782067e+20,1.3237191179739837e+20,1.3555655558351608e+20,1.3615493730191468e+20,1.7875419
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6e+20,1.495322464368323e+20,1.527577734459462e+20,1.577019886737435e+20,1.614616523365477e+20,1.6801411798584394e+20,1.6794403246
070763e+20,1.722803088530423e+20,1.7520319382179493e+20,1.778218438973422e+20,1.8084560954621523e+20,1.804187249840214e+20,2.3900
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3954329876342e+20,1.7673233255204153e+20,1.7876906038857826e+20,1.815655790317063e+20,1.903618433872602e+20,1.918214275433185e+
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674252e+20,2.7304736547060453e+20,2.9730173572629706e+20
12.360234375000001,2.127546996192567e+20,2.0547483117119734e+20,2.1967670712836253e+20,1.7567467826362096e+20,1.839745034827927e
+20,1.893148081177997e+20,1.9380983884358707e+20,1.9921545044519836e+20,2.0521625805952025e+20,2.0783703193885953e+20,2.15873505
48782216e+20,2.1911283722158414e+20,2.2405811435127744e+20,2.2662632408676447e+20,2.276048666839327e+20,2.3362903613996666e+20,3.
0624560444539555e+20,3.041865766690427e+20,3.322425557124316e+20
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476773561007768e+20,2.148556725205018e+20,2.191813298938764e+20,2.257651216491 0475e+20,2.3018210258553582e+20,2.3773222506612834
e+20,2.4105650895157068e+20,2.463092066801573e+20,2.490738682664052e+20,2.52964676813366e+20,2.5551642706946497e+20,3.33673468517
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e+20,2.5796570379259953e+20,2.645080813738837e+20,2.7040907020998112e+20,2.7449739250959873e+20,2.782007753719147e+20,3.655299944
46487e+20,3.638463489903717e+20,3.994577600238332e+20
1.704140625,2.649031088791974e+20,2.5815472233008834e+20,2.7413422206116518e+20,2.2113204367683682e+20,2.3282039784615916e+20,2.4
275502103423e+20,2.4787975958434805e+20,2.5404728579634263e+20,2.578005780477708e+20,2.6502416569534195e+20,2.6987599545818684e
+20,2.7613749998797783e+20,2.8248767332608518e+20,2.8906403176804152e+20,2.908756364026253e+20,2.9506218462762336e+20,3.88575920
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2.5050425012032337e+20,2.5602029951911916e+20,2.6622305293593113e+20,2.7179803788995514e+20,2.7925630585654326e+20,2.83251180789
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6105362428e+20,4.0360395597678313e+20,4.4364455981847957e+20
−5.399921875,2.8263156103299377e+20,2.8097711787901865e+20,2.976951703787667e+20,2.361027366067093e+20,2.470498832526205e+20,2.57
85420409351904e+20,2.629332808621089e+20,2.73157272316841e+20,2.7728647783945477e+20,2.8713667982679323e+20,2.922173494482271e+
20,2.9780932483258725e+20,3.028114137137168e+20,3.066316057845932 4e+20,3.142788922886701e+20,3.153349537242467e+20,4.15213198454
8531e+20,4.145797739738863e+20,4.5349423085487e+20
−8.951953125,2.8008034172784283e+20,2.736643304721832e+20,2.921462020211948e+20,2.3758090404594768e+20,2.461281524068885e+20,2.56
3409938917125e+20,2.6261417934235953e+20,2.714014175317974 4e+20,2.7811688512213038e+20,2.829819886586753e+20,2.9062449660422026
e+20,2.952761578596683e+20,3.024386861482192e+20,3.0875169291996633e+20,3.138169649639081e+20,3.16334203408387e+20,4.140132493123
6784e+20,4.142574867484489e+20,4.5128866061686846e+20
−12.503984375,2.6018083114766506e+20,2.5890442506866757e+20,2.7536124970199846e+20,2.2350645631630305e+20,2.32313870641765e+20,2.
4241096481992452e+20,2.4869370738763556e+20,2.5678858554087843e+20,2.637870499864966e+20,2.7130372255736503e+20,2.7564530845917
967e+20,2.8490243823759956e+20,2.879925727549729e+20,2.9159985349570042e+20,2.989460908122601e+20,3.000196736291207e+20,3.9727767
543133576e+20,3.942857668393429e+20,4.277850550016512e+20

A.66 "MITR_Cycle7_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,4.222541384174532e+19,4.5163199547026694e+19,4.962513003793801e+19,5.3129450197577335e+19,5.5978563288451604e+19,5.6
29399342946821e+19,5.907671436021394e+19,6.243764901570558e+19,6.203839764951994e+19,6.210811390968197e+19,6.514014234294012e+1
9,6.705194906490917e+19,6.670076238839709e+19,6.738570667638074e+19,6.642585033157259e+19,6.766196634132692e+19,6.82940843909403
7e+19,6.779214528588205e+19,6.4668749021973406e+19
37.224453125,5.898354971873454e+19,6.147222255732779e+19,6.619630774817829e+19,7.249529763057564e+19,7.575498199585774e+19,7.8425
67177142043e+19,8.335117945619264e+19,8.355916030963994e+19,8.341227102437587e+19,8.695243756005212e+19,8.495438391829499e+19,8.8
9470772612343e+19,8.947039841515672e+19,9.184757914181562e+19,9.204253311675325e+19,9.37279416742786e+19,9.608086545474709e+19,9.
184039189849996e+19,9.09617514031589e+19
33.672421875,6.981751045223874e+19,7.728164231664815e+19,8.410512128000677e+19,9.21449513340016e+19,9.629917797046067e+19,9.84751
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1.155457171644188e+20,1.1594191395219528e+20,1.1877368781857047e+20,1.2028031369861844e+20,1.2011321029152906e+20,1.204986262143
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1.6652932602956048e+20,1.761835906133384e+20,1.8450821518371675e+20,1.853482242462361e+20,1.850607345136091e+20,1.93457231517144
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e+20,1.9773364128996973e+20,2.0340617307686473e+20,2.096671604102305e+20,2.0946861281363503e+20,2.1768902235593615e+20,2.27304655
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A.67 "MITR_Cycle7_cladding_fluence_position_6.csv" File

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8813355645805e+19,5.35410935965507e+19,5.35915339366109e+19,5.374497876058358e+19,5.3574543506274845e+19,7.014923925006226e+19,6.
935546758279884e+19,7.53100824646445e+19
37.224453125,7.092920619267762e+19,6.926520592163846e+19,7.273178466114539e+19,5.789064376258544e+19,6.049017960400463 5e+19,6.10
8749942050721e+19,6.205276824397537e+19,6.488326774777557e+19,6.521935969786102e+19,6.635984233416993e+19,6.710742126895715e+19,
6.741590376974648e+19,6.867850512409592e+19,7.120742448943081e+19,7.165076853101273e+19,7.1596611534316495e+19,9.599752425165763
e+19,9.533595937044678e+19,1.0249424005141361e+20
33.672421875,8.620838162333945e+19,8.50848894173666e+19,9.017989471439657e+19,7.3038674309090705e+19,7.57767883479385e+19,7.58856
3329227897e+19,7.7142925137148 4e+19,8.05054374908469e+19,8.472543562556906e+19,8.424120836099098e+19,8.686569889696629e+19,8.7278
24778356405e+19,8.758035887297738e+19,8.883287215931477e+19,9.091950938496375e+19,9.090145705273167e+19,1.2067718621669997e+20,1.
208964689582249e+20,1.3174645158065165e+20
30.120390625,1.0808143687724171e+20,1.0305810995819205e+20,1.105854015480205e+20,8.902189069680358e+19,9.03779394180014e+19,9.317
764376681749e+19,9.647272535011969e+19,1.0003965381879905e+20,1.0108084862783154e+20,1.0106863675602747e+20,1.0434407315425359e
+20,1.0589922848095229e+20,1.0670149536338375e+20,1.1109245970336267e+20,1.1157403221320075e+20,1.1210710696499505e+20,1.4575877
804938004e+20,1.4576674231360009e+20,1.5953960990477943e+20
26.568359375,1.2807917338279985e+20,1.2510797187777908e+20,1.3127390523692963e+20,1.0394161233566784e+20,1.0786905649804078e+20,
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12394e+20,1.2831438465276486e+20,1.3125532195374953e+20,1.3197635334113665e+20,1.3418723308861818e+20,1.3589954989592556e+20,1.79
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7368587201793e+20,1.3520612795783458e+20,1.3800636325759866e+20,1.4051563743785745e+20,1.4581134219323227e+20,1.465716639507715
7e+20,1.4979347430324945e+20,1.5071148849234 54e+20,1.5511413375318036e+20,1.5651053474642638e+20,1.589114949332927e+20,2.05747739
9584705e+20,2.121717154783502e+20,2.2560052685522407e+20
19.464296875000002,1.731170875470939e+20,1.6397995268292254e+20,1.7562848553114475e+20,1.4115065477166819e+20,1.486604249802125e
+20,1.5199692073745895e+20,1.555245588359861e+20,1.596250930074078e+20,1.6313361687180693e+20,1.6672231432935437e+20,1.717482960
0314402e+20,1.7179873634320423e+20,1.7721071935619154e+20,1.7912798322942784e+20,1.8143231034375776e+20,1.797438863291105e+20,2.3
768709423554816e+20,2.3866935348935 24e+20,2.6005393387109253e+20
15.912265625,1.960345233157167e+20,1.883325488639955e+20,1.987731683055 1248e+20,1.5968243570979203e+20,1.6603738760643143e+20,1.6
975298234055144e+20,1.7767423953379757e+20,1.8057216980799406e+20,1.8414600063899746e+20,1.879162833207617e+20,1.91249062421372
04e+20,1.959501021149843e+20,1.9820611269304602e+20,2.0113643097507067e+20,2.0494653497793505e+20,2.0749669238119006e+20,2.722328
8671636898e+20,2.7230987460382936e+20,2.936296789699132e+20
12.360234375000001,2.173330896438804e+20,2.095031560136907e+20,2.2200492703535255e+20,1.7818235959103576e+20,1.8366867573674338e
+20,1.8965249292072914e+20,1.958651496330394e+20,2.0219939477297123e+20,2.0479468300680646e+20,2.118637639285089e+20,2.15850674
59705805e+20,2.188797497554111e+20,2.2335513529612236e+20,2.2583149051760506e+20,2.2925028367179176e+20,2.3195813350660345e+20,3.
059132291519461e+20,3.014845672946591e+20,3.294683370091196e+20
8.808203125,2.392337543470788e+20,2.341106086498047e+20,2.448719224639151e+20,1.949041287474198e+20,2.0097714569067e+20,2.0848903
970300635e+20,2.1492204138900206e+20,2.2057560708333044e+20,2.2589785938610536e+20,2.314155016377451e+20,2.3777841779860452e+20,
2.414143698905242e+20,2.4781285976489977e+20,2.5019736047237805e+20,2.551293638283713e+20,2.553656370002323e+20,3.35854615011691
13e+20,3.329465966694826e+20,3.5888195762655035e+20
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5638912398119e+20,2.311298500277199e+20,2.3769665135261218e+20,2.4307306065208335e+20,2.5040602419494293e+20,2.574575837353613e
+20,2.603087903261336e+20,2.67124076694691e+20,2.7117310862415646e+20,2.7588848399336473e+20,2.80361214779336e+20,3.651822215755
455e+20,3.67157359102114e+20,3.940033009350057e+20
1.704140625,2.7184369967148332e+20,2.6282655972156044e+20,2.7999167291952644e+20,2.2560211970806805e+20,2.350004824386566e+20,2.4
698723104075625e+20,2.5048301208240328e+20,2.5462018186923714e+20,2.6151988943852683e+20,2.671479694873511e+20,2.717375094818882
88e+20,2.7897755660884207e+20,2.8481642418402325e+20,2.876957711750396e+20,2.909637742599937e+20,2.9518005573807984e+20,3.8351542
685911646e+20,3.8523411507779994e+20,4.169844508173887e+20
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288981272969763e+20,2.5860337588115027e+20,2.672870786357277e+20,2.7266295698425083e+20,2.79661421429869e+20,2.8683669254117192
e+20,2.915472893518482e+20,2.968456488619631e+20,2.993023588983696e+20,3.0353828556153186e+20,3.0966386664863426e+20,4.0697708734
94415e+20,4.059045664344769e+20,4.372110271815358e+20
−5.399921875,2.908655483346133e+20,2.8374018661242257e+20,2.9748225904861774e+20,2.4046290579170407e+20,2.5440567768624418e+20,2.
581425104582843e+20,2.6683683223182174e+20,2.756283180288436e+20,2.814682475059208e+20,2.8855272600511532e+20,2.925030010582523
6e+20,3.0198737784241724e+20,3.0389773935332943e+20,3.100700441238560 4e+20,3.1379413407314405e+20,3.170775347355902e+20,4.1672375
39019196e+20,4.115655654420774e+20,4.512753868431685e+20
−8.951953125,2.882867195801662e+20,2.80586868932237e+20,2.9903051201299245e+20,2.4011672577360662e+20,2.5101502493230 16e+20,2.583
15069516385e+20,2.6485213758818917e+20,2.722711151846252e+20,2.7835103449019938e+20,2.8814920328463355e+20,2.932288110041715e+2
0,2.984645183024221e+20,3.055150159409444e+20,3.097493497512626e+20,3.1640641273731534e+20,3.1668463103406853e+20,4.176343347777
435e+20,4.1199988331754324e+20,4.497542123771419e+20
−12.503984375,2.697114006643 0616e+20,2.6243471792193474e+20,2.7773725519430867e+20,2.2369282009905185e+20,2.3486721375070798e+20
,2.42979613285235e+20,2.5163198993388025e+20,2.6047126131622232e+20,2.662437600229032e+20,2.7235500543440952e+20,2.7730718492642
686e+20,2.853431275244415e+20,2.9167896518695276e+20,2.957582613204544e+20,3.000010903459406e+20,3.001800208154174e+20,3.96437711
03159615e+20,3.933146575554466e+20,4.2557311335227366e+20

A.68 "MITR_Cycle7_cladding_fluence_position_7.csv" File

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40.776484375,5.6037094003109446e+19,5.402585181207678e+19,5.415805859812936e+19,4.349061691162579e+19,4.471281289883225e+19,4.577
147599405402e+19,4.691487886057695e+19,4.7264775535310455e+19,4.689210106490765e+19,4.9046593821711335e+19,4.9449532496150266e+
19,5.094707964499072e+19,5.10140325595338e+19,5.277912589107262e+19,5.232622473242667e+19,5.3053308960620995e+19,7.1315738482823
28e+19,7.039347668614332e+19,7.525114690941623e+19
37.224453125,7.508708306648352e+19,7.052302871745587e+19,7.1846689430825574e+19,5.569356873975198e+19,5.8800693687461356e+19,6.04
37084509204414e+19,6.093193079274254e+19,6.298511810866739e+19,6.1966754190399005e+19,6.551828508158631e+19,6.608268593931273e+
19,6.6854157666760065e+19,6.742705373965454e+19,6.952908854279559e+19,6.971067376701237e+19,7.039029098045529e+19,9.521277875051
025e+19,9.218848215068923e+19,9.910146349367902e+19
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9277166703e+19,7.64282651611373e+19,7.942123565502621e+19,8.097639098172491e+19,8.404316365738613e+19,8.456508843927237e+19,8.494
578026899e+19,8.735842137671241e+19,8.812299074183571e+19,8.8887560106959e+19,9.192247572574006e+19,1.1917512598480149e+20,1.1871
903912046753e+20,1.2914532288638845e+20
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6808664975344e+19,9.43876809773147e+19,9.745976316245595e+19,1.0018460342760368e+20,1.0022548665059983e+20,1.0108934384299958e+
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21154136e+20,1.512737655462798e+20,1.604900121017035e+20
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1.3051889298887036e+20,1.317082231123955e+20,1.34613055748916e+20,1.385766045757531e+20,1.4157913218670602e+20,1.448683733095801
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+20,1.491812878602028e+20,1.5219655829390777e+20,1.5581021044601137e+20,1.6087813724469317e+20,1.6490009067581053e+20,1.66282686
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4e+20,1.9170408738380998e+20,1.9510058059818064e+20,2.0026726427319094e+20,2.0435186991617252e+20,2.056495140330901e+20,2.7433651
437235408e+20,2.7209218471514836e+20,2.9500696572903116e+20
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854926e+20,3.378685119574638e+20,3.626331260741865e+20
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3255e+20,3.601036757579036e+20,3.883237186442252e+20
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e+20,2.7463490880513134e+20,2.7425421697541367e+20,2.8238519979312077e+20,2.8695243984783642e+20,2.8975108229475647e+20,3.8268077
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58154959382e+20,2.589023012648756e+20,2.6468648089241246e+20,2.7064746718563418e+20,2.8310835498429986e+20,2.839042504553553e+2
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133359550110292e+20,2.5599003531508302e+20,2.650730131825 5818e+20,2.674978661620846e+20,2.765261560819155e+20,2.820772482432794
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A.69 "MITR_Cycle7_cladding_fluence_position_8.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
40.776484375,5.563941174305574e+19,5.3441274818326266e+19,5.399027809856063e+19,4.177872486507682e+19,4.224447503666442e+19,4.455
225333215637e+19,4.387247683342904e+19,4.557051106023516e+19,4.57900592772341e+19,4.788211220255271e+19,4.819908991851007e+19,4.9
009639635730375e+19,4.907112375550904e+19,5.017725386548221e+19,5.106696836904964e+19,5.204413049375304e+19,6.870186696580802e+
19,6.889300930708885e+19,7.2178533773327e+19
37.224453125,7.25507303878766e+19,6.88346047028086e+19,7.077894707439297e+19,5.608275578463764 5e+19,5.678785864358469e+19,5.77048
1093078464e+19,5.885272688036558 4e+19,6.030593962504784e+19,6.216055128641983e+19,6.223116776250414e+19,6.399976537029976e+19,6.4
1256007449763e+19,6.665664391410321e+19,6.864983377290379e+19,6.92529940498344e+19,6.929334632188258e+19,9.253731692352673e+19,9.
098163064588003e+19,9.721393287353087e+19
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248157661451e+19,7.60104067650595e+19,7.603854716530364e+19,7.850162861308625e+19,7.934106206187785e+19,8.046189951311068e+19,8.2
11528076518982e+19,8.305347109030986e+19,8.500843248085428e+19,8.675632300167782e+19,8.957673443746596e+19,1.173651142030617e+2
0,1.14417805490701e+20,1.2438110002996394e+20
30.120390625,1.1222763283019157e+20,1.055503937081148e+20,1.1164836534592107e+20,8.674304922797775e+19,8.766318722086571e+19,8.91
663093546602e+19,9.286278985465212e+19,9.384133245182034e+19,9.53667545254309e+19,9.916092999985526e+19,9.945242207030852e+19,1.0
198718189607145e+20,1.0351897538105803e+20,1.0473166734629528e+20,1.0768800222477202e+20,1.0733704364814249e+20,1.4235485152173
74e+20,1.4387284028207592e+20,1.5367950429167816e+20
26.568359375,1.352406397694547e+20,1.2522637393918358e+20,1.3229704771373004e+20,1.0212629104444804e+20,1.0570808613967148e+20,1.
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17336e+20,3.434726992136279e+20,3.709387917537863e+20
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1388013567004e+20,2.3577673272463593e+20,2.3931445889117517e+20,2.4578462714353104e+20,2.5126669568165462e+20,2.539973764072304
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704517453941332e+20,2.4233397693246416e+20,2.4896767807680476e+20,2.544550561244084e+20,2.6140945164134233e+20,2.66520916417760
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36613543024699e+20,2.4718739954815307e+20,2.541529450349951e+20,2.61559710759627e+20,2.6818544763974756e+20,2.7124213224739678e
+20,2.7723763035223854e+20,2.8659404795793487e+20,2.9030645698636684e+20,2.962239053018524e+20,2.988239720942196e+20,3.962125878
296432e+20,3.936242019581321e+20,4.309548321612249e+20
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190435135529e+20,2.4457299708018983e+20,2.5297423393043002e+20,2.584154192455 5745e+20,2.628722215030887e+20,2.697448505740303e+
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358195e+20,3.930098917112933e+20,4.285915694916667e+20
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,2.2632208919355916e+20,2.350243752313167e+20,2.4176745227094573e+20,2.4564498704420648e+20,2.527889320495772e+20,2.615586488577
31e+20,2.6519619380249467e+20,2.702901371976286e+20,2.765962416070518e+20,2.8357506086759386e+20,2.850951734317244e+20,3.75930792
5669038e+20,3.7325957834750434e+20,4.016484636352906e+20

A.70 "MITR_Cycle7_cladding_fluence_position_9.csv" File

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921450266982e+19,4.319928412645694e+19,4.423819584641302e+19,4.377695875788343e+19,4.676106237094068e+19,4.635483180062414e+19,4.
743611340623079e+19,4.753656932559282e+19,4.868724622010338e+19,4.903735527521609e+19,4.849514816711616e+19,6.395675834351158e+
19,6.149155309193696e+19,6.542589961663391e+19
37.224453125,7.230224534421153e+19,6.769996252692771e+19,6.8853187985988665e+19,5.424938216118575e+19,5.617195554390203e+19,5.628
398619393052e+19,5.729332394608286e+19,5.8589906161104445e+19,6.028948014565977e+19,6.1140063564359434e+19,6.299520617667943e+1
9,6.330050297178075e+19,6.376667790412675 5e+19,6.323678885802047e+19,6.483282740771535e+19,6.307697262267179e+19,8.3609376732868
26e+19,8.187104332910877e+19,8.766796577939774e+19
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0097136476e+19,7.372997244338969e+19,7.418871406246366e+19,7.556069131210157e+19,7.68450616553191e+19,7.702877068332789e+19,7.995
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A.71 "MITR_Cycle8_cladding_fluence_position_0.csv" File

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9939539827481e+19,6.059273748486987e+19,6.126393367825016e+19,5.953852709837409e+19,5.8057077458210636e+19,5.526703344504193e+1
9,5.4476735208507924e+19,5.138736304160194e+19
37.224453125,6.442054231932895e+19,6.623484846198166e+19,6.861868709174531e+19,7.329341981344548e+19,7.597712592695316e+19,7.4414
59700911071e+19,7.457001393938452e+19,7.658710816456914e+19,7.887988991152923e+19,8.197070029178652e+19,8.351111667935286e+19,8.1
43253883340058e+19,8.107801924230752e+19,8.084098370434798e+19,8.120745844392364e+19,7.736329393259363e+19,7.916735280511838e+1
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5958839116e+19,9.667760035707761e+19,9.567624431124175e+19,1.0125201769752528e+20,1.0046549477103839e+20,9.919447372183562e+19,1.
0231539669413554e+20,1.0267225338192442e+20,1.0254640971368654e+20,1.0356754119310266e+20,1.0339585447429236e+20,1.005760574224
1884e+20,9.549377099229679e+19,9.369510541983939e+19
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1617258233381744e+20,1.1586875976335737e+20,1.1742382794944e+20,1.1908586325352482e+20,1.2346971732492653e+20,1.240809579992249
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9526612553e+20,1.6671499503153634e+20,1.6105562549420904e+20
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4e+20,1.8564547826789473e+20,1.886513441720914e+20,1.9304688372697235e+20,1.9513768638641062e+20,1.965929785212475e+20,1.99280639
7214712e+20,2.0026401810041592e+20,2.041426973217693e+20,2.053822598643202e+20,2.0580563391960633e+20,1.9829007027577e+20,1.9625
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48319392478196e+20,2.9786746830238666e+20,2.988364645478185e+20,2.963825130171795e+20,2.972903851951814e+20,2.885352614192018e+
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65793684696373e+20,3.0698214541618885e+20,3.0935070302909504e+20,3.1526805208631e+20,3.19918874511045e+20,3.165471630856138e+20,
3.2917287854325 34e+20,3.2872163910428 61e+20,3.3018502118922394e+20,3.305310912768782e+20,3.2812118503012244e+20,3.26638926394948
8e+20,3.1963842290754336e+20,3.0499111880799637e+20
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326782249817e+20,3.070684382172663e+20,3.157246848253447e+20,3.19489208272347e+20,3.255324009977999e+20,3.234487894051181e+20,3.3
2519421235037e+20,3.4025341353160055e+20,3.4222736135624 64e+20,3.4095903695707734e+20,3.3744080754648344e+20,3.369608038404903e
+20,3.2742814597146935e+20,3.095062098477033e+20
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416137227278 05e+20,3.0644191652611044e+20,3.137444448172869e+20,3.162999701658608e+20,3.1560513334051865e+20,3.2717376198495994
e+20,3.3104435366661987e+20,3.354839385053841e+20,3.43168492217997e+20,3.371217039591659e+20,3.350587666834089e+20,3.245786857692
2554e+20,3.20269439015422e+20,3.1258848083618726e+20
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,2.7050456041074 44e+20,2.8114464256025887e+20,2.8482377208952832e+20,2.9306653235911082e+20,2.9826926915740336e+20,3.02824809947
6154e+20,3.027304271964369e+20,3.039358297614871e+20,3.107017246959633e+20,3.10337676941418e+20,3.096185702657729e+20,3.057893272
1796245e+20,2.9468272461262324e+20,2.842493856323568e+20

A.72 "MITR_Cycle8_cladding_fluence_position_1.csv" File

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215107293364e+19,4.338498189856949e+19,4.423947082432106e+19,4.545132045902697e+19,4.61102092464747e+19,4.5453923502692614e+19,4.
739898147766729e+19,4.793913959995775e+19,4.825272667665725e+19,4.798131953200915e+19,4.906057330978774e+19,6.277638224742808e+
19,6.321411856997659e+19,6.9300990880366035e+19
37.224453125,6.91341835924046e+19,6.714896437230289e+19,6.9882691454626505e+19,5.447586035441945e+19,5.6504640713438044e+19,5.66
81010202620215e+19,5.837564475229227e+19,5.914752688416484e+19,6.07539766892458e+19,6.110193456699977e+19,6.195403294244616e+19,
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9.646136098101895e+19,9.75137343486993e+19,9.64432990453798e+19,1.0060763767700339e+20,1.0185975480351657e+20,1.3388197299150443
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A.73 "MITR_Cycle8_cladding_fluence_position_2.csv" File

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618407135488e+19,5.023709592331711e+19,5.006625126150691e+19,4.9844726933226906e+19,5.2641883283647 97e+19,6.7734914813772554e+1
9,6.778006965287041e+19,7.588456642083176e+19
37.224453125,7.266847941026294e+19,6.791340923655933e+19,7.020568136252638e+19,5.550804685287856 5e+19,5.644461133910797e+19,5.640
7425001027396e+19,5.900940619986554e+19,6.064507384200982e+19,6.166610443902226 4e+19,6.268872873623815e+19,6.294212706858724e+1
9,6.585222364009302e+19,6.701721848881741e+19,6.811049682838641e+19,6.849723474442441e+19,7.0140339654184804e+19,9.1436955472931
73e+19,9.071925914797659e+19,9.966735455696591e+19
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A.74 "MITR_Cycle8_cladding_fluence_position_3.csv" File

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A.75 "MITR_Cycle8_cladding_fluence_position_4.csv" File


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15214e+19,9.4956907988959e+19,1.0224489901935111e+20
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636e+20,2.738662240620317e+20,2.958916921071 58e+20
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37e+20,2.865021417418121e+20,2.9244451856708842e+20,2.976070447594749e+20,3.024641117461995e+20,3.059872516626337e+20,3.995730462
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32524087875528e+20,2.6035005263654398e+20,2.674271440066791e+20,2.765447028706356e+20,2.8100121987289232e+20,2.8853570320141854
e+20,2.9632624102929963e+20,2.999784706622 1353e+20,3.0687866130976526e+20,3.0921768197503366e+20,3.1438658296823395e+20,4.1656082
794523204e+20,4.157358224732443e+20,4.4731871063848026e+20
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2.559880951796922e+20,2.604690489184018e+20,2.7114205918092883e+20,2.7679650750278127e+20,2.7916740217211866e+20,2.8923427512393
22e+20,2.9494981528691707e+20,3.0267394893965427e+20,3.0792147247622495e+20,3.093451779913099e+20,3.154867673420179e+20,4.1398487
71830502e+20,4.10654574991234e+20,4.4770172992071015e+20
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59776354330906e+20,2.4628192970726018e+20,2.567822891144131e+20,2.628293189197162e+20,2.6617502688016594e+20,2.735352656531146e
+20,2.7830574159545156e+20,2.8692181612872142e+20,2.9013524697228457e+20,2.96443643610954e+20,2.981462466616433e+20,3.9407106190
749965e+20,3.8687391178870437e+20,4.233627404135708e+20
```

A.76 "MITR_Cycle8_cladding_fluence_position_5.csv" File

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55138162959e+19,5.848566503689514e+19,6.194987155848116e+19,6.267580974754493e+19,6.275850701524413e+19,6.570333874034545e+19,6.7
08761909096235e+19,6.849203442849731e+19,6.724833943296903 6e+19,6.617039852617696e+19,6.945716547554 6e+19,6.9827505413504795
e+19,6.613857805577967e+19,6.574693458255643e+19
37.224453125,5.831056256137555e+19,6.2825563712748044e+19,6.901806096173173e+19,7.2043612411174216e+19,7.678405350536146e+19,7.80
4428795442958e+19,7.831431251113432e+19,8.351992573612951e+19,8.501252152976546e+19,8.776616076747963e+19,8.803816286754241e+19,
8.98859074706126e+19,9.144160487503643e+19,9.160969606046846e+19,9.496792423573129e+19,9.34883622505914e+19,9.14892456922979e+19,
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4e+20,1.3961006665978175e+20,1.409053575592876e+20,1.451822445126871e+20,1.4568921471901691e+20,1.4809732319908354e+20,1.47408778
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A.77 "MITR_Cycle8_cladding_fluence_position_6.csv" File

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A.78 "MITR_Cycle8_cladding_fluence_position_7.csv" File

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2720051e+20,3.871315599882626e+20,4.146021703951878e+20

A.79 "MITR_Cycle8_cladding_fluence_position_8.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
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A.80 "MITR_Cycle8_cladding_fluence_position_9.csv" File

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37.224453125,7.184400517167641e+19,6.87570078775873e+19,7.117252615262142e+19,5.508996616617263e+19,5.697371980665448e+19,5.8016530973114106e+19,5.811852778613513e+19,5.8741133332284236e+19,6.0319427767104184e+19,6.0802850162151694e+19,6.323643037282496e+19,6.453476480523827e+19,6.444179896003682e+19,6.45799196443361e+19,6.533905217458105e+19,6.518127585443916e+19,8.515299557071523e+19,8.289153498201494e+19,8.817252622285816e+19
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26.568359375,1.305670765683215e+20,1.2047257947964808e+20,1.251697452126262e+20,9.914568335703551e+19,1.0183053696645323e+20,1.039597204382669e+20,1.0608996637688285e+20,1.1056826394858673e+20,1.1175928923396755e+20,1.1239836301555232e+20,1.1315749554579725e+20,1.1624608654008984e+20,1.1640598779383631e+20,1.191009348378759e+20,1.2135336445875662e+20,1.2003006205648919e+20,1.5882231870874688e+20,1.5697203277253755e+20,1.6722430618135285e+20
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−5.399921875,2.8317874558421074e+20,2.6579997609895323e+20,2.7539298885694e+20,2.1780569447195597e+20,2.2351964093473735e+20,2.3061107560670354e+20,2.338792234905852e+20,2.4176910196448153e+20,2.4439764483337722e+20,2.4892800327839393e+20,2.544995791896668e+20,2.5729439811312 28e+20,2.636946981301914e+20,2.7027083640304098e+20,2.7471779120407695e+20,2.7490372289447985e+20,3.602936485621084e+20,3.5626796184818536e+20,3.7933730352657354e+20
−8.951953125,2.7902662532081363e+20,2.632718363428751e+20,2.7551676623940823e+20,2.1662051275398788e+20,2.2373638416240702e+20,2.2947370489483906e+20,2.3526627390099076e+20,2.4020196343108574e+20,2.4409962289533144e+20,2.5095200253677964e+20,2.5446929888580118e+20,2.611670896075144e+20,2.6283303755352428e+20,2.6804549968561897e+20,2.7050032923233383e+20,2.7320749464460434e+20,3.558116323565965e+20,3.5192566002717616e+20,3.7526114963954106e+20
−12.503984375,2.567674145791809e+20,2.448194441538912e+20,2.5641467560081654e+20,2.0279888212283825e+20,2.095312030156263e+20,2.168909105551737 8e+20,2.2251560980656177e+20,2.2722977500837675e+20,2.301690894169458e+20,2.3799150124889594e+20,2.4061579425058246e+20,2.4852851576072828e+20,2.488855046063018e+20,2.5282194410883157e+20,2.5823149383275335e+20,2.5769707303119533e+20,3.388759115278991e+20,3.293477092448528e+20,3.540128760602989e+20

A.81 "MITR_Cycle1_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
top,1.7351741922925974e+19,1.7630798962727975e+19,1.8350908018827205e+19,1.8864825790376985e+19,2.0003287794031563e+19,2.045048464789598e+19,2.0937031723405165e+19,2.2324950871072793e+19,2.188764003348366e+19,2.193939623633663e+19,2.3230005968902427e+19,2.3112342990880883e+19,2.206209914422401e+19,2.304081475547734e+19,2.235887347593897e+19,2.1944630009658843e+19,2.235053820731471e+19,2.134526604253757e+19,2.084980216803499e+19
bottom,8.649818400925762e+19,9.107385879706411e+19,9.68895501753217e+19,1.0241350715173013e+20,1.0672633021268784e+20,1.1047991490199455e+20,1.15428738343329e+20,1.1906214008968125e+20,1.2227005544928344e+20,1.2539713808757593e+20,1.2986038367068309e+20,1.3215665325493877e+20,1.34492660747752e+20,1.3589376125045334e+20,1.390086317509940 2e+20,1.416466473488444e+20,1.4282928627620756e+20,1.4177807321226879e+20,1.3984913697784973e+20

A.82 "MITR_Cycle2_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
top,1.6423878479530918e+19,1.7192515945094017e+19,1.8381069249909336e+19,1.928549578846763e+19,1.9650686247655924e+19,2.0636668968166937e+19,2.16098272745593e+19,2.097254648079831e+19,2.1984118930864054e+19,2.2246714077103124e+19,2.312590547820214e+19,2.2193328267177554e+19,2.3306550451271324e+19,2.2929106865229267e+19,2.3049668177681113e+19,2.3093401202786198e+19,2.233536210096478e+19,2.2110590111573344e+19,2.075821525866169e+19
bottom,8.644738591818904e+19,9.326087303219564e+19,9.857404159124591e+19,1.0350306871356237e+20,1.084005765385058e+20,1.1390384585983718e+20,1.1739933593850642e+20,1.1988088961980695e+20,1.2367147912013121e+20,1.264666498102934e+20,1.2883768895517973e+20,1.3156391118233841e+20,1.3520242007283779e+20,1.4069603660934868e+20,1.4070568939416918e+20,1.4162349193455015e+20,1.4079906531263678e+20,1.4160339838247484e+20,1.3983122588767612e+20

A.83 "MITR_Cycle3_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
top,1.692584764926296e+19,1.7421721784418828e+19,1.8506496916828033e+19,1.872360554397125e+19,1.9597612999617294e+19,2.037473388423887e+19,2.14109475767809 4e+19,2.180400773077447e+19,2.16439085798743e+19,2.209584898739692e+19,2.2637389780724687e+19,2.306963779576266e+19,2.3419768534139666e+19,2.266988222832435e+19,2.272659631868013e+19,2.266732221730135e+19,2.3215558424073888e+19,2.1691564169687146e+19,2.1225051391725883e+19
bottom,8.602306578637803e+19,9.156292964015863e+19,9.801297587457955e+19,1.0375488367525257e+20,1.0759056788341191e+20,1.1165172998552042e+20,1.1641610742325566e+20,1.2139867656972075e+20,1.241798331603273e+20,1.2740662859286136e+20,1.2996467037661687e+20,1.3367550481642355e+20,1.3572075669987887e+20,1.3849580864881517e+20,1.3913167600214439e+20,1.3949519756741096e+20,1.400485537962295e+20,1.4049242032283342e+20,1.3842491603587044e+20

A.84 "MITR_Cycle4_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
top,1.6581950704007705e+19,1.6880647422116299e+19,1.7674618703733848e+19,1.887180790984937e+19,2.0452326868184547e+19,2.0387345711235613e+19,2.0725050814773252e+19,2.150285557219 2293e+19,2.1925626917553988e+19,2.1799012117801976e+19,2.310789014883093e+19,2.2359819193530946e+19,2.2335402031525892e+19,2.204889420316014e+19,2.2037473272544874e+19,2.1960283534593413e+19,2.2076068141520605e+19,2.2307440442778 17e+19,2.1653296796158915e+19
bottom,8.788701477343155e+19,9.088382758435824e+19,9.84566922326675e+19,1.0154900765304943e+20,1.0789648521137929e+20,1.0981953363187406e+20,1.1531398582080153e+20,1.1925440379566547e+20,1.2361621472771336e+20,1.2523089802159594e+20,1.2812413480659796e+20,1.3058081636446124e+20,1.3563930406404045e+20,1.3620444321689934e+20,1.3930660426712203e+20,1.4011670269041872e+20,1.4107940837624703e+20,1.3893936227406216e+20,1.3765568905542551e+20

A.85 "MITR_Cycle5_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
top,1.7247412768886639e+19,1.6957327434223856e+19,1.8490793201164755e+19,1.8570444551882863e+19,2.0063513454253326e+19,2.010605720241137e+19,2.101247539233405e+19,2.162679923726426e+19,2.2068978471592034e+19,2.196636137534047e+19,2.2709892159121023e+19,2.287356741245126e+19,2.300021384794024e+19,2.2457978020722258e+19,2.342506044412955e+19,2.2973820967138492e+19,2.2470977499326104e+19,2.213121839945287e+19,2.113872790422897e+19
bottom,8.535792486339889e+19,9.251413783481582e+19,9.633086353762366e+19,1.0165454394949244e+20,1.0604482240524571e+20,1.1113963320896 8e+20,1.150384919811365e+20,1.1885049060081567e+20,1.2201842414421343e+20,1.245344141394486e+20,1.2699111863377836e+20,1.3015984002436422e+20,1.3339749804390985e+20,1.371980728793614e+20,1.4029963029716071e+20,1.3981037719334327e+20,1.402643741354927e+20,1.3717955847044078e+20,1.3795972414846853e+20

A.86 "MITR_Cycle6_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
top,3.1393213609725227e+19,3.4807037927826457e+19,3.533573798216294e+19,3.6563534163428168e+19,3.801465545138186e+19,3.9463809117
25018e+19,3.94884043933172e+19,3.9762690912016515e+19,4.215177764806176e+19,4.192471405941112e+19,4.281742419953937e+19,4.3489367
14169013e+19,4.422998009461998e+19,4.417232876751891e+19,4.309918768216306e+19,4.30214666097913e+19,4.310587759725329e+19,4.23695
9341291122e+19,4.1463896967019586e+19
bottom,1.817728634898142e+20,1.9245882227320088e+20,2.0216923402666567e+20,2.151693131446454e+20,2.237048577509411e+20,2.34241474
01804893e+20,2.4102583496837346e+20,2.472395855139433e+20,2.54014108353841e+20,2.6107983926237217e+20,2.6697877027428424e+20,2.72
06900860911298e+20,2.7907177561091257e+20,2.8337890035576766e+20,2.8499235046576364e+20,2.8764667265891552e+20,2.87148864271319
2e+20,2.8840617478386483e+20,2.8447093061314298e+20

A.87 "MITR_Cycle7_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
top,3.0961751079557595e+19,3.3694243477022958e+19,3.4702121949396898e+19,3.697122647108721e+19,3.862359598224088e+19,3.8719265696
610755e+19,4.045726550766339e+19,4.08129757419563e+19,4.073994392296202e+19,4.151494734986362e+19,4.284113431634472e+19,4.2769677
307463524e+19,4.342991644737411e+19,4.312479698796402e+19,4.35661375221559e+19,4.505000235120956e+19,4.355353904125121e+19,4.2544
67631255661e+19,4.1760814578768904e+19
bottom,1.7985866931282723e+20,1.905606851388197e+20,2.0352255347708812e+20,2.1222534786452013e+20,2.215324756328482e+20,2.3128645
577076395e+20,2.4021956613723452e+20,2.4674321703068615e+20,2.5214481571856525e+20,2.6102280773107784e+20,2.654381815856379e+20,
2.700011938882996e+20,2.7546184795541933e+20,2.7988706437318617e+20,2.8555244375500685e+20,2.881371008531064e+20,2.8574142096857
697e+20,2.8740087712523954e+20,2.849146456592077e+20

A.88 "MITR_Cycle8_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19
top,3.1567711248256705e+19,3.4049356650171093e+19,3.5218290416866484e+19,3.6373634220646625e+19,3.727943479234538e+19,3.911388282
991924e+19,3.89927548995877e+19,4.0442547899864826e+19,4.121658491808099e+19,4.273728207269852e+19,4.329565228812926e+19,4.402734
3769725444e+19,4.461112130712817e+19,4.27459481360068e+19,4.528195338958216e+19,4.508204306553907e+19,4.32672906263931e+19,4.1905
53695108865e+19,4.177771251729163e+19
bottom,1.787365709531983e+20,1.90243724334694 2e+20,2.0009743222816853e+20,2.1162132686832815e+20,2.1862311210944374e+20,2.2837440
289109308e+20,2.3615810338979598e+20,2.448576492153959e+20,2.4954520164123414e+20,2.5780553562189205e+20,2.60039016483615e+20,2.6
823238542961803e+20,2.7291796829561353e+20,2.7721751743242237e+20,2.7869862643419988e+20,2.8305529280644956e+20,2.8462897112083
815e+20,2.8119996743454212e+20,2.8352207848919063e+20

APPENDIX B: Fast neutron fluence in MITR-DDE Side Plates

Appendices B.1-B.8 report the fast neutron fluence (E>0.1 MeV) for each cell position in the left and right side plates (see Figure 2.1) during eight irradiation cycles. The text under B.1-B.8 is generated in the comma-separated csv file format, which could be parsed using any spreadsheet software, text editor, or Python. Once the text is imported (copied) to spreadsheet software, the first line represents the plate position (left or right) as a column title. The fast neutron fluence data listed under the "left" and "right" side plates corresponds to axial node (z-coordinate) given in the first column of the spreadsheet (see Figure 2.1, Table 2.9).

B.1 "MITR_Cycle1_side_plates_fluence.csv" File

```
,left,right
0,1.779660577183048e+19,1.880020068567916e+19
1,2.6040915147236983e+19,2.8191114031911494e+19
2,3.6016319725001798e+19,3.813898177456777e+19
3,4.637603060625774e+19,4.895481800223306e+19
4,5.701743586828617e+19,5.955082469827173e+19
5,6.872170714759518e+19,7.2066223225254216e+19
6,8.03413212649151e+19,8.454654949941266e+19
7,9.27326179742125e+19,9.87390901322625e+19
8,1.0451810431112203e+20,1.1067202877879039e+20
9,1.1756367994379223e+20,1.2459487588130356e+20
10,1.2751043132826844e+20,1.3720135062981059e+20
11,1.382042123949121e+20,1.4714717171580338e+20
12,1.4638897845183517e+20,1.561943244668351e+20
13,1.5166377085269377e+20,1.6231661878606797e+20
14,1.5292990708859653e+20,1.6452235649020127e+20
15,1.4932593076391818e+20,1.60845816883818e+20
16,1.342569559298604e+20,1.4263429377284658e+20
17,1.0492055617093527e+20,1.075949217238627e+20
```

B.2 "MITR_Cycle2_side_plates_fluence.csv" File

```
,left,right
0,1.7856778444295449e+19,1.8385837777023621e+19
1,2.6602407284304314e+19,2.808634851941143e+19
2,3.696683298391491e+19,3.844283263748857e+19
3,4.6756533991360864e+19,4.907131592308631e+19
4,5.795974196912324e+19,6.046984898951659e+19
5,6.933700294216034e+19,7.212213449446797e+19
6,8.144990017605172e+19,8.505887626981727e+19
7,9.342495424475537e+19,9.81863105446066e+19
8,1.049376002744103e+20,1.1212756784129971e+20
9,1.181470975583786e+20,1.2518485147921026e+20
10,1.2972478892249796e+20,1.3732601696165521e+20
11,1.3876873760689825e+20,1.4848299358977645e+20
12,1.4669802859749674e+20,1.579816923596039e+20
13,1.51640717675695e+20,1.635757802088195e+20
14,1.5483247711104264e+20,1.6568408102067533e+20
15,1.499909486547585e+20,1.6175205273524558e+20
16,1.3434414218112013e+20,1.4400923599260082e+20
17,1.0552585439569027e+20,1.0921086542825515e+20
```

B.3 "MITR_Cycle3_side_plates_fluence.csv" File

```
,left,right
0,1.7802363310288214e+19,1.872103943801834e+19
1,2.6220081099686334e+19,2.8015170508840276e+19
2,3.641024248555584e+19,3.885762773034107e+19
3,4.647376283697941e+19,4.9247201491345605e+19
4,5.735742564879246e+19,6.000989376279239e+19
5,6.80372341984414e+19,7.165160712482255e+19
6,8.04359585201074e+19,8.473778434541376e+19
```

7,9.234872149914575e+19,9.797290654383097e+19
8,1.0505251840229795e+20,1.121463064959641e+20
9,1.176627521531356e+20,1.2514874942064555e+20
10,1.289300238894383e+20,1.3683752788353697e+20
11,1.3774102660689758e+20,1.4806416107893536e+20
12,1.457581601532032e+20,1.560699536370816e+20
13,1.516327920197405e+20,1.6250123900577307e+20
14,1.5307404259832306e+20,1.6432808318177316e+20
15,1.4964622892716304e+20,1.6041827751384128e+20
16,1.3375183402185032e+20,1.4296071640723307e+20
17,1.0405120441453132e+20,1.070882233672552e+20

B.4 "MITR_Cycle4_side_plates_fluence.csv" File

,left,right
0,1.7542270718828913e+19,1.8125449755958413e+19
1,2.604345910697171e+19,2.794315435404101e+19
2,3.6241634059326513e+19,3.855383402002915e+19
3,4.63078831055571e+19,4.87133573279679e+19
4,5.737255421353714e+19,5.970620630940466e+19
5,6.8338942411510596e+19,7.263957574668827e+19
6,8.073317338572374e+19,8.437493321265948e+19
7,9.2328288699345e+19,9.684835753166638e+19
8,1.0432258383206985e+20,1.1129500027465133e+20
9,1.1596816363986757e+20,1.2332331641468617e+20
10,1.2745781383082792e+20,1.3609442454765525e+20
11,1.3688257788366948e+20,1.4819739789817584e+20
12,1.4549367284501435e+20,1.5644560211488106e+20
13,1.5026133900831382e+20,1.6224618386123037e+20
14,1.5274676259790304e+20,1.6412489948712518e+20
15,1.493597823145997e+20,1.603514027596614e+20
16,1.3297016204302334e+20,1.422248210593150 6e+20
17,1.0466476724520218e+20,1.0658811603256659e+20

B.5 "MITR_Cycle5_side_plates_fluence.csv" File

,left,right
0,1.7654097225096489e+19,1.877971517831985e+19
1,2.6114905920864514e+19,2.829194367822389e+19
2,3.6709804101398045e+19,3.870308251698734e+19
3,4.638099019974167e+19,4.911488304062343e+19
4,5.691832179635648e+19,5.99195353257803e+19
5,6.83179233084171e+19,7.1802923006143005e+19
6,8.070116664684572e+19,8.467203209390196e+19
7,9.27195380412247e+19,9.813920978010707e+19
8,1.0412084102867208e+20,1.1053351270854962e+20
9,1.1673822628560095e+20,1.240734757307439e+20
10,1.2716637987823201e+20,1.357909695608975e+20
11,1.3636852592829214e+20,1.4676359527729175e+20
12,1.4525022744717913e+20,1.5526246483415774e+20
13,1.5011739231741637e+20,1.5951426277285013e+20
14,1.5173379393483833e+20,1.629550241105203e+20
15,1.486626308617366e+20,1.5931008572643895e+20
16,1.3177832344045527e+20,1.415391205758349e+20
17,1.0344136012177926e+20,1.0718242142045561e+20

B.6 "MITR_Cycle6_side_plates_fluence.csv" File

,left,right
0,3.4564725727522492e+19,3.710162432453636e+19
1,5.0045586320727286e+19,5.445644073764209e+19
2,6.959149775712087e+19,7.527763774244574e+19
3,8.835656236899148e+19,9.589807524199653e+19
4,1.0916180059844813e+20,1.1768699727494283e+20
5,1.305267291325302e+20,1.4072523645487656e+20
6,1.5330433979200396e+20,1.6732067721659535e+20
7,1.7604889973330346e+20,1.9306246514318442e+20
8,1.9951585394326913e+20,2.1874004024589202e+20
9,2.2007623356636614e+20,2.4489448999654274e+20
10,2.415846966479479e+20,2.6856919157788954e+20
11,2.6066440409683755e+20,2.9018908279503028e+20
12,2.760650944598141e+20,3.105521024153152e+20
13,2.865752228392322e+20,3.2507742089979036e+20
14,2.9227411095871e+20,3.3338542714256844e+20

15,2.8835901731440925e+20,3.319264740117328e+20
16,2.6222156507596246e+20,3.050628502796856e+20
17,2.120731890782809e+20,2.4664082689738382e+20

B.7 "MITR_Cycle7_side_plates_fluence.csv" File

,left,right
0,3.440369505763786e+19,3.65055718820808e+19
1,4.97670561317984e+19,5.410555638757796e+19
2,6.910074647965283e+19,7.47252181545834e+19
3,8.840666165874218e+19,9.623151243471931e+19
4,1.0861319140870768e+20,1.172404989312517e+20
5,1.3054990633133934e+20,1.4107896265402366e+20
6,1.5228161405594295e+20,1.6598780958124615e+20
7,1.7462361970647335e+20,1.9223062039609806e+20
8,1.971942508958184e+20,2.1774031791293268e+20
9,2.1951830660735097e+20,2.428163826929246e+20
10,2.401399523149825e+20,2.683808747603842e+20
11,2.585717054648767e+20,2.888220763444058e+20
12,2.7490331575202284e+20,3.095136323407657e+20
13,2.8492032644645274e+20,3.238820861417263e+20
14,2.9043851558784346e+20,3.3119526918290165e+20
15,2.849798446652351e+20,3.287512432783001e+20
16,2.5993684556234447e+20,3.033738084698707e+20
17,2.091865449410985e+20,2.4265794791860594e+20

B.8 "MITR_Cycle8_side_plates_fluence.csv" File

,left,right
0,3.4389257328199717e+19,3.6343908607952077e+19
1,5.026104483807428e+19,5.4119309988057375e+19
2,6.963451161438342e+19,7.468311271991892e+19
3,8.889757480170781e+19,9.526374065626615e+19
4,1.0879196125172847e+20,1.1790781923262836e+20
5,1.300210212261198e+20,1.4077969973489235e+20
6,1.5155728296780608e+20,1.6509212021671785e+20
7,1.7449627524077145e+20,1.9081861406430718e+20
8,1.967717116963797e+20,2.1664057676880973e+20
9,2.1906699811233625e+20,2.4223757325603163e+20
10,2.3837911905880713e+20,2.6651974617923122e+20
11,2.556041584600865e+20,2.8728847612078362e+20
12,2.7239060826128938e+20,3.065873637603557e+20
13,2.837391141632682e+20,3.219049164957851e+20
14,2.86785610458627e+20,3.2855654368487604e+20
15,2.8458226485996696e+20,3.27708666807142e+20
16,2.5807500590534492e+20,3.0020796459889846e+20
17,2.071532521887779e+20,2.4209332744891495e+20

APPENDIX C: Fast neutron fluence in NBSR-DDE Cladding

Appendices C.1-C.160 report the fast neutron fluence (E>0.1 MeV) for each cell position in the cladding (see Figure 2.2) during irradiation cycles 6-15. Fast neutron fluence in the top and bottom edges of the cladding (unfueled sections of the cladding, see Figure 2.2) is reported in C.161-C.180 for irradiation cycles 6-15.

The text under C.1-C.180 is generated in the comma-separated csv file format, which could be parsed using any spreadsheet software, text editor, or Python. Once the text is imported (copied) to spreadsheet software, the first line represents the plate number as a column title. The fast neutron fluence data listed under the plate number corresponds to axial coordinate (z-coordinate) given in the first column of the spreadsheet.

C.1 "NBSR_upper_Cycle6_cladding_fluence_position_0.csv" File


```
,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,4.592636160144383e+19,5.063358243085565e+19,5.327783021396597e+19,5.277737538494764e+19,5.3441984426678256e+19,5.6214
55626338039e+19,5.789282654859906e+19,5.870219302517223e+19,5.970570688023771e+19,6.213596150749828e+19,6.1887754590683865e+19,6.
2155088888567536e+19,6.397093261393298e+19,6.329824158142333e+19,6.3088020021270454e+19,6.233980727492544e+19,5.691616286289955e
+19
24.15442855,5.610913118366255e+19,5.910099437016962e+19,6.508813313929043e+19,6.460896088599027e+19,6.8563927973667545e+19,6.8743
25836906493e+19,7.0009347124552016e+19,7.244579794474318e+19,7.2106354332023054e+19,7.369913511477888e+19,7.75506527204843e+19,7.
3682342533937955e+19,7.816569221877006e+19,7.7025323319846e+19,7.471782515252945e+19,7.405743670598345e+19,6.881231449027704e+1
9
22.158714250000003,6.506559336500665e+19,6.598101352058046e+19,7.228577452731798e+19,7.411212484358837e+19,7.494187589690425e+19
,7.821840475862897e+19,7.77188479285861e+19,8.01325793746984e+19,8.184847581698843e+19,8.183976522692656e+19,8.387499010488662e+1
9,8.580038970814916e+19,8.81709274035423e+19,8.639764882672103e+19,8.733776395411972e+19,8.37190974827485e+19,8.049168916498088e+
19
20.163,7.0679344160135315e+19,7.562435511824631e+19,7.95856979986493e+19,8.13378735995475e+19,8.312201796397174e+19,8.90449498063
4788e+19,8.861022850233249e+19,9.048776476020354e+19,9.25792043740268e+19,9.451888216099897e+19,9.512323547147688e+19,9.458623208
415773e+19,9.746342080149982e+19,9.544920909956529e+19,9.52417713362363e+19,9.395942879929357e+19,8.556906517207322e+19
18.167285749999998,7.809897089489412e+19,8.03231347572889e+19,8.533648343743165e+19,8.953121625155409e+19,9.42144605083214e+19,9.
42018885226651e+19,1.0093328884263895e+20,1.0058396724118887e+20,1.0114611459982062e+20,1.066194183551889e+20,1.055588815651823
6e+20,1.0325910618905469e+20,1.0383202953539183e+20,1.0842978428969628e+20,1.0449116078337219e+20,1.0342344000156203e+20,9.743468
483428539e+19
16.171571450000002,8.3378935470848e+19,8.872310697354656e+19,9.124477789650797e+19,9.634271808013812e+19,1.0301934046260951e+20,
1.0210966750047858e+20,1.049266902864655e+20,1.1140305889741152e+20,1.1153147275090087e+20,1.08740491935202e+20,1.10263498197565
36e+20,1.1563622606768325e+20,1.1699489851753923e+20,1.1704967645504168e+20,1.1379532816801055e+20,1.122813018954017e+20,1.094561
9711863572e+20
14.17585715,8.873801375653904e+19,9.680698355044581e+19,9.812434804743107e+19,9.976678817352928e+19,1.0261075092877972e+20,1.0710
613379988283e+20,1.1365524032784033e+20,1.1663749492531008e+20,1.1698142853290747e+20,1.1975804136499916e+20,1.2044501058121844
e+20,1.2372001284468492e+20,1.2387446866846233e+20,1.2251220422270455e+20,1.2103140391218735e+20,1.1716282432594839e+20,1.1569639
199903839e+20
12.18014285,9.180423125821337e+19,9.884903322061929e+19,1.0420829110610543e+20,1.0724981363595487e+20,1.1255519158291399e+20,1.15
45932026951957e+20,1.1831405901247542e+20,1.2148399539581421e+20,1.2508137929146725e+20,1.2912147668201718e+20,1.28565615316213
58e+20,1.3274131055205651e+20,1.324701148614706e+20,1.3148590798437727e+20,1.3095249659295995e+20,1.2883591300782406e+20,1.2177722
530669337e+20
10.18442855,9.502804758007918e+19,1.0554990157542785e+20,1.0883657782557517e+20,1.1556618214759814e+20,1.1919948600226918e+20,1.2
111491781690424e+20,1.2767031033768986e+20,1.2704620104975206e+20,1.2802412193401718e+20,1.357783430870287e+20,1.33628533539801
2e+20,1.3506533190052133e+20,1.3770904088424638e+20,1.3789672267011545e+20,1.3608994873150988e+20,1.347025403144395e+20,1.3328908
992708105e+20
8.18871425,9.812524604640652e+19,1.1000667049058664e+20,1.152788224754541e+20,1.1804465931984037e+20,1.2378826076681912e+20,1.269
3395117782078e+20,1.3366624949677008e+20,1.3297119828977172e+20,1.3589867494973902e+20,1.3768389691293378e+20,1.424199435094575
3e+20,1.3939189096423984e+20,1.4581258363870798e+20,1.4108821102886507e+20,1.4281416505968014e+20,1.4172219830553282e+20,1.346756
00345176e+20
6.193,1.0459981865940165e+20,1.1182062842099581e+20,1.1550960821214477e+20,1.2096764598493041e+20,1.2849736739407938e+20,1.286060
2527010886e+20,1.3465584436771609e+20,1.3848491199903526e+20,1.3911440928082579e+20,1.4594638548605005e+20,1.4705541422073091e+
20,1.480234571162661e+20,1.5095183177520883e+20,1.495132374165378e+20,1.4870773233555908e+20,1.481024810261057e+20,1.39070407331
02871e+20
4.19728575,1.0679273215745076e+20,1.1400186793236408e+20,1.2008401499308751e+20,1.2848210141149674e+20,1.3047386313904503e+20,1.3
351628366786993e+20,1.3923114914763429e+20,1.4052785966818422e+20,1.443874592646687e+20,1.4778638538674727e+20,1.50615980158390
5e+20,1.5087909385819737e+20,1.5286905958779476e+20,1.5425826400281608e+20,1.5188036271582423e+20,1.5011310073213846e+20,1.453285
6219094037e+20
2.20157145,1.060402090160236e+20,1.146475291957127e+20,1.1779142360876343e+20,1.2396606456395825e+20,1.2985334584700903e+20,1.347
1780629702217e+20,1.3906950933205326e+20,1.4074248142331678e+20,1.4469637091222354e+20,1.4511214443785691e+20,1.504974442936310
8e+20,1.5153283711232505e+20,1.5330638508883896e+20,1.5437320787167366e+20,1.5165676097093717e+20,1.5237426215232176e+20,1.433996
603916736e+20
0.20585714999999993,9.939681259564384e+19,1.0531103384795813e+20,1.1148747080110383e+20,1.1513693863733299e+20,1.193467578342430
```

1e+20,1.2530318503840342e+20,1.2973391198327416e+20,1.3181277961144112e+20,1.3328549793117926e+20,1.3587712297432821e+20,1.4080713734954918e+20,1.4224932370412203e+20,1.4448803514991911e+20,1.3913955325213837e+20,1.4440721524212857e+20,1.4380555592857705e+20,1.3882704960868177e+20

C.2 "NBSR_lower_Cycle6_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,1.0414543117782391e+20,1.122130539732675e+20,1.2147142341015791e+20,1.2274478595734613e+20,1.2665377549748037e+20,1.3753752307993544e+20,1.3832955817628241e+20,1.40792769365942e+20,1.4334578045314659e+20,1.4559077789177181e+20,1.5173847887770308e+20,1.5388649242697 97e+20,1.5485722931944125e+20,1.594873120368619e+20,1.5386045045669167e+20,1.5216862038694 37e+20,1.4644926491230208e+20
−12.20157145,1.1680901272962105e+20,1.2262355609566038e+20,1.3017303348226926e+20,1.3922576115378158e+20,1.4005281821017113e+20,1.4489842068169977e+20,1.479543111951564e+20,1.5629133368323506e+20,1.5886050875199776e+20,1.6285301219684883e+20,1.6400604288132678e+20,1.662600203097065e+20,1.6748040091734314e+20,1.6908243108954613e+20,1.7062160133346755e+20,1.6780906854235786e+20,1.6084418848876698e+20
−14.197285749999999,1.2124692366629539e+20,1.2459466384677331e+20,1.35367957555248e+20,1.3845438003386997e+20,1.4313924068879308e+20,1.5019032864402715e+20,1.5555497452336598e+20,1.5556934250697317e+20,1.6005574538832183e+20,1.6177451542733328e+20,1.6864151359260013e+20,1.6768694068169664e+20,1.7236910733969342e+20,1.7038901959882598e+20,1.729097027229144e+20,1.6822753606491764e+20,1.6196309521217782e+20
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−18.188714249999997,1.2269090601881913e+20,1.2686031526183389e+20,1.3275927053156552e+20,1.388028036363446e+20,1.3878663965478648e+20,1.4752506768489131e+20,1.5228805425067855e+20,1.5614406185126124e+20,1.5727194856442656e+20,1.632984196886721e+20,1.6594751666624984e+20,1.6385068905857388e+20,1.7027587172791927e+20,1.7087304104659357e+20,1.6869629153010257e+20,1.6565925899513037e+20,1.5916672640262622e+20
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−22.180142850000003,1.0919577741575522e+20,1.1895253628402041e+20,1.257063865783805e+20,1.327179625786948e+20,1.3665568808604346e+20,1.3903358937303528e+20,1.4471702488865887e+20,1.5222339832444613e+20,1.5187317872402062e+20,1.5451239771286844e+20,1.5257900 59187244e+20,1.529426955037817e+20,1.5702410084720234e+20,1.5962650187805668e+20,1.5920803435549694e+20,1.6008986934938 89e+20,1.5117273952316955e+20
−24.17585715,1.0722107766874046e+20,1.1324036480118242e+20,1.2004719703509407e+20,1.2340661120225285e+20,1.2529240905069804e+20,1.3262636668319891e+20,1.3679936792211544e+20,1.3934160302161465e+20,1.4560514587537901e+20,1.4449432114274725e+20,1.4821652689 5879e+20,1.4831440878431194e+20,1.5167561894942163e+20,1.5309266133268185e+20,1.5316988924457055e+20,1.4910734187963436e+20,1.4358375018164086e+20
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−28.167285749999998,9.439136630648506e+19,1.0119011654961768e+20,1.078245329802429e+20,1.0772305909601706e+20,1.1488819292113332e+20,1.1636450323677327e+20,1.2199226281591895e+20,1.2509395127712357e+20,1.2496643542260965e+20,1.3019368745870462e+20,1.3149668 39720827e+20,1.3372551742914981e+20,1.3496834801117272e+20,1.3217018320367025e+20,1.3405418505416455e+20,1.3308883615555568e+20,1.2468625974226923e+20
−30.163,8.870954718901726e+19,9.398277677265527e+19,1.0080756898607594e+20,1.0409963323007597e+20,1.0666611430191232e+20,1.1005067244038368e+20,1.1239534776528388e+20,1.0990519660636078e+20,1.1799077938131336e+20,1.1809764125939193e+20,1.2308243357211537e+20,1.2619669401897627e+20,1.2492871946564074e+20,1.205051765125736e+20,1.2361135496865546e+20,1.1921295598690093e+20,1.1244563570790908e+20
−32.15871425,7.953936125151606e+19,8.491451371886766e+19,8.985736947943757e+19,9.221192279306771e+19,9.74203168506782e+19,9.861016549314956e+19,9.89244651345571e+19,1.0330759813372899e+20,1.0716809572918891e+20,1.0966992087479286e+20,1.0953252703154899e+20,1.1054097988097945e+20,1.1542609430742791e+20,1.0992315658586979e+20,1.0900450363398434e+20,1.116571926074639e+20,1.030516684257257e+20
−34.15442855,6.95552290426669e+19,7.581221650391047e+19,7.75712168970221e+19,8.390336687251086e+19,8.295921074972264e+19,8.973308642123527e+19,9.110379205736228e+19,9.026775501121826e+19,9.384628092838686e+19,9.404743269888767e+19,9.814230802694008e+19,9.947583650548344e+19,9.882299125033124e+19,9.79025423004949e+19,9.898732506283862e+19,9.472182992945068e+19,9.360112720808898e+19
−36.15014285,5.870201342537714e+19,6.21787062587297e+19,6.34858335673948 4e+19,6.601262308451629 5e+19,6.991640423059293e+19,7.552081583637692e+19,7.546235610307512e+19,7.505125217211407e+19,7.75057527717118e+19,7.639367084051441e+19,8.188978376985913e+19,8.053120111990068e+19,8.088564131551085e+19,8.10220473598817e+19,8.034010693792491e+19,8.107619669810135e+19,7.3029676878580834e+19

C.3 "NBSR_upper_Cycle6_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.080649141426101e+20,1.1925388879119424e+20,1.2490967320659819e+20,1.2962875315609353e+20,1.3241073888430478e+20,1.3551539674581487e+20,1.419545102216613 7e+20,1.4368038156621362e+20,1.4899969536893423e+20,1.4722234508206291e+20,1.5199874179207843e+20,1.5377503065745867e+20,1.5645193565804957e+20,1.571662723215697e+20,1.600193732896856e+20,1.548937689090934e+20,1.484265277637324e+20
24.15442855,1.2916331983223978e+20,1.3965759411488943e+20,1.46458652319206e+20,1.5243392460343542e+20,1.574618782068451e+20,1.599450737853076e+20,1.661788022026234e+20,1.681663139447354e+20,1.7054283666334096e+20,1.749880698681287e+20,1.7847961586315005e+20,1.7909205606352316e+20,1.852636913236088e+20,1.8463426837937786e+20,1.8459818004867993e+20,1.8455466176754424e+20,1.7900608092274288e+20
22.158714250000003,1.4449502256064406e+20,1.5887887589748303e+20,1.646036527098094 2e+20,1.674875349011677e+20,1.7551081995250264e+20,1.8094582869775476e+20,1.8233151445440483e+20,1.907124985482454e+20,1.8982568089241915e+20,1.973129480907413e+20,2.0136386321157997e+20,2.0264393752986416e+20,2.0427799591543505e+20,2.0700691056909055e+20,2.07353995396685e+20,2.0559734282889 04e+20,1.9723174934667103e+20
20.163,1.5968927120594898e+20,1.7404605829476454e+20,1.8442516834562828e+20,1.907305427135943e+20,1.9693932772586894e+20,1.988902204265375e+20,2.0351801812779696e+20,2.103169534891313e+20,2.1352244639229726e+20,2.1417256705560497e+20,2.221077541231774e+20,2.2584395777190068e+20,2.262027182358974e+20,2.2772532736490124e+20,2.276340451166654e+20,2.3228572480147492e+20,2.21342469228083 8e+20
18.167285749999998,1.7746118194242565e+20,1.9044873530770342e+20,1.9691491503157333e+20,2.043018778989851e+20,2.1108436122720715e+20,2.183471377801585e+20,2.2345947439211184e+20,2.269839244533577e+20,2.3364752857457492e+20,2.372478702724355e+20,2.3828647120 14911e+20,2.4404202923705978e+20,2.4700923301547056e+20,2.4944678746981512e+20,2.5204196301559025e+20,2.543643532381489e+20,2.4604864656601186e+20
16.171571450000002,1.906748180853108e+20,2.0465904623074515e+20,2.1519683879453052e+20,2.192519996013337e+20,2.255499440188619e+20,2.3721549691695655e+20,2.403944542828446e+20,2.4361639921912283e+20,2.4863904571507694e+20,2.5850071278902238e+20,2.615411546 5032e+20,2.6551723955603518e+20,2.71269082616385e+20,2.7435145062658163e+20,2.746157445778692e+20,2.7352566470649453e+20,2.6929165437843856e+20
14.17585715,1.9878938538488162e+20,2.1876427642616653e+20,2.2848795870626703e+20,2.3568704996975152e+20,2.466398583365627e+20,2.5204302443708137e+20,2.5979405487594547e+20,2.6287748430763327e+20,2.6966845900778422e+20,2.7669559998970816e+20,2.7958425857777648e+20,2.9111660307873615e+20,2.8866631156649817e+20,2.9250812665358732e+20,2.9183836969269405e+20,2.9618595212029926e+20,2.8776569543128747e+20
12.18014285,2.143408023619466e+20,2.3133787540990965e+20,2.4036898016705787e+20,2.5182596374214843e+20,2.5943211014747537e+20,2.6794046482024972e+20,2.7536510815059604e+20,2.825116590502704e+20,2.8583072405298564e+20,2.9321344123443295e+20,3.02211111214611 1e+20,3.066404230970321e+20,3.085987457481385e+20,3.121614069830645e+20,3.1523263006760455e+20,3.153647770432483e+20,3.113101469471907e+20
10.18442855,2.2627436418654762e+20,2.4028459715851423e+20,2.5605201340902103e+20,2.637255600790338e+20,2.7287076764647683e+20,2.7931253467605113e+20,2.8567257225080957e+20,2.968153750645302e+20,2.9919932773357355e+20,3.061489849466461e+20,3.146069220985925e+20,3.158758514912199e+20,3.2785930012590336e+20,3.3253008539755296e+20,3.3566711661454203e+20,3.3280711640673386e+20,3.259365350947493e+20
8.18871425,2.3661526306373127e+20,2.5365426226059326e+20,2.637834075502995e+20,2.740128571709161e+20,2.8275100959651673e+20,2.9219394589221745e+20,2.998659004299935e+20,3.063522471621945 6e+20,3.129282840103948e+20,3.246246181317317e+20,3.2603524729342296e+20,3.286893317319549e+20,3.4136695002182726e+20,3.426815205385726e+20,3.486711220129321e+20,3.485235844256672e+20,3.4418820834520977e+20
6.193,2.4210971141248575e+20,2.5902080931966848e+20,2.7374537895515524e+20,2.8362615161594072e+20,2.934044971028336e+20,2.9972154710720184e+20,3.066653665020733e+20,3.1798330386182786e+20,3.272834789669737e+20,3.370007927181275e+20,3.4288000635741104e+20,3.481271434987359e+20,3.548204674217043e+20,3.602342477371341e+20,3.654097389278087e+20,3.5909321963418596e+20,3.6132910400521876e+20
4.19728575,2.4781379050573542e+20,2.663929122862047e+20,2.807709278048425e+20,2.9500883568665323e+20,2.979479117955494e+20,3.1003856400083544e+20,3.19410915767377e+20,3.259970361197429e+20,3.3674127516354996e+20,3.438209565092841e+20,3.509372568964617e+20,3.587227835337 87e+20,3.6878558998029854e+20,3.707380748132038e+20,3.728842690682374e+20,3.752050671585594e+20,3.705815151432645e+20
2.20157145,2.4927908287421904e+20,2.6955170264376158e+20,2.8314745052344805e+20,2.932500602758764e+20,3.0254598969505776e+20,3.125838527365281e+20,3.1826829553219214e+20,3.305515957381154e+20,3.3611344435155567e+20,3.424793197445153e+20,3.534002854665 93e+20,3.601790538195962e+20,3.668315130151567e+20,3.7729235252083615e+20,3.796821430080806e+20,3.802431042661347e+20,3.6859718766562574e+20
0.20585714999999993,2.3434435178349167e+20,2.5690486557713164e+20,2.6804395341563337e+20,2.8197617190800307e+20,2.9350745498747168e+20,2.9932829044474393e+20,3.091225572540035e+20,3.187708786082349e+20,3.255385020355814e+20,3.307500815569538e+20,3.393284900481417e+20,3.492028941799805e+20,3.550141768438327e+20,3.5715453328066524e+20,3.682798226397825e+20,3.6655766627044904e+20,3.626065247697751e+20

C.4 "NBSR_lower_Cycle6_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.5034793431577143e+20,2.7647482431955635e+20,2.8949793530478644e+20,3.000962288935653e+20,3.138267773026233e+20,3.2270185310057765e+20,3.319107459574876e+20,3.3800914313468635e+20,3.482359392015751e+20,3.6094327729319864e+20,3.6987885411614654e+20,3.796009442640104e+20,3.836067489714767e+20,3.898686050583071e+20,3.961124169797885e+20,3.9720355827265425e+20,3.9080530952421566e+20
−12.20157145,2.696398006275241e+20,2.9193548975913104e+20,3.050329002487392e+20,3.1987475695899404e+20,3.316814789154068e+20,3.3708782928039 89e+20,3.4721060604115845e+20,3.593882948087158e+20,3.689044691873034e+20,3.781536960608757e+20,3.876820767866111e+20,4.014556127660592e+20,4.0612958230218224e+20,4.137755320134259e+20,4.189435932536627e+20,4.189738437661595e+20,4.1091659322710884e+20
−14.197285749999999,2.735681215661391e+20,2.96174276483897e+20,3.090917760307612e+20,3.2676762812229196e+20,3.325529059596119e+20,3.460674550952281e+20,3.5481091462828425e+20,3.6177065534552244e+20,3.739992923446533e+20,3.804299144485709e+20,3.9172290840328425e+20,4.008506025161239e+20,4.083357468714638e+20,4.171805721569214e+20,4.238680582616886e+20,4.198633149757134e+20,4.14610870726933e+20
−16.192999999999998,2.7388601730272792e+20,2.94599126991083e+20,3.110867177233111e+20,3.2344325601212124e+20,3.337141010708912e+20,3.435428640786121e+20,3.551786971749554e+20,3.590486399315592e+20,3.729304409031009e+20,3.795314211563424e+20,3.8917602753535 49e+20,3.9722001030576656e+20,4.072249692810124e+20,4.125984155797798e+20,4.192211549735891e+20,4.1695661222229626e+20,4.110779292937582e+20
−18.188714249999997,2.7220684850378467e+20,2.8950642667671536e+20,3.097748007602934e+20,3.1667191760955585e+20,3.257884667967388 4e+20,3.4016117520792106e+20,3.486084981449563e+20,3.5914788284097836e+20,3.651502213732312e+20,3.7452363456126386e+20,3.8604058845060 24e+20,3.8898231811322636e+20,4.030641970358433e+20,4.049322988602049e+20,4.102171164644644e+20,4.141539287750082e+20,4.041426012708157e+20
−20.18442855,2.6417188781604746e+20,2.8476452616516105e+20,3.0218351425584215e+20,3.10053423901711e+20,3.2026748291070643e+20,3.3230453333069 11e+20,3.428513479771509e+20,3.481372270029014e+20,3.5477907198355086e+20,3.6442208623032664e+20,3.7097954820243 24e+20,3.8188724615587116e+20,3.879522085560997e+20,3.9483764976895984e+20,3.979943172835346e+20,4.0265767260474645e+20,3.9032023990277636e+20
−22.180142850000003,2.545633697677329e+20,2.7573182927577616e+20,2.896895218839326e+20,3.032470585899389e+20,3.093820748085811e+20,3.1660717089859784e+20,3.294954813544564e+20,3.3398635568336214e+20,3.473475294135122e+20,3.555034921512365e+20,3.615811916093585e+20,3.7010812115822726e+20,3.730434822919046 4e+20,3.8074621805292297e+20,3.8336262202852036e+20,3.8266845237333144e+20,3.77295006074564e+20
−24.17585715,2.4291479961349613e+20,2.6394209006322116e+20,2.7933270168438227e+20,2.873267976447116e+20,2.952402255717162e+20,3.050392687776858e+20,3.132016000443568e+20,3.223537068514921e+20,3.242186244113804e+20,3.3338824467311906e+20,3.450760874225271e+20,3.507132969618365e+20,3.5811618115161496e+20,3.6017746168735944e+20,3.665831403862359e+20,3.676631367534449e+20,3.630974322094156e+20
−26.171571450000002,2.3294911323342153e+20,2.4809825146535412e+20,2.613166640049493e+20,2.7015034436475024e+20,2.8246283366167914e+20,2.8926176902301345e+20,2.9531505578683983e+20,3.054893114899184e+20,3.102243127617805e+20,3.1689163185821653e+20,3.2403499849341 76e+20,3.335490500290229e+20,3.432621180942123e+20,3.4355560113650545e+20,3.529889846387861e+20,3.517927626183e+20,3.4200486433798704e+20
−28.167285749999998,2.170277908667031e+20,2.3340021736714522e+20,2.450185370088852e+20,2.5189814040354426e+20,2.6241364311601617e+20,2.666396927828888e+20,2.7675344746097387e+20,2.8643573430292087e+20,2.908544319704308e+20,3.002358058196469e+20,3.053040934397189e+20,3.100847358356989e+20,3.18725237484117e+20,3.212169244345084e+20,3.243385650398763e+20,3.279500516633936e+20,3.189454824435232e+20
−30.163,2.015177693277916e+20,2.1750808409143247e+20,2.2644578375736263e+20,2.3531714458009813e+20,2.436386890704363e+20,2.473441114959172e+20,2.5939442968454083e+20,2.6105077792142508e+20,2.7036475150595536e+20,2.729917696964639e+20,2.8243311385992793e+20,2.8885418317042545e+20,2.9087566040025314e+20,2.970610941397232e+20,2.97665573678913e+20,2.996456554705872e+20,2.9470526914019446e+20
−32.15871425,1.8288398434052994e+20,1.9929621414688886e+20,2.048225051403768e+20,2.131806686721584e+20,2.2241715848783734e+20,2.26935629775512e+20,2.3263440176130608e+20,2.3762626703401782e+20,2.488125881288742e+20,2.496383740489613e+20,2.548876340332684e+20,2.615634445016334e+20,2.6425998579980996e+20,2.726935102574607e+20,2.762222060046711e+20,2.7527647945608797e+20,2.68105515862118e+20
−34.15442855,1.6408674044363663e+20,1.753049041832264e+20,1.8866820075635874e+20,1.9367704877292834e+20,2.0073444026734903e+20,2.0634989066609063e+20,2.1304109174607682e+20,2.161574252439892e+20,2.2027733276175037e+20,2.242003465929098e+20,2.26011662367496 8e+20,2.3133256830245405e+20,2.404788372913882e+20,2.4388122388115594e+20,2.443986668580743e+20,2.489633099806125e+20,2.374766066037707e+20
−36.15014285,1.38726727467182e+20,1.5158531812128935e+20,1.5751335714916414e+20,1.645654415361293e+20,1.6794235401011026e+20,1.73518531813680 6e+20,1.774866560582124e+20,1.8199504384172147e+20,1.8900838634426096e+20,1.9252116076910 46e+20,1.946885834539605e+20,1.971128701396662e+20,2.030610761758722e+20,2.032935274824263e+20,2.0898062383181808e+20,2.1198603878390894e+20,2.0169714955978998e+20

C.5 "NBSR_upper_Cycle6_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.1460645479234914e+20,1.2527692504252383e+20,1.2903966422852493e+20,1.3321741921755187e+20,1.3748698716556016e+20,1.
4293154870423231e+20,1.4394839049272005e+20,1.4621399466550408e+20,1.4697131889941432e+20,1.5104983097902203e+20,1.547807275202
8972e+20,1.57303726404669e+20,1.5967494201581896e+20,1.6264426863721195e+20,1.6280188972864245e+20,1.614337174265958e+20,1.583953
9840828034e+20
24.15442855,1.3537688124122443e+20,1.467914079566704e+20,1.5047347910934682e+20,1.587727337483716e+20,1.6144061666628803e+20,1.65
19698732334245e+20,1.6792165629103355e+20,1.741622839480416e+20,1.7606328983862778e+20,1.777727091500678e+20,1.824424330002263e
+20,1.853056174725078e+20,1.8682504233703827e+20,1.901334931248424e+20,1.9186838655206916e+20,1.898559314049159e+20,1.8695612789
119094e+20
22.158714250000003,1.5091768400337827e+20,1.641005389230211e+20,1.693482067750915e+20,1.752714694062563e+20,1.768816458082771e+2
0,1.8325495115167446e+20,1.8852756240878607e+20,1.9334535455695503e+20,1.9522300917473673e+20,2.0044095722505588e+20,2.020378658
5843774e+20,2.0958244981728005e+20,2.144092640481235e+20,2.119966529988201e+20,2.1664567912990186e+20,2.1651990068320482e+20,2.15
99131278062974e+20
20.163,1.6571389958951525e+20,1.800505196699996e+20,1.8892294191422623e+20,1.9321002331683796e+20,1.9869969526888245e+20,2.041851
215349625e+20,2.0965197292494805e+20,2.1155244810478866e+20,2.1724272871865382e+20,2.179379597953339e+20,2.273713432976145e+20,2.
2897727401367087e+20,2.310167954088475e+20,2.389583510053666e+20,2.383793455819636e+20,2.3823021586246202e+20,2.336650420291783
2e+20
18.167285749999998,1.823649492313749e+20,1.9446143925486204e+20,2.0353181660718147e+20,2.103498575553559e+20,2.1685743271737924e
+20,2.216481586175248e+20,2.2744670422348366e+20,2.340710357495296e+20,2.371974527516076e+20,2.4214102334647363e+20,2.4791834052
26102e+20,2.4905034654288386e+20,2.588780481291136e+20,2.56092878136429e+20,2.58895561583717e+20,2.623637563059338e+20,2.55232596
0178806e+20
16.171571450000002,1.9359585002885808e+20,2.113592693934058e+20,2.2014783933983434e+20,2.270502632965524e+20,2.3451630206505216e
+20,2.4516978957136904e+20,2.4744760009130086e+20,2.5248882146334946e+20,2.590436298817275e+20,2.621313049993797e+20,2.687991548
0656137e+20,2.699322222483262e+20,2.767959043206185e+20,2.8198572470142317e+20,2.851896254723525e+20,2.8668145337811393e+20,2.76
7683073618495e+20
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5734482478519858e+20,2.6529699459662886e+20,2.70138668728348e+20,2.753757223655072e+20,2.8195706632116303e+20,2.864792525840566
5e+20,2.978507917291124e+20,2.9956711028024476e+20,3.037655629883484e+20,3.084517388716191e+20,3.101388683317458e+20,3.0170162889
88761e+20
12.18014285,2.1977156542123426e+20,2.3681374888256966e+20,2.4896649424508584e+20,2.5337723125141242e+20,2.6600549344194786e+20,2.
698064438016291e+20,2.775611892157121e+20,2.8791164088631575e+20,2.9500777426516214e+20,2.997937237685977e+20,3.077862275966903
e+20,3.113287218232851e+20,3.152926003818525e+20,3.262868041868171e+20,3.3488166471111724e+20,3.323257617605134e+20,3.29096917584
54296e+20
10.18442855,2.337366879798285e+20,2.470235622056006e+20,2.612269738889501e+20,2.7032123322481967e+20,2.8202128232137546e+20,2.844
1213423011103e+20,2.9116012135987184e+20,3.041885394525575e+20,3.107826204661067e+20,3.160440867975616e+20,3.269358634286337e+2
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41e+20
8.18871425,2.3958193613139644e+20,2.6200658797417375e+20,2.688490416166438e+20,2.7945954155257047e+20,2.915634615264954e+20,2.998
8819028130687e+20,3.094669885278702e+20,3.163731274598071e+20,3.2521370705930034e+20,3.288644662779888e+20,3.465297041546085e+2
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990646e+20
6.193,2.507342917385371e+20,2.686319809217108e+20,2.803113322991899e+20,2.9432952593233992e+20,3.0263727194329357e+20,3.092738098
164873e+20,3.182465363916243e+20,3.2719485027246565e+20,3.3479568957033696e+20,3.435120828553698e+20,3.54910688248449e+20,3.59766
6915702982e+20,3.658926857062658e+20,3.731851820609685e+20,3.780995635648288e+20,3.7938706783355077e+20,3.757336550611345e+20
4.19728575,2.5705081103215985e+20,2.7426441406431022e+20,2.8689426838708244e+20,2.9679839232067233e+20,3.092801783454341e+20,3.19
8943932565796e+20,3.273800683226651e+20,3.323278846034957e+20,3.475162954305994e+20,3.530187044405373e+20,3.6282146262172585e+2
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354075826049175e+20,3.808263553755021e+20,3.887615424430746e+20,4.006208047632976e+20,4.021142248012958e+20,3.9748801923227294e
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9509e+20,3.6608851797137654e+20,3.7602023886373546e+20,3.8739973866997455e+20,3.9310540989546096e+20,3.950154378687216e+20,3.8111
240846735756e+20

C.6 "NBSR_lower_Cycle6_cladding_fluence_position_2.csv" File

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21618955116e+20,3.420850016605663e+20,3.623215330994109e+20,3.628373839440926e+20,3.815778417912113e+20,3.8789170753110625e+20,3.
9604342458286604e+20,4.046669434874263e+20,4.125501209019341e+20,4.21781834320903e+20,4.247914949589584e+20,4.178222014483002e+
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−12.20157145,2.8063241230025212e+20,3.002846312082382e+20,3.186015818804021e+20,3.3234486734735344e+20,3.425260222901243e+20,3.50
3916862500288e+20,3.619590576601953e+20,3.725891938937076e+20,3.801560677038633e+20,3.920243521067607e+20,3.980882530854983e+20,
4.111145483352017e+20,4.2390467730313216e+20,4.306823842346442e+20,4.383155968879945e+20,4.480461784077872e+20,4.379287087544832
5e+20
−14.197285749999999,2.8400401766677748e+20,3.0833020611088646e+20,3.236794222938942e+20,3.351809855716116e+20,3.409747547808604e
+20,3.554122099030007e+20,3.6261023974499405e+20,3.754582161841903e+20,3.844176749906883e+20,3.92517913100129e+20,4.015829833449
929e+20,4.188087927242911e+20,4.2232103643838926e+20,4.344838653050709e+20,4.457200732100097e+20,4.483460299790271e+20,4.34655815
5866315e+20
−16.192999999999998,2.8416800728715464e+20,3.053391203489257e+20,3.2101843861567e+20,3.3220582113201743e+20,3.444158832550538e+2
0,3.5536656877888274e+20,3.6414983161785575e+20,3.744663178007437e+20,3.861435463352405e+20,3.9386857194757226e+20,4.03833727616
9014e+20,4.136248101616877e+20,4.23876018922872e+20,4.330737668541252e+20,4.417312512464012e+20,4.51401862451946e+20,4.4017626876
191834e+20
−18.188714249999997,2.804243736879936e+20,3.013497676745716e+20,3.1614545254996306e+20,3.2837568168133054e+20,3.387436468065376e
+20,3.505514301844415e+20,3.6013341269547824e+20,3.711997931618386e+20,3.805870048292558e+20,3.9087801689635704e+20,3.9775921242
325274e+20,4.082497717306835e+20,4.173599523889198e+20,4.250494203812992e+20,4.330811968045631e+20,4.349885712240959e+20,4.307343
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5793421646855e+20,3.490442116670589e+20,3.615573096258084e+20,3.6547766990324e+20,3.79578654412697e+20,3.844213899659072e+20,3.96
488690898388 65e+20,4.092766970233369e+20,4.182717134497872e+20,4.22161823214722e+20,4.2343977469002396e+20,4.17950633448725e+20
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e+20,3.6545113436596214e+20,3.7187804149466084e+20,3.796975336197018e+20,3.865102674604206e+20,3.944321867593541e+20,3.8400478203
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458624e+20,3.433544617639393e+20,3.5459226180111466e+20,3.5956024509027634e+20,3.653736505971108e+20,3.686046176160635e+20,3.6269
67455965199e+20
−28.167285749999998,2.2540293714234258e+20,2.4070651220123232e+20,2.5207486708181484e+20,2.6031149785286384e+20,2.65709356845926
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3536375e+20
−30.163,2.046967266936797e+20,2.2164709719603367e+20,2.325532030172358e+20,2.4328258216016734e+20,2.4917718641107206e+20,2.587230
805914109e+20,2.6405513145202485e+20,2.7108068030171215e+20,2.7658574286537782e+20,2.8519334044757135e+20,2.8958337973482114e+2
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81413e+20
−32.15871425,1.8934326482470758e+20,2.0313325283726796e+20,2.132204719780752e+20,2.2083829401980443e+20,2.26453744418546e+20,2.32
78883858826324e+20,2.3997784634758214e+20,2.4527221674526163e+20,2.5304447561394795e+20,2.5754596415776476e+20,2.62250184206384
53e+20,2.6781150210907924e+20,2.735458317148257e+20,2.770628518256338e+20,2.829229598780773e+20,2.84717823619552e+20,2.8169754876
658552e+20
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2.3999695193442222e+20,2.4728254904943257e+20,2.500661269098805e+20,2.5460635733812304e+20,2.610942962025608e+20,2.5133399488101
682e+20
−36.15014285,1.4056192522531906e+20,1.5281072923278108e+20,1.6188482156031948e+20,1.6960347864370453e+20,1.754046778033912e+20,1.
7899812026155955e+20,1.841863485101275e+20,1.8913681834468583e+20,1.9311555680412874e+20,1.9623932025247888e+20,2.0280898857173
248e+20,2.0601819645011732e+20,2.11565185162682e+20,2.1191545425474984e+20,2.166042836917484e+20,2.179952765558541e+20,2.11715376
30367475e+20

C.7 "NBSR_upper_Cycle6_cladding_fluence_position_3.csv" File

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902870188140408e+20,1.4348295716886633e+20,1.4759649615768083e+20,1.4818346224226717e+20,1.518045016592045e+20,1.52515654058251
22e+20,1.555396438864366e+20,1.575462612153887e+20,1.5848402710278842e+20,1.621613218587548e+20,1.5707764362706164e+20,1.53676849
16953055e+20
24.15442855,1.3542093023310566e+20,1.4553256206820853e+20,1.5167341610505185e+20,1.557052256390505e+20,1.6446248365149117e+20,1.6
695788557710148e+20,1.6872621378129838e+20,1.742779788905731e+20,1.7447964897388485e+20,1.75995889573942e+20,1.8234796648751707
e+20,1.833886902595549e+20,1.8540592180341814e+20,1.8843468802831352e+20,1.8735575308259557e+20,1.8647636537720717e+20,1.81661226
78276597e+20
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,1.8538628550583252e+20,1.8868199923574322e+20,1.93803357930371e+20,1.9682097922960966e+20,2.0112928906204368e+20,2.029437891011
04e+20,2.078098759271187e+20,2.107845096559673e+20,2.1148239428637508e+20,2.1032756770404247e+20,2.0908411242720176e+20,2.0457572
464369264e+20
20.163,1.641345044107368e+20,1.8132847114530154e+20,1.8831633953205427e+20,1.9630884336014688e+20,1.9747640700037294e+20,2.038762
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2.3039533312579994e+20,2.359025185324478e+20,2.3405670655939956e+20,2.333880110199974e+20,2.330573782255152e+20,2.23717930525198
2e+20
18.167285749999998,1.8141922268279184e+20,1.9222555488382917e+20,2.0575231036659316e+20,2.068859085191035e+20,2.1752347470305362
e+20,2.2394772827802455e+20,2.2907598621234453e+20,2.3691723747795334e+20,2.4153017527833372e+20,2.450673623974765e+20,2.471461563
8782437e+20,2.52465470190545e+20,2.5367867495488894e+20,2.5755976263714927e+20,2.5814885156471787e+20,2.5559772501082404e+20,2.47
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1222472941e+20,3.6766101391046266e+20,3.732631965405653e+20,3.7881283880685784e+20,3.740990659648181e+20,3.652170909271714e+20
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916067521646e+20,3.321999833138163e+20,3.4267355987738925e+20,3.490723393365734e+20,3.5752390795957305e+20,3.6270735981143104e+
20,3.733953435162091e+20,3.8159907022103354e+20,3.862204993933463e+20,3.879511471346086e+20,3.86411555261747e+20,3.7919282770067
685e+20
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4471015471e+20,3.382198353006726e+20,3.4699407605697105e+20,3.5539735000212505e+20,3.6181311220516703e+20,3.7232383852092896e+2
0,3.776229853153184e+20,3.826610224228937e+20,3.8763431281951094e+20,3.92058848305222e+20,3.919006965030459e+20,3.88126812391388
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e+20,3.1686350418870206e+20,3.266307047499382e+20,3.340627780307224e+20,3.354081297707101e+20,3.4997773186849414e+20,3.6055691987
043294e+20,3.648848660004526e+20,3.7571401876354885e+20,3.7805869883742093e+20,3.828085600101586e+20,3.80097689521852e+20,3.69880
4462483832e+20

C.8 "NBSR_lower_Cycle6_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.6079179107759312e+20,2.8747433523197654e+20,3.044783075196318e+20,3.180952838291405e+20,3.2973483190070274e+20,3.3668183556004754e+20,3.4994111282705066e+20,3.613349418234199e+20,3.668373508333578e+20,3.754407027295869e+20,3.843072871556124e+20,3.957297745322417e+20,3.986311701782033e+20,4.050453402490086e+20,4.107212916727438e+20,4.109813399380667e+20,4.014343843362369e+20
−12.20157145,2.8211044172662905e+20,3.0807440353152795e+20,3.258319850778745e+20,3.353911470268523e+20,3.4468495360305136e+20,3.5701707919756585e+20,3.6581201767294114e+20,3.78527316425748e+20,3.8378347564974735e+20,3.975267611166987e+20,4.029055145229217e+20,4.168000525523568 6e+20,4.182966568548283e+20,4.288254273359393e+20,4.366555336758913e+20,4.3642361308008284e+20,4.245914170078832e+20
−14.197285749999999,2.8653656934457685e+20,3.1483618914067323e+20,3.316093022540111e+20,3.425302679760888e+20,3.5105507468197534e+20,3.628326075473825e+20,3.7343567753287146e+20,3.864481743031904e+20,3.9308471217638415e+20,3.9932215556891895e+20,4.0736030052112944e+20,4.203483845971528e+20,4.235119513514197e+20,4.359825924505247e+20,4.3839148852460926e+20,4.388526761624985e+20,4.3159043032029266e+20
−16.192999999999998,2.9023031613365548e+20,3.130938657630087e+20,3.280710537133807e+20,3.357514996230857e+20,3.5428444956869145e+20,3.632661982265029e+20,3.7310292189540706e+20,3.848146466283651e+20,3.904125835725033e+20,4.005693258209785e+20,4.0985729457897656e+20,4.150672819681123e+20,4.2563373291215775e+20,4.340046335018327e+20,4.365594750309455e+20,4.412965991457897e+20,4.3046160856449234e+20
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−20.18442855,2.7464174940440147e+20,3.0113429911187535e+20,3.1294473604350706e+20,3.276783277616683e+20,3.350127502652699e+20,3.488579321953683e+20,3.565903877581378e+20,3.638982747244615 4e+20,3.738363641457672e+20,3.8048723120909104e+20,3.924743948189933e+20,3.99290843634931e+20,4.0918382264286426e+20,4.124360180916393e+20,4.204922072091988e+20,4.194339699825576e+20,4.0722019288430235e+20
−22.180142850000003,2.6726009364444525e+20,2.8547514785346224e+20,3.049373723145389e+20,3.150288371413105e+20,3.212896318066497e+20,3.3125107250075986e+20,3.457033875237758e+20,3.496470990740119e+20,3.6031969216716885e+20,3.7030129986961015e+20,3.750787580011168e+20,3.819472164701191e+20,3.9262246311700385e+20,3.9453249109026446e+20,3.989931149066734e+20,3.993141949077356e+20,3.8838951421043894e+20
−24.17585715,2.5367230642594225e+20,2.7437374047789502e+20,2.8906222178268388e+20,3.0278268668757626e+20,3.092048174195649e+20,3.2100410942553994e+20,3.259853604833406e+20,3.354314810435146e+20,3.416317746838603e+20,3.4936104598215655e+20,3.586681203269945e+20,3.657987499043022e+20,3.750516917530934e+20,3.7534039839867653e+20,3.8019640172052565e+20,3.798323341490734e+20,3.731798749535129e+20
−26.171571450000002,2.4403566070811312e+20,2.617651145849452e+20,2.689514687905363e+20,2.8353699221068703e+20,2.9100621524366023e+20,2.9980433798350884e+20,3.068999406516097e+20,3.18600520458911e+20,3.231062546886923e+20,3.323984691326548e+20,3.391045301135166e+20,3.4602552894632906e+20,3.502149595717582e+20,3.539357726088603e+20,3.574113972815149e+20,3.631865916146693e+20,3.509722838056684e+20
−28.167285749999998,2.2546290745659054e+20,2.4429305541970838e+20,2.586466582440506e+20,2.6592960180533317e+20,2.731536364738588e+20,2.828205327041847e+20,2.874515146699175 7e+20,2.9869833678976745e+20,3.051714157533296e+20,3.133634668217518e+20,3.122638341569571e+20,3.192485182792365e+20,3.256472977384206e+20,3.310239283016614e+20,3.3844485665678885e+20,3.3887473236069024e+20,3.288113952034331e+20
−30.163,2.0977668995015403e+20,2.2638156775715016e+20,2.3817077625895964e+20,2.4643128901355867e+20,2.530327999775457e+20,2.600801079678008e+20,2.661604609796506e+20,2.7367904011196056e+20,2.8103734459911225e+20,2.8693354098225365e+20,2.942738013040564e+20,2.9565471066399644e+20,2.9980911438021886e+20,3.0449900523870854e+20,3.078923697458018e+20,3.087834330875924e+20,3.013439298563705e+20
−32.15871425,1.8876850508726908e+20,2.0465745409850845e+20,2.1546962411774696e+20,2.2517473152175296e+20,2.3196517551115833e+20,2.3536331641496 16e+20,2.400006669096411e+20,2.5069714198634814e+20,2.5435904613069337e+20,2.5785324567944246e+20,2.656955583665424e+20,2.7019120909215816e+20,2.7519156573679885e+20,2.775091795626475e+20,2.8017440892683613e+20,2.81234238285714e+20,2.7048946853116133e+20
−34.15442855,1.723775037107325e+20,1.838461629222253e+20,1.9437599482482727e+20,2.0127258096334412e+20,2.0621402871522797e+20,2.1453875747003947e+20,2.1726714141294946e+20,2.2014943147207105e+20,2.25235763257492e+20,2.3156342747677147e+20,2.374171670002683e+20,2.414118267820779e+20,2.4481898976855566e+20,2.455290807461113e+20,2.482670174824413e+20,2.4916604148541535e+20,2.44544081602357e+20
−36.15014285,1.4567001615130788e+20,1.5558528501055452e+20,1.647925857352278e+20,1.6943365120512623e+20,1.727038908192502e+20,1.782429188706315e+20,1.8370074817794258e+20,1.8600509423515232e+20,1.9216505385883566e+20,1.9778209638981 39e+20,2.016801668159321 2e+20,2.031491741596347e+20,2.059863538053839e+20,2.0910746370000627e+20,2.141895497994628e+20,2.1286808004302517e+20,2.0491113383488487e+20

C.9 "NBSR_upper_Cycle6_cladding_fluence_position_4.csv" File

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26.15014285,2.8120557990545396e+19,3.068362553621428e+19,3.1239651184334635e+19,3.2552045787023237e+19,3.22283122322333e+19,3.574
5226200892285e+19,3.5519886418328666e+19,3.5776166637358277e+19,3.6393807803037835e+19,3.8632504941022211e+19,3.848533885127908e
+19,3.883422809540843e+19,3.837208517817716e+19,3.8799307328350765e+19,3.7378117022822924e+19,3.5880716654233055e+19,3.353274617
373854e+19
24.15442855,3.482561062099062 8e+19,3.818193151804398e+19,3.874246820750158e+19,3.912994012283295e+19,4.102372834727956e+19,4.2640
37942039614e+19,4.430569666888033e+19,4.575979104063272 6e+19,4.53237060210083e+19,4.5632738888146305e+19,4.507321054910527e+19,4.
5794552594466726e+19,4.651525778693351e+19,4.6866800584790655e+19,4.629692338621125e+19,4.436816132363242e+19,4.159710823677963
5e+19
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19,4.829733139944029e+19,4.976904536794519e+19,4.960224298061654e+19,5.106928669456053e+19,5.311936923357375e+19,5.1927552112275
77e+19,5.364636500391213e+19,5.163842089809616e+19,5.0738229331481895e+19,5.144120878504708e+19,5.019764736605 7255e+19,4.74110444
5435964e+19
20.163,4.217330089323118e+19,4.67602869381573e+19,4.928774379279929e+19,5.134430100290832e+19,5.250905187618288e+19,5.31591725394
9055e+19,5.353544645809067e+19,5.53722363484644e+19,5.518011905857267e+19,5.875604806213763e+19,5.545502722477135e+19,5.803215860
51975e+19,5.821896878763366e+19,5.684336653514919e+19,5.658331826982611e+19,5.638960884769771e+19,5.288490122618655e+19
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,5.701531681670975e+19,5.838455054024753e+19,6.124083577283681e+19,6.278838830688184e+19,6.287860913362658e+19,6.286427994349654e
+19,6.5308202926787805e+19,6.3727746326518235e+19,6.3247983812534436e+19,6.4367783485660316e+19,6.055568820032236e+19,5.80629398
2843982e+19
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,6.249384384309756e+19,6.389492021136877e+19,6.549395168773286e+19,6.676606534483366e+19,6.8297165845766414e+19,6.78720665385750
4e+19,6.90030111373576e+19,6.9803853652403536e+19,6.886343421127604e+19,6.982455137148027e+19,6.601192537539676 6e+19,6.3723500640
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2510320172245e+19,7.404794748462511e+19,7.47808590242397e+19,7.696738729593571e+19,7.767270187678133e+19,7.8253299432421e+19,7.93
6142346914459e+19,7.972389890836021e+19,7.844594743305829e+19,7.816520144865849e+19,7.708361294921276e+19,7.283951911699118e+19
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516051256e+19,7.63761755253849e+19,7.902283001347906e+19,7.877976449201381e+19,8.0546500563974e+19,8.433895955172634e+19,8.408262
626162216e+19,8.441909687430547e+19,8.397064629430958e+19,8.385654348401476e+19,7.928340898954768e+19,7.726723886717557e+19
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6.193,6.555551413421751e+19,7.007929252934776e+19,7.398001650919376e+19,7.595479119341242e+19,7.902760641018906e+19,8.15235390465
4495e+19,8.402212523662864e+19,8.562805595268496e+19,8.82885089201636e+19,8.893916029421684e+19,8.908351361700841e+19,9.258248956
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9476032e+19,8.475715961922547e+19,8.781458422438096e+19,9.029406482762457e+19,9.077754231682725e+19,9.300758886965895e+19,9.24004
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+19,8.780609285245205e+19,8.811815077083973e+19,8.872687599599392e+19,8.779070224083088e+19,8.727432068540364e+19,8.344152768098
897e+19

C.10 "NBSR_lower_Cycle6_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,6.654528967468182e+19,7.0571792101224915e+19,7.53561494724238e+19,7.855739668962532e+19,7.944474505619708e+19,8.3036
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−12.20157145,7.343709941648867e+19,7.762812217415452e+19,8.267889633962317e+19,8.464942533787733e+19,8.895879659180247e+19,9.2894
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1.0397048074063567e+20,1.0345462989595402e+20,1.065752090798308e+20,1.037045946571115e+20,1.0212360734609634e+20,9.6968283163752
92e+19
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,9.465597644535987e+19,9.61202073923524e+19,9.761681169482393e+19,9.997794380180827e+19,1.0234226017326594e+20,1.019150380230923
4e+20,1.059654224331855e+20,1.069047804528219e+20,1.0910669933613905e+20,1.0732032696659324e+20,1.0369557257443702e+20,9.92620150
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9e+19
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01e+19,8.123377097947067e+19,8.219329600743824e+19,8.212854929648026e+19,7.974831160265585e+19,8.037348886092231e+19,7.5920294994
95118e+19
−30.163,5.462605704021087e+19,5.922519636121026e+19,6.00807020830486e+19,6.501206633076684e+19,6.459598910624993e+19,6.7621571066
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2832399420875e+19,6.326549726713783e+19,6.270453600908379e+19,6.680799149373269e+19,6.670291076611234e+19,6.713331718075929e+19,
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5.945180984956323e+19,5.980526320610436e+19,6.026485871175698e+19,5.9272960328310415e+19,5.9211928592571335e+19,5.52390279513295
26e+19
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1776031084e+19,4.548488287443405e+19,4.592733642300516e+19,4.598253034054311e+19,4.87420670042173e+19,4.866644072297539e+19,4.878
622213824768e+19,4.952778426301485e+19,5.025936902576557e+19,4.9503477771086833e+19,4.776476316627357e+19,4.4518564748923355e+19

C.11 "NBSR_upper_Cycle6_cladding_fluence_position_5.csv " File

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63409145443362e+20,2.4613074318316365e+20,2.4910760978678068e+20,2.5050579419155644e+20,2.579735536713994e+20,2.601089952350197
e+20,2.6610583239307536e+20,2.6843254773846658e+20,2.6555895101702626e+20,2.6934042470264658e+20,2.6481450986637813e+20,2.5864166
490913423e+20
24.15442855,2.3476746414781822e+20,2.536119726476383e+20,2.6373691109583805e+20,2.6633931212669243e+20,2.7406659331044042e+20,2.8
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e+20,3.0696298977810344e+20,3.113003248295274e+20,3.1990315501433926e+20,3.13294780554002e+20,3.1740492186463705e+20,3.0514184785
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4817e+20,3.565118792465129e+20,3.536724064861398e+20,3.571530505149843e+20,3.593333920273771e+20,3.500893905740939e+20,3.394139778
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3572131774e+20,3.5758319202422496e+20,3.612164958788959e+20,3.6328907751423476e+20,3.7750350329663416e+20,3.8150588473021524e+2
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C.12 "NBSR_lower_Cycle6_cladding_fluence_position_5.csv" File

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−30.163,3.6272782815457844e+20,3.8976837330333145e+20,3.9886330692669e+20,4.19471485414294e+20,4.3030314905617295e+20,4.417391660085299e+20,4.510083114331257e+20,4.640319905740783e+20,4.7691378587690985e+20,4.886003445434172e+20,4.9458371171684116e+20,4.992093044393846e+20,5.122329835803372e+20,5.180798549094927 6e+20,5.203796302856203e+20,5.215757649209199e+20,5.031766639129231e+20
−32.15871425,3.2999756149737364e+20,3.527851834983951e+20,3.705206632635343e+20,3.8751529387392716e+20,3.9506028126565886e+20,4.030479821522874e+20,4.1619917714775396e+20,4.238312704401043e+20,4.301486932323957e+20,4.415443002308572e+20,4.516198487354072e+20,4.578438796342517e+20,4.708657627772535e+20,4.7102560659488355e+20,4.690006189052436e+20,4.7638037448549243e+20,4.547134552058327e+20
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−36.15014285,2.47643871456797e+20,2.6919405086964823e+20,2.8071178572877103e+20,2.8837351298731118e+20,2.9602626025609685e+20,3.048176702257532e+20,3.0735092533549793e+20,3.177335894896518e+20,3.238776984796813e+20,3.3215724903333115e+20,3.436031439744179e+20,3.442946031855145e+20,3.492506595310235e+20,3.565774331717208e+20,3.600904051636816e+20,3.603616008542675e+20,3.414362724466569e+20

C.13 "NBSR_upper_Cycle6_cladding_fluence_position_6.csv " File

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26.15014285,1.177843507367461 2e+20,1.2668436993974172e+20,1.3048956598538743e+20,1.3348330530107603e+20,1.3847835483826117e+20,1.419640630150814e+20,1.4289864463800775e+20,1.4554158415088301e+20,1.4995762826466515e+20,1.5249124136395558e+20,1.5698742280031 69e+20,1.581411879611584e+20,1.6103674578891892e+20,1.631951463911004e+20,1.6349128298712134e+20,1.6067161679597555e+20,1.5705216 951127487e+20
24.15442855,1.3719615767699477e+20,1.4572414864735473e+20,1.5249654847141118e+20,1.578965303074565e+20,1.6254396430630158e+20,1.6 435209581641525e+20,1.7126460327729883e+20,1.7244225042169043e+20,1.762341786986972e+20,1.765435830633571e+20,1.837432050375871 7e+20,1.8432221046099014e+20,1.8675127354340583e+20,1.9049702998554912e+20,1.9303382734931293e+20,1.9505742742212284e+20,1.865522 5701382187e+20
22.158714250000003,1.5353674153270342e+20,1.6596704861514606e+20,1.7074503745739824e+20,1.7849023007806118e+20,1.808062517716731 8e+20,1.8590744345796975e+20,1.8975934204922452e+20,1.9533923482801373e+20,1.9848051173096728e+20,2.0269170149696432e+20,2.056201 633909494e+20,2.11866098155413e+20,2.124939289674073e+20,2.1458386788341187e+20,2.1653263774109817e+20,2.178265105387668e+20,2.09 4826761971153e+20
20.163,1.716430000388812e+20,1.8053930426665786e+20,1.8834234435858655e+20,1.9295315931598822e+20,2.0172740007228673e+20,2.042222 712871515e+20,2.095479536188188e+20,2.159143597225239 2e+20,2.1762165619098173e+20,2.2297599690290912e+20,2.2692607698209197e+20, 2.2920601034500604e+20,2.353802991588194e+20,2.4158536919587517e+20,2.3810656025874717e+20,2.3684665294879425e+20,2.334554112846 8316e+20
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10.18442855,2.414919641046571e+20,2.4972806416496053e+20,2.6293533177889902e+20,2.7025224082789722e+20,2.832090129699327e+20,2.88 455088689766 4e+20,2.9599861122711755e+20,3.083636408878566e+20,3.0924780498995505e+20,3.2119251174021274e+20,3.2563456068052727 e+20,3.35009035290051e+20,3.3959915252837594e+20,3.496348927268641e+20,3.5109275514491e+20,3.520475037761675e+20,3.44708304875855 9e+20
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C.14 "NBSR_lower_Cycle6_cladding_fluence_position_6.csv" File

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4.0340225978076325e+20,4.061704470295901e+20,4.162544819059239e+20,4.193676311393629 6e+20,4.179808839612218e+20,4.11803410882935
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−12.20157145,2.8780921371242316e+20,3.07964015696452e+20,3.217996448331303e+20,3.3343070153276365e+20,3.450591046786692e+20,3.555
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682e+20,4.2114498142623426e+20,4.241403128741595e+20,4.3262213200965606e+20,4.435844931698872e+20,4.446613052726229e+20,4.3474497
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,3.553670994896283e+20,3.645499875200059e+20,3.726231593814232e+20,3.826106049020657e+20,3.9323331118514025e+20,3.99164003766742
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020685836771779e+20,4.118394992136329e+20,4.212070745834645e+20,4.210133651613361e+20,4.269955366852578e+20,4.187780115010488e+
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799336e+20,3.467451727173047e+20,3.5295554986181604e+20,3.630045578289431e+20,3.711695426493419e+20,3.6819172465602e+20,3.5954273
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−32.15871425,1.920318454617008e+20,2.0556868444863034e+20,2.1603111608654656e+20,2.2365742950020465e+20,2.2636087003807344e+20,2.
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−36.15014285,1.4412246361726285e+20,1.5542235681166846e+20,1.6283797805934032e+20,1.700052266780914e+20,1.765128018401148e+20,1.7
96816757018373e+20,1.8667803546051894e+20,1.8830890958161645e+20,1.939413427242159e+20,1.9819498934985 75e+20,1.9995429547137984
e+20,2.0758538528174798e+20,2.1124888155832995e+20,2.1398628758391436e+20,2.153618898363988e+20,2.2248137448804975e+20,2.11038720
10308924e+20

C.15 "NBSR_upper_Cycle6_cladding_fluence_position_7.csv " File

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26.15014285,1.1189505359329701e+20,1.2229910704920193e+20,1.2757543328153241e+20,1.3067159977111357e+20,1.3482281922286263e+20,1.3605353744180996e+20,1.4244913263652074e+20,1.4431298877491791e+20,1.4728656108227535e+20,1.486802075001088e+20,1.5401119693923166e+20,1.52453560901021e+20,1.562391206490811e+20,1.5731540204107127e+20,1.5860025275606545e+20,1.56029489904586e+20,1.4481557185096067e+20
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22.158714250000003,1.53296329564966e+20,1.584707593341495e+20,1.7009014039738057e+20,1.7275536976156924e+20,1.7523272752183063e+20,1.8204121567658495e+20,1.8387800556695873e+20,1.894202178828134e+20,1.9332094186265942e+20,1.965933043197656e+20,1.9983223199990166e+20,2.0510961965372326e+20,2.050135610087774e+20,2.095325630071976e+20,2.073571796611584e+20,2.041824679812347e+20,1.9851978432613853e+20
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8.18871425,2.403944542828446e+20,2.5551652626675378e+20,2.654509007128405e+20,2.7975408601635475e+20,2.8527082421642268e+20,2.920177499246924e+20,3.0124840192217016e+20,3.070023678255022e+20,3.164415891459840 4e+20,3.257300886147275e+20,3.2867553325257045e+20,3.367683414115734e+20,3.4261942738134244e+20,3.4417281773358855e+20,3.5328246768107926e+20,3.4958341378454507e+20,3.385377310372614e+20
6.193,2.4967923877636925e+20,2.6290136629118334e+20,2.7689620865152883e+20,2.8425610527091717e+20,2.9692310934587834e+20,3.0465503419790236e+20,3.1054645418433374e+20,3.1899112356764117e+20,3.2973960829741276e+20,3.3907746386549316e+20,3.4457775003244875e+20,3.506941913749965e+20,3.533100646398483e+20,3.562369344015967e+20,3.664796517908522e+20,3.6027882743976093e+20,3.544627683791987e+20
4.19728575,2.5388989783162074e+20,2.703817342498132e+20,2.835401764751604e+20,2.9334824176380448e+20,3.062275301369886e+20,3.0769069966249e+20,3.1850923821067515e+20,3.2834330832585156e+20,3.372454503718294e+20,3.4681681866795496e+20,3.5171209458497526e+20,3.589053480302587e+20,3.6813334647400864e+20,3.7450665181740604e+20,3.76447461013909e+20,3.770100144041998e+20,3.621564820575426e+20
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C.16 "NBSR_lower_Cycle6_cladding_fluence_position_7.csv" File

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−12.20157145,2.783837908713259e+20,2.969554827013574e+20,3.111243981862456e+20,3.269268413459592e+20,3.366844891137754e+20,3.3825751576360714e+20,3.516064831466094e+20,3.620922660573301e+20,3.697329086611183e+20,3.8298369455619244e+20,3.877468234975691e+20,3.9981253229781385e+20,4.11507274286914e+20,4.1364497717001886e+20,4.1925352832906815e+20,4.2119911392228114e+20,4.0932127672596365e+20
−14.197285749999999,2.8361872166550287e+20,2.998000922975444e+20,3.153393029274615e+20,3.2875991626111405e+20,3.37468879595709e+20,3.4522362500979196e+20,3.5774733718345266e+20,3.65769030102551e+20,3.777763607207844e+20,3.796386247269449e+20,3.931340682757211e+20,4.025632060920372e+20,4.0945554654458964e+20,4.1816291774694785e+20,4.263608066335713e+20,4.217229254281461e+20,4.096052069748368e+20
−16.192999999999998,2.8294471901864513e+20,3.0229390209091804e+20,3.157442352263217e+20,3.2409762236139327e+20,3.379083080930304 3e+20,3.454210494071393e+20,3.5746924475278066e+20,3.617844538249069e+20,3.731358259616316e+20,3.836518593848491e+20,3.910940161697988e+20,3.99934595769292e+20,4.067112412793129e+20,4.155863170772673e+20,4.201366310096755e+20,4.2015308304278775e+20,4.102664725638012e+20
−18.188714249999997,2.799907830088733e+20,2.978067427372312e+20,3.125461722735936e+20,3.18957688790671e+20,3.347304121486335e+20,3.4170925845271164e+20,3.477222111998757e+20,3.590491706423047e+20,3.6976209775212397e+20,3.742614634529586e+20,3.8863629470712306e+20,3.924038102898342e+20,4.0621555744296246e+20,4.0520349205118465e+20,4.147302806446834e+20,4.154918505645582e+20,4.009689510123831e+20
−20.18442855,2.685905854835574e+20,2.8999733411635583e+20,3.041630653367707e+20,3.13013197729684e+20,3.20268013621452e+20,3.319643477427889e+20,3.451896595220763e+20,3.526509218938661e+20,3.5776538134880164e+20,3.664647918899766e+20,3.729384015642843e+20,3.850561200175936e+20,3.915016020223868e+20,3.9807339318462264e+20,4.013340800053266e+20,3.986608899799546e+20,3.9198295666860727e+20
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−26.171571450000002,2.3630585869907132e+20,2.5279769511726386e+20,2.632696795486001e+20,2.7630181261650464e+20,2.834526092021434 6e+20,2.9263178225730218e+20,2.967193164195843e+20,3.027179399766183e+20,3.1339106378052076e+20,3.2083481269770715e+20,3.28182502 9699477e+20,3.3526696071239185e+20,3.403999950434219e+20,3.458424337391117e+20,3.537632916165542e+20,3.5097387593790515e+20,3.420966772969685e+20
−28.167285749999998,2.2190979901508452e+20,2.3242689385979314e+20,2.4465128517295954e+20,2.5403265902217563e+20,2.63107282060459 57e+20,2.7191973399043817e+20,2.7890494882346313e+20,2.869940420072472e+20,2.9441019396566463e+20,3.015785040059068e+20,3.1101241 8218933e+20,3.1133774390595964e+20,3.2018203848067174e+20,3.27073317511733e+20,3.2492659254595374e+20,3.2494092173608387e+20,3.18 67800422776236e+20
−30.163,2.04412265734061e+20,2.1905457520398637e+20,2.275082666699683e+20,2.3867017507052903e+20,2.423973566365778e+20,2.51050595 34288922e+20,2.5999413282702056e+20,2.6143554321195414e+20,2.701954547781226e+20,2.7605503211982055e+20,2.8580631135869e+20,2.895 1173378417095e+20,2.9515902682764594e+20,2.9546683906006915e+20,2.989562622121083e+20,2.961827678558259e+20,2.885840514009368e+20
−32.15871425,1.8686802990742844e+20,1.9937529004797683e+20,2.0773080002603067e+20,2.1682505936190027e+20,2.2321640887064658e+20,2.2923466872526612e+20,2.3591790914406898e+20,2.3974061864431806e+20,2.491575501134864e+20,2.511323247977051e+20,2.5881701639337 447e+20,2.6222683293358005e+20,2.669989839576311e+20,2.7233103481824507e+20,2.7357767435955913e+20,2.753465332745015e+20,2.5948146 62468122e+20
−34.15442855,1.6682945357667664e+20,1.782434495813771e+20,1.87484715793766e+20,1.951683459679443e+20,2.0297085534912745e+20,2.057 7566163939767e+20,2.129062912167053e+20,2.1812105500255112e+20,2.1982463649578998e+20,2.2803579315105225e+20,2.2939971976713444 e+20,2.3293584546478257e+20,2.410732333264124e+20,2.429450501259929e+20,2.4631612478177273e+20,2.458522835901557e+20,2.3994335014 912092e+20
−36.15014285,1.4227028311526795e+20,1.5544146239850853e+20,1.5794588640679333e+20,1.663889636578641e+20,1.7036133358836036e+20,1.7565039368785842e+20,1.7765223781082628e+20,1.8507528900893596e+20,1.8955767196591273e+20,1.9109620241728 33e+20,1.9632423397176807e+20,1.974817141078285e+20,2.0352332523525256e+20,2.0449771016409576e+20,2.1156624658417317e+20,2.0791707949772128e+20,1.9953715682537184e+20

C.17 "NBSR_upper_Cycle7_cladding_fluence_position_0.csv " File

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24.15442855,5.279686051314456e+19,5.916556665570396e+19,6.145362556524849e+19,6.196499792715863e+19,6.325537762394629e+19,6.56863
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9331119e+19,9.37584874583702e+19,9.16346570585888e+19,9.253485928387687e+19,9.126810764948949e+19,8.810949389333404e+19
18.167285749999998,7.099783548575515e+19,7.826692353468549e+19,8.30871381048831e+19,8.712520092125261e+19,8.749507443038542e+19,
9.202995544095082e+19,9.536753155566653e+19,9.235338139015612e+19,9.768901114662892e+19,9.817594688126581e+19,9.915251356678288e
+19,1.0065105379810073e+20,1.0234634481518512e+20,1.0007966795450473e+20,9.877248807646667e+19,1.0286741995557143e+20,9.47269684
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58723798687e+20,1.1325388495213353e+20,1.18927315144443e+20,1.1777466099769192e+20,1.2164229630693851e+20,1.21565033441295e+20,1.
2457379317431868e+20,1.3038737461128376e+20,1.2595284548551347e+20,1.2716389598420438e+20,1.2335375862148316e+20,1.217006926588
7833e+20
10.18442855,9.419241727355398e+19,1.0288718487468953e+20,1.0641432453185608e+20,1.1059370651992e+20,1.1434454912528694e+20,1.1410
018285255405e+20,1.2045819797068135e+20,1.2488913347479385e+20,1.2719803538995382e+20,1.2701026865833186e+20,1.2830127723890965
e+20,1.3232613349568658e+20,1.3267112117483889e+20,1.3161369800202048e+20,1.3091563699498572e+20,1.310620770775426e+20,1.25324860
10080652e+20
8.18871425,9.961788757147271e+19,1.0423479299637823e+20,1.1103122995676162e+20,1.1930374701310136e+20,1.1730479496592979e+20,1.20
42675378117529e+20,1.2552879812988874e+20,1.2734717068875404e+20,1.299336798769819e+20,1.3188142281552932e+20,1.325848742550508
3e+20,1.3723232546404794e+20,1.3833646571841823e+20,1.402060473859077e+20,1.3685679200080403e+20,1.3509501898304966e+20,1.2819346
858917455e+20
6.193,9.957117048992083e+19,1.1028735027358946e+20,1.1502284521320356e+20,1.2089122938045066e+20,1.2058127951246228e+20,1.2554496
942734901e+20,1.2941170633118114e+20,1.3127949118784168e+20,1.3869582788420187e+20,1.3706162843530071e+20,1.3525134152516554e+2
0,1.4392005536928177e+20,1.4430367448125583e+20,1.4829439133228332e+20,1.4254279986891591e+20,1.4075856671580006e+20,1.3487670 64
673361e+20
4.19728575,1.0245954389740698e+20,1.0757775954358067e+20,1.146482101553741e+20,1.2066752643225035e+20,1.2699769098252943e+20,1.27
68407271917622e+20,1.2996602247190243e+20,1.3612369318260536e+20,1.3974336859746122e+20,1.4171716529302797e+20,1.40643570822749
29e+20,1.4218882813561902e+20,1.4892147831157578e+20,1.4588756322694727e+20,1.482315029532712e+20,1.4568182838703615e+20,1.387012
1831668864e+20
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0067056304547e+20,1.2922753322121701e+20,1.3766805209006064e+20,1.3665554918796517e+20,1.4159588056207598e+20,1.425149493010676
8e+20,1.4562882246758307e+20,1.4867621363342846e+20,1.495746190478876e+20,1.4687311396660896e+20,1.447636580534589e+20,1.39596030
10948995e+20
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8e+20,1.1707929520690055e+20,1.2349930029862556e+20,1.2768317431376177e+20,1.274576745547325e+20,1.317367795438014e+20,1.30889583
23796643e+20,1.3505728595564238e+20,1.362620476164321e+20,1.3309516853046362e+20,1.3934267978261247e+20,1.3587124126114236e+20,1.
3011246255445926e+20

C.18 "NBSR_lower_Cycle7_cladding_fluence_position_0.csv" File

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1.2213102885240429e+20,1.3010976733821588e+20,1.3218418544020203e+20,1.3580116563881455e+20,1.4113589698987291e+20,1.42095393972
51527e+20,1.4116734117937897e+20,1.4339179298557981e+20,1.49279942071945e+20,1.4787842962538876e+20,1.4870945463376344e+20,1.4354
182668979449e+20
−12.20157145,1.0933054850719046e+20,1.1864701265513174e+20,1.2349840189321108e+20,1.3171521781385437e+20,1.333592997223146e+20,1.
390938214828073e+20,1.4406380023559525e+20,1.46135523121338e+20,1.5119893603722972e+20,1.5174786174546425e+20,1.541430105804123
3e+20,1.5784444088798396e+20,1.5921540755044863e+20,1.6019287264138016e+20,1.590662722516484e+20,1.5946606266108274e+20,1.5292656
964923464e+20
−14.197285749999999,1.1253605902598067e+20,1.2601932748618342e+20,1.3103782013135218e+20,1.3657738791690723e+20,1.40671421390597
55e+20,1.41881573483874e+20,1.5068864176181694e+20,1.509770298998583e+20,1.5521301142903318e+20,1.5937622211963683e+20,1.60317750
99399e+20,1.643542865211549e+20,1.6357536902681883e+20,1.6545752837011073e+20,1.643012806017018e+20,1.5884526451969147e+20,1.5657
409563193875e+20
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7e+20,1.4606365068818127e+20,1.4597830217380767e+20,1.5126362122707078e+20,1.5506118091398955e+20,1.5798549053805406e+20,1.596044
1709490943e+20,1.5905010095418812e+20,1.647325152006422e+20,1.6546561401884085e+20,1.645690054152106e+20,1.593852061737814e+20,1.
5415379144538584e+20
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72e+20,1.4156982680505667e+20,1.4367029866406216e+20,1.4997440945732193e+20,1.50917735142504e+20,1.556056145951518e+20,1.56826547
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560602077348681e+20
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1.377057851174679e+20,1.4286712422353566e+20,1.481677161688446e+20,1.4904096623169887e+20,1.5037240305592733e+20,1.5688674071617
054e+20,1.5806275340369756e+20,1.5843289643445472e+20,1.586116791119321e+20,1.5667292022752926e+20,1.578830723208057e+20,1.493787
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1.2880348586559232e+20,1.3055268120754425e+20,1.3563945266421188e+20,1.365360612678421e+20,1.4112062409782711e+20,1.434717510674
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7556164496782e+20
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8e+20,1.2012848318357486e+20,1.2225950082667196e+20,1.2374276816594398e+20,1.2881965716305256e+20,1.293227641951497e+20,1.3694483
573142102e+20,1.3882879188554182e+20,1.3805436641827806e+20,1.3815498782469749e+20,1.3857364474783544e+20,1.3288584006889462e+2
0,1.315058893522854e+20
−28.167285749999998,9.100217964680957e+19,9.662260391966594e+19,1.0224392659793678e+20,1.052005788169218e+20,1.0863158909474126e
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72045442e+20,1.2946561065604868e+20,1.2924011089701944e+20,1.2537247558777284e+20,1.261298313521619e+20,1.2087775329923378e+20,1.
1995239572234086e+20
−30.163,8.102898114089866e+19,8.763639360207985e+19,9.58868098852239e+19,9.779681979636403e+19,1.0202651248763766e+20,1.056641560
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493749669882996e+20,1.1669028566243972e+20,1.187449388453078e+20,1.1703976536866433e+20,1.1429603523290612e+20,1.10953068685703
68e+20
−32.15871425,7.635583553704801e+19,8.102502815707505e+19,8.607478531066698e+19,8.995823255520803e+19,9.237045109303083e+19,9.4717
08603559811e+19,9.693345219306883e+19,1.0027552033485681e+20,1.0451240026944609e+20,1.057171619302358e+20,1.059588329867253e+20,
1.0850221871505914e+20,1.0635682658533073e+20,1.0667306529122034e+20,1.0900352893632735e+20,1.0394370964209346e+20,1.01486570833
54772e+20
−34.15442855,6.704332437293036e+19,6.9260139733108285e+19,7.625988583878377e+19,7.641638806198256e+19,8.363489588607885e+19,8.207
247902979296e+19,8.878868838666514e+19,8.781850037959072e+19,9.176133222202755e+19,9.270914993428193e+19,9.433616213986743e+19,9.
72874239263657e+19,9.44313931138001e+19,9.777975009348932e+19,9.679330094841319e+19,9.275407020500489e+19,8.9732373434013e+19
−36.15014285,5.645858130139713e+19,5.824173636801561e+19,6.125166402807808e+19,6.475787083908775e+19,6.830064275046592e+19,6.9434
34054297191e+19,7.427072641062979e+19,7.1912951240923234e+19,7.444735291511246e+19,7.67408920976852e+19,7.744191784258766e+19,7.6
19484128677693e+19,7.733716377126173e+19,8.001513063068153e+19,8.111073603361445e+19,7.813638522796458e+19,7.24378895245917e+19

C.19 "NBSR_upper_Cycle7_cladding_fluence_position_1.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0826089829766647e+20,1.1711609820844863e+20,1.2071275993021614e+20,1.2861861954597018e+20,1.3051039777370235e+20,1.336557511896679e+20,1.3784601607130196e+20,1.381810461194914e+20,1.4220512335440074e+20,1.436121433666068e+20,1.4536587434785836e+20,1.4887121250656947e+20,1.5081555487815382e+20,1.503594680138199e+20,1.5310820107162773e+20,1.5437711738373532e+20,1.4749127099271152e+20
24.15442855,1.2588900072229044e+20,1.3558947454229224e+20,1.4174585078437877e+20,1.4996444050850618e+20,1.493718992505356e+20,1.555505754324382 2e+20,1.561473642979928e+20,1.635020968297205e+20,1.644317919396725e+20,1.7121893790798574e+20,1.7269073393584 81e+20,1.7344893188959535e+20,1.7704134600377883e+20,1.7990423351540718e+20,1.7839314711739266e+20,1.7923045676239225e+20,1.7249375113413925e+20
22.158714250000003,1.423129063968452e+20,1.5595675290766e+20,1.607199138621885e+20,1.648087671127541e+20,1.7138512555471048e+20,1.764918117725965e+20,1.7816218345501167e+20,1.839012322519684e+20,1.864859014668435e+20,1.909395180186867e+20,1.9577966686067556e+20,1.9765976416755168e+20,2.0134562564858357e+20,1.9868078284056006e+20,2.0188188610606586e+20,1.9950269490806764e+20,1.9649963634616667e+20
20.163,1.551911216406407e+20,1.6483319085636223e+20,1.7563326408967678e+20,1.835077975994987e+20,1.8920702507535525e+20,1.9222601216569626e+20,1.9897439871480535e+20,2.0410869438198746e+20,2.0392923296156267e+20,2.0805684563133245e+20,2.120984442475259e+20,2.1458913514460463e+20,2.1914628713130826e+20,2.2088886814265177e+20,2.233131901712302e+20,2.2211589578348505e+20,2.1493106755511812e+20
18.167285749999998,1.6784527558437922e+20,1.8541815911041093e+20,1.9184638223787462e+20,1.9777498052326815e+20,2.033568678396162e+20,2.0874655091278748e+20,2.1774457662858222e+20,2.2061064984589858e+20,2.248874597320 57e+20,2.281570556698551e+20,2.312243592964643e+20,2.3560470461748317e+20,2.3908137142500216e+20,2.3836405669425106e+20,2.416474573566972e+20,2.437064851330501e+20,2.3514755558512532e+20
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14.17585715,1.9779621856118827e+20,2.0915113553516518e+20,2.20018639538876e+20,2.2645217217581977e+20,2.32784293181695e+20,2.400259331614982e+20,2.4630071146498926e+20,2.5349456585947224e+20,2.602275548310893e+20,2.6595332985434594e+20,2.6898930737502303e+20,2.7435722145932614e+20,2.8157974720500128e+20,2.8021042471010337e+20,2.8545781292920996e+20,2.845743105517342e+20,2.7644279678307913e+20
12.18014285,2.0326713712940384e+20,2.2358132039997135e+20,2.3065942748778984e+20,2.4409354837414376e+20,2.502743483598384e+20,2.570094611352474e+20,2.6220163045576176e+20,2.6892984086884673e+20,2.7380556342435178e+20,2.8233794515874854e+20,2.8484615743711137e+20,2.8982276067273 68e+20,2.9895670983122013e+20,2.992497947545174e+20,3.020309158201534e+20,3.036890756307645e+20,2.96371509665397e+20
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C.20 "NBSR_lower_Cycle7_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
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C.21 "NBSR_upper_Cycle7_cladding_fluence_position_2.csv " File

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C.22 "NBSR_lower_Cycle7_cladding_fluence_position_2.csv" File

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C.23 "NBSR_upper_Cycle7_cladding_fluence_position_3.csv " File

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C.24 "NBSR_lower_Cycle7_cladding_fluence_position_3.csv" File

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−34.15442855,1.6362793220439704e+20,1.790472786853315e+20,1.881982182741509e+20,1.9278616541583868e+20,1.9772400923225997e+20,2.0585764680470702e+20,2.091304284481931e+20,2.1415428631819076e+20,2.181406660357919e+20,2.269592303311619e+20,2.2871880177284 15e+20,2.3247315592616565e+20,2.3446050532453823e+20,2.363267979067663e+20,2.4317341038125577e+20,2.3795734826808135e+20,2.398379765259054e+20
−36.15014285,1.3791450874359425e+20,1.504162797652561e+20,1.5332589096030863e+20,1.616671303534246e+20,1.6926556937028534e+20,1.7244915125450703e+20,1.7756061603092506e+20,1.858763697785369e+20,1.872026852466466e+20,1.9054502146432103e+20,1.930633218106958 8e+20,1.971819083143496e+20,2.0067503460125665e+20,2.0319970635900758e+20,2.038092380473142e+20,2.046778737982459e+20,1.9579718824195965e+20

C.25 "NBSR_upper_Cycle7_cladding_fluence_position_4.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,2.7396007015022043e+19,2.845116583398699e+19,3.1845482149470814e+19,3.1669047149449654e+19,3.2209342834136785e+19,3.4
078874217147 64e+19,3.436394178113007e+19,3.4025460551778607e+19,3.658262650754723e+19,3.5301282584733303e+19,3.724111187325966e
+19,3.5756413737360863e+19,3.626756021500267e+19,3.719072462829425e+19,3.54848854225525e+19,3.428812198575534e+19,3.180597939893
9443e+19
24.15442855,3.359066482045954e+19,3.676325602005761e+19,3.6471232998656352e+19,3.7950834005454316e+19,3.940017080821617e+19,4.149
811728905761e+19,4.28192825329717e+19,4.293779078456581e+19,4.415228798302625e+19,4.4536059328242295e+19,4.3242450438529516e+19,
4.356882598626653e+19,4.589762993929937e+19,4.413120923039055e+19,4.294644528501825e+19,4.287046620435912e+19,4.148340994779795e
+19
22.158714250000003,3.6809448752533803e+19,3.925123906730153e+19,4.138730782620953e+19,4.336541867808685e+19,4.6176963233043366e+
19,4.827634328144441e+19,4.768539487631787e+19,4.891682941001957e+19,4.8975340204489425e+19,4.988029300026452e+19,5.028519619321
1064e+19,5.126442902661168e+19,5.0560972115603464e+19,5.101838635730743e+19,5.09854143034365e+19,4.818634709575803e+19,4.74617583
3701931e+19
20.163,4.288145688898277e+19,4.430785661079092e+19,4.600801464138904e+19,4.729668568728539e+19,4.9255310639371035e+19,5.064326951
2543805e+19,5.246236055549446e+19,5.31927897746612e+19,5.3494369913126494e+19,5.32172135182693e+19,5.453662662405499e+19,5.420478
228155356e+19,5.541853614868678e+19,5.5467914586851e+19,5.522898666024996e+19,5.357985301575486e+19,5.089796668230051e+19
18.167285749999998,4.432219228638697e+19,4.836410947314919e+19,5.0855543701555134e+19,5.2601682084250255e+19,5.352994362664266e+
19,5.692909159575329e+19,5.7365002324063175e+19,5.981056239056171e+19,6.125421801817994e+19,6.038611321819619e+19,5.980100527349
767e+19,6.149261499383295e+19,6.167420021804973e+19,6.202675164752326e+19,6.125527992007593e+19,6.141828186111264e+19,5.750676622
717978e+19
16.171571450000002,4.880830303624791e+19,5.085501275060712e+19,5.396160674736851e+19,5.680484907392075e+19,5.8953076609538015e+1
9,6.130784406392817e+19,6.14804031220289e+19,6.391852987525541e+19,6.437036913200535e+19,6.403799383855593e+19,6.665398915936321e
+19,6.817941123297377e+19,6.636462089270195e+19,6.812366138343352e+19,6.64442635349023e+19,6.504733159070827e+19,6.2119668063423
66e+19
14.17585715,5.278040017334782e+19,5.622823634439027e+19,5.84152232992117e+19,6.025178262835161e+19,6.332439576444085e+19,6.570942
742286713e+19,6.5603768184214675e+19,6.800950692961304e+19,6.850488416409916e+19,7.013224881972619e+19,7.070726869641266e+19,7.31
0875983422701e+19,7.3188402476427354e+19,7.084690879573726e+19,7.112618899438647e+19,7.061010467292824e+19,6.702193816632879e+1
9
12.18014285,5.376727870039968e+19,6.042274883360835e+19,6.3196967536920306e+19,6.555545164794647e+19,6.865567423333183e+19,6.8423
11771810682e+19,7.1222291115974885e+19,7.239834746579994e+19,7.369121302418552e+19,7.563077683723788e+19,7.51476114745558e+19,7.5
12743533853172e+19,7.747901708723384e+19,7.781139238068327e+19,7.641552233838526e+19,7.434746839591635e+19,7.0663730718676476e+
19
10.18442855,5.76501229831404e+19,6.208781100654352e+19,6.48896391591516e+19,6.910008017680974e+19,7.129980995438322e+19,7.1403876
34019167e+19,7.44026872945086e+19,7.555962941020556e+19,7.766325706619064e+19,7.970051585367541e+19,7.985077497196007e+19,8.16820
2479161994e+19,8.189387421987286e+19,8.232606829154673e+19,7.850215956403424e+19,7.906815327460468e+19,7.365086075213734e+19
8.18871425,6.123988234258387e+19,6.4990519839272034e+19,6.862487907834769e+19,6.943723402879119e+19,7.490496689131874e+19,7.63364
1064713291e+19,7.646118411991346e+19,7.845331207681804e+19,8.047995184534277e+19,8.009979096657312e+19,8.390033785237352e+19,8.47
0207378385696e+19,8.60082131159426e+19,8.429589630863522e+19,8.563070699191296e+19,8.287029301324908e+19,7.870869948280714e+19
6.193,6.364296633324222e+19,6.682973392315196e+19,6.8891416454244835e+19,7.454976070710521e+19,7.72384963077888e+19,7.73218556066
2517e+19,7.986564159850412e+19,8.328762045837887e+19,8.551389778335246e+19,8.558876186702078e+19,8.726975256839602e+19,8.86704011
6922606e+19,8.932665654095688e+19,8.672605879764168e+19,8.66825208199055e+19,8.33412465041271e+19,8.180626731345248e+19
4.19728575,6.511263855731256e+19,6.902627799503742e+19,7.153661407719224e+19,7.558352220286566e+19,7.847773582042615e+19,8.154981
800556737e+19,8.308798290193e+19,8.457358365444042e+19,8.488578281186576e+19,8.792972459676287e+19,8.939090160566516e+19,8.967442
94118984e+19,8.887375538231094e+19,8.979707908088691e+19,8.940258252652121e+19,8.798919110293914e+19,8.459641454520451e+19
2.20157145,6.21286942295397e+19,6.993208031232932e+19,7.314751925343117e+19,7.562546732775784e+19,7.799775616343207e+19,7.9407430
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7691426637e+19,9.010343777788423e+19,8.825572847883628e+19,9.19962779075124e+19,8.954487738058583e+19,8.341292488210742e+19
0.20585714999999993,5.977923628462958e+19,6.269999744959015e+19,6.656691320389082e+19,6.965757867221214e+19,7.097911558178984e+1
9,7.440215634356059e+19,7.528459681914038e+19,7.90479771385806e+19,8.043641386760659e+19,8.038969018418238e+19,8.174255319969222e
+19,8.41551943074146e+19,8.349840798473578e+19,8.458791933003648e+19,8.382281901396517e+19,8.293772378364536e+19,7.9390971450990
08e+19

C.26 "NBSR_lower_Cycle7_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,6.360155215929804e+19,6.769890062503169e+19,7.21572957354069e+19,7.544122734880106e+19,7.686205208565519e+19,8.05341
08842039e+19,8.203563812298947e+19,8.381591665164114e+19,8.712851961622743e+19,8.810759316434364e+19,8.879836034769461e+19,8.9305
4185030368e+19,9.018626612577259e+19,9.180725937002358e+19,9.142338183461793e+19,8.998928332406374e+19,8.697401289035876e+19
−12.20157145,7.095947039671375 5e+19,7.457365349976533e+19,7.90904532144208e+19,8.3086390049086e+19,8.60581225050548e+19,8.7169933
7901716e+19,9.041829169004959e+19,9.233396271044184e+19,9.268651413991537e+19,9.292066350798438e+19,9.661714400797629e+19,9.87717
429549696e+19,9.937490323190019e+19,9.828592283754748e+19,1.0071714722844996e+20,9.799071411045823e+19,9.222883442273739e+19
−14.197285749999999,7.170864218434498 6e+19,7.794147536294383e+19,8.033659508938215e+19,8.203085956445744e+19,8.626891003141172e+
19,8.885145544249483e+19,9.20674253345447e+19,9.442803324936287e+19,9.654121802241198e+19,9.79163809777379e+19,9.94715363044366e+
19,1.0091837763774282e+20,1.0121942682526014e+20,1.030262529013119e+20,1.0266998481520239e+20,1.0150614033718136e+20,9.509066003
246973e+19
−16.192999999999998,7.313105977404311e+19,7.805722266960834e+19,8.112718105095756e+19,8.298391651612156e+19,8.866402975785003e+1
9,9.119878958361295e+19,9.255908591239481e+19,9.411158648435352e+19,9.576921534401664e+19,9.838521066482393e+19,9.98532900360502
5e+19,1.005153858682091e+20,1.0084245165217851e+20,1.0065130931089768e+20,1.0388533153517963e+20,9.989045660241042e+19,9.46138660
8116367e+19
−18.188714249999997,7.038498147097527e+19,7.388978867873838e+19,7.915363637723305e+19,8.177441025657235e+19,8.473924035021713e+1
9,8.782406535811041e+19,9.041829169004959e+19,9.417052203958177e+19,9.635432328871518e+19,9.711252124246244e+19,1.00074165630419
2e+20,1.0086687539578662e+20,1.0037999337646852e+20,9.986178525121828e+19,1.0072139483603398e+20,9.547666137166738e+19,9.27948281
3330784e+19
−20.18442855,6.9156891928245985e+19,7.50313332169433e+19,7.798872999731603e+19,8.008014578149705e+19,8.53121364231116e+19,8.64043
025231523e+19,8.942913007392134e+19,9.533489746855077e+19,9.284686132621206e+19,9.384398720656035e+19,9.598265762511358e+19,9.671
855563904475e+19,9.845741999375224e+19,9.644086829323955e+19,9.820787304819116e+19,9.386787999922047e+19,9.33666623043063e+19
−22.180142850000003,6.85107246245272e+19,7.102106070668202e+19,7.3699708239353545e+19,7.781245428257928e+19,8.099231951016498e+1
9,8.540133618237599e+19,8.639262160229625e+19,8.671968738626567e+19,9.089030708282361e+19,9.203397542482056e+19,9.32716220846138
8e+19,9.312401772106924e+19,9.435954057707058e+19,9.185132829870776e+19,9.13718795926617e+19,8.864066791613794e+19,8.590839433771
816e+19
−24.17585715,6.428488602937699e+19,6.720670909623359e+19,7.226242402311136e+19,7.64033104665812e+19,7.88855061484919e+19,7.978918
46619918e+19,8.275507665753257e+19,8.438668892074361e+19,8.748478770233695e+19,8.808794797926756e+19,8.715294335983552e+19,8.8821
72218940672e+19,9.030944674570913e+19,9.004609507549998e+19,8.968026987232641e+19,8.691348448228649e+19,8.215350923344598e+19
−26.171571450000002,5.972720309172536e+19,6.464487077212253e+19,6.86493028219558e+19,7.157218779070839e+19,7.455560116753324e+19
,7.830995532085743e+19,7.787776124918356e+19,8.11362072170736e+19,8.050118988326286e+19,8.12546092784781e+19,8.504453714531844e+1
9,8.316603269128634e+19,8.497338971828612e+19,8.591158004340618e+19,8.369114317886061e+19,7.992510810468037e+19,7.86651615050709
6e+19
−28.167285749999998,5.708094356688195e+19,6.118997295347165e+19,6.42647098933529e+19,6.666513912927125e+19,6.902999465167344e+19
,7.195606532611404e+19,7.274240368010543e+19,7.502974036409929e+19,7.528565872103639e+19,7.68657687422912e+19,7.683338073446307e+
19,7.80264275146242e+19,7.999147697318067e+19,8.08133890406882e+19,7.847932867327015e+19,7.446958711395687e+19,7.093823235879366e
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−30.163,5.186997858280829e+19,5.764534442460838e+19,6.050132957391269e+19,6.1570664783189295e+19,6.299786093141945e+19,6.69125622
71040315e+19,6.826754909034215e+19,6.862063147076367e+19,6.9910842274409234e+19,7.101575119720199e+19,7.197730336403413e+19,7.248
3830568428315e+19,7.367475354479745e+19,7.484550038514249e+19,7.1430423887591784e+19,7.124140535010297e+19,6.699273586418866e+1
9
−32.15871425,4.800975281045246e+19,5.241781377095706e+19,5.482068538123621e+19,5.523004856214598e+19,5.840991378973168e+19,5.7454
20208332756e+19,6.057513175568501e+19,6.124997041059592e+19,6.174163098844604e+19,6.401834865347984e+19,6.520183831657694e+19,6.7
042114302352876e+19,6.469637301407877e+19,6.527723335119327e+19,6.441125235500153e+19,6.172092390147395e+19,5.8938740933941944e
+19
−34.15442855,4.103985352583683e+19,4.54872579515884e+19,4.902286031433563e+19,4.880771899020509e+19,5.056601614960948e+19,5.21669
3944802597e+19,5.423823219127771e+19,5.413469675641726e+19,5.592824905876898e+19,5.621655542353423e+19,5.769419191182459e+19,5.76
140183186762 55e+19,6.015939716339922e+19,5.729650965177087e+19,5.744464496626352e+19,5.626274815601042e+19,5.336003932328192e+1
9
−36.15014285,3.408184754245645e+19,3.744749250674815e+19,3.8858069890305835e+19,4.052637086402382e+19,4.156029164506867e+19,4.271
0065922967626e+19,4.538335085106435e+19,4.529792084353078e+19,4.608569276508177e+19,4.61610877996981e+19,4.730401281036783e+19,4.
686964183980715e+19,4.744652004481163e+19,4.832694290678903e+19,4.756492210621615e+19,4.582579227603465e+19,4.419810904983885e+
19

C.27 "NBSR_upper_Cycle7_cladding_fluence_position_5.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.922246192885063e+20,2.061840426183724e+20,2.1338386360984794e+20,2.1517169038462167e+20,2.2467951488584273e+20,2.32
27104063802242e+20,2.371960991200874e+20,2.4510386357815676e+20,2.5132082904621403e+20,2.5270706860072446e+20,2.526136344376207
3e+20,2.5594941374150748e+20,2.5250313057164224e+20,2.5675528339827737e+20,2.6089603395351953e+20,2.5930675477534127e+20,2.457075
920166733e+20
24.15442855,2.2849234746480732e+20,2.3821309404925516e+20,2.5006755349304348e+20,2.60468392976237e+20,2.7037960150855018e+20,2.69
97352226121466e+20,2.8128354802384077e+20,2.835717866144682e+20,2.8247393519799914e+20,2.92799308626378e+20,3.010278038174093e+
20,2.98945300066693e+20,3.0237181831744016e+20,3.045127184200963e+20,3.055593607279412e+20,3.0498348285727284e+20,2.902810782496
4905e+20
22.158714250000003,2.5450657464588612e+20,2.665182550372048e+20,2.8281982128256588e+20,2.89545284215207e+20,2.97955257299959e+20
,3.082321168359571e+20,3.158227441827223e+20,3.2137039761700756e+20,3.17150587385293e+20,3.308108417121442e+20,3.307362740627441e
+20,3.395918562330678e+20,3.435502304891748e+20,3.4796229947958356e+20,3.459858075677735e+20,3.348437836176512e+20,3.30455073168
0184e+20
20.163,2.8111105418426463e+20,2.956921740609364e+20,3.0365114762763e+20,3.200820842526731e+20,3.3075963260352e+20,3.3814722032661
75e+20,3.476766065577856e+20,3.587036346148571e+20,3.534659310485603e+20,3.598634760049238e+20,3.703820065974114e+20,3.7998775728
88086e+20,3.7979370171928537e+20,3.821843585271611e+20,3.8539076745136583e+20,3.77601592508005e+20,3.627725127369425e+20
18.167285749999998,3.0190554590733586e+20,3.26196631503482e+20,3.347566382924487e+20,3.493772880073567e+20,3.555520284209344e+20
,3.6334839060761084e+20,3.761839087639885e+20,3.882935153454833e+20,3.948680461684952e+20,3.99182188968728e+20,3.985281498270018e
+20,4.0785988686698886e+20,4.121677408293204e+20,4.170559646893926e+20,4.2406083170593047e+20,4.142439557421355e+20,3.9902676483
20265e+20
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,3.98589241395185e+20,4.0607745052470195e+20,4.1405618901051356e+20,4.228425939639239e+20,4.3300805122852907e+20,4.3871562082658
79e+20,4.5653818743862834e+20,4.499402980748404e+20,4.5223572390878354e+20,4.5766119420670234e+20,4.537360609509303e+20,4.3966433
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110002270039e+20,4.350159873298452e+20,4.483429332479322e+20,4.5040118005245 8e+20,4.6521768214771814e+20,4.7128102028990284e+20,
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2554888279017e+20,4.611371247552447e+20,4.741136925616925e+20,4.831094259766719e+20,4.981631071013492e+20,5.012311615917272e+20,
5.1047844852275505e+20,5.153711644098995e+20,5.213545444701975e+20,5.2105807068342596e+20,5.195936698578575e+20,5.01037106022204
05e+20
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06186222934e+20,4.917583749016701e+20,5.017702048404027e+20,5.1093304166247144e+20,5.188965072562372e+20,5.2495535337134975e+20,
5.389587985665242e+20,5.4379311810172885e+20,5.5215098367244226e+20,5.440509604556786e+20,5.525337043790018e+20,5.31493947977783
25e+20
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109496853633e+20,5.730685569372944e+20,5.790186959972573e+20,5.885525742554977e+20,5.827794210621832e+20,5.548875265648848e+20
6.193,4.136851475743419e+20,4.4498110018702606e+20,4.6495624617211055e+20,4.862870859276139e+20,4.9912979132730724e+20,5.13400961
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9442375134e+20,5.4719897302794345e+20,5.566717597180007e+20,5.6880382643485685e+20,5.775462095229588e+20,5.91106740848805e+20,6.0
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8796054708e+20,5.4834983036386555e+20,5.5581018892553434e+20,5.7308832185641245e+20,5.823248279224669e+20,6.046385232013885e+20,
6.156475831501708e+20,6.241590760467568e+20,6.274068116200265e+20,6.297246975893311e+20,6.268237465060425e+20,6.067102460871313e
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48e+20,5.895498042655473e+20,6.031570526729454e+20,6.04827188338425e+20,6.123010229813105e+20,5.981493408927502e+20,5.76472615052
6801e+20

C.28 "NBSR_lower_Cycle7_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
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−12.20157145,4.720087286756147e+20,5.0476459008679503e+20,5.260352366795296e+20,5.482123743354534e+20,5.713939292447426e+20,5.851071894910469e+20,6.009388897046457e+20,6.18766846749173e+20,6.257079269812843e+20,6.442582019790366e+20,6.570487998646914e+20,6.73259627163192e+20,6.794361643875986e+20,6.854878232593954e+20,6.99414903994341e+20,6.98790512231292e+20,6.722237657203207e+20
−14.197285749999999,4.7975028813200916e+20,5.1379446290752386e+20,5.3497886258047025e+20,5.520809080501145e+20,5.748258379279765 3e+20,5.96207886792104e+20,6.083049156977963e+20,6.21220391936061e+20,6.386036383004307e+20,6.566876408880788e+20,6.7077643459762 7e+20,6.819193569531637e+20,6.918629080803975e+20,7.042150841237961e+20,7.104877507275498e+20,7.040902057711863e+20,6.75847933162 2489e+20
−16.192999999999998,4.7811608868310804e+20,5.1460212937512255e+20,5.377890747168985e+20,5.51397221529711e+20,5.703212331798784e+20,5.885696439583723 4e+20,5.971655869639174e+20,6.139873299442504e+20,6.380322524568347e+20,6.490152586485978e+20,6.680237204077242e+20,6.82704563285401e+20,6.915457709690935e+20,7.058968990596636e+20,7.013599517166451e+20,6.948483092726452e+20,6.764696297090545e+20
−18.188714249999997,4.6852920450541453e+20,5.00189909716369e+20,5.20167750917697e+20,5.451524054938056e+20,5.583257240860199e+20,5.7756597444207686e+20,5.930041730841427e+20,6.082617922379024e+20,6.25977448605622e+20,6.366918315784617e+20,6.492587265159161e+20,6.628489052204395e+20,6.835939846457155e+20,6.956919119568224e+20,6.909636042605237e+20,6.887436444813954e+20,6.675197149702126e+20
−20.18442855,4.585389362966289e+20,4.948417022840937e+20,5.163144900950817e+20,5.379229371236529e+20,5.570077633430083 3e+20,5.661732953813205e+20,5.787473775620906e+20,5.967190794729313e+20,6.12049469465262e+20,6.227521731677137e+20,6.354924603501587e+20,6.473406309560459e+20,6.5540651476706e+20,6.637428186078265e+20,6.742254129837356e+20,6.659430134678368e+20,6.475553498501016e+20
−22.180142850000003,4.41046084471695e+20,4.701822704680193e+20,5.009427734536858e+20,5.172587141856782e+20,5.304652737782276e+20,5.43834444750794e+20,5.588629705238664e+20,5.683303667814369e+20,5.876128421919734e+20,6.049538635018636e+20,6.152037708754281e+20,6.211080912592534e+20,6.305763859222384e+20,6.456489335606192e+20,6.430983605889698e+20,6.386404729224236e+20,6.153520077688139e+20
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−26.171571450000002,3.9208927822157316e+20,4.2385779208226275e+20,4.4699891874790126e+20,4.598065863364308e+20,4.770056595908365 5e+20,4.9206652795882956e+20,5.120587436467887e+20,5.158068910359123e+20,5.298336947718627 7e+20,5.3688438046453806e+20,5.473651780296184e+20,5.567777715569068e+20,5.672657563653028e+20,5.754349567989798e+20,5.831576497416705e+20,5.7747972752228876e+20,5.5350667744286114e+20
−28.167285749999998,3.7122201565993075e+20,4.0054686679329145e+20,4.215839279782667e+20,4.344895217569722e+20,4.539750367911764e+20,4.6090084413124195e+20,4.7904593828707317e+20,4.824544884295312e+20,4.9541578334393316e+20,4.984964155101136e+20,5.133174096324459e+20,5.206241408682421e+20,5.324022358518015e+20,5.442773586201223e+20,5.3928042770490065e+20,5.3696972897891176e+20,5.2652307081958084e+20
−30.163,3.522342172253368e+20,3.7601770376231354e+20,3.8951354989831874e+20,4.07494235863304e+20,4.119188825295153e+20,4.282348232615077e+20,4.350204793569176e+20,4.5447275339078684e+20,4.5864045610846283e+20,4.689667279422561e+20,4.806648648439286e+20,4.8186962650471825e+20,4.9059583829535995e+20,4.923989379621793e+20,5.012644025920621e+20,5.022041346555864e+20,4.8154709896092746e+20
−32.15871425,3.1866260369782774e+20,3.3996738969631175e+20,3.578339781736606e+20,3.67062398590985e+20,3.8061394586268 66e+20,3.844366609012102e+20,3.962632697771503e+20,4.0821565541111464e+20,4.1558976705299526e+20,4.2991663819737524e+20,4.3566912806615705e+20,4.3978382486437 99e+20,4.452155840001998e+20,4.498064356680861e+20,4.5495340028752246e+20,4.485414808445276e+20,4.408556225238297e+20
−34.15442855,2.821972263303457e+20,3.081952822139643e+20,3.217773752697575e+20,3.3087103487491296e+20,3.402279272665049e+20,3.4706299565971e+20,3.549554872257335e+20,3.64311481211911e+20,3.814458692764757e+20,3.8041360145526214e+20,3.862514398384176e+20,3.924962558743231e+20,3.98499400853739e+20,4.018908812933223e+20,4.0720944134692045e+20,4.117850201227609e+20,3.96027887558562e+20
−36.15014285,2.388231113256729e+20,2.5084197896030732e+20,2.6628646644027438e+20,2.7446285411726695e+20,2.8441359248781 64e+20,2.934066306865524e+20,3.056240459177822e+20,3.129622213430845e+20,3.167741555166346e+20,3.221708768412906e+20,3.3059073238560165e+20,3.3807265267721745e+20,3.3939690225813016e+20,3.412000019249497e+20,3.41940287986464e+20,3.447037830413403e+20,3.307955688200984e+20

C.29 "NBSR_upper_Cycle7_cladding_fluence_position_6.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.123725824389962e+20,1.220236778208338e+20,1.2486532729454202e+20,1.2881241664199103e+20,1.3470331741007641e+20,1.397261133781780 6e+20,1.4296332130814799e+20,1.4513809639116538e+20,1.4545825981281074e+20,1.4806363611465797e+20,1.4910801662937848e+20,1.565153133049584e+20,1.5543854478240973e+20,1.5689388133088402e+20,1.602287842352864e+20,1.5504033157140804e+20,1.5323244359346025e+20
24.15442855,1.3672358576722511e+20,1.4271271246069093e+20,1.4791709365300933e+20,1.52482740854881e+20,1.552458095882849e+20,1.613634264111673e+20,1.6388119580659414e+20,1.6727715807001674e+20,1.707485153680557e+20,1.7275020044202436e+20,1.770031175355227e+20,1.7982671467699883e+20,1.8313560098494906e+20,1.841146745330653e+20,1.872223304317227e+20,1.8542134481609893e+20,1.821708631124289e+20
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6.193,2.4643982061336586e+20,2.6139193026005828e+20,2.767114579627683e+20,2.8397486693143957e+20,2.927982097853416e+20,3.01482974441815e+20,3.125766635493748e+20,3.190882459756749e+20,3.288386291847889e+20,3.3244909563120445e+20,3.389378471667404e+20,3.4654425044782114e+20,3.50431873289094e+20,3.579745624564144e+20,3.647171085450955e+20,3.6997086817557814e+20,3.580552670005108e+20
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C.30 "NBSR_lower_Cycle7_cladding_fluence_position_6.csv" File

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52115544054596e+20,3.309751757995501e+20,3.4277874632458895e+20,3.5147041334338644e+20,3.61352472387605e+20,3.7121647909959146e
+20,3.7368433910590616e+20,3.838966496397822e+20,3.931835126712901e+20,3.96154183225363e+20,3.964923989792404e+20,3.839380638137
264e+20
−12.20157145,2.738135276885718e+20,2.974955328223178e+20,3.043856832745435e+20,3.1843730011342406e+20,3.301314947431745e+20,3.427
214036222048e+20,3.519944619290646e+20,3.5794907681091033e+20,3.705288976219285e+20,3.7767815213677937e+20,3.8704784351617575e+
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9592e+20,4.008557538699231e+20,4.114445086259328e+20,4.209665829274058e+20,4.241422005474075e+20,4.2564054412267e+20,4.1712037426
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3e+20,3.43715874747813e+20,3.506262013360628e+20,3.63443357220838e+20,3.727902177094703e+20,3.779022134368363e+20,3.9197347546079
29e+20,4.0549254849882725e+20,4.108556840245983e+20,4.204956294365278e+20,4.2513932642775584e+20,4.25645322681202e+20,4.103247330
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−30.163,2.0267618872427728e+20,2.1574926296598964e+20,2.26977813614342e+20,2.356381545272073e+20,2.4103527091365056e+20,2.4709342
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01755082807e+20,2.9029955462404283e+20,2.9604444388142757e+20,3.013401486368024e+20,3.042741835754631e+20,2.9281944782326166e+2
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57e+20,2.6205349514126914e+20,2.678891770107623e+20,2.7117045386941638e+20,2.7580034613599637e+20,2.75710084474836e+20,2.70720207
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2653249116481e+20,2.064586832778456e+20,2.100489735882371e+20,2.197308641250588e+20,2.249198477398851e+20,2.2550548663553163e+2
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57325519169532e+20,1.8096932111709975e+20,1.835513355772349e+20,1.8588114833706893e+20,1.913510050033885e+20,1.9470289833812695
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7440343e+20

C.31 "NBSR_upper_Cycle7_cladding_fluence_position_7.csv " File

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69672663378869e+20,1.3829254581857188e+20,1.4088783405240707e+20,1.419911501223558e+20,1.4452644089906672e+20,1.458469159067484
2e+20,1.5014655668367095e+20,1.5168365967813758e+20,1.5083466911228192e+20,1.5307422021095555e+20,1.5320058653658012e+20,1.438330
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+20,1.752706245921912e+20,1.788195007286385e+20,1.8015855901950027e+20,1.7871384148998606e+20,1.7938761824300093e+20,1.702249977
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356108e+20,1.984126526118189e+20,1.9741765053526265e+20,2.006410537405845e+20,2.0028106899783896e+20,2.0059804671379633e+20,1.938
0718408884706e+20
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C.32 "NBSR_lower_Cycle7_cladding_fluence_position_7.csv" File

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−14.197285749999999,2.6701576270129853e+20,2.8612521827084884e+20,3.0262504993096794e+20,3.1361732740745934e+20,3.22305277769620 8e+20,3.3566984408178636e+20,3.4547226048380456e+20,3.506001847396106e+20,3.628938229896557e+20,3.6830421314979896e+20,3.774100219080382e+20,3.839449661760504e+20,3.9532483784458345e+20,4.0123644569964085e+20,4.0257815874524257e+20,4.051591113034817e+20,3.923939886116107e+20
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−30.163,1.9586621186519997e+20,2.096374866035353e+20,2.222486335204856e+20,2.2535257276250702e+20,2.3463625008832702e+20,2.449876697705796e+20,2.4569171072763065e+20,2.544985941021446e+20,2.6198447151802886e+20,2.668708130924939e+20,2.7237305776664157e+20,2.768011886729807e+20,2.8086243247425015e+20,2.844665275092897e+20,2.871026989661211e+20,2.8977125843078057e+20,2.780133496872699e+20
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C.33 "NBSR_upper_Cycle8_cladding_fluence_position_0.csv " File

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5e+19,8.359309484037641e+19,8.287578593142037e+19,8.161887735072712e+19,7.981967244826298e+19,7.8690854744169e+19,7.7790802851264
66e+19
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C.34 "NBSR_lower_Cycle8_cladding_fluence_position_0.csv" File

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7768942e+20,1.5505647805263944e+20,1.600524716816840 8e+20,1.556659209602487e+20,1.574097546486882e+20,1.5501063500206706e+20,1.48
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67e+20,1.5391849173843103e+20,1.5564614552666846e+20,1.5009104645730972e+20,1.5215937703313408e+20,1.501171140743019e+20,1.451633
6796245138e+20
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45e+20,1.3005673447383605e+20,1.3528374112243166e+20,1.3808376274072496e+20,1.407399630238892e+20,1.4262492139742401e+20,1.465746
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0758e+20,1.4200559077302467e+20,1.386338793475935e+20,1.4318222907104902e+20,1.413691813650787e+20,1.3926040103874932e+20,1.32261
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89215802e+20,1.3199832249807793e+20,1.322715830348231e+20,1.3412148495792023e+20,1.32294055118437e+20,1.311614621042959e+20,1.257
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4274954e+20,1.2284769005049348e+20,1.1938788805729578e+20,1.2515871912934806e+20,1.2336544685695797e+20,1.2116048601276108e+20,1.
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384436257877e+19,9.391353407256658e+19,9.5361635140647e+19,9.376971273743755e+19,8.948455605726816e+19,8.748436083896118e+19
−36.15014285,5.3828370652079374e+19,5.884998245644386e+19,6.102159472855774e+19,6.318448783222943e+19,6.422665318190814e+19,6.777
625362122703e+19,7.1311471927033e+19,7.308236200414362e+19,7.3850637598736e+19,7.133628110734276e+19,7.616768919599912e+19,7.5013
52298158865e+19,7.704203302524915e+19,7.802352374917032e+19,7.60161374641069e+19,7.815359216912764e+19,7.269898825769031e+19

C.35 "NBSR_upper_Cycle8_cladding_fluence_position_1.csv " File

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940208171959714e+20,1.3146698594987149e+20,1.3527539820272383e+20,1.415715764731668e+20,1.4109134147852619e+20,1.44362676762814
73e+20,1.431982131474915e+20,1.4671816566351877e+20,1.4714633978484657e+20,1.496133876997923e+20,1.4608227928234089e+20,1.4400993
77844504e+20
24.15442855,1.2464860524609698e+20,1.3494550226060896e+20,1.420071878621107e+20,1.4574282113900533e+20,1.4972175931362715e+20,1.5
28581613140233e+20,1.581210906192274e+20,1.6290803479700007e+20,1.619927196468167e+20,1.6506962350628395e+20,1.6932374058270206
e+20,1.7263066850486773e+20,1.7270450994762773e+20,1.7514818359292284e+20,1.7414415246474723e+20,1.7372129067743094e+20,1.6971366
589914697e+20
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20,1.7338980103511268e+20,1.7657348280821128e+20,1.807580083090786e+20,1.8420093198193895e+20,1.8729696024384763e+20,1.895403588
969088e+20,1.9273307163782693e+20,1.9378332007190272e+20,1.979274718342825e+20,1.9869776026595164e+20,1.9665676153872905e+20,1.95
52948426148643e+20
20.163,1.5446620481930772e+20,1.637431337036096e+20,1.723560208364726e+20,1.7928755425469232e+20,1.8355070229892995e+20,1.8937620
77759531e+20,1.9398040766372977e+20,1.9908237324838502e+20,2.0093265918459435e+20,2.0740626941102185e+20,2.0844961181088265e+20,
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e+20,2.20885254498429e+20,2.2697584544262652e+20,2.3231420989079405e+20,2.3704537456144584e+20,2.4177016443128838e+20,2.4621233813
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8.18871425,2.2146483013908487e+20,2.3538792634985434e+20,2.5080006978405043e+20,2.5948307972586534e+20,2.648830672485664e+20,2.73
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C.36 "NBSR_lower_Cycle8_cladding_fluence_position_1.csv" File

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−36.15014285,1.3041673751579574e+20,1.4195725192240253e+20,1.4852117182702564e+20,1.5451826569262052e+20,1.5896681419386e+20,1.6533258403985375e+20,1.6817521396941337e+20,1.735502335222603e+20,1.7516677676196312e+20,1.7869363531220537e+20,1.8408990420109836e+20,1.8972363442030592e+20,1.9315274602473636e+20,1.9617546407728615e+20,1.952787420961431e+20,1.9195534593854174e+20,1.867800701445276e+20

C.37 "NBSR_upper_Cycle8_cladding_fluence_position_2.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0667432311814909e+20,1.1578072608068146e+20,1.2143995550314424e+20,1.2906740467687191e+20,1.3025949242905502e+20,1.309192843132847e+20,1.382540238829781e+20,1.398827854909074 1e+20,1.3988862905832009e+20,1.4396478294535256e+20,1.4862954344086045e+20,1.5238323865339422e+20,1.538191625367057e+20,1.5319336959014968e+20,1.549034099084551e+20,1.5553026532181344e+20,1.5311899691398853e+20
24.15442855,1.2746839213940707e+20,1.37761570520111e+20,1.4208102930487072e+20,1.4727277333432053e+20,1.4976425798571924e+20,1.587553833002018e+20,1.5718293243279452e+20,1.6553020286508197e+20,1.6398378243433107e+20,1.711899635209459e+20,1.7294250251134344e+20,1.771684642175005e+20,1.8059438842152403e+20,1.818119753769624e+20,1.849590020453816e+20,1.867019788345584e+20,1.8132005324749645e+20
22.158714250000003,1.4540389422907092e+20,1.525824511788259e+20,1.56847192923267e+20,1.68926377998641e+20,1.7375847701551156e+20,1.788530053325507e+20,1.792790545202739e+20,1.850535615907865e+20,1.8754079637497597e+20,1.943894573826161e+20,1.9624930552004618e+20,1.997134785289526e+20,2.0127530472833683e+20,2.0466722999468668e+20,2.076219501718892e+20,2.061196221134338e+20,1.9757420162251706e+20
20.163,1.561815574716247e+20,1.6778847605337532e+20,1.770685923380841e+20,1.84693385344806e+20,1.9073563404949878e+20,1.9650482878599984e+20,1.9979953833993906e+20,2.0117383914871698e+20,2.079321904781614e+20,2.0973360294146487e+20,2.1809521667558333e+20,2.209994696796765e+20,2.2373585292900585e+20,2.2544270584690436e+20,2.291353092183058e+20,2.3066207401321408e+20,2.2305587417553243e+20
18.167285749999998,1.7050360996665868e+20,1.8493403407552748e+20,1.937073536955381e+20,1.9999821963196957e+20,2.069377215512065e+20,2.113304905453251e+20,2.177371653632075e+20,2.2150945374478156e+20,2.2305481170873018e+20,2.3076247712603 16e+20,2.3578369523371196e+20,2.3743795604489653e+20,2.421563711139207e+20,2.4974132161555625e+20,2.5006324905665384e+20,2.4930358529300775e+20,2.4338298903717 86e+20
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6.193,2.2970904129154898e+20,2.4963879356913412e+20,2.631570899282264e+20,2.7413343446281073e+20,2.8632683471943218e+20,2.8950945402572852e+20,2.989840017352587e+20,3.0958901412243826e+20,3.13226900453521e+20,3.201122165658407e+20,3.305807019689245e+20,3.374978920853131e+20,3.4781391350226656e+20,3.523432094804809e+20,3.5597259607714536e+20,3.5685072488924814e+20,3.4879935146140185e+20
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C.38 "NBSR_lower_Cycle8_cladding_fluence_position_2.csv" File

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−12.20157145,2.5796747083238123e+20,2.7772350978778995e+20,2.965073916190921e+20,3.085833892940592e+20,3.156604806641943e+20,3.26947596738452e+20,3.358393814069193e+20,3.4819587031769416e+20,3.52076530313103e+20,3.6454829687192774e+20,3.749934080053609e+20,3.8143833162812614e+20,3.8950511082460584e+20,4.000463752036473e+20,4.039817522393748e+20,4.0446729956802685e+20,3.9496725265524156e+20
−14.197285749999999,2.6120746334600184e+20,2.8659882622082154e+20,2.982535558086758e+20,3.080192194220367e+20,3.237012294240176e+20,3.311932140804517e+20,3.398337253501745e+20,3.5104540628146874e+20,3.5702443821142455e+20,3.703143042080219e+20,3.78730634982459e+20,3.872606497047424e+20,3.982008703680483e+20,4.0749904858839627e+20,4.090470627193507e+20,4.1113746615288024e+20,4.0351426684636175e+20
−16.192999999999998,2.5860813831416945e+20,2.8412805967206777e+20,2.981377469272249e+20,3.0579441393801586e+20,3.235854205425666e+20,3.331747146667453e+20,3.420670305686138e+20,3.5049929834508544e+20,3.606734804439315e+20,3.7004974997424865e+20,3.744430502017683e+20,3.8564410646503955e+20,3.954772367203466 6e+20,4.031556843005848e+20,4.070777805012834e+20,4.060971236427585e+20,3.999757211612942e+20
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−20.18442855,2.5540851953903632e+20,2.7549870430376906e+20,2.882132445269198e+20,2.9908334238127384e+20,3.097749458128411e+20,3.157066979700945e+20,3.2788787985848946e+20,3.326827925372794e+20,3.421780583494544e+20,3.545032044895613e+20,3.6423115053144054e+20,3.70293586105377e+20,3.771613715154586e+20,3.858810365619529e+20,3.923865207924494e+20,3.9321843229865214e+20,3.8228193026915415e+20
−22.180142850000003,2.443716143967208e+20,2.6437945798427515e+20,2.7942983147228735e+20,2.9144101867231396e+20,3.015774831996784e+20,3.11219369430571e+20,3.167771332734139e+20,3.24251055994209e+20,3.330573120850909 6e+20,3.433637713008236e+20,3.5082175701958414e+20,3.577973828100994e+20,3.6882791315160105e+20,3.6522721315859884e+20,3.806669807296548e+20,3.7660888877826 14e+20,3.709151291847238e+20
−24.17585715,2.3347973597291957e+20,2.476461370814163e+20,2.6613252820807385e+20,2.793660834641492e+20,2.8353095332917392e+20,2.9267454262978685e+20,3.0172994718580874e+20,3.1092134749252524e+20,3.151127790276075e+20,3.2600359498460647e+20,3.3281666335436944e+20,3.416840112863839e+20,3.498761615655352e+20,3.573569903205452e+20,3.621890893374156e+20,3.5804334387483233e+20,3.539321285833239e+20
−26.171571450000002,2.223328655165658e+20,2.381561836032528e+20,2.557293845133316e+20,2.647704457675224e+20,2.750152819087216e+20,2.7908399852813792e+20,2.888119445700171e+20,2.9660726349850857e+20,3.000932170768621e+20,3.103199912824222e+20,3.152593994463252e+20,3.249326284478858e+20,3.293243349752021e+20,3.369496592153252e+20,3.4291594154365334e+20,3.4010252945115696e+20,3.359886579926428e+20
−28.167285749999998,2.0991315983105396e+20,2.2762076279162392e+20,2.3941414429717866e+20,2.477051039889440 4e+20,2.54463455318388 5e+20,2.6274379034213086e+20,2.7001903177089532e+20,2.7593272199250955e+20,2.8252426603399256e+20,2.867236660700921e+20,2.958406937006475e+20,2.996783237905631e+20,3.1325771199078787e+20,3.156572932637874e+20,3.193557402026015e+20,3.231737146566745e+20,3.1313871570893e+20
−30.163,1.949297217515868e+20,2.0895109614156926e+20,2.2200509450805556e+20,2.2954914003780246e+20,2.3736570830233996e+20,2.4360504459885976e+20,2.480695301021337e+20,2.5586113039681706e+20,2.5898372032878333e+20,2.6913240322437418e+20,2.7072344726082172e+20,2.7602940647151904e+20,2.8742595662641388e+20,2.9095600257706302e+20,2.980118446111521e+20,2.985919514852091e+20,2.8849373576272763e+20
−32.15871425,1.7544726799777094e+20,1.9019961954773733e+20,2.005002351960574e+20,2.0896118957619115e+20,2.1522443137576275e+20,2.2046823627852536e+20,2.257152285816949e+20,2.3225418051646394e+20,2.3995600236635275e+20,2.4250857885888387e+20,2.4905974916187 94e+20,2.5360816954253522e+20,2.5895290879151653e+20,2.6683056889718637e+20,2.6863145012708868e+20,2.6811243509416405e+20,2.6053492186014456e+20
−34.15442855,1.580732796131238e+20,1.7213927760880294e+20,1.7892525307510725e+20,1.8762048138514865e+20,1.9095503344417417e+20,1.9467419848563314e+20,2.004088630510594e+20,2.0673797779237372e+20,2.1386234893521132e+20,2.1589484792801545e+20,2.2315149618773 96e+20,2.238341311082188e+20,2.3051492236109523e+20,2.3615715231472114e+20,2.4264669954318316e+20,2.422355248906922e+20,2.3669210434968034e+20
−36.15014285,1.3471122833070134e+20,1.4306221739679682e+20,1.5586600483134092e+20,1.5981785010250403e+20,1.6302437491185215e+20,1.6862835606059526e+20,1.7372766547824476e+20,1.7692356561956985e+20,1.8259235724325336e+20,1.8641192539759 99e+20,1.920908104558 3525e+20,1.9307040484755794e+20,2.0031589720585794e+20,2.0323449351178217e+20,2.022655237880825e+20,2.05174557892786e+20,1.9875407100647365e+20

C.39 "NBSR_upper_Cycle8_cladding_fluence_position_3.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0707965420322737e+20,1.1906852960040567e+20,1.2069038517412002e+20,1.247644141275479e+20,1.2858982584923706e+20,1.3249067271388968e+20,1.349779074980792e+20,1.38098903729842e+20,1.4262819970805634e+20,1.4367047964111485e+20,1.4433186522554797e+20,1.469056910541251e+20,1.4714846471845118e+20,1.4992522170626805e+20,1.5150351614108798e+20,1.4999587574862112e+20,1.4383091212826247e+20
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20.163,1.5891847195435522e+20,1.73218743879942e+20,1.7547595460143307e+20,1.854902354465327e+20,1.9095662714437763e+20,1.93972439 16271252e+20,1.999535960262729e+20,2.044988290065218e+20,2.0542370635792587e+20,2.0775528975557807e+20,2.1294119021761528e+20,2.1 80787484401477e+20,2.2157585791992542e+20,2.2249170430350996e+20,2.2259848221714134e+20,2.2240298832551772e+20,2.10105997555671 8e+20
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12.18014285,2.0955988961928846e+20,2.2573966531814785e+20,2.3348557954033218e+20,2.4792662831722408e+20,2.555009541508366e+20,2.6 0360677304567e+20,2.661771518137705e+20,2.711824329194163e+20,2.8243926868980836e+20,2.7947870494519322e+20,2.8676828967578878e +20,2.9288969215725303e+20,2.992113696309513e+20,3.0018299552165665e+20,3.0094903408611656e+20,3.004645492242667e+20,2.890382499 9890747e+20
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8.18871425,2.286837608273273e+20,2.463812703532755e+20,2.5839298878670327e+20,2.6925777430704583e+20,2.767210723598178e+20,2.8403 98749274767e+20,2.9382466294327902e+20,2.996491059534998e+20,3.066730739835198e+20,3.109271910599379e+20,3.135562651622348e+20,3. 2315087162042496e+20,3.2690456683295873e+20,3.31341959432774e+20,3.361660899486272e+20,3.3403212537620326e+20,3.243498654068231 e+20
6.193,2.3151098498825355e+20,2.5386528650869234e+20,2.66224431586473e+20,2.7726824276300343e+20,2.8617383949990068e+20,2.93043218 61018573e+20,3.006372000796409e+20,3.103959576587868e+20,3.146357314333738e+20,3.200798113283705e+20,3.304744552886942 4e+20,3.353 623338126857e+20,3.416611682501343e+20,3.454217694968831e+20,3.477113854558444e+20,3.4606084327846786e+20,3.3526830550068186e+2 0
4.19728575,2.403283969805597e+20,2.6116655837411325e+20,2.7298650154972548e+20,2.809390655649576e+20,2.907913202227062e+20,2.9917 31208260684e+20,3.0995344023562794e+20,3.155871704548354e+20,3.2523330655293735e+20,3.347726647374079e+20,3.4365169980424756e+2 0,3.458286942821648e+20,3.518401314495908e+20,3.556804177065122e+20,3.5797162736567694e+20,3.5790416072373076e+20,3.448772552607 0316e+20
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C.40 "NBSR_lower_Cycle8_cladding_fluence_position_3.csv" File

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−10.20585715,2.422376498242968e+20,2.621567774338589e+20,2.8114093425739504e+20,2.935181412708148e+20,3.028906921673239e+20,3.093627086935479e+20,3.186343252438382e+20,3.316575120730578e+20,3.366590745448955e+20,3.4166966798455276e+20,3.515463593787542e+20,3.591089980775415e+20,3.650965297419157e+20,3.631580590611153e+20,3.699987515677381e+20,3.6756464012366375e+20,3.550519685929504 6e+20
−12.20157145,2.565187973474422e+20,2.887651960307158e+20,2.985505152799193e+20,3.11806382338843e+20,3.210285941828263e+20,3.331662149323269e+20,3.420553434337885e+20,3.508265381201945e+20,3.598590996399 6686e+20,3.695636714121953e+20,3.754677994325888e+20,3.831313724775947e+20,3.889048170813051e+20,3.938670682814576e+20,3.9476644642960645e+20,3.9343411305951946e+20,3.7855373425987564 e+20
−14.197285749999999,2.632718363428751e+20,2.8953389076218144e+20,3.014292690807572e+20,3.16844068681959e+20,3.2696618990749234e+20,3.3750479811952804e+20,3.405673586771642e+20,3.513088980484397e+20,3.627091668371426e+20,3.7567869909284585e+20,3.800672182197552e+20,3.8567279306870175e+20,3.9296875260011105e+20,4.017601341557609e+20,3.999704088272827e+20,3.9779500804956894e+20,3.901489657068009 6e+20
−16.192999999999998,2.648437559768812e+20,2.8541895683686495e+20,3.018935670733632e+20,3.16190120365142e+20,3.281179039211879e+20,3.345601713769474e+20,3.412547746982537 5e+20,3.538853800440226e+20,3.641790896581277e+20,3.710574997362323e+20,3.767539154967757e+20,3.876511062545885e+20,3.890822490372896e+20,3.9958367091124476e+20,4.004501125885221e+20,4.01008438893132e+20,3.8782800697717175e+20
−18.188714249999997,2.6253660931568196e+20,2.810299064765545e+20,3.001697146866279e+20,3.135918578001119e+20,3.211465279978819e+20,3.3197518964694614e+20,3.384971421128783e+20,3.4691506658751886e+20,3.564188321341123e+20,3.611739023078158 5e+20,3.7420452640465145e+20,3.8087894285671406e+20,3.841757773442579e+20,3.903433971316222e+20,3.941369348492424e+20,3.954055202111912e+20,3.804635183370139e+20
−20.18442855,2.5788672335540627e+20,2.7755192139921814e+20,2.924338938990654e+20,3.014712365194481e+20,3.121304347135452e+20,3.1922505678591826e+20,3.3303925014945176e+20,3.4158307694016515e+20,3.474250506526239e+20,3.5828505507235614e+20,3.619707524095425 e+20,3.7080038277007514e+20,3.7862332583542646e+20,3.8489719230302106e+20,3.8615090312973766e+20,3.895534530641106e+20,3.7299012684962e+20
−22.180142850000003,2.470718737747719e+20,2.7128177356543155e+20,2.8318299545141995e+20,2.954343001487669e+20,3.0112965344250796 e+20,3.09057249487886e+20,3.189557214515347e+20,3.261013419304183e+20,3.384642056420069e+20,3.4587756775507073e+20,3.487945703607915e+20,3.5976719626156776e+20,3.612966172234818e+20,3.638364441143852e+20,3.687381347068066e+20,3.7007631164430615e+20,3.5916584005146463e+20
−24.17585715,2.3964204342627244e+20,2.549394404458199e+20,2.6713018453543564e+20,2.8048486100697347e+20,2.9128802345278243e+20,3.009065354140245e+20,3.08009657220816e+20,3.13845256132461e+20,3.2182278811754725e+20,3.284892360685925e+20,3.3276247754745204e+20,3.40228963000631e+20,3.473352722078294e+20,3.48522047626001e+20,3.5511412290088514e+20,3.5568891744093064e+20,3.4605021861044486e+20
−26.171571450000002,2.2386706757909017e+20,2.4138342651524583e+20,2.5841476935615047e+20,2.6989472315502602e+20,2.7642517535537663e+20,2.854757988107881e+20,2.8609202955612337e+20,2.9428896093588508e+20,3.043048354811882e+20,3.1332624109953633e+20,3.1704806230800105e+20,3.265348283857576e+20,3.284982670364121e+20,3.331975577029948e+20,3.376668243068791e+20,3.352258068285898e+20,3.237091979250348e+20
−28.167285749999998,2.1103990587489544e+20,2.2754479641525934e+20,2.423593022731604e+20,2.498714737988383e+20,2.5913034074750104e+20,2.6713655933624946e+20,2.72404800975465e+20,2.7711099767626272e+20,2.8466407417382922e+20,2.8934742783837743e+20,2.996496371869009e+20,3.031286847310396e+20,3.0920811977381295e+20,3.084373001087426e+20,3.12024188033315e+20,3.088776925982969e+20,2.971581525355022e+20
−30.163,1.9606124889603867e+20,2.0940530069955346e+20,2.2134689632402932e+20,2.2870076029616416e+20,2.402040883646903e+20,2.4165063691602484e+20,2.5359648240770987e+20,2.562558368138723e+20,2.6263223132788916e+20,2.6738942643519737e+20,2.7297162701449324e+20,2.7932889712606862e+20,2.8514909026908016e+20,2.874907671013543e+20,2.8703390637636433e+20,2.8504496852245456e+20,2.7749242325828924e+20
−32.15871425,1.8102468747645646e+20,1.940765609093381e+20,2.0435592722161212e+20,2.1195043992446845e+20,2.1902965622820818e+20,2.2395578355708245e+20,2.301759954511608e+20,2.3592022221780772e+20,2.3872619704268797e+20,2.441065289295464e+20,2.5070285407163974e+20,2.5337017697881955e+20,2.577331969024736e+20,2.6273635307451476e+20,2.6153842175491903e+20,2.5890775395241873e+20,2.524553930620373e+20
−34.15442855,1.5878088250345708e+20,1.7737458277714716e+20,1.829063161833337e+20,1.8677263287691146e+20,1.9598846992008095e+20,1.9855804588144892e+20,2.0286156766417404e+20,2.0849264171637585e+20,2.144807046141512e+20,2.1783703724262395e+20,2.2048895438117028e+20,2.304389559847306e+20,2.3000068842878095e+20,2.337596959753262e+20,2.310844045671292e+20,2.3233545922684007e+20,2.2550911002204825e+20
−36.15014285,1.3465385512337701e+20,1.4859607573658796e+20,1.5424999282503926e+20,1.62454361472417e+20,1.6450385993405802e+20,1.694491116653737e+20,1.7286653613497883e+20,1.7775866452617943e+20,1.8201012543559175e+20,1.8450107885358934e+20,1.8730758491187066e+20,1.9240636309611902e+20,1.9522614998942912e+20,1.9799546970962988e+20,2.003621145117581e+20,2.0144476818330403e+20,1.9266241759547385e+20

C.41 "NBSR_upper_Cycle8_cladding_fluence_position_4.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
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069070423708e+19,3.3397687710248358e+19,3.3470997919607206e+19,3.4731402287178334e+19,3.5684222515482972e+19,3.4872763495224644
e+19,3.6037439603908346e+19,3.544904548879337e+19,3.6760607632895357e+19,3.5833074114485514e+19,3.457946953444912e+19,3.237442593
2951077e+19
24.15442855,3.2372938479427863e+19,3.520988421159514e+19,3.7468476139929215e+19,3.914196760023545e+19,3.9414330965005615e+19,3.95
3773648409303e+19,4.015088607570163e+19,4.333812711258797e+19,4.209360662371126e+19,4.352007455248223e+19,4.257671027871807e+19,
4.469160357204077e+19,4.348750994499165e+19,4.320914364278848e+19,4.31906035970883e+19,4.2221793243409015e+19,4.035174542467687e
+19
22.158714250000003,3.6357613974782116e+19,3.959712837834172e+19,4.103613341537985e+19,4.3938952089289875e+19,4.512705559096434e+
19,4.665068610880584e+19,4.553456473298735e+19,4.635925146493434e+19,4.7663641958120776e+19,4.981646531628565e+19,4.866151077884
302e+19,5.172396508979896e+19,4.979107235971064e+19,4.995973896457612e+19,4.920342197135727e+19,4.788468817633976e+19,4.569520971
349545e+19
20.163,4.173746775157955e+19,4.371737463766974e+19,4.616885741396178e+19,4.83995064653953e+19,5.009132547804125e+19,5.02453300410
3496e+19,5.112685874690511e+19,5.371779029099932e+19,5.386494194311817e+19,5.288455070129379e+19,5.377091363111443e+19,5.53412395
6491711e+19,5.528439759099394 5e+19,5.643186173748035e+19,5.624168017986824e+19,5.1897997152016065e+19,4.964312385749006e+19
18.167285749999998,4.496269197664818e+19,4.696570751568844e+19,4.910115954163569e+19,5.205492349871 61e+19,5.335708281161771e+19,
5.503524912585407e+19,5.612958993222535 6e+19,5.697584474025909e+19,5.689722219688872e+19,5.876928870254524e+19,5.978447573214501
e+19,5.994703315289725e+19,5.908962244343936e+19,5.976641379650587e+19,5.840964368996594e+19,5.786937932099526e+19,5.559145049685
929e+19
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,5.853820217304451e+19,6.068438511369501e+19,6.317905716550062e+19,6.17797883868686e+19,6.401415607211017e+19,6.577997589753647e+
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8142e+19
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6.193,6.2119777763605316e+19,6.695187678047583e+19,6.950498450640807e+19,7.196247022013311e+19,7.4510796845455e+19,7.420321270618
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669327870878e+19,8.8052467474198e+19,8.86240746138366e+19,8.627602298074867e+19,8.646939193876768e+19,8.09350023655754e+19
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519596492e+19,7.866557327585785e+19,8.261423114661406e+19,8.429399116105387e+19,8.47566954534565e+19,8.519071314219696e+19,8.6467
79823856422e+19,8.678760074605719e+19,8.658148218641056e+19,8.574850821340563e+19,8.355132686624463e+19,8.078678824665424e+19
0.20585714999999993,5.663479289672007e+19,6.182813062637334e+19,6.332780251782292e+19,6.647482918624212 6e+19,7.133880220718171e+
19,7.195715788612161e+19,7.256063902982927e+19,7.48715043248366e+19,7.799621919040745e+19,7.862148090356231e+19,7.93869882346210
6e+19,7.906718572712 81e+19,8.111987158917598e+19,7.99840945775149e+19,8.167394802657658e+19,7.979125685289705e+19,7.5794787976037
24e+19

C.42 "NBSR_lower_Cycle8_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,5.922243079372714e+19,6.522271205972895e+19,7.06922911579808e+19,7.2057560998939165e+19,7.509143495291316e+19,7.7869
25440753233e+19,7.91149967332317e+19,8.166438582535587e+19,8.21074344819159e+19,8.46111375015411e+19,8.58802540968911e+19,8.61448
0833066435e+19,8.74569548315076e+19,8.653792104751617e+19,8.883391180729128e+19,8.51705262729532e+19,8.411071563765675e+19
−12.20157145,6.826774191512714e+19,7.190403454600649e+19,7.660120027898462e+19,8.031399051962974e+19,8.235977034746267e+19,8.4549
51442700757e+19,8.602581204880651e+19,8.675307057498238e+19,9.222477460683884e+19,9.543608051679732e+19,9.592109661204827e+19,9.6
01140629024396e+19,9.932683394682805e+19,9.74398929059393e+19,9.747442307701413e+19,9.391303435569707e+19,8.928758513187434e+19
−14.197285749999999,7.0157870356422795e+19,7.2895847305955615e+19,7.763232431061893e+19,8.302115593189582e+19,8.339248807930044e
+19,8.686409835582297e+19,8.96084501061696e+19,8.95978254381466e+19,9.399218813246859e+19,9.530645956691644e+19,9.734958322774 36e
+19,9.76693857352366e+19,9.935180191668216e+19,9.941608115822143e+19,9.912071538718143e+19,9.534683330540392e+19,9.3822724677501
39e+19
−16.192999999999998,7.172766505682432e+19,7.626174213564906e+19,7.937423863299342e+19,8.175150810314465e+19,8.665001129515906e+1
9,8.725933600627938e+19,8.916168281580152e+19,8.983634923526344e+19,9.145076754136167e+19,9.522518085654032e+19,9.61532456083513e
+19,9.62600235219827e+19,9.854963948094398e+19,9.956588897734605e+19,9.985806734797917e+19,9.653838982418586e+19,9.1653167467200
23e+19
−18.188714249999997,6.841489356724598e+19,7.523061810401475e+19,7.660916878000189e+19,8.220252526072195e+19,8.481725606118772e+1
9,8.733264621563824e+19,9.028311652563152e+19,9.21828071681479e+19,9.344501772928293e+19,9.250207844223972e+19,9.719233814100287e
+19,9.373932103352066e+19,9.68555361646731e+19,9.657876356267336e+19,9.616440150977549e+19,9.415580802002312e+19,9.2965313968043
47e+19
−20.18442855,6.800053151434811e+19,7.271629041636654e+19,7.78782853753519e+19,7.742248711716425e+19,8.167554172678005e+19,8.33388
3350578418e+19,8.572938381096418e+19,8.906871697060007e+19,9.296743890164808e+19,9.228958508177927e+19,9.184228655801003e+19,9.60
1299999044742e+19,9.509396620645599e+19,9.569319748295444e+19,9.441398745298258e+19,8.944748638562083e+19,8.823308683058938e+19
−22.180142850000003,6.494115835711885e+19,7.109390360925105e+19,7.496553263684035e+19,7.498306333907834e+19,7.842386207833409e+1
9,8.029274118358 37e+19,8.26052001787945e+19,8.453623359197879e+19,8.657085751838754e+19,8.833136500980233e+19,8.985281747069913e+
19,9.177535114946498e+19,9.179394431850529e+19,9.188584769690442e+19,9.159738796007937e+19,8.890509708304553e+19,8.4184025847015
6e+19
−24.17585715,6.189081616770916e+19,6.846164210654729e+19,7.020302519552063e+19,7.326399205295334e+19,7.591484672469739e+19,7.9231
86808148494e+19,8.12000878327498e+19,7.99830321107126e+19,8.170582203064566e+19,8.335105187401064e+19,8.639023816199617e+19,8.845
992349288091e+19,8.766413585795655e+19,8.713130875660196e+19,8.466904194226656e+19,8.623405554205775e+19,8.165747979114091e+19
−26.171571450000002,5.883409917748567e+19,6.419902529571076 5e+19,6.659116930109423e+19,7.025721100243804e+19,6.971269676625815e+
19,7.3755914182419276e+19,7.41075906939813e+19,7.849398488728602e+19,7.749898472693e+19,8.089515986048906e+19,8.186253588398524e+
19,8.082875568534518e+19,8.200384396869144e+19,8.147048563393572e+19,8.047495424017854e+19,8.022580577503866e+19,7.6058810976409
34e+19
−28.167285749999998,5.54347366435197e+19,5.88893474512054e+19,6.150939058568267e+19,6.418415076047853e+19,6.640789377769708e+19,
6.708202896375784e+19,6.946567323472288e+19,7.20049688922252e+19,7.542717446244067e+19,7.436683259374305e+19,7.692365895348335e+
19,7.652417143581771e+19,7.635948908146087e+19,7.660066904558348e+19,7.773272742343649e+19,7.471054060428781e+19,7.1096028542855
65e+19
−30.163,5.009987833579977e+19,5.544270514453697e+19,5.810312201750175e+19,6.056645129863945 4e+19,6.350576570720856 5e+19,6.4386550
6863171e+19,6.446251706268171e+19,6.711655913483 2665e+19,6.6964626382103446e+19,6.934508325266158e+19,6.902528074516862e+19,7.102
431203370026e+19,7.230299083027098e+19,6.980194397765154e+19,7.015946405662623e+19,6.827624164954554e+19,6.649129742167781e+19
−32.15871425,4.578620999511263e+19,5.076955116129087e+19,5.30104530173666e+19,5.609346606094709e+19,5.641061240143430 5e+19,5.8368
20748467615e+19,5.834695814863 01e+19,6.080444386235515e+19,6.2403456399820005e+19,6.278647568204995e+19,6.404815500978385e+19,6.3
18171333250639e+19,6.4367426283875656e+19,6.537358234565586e+19,6.435255174864343e+19,6.161563726591 2906e+19,5.9220837093523685
e+19
−34.15442855,4.204266134054086e+19,4.476613561822227e+19,4.574541126990423e+19,4.746543877615129e+19,4.891671530475602e+19,5.2049
770534724936e+19,5.353451476760217e+19,5.5747633116797714e+19,5.513033990466011e+19,5.522489945006501e+19,5.489765967495593e+19,
5.853820217304451e+19,5.6123746364812706e+19,5.520630628102473e+19,5.5261023321343295e+19,5.42601795935746e+19,5.11609108079189e
+19
−36.15014285,3.3555304660369883e+19,3.6201856341564617e+19,3.816359504533543e+19,3.977440096430584e+19,4.0066791828299416e+19,4.1
195397189044945e+19,4.1569438626795454e+19,4.2423608812506325e+19,4.394968300399312e+19,4.65137872613292e+19,4.527128545937686e
+19,4.522214636977039e+19,4.539421286840323e+19,4.575879835161324e+19,4.682084016719454e+19,4.405460160072047e+19,4.197062609134
477e+19

C.43 "NBSR_upper_Cycle8_cladding_fluence_position_5.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.856194106509026e+20,1.987691750984186e+20,2.065157517618059e+20,2.153931236726452e+20,2.2314149810272164e+20,2.2626
33199583661e+20,2.3061301646267466e+20,2.3870746098040527e+20,2.4111556946047196e+20,2.4037758623459112e+20,2.4717494208612686e
+20,2.5153902072394832e+20,2.5022215662417317e+20,2.5192194502872937e+20,2.5495118189988453e+20,2.5494039529974986e+20,2.3801532
080509682e+20
24.15442855,2.19643044125708e+20,2.341959654740765 6e+20,2.4671741046374764e+20,2.6095752040821057e+20,2.6173325673456275e+20,2.69
46455238109263e+20,2.7880125368100025e+20,2.8080396577267198e+20,2.8255049611114512e+20,2.9046426507661987e+20,3.02668404245662
5e+20,2.9673577417159016e+20,2.992733218532729e+20,3.0220817597324965e+20,3.034738037223851e+20,2.9994838324503485e+20,2.84949615
75776623e+20
22.158714250000003,2.4939877948055945e+20,2.673315022044602e+20,2.736299777997671e+20,2.9117977621888686e+20,2.9291462107388074e
+20,3.011924377939009e+20,3.080985585301279e+20,3.200312349291145 5e+20,3.223629383248939e+20,3.230694606337153e+20,3.29454229030
0996e+20,3.374785606469548e+20,3.370255234412984e+20,3.414408384297596e+20,3.400206027453604e+20,3.374102455127685e+20,3.21350795
67892336e+20
20.163,2.726807569879155e+20,2.970908330926899e+20,3.0500999535823205e+20,3.181345910721005e+20,3.2396474844489345e+20,3.34002578
7535551e+20,3.427325337958871e+20,3.487523555543815e+20,3.5077034866291075e+20,3.546103783108558e+20,3.6502843627426487e+20,3.682
185732640956e+20,3.730680489079775e+20,3.802114748471675e+20,3.7970810017421584e+20,3.7078578409614875e+20,3.56723653053908e+20
18.167285749999998,2.959744199787508e+20,3.145489454106649e+20,3.257283575669132e+20,3.464898661761331e+20,3.505132680263676e+20
,3.594562584213595e+20,3.6789767191008897e+20,3.743076090401208e+20,3.8125328064350834e+20,3.912983020189264e+20,3.9651092653400
916e+20,4.038557023423798e+20,4.0234378055683585e+20,4.136346542478092e+20,4.0796629587703634e+20,4.046422252688667e+20,3.9337831
807823e+20
16.171571450000002,3.2263080556157174e+20,3.43892093310365e+20,3.538301475677808e+20,3.700424075702006e+20,3.831274524169085e+20
,3.839499306771777e+20,4.004624177166792e+20,4.027932222291141e+20,4.176733371149012e+20,4.210612284405344e+20,4.304608515745609e
+20,4.3995845299314426e+20,4.4049598523318895e+20,4.51242135617361e+20,4.462901872721997e+20,4.3781731286641076e+20,4.2522036167
5797e+20
14.17585715,3.373428292619268e+20,3.648873115891694e+20,3.831202613501521e+20,3.9416124547133866e+20,4.0895237090601474e+20,4.196
841391566739e+20,4.213075224769428e+20,4.303565811065924e+20,4.3909732274905914e+20,4.542776646719281e+20,4.6385976112489955e+2
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31704e+20
12.18014285,3.591281660005964e+20,3.871634386339668e+20,4.050611049074299e+20,4.160724258782462e+20,4.3516830365000296e+20,4.4028
83431805965e+20,4.526336070347 3444e+20,4.651092089738331e+20,4.742652347214848e+20,4.797259510396652e+20,4.886509637677659e+20,4.
994070018687282e+20,5.036533267884127e+20,5.1227721359608716e+20,5.080254953763352e+20,5.0747088435274396e+20,4.820073169681493
e+20
10.18442855,3.793719178033518e+20,4.061559448210996 6e+20,4.251942940588049e+20,4.41691500081449e+20,4.544457558573601e+20,4.67084
05568182354e+20,4.783452662224265e+20,4.912757031338707e+20,4.972308052915 5706e+20,5.043625457472678e+20,5.1777927854811696e+20,
5.223447070551192e+20,5.352670540164624e+20,5.3904326294694396e+20,5.41689575513318e+20,5.3451918307379144e+20,5.136282352629554
e+20
8.18871425,3.93720792632506e+20,4.203088630811407e+20,4.459800725183277e+20,4.571927433583246e+20,4.687964284532036e+20,4.8165225
80470496e+20,4.993818331350806e+20,5.1083270806138493e+20,5.223357182216735e+20,5.2844093389790085e+20,5.371502146233081e+20,5.52
3422420296563e+20,5.6302457169636485e+20,5.610011852877 6844e+20,5.677482036720089e+20,5.6492481108675717e+20,5.436841976548888e
+20
6.193,3.975113836965005e+20,4.341633520707889e+20,4.571693723913661e+20,4.711649860661097e+20,4.876630909720984e+20,5.01898706499
83854e+20,5.102700070876926e+20,5.2635372677184075e+20,5.3632324194631614e+20,5.512590875994657e+20,5.563018231624273e+20,5.66931
11871180715e+20,5.801087485430044e+20,5.861204803513978e+20,5.912890595825971e+20,5.872638599656736e+20,5.572843026580276e+20
4.19728575,4.134566753455869e+20,4.44463656316061e+20,4.6924946565886e+20,4.7816099513679236e+20,4.953116893509291e+20,5.15745105
5393858e+20,5.193397400342 6696e+20,5.36161442944296e+20,5.518901037073444e+20,5.690246180212791e+20,5.828062974600182e+20,5.92116
0322595891e+20,5.9721270082 3224e+20,5.975596697942229e+20,6.083480676955892e+20,5.973493310915966e+20,5.735900465282811e+20
2.20157145,4.157587155909 9594e+20,4.47602556955252e+20,4.638067270075708e+20,4.889125388210318e+20,5.0117510540747067e+20,5.20051
6556431557e+20,5.268759779950281e+20,5.455403917613978e+20,5.61267254757757e+20,5.645634599822455e+20,5.785554781236109e+20,5.920
854702258742e+20,6.05734114929619e+20,6.177710617965742e+20,6.081682910266778e+20,6.01049134937791e+20,5.713284560333772e+20
0.20585714999999993,3.912551556183877e+20,4.216562892146307e+20,4.411782376917073e+20,4.637474007068301e+20,4.8524689254193077e+
20,4.8709589558168335e+20,5.046079409003 3165e+20,5.119239524416764e+20,5.346791843091225e+20,5.470325381133 6144e+20,5.6039623679
688186e+20,5.659540325162743e+20,5.7569702908792144e+20,5.8239281112152225e+20,5.907128753587366e+20,5.740709491176188e+20,5.4305
587819704386e+20

C.44 "NBSR_lower_Cycle8_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,4.141488155208954e+20,4.5021111642115487e+20,4.741789419204074e+20,4.983220496718486e+20,5.112524865832928e+20,5.314
9713726939274e+20,5.4477544203518024e+20,5.582056580861979e+20,5.785294105066187e+20,5.7706243288830267e+20,5.974527026762206e+
20,6.078312097724692e+20,6.178205003805247e+20,6.188227553097052e+20,6.235149263682897e+20,6.180398279165966e+20,5.8792813475730
076e+20
−12.20157145,4.554327297696831e+20,4.872064582330638e+20,5.1359407769586224e+20,5.235267386532108e+20,5.447125202010613e+20,5.668
6190469427626e+20,5.793428999334423e+20,5.931182871887696e+20,6.141800228350713e+20,6.301783486014866e+20,6.371887398056821e+20,
6.49019842386734e+20,6.611889251053388e+20,6.680276295907241e+20,6.663637965199502e+20,6.611682507884141e+20,6.3731458347392e+20
−14.197285749999999,4.559846441432407e+20,4.9351572042850533e+20,5.13892506966255e+20,5.386747207756758e+20,5.53333710358702e+20
,5.743756705714234e+20,5.819038185820835e+20,6.0469860131669e+20,6.19009723045373e+20,6.408777570517418e+20,6.427195690247379e+20
,6.549282026105033e+20,6.691584248381761e+20,6.77495567858937e+20,6.782173711846158e+20,6.719296821894435e+20,6.510890718459028e+
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−16.192999999999998,4.584583711074601e+20,4.904100784730629e+20,5.138214951820351e+20,5.385524726408161e+20,5.5435034742139526e+
20,5.701724920522776e+20,5.8403686875871596e+20,6.015641950942218e+20,6.14509014139179e+20,6.27818779822026e+20,6.47451290950482
9e+20,6.566306876650932e+20,6.646199628315107e+20,6.715126003175694e+20,6.873005873813585e+20,6.798533388717085e+20,6.44137108059
1033e+20
−18.188714249999997,4.472843522512791e+20,4.858374588993043e+20,5.170305087221015e+20,5.238817975743106e+20,5.500635727512057e+2
0,5.6324300034909215e+20,5.817546039468872e+20,5.8993713903238434e+20,6.060693984171386e+20,6.266565236575145e+20,6.344013025542
127e+20,6.45771277979507e+20,6.578891243474722e+20,6.619628636650019e+20,6.654208678915105e+20,6.706649533236526e+20,6.3593030312
33032e+20
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C.45 "NBSR_upper_Cycle8_cladding_fluence_position_6.csv " File

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C.46 "NBSR_lower_Cycle8_cladding_fluence_position_6.csv" File

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−34.15442855,1.6503881196901 72e+20,1.7541964386091103e+20,1.8014868359795825e+20,1.8749404783567467e+20,1.9355754587641348e+20,1.965770765285564e+20,2.012051819193849e+20,2.086318248674775e+20,2.153567084926494e+20,2.177472587978294e+20,2.2505065559685487e+20,2.3010109154159847e+20,2.3091122247835396e+20,2.3812377836578262e+20,2.4252185969391265e+20,2.4264669954318316e+20,2.3459691981554034e+20
−36.15014285,1.3602071866453882e+20,1.4676013310220971e+20,1.5618846350583965e+20,1.619635018097534e+20,1.6452935913731326e+20,1.7143539335227772e+20,1.7513862139170213e+20,1.7936617679806267e+20,1.815107660385097e+20,1.884752359276008e+20,1.92279398313243 9e+20,1.945339528677292e+20,2.0002743746903287e+20,2.0461782528837966e+20,2.0254389009028573e+20,2.0469591659834887e+20,1.956750422134018e+20

C.47 "NBSR_upper_Cycle8_cladding_fluence_position_7.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0684909890712779e+20,1.1440483157170006e+20,1.1988237917096917e+20,1.2535089580241874e+20,1.272171187406626e+20,1.2787956679189804e+20,1.3386762968967337e+20,1.375172031555815e+20,1.4003684317724125e+20,1.414717045937504e+20,1.428449429357260 3e+20,1.4243748691704311e+20,1.4712509044880051e+20,1.479208780837249e+20,1.4963676196944298e+20,1.4938920720450657e+20,1.4500281 301120184e+20
24.15442855,1.2651164078393393e+20,1.3473885246756119e+20,1.4330286612751827e+20,1.4653648384032509e+20,1.51303241148854e+20,1.55 31564702774837e+20,1.5638395739746327e+20,1.615470148232509 2e+20,1.630509365819097e+20,1.6598015755585693e+20,1.678968476672101 6e+20,1.7314755860418776e+20,1.7370269750839067e+20,1.7586216128406992e+20,1.7621277532882967e+20,1.7184391183776295e+20,1.683000 5381868385e+20
22.158714250000003,1.423593956070739e+20,1.500484678553351e+20,1.5750326617368864e+20,1.628835980605471e+20,1.6898587613956994e+ 20,1.7623933699888723e+20,1.7783569336934633e+20,1.8058695115390792e+20,1.826454805833685e+20,1.8604484311733446e+20,1.901098411 0294277e+20,1.9260716932175413e+20,1.98161214530789e+20,1.959953759542959e+20,1.9865313666025495e+20,1.9610427880153193e+20,1.924 1380036373512e+20
20.163,1.5814233995527343e+20,1.6934286498514348e+20,1.7683909950878686e+20,1.8220933796102344e+20,1.850673736592164e+20,1.918469 743247069e+20,1.9472254072513787e+20,1.993501148825652e+20,2.0385072425711747e+20,2.0759326356822698e+20,2.0950782874597563e+20, 2.1407696722927632e+20,2.1632620944975013e+20,2.164637989006483e+20,2.1822165022505733e+20,2.1737805158402936e+20,2.079683143494 3968e+20
18.167285749999998,1.652125252911936e+20,1.7879881952563328e+20,1.8933052170345408e+20,1.950689049026884e+20,2.0271335354525293e +20,2.0585931774686976e+20,2.119748766609214e+20,2.1621358796870615e+20,2.1923046245384326e+20,2.2723508734238825e+20,2.30733259 2889683e+20,2.3389569172602087e+20,2.394284875990097e+20,2.4018921382945813e+20,2.393753642588946e+20,2.3839205123336392e+20,2.26 85259929355944e+20
16.171571450000002,1.8406228006423852e+20,1.9373604029920025e+20,2.0034723997652586e+20,2.1409981026552585e+20,2.149811264780355 5e+20,2.2159498232236687e+20,2.29195869826037e+20,2.3414059032395157e+20,2.3698853258752 27e+20,2.4208412337136416e+20,2.4695818998 2692556e+20,2.5053392185007374e+20,2.586718863223076e+20,2.58406269621732e+20,2.6129246069018604e+20,2.5873935296425378e+20,2.477 9753860074432e+20
14.17585715,1.939835950641367e+20,2.0737439540695276e+20,2.1592884686568912e+20,2.2734664635662998e+20,2.3194765884399976e+20,2.3 93971448283418e+20,2.4695925229372788e+20,2.501992448073485e+20,2.5440130101045386e+20,2.5969079198571545e+20,2.635310782426368 6e+20,2.6928805461091146e+20,2.7305821805888088e+20,2.7568463599417197e+20,2.7724380602655044e+20,2.7546204919908965e+20,2.663625 5227077224e+20
12.18014285,2.031505586344003e+20,2.199518774126065e+20,2.3064348084417377e+20,2.3969676046659102e+20,2.4791653488260216e+20,2.54 50967262428863e+20,2.5860070104655338e+20,2.6622921268708334e+20,2.713917388794698e+20,2.7637948928287762e+20,2.832509933267672 8e+20,2.9098947028133557e+20,2.903594274675703e+20,2.9422361922754354e+20,2.996496371869009e+20,2.9481169460261775e+20,2.87866349 11596813e+20
10.18442855,2.145380778214755e+20,2.31530109390695e+20,2.4417505803829494e+20,2.511751205652631e+20,2.592652740313934e+20,2.65433 42505215892e+20,2.7207012393273983e+20,2.7987128642864385e+20,2.869016292594777e+20,2.8801084460108123e+20,2.94116310080511e+20, 3.031977450731892e+20,3.047638211397827e+20,3.103752395561419e+20,3.118929733832307e+20,3.149810331441221e+20,3.0113709071012404 e+20
8.18871425,2.26794694852834e+20,2.3772375961471582e+20,2.4982791265994388e+20,2.6182847519194753e+20,2.690049072080979e+20,2.7757 10458016596e+20,2.792215879790361e+20,2.8750882903699348e+20,2.9518727661723156e+20,3.068223505692433e+20,3.09398301331425e+20,3. 171665273564577e+20,3.209526278064617e+20,3.2646470557680566e+20,3.249113791118398e+20,3.234733302949237e+20,3.1269460458556765 e+20
6.193,2.326313562312812e+20,2.4781825670338924e+20,2.572662427428618e+20,2.728414748312112e+20,2.7978469538425625e+20,2.852059322 4300334e+20,2.8975382139025804e+20,2.990493434436002e+20,3.079703459491308e+20,3.166443249231262e+20,3.209324409372179e+20,3.2319 60264595228e+20,3.3161448216756453e+20,3.3148857985149174e+20,3.3824374378052924e+20,3.362468374256022e+20,3.247828206287612e+2 0
4.19728575,2.3671388491912757e+20,2.5526614898752782e+20,2.6445436189383747e+20,2.7410262292554396e+20,2.8252639096759717e+20,2.9 321161959835065e+20,2.99101404316913e+20,3.055457967062771e+20,3.1199815759665857e+20,3.220076573411478e+20,3.261071854978309e+ 20,3.3276938358166703e+20,3.4459517032469194e+20,3.4434017829213936e+20,3.4260676370418336e+20,3.473453656424512e+20,3.339837831 366985e+20
2.20157145,2.3375703980832044e+20,2.5238314531948072e+20,2.6644914331515988e+20,2.7917537067313594e+20,2.8469063584388678e+20,2.9 237492699153764e+20,3.0069829192077325e+20,3.0576253993394674e+20,3.1628149251014e+20,3.209855642773331e+20,3.285705147789687e+ 20,3.3481516340949995e+20,3.4338448940346855e+20,3.443444281593486 4e+20,3.4902831305729796e+20,3.464258006250587e+20,3.368832550 401813e+20
0.20585714999999993,2.226691362594944e+20,2.3820877570996676e+20,2.5132439715098657e+20,2.6259663869001207e+20,2.711266534122954 4e+20,2.7758273293648488e+20,2.868777237564259e+20,2.9601440702282387e+20,3.009150351484429e+20,3.0782106936340736e+20,3.14509297 8838998e+20,3.2413152847894997e+20,3.233224600089968e+20,3.336108572890904e+20,3.337553527742035e+20,3.322376189471148e+20,3.2087 9848830504e+20

C.48 "NBSR_lower_Cycle8_cladding_fluence_position_7.csv" File

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−10.20585715,2.321994634761454e+20,2.535322031661706e+20,2.6583663120363268e+20,2.794420498405138e+20,2.8761560695062482e+20,2.995077978687936e+20,3.0870929161013205e+20,3.146304190993623e+20,3.177551339649331 4e+20,3.3317099603293726e+20,3.3765035607144346e+20,3.4196662745579625e+20,3.514677368353839e+20,3.572066512680193e+20,3.617290412120188e+20,3.595616089353222e+20,3.429435656805132e+20
−12.20157145,2.5032620959022363e+20,2.7223215012009093e+20,2.8684159988514764e+20,3.02237275083908e+20,3.1151261026800645e+20,3.1396903351492916e+20,3.286193882518746e+20,3.356237006460519e+20,3.4304981236074335e+20,3.508817863939142e+20,3.610448125913361e+20,3.689607215018889e+20,3.764404877900965e+20,3.7627368050213506e+20,3.8647070563723064e+20,3.886152948776777e+20,3.6651014182237884e+20
−14.197285749999999,2.6165688680337572e+20,2.7761726310755975e+20,2.9521808815449833e+20,3.053173663437822e+20,3.14657512002821e+20,3.226685116921798e+20,3.317132915801786e+20,3.3840098886727e+20,3.456268255897274e+20,3.5781597597913966e+20,3.618777865643411e+20,3.696826676940532e+20,3.795221727501741e+20,3.791519030695718e+20,3.842591809882386e+20,3.8753104750592826e+20,3.7198556448804327e+20
−16.192999999999998,2.5677007074618665e+20,2.744500495698968e+20,2.9488234864497086e+20,3.027127289779383e+20,3.101627461956815e+20,3.2457563960231225e+20,3.2784166255258934e+20,3.3872397877516986e+20,3.4923336915014235e+20,3.575376096769365e+20,3.623288037219184e+20,3.700162822699761e+20,3.769722524246488e+20,3.802005578034441e+20,3.8828805510256866e+20,3.857981641513733e+20,3.753280850480861e+20
−18.188714249999997,2.546998541819007e+20,2.739905326779011e+20,2.9154142178513148e+20,3.0189834817397365e+20,3.0940892599944806e+20,3.196803238107048e+20,3.244423000186234e+20,3.3755792145964315e+20,3.4496119013808505e+20,3.5335149047586574e+20,3.635129229730842e+20,3.684140823321044e+20,3.7110052964172556e+20,3.745232664453422e+20,3.800284381814711e+20,3.766545748507604e+20,3.667327286174611e+20
−20.18442855,2.4978382028764833e+20,2.6588125480932934e+20,2.8340983211371145e+20,2.9389691068583556e+20,2.990971544497038e+20,3.129681897871605e+20,3.166783238607998e+20,3.275696710512e+20,3.336374189591479e+20,3.4545683090135 9e+20,3.5062520066115 82e+20,3.567593527442501e+20,3.61691323640537e+20,3.6722890061413627e+20,3.7070582322467026e+20,3.714548623202933e+20,3.557367284470342e+20
−22.180142850000003,2.4419311997393407e+20,2.6331805364877525e+20,2.7325530565070792e+20,2.7980435102009888e+20,2.8715980869243712e+20,2.965185475205163e+20,3.047956951438518e+20,3.184011137807329e+20,3.196526996738449e+20,3.31890192302762e+20,3.3832396002410306e+20,3.4280438252941155e+20,3.4810768557310306e+20,3.512621495091384e+20,3.579695024320723e+20,3.561346222644964e+20,3.4157617090595016e+20
−24.17585715,2.3139464487340143e+20,2.4957610802779822e+20,2.5916593338537814e+20,2.7166957394827182e+20,2.8243714375620375e+20,2.8512199736562146e+20,2.9164341859815254e+20,3.047468216709459e+20,3.111588088228398e+20,3.1578160187965676e+20,3.2337770828271655e+20,3.320219381862475e+20,3.350924672449009e+20,3.4055301537533316e+20,3.414194570526106e+20,3.450706242187222e+20,3.291022794135209e+20
−26.171571450000002,2.2167254239893494e+20,2.394348623998235e+20,2.489216284775801e+20,2.560359061857958e+20,2.6561404440855036e+20,2.7306034299248548e+20,2.766281065146164e+20,2.8573557194395107e+20,2.8953769939598954e+20,2.934400499608456e+20,3.041013730885473e+20,3.116512621857068e+20,3.1746986162851506e+20,3.2201934447597316e+20,3.2006440555973706e+20,3.201881829422052e+20,3.1344470614799306e+20
−28.167285749999998,2.048744110211356e+20,2.2158223272073924e+20,2.308549117378319e+20,2.432979916929944e+20,2.4812743454285916e+20,2.5459201380146705e+20,2.6426577403642882e+20,2.6880941331647427e+20,2.750561868806102e+20,2.79207244677205e+20,2.8707799874865988e+20,2.9106171802389204e+20,2.992554620032468e+20,3.034755801419913e+20,2.988336626827328e+20,3.0414280929383704e+20,2.928434748513529e+20
−30.163,1.9193940893650718e+20,2.0425924274260263e+20,2.1543958090322896e+20,2.2241095682653503e+20,2.2975950846465832e+20,2.4017274559402243e+20,2.455084538751841 7e+20,2.4906240532888515e+20,2.584673614628644e+20,2.5930936640368892e+20,2.668390686316048e+20,2.6447401752968007e+20,2.74563202284342e+20,2.790951544295621e+20,2.792449622486867e+20,2.7715137141475017e+20,2.695090477057903e+20
−32.15871425,1.738238187238531e+20,1.8862345004652197e+20,1.991949947294291e+20,2.0443561223178482e+20,2.0998487634020932e+20,2.1601278174307104e+20,2.216050757569888e+20,2.2930424143987183e+20,2.325256407844522e+20,2.3638345774361153e+20,2.4197840792453507e+20,2.412527430985626e+20,2.462904294416786e+20,2.549745018502959e+20,2.5900921953203857e+20,2.5666541776615988e+20,2.406476682546515e+20
−34.15442855,1.5728599171261786e+20,1.6675044598752608e+20,1.766234187479195e+20,1.8203881203925392e+20,1.8951751586065927e+20,1.9295618966631042e+20,1.996008570479085e+20,2.054667362634191e+20,2.0978035148076614e+20,2.1276269579482848e+20,2.19374426705555 2e+20,2.1922196271942487e+20,2.266172628968495e+20,2.296846045550960 3e+20,2.283921136900954e+20,2.29111403715254e+20,2.188203502681546e+20
−36.15014285,1.3605099896840443e+20,1.4461873126216956e+20,1.4942373737558137e+20,1.5544739291123384e+20,1.6196615797675914e+20,1.663674267052961 3e+20,1.706868854900558e+20,1.714247686842547e+20,1.768024444041074e+20,1.817004163627207e+20,1.855593294726356e+20,1.8985909893759946e+20,1.9224221197516335e+20,1.9598953238688327e+20,1.9645967394690197e+20,1.9547529845456903e+20,1.856474805332734e+20

C.49 "NBSR_upper_Cycle9_cladding_fluence_position_0.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,4.701179762731971e+19,5.003060804042233e+19,5.165517913698438e+19,5.413256154370494e+19,5.4558105966536516e+19,5.4629
87532051081e+19,5.909651283445478e+19,5.9640524537579905e+19,6.2012071070306394e+19,6.419567758809219e+19,6.34466926100165e+19,6.
261732595548958e+19,6.2751773878601425e+19,6.2438285340441715e+19,6.3752812827834835e+19,6.307109965755104e+19,5.79239929754588
e+19
24.15442855,5.661932181267815e+19,6.105103157435465e+19,6.447730053308731e+19,6.360123595224112e+19,6.749429278922258e+19,6.99993
3032034124e+19,7.0269183091284574e+19,7.32399558834604e+19,7.475199263299078e+19,7.418482335164994e+19,7.46419462902302e+19,7.434
663932174397e+19,7.886112307167083e+19,7.920552027827546e+19,7.623139824958082e+19,7.535839582783755e+19,7.452682824645544e+19
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,7.773262174977907e+19,7.852696495956653e+19,8.419109806768955e+19,8.484592165335097e+19,8.564007347819451e+19,8.551883111621393e
+19,8.94082516392607e+19,8.801095016361722e+19,8.825247796285871e+19,8.700550936067339e+19,8.392813515472775e+19,8.2294377580856
98e+19
20.163,7.0306024692991386e+19,7.3562535206456844e+19,7.801874224095658e+19,8.343541461657625e+19,8.388966678099753e+19,8.55125154
1306419e+19,9.113282136902692e+19,8.981676280208237e+19,9.361594532406544e+19,9.281519071865628e+19,9.754412129825831e+19,9.54721
878952565e+19,9.988284531310055e+19,9.700824345525027e+19,9.97249527343571e+19,9.251213265994084e+19,8.89171578731326e+19
18.167285749999998,7.821806966006117e+19,8.478850617017156e+19,8.826463090679834e+19,8.916079090675735e+19,9.18341514960637e+19,
9.323202712653896e+19,9.632882690429364e+19,9.754890592185659e+19,1.0382537515808827e+20,1.0339475903424253e+20,1.04859810780037
71e+20,1.022416647470556e+20,1.0792675450653909e+20,1.0582439089745219e+20,1.0319954639143245e+20,1.0385791059855661e+20,9.744460
112741394e+19
16.171571450000002,8.350096395987277e+19,8.896500410911547e+19,9.235519700591704e+19,9.488779396896183e+19,9.852783991006593e+19
,1.033660512926528e+20,1.0599376657283152e+20,1.0877267595871609e+20,1.1047217426082729e+20,1.1113340924211043e+20,1.14709436919
46946e+20,1.1589315279768542e+20,1.1181665349194567e+20,1.1472666156442326e+20,1.1405872811010254e+20,1.125745378699142e+20,1.073
4015965338922e+20
14.17585715,8.894940623618507e+19,9.387546330803646e+19,1.0189047337494138e+20,1.0486555232835566e+20,1.0747317218942157e+20,1.07
31719346011745e+20,1.0824253966402598e+20,1.1777446679653152e+20,1.199485997595927e+20,1.1814288281359953e+20,1.163763997811123
2e+20,1.2316099604348199e+20,1.2179737831797044e+20,1.2310070978614357e+20,1.2253229650266718e+20,1.1582808191674873e+20,1.145879
0748007298e+20
12.18014285,9.286648188362991e+19,9.84436305347361e+19,1.043545545280587e+20,1.1086642724532607e+20,1.1228458967985807e+20,1.1984
620881458936e+20,1.1841369250926253e+20,1.2185575072586952e+20,1.2496575606475548e+20,1.2565187108874972e+20,1.2668917748485813
e+20,1.2702888576033645e+20,1.2867862397702549e+20,1.2526335965256889e+20,1.294164129358812e+20,1.268241038703298e+20,1.205744285
2624851e+20
10.18442855,9.41260818921147e+19,1.0492201088681543e+20,1.1011524134039514e+20,1.1426925154842711e+20,1.1479843091839759e+20,1.23
0145865613744e+20,1.2184043993035504e+20,1.2833508800266854e+20,1.2828532791724635e+20,1.3466323117376171e+20,1.337493680664890
7e+20,1.3496561938517338e+20,1.3650818203326082e+20,1.3715506314374909e+20,1.3538858011126188e+20,1.3316373013805885e+20,1.266375
0354999665e+20
8.18871425,1.044531177741834e+20,1.1088939343859786e+20,1.1361854273906021e+20,1.1859550820599741e+20,1.2274473379043108e+20,1.23
63180300555333e+20,1.2817910927336441e+20,1.3191015875530793e+20,1.3412926718019299e+20,1.408832418515336e+20,1.403138716433375
7e+20,1.4127175328771444e+20,1.4279709129084805e+20,1.4027368080511199e+20,1.4219805641634262e+20,1.39223934387648e+20,1.33690995
65859e+20
6.193,1.0170387305460823e+20,1.1256783939687658e+20,1.1809695042705598e+20,1.2480977733545137e+20,1.2804705366205173e+20,1.304623
3165446655e+20,1.3125657917178202e+20,1.3555891271136086e+20,1.3585173167557598e+20,1.4058085364012194e+20,1.4626211570072682e+
20,1.4389946856789316e+20,1.467051718459281e+20,1.462247956366602e+20,1.492534623743753e+20,1.4388894239597693e+20,1.37641180901
33497e+20
4.19728575,1.0443780697866889e+20,1.1492378805667268e+20,1.1847397876660093e+20,1.24676764799419e+20,1.225275118790689e+20,1.3184
221710021222e+20,1.3462495418497542e+20,1.3611201519932273e+20,1.4306598713707173e+20,1.4398080716906403e+20,1.4590518278029468
e+20,1.4791663854101417e+20,1.457434625026726e+20,1.4528318171251748e+20,1.489204525719346e+20,1.4570805628804532e+20,1.433080890
91145e+20
2.20157145,1.044129269359578e+20,1.113353203579581e+20,1.2184809532811228e+20,1.2389591422817873e+20,1.2534948287733803e+20,1.302
8817135548891e+20,1.3390438987107354e+20,1.3765936247100845e+20,1.435004309597961e+20,1.4257317090644827e+20,1.464707252896121e
+20,1.4686115057523225e+20,1.455913114722471 1e+20,1.4559896687000437e+20,1.50158713159171e+20,1.478888877241441e+20,1.4206600080
502994e+20
0.20585714999999993,9.359479728776102e+19,1.0151727273427508e+20,1.0753919999507794e+20,1.1409030662585121e+20,1.148510617779787
2e+20,1.1660414786439073e+20,1.2234952388121266e+20,1.2726620309081144e+20,1.3273215708949347e+20,1.3261062765009699e+20,1.352699
2144602438e+20,1.357034083440291e+20,1.3747563292483427e+20,1.3857035480412214e+20,1.3698664439308943e+20,1.3387663905420349e+2
0,1.310355295615412e+20

C.50 "NBSR_lower_Cycle9_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,1.002015012447464e+20,1.0804637009649623e+20,1.1580033109987867e+20,1.1977539638533476e+20,1.2549684928416524e+20,1.261035395564279e+20,1.3239053496457581e+20,1.3571106374178637e+20,1.4308799640562385e+20,1.4189566820493096e+20,1.4153107988674155e+20,1.474707116216539e+20,1.4663053171779484e+20,1.480745311197576e+20,1.458573365443118e+20,1.4516547997199966e+20,1.4041434873890159e+20
−12.20157145,1.1246353460243397e+20,1.2042323442054269e+20,1.2427198564300399e+20,1.3033984529035043e+20,1.3566226058108382e+20,1.4471572535376075e+20,1.4127653791131271e+20,1.505768892616612e+20,1.501367038906189e+20,1.556495472005641e+20,1.5559500249154365e+20,1.6024278585491875e+20,1.6232218327073387e+20,1.630733691756648e+20,1.6054708791576976e+20,1.620379766289957e+20,1.513357305643494e+20
−14.197285749999999,1.1745198316600703e+20,1.2474087875563605e+20,1.3553977421696773e+20,1.3608330745773303e+20,1.422573857489613 8e+20,1.4513581530569028e+20,1.500993838265523e+20,1.5334335862619023e+20,1.5840453346845724e+20,1.5947533222975367e+20,1.6268294389004465e+20,1.6080067296647892e+20,1.6511257575325434e+20,1.6674700317442884e+20,1.6451067010458993e+20,1.6526855448255847e+20,1.603002013380982e+20
−16.192999999999998,1.1862612979702643e+20,1.2423083788005876e+20,1.318115955091832e+20,1.3951771027658277e+20,1.431779473292716 1e+20,1.4294062999879664e+20,1.5178835595674724e+20,1.531069982204349e+20,1.5443233895716012e+20,1.6014900723239236e+20,1.6387718594017686e+20,1.6376139804909833e+20,1.699239932436908e+20,1.6919385968259236e+20,1.6432215593481745e+20,1.634178620747414e+20,1.558552860152904e+20
−18.188714249999997,1.1530464409509624e+20,1.2541646760771404e+20,1.3233311948139638e+20,1.3446801853095164e+20,1.406105183064313 2e+20,1.4415879516692026e+20,1.4969556159485693e+20,1.5174433741964303e+20,1.5492228441362 46e+20,1.6309059382061865e+20,1.6188965329744885e+20,1.645958364046394e+20,1.658628047334656e+20,1.6294514126323073e+20,1.631700185723502e+20,1.5644953626619753e+20,1.545002806122558e+20
−20.18442855,1.1266927341716028e+20,1.1845388334748813e+20,1.2917335405708827e+20,1.3158480435062445e+20,1.3566226058108382e+20,1.4158849536992097e+20,1.4754343790034783e+20,1.4905059433380794e+20,1.525921727212593e+20,1.5537778058018148e+20,1.587375432708 9794e+20,1.6101980872728042e+20,1.6160736050514995e+20,1.5941121827353664e+20,1.5634236069759592e+20,1.590810792452549e+20,1.4969556159485693e+20
−22.180142850000003,1.0961285586257512e+20,1.1438312559006633e+20,1.2475044800283261e+20,1.2794370579232874e+20,1.31451791814592 1e+20,1.3686607187841262e+20,1.4390233934205213e+20,1.4471955305263938e+20,1.4697023999327316e+20,1.5288212091131544e+20,1.557394981242119e+20,1.557261011781367e+20,1.558533721658511e+20,1.5532323587116102e+20,1.549873552945613e+20,1.494946074037289e+20,1.458200164802451 9e+20
−24.17585715,1.0400049238178555e+20,1.126300395036543e+20,1.131199849601188e+20,1.2245095790149632e+20,1.2652171565891812e+20,1.3323358564259383e+20,1.3351683535961237e+20,1.3777132266320835e+20,1.418679173880609e+20,1.3982392618687308e+20,1.4470998380544282e+20,1.528514993202864e+20,1.4630134961423276e+20,1.4862189205940152e+20,1.5250222179761154e+20,1.4659512550316755e+20,1.3895695239086363e+20
−26.171571450000002,9.962256178935379e+19,1.0277945643950293e+20,1.1162431162329457e+20,1.1576683873469065e+20,1.197103255043980 5e+20,1.2304712200184278e+20,1.2671884215116751e+20,1.3291205893678901e+20,1.3076854756475684e+20,1.360345042970305e+20,1.3746989137651633e+20,1.3788711055428688e+20,1.404985581142314e+20,1.4106888524714711e+20,1.3893207234815256e+20,1.3782491044750916e+20,1.3239149188929547e+20
−28.167285749999998,9.287059665992443e+19,9.851731373814968e+19,1.0697461441048016e+20,1.0962625280865031e+20,1.1115733236010187e+20,1.1945578352896924e+20,1.1876584080609639e+20,1.2403371138780937e+20,1.2561263717524375e+20,1.2903842767161 66e+20,1.265198018094788e+20,1.2672458369948544e+20,1.3064510427592108e+20,1.3096280328284725e+20,1.3129868385944696e+20,1.2471408486348564e+20,1.2219928670022646e+20
−30.163,8.56796901615883e+19,9.292695952591223e+19,9.753742282522072e+19,1.018321009670423e+20,1.0365504255798927e+20,1.074272398 0287803e+20,1.1178220420203802e+20,1.1467881532844042e+20,1.1578884800324277e+20,1.1916966303779173e+20,1.1989596890001156e+20,1.2222512366765721e+20,1.185639296902487e+20,1.2293420488492322e+20,1.1840603711150526e+20,1.1896583807250475e+20,1.1179081652451493e+20
−32.15871425,8.154108644154281e+19,8.682197119944314e+19,8.9700496448644e+19,9.679293539332738e+19,9.672212296407278e+19,9.796421125018785e+19,1.0261008076412361e+20,1.0302060146885657e+20,1.068061956598205e+20,1.0883296221605449e+20,1.0708178997908177e+20,1.0783584665817167e+20,1.0721193174095516e+20,1.1129034489613422e+20,1.1211999862807701e+20,1.0748465528605747e+20,1.0161487905568011e+20
−34.15442855,7.233815002765544e+19,7.892753364721503e+19,7.998866746884293e+19,8.304719025781131e+19,8.573069424914604e+19,8.9090 8397097504e+19,9.143109480414408e+19,9.253241946399757e+19,9.467727053063725e+19,9.832401494477893e+19,9.913931480592689e+19,9.963404488598967e+19,9.900343149573556e+19,1.015144019601161e+20,9.856515997413256e+19,9.524865028074458e+19,9.457497527810589e+19
−36.15014285,5.572354458260704e+19,6.176958634634526e+19,6.604713553568498e+19,6.677401555273659e+19,6.883044677527996e+19,6.952727935613434e+19,7.205078553434238e+19,7.453390948938088e+19,7.73126274903215e+19,7.999086839569813e+19,8.245619355095102e+19,8.283264773566416e+19,7.922982616615476e+19,8.26563822023033e+19,8.152520149119651e+19,7.755654760136214e+19,7.388128683057467e+19

C.51 "NBSR_upper_Cycle9_cladding_fluence_position_1.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.144869912523111e+20,1.221703555239664e+20,1.2829906291480465e+20,1.3305634690223025e+20,1.3684939299158634e+20,1.39
5192857287043e+20,1.4078042972441987e+20,1.4520857209502645e+20,1.456785320324433e+20,1.5284782460360288e+20,1.544273650430798e
+20,1.5773687834484094e+20,1.5565966673216687e+20,1.60439572136556e+20,1.6123584601367144e+20,1.573811565630898e+20,1.5085996758
79347e+20
24.15442855,1.309361545973255e+20,1.4284067527437132e+20,1.4977753278623085e+20,1.549567062000259e+20,1.57963092514476e+20,1.6313
887271572652e+20,1.6642350245882783e+20,1.7052759303143214e+20,1.7657938760459446e+20,1.7588377903296663e+20,1.8133158177320328
e+20,1.8356148795038102e+20,1.869185062277655e+20,1.8629302404872454e+20,1.893503085513426e+20,1.8575463432499308e+20,1.804674436
451973e+20
22.158714250000003,1.4814370094604154e+20,1.5555221500159024e+20,1.659953921427934e+20,1.7242383330839813e+20,1.7634695254529442
e+20,1.8219176115324066e+20,1.8604418846212597e+20,1.9300536399722126e+20,1.957391622372612e+20,2.0139338540728985e+20,2.04260650
0074144e+20,2.059549941379811e+20,2.0557721647469052e+20,2.084733233814436e+20,2.102491046130789e+20,2.112631096284681e+20,1.9939
76108956844e+20
20.163,1.6402280458357555e+20,1.7762732474542893e+20,1.8627436137972967e+20,1.8849635006097018e+20,1.945651106968551e+20,2.013447
4936081832e+20,2.067229912438922e+20,2.1042611720081834e+20,2.158066212255886e+20,2.1619344745566457e+20,2.2303699162254967e+20,
2.267327656189627e+20,2.2784064951475048e+20,2.338104414514201e+20,2.3110209230546413e+20,2.287036565719083e+20,2.24977343662594
46e+20
18.167285749999998,1.793697393870431e+20,1.93648377774409e+20,2.009403915325956e+20,2.0772851322792008e+20,2.1442784586166282e+2
0,2.16808749997072e+20,2.2409510840101772e+20,2.2997271806356105e+20,2.362835278609555e+20,2.4014161052408172e+20,2.401631008701
9705e+20,2.4803591950892175e+20,2.4914041019216498e+20,2.5042361006941995e+20,2.5253475380753927e+20,2.552787316852127e+20,2.4185
687946533965e+20
16.171571450000002,1.928815117393461e+20,2.060850672855213e+20,2.1346134582189697e+20,2.2004757137082217e+20,2.304658649533655e+
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66e+20,2.6796934660174037e+20,2.7220237925103672e+20,2.7082812817050367e+20,2.7613624366099068e+20,2.6875091655782954e+20,2.66987
01157010008e+20
14.17585715,2.0244641236694113e+20,2.22226579359832e+20,2.3011523299043115e+20,2.3558452607678313e+20,2.468256737013741e+20,2.541
7028225400085e+20,2.5918940914277905e+20,2.634122621544418e+20,2.6895337823965295e+20,2.7260390940213894e+20,2.8138045364855574
e+20,2.8082000804328486e+20,2.877806180429561e+20,2.9244571975625543e+20,2.9273697049966058e+20,2.9007782293560027e+20,2.82907399
29359254e+20
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722849519618 55e+20,2.748915001925737e+20,2.8056834477956583e+20,2.835210052287276e+20,2.9138816851321147e+20,2.962297172788261e
+20,2.977074613419672e+20,3.064189690146139e+20,3.0986025207018757e+20,3.1192898065150024e+20,3.092862336147385e+20,2.9544079536
22238e+20
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24385704270238e+20,2.8927193495627534e+20,2.972369358691263e+20,3.04194152656253e+20,3.113023674016111e+20,3.086053289641369e+2
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4e+20
8.18871425,2.292935100192317e+20,2.523277678423232e+20,2.6213019334803505e+20,2.7615773400710595e+20,2.834491822298685e+20,2.9155
556669987414 2e+20,2.9836122029221256e+20,3.089582230687 6765e+20,3.1280669162968436e+20,3.201999362287827e+20,3.238629091705987e+
20,3.342840304302624e+20,3.3604397667002325e+20,3.444467020011181e+20,3.431663298009836e+20,3.3590768263281816e+20,3.28504258396
0863e+20
6.193,2.35383761001232e+20,2.567988909051604e+20,2.7222217299087976e+20,2.82439701497872e+20,2.93094388887684e+20,3.0293074651884
107e+20,3.100740244604927e+20,3.156383274980914e+20,3.241909197165695e+20,3.2882378591069576e+20,3.363239167049467e+20,3.42498998
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8117739607124e+20,3.174751865555282e+20,3.208322048329127e+20,3.286303727956578e+20,3.361101443146415e+20,3.4217947048595056e+2
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867564e+20,3.34339452901823e+20,3.369805033323125e+20,3.451700218085262e+20,3.466545522967565e+20,3.4254141315736666e+20,3.306764
7996000705e+20

C.52 "NBSR_lower_Cycle9_cladding_fluence_position_1.csv" File

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−10.20585715,2.42374909913804e+20,2.680931988596156e+20,2.7988291584557154e+20,2.922189400511962e+20,3.0113008172854595e+20,3.077547636863091e+20,3.231316718672532e+20,3.299180969563055e+20,3.385362912839777e+20,3.475333943457887e+20,3.5354220822672035e+20,3.557291337116674e+20,3.592303635221943e+20,3.673180856220723e+20,3.72908403289179e+20,3.6873418632398786e+20,3.5771246644394295e+20
−12.20157145,2.62813925675757e+20,2.842810848387015e+20,2.9688234515822327e+20,3.1103147593347314e+20,3.206105149466703e+20,3.3644663789197374e+20,3.469684244571251e+20,3.497994947901081e+20,3.611158586261026e+20,3.692771003311121e+20,3.744438319655772e+20,3.827447609203362e+20,3.9269309456546274e+20,3.9837616004211986e+20,3.9911983912479516e+20,3.995321144489551e+20,3.908782913895653e+20
−14.197285749999999,2.7068222003108902e+20,2.89489100559125e+20,3.069319096442615e+20,3.18073523034213e+20,3.3098865551410345e+20,3.3718013733701544e+20,3.475152972122179e+20,3.538843571582934e+20,3.654450322974939e+20,3.7229253521234756e+20,3.832265971016589e+20,3.86019211025804e+20,3.9365676692810806e+20,4.012711358780246 6e+20,3.993183420586499e+20,3.979350424113314e+20,3.8811169209492826e+20
−16.192999999999998,2.6675683865249638e+20,2.8880536823140296e+20,3.0367555667236454e+20,3.1853443440484444e+20,3.28794378068643 2e+20,3.3858096858248066e+20,3.4590974214323293e+20,3.5499902747917025e+20,3.671755706952022e+20,3.75029726664932e+20,3.778076366680507e+20,3.920031413480757e+20,3.944586961594644e+20,4.0079438951551874e+20,4.03266344854206e+20,4.023908960177183e+20,3.870088980179574e+20
−18.188714249999997,2.665956610566314e+20,2.900981822108674e+20,3.028894624328827e+20,3.0915333279007795e+20,3.2128689531387924e+20,3.3366476914088634e+20,3.444110732694006e+20,3.522918094040625e+20,3.622424051908854e+20,3.662769349063269e+20,3.78041768333623e+20,3.8283468105276614e+20,3.854344473972972e+20,3.885573340091094e+20,3.9448075204100384e+20,3.930618236619678e+20,3.840664172064291e+20
−20.18442855,2.601807927412048e+20,2.8394741893848977e+20,2.9476271838874262e+20,3.039504068884712e+20,3.1749498029537126e+20,3.2916536930684476e+20,3.3321799615585714e+20,3.4328678884631426e+20,3.544951354163081e+20,3.630760044059905e+20,3.682862822681103 6e+20,3.745241379957976e+20,3.7782912701416604e+20,3.868443272095478e+20,3.869076671770456e+20,3.864122581455448e+20,3.748351824790458e+20
−22.180142850000003,2.4846855410834883e+20,2.705176492226795e+20,2.7933547755505467e+20,2.983844072446002e+20,3.0765466391624555e+20,3.191870622842416e+20,3.296421156693507e+20,3.369635372695899e+20,3.424633692688426e+20,3.52810970923375e+20,3.5794772918036 34e+20,3.592371499472834e+20,3.6961302837302015e+20,3.742583363464764e+20,3.7090244913994007e+20,3.7477297358239616e+20,3.586246750829964e+20
−24.17585715,2.412087758693352e+20,2.5768338830843353e+20,2.7147396962481173e+20,2.8429296108260734e+20,2.9553184656550186e+20,3.0346178428205944e+20,3.0843001298266974e+20,3.192294774410482e+20,3.2431364090359644e+20,3.316288416141707e+20,3.4073113426486205e+20,3.441758105329801e+20,3.5111832339908054e+20,3.56306545379661e+20,3.582565115219153e+20,3.5442896777168984e+20,3.4058805380256786e+20
−26.171571450000002,2.2799843389807097e+20,2.4476825582854308e+20,2.580809597115672e+20,2.723918336181061e+20,2.7799628967081506e+20,2.8651099101587925e+20,2.9543344340171063e+20,3.0419528372710113e+20,3.0718187630170826e+20,3.132998385194889e+20,3.195371287117519e+20,3.250850312220521e+20,3.339045561606995e+20,3.375590460711542e+20,3.373695917040848e+20,3.414900828039877e+20,3.2881134413136586e+20
−28.167285749999998,2.1213912400037993e+20,2.3016613117859902e+20,2.4363831605121588e+20,2.5098179353299445e+20,2.6331894880946736e+20,2.731208087797551e+20,2.751482532751095e+20,2.8372403244597513e+20,2.901733984222711e+20,2.9526038956193985e+20,2.988984789450959e+20,3.0698337336785345e+20,3.114816421310469e+20,3.1643629798147916e+20,3.196717261426848e+20,3.203814730999149e+20,3.0837741818822964e+20
−30.163,1.9969168931620998e+20,2.1517378708603447e+20,2.2576626557919684e+20,2.3265844579255347e+20,2.4232344619021202e+20,2.5112317738901638e+20,2.5588272351813835e+20,2.6358248831709223e+20,2.681520145437207e+20,2.7452559877318892e+20,2.7899219755263345e+20,2.823226356650858e+20,2.9034588672661786e+20,2.9646384894439843e+20,2.957716335853151e+20,2.973805818668445e+20,2.82369009569861e+20
−32.15871425,1.8434984433155926e+20,1.9736111783354463e+20,2.0951447409718903e+20,2.1525635525795126e+20,2.249049551283113e+20,2.3004567213326837e+20,2.3387095374179746e+20,2.385722497222384e+20,2.4505894103652414e+20,2.5225255163091953e+20,2.5843894363501468e+20,2.6230268165238175e+20,2.640773318131689e+20,2.7179971802908626e+20,2.7258581226856818e+20,2.722029447864608e+20,2.57383088998243e+20
−34.15442855,1.6629569145298397e+20,1.796672110201132e+20,1.907143799942421e+20,1.9778357279533813e+20,2.0101391113772702e+20,2.0553084256991533e+20,2.10456090578295e+20,2.161877921014237e+20,2.198552893266323e+20,2.2494963242681424e+20,2.307514603425298e+20,2.369469009134104e+20,2.399968334555153e+20,2.4152434463597612e+20,2.4198864921915212e+20,2.4472188192376786e+20,2.3451509858983333e+20
−36.15014285,1.3731709078730688e+20,1.490966281356293e+20,1.580139907026439e+20,1.6422639733624714e+20,1.685075004965909e+20,1.72696986918232 5e+20,1.7812838913117064e+20,1.831497781616452e+20,1.8765370227907953e+20,1.926185377671453e+20,1.9589072573091666e+20,2.01703864355114e+20,2.0058975956966123e+20,2.0139338540728985e+20,2.0533516731318098e+20,2.0614218636335412e+20,1.964545645487321e+20

C.53 "NBSR_upper_Cycle9_cladding_fluence_position_2.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
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5564439024629e+20,1.4562763384427543e+20,1.5199273504238233e+20,1.552587021164887e+20,1.5737889442139344e+20,1.579732721521096e
+20,1.628730710664053e+20,1.6423092161963983e+20,1.6772084072168492e+20,1.6640314318356064e+20,1.6501531925284946e+20,1.60381321
98787495e+20
24.15442855,1.3592361150235481e+20,1.470844530967253e+20,1.5315264819718616e+20,1.6065617220398157e+20,1.624913346551461e+20,1.71
05014776328913e+20,1.73564518258783e+20,1.745293216922766 2e+20,1.7990530143365415e+20,1.8397998416420592e+20,1.8686704250417355
e+20,1.916718314672225e+20,1.926569941759833e+20,1.932434544107622e+20,1.961842386160182e+20,1.9716770471850666e+20,1.88552903603
37895e+20
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20,1.8998823250971348e+20,1.9265416649886286e+20,1.9558646767275747e+20,2.0239098989538053e+20,2.0720652403148718e+20,2.08625452
41052312e+20,2.1308187155233414e+20,2.1771247560476413e+20,2.186812377862263e+20,2.1773622809257578e+20,2.1656952851268292e+20,2.
1167029513381128e+20
20.163,1.6666498608491325e+20,1.8370739608979564e+20,1.9214235694006347e+20,1.9798207572919294e+20,2.057587533458227e+20,2.099708
611844278e+20,2.0985153320994528e+20,2.1392338826337662e+20,2.2072168959633467e+20,2.2832983765658635e+20,2.31533030298619e+20,2.
3675631547549283e+20,2.4084513659164682e+20,2.429398798024676e+20,2.4364170926376042e+20,2.44935654314073e+20,2.372019573896739
6e+20
18.167285749999998,1.8107030440727478e+20,1.9665702623055548e+20,2.0838566539071e+20,2.1200735424656753e+20,2.1995086481330314e+
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833092636901396e+20
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3e+20,2.4015405230341164e+20,2.497336568520329e+20,2.5640188503745085e+20,2.590254038697936e+20,2.6679529506133433e+20,2.68854975
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7577316991872637e+20,2.862502791853748e+20,2.8993757015042654e+20,3.008326100954758e+20,3.038701008582507e+20,3.095542974057561
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2865546904653e+20,2.977934227264286e+20,3.0312585624015136e+20,3.162683339605 2505e+20,3.20808452345101e+20,3.281355292995811e+2
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6.193,2.478787006610253e+20,2.6868983873202805e+20,2.804331818132089e+20,2.9162569339132828e+20,3.0382429248889966e+20,3.10289493
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7020521204557e+20,3.668175867717547e+20,3.6613555105030 5e+20,3.7370241502459816e+20,3.7552004587761 6e+20,3.6218189290050794e+20
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4e+20,3.0664574871967315e+20,3.1541098225760816e+20,3.2880795091882135e+20,3.3170349229015033e+20,3.388823989635194e+20,3.4496925
67329751e+20,3.564383151334734e+20,3.611797641290245e+20,3.663860832431758e+20,3.726380773564651e+20,3.6800181794979434e+20,3.572
413754356779e+20

C.54 "NBSR_lower_Cycle9_cladding_fluence_position_2.csv" File

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5155157614639e+20,3.376687599434272e+20,3.483274060812078e+20,3.604315607629565e+20,3.66800055173608e+20,3.776000851674105e+20,3.
8332726240714654e+20,3.906351111572076e+20,3.9250929555263416e+20,3.9978547431894634e+20,3.9750919423699334e+20,3.8335893239089
54e+20
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464e+20
−14.197285749999999,2.762442609269914e+20,3.0070027480623926e+20,3.118187012438032e+20,3.2749477766408975e+20,3.431584123050464e
+20,3.5297271405466406e+20,3.623764370863941e+20,3.689445655017485e+20,3.822804563371602e+20,3.8883840511488105e+20,3.9867985256
48549e+20,4.145165410455823e+20,4.2151843513121204e+20,4.263458455112245e+20,4.303487052429171e+20,4.319135417613678e+20,4.240497
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C.55 "NBSR_upper_Cycle9_cladding_fluence_position_3.csv " File

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C.56 "NBSR_lower_Cycle9_cladding_fluence_position_3.csv" File

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−28.167285749999998,2.248992997740704e+20,2.4454543487145253e+20,2.51862332188299e+20,2.6291459098124465e+20,2.746833831565094e+20,2.8237240278240556e+20,2.948124855060624e+20,2.988340079067499e+20,3.059139458809037e+20,3.0779830991396386e+20,3.156541624899659e+20,3.219344333744596e+20,3.2707458484399256e+20,3.27516268010205e+20,3.2951939448232365e+20,3.2807218933208325e+20,3.1693001040670766e+20
−30.163,2.108842008943294e+20,2.3113885210802982e+20,2.4041702627561243e+20,2.4469077747544306e+20,2.577110995442138e+20,2.6215281476499852e+20,2.69792067273575e+20,2.7374515988794792e+20,2.8188830445938652e+20,2.850434265903717e+20,2.9202326479446193e+20,2.992394968058208e+20,3.031513053342353e+20,3.0351946889531634e+20,3.0706424493349804e+20,3.0423487120678735e+20,2.9032552745135066e+20
−32.15871425,1.90984705926956e+20,2.0811986374138878e+20,2.198649034288418e+20,2.2550272607157194e+20,2.3223655636618402e+20,2.3899583575488004e+20,2.4504423711549784e+20,2.5122723590704854e+20,2.5894622891042138e+20,2.6344789088615934e+20,2.679455941139287e+20,2.7042320480685687e+20,2.7450184628537727e+20,2.7889153224714587e+20,2.8210433899138797e+20,2.798116583821365e+20,2.6747054435769503e+20
−34.15442855,1.7469615464238245e+20,1.8865865872768336e+20,1.9874724515798355e+20,2.0548390312971606e+20,2.0979102091956788e+20,2.1548992138809947e+20,2.2261397112533208e+20,2.274651339931562e+20,2.3281679571129798e+20,2.3768379357099662e+20,2.4094636743255848e+20,2.4093901547204536e+20,2.445952019887723e+20,2.4740760965276028e+20,2.487677223476912e+20,2.4729224042624644e+20,2.3825215667220473e+20
−36.15014285,1.4579616340065354e+20,1.5820966595937826e+20,1.6734758734178 71e+20,1.7317316774531432e+20,1.756394677297607e+20,1.801586613036454e+20,1.8601647722634566e+20,1.9127878434748157e+20,1.9219438619907955e+20,1.9581268184239256e+20,2.03837063974772 74e+20,2.0512874688338903e+20,2.0749551263319595e+20,2.093821388079525e+20,2.070430842939258e+20,2.0785688976918797e+20,2.01987763138006e+20

C.57 "NBSR_upper_Cycle9_cladding_fluence_position_4.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,2.9383637136408707e+19,3.1269754329283543e+19,3.2178795969962086e+19,3.3603832131578245e+19,3.4837660766310347e+19,3.5490684520504394e+19,3.6458881166542512e+19,3.603953664958149e+19,3.856510474647228e+19,3.7607653273491825e+19,3.85737008849184 2e+19,3.7857110949056905e+19,3.839329508463444e+19,3.930934936457168e+19,3.95468176891461e+19,3.7035896959739175e+19,3.5304170937 640276e+19
24.15442855,3.5348113040091894e+19,3.849260310510424e+19,4.07825691443201e+19,4.216004377677047e+19,4.302107145994397e+19,4.44571 9211587224e+19,4.580062151579254e+19,4.56172183777609e+19,4.542753779652189e+19,4.608746108288981e+19,4.6280930751470215e+19,4.66 8257400965729e+19,4.662986610813231e+19,4.6455850858140525e+19,4.710910082650422e+19,4.4197950677470446e+19,4.3567265572527866e +19
22.158714250000003,4.09728718145256e+19,4.336842331741863e+19,4.520522582131302e+19,4.547962360908037e+19,4.702246079953398e+19, 4.919134570445267e+19,4.905539098850199e+19,5.058827475549165e+19,5.1769508595783614e+19,5.259688692122389e+19,5.409810070446466 e+19,5.395943141847836e+19,5.191694368084327e+19,5.318606172603844e+19,5.356813745855209e+19,5.271502727131582e+19,4.798963948180 875e+19
20.163,4.494609748999605e+19,4.724460311411563e+19,4.93824401242519e+19,5.065438584656751e+19,5.433890568804119e+19,5.48434763934 1222e+19,5.514349293589075e+19,5.728421417668986e+19,5.79668154335637e+19,5.7575464920095015e+19,5.967869116227713e+19,5.88134219 6342296e+19,5.953447962913476e+19,6.042123917410425e+19,5.712077443912852e+19,5.6138609068115436e+19,5.435213921696484e+19
18.167285749999998,4.593685899945526e+19,5.148210349326225e+19,5.6040601779121046e+19,5.664402807662261e+19,5.709136659707596e+1 9,5.844695500861414e+19,6.1349848340456235e+19,6.146465203154603e+19,6.226827786917464e+19,6.318896953958939e+19,6.4339834127607 824e+19,6.4175828854622405e+19,6.572426484577449e+19,6.362047306816829e+19,6.3341664104093065e+19,6.095284247274667e+19,5.7913089 568275366e+19
16.171571450000002,5.165102892443725e+19,5.493062540226414e+19,5.742842570983222e+19,6.023970230297211e+19,6.2259229302389236e+1 9,6.3091131911222215e+19,6.5183047444922565e+19,6.739881523649812e+19,6.639612092959064e+19,6.849821610092458e+19,6.908297972943 124e+19,7.010264009906134e+19,7.094359127467973e+19,6.9196652349672866e+19,7.0381449063136575e+19,6.547825693629634e+19,6.3613121 10765515e+19
14.17585715,5.598048536354052e+19,5.9081485754440516e+19,6.098960227531238e+19,6.458584203708597e+19,6.804069794283764e+19,6.8334 77636336324e+19,6.946471614069042e+19,7.131967233169805e+19,7.179132887538717e+19,7.530104171727536e+19,7.422652441150875e+19,7.5 044428863473954e+19,7.404442200495251e+19,7.466933864856941e+19,7.358237956347287e+19,7.439166075534234e+19,6.94709370303554e+19
12.18014285,5.83983189621426e+19,6.320706667316019e+19,6.612579499687673e+19,6.842130328324865e+19,6.891105696050858e+19,7.192140 202292735e+19,7.449798141507084e+19,7.635859296031932e+19,7.67267565214004e+19,7.951767383927313e+19,7.88633493536037e+19,8.03869 0178609591e+19,7.722951751341436e+19,7.951654276842496e+19,7.997971628075277e+19,7.83204353464795e+19,7.432662418157227e+19
10.18442855,6.083973538792914e+19,6.585490352873873e+19,6.762785708325363e+19,7.123993183690168e+19,7.345965837644585e+19,7.52196 0461620671e+19,7.826275073322256e+19,7.936441373934538e+19,8.173174502457642e+19,8.352505785435847e+19,8.28119176845839e+19,8.250 087320133568e+19,8.490326768286017e+19,8.260945600276052e+19,8.215533105721811e+19,8.203487201188741e+19,7.858454038952845e+19
8.18871425,6.4066680517773476e+19,6.892462981068669e+19,6.998896747881972e+19,7.440353699924818e+19,7.668660350629018e+19,7.94503 751238067e+19,8.1296848283453e+19,8.089927688031937e+19,8.466574280474334e+19,8.608297457750707e+19,8.75703327428577e+19,8.704947 461727294e+19,8.688433827343932e+19,8.751095152332849e+19,8.702119784606854e+19,8.531893621956462e+19,7.876381511896424e+19
6.193,6.487426510337069e+19,6.925659910462617e+19,7.386797495263715e+19,7.648583843073905e+19,7.934348892865413e+19,7.97744269218 0895e+19,8.303077989370583e+19,8.696125109111525e+19,8.740123765105548e+19,8.694485056381672e+19,8.997442383065445e+19,9.32811094 5529514e+19,9.107382469508091e+19,9.119258713413932e+19,8.703081194827804e+19,8.657725253815971e+19,8.262585653005905e+19
4.19728575,6.721049194027692e+19,7.149102956519662e+19,7.491308441635119e+19,7.768138031726041e+19,8.171138574930928e+19,8.333390 688101684e+19,8.428796514145277e+19,8.861204899402721e+19,8.91736256701463e+19,8.695559573687439e+19,8.994331938232962e+19,9.2014 87564076281e+19,9.341853456334845e+19,9.214268664660663e+19,9.038726469023846e+19,8.741594157208175e+19,8.457016731807251e+19
2.20157145,6.771947382195584e+19,7.061840840582932e+19,7.516700982176658e+19,7.595423513209663e+19,8.306810523169563e+19,8.230293 580290499e+19,8.561414571093839e+19,8.82450165037943e+19,8.824218882667387e+19,9.155226766385909e+19,9.117731767768895e+19,9.1292 12136877875e+19,9.370186781081638e+19,9.15234253572306e+19,9.14504712875233e+19,8.90763535772032e+19,8.473813133902656e+19
0.20585714999999993,6.068930296512181e+19,6.5456766590181e+19,6.881152272586916e+19,6.960779660298462e+19,7.397995096660653e+19, 7.664645049117994e+19,7.662269800336826e+19,7.92677071818264e+19,8.159205777482677e+19,8.05571279487463e+19,8.371168454430741e+1 9,8.627073233830419e+19,8.504634814515434e+19,8.403686741315784e+19,8.454811143653309e+19,8.41759891274834e+19,7.844994295859559e +19

C.58 "NBSR_lower_Cycle9_cladding_fluence_position_4.csv" File

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−10.20585715,6.522715920800141e+19,7.086837506327608e+19,7.424914582847226e+19,7.748570506052608e+19,8.013693512864915e+19,8.1388
4650221552e+19,8.480882326703751e+19,8.47777188187127e+19,8.731697287286641e+19,8.779258816452416e+19,9.007565467156613e+19,9.072
54548738429e+19,9.316065040996447e+19,9.17123141888759e+19,8.990655957976392e+19,9.068360525246041e+19,8.523976126019233e+19
−12.20157145,7.053979898188113e+19,7.626527961534485e+19,8.149195800476325e+19,8.673220924435975e+19,8.730962091235328e+19,9.0850
43820256628e+19,9.168347188224742e+19,9.363796230789448e+19,9.879507984015008e+19,1.0029374871398244e+20,1.0094637659337964e+20,
1.0073316973849857e+20,9.977062844670134e+19,1.0200788658439222e+20,1.0026264426565761e+20,1.0091301000335847e+20,9.531251269861
812e+19
−14.197285749999999,7.57902298591112e+19,7.768307692353269e+19,8.523127822883103e+19,8.561188356924203e+19,9.076673895980129e+19
,9.269238707881986e+19,9.505123533268961e+19,9.787495370515941e+19,9.919660999125234e+19,1.01554892709698e+20,1.0316044777868291e
+20,1.0289351505851354e+20,1.0348676171838151e+20,1.0572571646234467e+20,1.0434920324011524e+20,1.0149211827762425e+20,9.5645613
06340575e+19
−16.192999999999998,7.49407956521315e+19,8.089984241574343e+19,8.407362721572353e+19,8.855153670364982e+19,9.030526205374574e+19
,9.28332053994177e+19,9.438616567396249e+19,1.0102498601732782e+20,1.0052901145040293e+20,1.0073712848646719e+20,1.0402232376499
256e+20,1.0226294306065578e+20,1.0356367453605744e+20,1.0491417312877884e+20,1.0243373475873025e+20,1.0024228499039047e+20,9.4424
62208280044e+19
−18.188714249999997,7.252765599754934e+19,7.869764747434598e+19,8.425742622855202e+19,8.749794420857446e+19,9.00309773730632e+19
,9.234854154097454e+19,9.568576607851599e+19,9.594421576732403e+19,9.605110196247663e+19,9.996460709716342e+19,1.004549263098474
3e+20,1.0300888428502743e+20,1.0330296270555301e+20,1.022606809189594e+20,1.0012635022845249e+20,9.668563270830301e+19,9.45235907
820158e+19
−20.18442855,7.197060360482297e+19,7.449685034422265e+19,8.035127305437838e+19,8.480882326703751e+19,8.782029940030444e+19,8.8525
52207414179e+19,9.310918668637248e+19,9.544484798785464e+19,9.729980417886224e+19,9.746154731015132e+19,1.002682996198985e+20,1.0
165782015688194e+20,1.0118107379437602e+20,1.0034860565011895e+20,9.852814711998069e+19,9.69293784760848e+19,9.366963229164338e
+19
−22.180142850000003,6.95399323520941e+19,7.428590563103795e+19,7.950466652451912e+19,8.026983595330976e+19,8.183523800718447e+19
,8.668922855212907e+19,9.099917401910134e+19,9.42617478806632e+19,9.329015802208054e+19,9.445120224773256e+19,9.537132838272323e+
19,9.604601214365984e+19,9.66861982437271e+19,9.554212008079771e+19,9.691976437387531e+19,9.369281924403098e+19,8.89377973983017
2e+19
−24.17585715,6.63570989853286e+19,7.123314541181262e+19,7.70711675946698e+19,7.74896638084947e+19,8.073187839478941e+19,8.3752968
63026582e+19,8.528896284208796e+19,8.646075224079766e+19,8.692109807600504e+19,9.103197507369841e+19,9.435562676106174e+19,9.2605
29462351036e+19,9.315838826826811e+19,9.287562055622428e+19,9.449022419199461e+19,8.953387173529013e+19,8.521657430780474e+19
−26.171571450000002,6.292712663823679e+19,6.77534059474011e+19,7.12393663014776e+19,7.420446852996933e+19,7.773284404085239e+19,
7.8742890308273e+19,7.91800491910928e+19,8.280456572407076e+19,8.48834739430171e+19,8.797186289395995e+19,8.534155763652811e+19,8
.659082538833782e+19,8.786497669880739e+19,8.838074500557534e+19,8.69182703988846e+19,8.58748575414428e+19,8.088061421132446e+19
−28.167285749999998,5.968208437482164e+19,6.330207662440693e+19,6.589279440215261e+19,6.960383785501601e+19,7.138923318886082e+1
9,7.452229943830 66e+19,7.555383605184255e+19,7.81230634834729e+19,7.707286420094208e+19,8.163334186078518e+19,7.89221650377088e+1
9,8.241830502941886e+19,8.10847724994201e+19,8.197605632778232e+19,7.96245600344257e+19,7.658424159453032e+19,7.3213084931543605e
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713333e+19,6.92961865843123e+19,6.981704470989706e+19,7.1860324197125865e+19,7.323061652969034e+19,7.559794781492139e+19,7.368474
147523274e+19,7.540849344785202e+19,7.484239248834024e+19,7.356597903617433e+19,7.292239972356254e+19,6.832516226115375e+19
−32.15871425,4.8604037166537605e+19,5.294095867323884e+19,5.6615185769994125e+19,5.8402843245535306e+19,6.0038937227420975e+19,6.
248374686575204e+19,6.3742063184347144e+19,6.737449721326235e+19,6.638594129195708e+19,6.742652647227841e+19,6.832176904860922e
+19,6.878381149008887e+19,6.80469188325026e+19,6.732529563136672e+19,6.8001110463151505e+19,6.846654611717567e+19,6.168577638236
433e+19
−34.15442855,4.348254836599953e+19,4.885208100354247e+19,4.986303212764161e+19,5.209684049924556e+19,5.336041629728468e+19,5.6188
94172085925e+19,5.421488376953877e+19,5.648533883662358e+19,6.007965577795529e+19,5.797869167746954e+19,5.99869079684049e+19,5.89
1125959179014e+19,5.881738071139158e+19,6.006382078608083e+19,5.878571072764267e+19,5.854875138494991 6e+19,5.503179968963342e+1
9
−36.15014285,3.5228954726236615e+19,3.891884715423913e+19,4.120417580297746e+19,4.138412917492217e+19,4.267716936855625e+19,4.589
551835995445e+19,4.574140995689056e+19,4.906410023403294e+19,4.830068396505 6975e+19,4.903537103448928e+19,5.010626891354172e+19,
4.836249698690975e+19,4.966588647880465e+19,5.0229216314738385e+19,4.875260332244544e+19,4.605482968891996e+19,4.494230840265467
e+19

C.59 "NBSR_upper_Cycle9_cladding_fluence_position_5.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,2.0177714638692057e+20,2.166200057135236e+20,2.3069828218912057e+20,2.310552151095527e+20,2.3921778296822877e+20,2.46
74495281305243e+20,2.505439439500916e+20,2.5452379385914596e+20,2.5734672178213473e+20,2.608500231807998e+20,2.6435619535362386
e+20,2.6942885329252673e+20,2.67415483682368e+20,2.7178001732872454e+20,2.7531489724313823e+20,2.6796380154673155e+20,2.556070326
4179795e+20
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C.60 "NBSR_lower_Cycle9_cladding_fluence_position_5.csv" File

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C.61 "NBSR_upper_Cycle9_cladding_fluence_position_6.csv " File

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191395913233e+20,3.0093949629062834e+20,3.076806785457536e+20,3.146769172771423e+20,3.1860682293912764e+20,3.309960074746166e+2
0,3.311945104084714e+20,3.4105405499201605e+20,3.4590012804102344e+20,3.4549633574822484e+20,3.456807002964775e+20,3.33556186339
46156e+20
8.18871425,2.4799293881669104e+20,2.6494882190168808e+20,2.74236044636056e+20,2.8619316010754195e+20,2.9510826053286022e+20,3.045
956828073553e+20,3.108471113852205e+20,3.218004014789508 5e+20,3.2836344007548836e+20,3.345611427880653e+20,3.408493311684963e+2
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6.193,2.5345318333625765e+20,2.718641890674323e+20,2.854443912060499e+20,2.9328780200272196e+20,3.0536141777157004e+20,3.13792419
8738692e+20,3.237243529916971e+20,3.365523930162781e+20,3.4167105413969576e+20,3.438149989324122e+20,3.526034194227348e+20,3.6360
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14040499043e+20,3.2908789095374474e+20,3.3428629257195874e+20,3.4285415424688724e+20,3.500997941002986e+20,3.615586728631633e+2
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5e+20,3.102781827485883e+20,3.186701629066255e+20,3.317538249428941e+20,3.349230854594815e+20,3.3943718921454944e+20,3.4970957465
76781e+20,3.540506245729752e+20,3.630104022967964e+20,3.6846386039127396e+20,3.715081375791379e+20,3.669312593919 9626e+20,3.53556
91214774665e+20

C.62 "NBSR_lower_Cycle9_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.5659473261706478e+20,2.8100550366238563e+20,2.9521853994055736e+20,3.0940273391210063e+20,3.1930356258160365e+20,3.3263832234616724e+20,3.385097111190456e+20,3.5147291410998356e+20,3.625828575161861e+20,3.641595702785426e+20,3.77406672052372 5e+20,3.874805545616464e+20,3.9123288210046825e+20,3.96461257096159e+20,3.994580293083995e+20,3.9431052587835346e+20,3.7525424422 82948e+20
−12.20157145,2.8035344131841252e+20,2.9821474661737385e+20,3.1716979742652085e+20,3.23994678924411e+20,3.392364241389983e+20,3.50 0992285648745e+20,3.6659024153127145e+20,3.733121955819777e+20,3.793820872887108e+20,3.9172150470688e+20,3.948715370190485e+20,4. 0738683595410886e+20,4.20570597760441e+20,4.243856997313366e+20,4.2856161330280006e+20,4.271211945776487e+20,4.0748184590535564 e+20
−14.197285749999999,2.7979525785483798e+20,3.021582251295373e+20,3.200466761288549e+20,3.310197599624283e+20,3.474276392214843e+ 20,3.546376503431782e+20,3.644310272821046e+20,3.7587689873021534e+20,3.845182800102752e+20,3.926676454713788e+20,4.030310821177 855e+20,4.1517199660209996e+20,4.2337395685764366e+20,4.243851341959125e+20,4.3206962953841594e+20,4.3032325614883314e+20,4.14066 3748480086e+20
−16.192999999999998,2.8651664637012014e+20,3.0138570374023355e+20,3.208898894461697e+20,3.3412398390524556e+20,3.423802355615017 e+20,3.556804976651959e+20,3.669414390296299e+20,3.777188476064689e+20,3.855175811046381e+20,3.9082173784715644e+20,4.05238932413 4239e+20,4.144486767946919e+20,4.179414235738574e+20,4.245621467836519e+20,4.292815398976637e+20,4.2572714975727256e+20,4.1206437 94467381e+20
−18.188714249999997,2.8388464450641602e+20,3.038746251416435e+20,3.1696054931960843e+20,3.277436132506883e+20,3.407271755168934e +20,3.5286300018239105e+20,3.633401094490396e+20,3.6670617629320944e+20,3.7876678474730 34e+20,3.871548061573719e+20,3.9321904250 986424e+20,4.085292175107661e+20,4.134872665737428e+20,4.1228154504958784e+20,4.221925533567 2454e+20,4.205310102807549e+20,4.0793 71019217463e+20
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−22.180142850000003,2.6603351884508827e+20,2.8755044512535242e+20,2.998508405992596e+20,3.1001860198893204e+20,3.227748190146539 e+20,3.308704586104692e+20,3.4178811997248186e+20,3.525004919755508e+20,3.606820929558274e+20,3.677258366628395e+20,3.76509732869 7694e+20,3.807987535260505e+20,3.930013113715904e+20,3.93527259315992e+20,3.993200386649222e+20,4.0170716368999626e+20,3.82554175 4824186e+20
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−26.171571450000002,2.4350145647858667e+20,2.6069712658339683e+20,2.7406468740255747e+20,2.8246288845025958e+20,2.92251175570369 22e+20,2.970356052581511e+20,3.086053289641369e+20,3.1638935854127985e+20,3.215640076716822e+20,3.289459415622988e+20,3.409669625 3670654e+20,3.4938382625340365e+20,3.525502590928706e+20,3.611254727283122e+20,3.587078087903372e+20,3.637020521204557e+20,3.4672 12854767988e+20
−28.167285749999998,2.280221863858826e+20,2.4179749824581045e+20,2.512962312287872e+20,2.668795598395234e+20,2.7433444779984726e +20,2.837704063507503e+20,2.8874824915357014e+20,2.977204686567212e+20,3.0283290889047396e+20,3.115930526095922e+20,3.1550033685 4614e+20,3.211964096460252e+20,3.2904943454490676e+20,3.314043240508079e+20,3.3352847510368125e+20,3.373554533184826e+20,3.274829 0142018385e+20
−30.163,2.1286018166609176e+20,2.3016782778487128e+20,2.406613375788183e+20,2.4354500270624142e+20,2.5413069477431473e+20,2.60951 6175242363e+20,2.6821761665291485e+20,2.7251681694682946e+20,2.7905214430758676e+20,2.8798421079562768e+20,2.937498444442016e+2 0,2.996455512403158e+20,3.036076924214741e+20,3.117836380475098e+20,3.135531983894801e+20,3.135531983894801e+20,3.04319701520400 45e+20
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−36.15014285,1.4752217751496915e+20,1.5863834181083 67e+20,1.6522682950145825e+20,1.7180513755444622e+20,1.787142838305255e+20,1.8 261138843791373e+20,1.8574728236447993e+20,1.938214316141798e+20,1.9798886215428198e+20,1.9995070454044217e+20,2.05452798681391 27e+20,2.0964567831557734e+20,2.1359311557570937e+20,2.1331430661163418e+20,2.1717974123527353e+20,2.1646377338837847e+20,2.08243 15046383986e+20

C.63 "NBSR_upper_Cycle9_cladding_fluence_position_7.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.1589687106456168e+20,1.2466606335046535e+20,1.306573456332503e+20,1.3234433780330385e+20,1.3542141604576494e+20,1.3981449522007808e+20,1.405536500193607e+20,1.4634020847862589e+20,1.491232083005614e+20,1.532063740624745e+20,1.524847508613386e+20,1.5694117000314955e+20,1.5870168177833453e+20,1.592129258017098e+20,1.5936675143706165e+20,1.5731103017050292e+20,1.4987932916256666e+20
24.15442855,1.3301619388712003e+20,1.456836218512601e+20,1.496887437246491e+20,1.5447769769582363e+20,1.5946685120712517e+20,1.6278371646939945e+20,1.694904010636553e+20,1.710755968573731e+20,1.79259459979346e+20,1.7706801021100622e+20,1.836802503894394e+20,1.8340031035451605e+20,1.876129837285452e+20,1.8597632421123542e+20,1.8743597114080575e+20,1.8353773546256935e+20,1.793646495682263e+20
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20.163,1.6817440013180328e+20,1.785893005018021e+20,1.8589036282677412e+20,1.9122958276558596e+20,1.973905256755972e+20,2.0115529499374892e+20,2.0853779441978956e+20,2.125542270016603e+20,2.1700103204126183e+20,2.2093772412833618e+20,2.2666207369095175e+20,2.277275424299329e+20,2.300123055432472e+20,2.322026242407388e+20,2.309946405748875e+20,2.273746483253022e+20,2.204864268599142e+20
18.167285749999998,1.81174928460731e+20,1.951227286250056e+20,2.0482109561268532e+20,2.121821046926106e+20,2.121097161583274e+20,2.170049907892304e+20,2.2419916691904987e+20,2.2978382923191576e+20,2.359131021581781e+20,2.4241166971636697e+20,2.4201862259662874e+20,2.4737028431477054e+20,2.4785947245660635e+20,2.5050391609964036e+20,2.5402833286255483e+20,2.527287324580013e+20,2.4130774456855052e+20
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C.64 "NBSR_lower_Cycle9_cladding_fluence_position_7.csv" File

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−28.167285749999998,2.186620095818073e+20,2.3356613014821418e+20,2.4358968000474435e+20,2.5565141952968648e+20,2.658548096510764 6e+20,2.7167869344833153e+20,2.7689801987723674e+20,2.8526624754746225e+20,2.910228326292508e+20,2.9376341729437975e+20,3.0293074 651884107e+20,3.086403921604304e+20,3.165007690198251e+20,3.17035199995588e+20,3.168451800930945e+20,3.1632488750293385e+20,3.057 8161059166716e+20
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C.65 "NBSR_upper_Cycle10_cladding_fluence_position_0.csv" File

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C.66 "NBSR_lower_Cycle10_cladding_fluence_position_0.csv" File

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−24.17585715,1.0314996588987384e+20,1.0771585064436818e+20,1.1646880381923156e+20,1.1925731294543122e+20,1.2525810805646051e+20,1.2715027651327798e+20,1.310006193033135e+20,1.3302288629912352e+20,1.3642553705792012e+20,1.3988462762407377e+20,1.3763660139883564e+20,1.4173566198400393e+20,1.4264348193623838e+20,1.4310360985723394e+20,1.454769515869428e+20,1.4302803791178976e+20,1.3577121793534018e+20
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−28.167285749999998,9.483351244546838e+19,9.81832628424541e+19,1.0162704769813805e+20,1.04920645269838e+20,1.105732354676815e+20,1.1704755099636734e+20,1.1938836809132807e+20,1.1738523323360525e+20,1.225767389035488e+20,1.2219026971418871e+20,1.2678963571033552e+20,1.2606739749754626e+20,1.2634959653433146e+20,1.2729185433512277e+20,1.2785146937417141e+20,1.2408817781243234e+20,1.2022348591883143e+20
−30.163,8.496152051388857e+19,9.104506212214473e+19,9.513560890586422e+19,9.678948658280645e+19,1.0438877183101572e+20,1.0420510330537923e+20,1.0759914460203665e+20,1.1233913181312388e+20,1.1272560100248397e+20,1.1823087373705652e+20,1.1752298462783262e+20,1.1634157510095213e+20,1.1734983877814406e+20,1.1528261125782931e+20,1.1738810305431832e+20,1.1433270060180315e+20,1.0815301999965915e+20
−32.15871425,7.361453639648122e+19,8.098595788007275e+19,8.73524638106376e+19,9.14907452788845e+19,9.517195996822977e+19,9.744189249157769e+19,9.78073163290419e+19,9.679426961732823e+19,9.992141758767012e+19,1.0522293305161472e+20,1.0457531017736526e+20,1.0238775501871016e+20,1.0891830552314446e+20,1.0764601834035012e+20,1.0799039682591854e+20,1.0760584085036717e+20,1.0202595277725476e+20
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C.67 "NBSR_upper_Cycle10_cladding_fluence_position_1.csv" File

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97853486951717e+20,1.405900752231258e+20,1.4378598519234847e+20,1.4686543355646594e+20,1.4900131678019197e+20,1.534568211968635
e+20,1.507030130489626e+20,1.5043447294010343e+20,1.567714541615818e+20,1.539000537133805e+20,1.5220627231097625e+20,1.4374471481
772381e+20
24.15442855,1.3076603001966164e+20,1.4035884805570812e+20,1.4616496788304175e+20,1.5406174312629358e+20,1.5882253524566812e+20,1.
6200035409176766e+20,1.6495203389882794e+20,1.6755263284777973e+20,1.713579875272133e+20,1.7327508123066882e+20,1.7511415696565
58e+20,1.795086038417867e+20,1.811481118748215e+20,1.838979625895392e+20,1.8209563444839612e+20,1.7986986095665198e+20,1.75261712
68862896e+20
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00246622e+20,1.994274957479645e+20,2.0650677836508932e+20,2.042990959964683e+20,2.0445287054301502e+20,2.0543996744842153e+20,1.9
689756524871247e+20
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964769405e+20,2.0203318282529466e+20,2.1035340341914753e+20,2.13064245149631e+20,2.1858825652576312e+20,2.196137970678063e+20,2.1
97404349296683e+20,2.2492523774724145e+20,2.2651838727727327e+20,2.2720754599874544e+20,2.2442660116615987e+20,2.16631588490475
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4994e+20,3.2286322881716775e+20,3.25052254715068e+20,3.2684384125185674e+20,3.276398506692751e+20,3.254621317233267e+20,3.1632328
780815165e+20

C.68 "NBSR_lower_Cycle10_cladding_fluence_position_1.csv" File

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−18.188714249999997,2.5371839089291844e+20,2.7773548753408567e+20,2.8952863841998442e+20,3.018769606467245e+20,3.09518198775862e+20,3.2092125981941786e+20,3.338733732803929e+20,3.452357292969218e+20,3.512651614253067e+20,3.544565486137486e+20,3.67696424001 9011e+20,3.667353330859842e+20,3.779263887804408e+20,3.8086393489756114e+20,3.867740786828529e+20,3.844674604846521e+20,3.6526316794183844e+20
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−30.163,1.9305885506102067e+20,2.116350464229027e+20,2.2089317868117102e+20,2.306363791781786e+20,2.370513783681255e+20,2.4343189135461895e+20,2.512534753673902e+20,2.5351430041018548e+20,2.6518081343422218e+20,2.6894037495825026e+20,2.737803157412886e+20,2.7983518851156538e+20,2.8388646939595424e+20,2.8498550512568513e+20,2.8620326385091173e+20,2.8396561805961796e+20,2.712006346534479e+20
−32.15871425,1.7764352211731022e+20,1.9353996586657674e+20,1.9987807778325027e+20,2.0767195976373925e+20,2.1607585180203667e+20,2.1982467172171045e+20,2.2740767904829525e+20,2.2998396805055024e+20,2.376359477840421e+20,2.4286937049500874e+20,2.5008942466393 55e+20,2.5027938145672852e+20,2.5579604331409185e+20,2.6099215308361702e+20,2.6531988894323134e+20,2.618198219665008e+20,2.5024772199126296e+20
−34.15442855,1.582272242253972e+20,1.714829293462825e+20,1.8177847444614352e+20,1.8786896411506908e+20,1.9537225743039257e+20,1.9855346836207767e+20,2.0053161960607384e+20,2.0920574779602318e+20,2.13369532852334e+20,2.2007059792666562e+20,2.223659091729143 6e+20,2.258382741173626e+20,2.320565323433453e+20,2.3574825215566165e+20,2.3637126520821488e+20,2.3580535226301904e+20,2.295282978868861e+20
−36.15014285,1.3197191644533865e+20,1.4636566628018912e+20,1.540572203455128e+20,1.5723164710602677e+20,1.639197091856136e+20,1.6961558623141156e+20,1.7283184871415182e+20,1.7842087506400775e+20,1.8164052963232296e+20,1.8531868110230353e+20,1.898482460542604e+20,1.8958705546417003e+20,1.9651143283955296e+20,1.9785922151222706e+20,1.9771053509405875e+20,1.9659849636958306e+20,1.9050857204825512e+20

C.69 "NBSR_upper_Cycle10_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.1494308145808289e+20,1.2454777179367891e+20,1.3012888327716846e+20,1.356585481292766e+20,1.3761973894534478e+20,1.4206789384324751e+20,1.4455203118709406e+20,1.472555233988087e+20,1.5157477904445905e+20,1.5296270739656265e+20,1.5594548132149084e+20,1.5948286123966284e+20,1.6013301097690078e+20,1.6251142831999648e+20,1.6119077633200705e+20,1.6372070683125904e+20,1.5455416088380162e+20
24.15442855,1.3518026406170848e+20,1.4701920810301266e+20,1.5460730355797583e+20,1.5872246872089323e+20,1.6403843218110926e+20,1.6824179156925193e+20,1.6888854922090427e+20,1.727815327779656e+20,1.7472915525169142e+20,1.807461497329292e+20,1.81529156155602 7e+20,1.8663876774269537e+20,1.8680554528398682e+20,1.8986577182978595e+20,1.899709264829392e+20,1.9050122252948632e+20,1.8883853524494998e+20
22.158714250000003,1.511219356187829e+20,1.6179343687104673e+20,1.7045456206625135e+20,1.7588642178397497e+20,1.8233251509178977e+20,1.8653248239434688e+20,1.893823996338395e+20,1.979134948815965e+20,1.982979312479633e+20,2.0280940507679703e+20,2.0344146369 09118e+20,2.081315873605866e+20,2.138806070805628e+20,2.149321536120955e+20,2.1884209759708473e+20,2.157507769334177e+20,2.1083112213911808e+20
20.163,1.6248768372089733e+20,1.7839373837932305e+20,1.8940727492813377e+20,1.9364794725771798e+20,2.0064921190637424e+20,2.0575769279827172e+20,2.105993296241028e+20,2.1256447787335418e+20,2.1761642400549175e+20,2.197517418816202e+20,2.2850841082081893e+20,2.27800595628626e+20,2.3541243568268827e+20,2.3509866776602132e+20,2.3575899376001604e+20,2.3434505941842287e+20,2.291896546759248e+20
18.167285749999998,1.768520354806727e+20,1.9553281614811044e+20,2.0215190582079033e+20,2.0949633646118866e+20,2.1722746485834423e+20,2.220289620047458e+20,2.2659697059333938e+20,2.343071811293838e+20,2.3772979548524338e+20,2.4196990246722997e+20,2.4664476175176958e+20,2.5028955771348523e+20,2.5574007390192963e+20,2.5762494279232206e+20,2.6079145468646965e+20,2.602656814207033e+20,2.532423682157407e+20
16.171571450000002,1.9156633740336136e+20,2.0972360619542315e+20,2.1908293567366155e+20,2.2231672393192335e+20,2.330068816549079e+20,2.3937778373224212e+20,2.4560395682459127e+20,2.5216481569471857e+20,2.5946514922250427e+20,2.6588014841245113e+20,2.6818337452506628e+20,2.6944523036290548e+20,2.767721352277783e+20,2.7616834399354343e+20,2.7714187255660754e+20,2.7838620261892 12e+20,2.7187622503257778e+20
14.17585715,2.039559300047262e+20,2.2011469503927832e+20,2.310389066676685e+20,2.38424042335094e+20,2.4701280303780603e+20,2.5746042664142098e+20,2.644933507555427e+20,2.7138324192747215e+20,2.7580934826906852e+20,2.806905594267315e+20,2.868398452458073e+20,2.930954164132315e+20,2.976193278892124e+20,2.998309676910166e+20,3.005947522953718e+20,2.987449349560304e+20,2.9308524015647474e+20
12.18014285,2.1319201370668817e+20,2.3056175329529564e+20,2.4282583872999365e+20,2.5186348542519873e+20,2.640936500040408e+20,2.7037635785614973e+20,2.7882434700705985e+20,2.8371234233589393e+20,2.9200655694025733e+20,2.939106476489681e+20,3.02308320863691 9e+20,3.105884017781154e+20,3.149873714350271e+20,3.18040813809655e+20,3.2095574602287133e+20,3.1727759455289744e+20,3.0911567128637204e+20
10.18442855,2.263058165805763e+20,2.439859320002652e+20,2.5673847775928805e+20,2.699105114357288e+20,2.7637017308588596e+20,2.7835171641546772e+20,2.9017878815722232e+20,2.967763946211935e+20,3.050621290115928 6e+20,3.1289784671430404e+20,3.1576642042451735e+20,3.215759323374366e+20,3.314491627818918e+20,3.350379893314452e+20,3.361098883764915e+20,3.368640620716875e+20,3.242641601640161e+20
8.18871425,2.3554924980130703e+20,2.5334186939291802e+20,2.6039344997776055e+20,2.767065549064569e+20,2.853744642728327e+20,2.968165343006229e+20,3.0524756302360504e+20,3.107342614582956 5e+20,3.191358921062027e+20,3.2564926177813175e+20,3.292420457608684e+20,3.427753365521743e+20,3.47133601182059e+20,3.490631325326617e+20,3.523766348021848e+20,3.4963243756344305e+20,3.3888574508069867e+20
6.193,2.3901483057458414e+20,2.5847692162190256e+20,2.7417775510240012e+20,2.8390512586667498e+20,2.939400457240432e+20,3.031048956287079e+20,3.124794894920814e+20,3.19546334462059e+20,3.2283496143728784e+20,3.3317799573534717e+20,3.416915652075786e+20,3.445940597736478e+20,3.521047026077401e+20,3.6456892109198787e+20,3.667726460274257e+20,3.630911024718661e+20,3.544090594155504e+20
4.19728575,2.4339627445597028e+20,2.632563702119978e+20,2.7781915897853018e+20,2.9010359692674174e+20,3.008683805326093e+20,3.091428079710567e+20,3.1815897145755304e+20,3.259381544005044e+20,3.334126149883504e+20,3.391045346009692e+20,3.513567477361176e+20,3.519062656009831e+20,3.608930310124043e+20,3.6598455147637375e+20,3.746106251205273e+20,3.717720148329867e+20,3.6131138823462696e+20
2.20157145,2.4356983616843288e+20,2.633309960948808e+20,2.7926192604760084e+20,2.920223866729901e+20,3.0227383466023846e+20,3.110559442413291e+20,3.222616989733233e+20,3.2692355526311805e+20,3.3692964239300887e+20,3.425475014703423e+20,3.5046689061749804e+20,3.5754278114903726e+20,3.6163194032246515e+20,3.70893464666319e+20,3.785188730627237e+20,3.714164111940974e+20,3.585259206212606e+20
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C.70 "NBSR_lower_Cycle10_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.4548410313390047e+20,2.6753831291620668e+20,2.7936255791997334e+20,2.9318926411443277e+20,3.058807523329151e+20,3.1308949954988986e+20,3.272096211475028e+20,3.320195985078684e+20,3.4231005547935105e+20,3.51186578109240 6e+20,3.575184712023405 4e+20,3.6883446871586636e+20,3.705893076588112e+20,3.761698537947032e+20,3.789253579853969e+20,3.743861821242808e+20,3.581878427578968e+20
−12.20157145,2.6128330709638008e+20,2.8944892440872305e+20,3.009170004260028e+20,3.1760040803112596e+20,3.270360594350402e+20,3.355275803509653e+20,3.4964657125338304e+20,3.615578797871798e+20,3.677631350184177e+20,3.7727623904320284e+20,3.8343739716182816e+20,3.970091315897708e+20,4.0130294659352927e+20,4.070480088803224e+20,4.101664662286741e+20,4.117239988600571e+20,3.957421876235533e+20
−14.197285749999999,2.6525035118872678e+20,2.901940525423575e+20,3.096018702203065e+20,3.222181672083082e+20,3.3461737071883216e+20,3.4227443858070235e+20,3.5025092783521556e+20,3.606380592458875e+20,3.7437091773914566e+20,3.8042070238104414e+20,3.902040425574812e+20,3.998449151393235e+20,4.0524907282476486e+20,4.099256281520972e+20,4.159663672324341e+20,4.144857218743244e+20,3.98302081545478e+20
−16.192999999999998,2.6822860233287415e+20,2.9048803329310854e+20,3.0798158400559e+20,3.212361584312801e+20,3.319178359403007e+20,3.426017748397117e+20,3.537396878599941e+20,3.6727184795610494e+20,3.7734803818809786e+20,3.822631702016167e+20,3.861414547211404e+20,3.9816979020764e+20,4.095185778818265e+20,4.1321934325570444e+20,4.140752795184681e+20,4.1122083949819475e+20,3.9748515426694857e+20
−18.188714249999997,2.611153988598934e+20,2.8504543197103056e+20,2.9778667077810147e+20,3.1497493378787987e+20,3.2884743313774484e+20,3.407129485161361e+20,3.487408844020307e+20,3.6064653945985145e+20,3.7176240392382756e+20,3.762049053457543e+20,3.856756083007195e+20,3.9605652088781786e+20,3.987928032601932e+20,4.047136886498393e+20,4.074935027872297e+20,4.079299511325755e+20,3.944153168119903e+20
−20.18442855,2.6083385575628947e+20,2.8131979130285838e+20,2.9544613172404486e+20,3.093203271167026e+20,3.2111630474058924e+20,3.336477995889512e+20,3.3740792646057696e+20,3.4913493167755664e+20,3.5816749024438334e+20,3.652642986370336e+20,3.78587280122033 1e+20,3.845341715011687e+20,3.902147841618356e+20,3.9490829991709586e+20,3.998279547113955e+20,3.954736475146941e+20,3.875141186881089e+20
−22.180142850000003,2.507525773959182e+20,2.7021240705284627e+20,2.872169320934053e+20,2.9869857645302736e+20,3.117377534440325e+20,3.198872390634108e+20,3.2785242136597194e+20,3.327426780851965e+20,3.449841496159906e+20,3.5472791546059574e+20,3.604384915439353e+20,3.714588122639172e+20,3.76931377008668e+20,3.8429389877218 95e+20,3.8451438433525275e+20,3.839049396250419e+20,3.758006818134715e+20
−24.17585715,2.409217480212829e+20,2.6249993512641144e+20,2.7559282013918842e+20,2.840973440498583e+20,2.985385830829071e+20,3.083055281790138e+20,3.179837137022976e+20,3.2328837021056165e+20,3.3061357903264167e+20,3.3912545246208026e+20,3.469543859936203e+20,3.536362292496336e+20,3.5876449730744707e+20,3.6471591146736334e+20,3.719467072406445e+20,3.6604165158373124e+20,3.59538458168559e+20
−26.171571450000002,2.2702098129153806e+20,2.5016065846123286e+20,2.626429680686038e+20,2.7219112364444002e+20,2.8273429099204906e+20,2.9625570948380544e+20,2.9987732619401966e+20,3.082048963066413e+20,3.148884356054474e+20,3.255735052000536e+20,3.300227907931515e+20,3.360154753276925e+20,3.4339948029992285e+20,3.435572122796527e+20,3.510825541512825e+20,3.471635646047317e+20,3.385962871107284e+20
−28.167285749999998,2.186300922479854e+20,2.3170601683283444e+20,2.4362524023299755e+20,2.5483495239817488e+20,2.6610289536590483e+20,2.7448982697627432e+20,2.831334263959534e+20,2.8751204353935157e+20,2.9389538326383297e+20,3.03120160013843e+20,3.042316333907211e+20,3.1336765056790816e+20,3.224708775844346e+20,3.271429101309862e+20,3.278088896009569e+20,3.288361261857928e+20,3.173680501685131e+20
−30.163,2.0185453298445363e+20,2.1812523684333017e+20,2.3178573084409577e+20,2.3764047056482284e+20,2.4513189158059678e+20,2.560657141181462e+20,2.570403733764055e+20,2.6736248981335366e+20,2.7127186845074532e+20,2.7783442336366533e+20,2.862920234237346e+20,2.8894180761367744e+20,2.9625457878861034e+20,3.042073234440244e+20,3.042610314657962e+20,3.0544656537795966e+20,2.9257868870902674e+20
−32.15871425,1.8528080281326446e+20,1.9886667093114706e+20,2.0963315057980745e+20,2.2166544349949015e+20,2.239579280077509e+20,2.3009025339889874e+20,2.3669916681482184e+20,2.445495835550706e+20,2.488224806977179e+20,2.541944135700961e+20,2.618661804695038 5e+20,2.6216694539142608e+20,2.6864639420749916e+20,2.7286275659038668e+20,2.7684958784864928e+20,2.7613385779008992e+20,2.677090478906814e+20
−34.15442855,1.646478768913201e+20,1.7932656191536005e+20,1.8523500965785898e+20,1.969919782975114e+20,1.9997927500322036e+20,2.054693655234966e+20,2.0882696490563237e+20,2.1865101010909653e+20,2.2339936458132387e+20,2.2752244461060768e+20,2.306719960768273e+20,2.379050532404988e+20,2.394026590265365e+20,2.4277213070822185e+20,2.4858107727354348e+20,2.472672094567252e+20,2.3779537580656476e+20
−36.15014285,1.3773337381246203e+20,1.5221362182974502e+20,1.5941106209476785e+20,1.6460604116909782e+20,1.6692227027645773e+20,1.7515995012106125e+20,1.785689961345785e+20,1.8346603702497418e+20,1.9150527986282077e+20,1.9222327131177705e+20,1.9869650130427003e+20,1.989774790602763e+20,2.0612686477950334e+20,2.0802190992665254e+20,2.085120662937702e+20,2.0899996127049744e+20,2.025973997276977e+20

C.71 "NBSR_upper_Cycle10_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.129784985564291e+20,1.2340068151815216e+20,1.296121555729637e+20,1.3469123838978605e+20,1.3874817275015081e+20,1.44
0635708627692e+20,1.445367668019588e+20,1.4744039206322333e+20,1.5180883294986473e+20,1.5185236471487978e+20,1.559234327651844
8e+20,1.5744987127869964e+20,1.571869846458165e+20,1.5747078913981078e+20,1.566770411127829e+20,1.5508671832073916e+20,1.51768127
92283765e+20
24.15442855,1.3590447433423179e+20,1.4566067782598415e+20,1.543393287967143e+20,1.595936693687921e+20,1.631174809446217e+20,1.679
7890493636877e+20,1.719708243230097e+20,1.7521931161880907e+20,1.7885223528097517e+20,1.798953015985439e+20,1.8002985432677224e
+20,1.8154668193112824e+20,1.8499925970966053e+20,1.876043814393931e+20,1.8688921672843133e+20,1.85675415436388e+20,1.7710926863
75799e+20
22.158714250000003,1.5224415060001535e+20,1.6469027796113998e+20,1.7171359116610257e+20,1.767570570842762e+20,1.8240148749869675
e+20,1.8835403235380832e+20,1.9198808671116965e+20,1.9811306258354867e+20,1.9708695669390793e+20,2.023995280685383e+20,2.00616987
09331113e+20,2.0668542820593035e+20,2.0551007055052367e+20,2.126475839702011e+20,2.0886032041389067e+20,2.0927189346494215e+20,1.
987677351015674e+20
20.163,1.6657118941834923e+20,1.801774100497454e+20,1.8914495364284822e+20,1.9454176180952075e+20,2.0032865981853608e+20,2.022909
8132979945e+20,2.0823504597094705e+20,2.1494063382605945e+20,2.197613527907794e+20,2.2357801442216493e+20,2.2133641119768802e+2
0,2.2807874664664433e+20,2.3228040999199425e+20,2.309394054904913e+20,2.2768922215189915e+20,2.300653781046044e+20,2.21269700181
17144e+20
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2e+20,2.206246385723119e+20,2.257065481271222e+20,2.343642812367412e+20,2.379050532404988e+20,2.4239843594620944e+20,2.4640505437
038795e+20,2.529964420107855e+20,2.5630824823751583e+20,2.5634669187415248e+20,2.5688772952505398e+20,2.4908762872099147e+20,2.42
1943454634765e+20
16.171571450000002,1.946293906871485e+20,2.1016740405953777e+20,2.1950468498146984e+20,2.2458320245069457e+20,2.32218787103856e+
20,2.3669068660085785e+20,2.442284661196348e+20,2.538229801984741e+20,2.549174931474242e+20,2.6298330732235579e+20,2.658869325836
2234e+20,2.6975447549879173e+20,2.714132053501449e+20,2.7378088108888618e+20,2.710095471654597e+20,2.6950402651305574e+20,2.58897
54023451564e+20
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5769956867520502e+20,2.616960108426268e+20,2.6580552252956823e+20,2.7411500151906674e+20,2.7777166978033195e+20,2.8566279154760
778e+20,2.8643505636592694e+20,2.9343971310016882e+20,2.9385807032239145e+20,2.921456324492665e+20,2.9168543950482155e+20,2.82449
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85906480434738e+20,2.8035530830135575e+20,2.8848048397403727e+20,2.9180585854310996e+20,2.9842551356338746e+20,3.00339780528855
e+20,3.060746665588912e+20,3.084072907465815e+20,3.116178997533417e+20,3.0691929586970296e+20,3.0429099488846886e+20,2.9300835288
320134e+20
10.18442855,2.2537242769694166e+20,2.4464343125627195e+20,2.5269737313165607e+20,2.638482891466832e+20,2.7780276389819983e+20,2.8
434270490721593e+20,2.8775966578709958e+20,3.019877687758538e+20,3.02990695413993e+20,3.125812520596491e+20,3.166591042811249e+
20,3.174398493134081e+20,3.2516645492978287e+20,3.302037020243829e+20,3.278919956978039e+20,3.28845737094952e+20,3.1119275835994
79e+20
8.18871425,2.3238556464514748e+20,2.5529005721424147e+20,2.6616451825404304e+20,2.7612594292372354e+20,2.892764933914556e+20,2.97
29821045377656e+20,3.0607975468726963e+20,3.1575511347256544e+20,3.18866786649746e+20,3.243218256189711e+20,3.296117830896976e+
20,3.3386771980441695e+20,3.379512255018689e+20,3.40349995358478e+20,3.433661247916645e+20,3.3683749073460036e+20,3.231854769477
987e+20
6.193,2.389509462960555e+20,2.6391669620599256e+20,2.7725154999054162e+20,2.8904978900481868e+20,2.959639901234448e+20,3.05664224
203035e+20,3.1147034403036856e+20,3.2323522753638734e+20,3.2939016683143915e+20,3.363462036722875e+20,3.417509267053264e+20,3.423
880734478196e+20,3.501666910431734e+20,3.513454407841657e+20,3.51931706242875e+20,3.509983173592404e+20,3.34651291574688e+20
4.19728575,2.4305141242143536e+20,2.6335756743196797e+20,2.811722355798852e+20,2.9088943008740327e+20,3.0028663785468074e+20,3.10
63589097631364e+20,3.1962661382091794e+20,3.286891358104173e+20,3.3365062632693924e+20,3.435973519590822e+20,3.438336672548782e
+20,3.541874431572919e+20,3.587950260777173e+20,3.630464400116559e+20,3.669823899861346e+20,3.557331034891254e+20,3.435555162368
5997e+20
2.20157145,2.4003415229305373e+20,2.6746877516170215e+20,2.8427768993349216e+20,2.9502777450182214e+20,3.0359335595303266e+20,3.1
269092949358313e+20,3.212836476294783e+20,3.3356186675411635e+20,3.383062637931605e+20,3.437940929230463e+20,3.5030237446659696
e+20,3.5817879719633525e+20,3.602089604193104e+20,3.6457061713478065e+20,3.6187503978943244e+20,3.603372943239652e+20,3.431484659
6658915e+20
0.20585714999999993,2.2775141038763495e+20,2.4905031577954995e+20,2.629725657180035e+20,2.7836302336741966e+20,2.878817808681807
4e+20,2.965219882022742e+20,3.026729700641428e+20,3.103944875521392e+20,3.156448706910338e+20,3.2360383417002145e+20,3.3062318994
180086e+20,3.322920960499107e+20,3.407643951475175e+20,3.46013647591217e+20,3.4405189142755115e+20,3.3701557522784374e+20,3.22406
993305906e+20

C.72 "NBSR_lower_Cycle10_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.4371739189140606e+20,2.6875720233662847e+20,2.804033628471516e+20,2.9388011887869778e+20,3.043543138193999e+20,3.1258973227361305e+20,3.233528198366878e+20,3.2784789858519115e+20,3.438053998749983e+20,3.473145124132904e+20,3.505799601370177e+20,3.560711813524891e+20,3.622611721985919e+20,3.63447271458353e+20,3.598403537856764e+20,3.604780658757672e+20,3.4910836034046 96e+20
−12.20157145,2.6562969942671514e+20,2.8810452782163447e+20,2.9927523100257753e+20,3.1838567584418995e+20,3.274029700258814e+20,3.379602710634304e+20,3.5040865981494546e+20,3.5708032681420194e+20,3.647147807721682e+20,3.749277851227799e+20,3.827895088149806 4e+20,3.8877145174516734e+20,3.9613171211829844e+20,3.986237643285114e+20,3.986011504246074e+20,3.932523968037307e+20,3.732351344 1557086e+20
−14.197285749999999,2.6941922437341594e+20,2.932124433659343e+20,3.102333634868237e+20,3.2343196850035157e+20,3.354953555379022e+20,3.450265506858105e+20,3.538765019786129e+20,3.644230614118075e+20,3.725940302398945e+20,3.8217441062879376e+20,3.88697956557 47954e+20,3.934056060026799e+20,3.993462785582419e+20,4.055860199929333e+20,4.0301651516184946e+20,3.979165144839159e+20,3.810821 59070234e+20
−16.192999999999998,2.7171792770525027e+20,2.934035308539225e+20,3.081393159853199e+20,3.251641935393925e+20,3.364112186460113e+20,3.438308405168902e+20,3.543480018750097e+20,3.670191375799785e+20,3.7225369098614034e+20,3.8006001061377645e+20,3.86468790980 14984e+20,3.952441163900692e+20,4.046464122857250 5e+20,4.016545927992354e+20,3.992252941723559e+20,3.9654215447415475e+20,3.83880 0643307476e+20
−18.188714249999997,2.6953625132611876e+20,2.917007038899567e+20,3.110050629575452e+20,3.194575748892361e+20,3.322372573329437e+20,3.413958884140347e+20,3.499541203464765e+20,3.592382585942343e+20,3.690170759898906e+20,3.7591431668058875e+20,3.815135192872 015e+20,3.9219971957700284e+20,3.945227328555339e+20,3.988244627256587e+20,4.0015472562280727e+20,3.973743461378193e+20,3.7937820 1411073e+20
−20.18442855,2.627746940588442e+20,2.8798015135016285e+20,3.0087007657540216e+20,3.117304039252638e+20,3.228163049665671e+20,3.36 50054356643185e+20,3.39477664015384e+20,3.490252542436226e+20,3.631363302796 74e+20,3.722203354778821e+20,3.77930346213624e+20,3.7 83882777676785e+20,3.8367880058600266e+20,3.8594527910477385e+20,3.8406945577594303e+20,3.876288842504213e+20,3.690690879688696 e+20
−22.180142850000003,2.54354972287814e+20,2.7730299662192302e+20,2.8777210343424675e+20,3.028923249320109e+20,3.1575680951535816e+20,3.2376778497332485e+20,3.367815213224381e+20,3.426707472466187e+20,3.462923639568328e+20,3.532710147015852e+20,3.58705135809 6992e+20,3.606996821340257e+20,3.68017541437337e+20,3.7551574662428203e+20,3.756559528284864e+20,3.696067335341855e+20,3.59261437 84573587e+20
−24.17585715,2.5012108412940098e+20,2.6604749130134022e+20,2.801240811339381e+20,2.8947775713620056e+20,3.0152192235543285e+20,3.0963918316174796e+20,3.185185325296255e+20,3.2306392721431514e+20,3.341877065446576e+20,3.4066037118955946e+20,3.46511718824700 94e+20,3.523935952301127e+20,3.528537881745577e+20,3.573149460672052e+20,3.5952262843582625e+20,3.548828207023377e+20,3.390909662 5862685e+20
−26.171571450000002,2.3441629321572024e+20,2.5560269443571322e+20,2.6721493409038057e+20,2.7640635533213224e+20,2.86391524600911 9e+20,2.956914925814025e+20,3.0162538096579333e+20,3.1302335388097085e+20,3.143564435161075e+20,3.25644738997351e+20,3.3000469967 002835e+20,3.287756339928498e+20,3.377861440033702e+20,3.410052332240944e+20,3.422071622165882e+20,3.3699522271433025e+20,3.23446 6675378891e+20
−28.167285749999998,2.1560604794843254e+20,2.346831372817866e+20,2.4941553032759847e+20,2.5849953552580647e+20,2.668480234995393 2e+20,2.7567705693123058e+20,2.8313455709114856e+20,2.9091600142449036e+20,2.966751974012234e+20,3.051231865521334e+20,3.06677892 4455285e+20,3.1075178723382126e+20,3.1510439838772986e+20,3.179933246114567e+20,3.177609667488439e+20,3.161542488764698e+20,3.089 189303224078e+20
−30.163,2.051889531150879e+20,2.2315230768117347e+20,2.3548480017518087e+20,2.4048473432833943e+20,2.4663571619020803e+20,2.54729 232397424e+20,2.6036065981709982e+20,2.686503516406824e+20,2.7459837371501314e+20,2.7442311595975773e+20,2.852297352878476e+20,2.8748547220226438e+20,2.904750302983638e+20,2.9498254669401434e+20,2.974972128081312e+20,2.980331623306543e+20,2.825833431834903 2e+20
−32.15871425,1.8484718120590662e+20,2.0153228485382265e+20,2.1690012859933488e+20,2.2076654081930904e+20,2.300670741473972e+20,2.3481090583884382e+20,2.398108399920024e+20,2.451256727570232e+20,2.5202234810012382e+20,2.5263179281033462e+20,2.55850882031058 9e+20,2.6252933320148658e+20,2.6919534672476704e+20,2.6930163207311553e+20,2.722810139124581e+20,2.6875550629383566e+20,2.5931024 398076232e+20
−34.15442855,1.653166830992788e+20,1.801706258785742e+20,1.934777776308409e+20,1.968738206496134e+20,2.0490119118791043e+20,2.093 3125496268997e+20,2.1263118888987073e+20,2.1896647406855632e+20,2.21419517294535e+20,2.2846092162262067e+20,2.333720962029564e+20,2.3791749088764602e+20,2.4090365689815974e+20,2.403439627765375e+20,2.4733674721122982e+20,2.429298626879517e+20,2.2746195241 76647e+20
−36.15014285,1.4168402282447831e+20,1.525726175542199e+20,1.5824135791533716e+20,1.6568472438531523e+20,1.7122456549418012e+20,1.7722516489508753e+20,1.8324102868113008e+20,1.845611153215219e+20,1.9049104627272958e+20,1.926077076781373e+20,1.95813228556519 15e+20,1.949047149671788e+20,2.0175559715487393e+20,2.05025567659382e+20,2.036037184514225e+20,1.9995100762334046e+20,1.941443224 484092e+20

C.73 "NBSR_upper_Cycle10_cladding_fluence_position_4.csv" File

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24.15442855,3.4362561933896204e+19,3.725001825386932e+19,3.973251608968282e+19,4.157978936483498e+19,4.028491722729604e+19,4.350349763518242e+19,4.535614171251176e+19,4.5311253113262465e+19,4.571219762947912e+19,4.559941078375827e+19,4.585664394066546e+19,4.668832679149219e+19,4.651374745335385e+19,4.677482497392471e+19,4.569619829246709e+19,4.522057135860771e+19,4.2169503443889684e+19
22.158714250000003,3.88660643616038e+19,4.214231022444521e+19,4.477225071371233e+19,4.639078435087625e+19,4.704104715763371e+19,4.909574646634465e+19,4.941839034029396e+19,4.974510471694596e+19,5.15864983770831e+19,5.24214602439759e+19,5.2165640456062706e+19,5.165377474119728e+19,5.363124756807632e+19,5.104568686522064e+19,5.142593965936521e+19,5.079682085275791e+19,4.7212743223024 3e+19
20.163,4.388510726356121e+19,4.650939427685235e+19,4.81684068020043e+19,5.106931839480026e+19,5.142497856844929e+19,5.432577709172572e+19,5.394156686439796e+19,5.585758640741808e+19,5.678577409315481e+19,5.764171035591851e+19,5.655285088294436e+19,5.741500596928163e+19,5.963003785667141e+19,5.799787934240539e+19,5.561239515433972 5e+19,5.625078566154763 5e+19,5.416312658789672e+19
18.167285749999998,4.733791118113254e+19,5.057514805973965e+19,5.539547128114129e+19,5.339674138459257e+19,5.768015399255518 4e+19,5.749076254735978 5e+19,5.819688169675995 5e+19,6.165172086568264e+19,6.231091616448214e+19,6.182245584015728e+19,6.420652665922895e+19,6.556279554586706e+19,6.2602070177245225e+19,6.272870803910722e+19,6.266538910817623e+19,6.133682225382043e+19,5.6928807035347165e+19
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14.17585715,5.393195595523881e+19,5.814430437018332e+19,6.004387229811331e+19,6.102588107514141e+19,6.440157158040033e+19,6.315328408490348e+19,6.790955342349724e+19,6.877849268100569e+19,7.087706296329025e+19,7.166006938596378e+19,7.196309569827641e+19,7.347313913146125e+19,7.274610212094994e+19,7.288008950158072e+19,6.9892792795871764e+19,7.006522381313922e+19,6.647300517800018e+19
12.18014285,5.4403173177836904e+19,5.9393157213277766e+19,6.362648002409319e+19,6.545142206914022e+19,6.750702593400731e+19,6.932179172229757e+19,7.11190317350623e+19,7.386379432140161e+19,7.536140010743928e+19,7.692006343401754e+19,7.662325594527849e+19,7.868790537170716e+19,7.567516802410627e+19,7.766519156765198e+19,7.631570685218505e+19,7.536592288822005e+19,7.317180886194139e+19
10.18442855,5.884363934841231e+19,6.280446461718538e+19,6.623499383941133e+19,6.966382701884496e+19,7.239558661043905e+19,7.288291623956871e+19,7.504367475758907e+19,7.863645874032571e+19,7.792072868176639e+19,8.059369212321071e+19,8.063043971705461e+19,8.204550475384293e+19,8.131564100534364e+19,8.162375544603466e+19,8.076386175008778e+19,7.797217531314782e+19,7.619528781389663e+19
8.18871425,6.080426481688291e+19,6.677716218550802e+19,6.723000561118419e+19,7.0884412482059035e+19,7.3392859772602294e+19,7.82005757425775e+19,7.90197644114973e+19,7.960998730338984e+19,8.249326005114072e+19,8.391228252111223e+19,8.319711781015049e+19,8.538953357936363 4e+19,8.532621686270535e+19,8.377151096931028e+19,8.313097214123149e+19,8.014537147831534e+19,7.93080916862724e+19
6.193,6.275019124781595e+19,6.787506722004374e+19,7.233170233191045e+19,7.451281336344435e+19,7.581593957590824e+19,7.7849494884469e+19,8.26249860413811e+19,8.290426775459462e+19,8.566938285444746e+19,8.560662927111407e+19,8.606116873958303e+19,8.916153496481163e+19,8.933905411045746e+19,8.629465729739108e+19,8.798052383342895e+19,8.510233921405647e+19,8.043709083867603e+19
4.19728575,6.44100517943643e+19,6.918836968926439e+19,7.3382683515845526e+19,7.407071154212256e+19,7.86070606652506e+19,8.054846431540286e+19,8.299133128462474e+19,8.485415161871085e+19,8.408527888597728e+19,8.75570784828279e+19,9.00598722973952e+19,8.995076021105874e+19,9.017576855490281e+19,8.880027784994636e+19,8.735242265249735e+19,8.588704167952278e+19,8.0065657467054e+19
2.20157145,6.446093307814813e+19,6.887629781539018e+19,7.215192179587422e+19,7.341773506689661e+19,7.669109765699027e+19,7.831873339047551e+19,8.244237876735689e+19,8.503506284994227e+19,8.455055995880061e+19,8.507181044378616e+19,8.69889041472417e+19,8.882911057742388e+19,8.864763399859485e+19,8.785388597156697e+19,8.616123526435791e+19,8.656545879664063e+19,8.193808871029927e+19
0.20585714999999993,5.867290437393765e+19,6.331271210742617e+19,6.584264260667818e+19,6.892378701358843e+19,7.238654104887747e+19,7.355907196629617e+19,7.546938149858053e+19,7.939289382591211e+19,8.048797212245985e+19,8.055411779137885e+19,7.994693447155837e+19,8.22405496750143e+19,8.32587406982887e+19,8.324121492276316e+19,8.1709122933272e+19,8.17373903131519e+19,7.505780844752903e+19

C.74 "NBSR_lower_Cycle10_cladding_fluence_position_4.csv" File

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−10.20585715,6.221367637769525e+19,6.818883513671076e+19,7.140679366223977e+19,7.3233431750079594e+19,7.40865412748553e+19,7.7591
69637996421e+19,8.184424100909797e+19,8.272900999933918e+19,8.354932936345416e+19,8.387440423207315e+19,8.700642992276724e+19,8.7
03808938823274e+19,8.92638628799769e+19,8.929043421706402e+19,8.695781002937378e+19,8.393489642501615e+19,8.380373578237338e+19
−12.20157145,6.966326167124688e+19,7.426575646329394e+19,7.798574365549017e+19,8.017363885819524e+19,8.56914314107538e+19,8.56846
4723958263e+19,8.89817544287754e+19,9.299515702412511e+19,9.276449520430504e+19,9.370523360670846e+19,9.691018913749272e+19,9.963
346851512331e+19,9.665691341376872e+19,9.702099726662197e+19,9.629678699409865e+19,9.57178710541581e+19,8.91044348574542e+19
−14.197285749999999,7.382930811794812e+19,7.747749616524938e+19,7.956532484317959e+19,8.298737385144156e+19,8.678085623132555e+1
9,9.127423893703614e+19,9.49868766104636e+19,9.558331832592972e+19,9.651840325235717e+19,9.895392070281026e+19,9.757503791226823e
+19,1.0129389440926926e+20,1.015703493844948e+20,1.0104005333794767e+20,1.0067427344230162e+20,9.817374101812473e+19,9.095312150
160037e+19
−16.192999999999998,7.1638586177255055e+19,7.682847712320663e+19,8.07943905203581e+19,8.479252873057264e+19,8.826206693703287e+1
9,9.036629069529342e+19,8.987839571856615e+19,9.43254199212737e+19,9.644943084545017e+19,9.976462915776609e+19,1.007494646727821
7e+20,9.904946444680436e+19,1.014979848920022e+20,1.0009140006917787e+20,9.887420669154892e+19,9.649805073884363e+19,9.1107461395
74467e+19
−18.188714249999997,7.25697136704993e+19,7.642029615774071e+19,8.242146090624575e+19,8.26883049723121e+19,8.709971227637095e+19,
8.87590074753217e+19,9.485458527262561e+19,9.42270494392916e+19,9.43582100819344e+19,9.741674058494073e+19,9.952944455716523e+19,
9.924224797758536e+19,9.827889567127801e+19,9.94446424175255e+19,9.792724946557192e+19,9.525258998133477e+19,9.103000877487371e+
19
−20.18442855,6.887290572980459e+19,7.516918192425588e+19,7.695907241825182e+19,8.231291416750688e+19,8.488807247456675e+19,8.8794
62437397039e+19,8.889864833192845e+19,9.079821625985845e+19,9.327387338974103e+19,9.550360431466837e+19,9.751115363373962e+19,9.5
25767810971314e+19,9.705774486046587e+19,9.579193158944345e+19,9.8845939311669e+19,9.467932751737019e+19,8.872339057667301e+19
−22.180142850000003,6.694846250758028e+19,7.373206833116123e+19,7.500975390173318e+19,7.843406430038555e+19,8.271261491900881e+1
9,8.393150433943057e+19,8.675315419904324e+19,8.842997517351954e+19,9.064161497532375e+19,9.168581198808764e+19,9.40642293311833
2e+19,9.262202760971028e+19,9.449898163373634e+19,9.286003894829913e+19,9.339881520881023e+19,9.168863872607563e+19,8.67197986907
8495e+19
−24.17585715,6.499575190547604e+19,7.068767151809485e+19,7.1742044787615515e+19,7.419904544677736e+19,8.074181319378144e+19,8.160
001084693555e+19,8.156835138147004e+19,8.644108232516903e+19,8.689505644604039e+19,8.707257559168623e+19,8.902019806541206e+19,8.
82541520706665e+19,8.960476748132863e+19,8.951657325610331e+19,8.876522629889527e+19,8.509612039048289e+19,8.132468656690522e+1
9
−26.171571450000002,6.307074333565414e+19,6.49171685894099e+19,7.011440905413026e+19,7.130672713746489e+19,7.371849998881887e+19
,7.701786856840202e+19,8.005774260068762e+19,8.09792591847727e+19,8.393715781540654e+19,8.310835823732757e+19,8.487846156540758e+
19,8.451381236495673e+19,8.557101237246539e+19,8.346452722381444e+19,8.077177661645416e+19,8.176056956465344e+19,7.6719930384467
77e+19
−28.167285749999998,5.956671892574042e+19,6.196379273955683 5e+19,6.465258591373394e+19,6.711184796328616e+19,7.170925462695482e+
19,7.144523729887645e+19,7.445853999407492e+19,7.476891582515636e+19,7.716372824858237e+19,7.742548418627035e+19,7.8852421522608
23e+19,7.957437040474115e+19,7.852338922080607e+19,7.776130065924368e+19,7.88908651592449e+19,7.55745361517338e+19,7.152608200533
3e+19
−30.163,5.369015678774604e+19,5.874187678084463e+19,5.966169732213692e+19,6.22962171269446e+19,6.551078356688803e+19,6.4647497785
35554e+19,6.782192454586951e+19,6.9597681349925495e+19,7.044683344151802e+19,7.289931131989906e+19,7.328148629587545e+19,7.202980
671479302e+19,7.048584242575229e+19,7.273196843101e+19,7.219036543251089e+19,7.117273975683412e+19,6.609535298280458e+19
−32.15871425,4.840760537054811e+19,5.170669127633248e+19,5.592865060043616e+19,5.681064938744913e+19,5.783166714871151e+19,5.9366
02052859306e+19,6.054703165997572e+19,6.2916403441509835e+19,6.469611767874899e+19,6.569169479811945e+19,6.47255157538241e+19,6.4
42418548430426e+19,6.69411129888115e+19,6.496748452559614e+19,6.402787681838791e+19,6.388427852859797e+19,6.050293454736306e+19
−34.15442855,4.323393990064759e+19,4.574583581153621e+19,5.063089133286 2845e+19,5.146093467565653e+19,5.198874319277422e+19,5.346
102140643949e+19,5.4901414015600206e+19,5.64065954594457e+19,5.7383346503816135e+19,5.7849192924237046e+19,5.751168040847093e+1
9,5.997094245802314e+19,5.999468705712227e+19,5.819348961117436e+19,5.806402501132438e+19,5.742009409766002e+19,5.10497008331635
9e+19
−36.15014285,3.422506939816032e+19,3.866564863825524e+19,3.85231810436605e+19,3.9778196175568765e+19,4.288958667895052e+19,4.3909
7564188165e+19,4.565475831356314e+19,4.6086514273848885e+19,4.779126341964654e+19,4.7983707741868966e+19,4.738681374832478e+19,4.
750389723578737e+19,4.929407040358211e+19,4.95775922237776e+19,4.717203819599722e+19,4.431115321311122e+19,4.173791708788321e+1
9

C.75 "NBSR_upper_Cycle10_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.9296100510539848e+20,2.1370885225399e+20,2.2221882727514682e+20,2.3144338765385807e+20,2.332695502342749e+20,2.4155
66358467166e+20,2.4526157438728995e+20,2.5475111487850797e+20,2.5476163755445587e+20,2.563763900090099e+20,2.656851317953045e+2
0,2.6472182864261743e+20,2.6020473084024524e+20,2.6669626529320953e+20,2.643793633708577e+20,2.607643458792939e+20,2.51506304258
93018e+20
24.15442855,2.3328198612403152e+20,2.4839446199905796e+20,2.6301045889072336e+20,2.7319736581521736e+20,2.7948896942517105e+20,2.
8656116426907986e+20,2.9215540144575755e+20,2.965490969574677e+20,3.0445258320126234e+20,3.0780453379412813e+20,3.0748311387426
43e+20,3.120164739940105e+20,3.145036519453377e+20,3.207282930719865e+20,3.179703953667263e+20,3.153779906559197e+20,3.0239205192
92781e+20
22.158714250000003,2.6120247184148922e+20,2.792125100298120 4e+20,2.932870674136115e+20,3.011197647464838e+20,3.164340846783295e+
20,3.218790911779276e+20,3.240314567127301e+20,3.322391439521101e+20,3.378907775430493e+20,3.4541927388033614e+20,3.445535446318
9336e+20,3.4698619598967236e+20,3.537867144727438e+20,3.584903506214655e+20,3.612654672510042e+20,3.509781166015527e+20,3.3734455
50006617e+20
20.163,2.8757421098079355e+20,3.10698269679807e+20,3.2404389260248665e+20,3.3423366934769376e+20,3.419047001137298e+20,3.50068383
4355095e+20,3.5210212971417174e+20,3.656888175767493e+20,3.738410216156768e+20,3.790124385406288e+20,3.7763970763287704e+20,3.853
930065926878e+20,3.9475436175872197e+20,3.977877622524369e+20,3.8879278753077133e+20,3.867877394592398e+20,3.6935836166186146e+
20
18.167285749999998,3.1078340769429475e+20,3.3267057366597494e+20,3.500951684288315e+20,3.6124442189910835e+20,3.748291965478772e
+20,3.771824495325945e+20,3.851949889634 86e+20,3.967747155407232e+20,4.0582326024903275e+20,4.1331923195157165e+20,4.19673971617
21286e+20,4.272168170580651e+20,4.292476935160143e+20,4.329937661534749e+20,4.2852832512393806e+20,4.226471058759533e+20,4.025009
644702022e+20
16.171571450000002,3.3214443986857886e+20,3.547385383425787e+20,3.739634672994343e+20,3.8687287747372745e+20,4.024693964423584e+
20,4.053851342868375e+20,4.176077007038024e+20,4.254308319676312e+20,4.3868749044820574e+20,4.4802493044163104e+20,4.46547929381
3045e+20,4.592430596090215e+20,4.6213488228089174e+20,4.69774345019084e+20,4.6316706113069266e+20,4.541242560638091e+20,4.3963070
48559015e+20
14.17585715,3.5089010876635185e+20,3.8222185470474545e+20,3.9097672109341757e+20,4.144107204294425e+20,4.266141547083204e+20,4.36
7551445014053e+20,4.523401841871839e+20,4.522665254555485e+20,4.679931429631719e+20,4.7866409298127035e+20,4.8363844888392506e+
20,4.9591649850134266e+20,4.975513397008882e+20,4.9721078764293725e+20,4.9157445976246786e+20,4.8739121776971625e+20,4.652601170
374249e+20
12.18014285,3.6963290784341184e+20,3.9799247612996924e+20,4.130475555907343e+20,4.327326124685856e+20,4.410350037914969e+20,4.591
875764085688e+20,4.706496403365703e+20,4.874945313153866e+20,4.943830576336589e+20,5.054127352408912e+20,5.0710114642708064e+20,
5.1340996896131285e+20,5.220777841216885e+20,5.318830048913443e+20,5.2766054201551395e+20,5.0919228912000434e+20,4.8677803274402
35e+20
10.18442855,3.910723817838532e+20,4.210428760973472e+20,4.3745442414849e+20,4.5526548810070655e+20,4.721562962109321e+20,4.843339
021033922e+20,4.9433905371605854e+20,5.1081756425050626e+20,5.1781323054206655e+20,5.331524222534255e+20,5.4592790746110866e+20,
5.430064299752034e+20,5.523534360376723e+20,5.548205252440081e+20,5.545536319176926e+20,5.500422737567466e+20,5.1994168090426e+2
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843450834865e+20,5.2248816848365755e+20,5.260936199061777e+20,5.441180071980658e+20,5.518751325854939e+20,5.650916135760856e+20,
5.649682112854235e+20,5.745381067566075e+20,5.8012564768495475e+20,5.824368099658805e+20,5.719016981282005e+20,5.432006211767879
e+20
6.193,4.218196409036849e+20,4.539798084212512e+20,4.7314829757074984e+20,4.939076240021938e+20,5.07469440085258e+20,5.19206050194
8098e+20,5.3647758785296925e+20,5.40500119885789e+20,5.6448990783324514e+20,5.709709196102616e+20,5.741698130984302e+20,5.8530950
049966345e+20,5.903355131751533e+20,6.004956351063253e+20,6.02093168636601e+20,5.918842597533068e+20,5.62882808233926e+20
4.19728575,4.183567239099138e+20,4.5251237342996804e+20,4.838508156166922e+20,5.001772256533472e+20,5.155680741375415e+20,5.31465
92428104476e+20,5.488541679815357e+20,5.592649209216491e+20,5.7249766422961475e+20,5.85252104085402e+20,5.972795226938783e+20,5.9
922252611373397e+20,6.053188471180916e+20,6.206408199051722e+20,6.186376850474494e+20,6.028641938015124e+20,5.738187383645313e+2
0
2.20157145,4.168864190979175e+20,4.5214407977179074e+20,4.835475712280111e+20,5.00836327810449e+20,5.205596489644746e+20,5.307130
7464731604e+20,5.448182434520549e+20,5.5981783971236735e+20,5.659171653345453e+20,5.811980040247385e+20,5.948908752536998e+20,6.0
41680490121505e+20,6.143616521849751e+20,6.193474873704 82e+20,6.168067394325107e+20,6.05577130982268e+20,5.776566452648102e+20
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+20,5.07549795065224e+20,5.1847137609226394e+20,5.2741086761347685e+20,5.3282239287142246e+20,5.4923394092256526e+20,5.650485662
6538945e+20,5.6609126779113826e+20,5.740961543667947e+20,5.849976466488432e+20,5.8281084326548387e+20,5.720432759500453e+20,5.403
021022565872e+20

C.76 "NBSR_lower_Cycle10_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,4.235310106555788e+20,4.636061436997922e+20,4.846352332782647e+20,5.0475650290450255e+20,5.162195234394084e+20,5.353
8418616128956e+20,5.506258039684042e+20,5.643531130459223e+20,5.826663956229260 6e+20,5.930781051699439e+20,5.978171357741267e+2
0,6.058564601983401e+20,6.242960148867189e+20,6.10472088511861e+20,6.094073850273121e+20,5.989430621005545e+20,5.769439731210644
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C.77 "NBSR_upper_Cycle10_cladding_fluence_position_6.csv" File

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C.78 "NBSR_lower_Cycle10_cladding_fluence_position_6.csv" File

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5516396e+20,3.3708963576312916e+20,3.4695099390803476e+20,3.4995355499887886e+20,3.468780640679446e+20,3.44931572289414e+20,3.399
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+20,2.7395953092972724e+20,2.8261443730135823e+20,2.8705580802808978e+20,2.961256795363579e+20,3.0192897262570353e+20,3.05873402
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37021165875226e+20
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472952613e+20

C.79 "NBSR_upper_Cycle10_cladding_fluence_position_7.csv" File

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24.15442855,1.3250899666305694e+20,1.4606603205346206e+20,1.479791683237344e+20,1.5367278397914602e+20,1.6145988178846378e+20,1.6 26239324919185e+20,1.6716028161504654e+20,1.6805579220964213e+20,1.713856895594956e+20,1.735690619814199e+20,1.7791884639734055 e+20,1.810259967937403e+20,1.808360400009473e+20,1.8291708450770633e+20,1.7961715058052556e+20,1.7903145046941378e+20,1.721392979 07094e+20
22.158714250000003,1.513034121976119e+20,1.5923750038230522e+20,1.664219376525833e+20,1.7298844999868652e+20,1.745742500099495e+ 20,1.8174851102347084e+20,1.853593861293306e+20,1.905170522622191e+20,1.9327086041011998e+20,1.964498099514147e+20,1.98463012746 46194e+20,2.048011246631355e+20,2.0493567739136387e+20,2.0753005751674205e+20,2.024888529889588e+20,2.037970673298011e+20,1.92053 66703249107e+20
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C.80 "NBSR_lower_Cycle10_cladding_fluence_position_7.csv" File

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074232399342882e+20,1.7483600594763747e+20,1.7734784532376633e+20,1.8189889348443193e+20,1.8514738078023126e+20,1.8845070679299
762e+20,1.9185353398294125e+20,1.961043825692822e+20,1.9881296290937527e+20,1.9643341487108437e+20,1.9492280609030196e+20,1.85149
07682302407e+20

C.81 "NBSR_upper_Cycle11_cladding_fluence_position_0.csv" File

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23751996545e+19,5.6921296374613926e+19,5.691651206729098e+19,5.8764881358433526e+19,5.981733328333267e+19,6.006439491348902e+19,
6.002210163675428e+19,6.023328096198859e+19,6.265988163618102e+19,6.101589795387423e+19,5.895041679641693e+19,5.551662374460034e
+19
24.15442855,5.573746737062699e+19,5.835381367324699e+19,6.12830536747869e+19,6.483616735123748e+19,6.6541581539570745e+19,6.84571
2250552728e+19,6.881039575825276e+19,6.957837276973022 4e+19,7.385755292550934e+19,7.2097980378280485e+19,7.3368788089398346e+19,
7.163543354629916e+19,7.006981681794206e+19,7.339959902855805e+19,7.150941489141306e+19,7.213414974164186e+19,6.604209985505637e
+19
22.158714250000003,5.9852545785229476e+19,6.590249376737314e+19,7.117891494154475e+19,7.256598132060993e+19,7.179609058620331e+1
9,7.698304521343604e+19,7.890049990232175e+19,7.88542834935822e+19,8.04991283512071e+19,7.983066493204668e+19,8.308380253934926e+
19,8.49590596376467e+19,8.411472508129521e+19,8.599505354535495e+19,8.604117426794804e+19,8.087498353449684e+19,7.92443959126942
8e+19
20.163,6.695618961217625e+19,7.259746206279484e+19,7.717757514918655e+19,7.891131243687158e+19,8.178974309464192e+19,8.4032913426
07303e+19,8.956625190348621e+19,8.896524721757918e+19,8.819028511741025e+19,8.86429762763063e+19,9.455245699544862e+19,9.17902850
0562559e+19,9.504514496356442e+19,9.311754754315407e+19,9.322806504231389e+19,8.795106975126351e+19,8.72638518473972e+19
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,9.00527202720822e+19,9.266485638425802e+19,9.735491285293077e+19,9.839884871079508e+19,9.969348227238119e+19,9.96121490478913e+1
9,1.0183302450719752e+20,1.0154596606782143e+20,1.0273725859123213e+20,9.778837109638865e+19,9.61980673422452e+19,9.494553568510
091e+19
16.171571450000002,7.704447571946252e+19,8.410860116792186e+19,8.814158086886277e+19,9.093437242555258e+19,9.211303437763074e+19
,9.72716659055117e+19,1.0194784788294795e+20,1.0473422846782505e+20,1.0680774726158487e+20,1.054891921633841e+20,1.0846885876410
776e+20,1.0879036421620895e+20,1.068938647933977e+20,1.1072992240492664e+20,1.1155282326447138e+20,1.0403284901428281e+20,1.01789
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7318427193e+20,1.0807750242509169e+20,1.0916641077179161e+20,1.1306753496291244e+20,1.1233362221957428e+20,1.1350864809808701e+
20,1.132550798099715e+20,1.1522908501141429e+20,1.1738010958380566e+20,1.1660600865895486e+20,1.1191164631369146e+20,1.087731407
098464e+20
12.18014285,8.790581020398856e+19,9.405441060313113e+19,9.57588679299998e+19,1.0498971047886974e+20,1.0503276924477614e+20,1.0906
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+20,1.1891682909593225e+20,1.2175105275403869e+20,1.2082768144071231e+20,1.2079323442798717e+20,1.2115110061574269e+20,1.1747962
317612271e+20
10.18442855,8.991761143328254e+19,9.410732504212278e+19,1.0327788531872376e+20,1.1039693461525037e+20,1.1280439806015105e+20,1.12
88764500757011e+20,1.1861541773458735e+20,1.2013204315595761e+20,1.2693437130770596e+20,1.2488094660470247e+20,1.26892269403264
16e+20,1.2734295115308458e+20,1.2943273659174242e+20,1.2762235470074397e+20,1.2669132849570 09e+20,1.2375567752234824e+20,1.198717
7683758998e+20
8.18871425,9.581245217201666e+19,1.0113929994537201e+20,1.0624319899747859e+20,1.108102987679519e+20,1.1467793280781222e+20,1.177
0831306615898e+20,1.20703289450316e+20,1.2702527314684173e+20,1.2770273106376927e+20,1.2913993698357882e+20,1.3037428827289592e
+20,1.3204879583592302e+20,1.3515668187290132e+20,1.3158184744120461e+20,1.3301905336101415e+20,1.2789601707961582e+20,1.2582154
142439139e+20
6.193,9.775105349926976e+19,1.0104935496770083e+20,1.0938266146278826e+20,1.1431049800541086e+20,1.1851111983494739e+20,1.2270695
735716099e+20,1.2727692771202811e+20,1.2650761109450021e+20,1.2848735746473055e+20,1.3223442696005288e+20,1.3283820654420722e+2
0,1.3422278508346448e+20,1.3785216061864265e+20,1.3789521938454905e+20,1.3551359119919224e+20,1.3644270368130612e+20,1.310670559
7325686e+20
4.19728575,9.831464490191143e+19,1.039381197292887e+20,1.1195757566399164e+20,1.1597735067672125e+20,1.1990143954299218e+20,1.228
1316897973017e+20,1.2928442306473144e+20,1.3255880299654785e+20,1.316641375271591e+20,1.3875926528707102e+20,1.3431273006113566
e+20,1.3848751663112836e+20,1.4068159996942613e+20,1.4397990143785723e+20,1.4055338053317149e+20,1.3658814662392334e+20,1.2943560
717613616e+20
2.20157145,9.634925145364991e+19,1.0497440069543636e+20,1.0906306973361622e+20,1.1150689391417118e+20,1.1788437557564296e+20,1.23
19974101142325e+20,1.2501203662535087e+20,1.265678933667692e+20,1.2913706639918504e+20,1.3217318782631931e+20,1.353825011785438
2e+20,1.3914775104169332e+20,1.3523323079006829e+20,1.3791531347530537e+20,1.3938600954637546e+20,1.3534518358142493e+20,1.319741
6064168524e+20
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20,1.1210397546807342e+20,1.1189537966879348e+20,1.147678777854834e+20,1.1994545517036316e+20,1.220476798080606e+20,1.2406952808
273276e+20,1.2681476362463261e+20,1.2817159318141687e+20,1.2714775141430886e+20,1.2882991386905264e+20,1.239422655079427e+20,1.20
24973711610182e+20

C.82 "NBSR_lower_Cycle11_cladding_fluence_position_0.csv " File

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−10.20585715,9.293574386488256e+19,1.0291714854657448e+20,1.0340706161644298e+20,1.1124758445726813e+20,1.1938282062918607e+20,1.188020057201818e+20,1.2458623327360968e+20,1.219615622762478e+20,1.3019344145608899e+20,1.29285379926196e+20,1.3076277402751821e+20,1.3236934442655962e+20,1.3331089610771315e+20,1.3461796886833888e+20,1.335711624260808e+20,1.3234733661287414e+20,1.2886149029738403e+20
−12.20157145,1.0458017377202654e+20,1.1218243810816955e+20,1.1561757076603657e+20,1.2525316571442677e+20,1.267793597504429e+20,1.2852371820038483e+20,1.3463519237470144e+20,1.3670201313820918e+20,1.3781197243713e+20,1.474246027103701e+20,1.4862642070989128e+20,1.484551425077302e+20,1.5020045781913672e+20,1.5093819800833324e+20,1.4743704190940974e+20,1.4733370087123436e+20,1.4165472807891101e+20
−14.197285749999999,1.0690343340804357e+20,1.1518889682990164e+20,1.1693421214130817e+20,1.2443600602366955e+20,1.31825847114674 27e+20,1.3331568041503608e+20,1.3971708361312253e+20,1.4109400726066309e+20,1.468983289048473e+20,1.5174483222298e+20,1.530834814119371e+20,1.4948568230509032e+20,1.5596172069741455e+20,1.5425755042898523e+20,1.5645354749021222e+20,1.5025978322994112e+20,1.4353113341096596e+20
−16.192999999999998,1.1120930999868467e+20,1.167772868611159e+20,1.2447428048225305e+20,1.2987002228105863e+20,1.336926838320833e+20,1.3719288306954225e+20,1.4137532453125164e+20,1.461385809019653e+20,1.4456358693125525e+20,1.4740068117375545e+20,1.536528139833663e+20,1.541905701264642e+20,1.5061956314062583e+20,1.5589378353342887e+20,1.539102097173402e+20,1.4971054474926824e+20,1.443923087290942e+20
−18.188714249999997,1.0965823756458928e+20,1.1942205194923413e+20,1.2230698926496367e+20,1.248130094407168e+20,1.298307909610105 6e+20,1.3279036347097788e+20,1.4319814562128971e+20,1.4633186691781186e+20,1.4647252555310614e+20,1.4849915813510123e+20,1.5159651869596906e+20,1.5306338732118078e+20,1.5043201829356677e+20,1.5480487518672896e+20,1.518692242133763e+20,1.4930866293414173e+20,1.4335124345562363e+20
−20.18442855,1.0687185697971221e+20,1.1551327286639659e+20,1.1784897170145323e+20,1.2857443185800795e+20,1.299972848558487e+20,1.3352140562992228e+20,1.3496435271851934e+20,1.4149780279871876e+20,1.4394067011780916e+20,1.4910102399632638e+20,1.469901876054 4764e+20,1.4794704907003455e+20,1.5504409055287568e+20,1.4855561296151183e+20,1.4844557389308435e+20,1.4858527566691405e+20,1.4198006097687059e+20
−22.180142850000003,1.0186460093552887e+20,1.0974722568079588e+20,1.16552424416938e+20,1.1927852272954609e+20,1.2254046346232286e+20,1.2785200145224481e+20,1.3325539814276712e+20,1.3513371719775122e+20,1.4104712104889834e+20,1.4202886091156452e+20,1.4578836960592649e+20,1.4694043080928915e+20,1.4540562502009173e+20,1.4286707155454267e+20,1.4552236211877133e+20,1.4140307351372467e+20,1.3641399783736851e+20
−24.17585715,9.953272954633059e+19,1.03566857481029e+20,1.1032804058980011e+20,1.1562331193482407e+20,1.202640900380706e+20,1.2171947632570732e+20,1.287677178738545e+20,1.3025181000542881e+20,1.296011442095097e+20,1.3520452494613065e+20,1.4081651743593288e+20,1.3985678538695223e+20,1.3921281762128522e+20,1.3894585327266547e+20,1.3787225470939895e+20,1.361432060428904e+20,1.3115795781239259e+20
−26.171571450000002,9.167718398051141e+19,1.0004560729134916e+20,1.020674555660213e+20,1.0952044951368888e+20,1.0984099810432537e+20,1.1765090137828375e+20,1.2123817500902008e+20,1.223500480308701e+20,1.2452786472426989e+20,1.2405708888369314e+20,1.3087855426473322e+20,1.2853807112235365e+20,1.3346208021911788e+20,1.3313004929090624e+20,1.3131583995404943e+20,1.2731520217061156e+20,1.253746871204293e+20
−28.167285749999998,8.789337100494892e+19,9.148973481959883e+19,9.733960306949738e+19,1.0294202694465375e+20,1.0602025027622984e+20,1.0981803342917 53e+20,1.1376413010913172e+20,1.149793441691571e+20,1.156816804841639e+20,1.190527034239036e+20,1.231251058171855e+20,1.2216537376820 48e+20,1.2333944278525295e+20,1.2507901692787196e+20,1.2295382761502441e+20,1.2077505406016001e+20,1.1613331909544893e+20
−30.163,8.158172141224074e+19,8.520305931111637e+19,8.997645841335463e+19,9.629662407309766e+19,9.972601556217715e+19,1.0065034373696809e+20,1.0458208749495571e+20,1.0637524587959157e+20,1.1013858201981192e+20,1.1268861782293602e+20,1.1475065427912081e+20,1.1417940798476244e+20,1.1266373942485677e+20,1.1530946137443957e+20,1.1437173713914443e+20,1.1326464842461736e+20,1.0784115764333873e+20
−32.15871425,7.539312420387842e+19,7.835307945843158e+19,8.350386472230293e+19,8.697277458986985e+19,9.101015585354788e+19,9.238315636908363e+19,9.312223616433055e+19,9.787927293552438e+19,9.84141584942285e+19,1.0042643815425475e+20,1.0380894343156947e+20,1.0288174467238478e+20,1.0420891152376681e+20,1.0402615098403072e+20,1.046710756111623e+20,1.0312670120731904e+20,9.861509940179173e+19
−34.15442855,6.603903789836969e+19,7.037103680699402e+19,7.645026474995404e+19,7.798143446558602e+19,7.885600584421845e+19,8.348893768345536e+19,8.413596740580903e+19,8.658859471183821e+19,8.947994299938048e+19,9.057009526598435e+19,9.1600539377198e+19,9.349072351434298e+19,9.189065977326074e+19,9.249463072970801e+19,9.023519375337893e+19,8.922149471779555e+19,8.590673523217359e+19
−36.15014285,5.6466978551228056e+19,5.874517001226304e+19,5.9844986579659235e+19,6.431879235733534e+19,6.5535824454143435e+19,6.930471039085836e+19,6.98121340255288e+19,7.072239633679032e+19,7.393180537516129e+19,7.265554355369527e+19,7.5582869832306e+19,7.575921940022937e+19,7.787302206164833e+19,7.675167611129892e+19,7.648853920853752e+19,7.497727221136895e+19,7.261669497823304e+19

C.83 "NBSR_upper_Cycle11_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0732983165151823e+20,1.1616008355875838e+20,1.2137736848326669e+20,1.2512831698795602e+20,1.3069621069776622e+20,1.
3418929209584643e+20,1.3749123515613302e+20,1.4052117365879025e+20,1.4188402393714111e+20,1.4494563033008369e+20,1.488368223488
9378e+20,1.4883738784693 46e+20,1.507640396719725e+20,1.540003849595252e+20,1.4985076033606517e+20,1.481717966529018e+20,1.4591998
345439845e+20
24.15442855,1.2633508980704344e+20,1.3798604594184133e+20,1.4326270816063465e+20,1.5149579413677998e+20,1.554084750811408e+20,1.5
977864394051656e+20,1.6270509630170563e+20,1.661970467037042e+20,1.668993952703896e+20,1.6967825264292744e+20,1.740178846080994
2e+20,1.756923243069363e+20,1.7701615522047132e+20,1.763352955793367e+20,1.787064288644509e+20,1.7505331152082158e+20,1.701153826
2847318e+20
22.158714250000003,1.4277637984553851e+20,1.54573799972906e+20,1.627689975803171e+20,1.7157945705612893e+20,1.7532361958432853e+
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3527e+20,1.9758558095687562e+20,2.0008508229725356e+20,1.9927528910281435e+20,2.0062060894190014e+20,1.9891280485865547e+20,1.911
7113667997362e+20
20.163,1.5699356608952982e+20,1.6858853791828754e+20,1.7617186664554276e+20,1.8308960417876522e+20,1.882339398560091e+20,1.925481
2440934472e+20,1.9619502127452507e+20,2.0313933721566557e+20,2.0685465934378392e+20,2.1064066872700345e+20,2.1437521778850926e+
20,2.1674013059517458e+20,2.1854293834927522e+20,2.2265184711379694e+20,2.230765361424449e+20,2.2271744488653082e+20,2.134596764
6043875e+20
18.167285749999998,1.676910925275229e+20,1.8144965986041768e+20,1.8960810009517615e+20,1.974911427840604e+20,2.029312339366477e+
20,2.069247811008443e+20,2.1129438446217917e+20,2.18899767613026e+20,2.2657696901505558e+20,2.2873830252702944e+20,2.32699616302
89992e+20,2.3872217043752095e+20,2.3768504703067632e+20,2.395890789340819e+20,2.4149593832769146e+20,2.3969426156967245e+20,2.297
2792409844607e+20
16.171571450000002,1.799041537148854e+20,1.9749227378014205e+20,2.0405148555549127e+20,2.1634088897836306e+20,2.2290179724783472
e+20,2.2334458221378857e+20,2.3288170667204054e+20,2.389421491753958e+20,2.4428214717475983e+20,2.4590343005776062e+20,2.52989686
0071443e+20,2.5483603711038728e+20,2.5849537493246545e+20,2.588714311296038e+20,2.6086650821757963e+20,2.61364146493492e+20,2.485
0754853568833e+20
14.17585715,1.8972515818962374e+20,2.0689198221447735e+20,2.190959954331869e+20,2.2571967398518843e+20,2.3214938670919228e+20,2.4
282372772751216e+20,2.470406466178285e+20,2.4885815732099023e+20,2.571506205914205e+20,2.61482901082062e+20,2.6429116435272185e
+20,2.6848715981552825e+20,2.7488350815512442e+20,2.766993223641637e+20,2.7766519301786626e+20,2.679872595474527e+20,2.596993202
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C.84 "NBSR_lower_Cycle11_cladding_fluence_position_1.csv " File

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C.85 "NBSR_upper_Cycle11_cladding_fluence_position_2.csv" File

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7322151241931e+20,1.4492131391432889e+20,1.4712675627348589e+20,1.4712505977936347e+20,1.4937970046807088e+20,1.554684178734666
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C.86 "NBSR_lower_Cycle11_cladding_fluence_position_2.csv " File

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e+20,2.8390263640799494e+20,2.8749241797105353e+20,2.9681861166008356e+20,3.047225777764778e+20,3.152442343237791e+20,3.204790496
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63131329597e+20
−30.163,1.9298808188509448e+20,2.1281105020763023e+20,2.246169528056099e+20,2.3147700953866977e+20,2.4099660355765666e+20,2.47311
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e+20,2.5714666210513486e+20,2.6354809992709826e+20,2.6481198804830742e+20,2.709436833048047e+20,2.680177964416564e+20,2.595947031
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−36.15014285,1.3658813478496028e+20,1.4642836619308619e+20,1.535881368877751e+20,1.60997292218461e+20,1.6531656625416385e+20,1.72
05730290061296e+20,1.7683406485133066e+20,1.8013544241357645e+20,1.8563660735457116e+20,1.8903638157591776e+20,1.92999391845910
68e+20,1.99006112035389e+20,1.9804419986797212e+20,2.0393103447279885e+20,2.0299400421917757e+20,2.0188506256115014e+20,1.9328214
086631542e+20

C.87 "NBSR_upper_Cycle11_cladding_fluence_position_3.csv" File

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C.88 "NBSR_lower_Cycle11_cladding_fluence_position_3.csv " File

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−28.167285749999998,2.1182538712249932e+20,2.284436780477677e+20,2.4549344397817378e+20,2.5097538198578092e+20,2.605221199107267e+20,2.6815747445773638e+20,2.7537944993691432e+20,2.8153319961700313e+20,2.8959267769461998e+20,2.916867169397375e+20,2.95071222 7139823e+20,3.0297745082253974e+20,3.0730351083473234e+20,3.0797701900133637e+20,3.074776842313016e+20,3.006379854277109e+20,2.90 09879844114445e+20
−30.163,1.9797464360895254e+20,2.162662432369762e+20,2.250037534655236e+20,2.342914932877786e+20,2.4151855824932386e+20,2.4604480 4567963e+20,2.5386846996256224e+20,2.6316978173779667e+20,2.6795898464541226e+20,2.7431292063194766e+20,2.7626162688057714e+20,2. 7812720491720763e+20,2.82857596028579e+20,2.852824516275701e+20,2.8328228505722693e+20,2.7945386332094667e+20,2.721792965239734 3e+20
−32.15871425,1.8229790692163194e+20,1.9689001836667994e+20,2.0643279780534003e+20,2.1770034626846904e+20,2.2050295455872087e+20, 2.270378499183153e+20,2.312694717576927e+20,2.4320091492073207e+20,2.4606120401114648e+20,2.5093127313859776e+20,2.5577476385813 103e+20,2.5920903345996703e+20,2.6103728862590412e+20,2.6537465859891285e+20,2.616983558356104e+20,2.5818887499434675e+20,2.481111 3440908086e+20
−34.15442855,1.6231037866922063e+20,1.7525858730963544e+20,1.8455933358682905e+20,1.914572786886232e+20,1.9591227225412033e+20,2. 0229731063290026e+20,2.0838433154417358e+20,2.1538293529723178e+20,2.1493053686458417e+20,2.1912087734698 25e+20,2.2520111228176 61e+20,2.2873943352311103e+20,2.3351393348166556e+20,2.3177785449638045e+20,2.3080067388186165e+20,2.313769163854465e+20,2.198469 7683138183e+20
−36.15014285,1.3553008795060573e+20,1.4864568401110018e+20,1.537261184097326e+20,1.617358326597582e+20,1.689917380213847e+20,1.72 71554262011517e+20,1.7381713280361205e+20,1.7901632179081 45e+20,1.8071338141128376e+20,1.8721660888059285e+20,1.893796388866891 e+20,1.909568129225068e+20,1.9491756120033647e+20,1.9502613682417186e+20,1.940755346175711e+20,1.9149460155931663e+20,1.828159031 2701343e+20

C.89 "NBSR_upper_Cycle11_cladding_fluence_position_4.csv" File

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70022941415002e+19,3.4378944442776044e+19,3.629151536659781e+19,3.6909095776965845e+19,3.740034392501705e+19,3.702971650907051e
+19,3.83156025040672e+19,3.755529038819885e+19,3.5684113920964338e+19,3.669448927047865e+19,3.52386711142187e+19,3.3383385141930
94e+19
24.15442855,3.4020079386078335e+19,3.6036815049017217e+19,3.8700310821229896e+19,3.913286027264508e+19,4.031039684302266e+19,4.22
07586220134425e+19,4.349844859789024e+19,4.294708800810099e+19,4.3768587011984925e+19,4.5436014535115784e+19,4.332568894642293e
+19,4.429563118601936e+19,4.60224925532393e+19,4.416307844525363e+19,4.395559721408062e+19,4.265710061277388e+19,4.1026995960336
45e+19
22.158714250000003,3.827986302788805e+19,3.972210923116856e+19,4.287210296788965e+19,4.3430362633776775e+19,4.6411724854728475e+
19,4.8628986122938384e+19,4.87165252196557e+19,4.867026747991748e+19,4.910208178387961e+19,5.043292487312065e+19,4.9138443307903
66e+19,5.072862379865993e+19,5.223426233231519e+19,5.018071274691962 5e+19,5.0401030783619e+19,4.814294055686263e+19,4.54426874119
9733e+19
20.163,4.271749580353233e+19,4.546106609832364e+19,4.780562097551977e+19,4.720992533933107e+19,5.056055778093135e+19,4.9435725627
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,5.6148752870406636e+19,5.915561905299883e+19,5.86353608554541e+19,6.00541954398451e+19,6.01610745695581e+19,6.067002280628664e+1
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+19,7.818971760860535e+19,7.829772773439996e+19,7.791658205489437e+19,7.869357636296661e+19,7.543970063614883e+19,7.081109917212
3165e+19

C.90 "NBSR_lower_Cycle11_cladding_fluence_position_4.csv " File

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,6.7464481766612615e+19,7.2198265866228834e+19,7.244199552181774e+19,7.406893338522662e+19,7.480577733240139e+19,7.5673251327003
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81e+19
−34.15442855,4.2025608950601925e+19,4.465551413919053e+19,4.646154523212378e+19,4.930933681583628e+19,4.908364654774922e+19,5.162
1714854510354e+19,5.32595102803028e+19,5.3474342986006315e+19,5.436562444812615e+19,5.526567112987852e+19,5.7218788263226335e+1
9,5.69778860978415e+19,5.69903270547393e+19,5.623091973573625e+19,5.476571431199886e+19,5.455687588552792e+19,4.992612552894719e+
19
−36.15014285,3.464846080902584e+19,3.6345068031062467e+19,3.8367684873625756e+19,4.0182028787758916e+19,4.1695810493201826e+19,4.
1820446261396 24e+19,4.308704877320133e+19,4.542775826371996e+19,4.470341182324708e+19,4.570660534764311e+19,4.578537922472788e+
19,4.6124395300193165e+19,4.552451497850247e+19,4.770886426073728e+19,4.578130763883405e+19,4.371921903302226e+19,4.162925137379
8556e+19

C.91 "NBSR_upper_Cycle11_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.9193014315123655e+20,2.0603236741631846e+20,2.1546510773421625e+20,2.22118165497489e+20,2.3042372301010344e+20,2.37
16194144372448e+20,2.3971484783124234e+20,2.4421018299187164e+20,2.511531697789143e+20,2.5588772030569035e+20,2.576320787556322
5e+20,2.6501809240077864e+20,2.580253488175775e+20,2.6063758061589977e+20,2.6490901019381573e+20,2.5575471656211277e+20,2.4666740
32329309e+20
24.15442855,2.2198133430805312e+20,2.4519766402332538e+20,2.5751629851841728e+20,2.5739286338948556e+20,2.7314184623512152e+20,2.
7793380844977278e+20,2.819497560166441e+20,2.9151262949372566e+20,2.932474193290217e+20,3.0109846764595734e+20,3.02479218739356
24e+20,3.086481046015481e+20,3.1160672025005084e+20,3.17267512674547e+20,3.091246216109123e+20,3.075209217962647e+20,2.9459085282
530175e+20
22.158714250000003,2.5637571965262968e+20,2.7324518727329694e+20,2.829860369827917e+20,2.957228199379081e+20,2.9847475351006e+20
,3.144208498174009e+20,3.1987878761140466e+20,3.230833166563062e+20,3.315716347086567e+20,3.3628226369881814e+20,3.4425100597589
79e+20,3.4144740188465824e+20,3.506858993252449e+20,3.517882037324490 3e+20,3.402876857895789e+20,3.44340950953569e+20,3.289613166
332636e+20
20.163,2.8116130216982446e+20,3.0174817658041185e+20,3.104747531374445e+20,3.221321963605068 3e+20,3.328777506078179e+20,3.3836343
73842946e+20,3.460881799879048e+20,3.568614832176888e+20,3.6567322044506964e+20,3.790501437199947e+20,3.7791721974592386e+20,3.74
5682046198696e+20,3.781286861295975e+20,3.875528146943139 5e+20,3.800931227163944 6e+20,3.7828274082539595e+20,3.594096052978838e
+20
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0,3.6932173320953954e+20,3.8197239863284315e+20,3.879173789123216e+20,3.9625164226887365e+20,4.0119670231785885e+20,4.1306656878
605946e+20,4.163380781334821e+20,4.139086068748959e+20,4.140674458780173e+20,4.2005261433900846e+20,4.10044800280894e+20,3.914883
858981599e+20
16.171571450000002,3.2076484132761213e+20,3.483004436940297e+20,3.585560848714723e+20,3.7952474700642976e+20,3.875480303869911e+
20,3.9769650308039986e+20,4.0750146250802194e+20,4.172011671745394e+20,4.192612899077951e+20,4.271716636355351e+20,4.33749129343
1055e+20,4.4754994224684256e+20,4.4820347862715544e+20,4.5030187581899447e+20,4.609823634867136e+20,4.442736485920969e+20,4.24327
87136278286e+20
14.17585715,3.3676547873843446e+20,3.647670726381059e+20,3.820288534592538e+20,3.997652375668368e+20,4.093032326458391e+20,4.2194
05020086384e+20,4.377057514991724e+20,4.428067799668852e+20,4.48892418881658e+20,4.5831080627758694e+20,4.714580828010111e+20,4.6
64919717998051e+20,4.82558632651684e+20,4.8315092989826315e+20,4.878806961177163e+20,4.7668541698204946e+20,4.548861990958304e+
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12.18014285,3.53247417465944e+20,3.819417790659763e+20,4.0807462252530945e+20,4.2246868953709045e+20,4.354408604124952e+20,4.4820
347862715544e+20,4.582648769272868e+20,4.693797796999284e+20,4.7935984477556985e+20,4.8319781611002796e+20,4.90399155492509e+20,
5.007074240505039e+20,5.0856612725915615e+20,5.1213139307620696e+20,5.007744043530249e+20,4.998701702689904e+20,4.83093518210388
e+20
10.18442855,3.733625591744901e+20,3.979873889656343e+20,4.263640725594237e+20,4.495698767985855e+20,4.631927134699094e+20,4.71121
2675654765e+20,4.797827775429173e+20,4.86440619613513e+20,5.0126910173021635e+20,5.105363050147407e+20,5.148450521897755e+20,5.32
7220949326528e+20,5.2397446742339925e+20,5.381360170992855e+20,5.321049192879942e+20,5.2989552616626304e+20,5.0398084712085566e
+20
8.18871425,3.920969497896372e+20,4.211740559755044e+20,4.48600576134959e+20,4.569826825647403e+20,4.738885109210619e+20,4.8234046
82377581e+20,5.0174561873958063e+20,5.1372743799913795e+20,5.214722746935044 3e+20,5.345678806978409e+20,5.442006050618373e+20,5.4
35202765605161e+20,5.525205154964205e+20,5.578588456073509e+20,5.576693870373627e+20,5.448503139962918e+20,5.1886578406396966e+
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6.193,4.018425838064549e+20,4.372694226713208e+20,4.541408040149172e+20,4.7625769990737913e+20,4.863219687919042e+20,5.1028847789
541255e+20,5.1838543960874707e+20,5.32014974310323e+20,5.42693548255113e+20,5.5126032894755956e+20,5.640631353437324e+20,5.683173
414152859e+20,5.6391195123232775e+20,5.820100289735246e+20,5.8279561223595046e+20,5.6819677687074796e+20,5.381934287871607e+20
4.19728575,4.104007527457203e+20,4.41241354610821e+20,4.6175837813449366e+20,4.849699235424429e+20,5.001849776908394e+20,5.120873
77448836e+20,5.2570734353576604e+20,5.415625380039712e+20,5.494355941345924e+20,5.629608309365284e+20,5.726452258196125e+20,5.794
083226513128e+20,5.782753986772419e+20,5.8624127036992795e+20,5.879664915905781e+20,5.75342618288283e+20,5.461994886613594e+20
2.20157145,4.0430554521630166e+20,4.392616082405907e+20,4.629783765018419 5e+20,4.7801162697196687e+20,4.998462487323756e+20,5.143
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0,5.815277707953728e+20,5.926962578100313e+20,5.907366055305573e+20,5.8826885981338757e+20,5.736614126950038e+20,5.4472496514443
103e+20
0.20585714999999993,3.820317240436475e+20,4.1303499235772806e+20,4.3549157407011825e+20,4.508740789748175e+20,4.627898747933183e
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937e+20,5.369150618704726e+20,5.4486083947240233e+20,5.486624500712061e+20,5.48672018685852e+20,5.322245269710676e+20,5.035713104
140125e+20

C.92 "NBSR_lower_Cycle11_cladding_fluence_position_5.csv " File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,3.995891750573527e+20,4.3376922343386186e+20,4.646031272687105e+20,4.8140944203271504e+20,4.99122861465148e+20,5.104865482185821e+20,5.2411416919722885e+20,5.485361443578806e+20,5.4836486615571956e+20,5.5979649007313945e+20,5.653233218925935e+20,5.766449067415858e+20,5.841400025936951e+20,5.8291139247316546e+20,5.8868700827341206e+20,5.755062415987274e+20,5.376939471026464e+20
−12.20157145,4.379258296360274e+20,4.7705667923030914e+20,5.050506182382639e+20,5.2997207508343e+20,5.378326920150114e+20,5.5449069325200504e+20,5.77153957040746e+20,5.869847517279119e+20,5.961266061605753e+20,6.047661083243306e+20,6.250659242955419e+20,6.275107053375615e+20,6.349072444588183e+20,6.38894486181752e+20,6.379414521630234e+20,6.21890101094578e+20,6.000439969965942e+20
−14.197285749999999,4.499258292634119e+20,4.912565033647789e+20,5.1294376845964136e+20,5.2929078972064406e+20,5.522171904121465e+20,5.702715252259724e+20,5.8853391043907813e+20,5.936722565039098e+20,6.044359911190481e+20,6.209619454739287e+20,6.332403917875079e+20,6.487463318211388e+20,6.440328322465837e+20,6.600736578389187e+20,6.490592255200587e+20,6.422042699877581e+20,6.005032904995958e+20
−16.192999999999998,4.515228310478074e+20,4.915789656783447e+20,5.1244045932926855e+20,5.2488348581475674e+20,5.5359602778261625e+20,5.601811483819034e+20,5.8297932963715115e+20,5.975877336169995e+20,6.041527601255303e+20,6.267595690878607e+20,6.339455986869084e+20,6.458575670595509e+20,6.52817777352956e+20,6.579848292617254e+20,6.601119322975021e+20,6.395317559171668e+20,6.090413653481048e+20
−18.188714249999997,4.444879855601644e+20,4.814716380279132e+20,5.040717489599914e+20,5.284927672591786e+20,5.4295381457348054e+20,5.617513580452905e+20,5.771587413480689e+20,5.947362864525305e+20,6.007195411905924e+20,6.130247796251802e+20,6.278599597721357e+20,6.361061918739456e+20,6.441840163579884e+20,6.456499281217356e+20,6.429735866052858e+20,6.326777572463308e+20,6.065219491118475e+20
−20.18442855,4.407256062814087e+20,4.7619167646632254e+20,5.004280205028445e+20,5.167358104437993e+20,5.395043289936448e+20,5.4701664835211664e+20,5.64008594240251e+20,5.775462702412267e+20,5.931545944515683e+20,6.01357767787472e+20,6.14817938009816e+20,6.184099959478754e+20,6.238755886335957e+20,6.294866242619334e+20,6.258792565404408e+20,6.209782121188266e+20,5.7848590819945107e+20
−22.180142850000003,4.218323766631402e+20,4.585241863841899e+20,4.791292411626044e+20,5.0081841998039594e+20,5.217851683924243e+20,5.324838364279706e+20,5.520688768851355e+20,5.603753912592145e+20,5.68456086327651e+20,5.812368849101383e+20,5.953993914474892e+20,5.933411824371627e+20,6.125377371397055e+20,6.152045100415091e+20,6.156475368996129e+20,6.118363576861631e+20,5.703525857504623e+20
−24.17585715,4.051427989978152e+20,4.473834483520044e+20,4.687396393801197e+20,4.747305490098984e+20,4.921215061287655e+20,5.0496354384498636e+20,5.312982850733474e+20,5.397110110699955e+20,5.5052067503543386e+20,5.596950627578933e+20,5.630191994858681e+20,5.684627843579031e+20,5.875894881735309e+20,5.831037216275475e+20,5.884343968467612e+20,5.760325154042501e+20,5.3873118493025855e+20
−26.171571450000002,3.9140322522781175e+20,4.176413234482495e+20,4.352389626434673e+20,4.573376781681021e+20,4.666230618204534e+20,4.880950330857838e+20,4.970962288831529e+20,5.152813810176271e+20,5.22001419083421e+20,5.2886498636890296e+20,5.372882378416615e+20,5.4722907159725493e+20,5.548858770368794e+20,5.485753756779287e+20,5.5257505659990206e+20,5.376164413240148e+20,5.108941712024962e+20
−28.167285749999998,3.681495779154206e+20,3.962631246064486e+20,4.164529015092325e+20,4.280452781527029e+20,4.4137722893879234e+20,4.557119705397689e+20,4.6696561822477556e+20,4.7609694718132847e+20,4.8918681201687744e+20,5.0160304638135724e+20,5.0602661693214255e+20,5.137379634752485e+20,5.221516463333611e+20,5.220262974815002e+20,5.2294871193336204e+20,5.064782555434275e+20,4.855373423909429 6e+20
−30.163,3.398800628056649e+20,3.65885643690208e+20,3.8980335285902246e+20,3.976534443144934e+20,4.175925235135555e+20,4.2163047889411224e+20,4.3585805201105505e+20,4.507181105560898e+20,4.5363845174600906e+20,4.644318490665495e+20,4.685855846843212e+20,4.756682732451936e+20,4.8326671013547816e+20,4.866492154127929e+20,4.8382934467665533e+20,4.6858462782285663e+20,4.550536498521331e+20
−32.15871425,3.076788039379215e+20,3.351235044652033e+20,3.528484062352113e+20,3.626916401214169e+20,3.791075554078699e+20,3.878102104282879e+20,3.947378874318972e+20,4.12015934897943e+20,4.214467614929117e+20,4.229940064811486 4e+20,4.303532280052865 4e+20,4.4104137056472236e+20,4.391850593234238e+20,4.3907502025499627e+20,4.451472631092648e+20,4.3664650585787466e+20,4.11285849600463 2e+20
−34.15442855,2.7582010147450028e+20,3.000803670476369e+20,3.142438304464523e+20,3.2464778515090584e+20,3.3746016016172456e+20,3.489587643816655e+20,3.594698875701527e+20,3.6332029810365045e+20,3.665200428412291e+20,3.760101948470021e+20,3.87492532422045e+20,3.9411018631112806e+20,3.9450537009600247e+20,3.925543295697098e+20,3.966736181747565e+20,3.918271148566238e+20,3.695236309785674e+20
−36.15014285,2.3414400038441737e+20,2.5385821713930145e+20,2.640851524728064e+20,2.7802088284305018e+20,2.9079976770260838e+20,2.940243908382663e+20,3.0345043312591195e+20,3.085935634980666e+20,3.080548504935042e+20,3.19954379667107e+20,3.2328999873265703e+20,3.289670578020511e+20,3.3077456910865583e+20,3.33919772742753e+20,3.302473384416684e+20,3.291699124325435e+20,3.084337676334806e+20

C.93 "NBSR_upper_Cycle11_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.134015841156897e+20,1.2465273313563522e+20,1.2648042280353151e+20,1.3179440789301826e+20,1.3400720172670578e+20,1.4074228339274677e+20,1.4565702686542204e+20,1.4478898737277947e+20,1.476758548711119e+20,1.4999835532471645e+20,1.5412366353242168e+20,1.583739468071458e+20,1.560446603770515e+20,1.61332632556661e+20,1.632626773699438e+20,1.5908817083268815e+20,1.5338455759308367e+20
24.15442855,1.3415479671535706e+20,1.4297826264610749e+20,1.5219192222501647e+20,1.5806122639057814e+20,1.6088249611617665e+20,1.6588828477342224e+20,1.683968340824531e+20,1.7111914165091e+20,1.7140358716543717e+20,1.783388551379247e+20,1.8454067215148237e+20,1.8445188895907525e+20,1.8580003628836507e+20,1.8753046029324213e+20,1.8841263723690492e+20,1.868546901344748e+20,1.7755450935532195e+20
22.158714250000003,1.5117063276331454e+20,1.586436893726119e+20,1.6744396988368912e+20,1.741536041378937e+20,1.79023673265345e+20,1.8421268328781287e+20,1.8744393909299826e+20,1.917366347207831e+20,1.97346940783654e+20,2.0005963488541714e+20,2.031472541882369e+20,2.0272030316742574e+20,2.0796699399005615e+20,2.121759959078012e+20,2.0955095400236355e+20,2.082831073948687e+20,2.009898791625487e+20
20.163,1.6638875053953863e+20,1.7696299840463523e+20,1.844711158924628e+20,1.9141882482184815e+20,1.9567419757893956e+20,2.0206884942441323e+20,2.079251471350363e+20,2.131022816986471e+20,2.178507687473244e+20,2.1572053762759505e+20,2.2384900646619064e+20,2.2435682370683755e+20,2.3265267996551273e+20,2.348213649520171e+20,2.310675889571237e+20,2.342869693034521e+20,2.2319981471534134e+20
18.167285749999998,1.7446519355837972e+20,1.8892893694816397e+20,1.9993748730860228e+20,2.051604272135187e+20,2.1579970735330833e+20,2.1941719832036665e+20,2.263258878849362e+20,2.3020690093901165e+20,2.3804979326699848e+20,2.4064260178410996e+20,2.40536288 1524378e+20,2.4901140729004956e+20,2.540143684570911e+20,2.5731461502325522e+20,2.5620567336522783e+20,2.5210468157327743e+20,2.4000811298232173e+20
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C.94 "NBSR_lower_Cycle11_cladding_fluence_position_6.csv " File

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−12.20157145,2.5602527949020958e+20,2.808314165483586e+20,2.936456021531016e+20,3.072130311482028e+20,3.208183485120382e+20,3.299833752594376e+20,3.381061891176251e+20,3.4646877414511575e+20,3.593270685970419e+20,3.628354184422239e+20,3.697118746184673e+20,3.784007520155051e+20,3.8587663611500646e+20,3.8822628047456993e+20,3.949574036543253e+20,3.871518341970319e+20,3.6997539670548454e+20
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−16.192999999999998,2.6344687577779336e+20,2.868092963377557e+20,3.0227566775389513e+20,3.138734670728569e+20,3.234077640409049e+20,3.311302052861992e+20,3.404965493361268e+20,3.504363083994351e+20,3.6331778827103443e+20,3.663341548207123e+20,3.806514342179268e+20,3.8841854980844506e+20,3.9053068499086855e+20,3.980071345884108e+20,3.998868500760615e+20,3.9453893510412626e+20,3.808719784538425e+20
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−28.167285749999998,2.140014235835342e+20,2.3288566515832622e+20,2.4380430133027583e+20,2.514634067949995e+20,2.604169372751361e+20,2.6480972605614418e+20,2.7976432174535108e+20,2.8539328924356873e+20,2.8657800763906458e+20,2.970035295194283e+20,3.020415515 65e+20,3.05392692954837e+20,3.11649928776394e+20,3.1858576224692234e+20,3.187644596278181e+20,3.1665232444539476e+20,3.051546182796562e+20
−30.163,2.016814832664587e+20,2.13500392319377e+20,2.2663860830150377e+20,2.3257011725155454e+20,2.4169895212434204e+20,2.4952261751894134e+20,2.5715344808162453e+20,2.6086707371562045e+20,2.6605438724396587e+20,2.7219965445344254e+20,2.8173112393128647e+20,2.8167683611936876e+20,2.8777120850517262e+20,2.9035949303795764e+20,2.971799649081609e+20,2.9338716954845164e+20,2.840615413574624e+20
−32.15871425,1.8011169149586244e+20,1.9501426136531485e+20,2.056297905873906e+20,2.1257919601089836e+20,2.1994650448656436e+20,2.2430819087532792e+20,2.328545627660817e+20,2.4137379075087663e+20,2.464389567024072e+20,2.495005630953498e+20,2.5696004775166768e+20,2.6071099625635702e+20,2.6024672236485244e+20,2.6612563999710786e+20,2.712049433996587e+20,2.6764513323276295e+20,2.5851290537173054e+20
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−36.15014285,1.4268929314725385e+20,1.4568473626942169e+20,1.5832078999130967e+20,1.6323270597378087e+20,1.665968538185565e+20,1.721839744617542 5e+20,1.8051715359112285e+20,1.819789660266154e+20,1.8712160520973684e+20,1.9100657675009802e+20,1.9521388217372063e+20,1.9658860791092848e+20,2.0091184043291702e+20,2.0234198497812418e+20,2.0286733265803624e+20,2.031941905256241e+20,1.901764256261897e+20

C.95 "NBSR_upper_Cycle11_cladding_fluence_position_7.csv" File

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18.167285749999998,1.7001868246349473e+20,1.8280233117403398e+20,1.9207989203155442e+20,1.979005633656065e+20,2.0610593993775215e+20,2.1034321675753767e+20,2.1627585670366996e+20,2.2038137247994685e+20,2.2439923605989825e+20,2.2926478120302307e+20,2.3144986563271095e+20,2.398656074760377e+20,2.3995665266060807e+20,2.4164692630458758e+20,2.3958794793800027e+20,2.3883753203784607e+20,2.2813208862728166e+20
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6.193,2.282293542903009e+20,2.4428440916692307e+20,2.570149010616262e+20,2.6801836193969724e+20,2.773089292521563e+20,2.8419726088725666e+20,2.926056512560529e+20,3.0026362572469495e+20,3.1034080081192006e+20,3.1333171994976145e+20,3.178805861900329e+20,3.19242230547230 22e+20,3.250952101946804e+20,3.284548340551295e+20,3.283270314979066e+20,3.284848054512924e+20,3.149835397269659e+20
4.19728575,2.3130227064405964e+20,2.4760840665080123e+20,2.6196187792262762e+20,2.7475344360573823e+20,2.828649475031095e+20,2.857422015347482e+20,3.00643640408119e+20,3.076603400984831e+20,3.1271532708527905e+20,3.216807330242727e+20,3.238058746616348e+20,3.316193610914994e+20,3.353759645765969e+20,3.379371052034231e+20,3.379359742073414e+20,3.3318126668021524e+20,3.157339556271201e+20
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C.96 "NBSR_lower_Cycle11_cladding_fluence_position_7.csv " File

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65661186673715e+20,2.930778421201289e+20,2.9970661015449764e+20,3.089841710120181e+20,3.2077537066093674e+20,3.229022087924212e
+20,3.2595759470691484e+20,3.295835681445853e+20,3.32273642324716e+20,3.330670360759717e+20,3.249102923353357e+20,3.107355184444
0506e+20
−12.20157145,2.444908159518185e+20,2.6396317548905238e+20,2.7917790027703163e+20,2.9174665973206335e+20,3.017067767248408e+20,3.1
08898994095461e+20,3.1460691803178684e+20,3.221229524921857e+20,3.306325670102869e+20,3.430452490060551e+20,3.491379248977365e+
20,3.608544788052683e+20,3.581231232681585e+20,3.621047949734981e+20,3.639053407354355e+20,3.551666995108065e+20,3.3905679132422
58e+20
−14.197285749999999,2.5156858943059005e+20,2.6812467557136944e+20,2.862749006891907e+20,2.970849612373049e+20,3.103102639177163e
+20,3.1626043430311376e+20,3.2855323071423034e+20,3.372483285897171e+20,3.429943541823823e+20,3.508412049966547e+20,3.6039585989
41718e+20,3.6418526226563616e+20,3.7052279880898806e+20,3.685175427562777e+20,3.703605008712758e+20,3.6801198750779395e+20,3.4879
240559480196e+20
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26924e+20,3.624412663077797e+20,3.706189334759257e+20,3.7346225762511585e+20,3.697039576458959e+20,3.637978961076817e+20,3.437102
747020471e+20
−18.188714249999997,2.526571731591483e+20,2.732429983387361e+20,2.8366003774848762e+20,2.9342166492894105e+20,3.043091987086461e
+20,3.14757906008683e+20,3.222954293946326e+20,3.33285883817765e+20,3.388051446960656e+20,3.476789399524481e+20,3.54598939477833
74e+20,3.595029384877337e+20,3.6128878130061e+20,3.634297568831147e+20,3.657222859405563e+20,3.644086339917559e+20,3.462029900659
353e+20
−20.18442855,2.4793865750663397e+20,2.6603459481253754e+20,2.764391932653913e+20,2.9018249215118426e+20,2.991428086078106e+20,3.0
97905712182124e+20,3.149117214757831e+20,3.246117093697883e+20,3.313411360554212e+20,3.3699102698114875e+20,3.432737102145422e+
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391784e+20,3.396330338278107e+20,3.4446634558260937e+20,3.46197900583568e+20,3.438290292906171e+20,3.436672968509456e+20,3.292261
7338279376e+20
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19e+20,3.288619926445123e+20,3.2697831867057596e+20,3.2922617338279376e+20,3.304555661235135e+20,3.240507353133053e+20,3.09839204
049722e+20
−26.171571450000002,2.1885452776976122e+20,2.3091264249394187e+20,2.4540013680144017e+20,2.5143286990079574e+20,2.60810523911539
52e+20,2.724043647442156e+20,2.76158140739109e+20,2.8261047338474527e+20,2.924976411302583e+20,3.0164287544622934e+20,3.016129040
5006646e+20,3.078509129382359e+20,3.103809511728175e+20,3.113677452540301e+20,3.18254380395008e+20,3.096757751159281e+20,2.998649
496059243e+20
−28.167285749999998,2.0427202979140698e+20,2.241752988357377e+20,2.305773021557419e+20,2.416090379358534e+20,2.450105086513224e+
20,2.579083879661052e+20,2.608472812841921e+20,2.701033532161618e+20,2.7442828223227278e+20,2.8165817468402208e+20,2.88659040429
24353e+20,2.9176249367720598e+20,2.973564002968934e+20,2.987090716105097e+20,3.034609516474318e+20,2.9333684022281958e+20,2.80077
60765995955e+20
−30.163,1.895509847930544e+20,2.0299909370154484e+20,2.1340256115831695e+20,2.2297870498138486e+20,2.3330583020264764e+20,2.39391
7201178394e+20,2.4157567355144566e+20,2.4997049196726246e+20,2.5585393358384434e+20,2.6001656466224297e+20,2.7020570836154837e+
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63143e+20
−32.15871425,1.7744989221777218e+20,1.89026768109224e+20,1.993307079108137e+20,2.0506259605245867e+20,2.1180502919303017e+20,2.17
5250418758181e+20,2.2263262018040942e+20,2.2714699104019156e+20,2.3230094018412924e+20,2.359303066100445e+20,2.4530909161686984
e+20,2.457965509280476e+20,2.518400284901786e+20,2.5117443729614586e+20,2.5792818039753353e+20,2.5445545692892247e+20,2.411572050
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−34.15442855,1.5698791110912174e+20,1.705904009827531e+20,1.807524007760996e+20,1.8480984921890767e+20,1.8880169988898187e+20,1.9
936294129913984e+20,2.0045322152182048e+20,2.059538209647744e+20,2.1230323296698332e+20,2.1436108033748907e+20,2.221655187987000
77e+20,2.2454117606814145e+20,2.2721880929137436e+20,2.2667140718787078e+20,2.2862916140515318e+20,2.3174392461393183e+20,2.16707
33170880764e+20
−36.15014285,1.3342078025838635e+20,1.427876898063547e+20,1.5304752076076122e+20,1.577626434250307e+20,1.6084573874352403e+20,1.6
639214352777835e+20,1.6861850931445046e+20,1.7573190916979297e+20,1.7562389904399837e+20,1.8150790615862107e+20,1.86026235504688
9e+20,1.8646619298043865e+20,1.890663529720807e+20,1.9323520452892823e+20,1.920499206353915e+20,1.8771198516434197e+20,1.80548821
4814082e+20

C.97 "NBSR_upper_Cycle12_cladding_fluence_position_0.csv" File

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24.15442855,5.340155732382319e+19,5.6895523298303074e+19,5.735690561651335e+19,6.154466807281649e+19,6.6306382475175395e+19,6.743246164352096e+19,6.8142722390641345e+19,6.6700663340052595e+19,7.0491273257298805e+19,7.176466931106463e+19,7.049663371576763e+19,7.396130861361426e+19,7.491853334019159e+19,7.279234577751804e+19,7.289783194238686e+19,7.240342537110967e+19,6.6585509205445345e+19
22.158714250000003,6.063817625674776e+19,6.483876552438704e+19,6.893788897100912e+19,6.819154085169678e+19,7.290022500420331e+19,7.364753034824222e+19,7.774119761392281e+19,7.891236206689016e+19,7.790191564551514e+19,8.046909663972286e+19,7.916861112619491e+19,8.317019337317877e+19,8.288178156306103e+19,8.369025356712822e+19,8.29810457672071e+19,7.80983381594088e+19,7.602077761284538e+19
20.163,6.9248508394783465e+19,6.832966837974189e+19,7.508891934152235e+19,7.847347452975446e+19,8.087247113950256e+19,7.973921278570766e+19,8.511919863896285e+19,8.785274529064973e+19,8.840908430173648e+19,9.06220921471106e+19,8.775941587980845e+19,9.083076713750444e+19,8.847388841572576e+19,9.152925402048794e+19,9.267543490809163e+19,8.85933500616026e+19,8.535333580708368e+19
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16.171571450000002,7.64109424113983e+19,8.172219952928526e+19,8.422783097357405e+19,8.826530914780457e+19,9.204864415712877e+19,9.439432334960651e+19,9.884302526637464e+19,9.929579256204571e+19,1.037459303159037e+20,1.0501712475279838e+20,1.028279518031160 6e+20,1.0596477723210994e+20,1.0875891620898916e+20,1.0606720027785373e+20,1.0363297779816758e+20,1.0378996265332626e+20,9.755555800912812e+19
14.17585715,8.127929164829793e+19,8.636914268692754e+19,9.198776466451846e+19,9.450631864261606e+19,9.860084741055055e+19,1.032060 5557011408e+20,1.0185541148091348e+20,1.0124374488063056e+20,1.0893504555867939e+20,1.1256962784549349e+20,1.0940791457360858e+20,1.1532547783330962e+20,1.14624789333455e+20,1.1413947639708031e+20,1.1282712129694278e+20,1.064012717074292e+20,1.0066653837050444e+20
12.18014285,8.563380265197083e+19,8.96611342440996e+19,9.37063659386154e+19,9.809543275491775e+19,1.0320701279484068e+20,1.0538948517143698e+20,1.0772702795373878e+20,1.1166696492833107e+20,1.160309524567971e+20,1.1699296330700731e+20,1.2087450957327835e+20,1.1905195369387514e+20,1.1958129896767239e+20,1.2200212030118647e+20,1.1885763707437995e+20,1.181043012145636e+20,1.1209005825747825e+20
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C.98 "NBSR_lower_Cycle12_cladding_fluence_position_0.csv" File

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18440825232276e+20,1.1379678994496563e+20,1.17081985206579e+20,1.2039398276053656e+20,1.2291722713979437e+20,1.2674708327083026
e+20,1.283408624405815e+20,1.2840595372198878e+20,1.3227792774099404e+20,1.3067840522288333e+20,1.2334032046894154e+20,1.26130630
54691446e+20
−12.20157145,9.741676042377444e+19,1.0789454228088983e+20,1.1368479465195607e+20,1.1927498705516765e+20,1.2483454826712873e+20,1.
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61e+20,1.4187219117547859e+20,1.4650228717793311e+20,1.4783953012096162e+20,1.4147590013867557e+20,1.391823896937963e+20,1.369999
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C.99 "NBSR_upper_Cycle12_cladding_fluence_position_1.csv" File

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C.100 "NBSR_lower_Cycle12_cladding_fluence_position_1.csv" File

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C.101 "NBSR_upper_Cycle12_cladding_fluence_position_2.csv" File


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C.102 "NBSR_lower_Cycle12_cladding_fluence_position_2.csv" File

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−36.15014285,1.321420070924718e+20,1.460427001946668e+20,1.506645731657287e+20,1.5823098087539248e+20,1.6473271723480672e+20,1.6897216840312226e+20,1.7130290483406166e+20,1.7809089183280975e+20,1.8276141609830387e+20,1.8848133747044673e+20,1.88755708142535 44e+20,1.936649631783831e+20,1.9178566550275667e+20,1.9654783522985574e+20,1.9882313181365145e+20,1.9464817189609492e+20,1.8946341476270663e+20

C.103 "NBSR_upper_Cycle12_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0881767140129895e+20,1.1998257776076992e+20,1.2377059017377241e+20,1.2924612364830219e+20,1.3285254227627277e+20,1.3286442224351784e+20,1.3761358057791986e+20,1.3971350621666918e+20,1.4436309911130292e+20,1.4278023490407758e+20,1.483389281493229e+20,1.4674192112394863e+20,1.4952296488474916e+20,1.517151016978293e+20,1.497871527277707e+20,1.5035795686826046e+20,1.4216473945823727e+20
24.15442855,1.281910694143939e+20,1.4125733624580357e+20,1.4803740326638826e+20,1.5353669667540818e+20,1.581517810937586e+20,1.6002146165304377e+20,1.6563106904362513e+20,1.66342735652878e+20,1.722046509192358e+20,1.7516049991235682e+20,1.7444487331402223e+20,1.753890478536431e+20,1.793824139861677e+20,1.78758432849438e+20,1.774244822416334e+20,1.7670828993057286e+20,1.6873174049459087e+20
22.158714250000003,1.4219189366908317e+20,1.560631697095284e+20,1.6055549446634635e+20,1.6911472801006322e+20,1.7370605249392345e+20,1.803526113111825e+20,1.8602840709070075e+20,1.8582192194572678e+20,1.8914322135981546e+20,1.9318014737223864e+20,1.948393827974681e+20,1.9840507010916984e+20,2.013445134323808e+20,2.0101470291400566e+20,1.9968754085891247e+20,1.9802038545551963e+20,1.8945436335909133e+20
20.163,1.589635788555057e+20,1.6995990282264173e+20,1.812085346655542e+20,1.839499785355378e+20,1.900404417431819e+20,1.922421956726033e+20,2.0141579323670856e+20,2.0734276546655132e+20,2.0831975134427765e+20,2.1296255568619995e+20,2.1437287751200866e+20,2.1699382457136362e+20,2.1861176296759827e+20,2.207099914681697e+20,2.197024571032417e+20,2.1595913599558975e+20,2.083774540423252e+20
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16.171571450000002,1.8264431356403096e+20,1.9479695434302138e+20,2.064670421667712e+20,2.1483619623456676e+20,2.2312671623345445e+20,2.2885512329648662e+20,2.3314039719560 46e+20,2.4089801580664157e+20,2.426059025263032e+20,2.4743878634414676e+20,2.5164825473798662e+20,2.590285429608108e+20,2.608484408002118e+20,2.5773079796746735e+20,2.6097459473810004e+20,2.5532482174397576e+20,2.4609352148182303e+20
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12.18014285,2.072895884703115e+20,2.2382310859910644e+20,2.317046182971278e+20,2.4094270711199203e+20,2.4986569393849873e+20,2.5320226759618822e+20,2.6182429525248616e+20,2.659223182393127e+20,2.7312327552800876e+20,2.7661768303624e+20,2.8352786398379462e+20,2.886288956337414e+20,2.933316655246151e+20,2.8995436055065674e+20,2.9051102187299756e+20,2.828275116290609e+20,2.7375857091925754e+20
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4.19728575,2.2847666148282198e+20,2.5012931606879434e+20,2.616653299764925e+20,2.723725747406649e+20,2.8541395021213254e+20,2.955656650794159e+20,3.033945634939234e+20,3.075208721170477e+20,3.1694451470602555e+20,3.2151038211721776e+20,3.213327483212676e+20,3.2847656862464206e+20,3.3363417154718446e+20,3.310415101241274e+20,3.2879563060208134e+20,3.278803074114843e+20,3.105089667355481e+20
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C.104 "NBSR_lower_Cycle12_cladding_fluence_position_3.csv" File

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−12.20157145,2.440603499447365e+20,2.6496400088154296e+20,2.8283430018177237e+20,2.966903019786168e+20,3.041158472195175e+20,3.133431874925886e+20,3.2296256668474724e+20,3.2977544504343744e+20,3.3889756274948067e+20,3.436269211384742e+20,3.4991142381112125e+20,3.5474770190532064e+20,3.60796302371244e+20,3.6051231458281397e+20,3.577963277854984e+20,3.493971909432271e+20,3.3590946813098013e+20
−14.197285749999999,2.4743652349324293e+20,2.712422807142045e+20,2.891193685671454e+20,3.00818307739919e+20,3.12253624782397e+20,3.1781344945309424e+20,3.307003853503758e+20,3.397823374528761e+20,3.522591316238394e+20,3.5181391570851194e+20,3.601479955872982e+20,3.63029170500593e+20,3.683581843791001e+20,3.713343990303555e+20,3.7031159042182676e+20,3.5994037901687226e+20,3.4300746570355225e+20
−16.192999999999998,2.4884401675542187e+20,2.745092717066014e+20,2.8720386527705783e+20,3.031756326689783e+20,3.1148991260235615e+20,3.2079645265706117e+20,3.2624483192074496e+20,3.384998667031335e+20,3.482663312040408e+20,3.50431879519001e+20,3.58085972701188e+20,3.638800024404311e+20,3.690709824138048e+20,3.704281272433737e+20,3.6915470789724635e+20,3.5894133034283375e+20,3.456668812282722e+20
−18.188714249999997,2.5040481816633498e+20,2.7014196946221974e+20,2.8757440711255913e+20,2.9843269717456185e+20,3.082529043844349e+20,3.2122582861606185e+20,3.2717938934402456e+20,3.3473391708644324e+20,3.411343908679113e+20,3.4879640402826166e+20,3.546503993164561e+20,3.603324179359599e+20,3.631247759512796e+20,3.646176918350779e+20,3.6798707683087286e+20,3.547064048763258e+20,3.3861696923740635e+20
−20.18442855,2.4097947843917922e+20,2.6520782306643008e+20,2.8323539050447526e+20,2.938854982833261e+20,3.012284494662372e+20,3.118774258196361 4e+20,3.1858225304766 87e+20,3.303123064203698e+20,3.386231920773919e+20,3.436523782111423e+20,3.455752357666673e+20,3.503690854064198e+20,3.5200116662080347e+20,3.550362153955582e+20,3.562564577454457e+20,3.4767346426723874e+20,3.362720899883181e+20
−22.180142850000003,2.338475381030498e+20,2.5622713354187584e+20,2.747774195387046e+20,2.8135326426521915e+20,2.9064905577813092e+20,2.9986564750940884e+20,3.097441231300554e+20,3.1884361232706044e+20,3.239395525624736e+20,3.2764327377930866e+20,3.3193477051841217e+20,3.3973255473299194e+20,3.454241904688 37e+20,3.467626667784493e+20,3.454654874978318e+20,3.4286886608569296e+20,3.261826035208898e+20
−24.17585715,2.2584836015802958e+20,2.4847008064356486e+20,2.6293648647171604e+20,2.7467898552438823e+20,2.841495822696205e+20,2.911033230970737e+20,3.0247414888879266e+20,3.083479441223956e+20,3.152258794445707e+20,3.205503676212702e+20,3.217479814621194e+20,3.2569495915111553e+20,3.332070584390875e+20,3.3415971866959774e+20,3.3375749692144294e+20,3.233625255819983e+20,3.1484458906727626e+20
−26.171571450000002,2.171624069636985e+20,2.4133191746744992e+20,2.5042688096264723e+20,2.6136154224265403e+20,2.698958844264288 4e+20,2.786842316241579e+20,2.831686364028125e+20,2.9358057912403575e+20,2.9549664812684935e+20,3.028531764151833e+20,3.0695742224199536e+20,3.1133660445362205e+20,3.1258796100343706e+20,3.162962079220798e+20,3.14486492911746e+20,3.1069338908420984e+20,2.9308501477609816e+20
−28.167285749999998,2.0366054133330264e+20,2.2249368369310948e+20,2.343595081200401e+20,2.4422101235890805e+20,2.553525416675476e+20,2.56935405874773e+20,2.669558753896346e+20,2.7428638089257465e+20,2.7950055508771268e+20,2.8137589277425736e+20,2.8750821872362222e+20,2.9401221793394033e+20,2.9236316533777806e+20,2.983353945856974e+20,2.927552042568657e+20,2.929537694236763e+20,2.7617303283363853e+20
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−34.15442855,1.5672392217344524e+20,1.7086843746052735e+20,1.785213992172624e+20,1.8900009604014853e+20,1.9536153564352564e+20,2.0133942201870447e+20,2.0594092933163198e+20,2.080402892576553e+20,2.1318205222387093e+20,2.1869322560013592e+20,2.1954405753997 4e+20,2.240007423950567e+20,2.2701881978803287e+20,2.2901465428520626e+20,2.2568034347842057e+20,2.248198944222412e+20,2.1446452297361356e+20
−36.15014285,1.3346577487120925e+20,1.4491297188093236e+20,1.528844299023807e+20,1.5700904138732713e+20,1.6359280609200505e+20,1.660293308026982 6e+20,1.7069702650456264e+20,1.724111360642098e+20,1.8021288026787135e+20,1.8208595510351223e+20,1.8682210204521 73e+20,1.900687273794797e+20,1.9030349816075154e+20,1.9277679419863187e+20,1.918026368845354e+20,1.873861176329956e+20,1.7840599382116732e+20

C.105 "NBSR_upper_Cycle12_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,2.7006899252057137e+19,2.9553851086857007e+19,3.033425179231354e+19,3.2329068006580187e+19,3.2281944136508035e+19,3.2
791821016412328e+19,3.468939121308711e+19,3.496642073498785e+19,3.5592777865166516e+19,3.6242046360746414e+19,3.704716871232723
e+19,3.757633639618663e+19,3.708812631368646e+19,3.732957250512455e+19,3.651290961393423e+19,3.407927003814338e+19,3.340754274734
303e+19
24.15442855,3.360656048433441e+19,3.428728260747747e+19,3.652145187609616 4e+19,3.903231123898003e+19,3.894128806137368e+19,4.0726
90370958173e+19,4.152325751391023e+19,4.362295686756915e+19,4.429072416928782e+19,4.4458175136170844e+19,4.350658975983997e+19,4.
406602307453802e+19,4.497331314442653e+19,4.512096416590108e+19,4.248242344076892e+19,4.222615557591078e+19,3.962868559468302e+
19
22.158714250000003,3.815879421883111e+19,4.004216502608439e+19,4.188870793487792e+19,4.536772805696317e+19,4.640264291782739e+19
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20.163,3.985310383306983e+19,4.426967965588224e+19,4.589220032519713e+19,4.618105324307036e+19,5.008797847106885e+19,5.0197387312
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18.167285749999998,4.361220832577598e+19,4.762277212516965 6e+19,5.057862111829064e+19,5.2822124646887604e+19,5.414549642671688e+
19,5.4743907348233314e+19,5.602236153762166e+19,5.840135326408513e+19,5.823220515902424e+19,5.909208850247763e+19,5.939191624723
44e+19,6.005889155113671e+19,6.037851924130195e+19,5.923295097124071e+19,5.7906354628873495e+19,5.786788616350847e+19,5.500849119
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16.171571450000002,4.727214337762202e+19,5.153224306043495e+19,5.365672063149074e+19,5.60284146637894e+19,5.701648851094443e+19,
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12.18014285,5.446976296123495e+19,5.836288479872012e+19,6.026198241975497 5e+19,6.275507840304381e+19,6.488159253991306e+19,6.8206
851943083426e+19,6.774975606051084e+19,7.007709821509452e+19,6.928510039875587e+19,7.069994792637226e+19,7.168259093135813e+19,7.
117005520164184e+19,7.50078503345285e+19,7.303577577184529e+19,7.2919238950298305e+19,6.897565553765786e+19,6.766093916253572e+
19
10.18442855,5.664198668636145e+19,6.040397631396998e+19,6.326082558004864e+19,6.5297957106216804e+19,6.700923810223421e+19,7.0146
68088038713e+19,7.092849586765855e+19,7.350644875984082e+19,7.397825317328826e+19,7.510685006157082e+19,7.512721571970525e+19,7.7
04894184977834e+19,7.419888113641118e+19,7.686565092656855e+19,7.510175864703721e+19,7.3031250070037635e+19,6.842125706622081e+
19
8.18871425,5.7544864196987494e+19,6.242130789472967e+19,6.645993104533077e+19,6.725871741438089e+19,6.916007788631956e+19,7.26844
681690265e+19,7.593731634327447e+19,7.607365311022991e+19,7.851583494818269e+19,7.639610936402493e+19,8.112999345482616e+19,8.148
29981958228e+19,7.92693642991563e+19,8.040022403834269e+19,7.743079793979875e+19,7.792919085136614e+19,7.237106331884678e+19
6.193,6.01443141727561e+19,6.305038044599294e+19,6.790815562377857e+19,6.987570448465411e+19,7.120173511429539e+19,7.319304390966
11e+19,7.595598486323103e+19,7.56646428093636e+19,7.912454184131155e+19,7.983847130146823e+19,8.145244970862117e+19,8.33441930642
186e+19,8.252277818613024e+19,8.168722048989299e+19,7.838119531940513e+19,7.713153590776796e+19,7.470519402614194e+19
4.19728575,5.958652142496331e+19,6.327496839819754e+19,6.722081466174181e+19,7.012292094589697e+19,7.364221981407029e+19,7.483191
367675612e+19,7.890504530364054e+19,8.002119651195206e+19,8.182468868230033e+19,8.140096985055917e+19,8.159840359191786e+19,8.232
98701465792e+19,8.318975349003256e+19,8.104061084412509e+19,8.242038418273218e+19,8.162612351548971e+19,7.550454610791801e+19
2.20157145,5.8958014586426016e+19,6.446183369725357e+19,6.688591272797577e+19,7.004824686607076e+19,7.374404810474241e+19,7.37576
2521016535e+19,7.755129475042743e+19,7.75258376777594e+19,7.939155824796285e+19,8.092181117167428e+19,8.141680980688593e+19,8.125
841024361821e+19,8.19813911073902e+19,8.048451523451016e+19,8.177151168606044e+19,7.907928482323504e+19,7.569971699837288e+19
0.20585714999999993,5.3316896997008876e+19,5.855239856191543e+19,5.991633194419575e+19,6.068060983696254e+19,6.610409774070434e+
19,6.800149822356133e+19,7.0442548636062196e+19,7.127244920503991e+19,7.314043262614717e+19,7.31749411024305e+19,7.2351829086164
28e+19,7.50231245781293e+19,7.529410097386232e+19,7.326941512766517e+19,7.247176018406698e+19,7.458526292823923e+19,7.05370226612
9689e+19

C.106 "NBSR_lower_Cycle12_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,5.705665411448732e+19,6.149636758779135e+19,6.42847656140293e+19,6.8108983641493004e+19,6.9346197373159145e+19,7.364674551587795e+19,7.454226876106655e+19,7.437990920871715e+19,7.715359870408024e+19,7.813001886908057e+19,7.903742208151427e+19,7.754111192136022e+19,7.938250684434753e+19,7.796879074218308e+19,8.136363281064604e+19,7.779059123350689e+19,7.515663278145497e+19
−12.20157145,6.312675166399703e+19,6.919684921350672e+19,7.07717934425687e+19,7.574440830372341e+19,7.819507583256555e+19,7.953298642945 19e+19,8.202721383819264e+19,8.456047542502438e+19,8.533154187050263e+19,8.735170201489211e+19,8.674016655813349e+19,8.688046331417061e+19,8.826023665277772e+19,9.145538212897815e+19,8.704112572834216e+19,8.86488812955096e+19,8.284693157810312e+19
−14.197285749999999,6.579861286868801e+19,7.096583290757166e+19,7.405971580582594e+19,7.774307136452657e+19,8.011793338809057e+19,8.22885731175844e+19,8.418427646226352e+19,8.744504461467487e+19,9.071825844705727e+19,9.156626182326557e+19,9.280800125673936e+19,9.266770450070223e+19,9.317854309224065e+19,9.158323320504425e+19,8.812163703491845e+19,9.082404672681107e+19,8.463232094122079e+19
−16.192999999999998,6.620875459500623e+19,7.037522882167342e+19,7.563409432216198e+19,7.892541096177497e+19,8.302739393768315e+19,8.243452700088109e+19,8.635378476630542e+19,8.838073346340638e+19,8.764021550512975e+19,9.132979390381588e+19,9.150799341249208e+19,9.319777732492316e+19,9.227679700706651e+19,8.926946815588352e+19,8.892042340396856e+19,8.966886134040858e+19,8.489594307151638e+19
−18.188714249999997,6.5590996298262086e+19,6.942992285660067e+19,7.407668718760464e+19,7.681303964305462e+19,7.992163107218378e+19,8.415089941143211e+19,8.435681884368015e+19,8.70767656300774e+19,8.8966246134771e+19,8.971128979485529e+19,9.022665408820137e+19,9.182592396447944e+19,9.092530930475724e+19,9.132017678747464e+19,8.951159320259276e+19,8.612749967592296e+19,8.474998918821968e+19
−20.18442855,6.474751862386145e+19,7.002222408067677e+19,7.557922018774422e+19,7.713832446047941e+19,7.701160480986523e+19,8.019543603154656e+19,8.302060538497167e+19,8.295950841056841e+19,8.647880727874173e+19,8.742977037107405e+19,8.941146205009851e+19,8.961964433325038e+19,9.20601290330253e+19,9.2206082916322e+19,8.960041010056788e+19,8.682785203065668e+19,8.266986349487886e+19
−22.180142850000003,6.40924232872042e+19,6.6443525376278045e+19,7.1131586736276816e+19,7.38215507481984e+19,7.634180094233315e+19,7.900800501976456e+19,7.999291087565426e+19,8.242490988453983e+19,8.25838751605335e+19,8.479581191902213e+19,8.166911768266238e+19,8.778277511207071e+19,8.83648935070796e+19,8.722102237519621e+19,8.424876771302253e+19,8.243056701179938e+19,8.143604403956844e+19
−24.17585715,5.977207519907694e+19,6.5219323037308895e+19,6.911651800642095e+19,7.2264709326367015e+19,7.39997502568746e+19,7.712757591868626e+19,7.678871399583852e+19,7.928916424456479e+19,8.100270809148601e+19,8.216751059422976e+19,8.146263253768838e+19,8.385446594303107e+19,8.330402746067572e+19,8.311507941020634e+19,8.167873479900363e+19,7.924277580103637e+19,7.718754146763761e+19
−26.171571450000002,5.701535708549253e+19,6.299267774794541e+19,6.502415214685402e+19,6.878897033809231e+19,7.0270006254645576e+19,7.28445648704721e+19,7.2774982205179494e+19,7.50089817599804e+19,7.65267890037208e+19,8.009530487905232e+19,7.84241894865778e+19,7.821487577797403e+19,7.823128144702675e+19,7.682491961029971e+19,7.743475792888046e+19,7.362015701775801e+19,7.163507106237781e+19
−28.167285749999998,5.357876541785398e+19,5.803703426856937e+19,6.097760901808953e+19,6.298984918431562e+19,6.594286961380683e+19,6.815876636137716e+19,6.8562685247709856e+19,7.110839251451262e+19,7.3664282610382586e+19,7.196601300706217e+19,7.453095450654743e+19,7.47883537968575e+19,7.529919238839591e+19,7.494222765831758e+19,7.1467054382768824e+19,7.097092432210526e+19,6.669017612479494e+19
−30.163,4.779259908550169e+19,5.509911836886118e+19,5.624038722220517e+19,5.805853135215569e+19,6.026254813248094e+19,6.269680999227033e+19,6.34277108342057e+19,6.717725478184319e+19,6.842861133165824e+19,6.907239241379635e+19,6.924493479521299e+19,6.911651800642095e+19,6.969637355052602e+19,6.81180350451083e+19,6.743635121033112e+19,6.799470967084986e+19,6.350804204129148e+19
−32.15871425,4.462143982888178 5e+19,4.936465817966129e+19,5.191144030064337e+19,5.333268038206305e+19,5.6219286137527e+19,5.786222903624892e+19,6.092499773457561e+19,6.0475256117440455e+19,6.169040705279431e+19,6.206604030282921e+19,6.249145627274825e+19,6.042151340847462e+19,6.241338791656629e+19,6.249032484729633e+19,6.122821975568811e+19,5.9948577569575264e+19,5.728576776849958e+19
−34.15442855,3.945116494127798e+19,4.367319215763405e+19,4.532665731305874e+19,4.713699460739108e+19,5.001958380250075e+19,5.043934264516023e+19,5.23781532995572e+19,5.39489678257197e+19,5.599169990787484e+19,5.40097253724874e+19,5.632343385037554e+19,5.541139179358 9e+19,5.483119682184835e+19,5.549698412902616e+19,5.463670478666461e+19,5.299251731994561e+19,4.9737972007519764e+19
−36.15014285,3.346479287520979e+19,3.523479485218145e+19,3.812581316690786e+19,4.09531887999642e+19,4.016481154507168e+19,4.0452363323675206e+19,4.185159718005519e+19,4.302618651295799e+19,4.378277071265177e+19,4.61971760557601e+19,4.614614876787886e+19,4.520616050243009e+19,4.578222577127125e+19,4.4524024097472135e+19,4.572316536268143e+19,4.159640416937636e+19,4.044257649351616e+19

C.107 "NBSR_upper_Cycle12_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.890825146902723e+20,2.076823483523963e+20,2.10538706936503e+20,2.2283521577411536e+20,2.2819184534404206e+20,2.2696755491874967e+20,2.3089026184826356e+20,2.3942009138679408e+20,2.4588805886427706e+20,2.476235072935617 8e+20,2.5023385912293815e+20,2.5293801897551 91e+20,2.5098815220748108e+20,2.582391295113043e+20,2.5803141174563704e+20,2.4733538265086198e+20,2.383049245803315e+20
24.15442855,2.2366991573569074e+20,2.4265359651317232e+20,2.504166890457144e+20,2.586832817844362e+20,2.668292642076092e+20,2.7051553662965852e+20,2.8141928349010084e+20,2.846068418296033e+20,2.886511162993925e+20,2.9849425816278716e+20,2.952387368676977e+20,2.9471992106589277e+20,3.059519960075511e+20,3.024093072944884e+20,3.059491243333713e+20,2.932534527847763e+20,2.7743818585226573e+20
22.158714250000003,2.436089067902965e+20,2.678898692046569e+20,2.767690857683882e+20,2.8755605121218802e+20,2.983688617236055e+20,3.109850836198947e+20,3.1516336955140466e+20,3.148101536272977e+20,3.222305597077251e+20,3.317415445909974e+20,3.2750678240061935e+20,3.3488124169417104e+20,3.3497696416682875e+20,3.369000286425226e+20,3.353349662145687e+20,3.347434013335439e+20,3.18503126622433e+20
20.163,2.7052032275329142e+20,2.9773517895461136e+20,3.054283940821133e+20,3.099484092410114e+20,3.302750763098809e+20,3.2889858715306276e+20,3.405125947606254e+20,3.520739550082264e+20,3.4988191038436436e+20,3.595776396398661e+20,3.6304279315007596e+20,3.7085087524476715e+20,3.744356818457993e+20,3.7341049416363495e+20,3.7492195200690055e+20,3.664706148959494e+20,3.437298270666518e+20
18.167285749999998,2.9261977001578208e+20,3.1598945449044096e+20,3.33649293471066e+20,3.418680249734589e+20,3.4887012384837206e+20,3.662092925455938e+20,3.674354974203393e+20,3.781162109194891e+20,3.8623347660086483e+20,3.932231315543324e+20,3.904873832857744e+20,3.9917802657837004e+20,4.039488346156314e+20,4.0331610907136387e+20,4.050678303210003e+20,4.017050998565342e+20,3.809266227167201e+20
16.171571450000002,3.1555870336348114e+20,3.364491757963047e+20,3.5676914229208816e+20,3.675876961518651e+20,3.802039180481543e+20,3.826735578427238e+20,3.975785040602593e+20,4.062825484990269 4e+20,4.0582308063026984e+20,4.187867751023066e+20,4.279043406229556e+20,4.3860419861663685e+20,4.3778098535178043e+20,4.383218173222966e+20,4.360857403610119e+20,4.320194497225115e+20,4.0892257429492715e+20
14.17585715,3.4176560192771516e+20,3.599586150810438e+20,3.802297631157718e+20,3.8712752449548806e+20,4.0077754909648074e+20,4.167268274907121e+20,4.2339868383495605e+20,4.3170548001219405e+20,4.358770653706181e+20,4.471043541886435e+20,4.5972344775911244e+20,4.639830977923815e+20,4.627061600071274e+20,4.637849522739801e+20,4.649479803167715e+20,4.52316442824857e+20,4.3746414396728325e+20
12.18014285,3.554663594392164e+20,3.7787690473784476e+20,3.972999516648253e+20,4.0814722226639956e+20,4.250163936228718e+20,4.331595043718651e+20,4.45959513415657e+20,4.55935709516046e+20,4.660583609996012e+20,4.710627318701474e+20,4.828691416477522e+20,4.91913000086445474e+20,4.9281949268052345e+20,4.955877865897851e+20,4.9188428412265746e+20,4.7855492980506814e+20,4.548339438557554e+20
10.18442855,3.652645117404619e+20,3.980437152773759e+20,4.129371747981925e+20,4.2445928883200374e+20,4.3951834823051824e+20,4.526141397148227e+20,4.6646709595784964e+20,4.777384171132977e+20,4.863323807085089e+20,5.026846507126293e+20,5.052366118336845e+20,5.0924834066277014e+20,5.14699735480628e+20,5.164217827637405e+20,5.089343709524527e+20,5.0067064988791066e+20,4.883061780947114e+20
8.18871425,3.811056237405901e+20,4.136292482755079e+20,4.315169067410584e+20,4.447371374398178e+20,4.672405335369242e+20,4.722898939696196e+20,4.890556850556214e+20,4.9705042597199525e+20,5.089401143008122e+20,5.132562405929493 5e+20,5.194772440909754 5e+20,5.280185603262249e+20,5.4137854583506456e+20,5.3800241422442634e+20,5.3076770974095494e+20,5.276222692894218e+20,4.946966103693416e+20
6.193,3.919729960614224e+20,4.169134863123947e+20,4.3913162944098094e+20,4.5912422508027504e+20,4.792489177318367e+20,4.939183866666342e+20,5.076641337402847e+20,5.120999131232439 5e+20,5.1630500134709815e+20,5.2517081676465734e+20,5.3781671262747034e+20,5.5240386023578226e+20,5.599046731932421e+20,5.561083199276365e+20,5.559187894317741 5e+20,5.428938325772365e+20,5.1164331692866653e+20
4.19728575,3.933178968022636e+20,4.254203424574874e+20,4.485229612334311e+20,4.663445711928478e+20,4.805727595286931e+20,5.046756781439102e+20,5.1463081530031433e+20,5.224599563389903e+20,5.31331515104909e+20,5.407276330209921e+20,5.543278819362027e+20,5.627265716871921e+20,5.570942613960111e+20,5.553578557419999e+20,5.612763762264275e+20,5.5073254586317826e+20,5.188952514572164e+20
2.20157145,3.867159178630598e+20,4.166760945802035e+20,4.494935871061805e+20,4.639170492862477e+20,4.918124922681641 5e+20,5.0507388363016634e+20,5.119965328527736e+20,5.2368328953955615e+20,5.303608892321596e+20,5.4609096316400475e+20,5.5585465537509347e+20,5.599898661939076e+20,5.5211860726726223e+20,5.6110503300037004e+20,5.614601633739303e+20,5.455175855527849e+20,5.041329317239 4086e+20
0.20585714999999993,3.5772636701866544e+20,3.8920278770270765e+20,4.090728585769998e+20,4.289448439007451e+20,4.4522915094927855e+20,4.6404340295015596e+20,4.704443646967785e+20,4.8794913327169805e+20,4.963975987084696e+20,5.044373291869925e+20,5.1621502222279986e+20,5.2099731695678017e+20,5.195136186305853e+20,5.219545416833575e+20,5.179734440455224e+20,5.0597558932260225e+20,4.823283096772359e+20

C.108 "NBSR_lower_Cycle12_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,3.82645798325653e+20,4.1685509560407346e+20,4.345398224275896e+20,4.5362305457663515e+20,4.7766183913517154e+20,4.88
11281869994276e+20,5.0750810610985245e+20,5.104142403797413e+20,5.176499020879393e+20,5.212682115544015e+20,5.3893187943393296e
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99775e+20
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C.109 "NBSR_upper_Cycle12_cladding_fluence_position_6.csv" File

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C.110 "NBSR_lower_Cycle12_cladding_fluence_position_6.csv" File

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−24.17585715,2.3248756470985117e+20,2.4734770659526784e+20,2.6268587573411748e+20,2.7037560881803965e+20,2.8095613393159792e+20,2.9366034461839557e+20,3.003600804318945e+20,3.10188773332657e+20,3.175062674429 0004e+20,3.24747956047865e+20,3.3197154184559926e+20,3.3703353931745506e+20,3.4398162301764875e+20,3.439301431595867e+20,3.48496010570779e+20,3.4459598703803715e+20,3.2504721807 98958e+20
−26.171571450000002,2.2161513182969954e+20,2.3747941380368843e+20,2.516075234217178e+20,2.5932214786558204e+20,2.670837264657007 2e+20,2.7845002655561207e+20,2.847780891081578e+20,2.9297300365635882e+20,2.975173739839647e+20,3.062412299309348e+20,3.106130578 771241e+20,3.1966729005605264e+20,3.225750534674673e+20,3.279391415349837e+20,3.2843583730837324e+20,3.2228597426450366e+20,3.069 302680311494e+20
−28.167285749999998,2.0807706058484246e+20,2.2473843178970356e+20,2.372610486914693e+20,2.438035163671524e+20,2.5252737231412252e+20,2.5976114094092403e+20,2.6739543417770256e+20,2.750812072725431e+20,2.792918070918348e+20,2.8526856204156178e+20,2.926460217 0075618e+20,2.995058542157007e+20,3.027575709644967e+20,2.981798235860594e+20,3.070015478346199e+20,3.004737886898117e+20,2.92965 08367819548e+20
−30.163,1.954073583743281e+20,2.0930635433834522e+20,2.1664081983036696e+20,2.269243457627982e+20,2.317000925953202e+20,2.4475504 51722107e+20,2.470535359777706e+20,2.5516133076617445e+20,2.6101249749073907e+20,2.6783555867849644e+20,2.7756185757586086e+20,2.785337520390536e+20,2.858744403710608e+20,2.8261763220773116e+20,2.916995843102315e+20,2.81212967509182e+20,2.7157831407342243e+20
−32.15871425,1.757669439545818e+20,1.893893063956064e+20,2.011691424881917e+20,2.060246548150735e+20,2.152768864480866e+20,2.2480 575160409237e+20,2.287911977584536e+20,2.3153716733024482e+20,2.3628689137737282e+20,2.4475278232130686e+20,2.5206405359156435e+20,2.5524392482416404e+20,2.5573779203392376e+20,2.6243922498560056e+20,2.6174679260903018e+20,2.5971022679558803e+20,2.5019324 160682738e+20
−34.15442855,1.5958190286497582e+20,1.700634282514917e+20,1.8085326707365374e+20,1.8706535851737845e+20,1.9442188680571244e+20,1.9854480115248103e+20,2.0038902463909813e+20,2.074491194590311 2e+20,2.1264405942148665e+20,2.180562330707093e+20,2.2190703959629 293e+20,2.3066540401954636e+20,2.293506876444242e+20,2.3250510180435585e+20,2.3580716698576195e+20,2.365052564895919e+20,2.224659 6376953763e+20
−36.15014285,1.3579651130214867e+20,1.4632385941946702e+20,1.5359553079890762e+20,1.6073369397502257e+20,1.6645418105989133e+20,1.6878604891628267e+20,1.7334229921113365e+20,1.7794267509860922e+20,1.82566245207849e+20,1.877702365739198e+20,1.9163066021584473e+20,1.9398572229400025e+20,1.9481562286297793e+20,1.947307659540845e+20,1.997169579206622e+20,1.9763909507822518e+20,1.8699011 872482623e+20

C.111 "NBSR_upper_Cycle12_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0572095993941488e+20,1.1458680978059996e+20,1.2103932913285603e+20,1.2447490251758784e+20,1.289813700925547e+20,1.3147163751221376e+20,1.3447444066158908e+20,1.3949796966807988e+20,1.4143779860538358e+20,1.4361692402576676e+20,1.4428163647876525e+20,1.4536441063624534e+20,1.466740355968339e+20,1.4830951108757317e+20,1.4624861962691484e+20,1.4474778376495314e+20,1.4104349683539214e+20
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20.163,1.538704671837223e+20,1.6277308835209455e+20,1.7644693065118112e+20,1.8098734098970532e+20,1.8853338304123467e+20,1.9320221016855095e+20,1.9468833749963778e+20,1.9882709180273315e+20,2.0290305199324732e+20,2.088498241684986e+20,2.1183791878699906e+20,2.1343945151418096e+20,2.1675283095010623e+20,2.1653672868879103e+20,2.1822142118668848e+20,2.1262595661425603e+20,2.0351458745000598e+20
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6.193,2.163726719982637e+20,2.3581678410210325e+20,2.4968636300437068e+20,2.650703548740227e+20,2.6774391321689152e+20,2.741930382927919e+20,2.8168477192262943e+20,2.911655514969289e+20,2.947442502013276e+20,3.000879726107096e+20,3.081210933192871e+20,3.146590352931626e+20,3.185454817204815e+20,3.182320768703018e+20,3.169207547715354e+20,3.139434086948281e+20,2.959198012458645e+20
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C.112 "NBSR_lower_Cycle12_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.1377039345886537e+20,2.344817020688467e+20,2.5014572173784705e+20,2.559618142734024e+20,2.656915074471226e+20,2.76335595809871387e+20,2.8363478368900032e+20,2.9002903463048292e+20,2.9394602954500345e+20,3.007006394929201e+20,3.010344100012342e+20,3.066355317009263e+20,3.117778603798679e+20,3.158379806140553e+20,3.1555172997472145e+20,3.0825233867170906e+20,2.953093972145578e+20
−12.20157145,2.341083316697156e+20,2.516205348144148e+20,2.683678943536212e+20,2.8116657906565356e+20,2.9336334543726857e+20,3.00521308558792e+20,3.121059737609225e+20,3.141448024252685e+20,3.239169240534353e+20,3.295847998547902e+20,3.358919310364756e+20,3.402688503971984e+20,3.429548544200383e+20,3.5101343220128396e+20,3.4712698577396 5e+20,3.3836579278708185e+20,3.2391579262798345e+20
−14.197285749999999,2.430313184962222e+20,2.6105549165791176e+20,2.7625166690254638e+20,2.8920026548695715e+20,2.963882113829562e+20,3.078246598508862e+20,3.1786832358751194e+20,3.226299276018851e+20,3.329513562869553 6e+20,3.4120963066046356e+20,3.48160542924287e+20,3.4972587003700766e+20,3.575151685606982e+20,3.606356399570724 5e+20,3.5525344908232543e+20,3.450836314078114e+20,3.299281874794456e+20
−16.192999999999998,2.4344655163707405e+20,2.5920504533130913e+20,2.7921600158655673e+20,2.898593208126961e+20,3.000517669962483 6e+20,3.0605058474228764e+20,3.175600101518658e+20,3.251807262832214e+20,3.325146260625172e+20,3.385734093575078e+20,3.4341364744 07888e+20,3.447617408667423e+20,3.5140264255674175e+20,3.5506223818095224e+20,3.554633285036552e+20,3.463355536703524e+20,3.344787806470371e+20
−18.188714249999997,2.4264041100258653e+20,2.635355762485036e+20,2.7371444532663294e+20,2.8750086445818482e+20,2.954134883561337e+20,3.0410623010317625e+20,3.122988818004735e+20,3.196887871396389e+20,3.280749125892132e+20,3.3590663956735 04e+20,3.404510098949563e+20,3.4378588641446796e+20,3.463604450302945e+20,3.549915240902077e+20,3.534822025373566e+20,3.4382265774165505e+20,3.316100514137133e+20
−20.18442855,2.377011731922632e+20,2.581861967118621e+20,2.6987721590647226e+20,2.78515649231823e+20,2.8865661555731327e+20,2.963706742884516e+20,3.0632212685074655e+20,3.142132536651091e+20,3.237766272973982e+20,3.255444795660112 4e+20,3.342881354583898e+20,3.384444268559898e+20,3.3948081256994144e+20,3.4546944748691345e+20,3.449376775245147e+20,3.4103765399177285e+20,3.236170963086786e+20
−22.180142850000003,2.3076836373567036e+20,2.490414504967806e+20,2.5739306747007153e+20,2.6767942196613254e+20,2.778769595642184 7e+20,2.9096528919194036e+20,2.9564486486104985e+20,3.035512659190133e+20,3.071995472887046e+20,3.185958301530916e+20,3.207359213953838e+20,3.3002549006831007e+20,3.311348527239101e+20,3.3202981025637276e+20,3.335261204165268e+20,3.249719782873436e+20,3.119792541103083e+20
−24.17585715,2.2107713902731523e+20,2.3758746493434606e+20,2.5151644367283883e+20,2.6178356393621733e+20,2.7019740930936344e+20,2.7909663620137995e+20,2.8353804681286186e+20,2.9436013126040335e+20,2.9327735710292325e+20,3.0183715635936605e+20,3.097712773409013e+20,3.130484511623654e+20,3.1663733269583135e+20,3.2146965080094894e+20,3.212382742960329e+20,3.191943542171532e+20,3.033685407085294e+20
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−28.167285749999998,1.959945681838706e+20,2.156349826036169e+20,2.2777970340444393e+20,2.3536704248496803e+20,2.4241186306130028e+20,2.51669186108847e+20,2.5244477825613293e+20,2.58458870245773e+20,2.64949858063394e+20,2.682145862048871e+20,2.76097227328360 35e+20,2.856334467498035e+20,2.8581560624756136e+20,2.897688067765431e+20,2.850694311620252e+20,2.829802540650691e+20,2.6817272346316633e+20
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−32.15871425,1.707524663517063e+20,1.818268586750243e+20,1.932633071429542e+20,1.9774544907070492e+20,2.068862352967047e+20,2.143180033775909e+20,2.168546592407784e+20,2.2464961489172844e+20,2.270380540207154e+20,2.3054660434709558e+20,2.3580660127303605e+20,2.3952672815892382e+20,2.4193383580786734e+20,2.4529699796367675e+20,2.464929146663 4805e+20,2.4352518570598197e+20,2.3115757409112824e+20
−34.15442855,1.523011800819199e+20,1.6764670348620694e+20,1.7276923221974008e+20,1.806846846813188e+20,1.8552492276459975e+20,1.902520183026895e+20,2.0054629277691396e+20,2.0311575997820686e+20,2.0675272709337907e+20,2.0648853925035757e+20,2.13259554867326 94e+20,2.1182264454339825e+20,2.1843865487345562e+20,2.2145899511733567e+20,2.2004754186607496e+20,2.191118530173435e+20,2.0782644984724387e+20
−36.15014285,1.2679715325763792e+20,1.4065315505448236e+20,1.4676907533479453e+20,1.5004511773080673e+20,1.608666364656223e+20,1.6040671201941992e+20,1.6594390818107887e+20,1.7175490930210066e+20,1.7241792461692125e+20,1.7369756680303414e+20,1.797003445381 5506e+20,1.7825211995970725e+20,1.830504953012675e+20,1.851526837909206e+20,1.8765087118874306e+20,1.8317495210097784e+20,1.735521786324 6342e+20

C.113 "NBSR_upper_Cycle13_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,4.2747347045802025e+19,4.607155532019469e+19,4.809023858027453e+19,5.317880706001078e+19,5.327814152621097e+19,5.3374
60228557798e+19,5.357039751114518e+19,5.768602464739509e+19,5.498713497990198e+19,5.538399389356388e+19,5.7900211596694634e+19,5.
755038568486909e+19,5.70650166007453e+19,5.720994721536525e+19,5.786563132446872e+19,5.433748565514819e+19,5.401093676867138e+1
9
24.15442855,5.277955339065456e+19,5.391141072201564e+19,5.687592669112218e+19,5.991343481379143e+19,6.0879000309739274e+19,6.1322
41327409866e+19,6.510555252974987e+19,6.50951113949226 4e+19,6.954868645475429 6e+19,6.896886820604672e+19,7.087643480191037e+19,7.
060190000911408e+19,6.92819106704076e+19,7.10730921395275e+19,6.999736788164117e+19,6.787685960943905e+19,6.519751114841157e+19
22.158714250000003,5.978038218741528e+19,6.321197130714104e+19,6.296741885563757e+19,7.030715347825778e+19,7.212419830887629e+19
,7.22969080895503e+19,7.420686944110823e+19,7.518613293962756e+19,7.653734989258791e+19,7.98890499623513e+19,7.862950425736934e+1
9,8.13208264968684e+19,8.106956872942087e+19,8.09085453565351e+19,7.90985890027718e+19,8.005754497300333e+19,7.540759994623782e+1
9
20.163,6.621001385596798e+19,6.775721761495106e+19,7.021730224460708e+19,7.36573209043768e+19,7.742082316333462e+19,7.86789320149
e+19,7.847240828382228e+19,8.226292338701196e+19,8.614233182157464e+19,8.564288157396828e+19,8.666572962612412e+19,8.820038486526
9e+19,8.982996822013654e+19,8.930580409376486e+19,8.428361824187503e+19,8.394490399647112e+19,8.32837598443846e+19
18.167285749999998,6.975521018583204e+19,7.47132165851142e+19,7.725797977612121e+19,8.258439539141681e+19,8.304897799611395e+19,
8.614376867499123e+19,8.882675716467407e+19,9.131529149197853e+19,9.410958822633315e+19,9.470980979355628e+19,9.524000870427764e
+19,9.495053063594883e+19,9.573898000074514e+19,9.790336019726711e+19,9.461344482441706e+19,9.376244444088525e+19,8.963302351183
607e+19
16.171571450000002,7.39535042886497e+19,7.89054759035822e+19,8.339343965518422e+19,8.850308198503046e+19,9.331979779834803e+19,9.
190823300189096e+19,9.57600538508551e+19,9.724336552791387e+19,1.0025405238680681e+20,9.897908445515347e+19,1.0138916658591215e
+20,1.0538361908402972e+20,1.0262007101279012e+20,1.0456844424568488e+20,1.0032110554624762e+20,1.007100138710044e+20,9.42506872
3184219e+19
14.17585715,7.756996854797658e+19,8.38707623601751e+19,8.978638366650001e+19,9.523023810104484e+19,9.428277695814604e+19,9.810835
12847005e+19,1.0004522969026252e+20,1.0075216157122434e+20,1.0562309465346123e+20,1.0614323559026645e+20,1.075168674565256e+20,1.
1029191035509791e+20,1.0840771657481082e+20,1.0879470909501212e+20,1.0880716182462258e+20,1.0403872428610236e+20,1.035492362221
8436e+20
12.18014285,8.345378749868098e+19,8.860787649421366e+19,9.676154068221767e+19,1.0013718830892422e+20,1.0016113586586737e+20,1.043
605794514183e+20,1.0489604682466717e+20,1.0443338002452549e+20,1.1066166063430019e+20,1.1459384948436555e+20,1.175211988450963e
+20,1.1316274348144286e+20,1.1977514290458568e+20,1.1356889404719867e+20,1.1259183372391812e+20,1.1301235282383987e+20,1.0865198
165563095e+20
10.18442855,8.6423571930315e+19,9.246467843502201e+19,9.71993020231385e+19,1.0304154801498956e+20,1.0760116285696547e+20,1.076787
529414613e+20,1.0987618076656483e+20,1.115017409318659e+20,1.166150232903674 8e+20,1.1937569765477389e+20,1.222149200059539e+20,1.
2112003770251302e+20,1.226938711448169e+20,1.186371549986471 3e+20,1.252294384739577e+20,1.185289120412640 9e+20,1.13370608275709
4e+20
8.18871425,9.058163413746816e+19,9.41237651800435e+19,1.0271681914284043e+20,1.033998034668591e+20,1.0940010333453497e+20,1.13440
5351419834e+20,1.1388117018973736e+20,1.2115260637995568e+20,1.186946291353107e+20,1.2343241380094368e+20,1.2333566567089332e+2
0,1.27256359693626e+20,1.2680135611170613e+20,1.2420448303679085e+20,1.2587602251142277e+20,1.2190264386341518e+20,1.18488680145
59961e+20
6.193,9.277494298277747e+19,9.570545342102472e+19,9.951550973068003e+19,1.076682160164063e+20,1.0813950393704753e+20,1.1361774706
336272e+20,1.1689856236457437e+20,1.2058169662243098e+20,1.2317378018595765e+20,1.2215169845562397e+20,1.2623619376784777e+20,1.2
96664418243847e+20,1.2794509143131102e+20,1.2978522170682273e+20,1.290447632461405e+20,1.2708968469730168e+20,1.173669765783824
1e+20
4.19728575,8.972958006143086e+19,9.737747184679551e+19,1.0233557403630548e+20,1.0968555821329732e+20,1.1303534247850528e+20,1.159
282073572379e+20,1.1719455416839171e+20,1.1788232800379902e+20,1.2532906031084121e+20,1.2788378568553654e+20,1.2452921190893996
e+20,1.2752648813594472e+20,1.3088776722848539e+20,1.2856485420499976e+20,1.2938290275017779e+20,1.2245631137994082e+20,1.2005293
456512623e+20
2.20157145,8.782201346556723e+19,9.596552388942735e+19,1.0362682630668018e+20,1.0589992841172405e+20,1.1220579910599454e+20,1.135
1046200825741e+20,1.172443650868335e+20,1.2007879792662482e+20,1.2014010367239928e+20,1.2733203397356634e+20,1.2886946712931664
e+20,1.2456465429321584e+20,1.2637317379356259e+20,1.295514935510576e+20,1.2614136144235289e+20,1.2189689644974883e+20,1.21573125
47987743e+20
0.20585714999999993,8.100855035432971e+19,8.679351378997279e+19,9.16293876488449e+19,9.751062026339944e+19,1.0243998538457761e+2
0,1.0490754165199988e+20,1.048280357629486e+20,1.095255885329171e+20,1.1107068490688918e+20,1.0992120217361793e+20,1.14352458110
37857e+20,1.15235644010442e+20,1.1825207828300127e+20,1.1777887455780461e+20,1.1493486269523601e+20,1.1543584558648674e+20,1.1399
995007217541e+20

C.114 "NBSR_lower_Cycle13_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,8.816082350119893e+19,9.537555187657589e+19,9.88047462406073e+19,1.0143706169979847e+20,1.0529644997675665e+20,1.078626701787847e+20,1.1135039237198519e+20,1.1485152519707384e+20,1.1817831980761639e+20,1.2052613829032288e+20,1.193862345798289e+20,1.2166891570765005e+20,1.2546029292288968e+20,1.2403206062680016e+20,1.2216894069662304e+20,1.2334141308455969e+20,1.2083745653058383e+20
−12.20157145,9.31726640084893e+19,1.0361628938162517e+20,1.0870658208546134e+20,1.1543680348876444e+20,1.1879808258130511e+20,1.241067770044628e+20,1.2471408704854111e+20,1.2932159700440333e+20,1.3128721247829716e+20,1.3694937284193575e+20,1.3438794215129634e+20,1.3657004353995624e+20,1.3829043603075218e+20,1.371754377794791e+20,1.3450097462006799e+20,1.3364652578833637e+20,1.2966165231299607e+20
−14.197285749999999,9.728934483724472e+19,1.076605527981845e+20,1.1640332688699002e+20,1.2120529100523063e+20,1.2126563884872737e+20,1.295514935510576e+20,1.2676687162970798e+20,1.3611120834892548e+20,1.3707102643120695e+20,1.3367813656350133e+20,1.4047732693080072e+20,1.3949643499840928e+20,1.4023593555681377e+20,1.408336665781148e+20,1.3793601218799357e+20,1.3878088199694793e+20,1.3517725362814258e+20
−16.192999999999998,1.0107497463881803e+20,1.0872957174012677e+20,1.1578356411330129e+20,1.1858255456881675e+20,1.22715902897204 6e+20,1.2895759413886743e+20,1.3385630638715838e+20,1.34570901486342e+20,1.352404751784725e+20,1.4351770876030317e+20,1.424994586390804e+20,1.4272552357662374e+20,1.4582433744506744e+20,1.4595078054572728e+20,1.4405509193810747e+20,1.387578923422825e+20,1.3645988477801773e+20
−18.188714249999997,1.0250033322807435e+20,1.102248571956571e+20,1.1573183739030408e+20,1.1941592955043842e+20,1.237140370705951e+20,1.2929956525201564e+20,1.3209568200069792e+20,1.3588993292277075e+20,1.3259379118511548e+20,1.3907112638709891e+20,1.4251957458691262e+20,1.3778945313950147e+20,1.448108768352333e+20,1.4435778905786889e+20,1.4222454068537301e+20,1.3621657759947534e+20,1.3346644016012388e+20
−20.18442855,9.958639449923174e+19,1.0403489267699147e+20,1.1282651978196102e+20,1.1803080285684656e+20,1.2148499847032665e+20,1.2465373920504436e+20,1.2736556055328676e+20,1.304059423827892e+20,1.3540619227251909e+20,1.3609492401020412e+20,1.3599147056420 97e+20,1.3621657759947534e+20,1.3896288342971585e+20,1.4034896802558545e+20,1.4108272117032359e+20,1.3412739273175486e+20,1.30450 0058875646e+20
−22.180142850000003,9.567633319178184e+19,1.0241220621852356e+20,1.0944799844842127e+20,1.1173834279446422e+20,1.18672597382923e+20,1.221497826510685e+20,1.2561355728732586e+20,1.2572658975609753e+20,1.2930243895884882e+20,1.3229779938129812e+20,1.30927041 22187216e+20,1.3444350048340441e+20,1.3858259622545863e+20,1.3446649013806984e+20,1.3675683448411279e+20,1.3439177376040722e+20,1.2874877144232316e+20
−24.17585715,9.290607980459816e+19,1.0103282693859806e+20,1.0301472675121324e+20,1.0960126281285745e+20,1.1267325541752485e+20,1.130947324197243e+20,1.1827411003538896e+20,1.2443629538800055e+20,1.2487980414258769e+20,1.2669311315432309e+20,1.2508958474140 97e+20,1.2798915493608643e+20,1.2941738723217593e+20,1.2804567117047226e+20,1.306636180954975e+20,1.2694216774653187e+20,1.2256742804415706e+20
−26.171571450000002,8.952516371536413e+19,9.614560951763986e+19,1.0064583441839676e+20,1.0307507459470998e+20,1.0844890637275303e+20,1.1241845341164972e+20,1.1066549224341106e+20,1.1611404039911676e+20,1.2098497348135364e+20,1.2136238697877771e+20,1.2339026610072371e+20,1.2527924939239945e+20,1.2050123283110201e+20,1.2144189286782897e+20,1.1938144506844027e+20,1.2199556038435462e+20,1.1210809307366647e+20
−28.167285749999998,8.29711963311626e+19,8.96640595456344e+19,9.21406200944673e+19,9.64502224419567e+19,1.0267179773578733e+20,1.0455579073206298 9e+20,1.0770844791207079e+20,1.100150765968351e+20,1.107660719825723e+20,1.134817249399256e+20,1.1565328940353054e+20,1.1593108106407109e+20,1.1500766326834319e+20,1.120132607481716e+20,1.1334378701193306e+20,1.1463695508686322e+20,1.06001466053163e+20
−30.163,7.701879157737302e+19,8.037815486535821e+19,8.745666953684253e+19,9.081814020983874e+19,9.11248605191666e+19,9.392250991149326e+19,9.910840126264648e+19,1.0171293755578353e+20,1.028068619569467e+20,1.1013098277243996e+20,1.0448031723613405e+20,1.0827744186504007e+20,1.0670935583640255e+20,1.0861845507591053e+20,1.0583766476367186e+20,1.0215357260353754e+20,1.0032589505763626e+20
−32.15871425,7.2695203856628785e+19,7.561096259979906e+19,7.978665020886239e+19,8.057375851046987e+19,8.751740054125036e+19,9.046429110844673e+19,9.30650915827007e+19,9.572145038906275e+19,9.494717797797678e+19,9.602204012381318e+19,9.806716148675826e+19,1.0003086115609662e+20,9.945803559401647e+19,9.698185820609467e+19,9.463298603088265e+19,9.788899166310121e+19,8.74784139185469e+19
−34.15442855,6.124760110620825e+19,6.68872507663203e+19,7.267930267881853e+19,7.460823049547545e+19,7.727474306598142e+19,7.931258437161578e+19,7.967850304170711e+19,8.396348730065899e+19,8.494236763826723e+19,8.675251557248611e+19,8.647539444353997e+19,8.870232565879746e+19,8.830786150082988e+19,8.777335202985876e+19,8.510147520659753e+19,8.565475956221208e+19,8.059971766219625e+19
−36.15014285,5.088175739802375e+19,5.6746130932490306e+19,5.9018658296167555e+19,6.08250704115033e+19,6.329071087437011e+19,6.560021326596759e+19,6.676818351319894e+19,6.5908274638484275e+19,6.7636713508413145e+19,6.894779435593676e+19,7.238484351864554e+19,7.203999869866417e+19,7.343786549254977e+19,7.1736247886397235e+19,7.263974131474844e+19,7.263207809652664e+19,6.7213416491886e+19

C.115 "NBSR_upper_Cycle13_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.024194934222178e+20,1.1084155880934194e+20,1.1685198216183885e+20,1.2046718076116496e+20,1.2547501771851524e+20,1.2
882980426715392e+20,1.33547791289826e+20,1.3479324022789638e+20,1.3618304802015219e+20,1.3780156552648999e+20,1.417286924734874
e+20,1.4256993661983857e+20,1.4296112080902339e+20,1.458556573637297e+20,1.4273297720809506e+20,1.4170038681580395e+20,1.35648637
20308924e+20
24.15442855,1.214612754590066e+20,1.3246934573208776e+20,1.3739113349008045e+20,1.445139691895357e+20,1.465672615978908e+20,1.482
4635321167115e+20,1.5533239155613798e+20,1.5727132910745207e+20,1.6020945637499086e+20,1.6565150212120474e+20,1.664610439309505
e+20,1.671047145866714e+20,1.6996811491792595e+20,1.7085917702179992e+20,1.730205970425057e+20,1.70882953774254e+20,1.64826675256
30994e+20
22.158714250000003,1.3794365992806077e+20,1.4804085413688954e+20,1.5254881317955066e+20,1.5920460552722953e+20,1.653729744495999
e+20,1.6961202974426854e+20,1.739507209539828e+20,1.7403224124811107e+20,1.7805051241284908e+20,1.842856826873523e+20,1.878567244
606923e+20,1.8857851873161943e+20,1.9087297534343728e+20,1.9205671794775782e+20,1.9050330345409187e+20,1.9178271918138235e+20,1.8
12983035754444e+20
20.163,1.4973466469266525e+20,1.623442690774742e+20,1.683263867722877e+20,1.7382844051279046e+20,1.7554772616048126e+20,1.8315572
08326303e+20,1.8822752857634508e+20,1.8984547996952923e+20,1.9534017423903428e+20,1.9843115205806352e+20,1.9923503273627258e+20,
2.054300089768654e+20,2.0683566793742392e+20,2.087836632991967e+20,2.1221544123673428e+20,2.104876638917385e+20,2.00636728904755
4e+20
18.167285749999998,1.6131847204302714e+20,1.7423038085189498e+20,1.81275092936144e+20,1.8999493384209767e+20,1.927694544082263e+
20,1.9741271449461414e+20,2.03959247003635e+20,2.0979757195741676e+20,2.170172130061493 2e+20,2.1590083786711525e+20,2.2181105919
141288e+20,2.247180502354999e+20,2.259295323843502e+20,2.2990138227048736e+20,2.2739859601811954e+20,2.261180480645218e+20,2.1786
921330242018e+20
16.171571450000002,1.6634272628183378e+20,1.8403659289973824e+20,1.9388130064203094e+20,2.001878011738964e+20,2.1129097845679387
e+20,2.144787616251004e+20,2.176093673648864e+20,2.2435630393030587e+20,2.2865989612449264e+20,2.3392701290622304e+20,2.372472665
524879e+20,2.444272796804637e+20,2.4260779200457356e+20,2.46207139435597e+20,2.442693341105902e+20,2.4337204476202585e+20,2.33656
97693192323e+20
14.17585715,1.771753014772778e+20,1.9578287472520295e+20,2.0578722637683012e+20,2.158787594541222e+20,2.1843419422978117e+20,2.27
93300683518253e+20,2.3286158795101923e+20,2.378043218956976e+20,2.425698624232778e+20,2.4942436048789425e+20,2.5083964337206516
e+20,2.551471983583277e+20,2.5765281517646378e+20,2.6120687355519372e+20,2.5830667586895076e+20,2.5443785857678125e+20,2.43721902
69099295e+20
12.18014285,1.8786068725276797e+20,2.0770012272307546e+20,2.1617143995456874e+20,2.2758541335883016e+20,2.3149951970329317e+20,2.
3713347780860058e+20,2.4599711145558606e+20,2.518654404065122e+20,2.6150747963979163e+20,2.5818779210668042e+20,2.6739788700371
09e+20,2.7302901454325e+20,2.7378987062178025e+20,2.742778601602424e+20,2.73812515147927e+20,2.7092647029052572e+20,2.60053134948
01767e+20
10.18442855,1.9376694578498994e+20,2.135282576400912e+20,2.236469641487594e+20,2.360725817586261e+20,2.4364151462317202e+20,2.487
3879745880192e+20,2.5829818417164573e+20,2.636451229080433e+20,2.674799734109928e+20,2.7441316120396913e+20,2.830475190237189e+
20,2.8442940123182337e+20,2.895340435384509e+20,2.9020092483347222e+20,2.8880319145706507e+20,2.8269426441582982e+20,2.713578485
1362097e+20
8.18871425,2.0096167785496104e+20,2.19276004489286e+20,2.3365697693192323e+20,2.4165728801956443e+20,2.5162993733458615e+20,2.616
524046071307e+20,2.6547593284700678e+20,2.7361097886522107e+20,2.7618962428018042e+20,2.8141654702800034e+20,2.849309774859735e
+20,2.9028640791967613e+20,2.9773928758772e+20,3.00846116575052e+20,3.000150624654668e+20,2.948628666539311e+20,2.82167213069764
6e+20
6.193,2.0779353139343075e+20,2.2636713785213583e+20,2.377329916383354e+20,2.4814947366583314e+20,2.571597306196187e+20,2.64783010
3469167e+20,2.7233892260892826e+20,2.771378638125749e+20,2.8824387166124075e+20,2.891858839489449e+20,2.9873734507763735e+20,2.99
38554463858767e+20,3.0598585789720704e+20,3.0408937883241795e+20,3.093163015802379e+20,3.007849763544558e+20,2.89576502024976e+
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4.19728575,2.0783089486157287e+20,2.2853988013591495e+20,2.4244588364262444e+20,2.5308824481843583e+20,2.6473375850254762e+20,2.7
04067784154582e+20,2.7504947238869238e+20,2.8461508634622657e+20,2.8817537196964685e+20,2.93596471529175e+20,3.011167186625054e
+20,3.0398351567268204e+20,3.096361555120605e+20,3.099820506489519e+20,3.0686616385116124e+20,3.059264160160718e+20,2.9008826831
589222e+20
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01061891713e+20,2.7293220919397273e+20,2.8212758514900782e+20,2.9060286517257673e+20,2.9224346109190763e+20,2.994432881802618e+
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599716e+20
0.20585714999999993,1.8867702242035774e+20,2.073264880416544e+20,2.235609149494018e+20,2.3220150001384194e+20,2.3887597409559177
e+20,2.4613297861246642e+20,2.5240211567618982e+20,2.6095891599388698e+20,2.6324771147416814e+20,2.696623396183843e+20,2.73832329
1083054e+20,2.8066871155200442e+20,2.848166226289325e+20,2.8357004146455475e+20,2.8133106394179646e+20,2.7990162822878383e+20,2.6
57420060292309e+20

C.116 "NBSR_lower_Cycle13_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,1.9999815326741748e+20,2.1912541839041026e+20,2.3496073552482143e+20,2.454666634305988e+20,2.5426745851752704e+20,2.605405583733261e+20,2.683563165728714e+20,2.744997765164804e+20,2.7959366267318823e+20,2.8739583415704555e+20,2.879166582584204e+20,2.9283674767695204e+20,2.9615530298375596e+20,2.9839541273282164e+20,2.97169211641976e+20,2.9661498686453475e+20,2.8143013374368835e+20
−12.20157145,2.214532756782945e+20,2.4091681201456624e+20,2.5683987668779604e+20,2.6868522831515268e+20,2.8045472077991782e+20,2.8976162102622562e+20,2.963024924037097e+20,3.0447263743745144e+20,3.1328022588222374e+20,3.1819918307444805e+20,3.2463872019742556e+20,3.295004999611294e+20,3.3133583880532224e+20,3.310148526471923e+20,3.2991715924222935e+20,3.281774935210065e+20,3.097929688556267e+20
−14.197285749999999,2.3013971590818172e+20,2.494741784454171e+20,2.656078372118115e+20,2.7560256493982633e+20,2.86877840621439e+20,2.9500213048973355e+20,3.0559523982117575e+20,3.1497007364592376e+20,3.178261145061806e+20,3.222853878176263e+20,3.3177740706518355e+20,3.3776122309945806e+20,3.381054198968884e+20,3.363606591572826e+20,3.385050957833783e+20,3.3287283601753185e+20,3.219451538122715e+20
−16.192999999999998,2.2806038229475787e+20,2.507117017993361e+20,2.659610918197006e+20,2.7656722175367725e+20,2.8730752050507324e+20,2.974505698793492e+20,3.039133176416271e+20,3.109410463312661e+20,3.187381227967404e+20,3.258575618172735e+20,3.295061610926661e+20,3.368424214510543e+20,3.380176723580698e+20,3.377957560018319e+20,3.4062971844909564e+20,3.342666066018633e+20,3.2172493579549455e+20
−18.188714249999997,2.2699212677378566e+20,2.484540425425067e+20,2.6226267458678525e+20,2.7505909631230473e+20,2.844690291525801e+20,2.9441846782830146e+20,2.990028521467078e+20,3.088305764943905e+20,3.159788872857608e+20,3.236225470865909e+20,3.267837229366675e+20,3.3555791070538085e+20,3.327913157234036e+20,3.380839075970491e+20,3.3795426768485895e+20,3.317853326493349e+20,3.1928385587687666e+20
−20.18442855,2.228776163729241e+20,2.4324976432083357e+20,2.5897808606920144e+20,2.6806590052503952e+20,2.806692776651581e+20,2.836402394560965e+20,2.9685501884169 01e+20,3.0081667869106124e+20,3.0969559739319576e+20,3.14607761227576e+20,3.1818786081137466e+20,3.236400965943546e+20,3.331128679946872e+20,3.313930162338428e+20,3.3371521239019035e+20,3.252886180978369e+20,3.0767513954775335e+20
−22.180142850000003,2.1930374403381576e+20,2.4165842024587176e+20,2.5135424022874967e+20,2.5979725180255955e+20,2.7006767663641084e+20,2.816905457943758e+20,2.83981039614118e+20,2.943963894153084e+20,2.965164831757964e+20,3.042241137629911e+20,3.11912496502 96095e+20,3.1403202415029525e+20,3.199196009484462e+20,3.207999069024005e+20,3.184437439568328e+20,3.134970472200787e+20,3.0193701662217083e+20
−24.17585715,2.0878309718604307e+20,2.2704534141023053e+20,2.436233990022546e+20,2.5150142964870342e+20,2.602144771968131e+20,2.6933116342348828e+20,2.7456714398176693e+20,2.8112782931962947e+20,2.878272123801 4083e+20,2.9472077225236034e+20,2.9916589273496 43e+20,3.043848898986328e+20,3.0822426930681165e+20,3.1111880586151795e+20,3.087162216373494e+20,3.094153713821298e+20,2.946760493132206e+20
−26.171571450000002,2.0017138389244006e+20,2.1614822931526835e+20,2.2915128234187676e+20,2.3761014508398933e+20,2.47941710138436 88e+20,2.5140179373365785e+20,2.6030109250932438e+20,2.6764697679132495e+20,2.7723523527500585e+20,2.8119802735068437e+20,2.8510647256361068e+20,2.877513532175492e+20,2.916003565493404e+20,2.96633102485452e+20,2.925553894395789e+20,2.91358626232724e+20,2.785961712964 2462e+20
−28.167285749999998,1.8757253565755077e+20,2.0486049914427503e+20,2.1808716686658247e+20,2.2540248103828495e+20,2.3101266239113835e+20,2.427657375744471e+20,2.4793095398851713e+20,2.542493428966097e+20,2.5658455965549163e+20,2.6363549898443096e+20,2.6895583040260624e+20,2.720071803008786e+20,2.7331546779900618e+20,2.7837991607172327e+20,2.750273939756993e+20,2.7050075319896713e+20,2.6035826993784485e+20
−30.163,1.778670917510605e+20,1.9406132462489747e+20,2.061167042322651e+20,2.091867358646086e+20,2.1692323822264035e+20,2.2033010718141645e+20,2.328474351221775e+20,2.347971288234113e+20,2.406507388323421e+20,2.4229926033582434e+20,2.5130725283699524e+20,2.515529459456873e+20,2.5446389978185e+20,2.5845386528890454e+20,2.5842555963122113e+20,2.5530118113612546e+20,2.4310257490087972e+20
−32.15871425,1.6261430505177403e+20,1.8031779559329084e+20,1.8659259378855087e+20,1.9232109279052094e+20,2.0012892540591492e+20,2.0754840439789237e+20,2.1216222660028947e+20,2.1743104172148084e+20,2.2234660223478317e+20,2.2678549547269672e+20,2.2710195272559732e+20,2.333342924343323e+20,2.368600451553788e+20,2.3384945540417044e+20,2.3812360971436655e+20,2.3467654672167997e+20,2.2676737985177934e+20
−34.15442855,1.4602435908352285e+20,1.595340833826645e+20,1.6728360634323062e+20,1.7432775231432596e+20,1.825329963635951e+20,1.8224937367360723e+20,1.880933597589257e+20,1.956571976050886e+20,1.9931938359616917e+20,2.030059124528575e+20,2.0532301359082206e+20,2.0847909442252315e+20,2.107107124742838e+20,2.1044294095259864e+20,2.108341251417835e+20,2.1030254489048895e+20,2.0455932694652345e+20
−36.15014285,1.2035905314881431e+20,1.3125503301746916e+20,1.4146148706495591e+20,1.458879258134888e+20,1.5016094789737755e+20,1.56684269767098e+20,1.5934953049456863e+20,1.637810642614845e+20,1.663766930710539e+20,1.725863882536421e+20,1.744466360765963e+20,1.7585569171607683e+20,1.8152305049745074e+20,1.8120772547085746e+20,1.8262470669448937e+20,1.7872362095256073e+20,1.7017078342693922e+20

C.117 "NBSR_upper_Cycle13_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0617452197050001e+20,1.1661195018468349e+20,1.215705352976646e+20,1.2938629349720991e+20,1.3181944783167649e+20,1.3415466459055846e+20,1.372292251281313e+20,1.4139864850489875e+20,1.4363479546188874e+20,1.4627344887113694e+20,1.4797461889791033e+20,1.49982622253972e+20,1.5093029567321283e+20,1.5248314405372513e+20,1.5101238208049473e+20,1.5022152200482003e+20,1.4474437724307872e+20
24.15442855,1.2529046483041935e+20,1.361847463596132e+20,1.4179889050454226e+20,1.4777817763358743e+20,1.527056265231168e+20,1.56455560053016e+20,1.611050473840941 8e+20,1.6465514297074844e+20,1.6788028960719707e+20,1.7308909673409957e+20,1.7491481165468005e+20,1.7422075692828262e+20,1.7882325686760636e+20,1.7941427900003615e+20,1.8217974175570605e+20,1.7929992414299514e+20,1.743826652902318e+20
22.158714250000003,1.3938328567783938e+20,1.5228273999732654e+20,1.5757419964466466e+20,1.6606646306284365e+20,1.6790406635965114e+20,1.7912895797058725e+20,1.8202632509106192e+20,1.8589004736484842e+20,1.8754139943409903e+20,1.8927313957117 05e+20,1.9397301 09729252e+20,1.95810048156579e+20,1.9914898353691496e+20,1.9938335438253366e+20,2.0321933711179045e+20,2.0092771106574094e+20,1.942775798495988e+20
20.163,1.5269770093896545e+20,1.669190294722682e+20,1.7526466958364708e+20,1.8475046158651402e+20,1.898013231435431e+20,1.9281644179998078e+20,1.9679565115711563e+20,2.012039742847311e+20,2.0740178109109223e+20,2.096962377029101e+20,2.149701478424845e+20,2.182117117603895e+20,2.1920354200561648e+20,2.2107794265741237e+20,2.203063304289624e+20,2.2110058718355908e+20,2.1438252238897673e+20
18.167285749999998,1.6690431053027282e+20,1.7924614339339662e+20,1.8778256363756175e+20,1.958247670985744e+20,2.0392811078018325e+20,2.0778617192243308e+20,2.1368280653104264e+20,2.1793941133347504e+20,2.2434837834615446e+20,2.2840005018695886e+20,2.3310558272025028e+20,2.3679380991639958e+20,2.3848026100117763e+20,2.4164200296441538e+20,2.4343771388785138e+20,2.4084378341774305e+20,2.3065374665171262e+20
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6.193,2.1807471237720175e+20,2.3959663614021163e+20,2.527151762501684e+20,2.583723449947763e+20,2.6875372800674657e+20,2.783187758511271e+20,2.8644872685095836e+20,2.9461547520577805e+20,3.02544456036057e+20,3.089919187431859e+20,3.168824038790155e+20,3.26741264450149 9e+20,3.277223385454572e+20,3.292004599896852e+20,3.2915686927685275e+20,3.256769717212534e+20,3.120030746075479e+20
4.19728575,2.179145023547136e+20,2.3875595810701407e+20,2.5028145580254813e+20,2.666104236069582e+20,2.747624530197825e+20,2.847124578086575e+20,2.9194002444154143e+20,2.9998222790255405e+20,3.102271776444903e+20,3.130373633393e+20,3.1856206160594954e+20,3.2358574973160246e+20,3.313805617444621e+20,3.3579624434307524e+20,3.349629257808754e+20,3.281316383555594e+20,3.117081296544867e+20
2.20157145,2.148908920009709e+20,2.335465848669579e+20,2.517176848734048e+20,2.6101835787502217e+20,2.712701009748024e+20,2.834885211704265e+20,2.9258199675780134e+20,2.9706391459539367e+20,3.07001464894888e+20,3.113537428202903e+20,3.205536476805547e+20,3.292163111579879e+20,3.3020870751636855e+20,3.349114094838916e+20,3.335850063648466e+20,3.2538768789972884e+20,3.0971258078780575e+20
0.20585714999999993,1.9876006380034484e+20,2.2037256566794165e+20,2.31904290608166e+20,2.4564215850823595e+20,2.5865087266638112e+20,2.668198854738155e+20,2.691058503883283e+20,2.783255692089711e+20,2.8294505254290493e+20,2.9056153891235896e+20,2.9370912804675505e+20,3.016109354456579e+20,3.0508970077494995e+20,3.061409729013121e+20,3.0322775461253474e+20,2.96392504395143e+20,2.8247631085166753e+20

C.118 "NBSR_lower_Cycle13_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.1182086036862747e+20,2.348226039153264e+20,2.5177825898084732e+20,2.602625968148749e+20,2.687565585725149e+20,2.81022532273047 18e+20,2.8482001930785453e+20,2.991172070037488e+20,2.9938158184651194e+20,3.0883453928646613e+20,3.134772332597003 6e+20,3.200243318818748e+20,3.216168081831439e+20,3.240748714963719e+20,3.2609985824704365e+20,3.1832882298663816e+20,2.991834422 4272803e+20
−12.20157145,2.3523869708327263e+20,2.568998846820849e+20,2.7374797824840886e+20,2.8403991538209954e+20,2.954940828202714e+20,3.0 838674378191444e+20,3.143054568035171e+20,3.265663354856663e+20,3.3176778314157117e+20,3.383601708160392e+20,3.47991453899399e+ 20,3.5349577209251634e+20,3.547967001196462e+20,3.595588439683044e+20,3.621833445487109e+20,3.473953367485861e+20,3.344177588138 928e+20
−14.197285749999999,2.388674823982868e+20,2.6316562506688627e+20,2.7910906981364813e+20,2.9017035472317412e+20,3.048145697822671 5e+20,3.1482514867858466e+20,3.1911911694915915e+20,3.295837185947 1866e+20,3.389070361224828e+20,3.437637208678037e+20,3.51660999 36147715e+20,3.571149334839181e+20,3.6340784730009554e+20,3.6544925133222366e+20,3.658330760504108e+20,3.6110320065151166e+20,3.3 98881102177901e+20
−16.192999999999998,2.451666234591546e+20,2.6673496850076523e+20,2.8071852950952726e+20,2.9315999828769672e+20,3.053993646700065 6e+20,3.1378633104160326e+20,3.199847039611181e+20,3.291415842217037e+20,3.4334536324724274e+20,3.4486594317799594e+20,3.53995650 0072055e+20,3.592678618073189e+20,3.671860864876781e+20,3.662820037812698e+20,3.6675697271719756e+20,3.5920728769987635e+20,3.495 2731888530116e+20
−18.188714249999997,2.404197646656454e+20,2.6237306665175056e+20,2.7743733767086545e+20,2.8668706048865272e+20,2.962838106696387 e+20,3.0564392555239126e+20,3.177128918754469e+20,3.254108985390292e+20,3.3658370773982786e+20,3.468252608028421e+20,3.4935635271 289333e+20,3.517917714999746e+20,3.622507120139974e+20,3.6094751953425295e+20,3.6268661914232206e+20,3.5944562133757074e+20,3.431 143890805461e+20
−20.18442855,2.3521265587820387e+20,2.5708500368333446e+20,2.7381874239261734e+20,2.8286409836193032e+20,2.9498231652935513e+20, 3.0334890282741976e+20,3.1109276465644916e+20,3.1846016123828915e+20,3.243856676177359e+20,3.3520408998433817e+20,3.424741151037 4715e+20,3.468020501635417e+20,3.535303049948901e+20,3.6052010410323326e+20,3.567809267232538e+20,3.5262169338325244e+20,3.382712 910509133e+20
−22.180142850000003,2.266105665082132e+20,2.4901166399887003e+20,2.6742732488770165e+20,2.7557992041367958e+20,2.860705632643078 6e+20,2.9700277437479754e+20,3.0389690036017077e+20,3.101145211269102e+20,3.1813804285385186e+20,3.196671144819101e+20,3.26759946 18422093e+20,3.344970146554063e+20,3.4402526514479845e+20,3.4507200836593123e+20,3.4626197821494205e+20,3.382520432036885e+20,3.2 77353591479915e+20
−24.17585715,2.2030010318427206e+20,2.3832401377076517e+20,2.5153539643792356e+20,2.6190149439474483e+20,2.7417482756627477e+20, 2.823642204472412e+20,2.9126125477029305e+20,2.9787741919721515e+20,3.076507966821456e+20,3.107814024219317e+20,3.20908600627904 8e+20,3.2007584817885866e+20,3.270769695502752e+20,3.326718658479796e+20,3.347851662506235e+20,3.2844017002430864e+20,3.133917501 734964e+20
−26.171571450000002,2.106297582933092e+20,2.2678889215161876e+20,2.4453937008489002e+20,2.5135707079451804e+20,2.594394682894411 4e+20,2.6923718863997934e+20,2.758946793271192e+20,2.8713768655897274e+20,2.8693105525788377e+20,2.9389254870854356e+20,3.0279014 91447491e+20,3.051491426560851e+20,3.095552013310859e+20,3.123433086129026e+20,3.170205354885105e+20,3.109178356919657e+20,2.9861 61968627523e+20
−28.167285749999998,1.9619726955368815e+20,2.1486881358797786e+20,2.2769523931064182e+20,2.3779130129316325e+20,2.45658009676538 7e+20,2.5050846717716916e+20,2.565387044900445e+20,2.6267367273634847e+20,2.7148465786004272e+20,2.758641092168211e+20,2.84041047 60840684e+20,2.9014431351810536e+20,2.9289449121862625e+20,2.9526084420096e+20,2.9832351636230577e+20,2.9434940202355393e+20,2.81 46749721183053e+20
−30.163,1.858045642786445e+20,2.0160817907645032e+20,2.1272041416980644e+20,2.2178954689157353e+20,2.294807601973118e+20,2.347909 0157872094e+20,2.4084718009666503e+20,2.4645283254428906e+20,2.571325571882426e+20,2.583553616001662e+20,2.6528005769583755e+20, 2.6906282578864954e+20,2.737355237590281e+20,2.7269783834835406e+20,2.7770284473993593e+20,2.767410184918534e+20,2.6652210995498 595e+20
−32.15871425,1.6838639476657655e+20,1.8647144557366585e+20,1.9718796757260783e+20,1.990408559245643e+20,2.0956659779072007e+20,2. 1415268044858737e+20,2.217697329311951e+20,2.2821832786463136e+20,2.3183069589818917e+20,2.3864273547628046e+20,2.4449011824052 086e+20,2.4597786360836135e+20,2.5200640258177568e+20,2.532546820856144e+20,2.562952758339671e+20,2.509313537029594e+20,2.4143537 166332648e+20
−34.15442855,1.5187627075299266e+20,1.677359307530116e+20,1.7687752595844822e+20,1.8164533093864312e+20,1.8622292189920543e+20,1. 9385865611588 42e+20,1.9992342633113328e+20,2.042366424489325e+20,2.0765483367078203e+20,2.120733468351635e+20,2.154915380570130 4e+20,2.2416495768436595e+20,2.2484938848715096e+20,2.2696382111610228e+20,2.283875956975782e+20,2.2477183098509843e+20,2.1716723 29918714e+20
−36.15014285,1.2616963855806631e+20,1.371097752527073e+20,1.4508914015366275e+20,1.5353498229324094e+20,1.5778139705890724e+20,1. 6228256274372438e+20,1.6617572290150164e+20,1.704951662639912e+20,1.7827412710854807e+20,1.81828185487278e+20,1.833538604364142 e+20,1.8554811502003274e+20,1.914685263810964e+20,1.9134737816621138e+20,1.923058077353719e+20,1.9068955468164874e+20,1.827039625 3600295e+20

C.119 "NBSR_upper_Cycle13_cladding_fluence_position_3.csv" File

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26.15014285,1.058116434389986e+20,1.1511741145899903e+20,1.2225949500567898e+20,1.2628399341510731e+20,1.275973759316179e+20,1.33
66950561786467e+20,1.3658385613294936e+20,1.3855902492609828e+20,1.405336276060935e+20,1.4216459960181203e+20,1.445677499391341
9e+20,1.4854469484365437e+20,1.485724343881841e+20,1.4610021824611444e+20,1.4562807987595502e+20,1.4580131050097752e+20,1.3969747
847812532e+20
24.15442855,1.2819519142189169e+20,1.3695069745652648e+20,1.4246690402587093e+20,1.499073292045341e+20,1.521525339719828e+20,1.57
81140105605172e+20,1.6236464915100628e+20,1.6557054794023014e+20,1.652608840451736e+20,1.7030268779174396e+20,1.700552963435909
e+20,1.709831558024533e+20,1.7254392976711696e+20,1.7635613574391965e+20,1.743051077881792e+20,1.7080143348012578e+20,1.650282115
390159e+20
22.158714250000003,1.4092877458735402e+20,1.5289017941121266e+20,1.6158454522525126e+20,1.6614798335697188e+20,1.696069347258855
3e+20,1.7484687807623984e+20,1.8125301452315096e+20,1.848585891988647e+20,1.8720456210766637e+20,1.8982057099076783e+20,1.9041046
089689026e+20,1.9532432307073158e+20,1.9852116404949677e+20,1.9799694326919984e+20,1.949212505053197e+20,1.979358030486037e+20,1.
8541734288153533e+20
20.163,1.5344553641496134e+20,1.668861949093554e+20,1.7420094296790424e+20,1.8158192626543224e+20,1.869260344360615e+20,1.9079711
618084576e+20,1.9554680554012325e+20,1.9894405057528706e+20,2.040594490318343e+20,2.0679717224297447e+20,2.1009647970255362e+20,
2.1160290680446514e+20,2.125120845292565e+20,2.1725894332276566e+20,2.141815522194245e+20,2.1394265246857645e+20,2.0155439832685
183e+20
18.167285749999998,1.683745063903495e+20,1.7856907206160928e+20,1.882841398917119e+20,1.9633426893687593e+20,2.0279701669915387e
+20,2.0542378173217505e+20,2.1324689940271802e+20,2.158900817171956e+20,2.1817661274486207e+20,2.2806321286052623e+20,2.29385087
0743418e+20,2.2889539919641874e+20,2.330670870258008e+20,2.3340675491800184e+20,2.3322616482198163e+20,2.3020425280769992e+20,2.2
150365974897095e+20
16.171571450000002,1.7516616589490877e+20,1.895171343404016e+20,2.020599373730777e+20,2.0929882626903494e+20,2.161923861412545e+
20,2.2185634824370638e+20,2.2755654158799307e+20,2.3582632253678033e+20,2.3996347746378875e+20,2.4085963458604568e+20,2.46816843
30209785e+20,2.474056009819129e+20,2.5198658862139726e+20,2.5118497239580282e+20,2.5156766488768266e+20,2.4307087256427436e+20,2.
3527436221195374e+20
14.17585715,1.8819412790027865e+20,2.0166931929704648e+20,2.12678521796435e+20,2.221818633070657e+20,2.2775128451285497e+20,2.336
6660085553558e+20,2.4173031661638762e+20,2.514221738071899e+20,2.557297287934524e+20,2.5847367924928292e+20,2.614344510429684e+
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261147e+20
12.18014285,1.967073375051434e+20,2.1330520905754588e+20,2.2137345372362726e+20,2.3285705904578986e+20,2.4452804782181666e+20,2.5
047959540633207e+20,2.5499095112791523e+20,2.639106299771138e+20,2.671867267973926e+20,2.7158259543562735e+20,2.775896221092023
e+20,2.7955969588396812e+20,2.844843142077292e+20,2.830401595527212e+20,2.8145730717506447e+20,2.7612678572012326e+20,2.635018962
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10.18442855,2.0123624273449022e+20,2.245731252681608e+20,2.36047106666711e+20,2.5048921932994445e+20,2.5514550001886665e+20,2.618
51676437222e+20,2.6753205582113027e+20,2.765253293803058e+20,2.821383412989275e+20,2.8783627019059954e+20,2.8614868687951417e+2
0,2.956644828795255e+20,2.933439850626389e+20,2.9616945581259765e+20,2.9590960987506396e+20,2.864209873064286e+20,2.743248475519
969e+20
8.18871425,2.1061164267239185e+20,2.263767617757482e+20,2.4127063273560896e+20,2.5171938321286575e+20,2.6269688337564887e+20,2.70
57208345632935e+20,2.7891828968086192e+20,2.8381120566801754e+20,2.9156412530750564e+20,2.916428150358655e+20,2.992904376287713
e+20,3.030369744797485e+20,3.059507588816796e+20,3.045977484444122e+20,3.034830716448392e+20,2.9815651298197366e+20,2.85825436268
7695e+20
6.193,2.160989774708992e+20,2.371317794691396e+20,2.502276750529497e+20,2.5954136865710142e+20,2.6900225168120703e+20,2.748790723
2943813e+20,2.8718354172441988e+20,2.9081742205781705e+20,3.0013394622773717e+20,3.0602718415742475e+20,3.08465999623428e+20,3.10
39418102482246e+20,3.165070708581334e+20,3.1423695711192325e+20,3.1424827937499664e+20,3.109585958390298e+20,2.9034754814027232
e+20
4.19728575,2.1744066564509322e+20,2.445846591371835e+20,2.5462750648326013e+20,2.635511481245344e+20,2.7782172850220627e+20,2.837
240242423526e+20,2.9040302722933185e+20,2.976056848834543e+20,3.037451820349876e+20,3.079927290269613e+20,3.1089009614743595e+2
0,3.169084450840842e+20,3.19192711659136e+20,3.2048967689419024e+20,3.1687051550278844e+20,3.0978674161093634e+20,2.949081557062
246e+20
2.20157145,2.1491749931919334e+20,2.386795328312689e+20,2.5125290597424303e+20,2.6279199038546513e+20,2.7177054500264523e+20,2.81
1561349773129e+20,2.8852239933284555e+20,2.9255878611850094e+20,2.998910836848135e+20,3.024425556683968e+20,3.110140749280893e+
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6274e+20
0.20585714999999993,1.9890272431506925e+20,2.1907276986711907e+20,2.3211148802240863e+20,2.3900901068670383e+20,2.46803822699563
48e+20,2.5685006672456208e+20,2.6496643100870528e+20,2.67670753543779e+20,2.7750640347561306e+20,2.8255160390110054e+20,2.86200769
28965165e+20,2.8755151527430432e+20,2.8839842055219213e+20,2.8911851648365832e+20,2.8695256755772323e+20,2.8560295379937788e+20,
2.6844406411169e+20

C.120 "NBSR_lower_Cycle13_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.1132154856709194e+20,2.3227452861066514e+20,2.444204863226197e+20,2.5664003874455113e+20,2.678887071079414e+20,2.7484170886129605e+20,2.8144428657253014e+20,2.8407897718970263e+20,2.91699992464386e+20,2.9653856158878944e+20,3.054089885936189 4e+20,3.061177622620117e+20,3.1036983815921474e+20,3.077979861020994e+20,3.075211567699555e+20,2.979136504390499e+20,2.8689425790289537e+20
−12.20157145,2.3411326413377998e+20,2.5763356732923907e+20,2.727091606114274e+20,2.8421484434658306e+20,2.9281580149026636e+20,3.022393210462298e+20,3.1553562068643847e+20,3.214452758975824e+20,3.2596625554277794e+20,3.336693572247432e+20,3.347998851926189 e+20,3.399107547439368e+20,3.433136609106373e+20,3.397250696295336e+20,3.41078080066801e+20,3.310561789074101e+20,3.1781082945103154e+20
−14.197285749999999,2.412802566592213e+20,2.6025863402279925e+20,2.78841298291963e+20,2.900129752664543e+20,3.005726839218302e+20,3.0793668382474817e+20,3.213529994535344 5e+20,3.283914842930931e+20,3.357888848720775e+20,3.433300781920937e+20,3.463525563195 2904e+20,3.4824620481854964e+20,3.562657637534156e+20,3.544298587960691e+20,3.4890855720834164e+20,3.4218256682960794e+20,3.27799 32993435605e+20
−16.192999999999998,2.4357641161050017e+20,2.634260371175737e+20,2.8012297847186814e+20,2.9372214864928942e+20,3.028309092918132 e+20,3.0776005652080355e+20,3.1907099733109735e+20,3.280631386639655e+20,3.361964863427187e+20,3.396707227667815e+20,3.4609044592 93806e+20,3.495171288485351e+20,3.560121450605722e+20,3.555111349195757e+20,3.512647201539094e+20,3.4142227686423134e+20,3.256447 032714943e+20
−18.188714249999997,2.4283480337919463e+20,2.642276533431681e+20,2.8304978347633354e+20,2.8868543992110206e+20,3.006338241424264 e+20,3.077368458815032e+20,3.1711224581940485e+20,3.258354834042805e+20,3.350880367878361e+20,3.344132299086634e+20,3.43799385996 48474e+20,3.465699437705377e+20,3.531849759711523 6e+20,3.480899575881372e+20,3.4711737519013495e+20,3.4045252503199744e+20,3.2964 31604758539e+20
−20.18442855,2.3759316168937934e+20,2.5919717185967106e+20,2.7265028484344585e+20,2.811957628980697e+20,2.916071499071844e+20,3.0 30969824740373e+20,3.117426625568605e+20,3.1624439435483126e+20,3.2166096500913006e+20,3.302851327921138e+20,3.3550582829524333 e+20,3.403851575667109e+20,3.395257977994423e+20,3.4021419139430305e+20,3.403376040618027e+20,3.347721456480892e+20,3.20029426900 25787e+20
−22.180142850000003,2.314412100484653e+20,2.5201093148700495e+20,2.6069171058535553e+20,2.7452921440047114e+20,2.884793747331667 4e+20,2.943912943969254e+20,2.995882131476009e+20,3.056812890205334e+20,3.130396277919147e+20,3.1926404191649825e+20,3.2525408519 54631e+20,3.251753954671032 4e+20,3.280676675691948e+20,3.3257109770662655e+20,3.277529086557552e+20,3.243324529812 91e+20,3.126942 98768177e+20
−24.17585715,2.187483870300671e+20,2.375439098450102e+20,2.5089172578220265e+20,2.658422080574302e+20,2.7373835432479647e+20,2.82 99260604781304e+20,2.8642381787219696e+20,2.9615077407852665e+20,3.0216968912832856e+20,3.13033966660378e+20,3.142341265461549e +20,3.135610180064432e+20,3.214899988367222e+20,3.228305547846089e+20,3.190443900128749e+20,3.114737588088679e+20,2.967072633085 826e+20
−26.171571450000002,2.10555031357025e+20,2.2980118024228805e+20,2.4271308905115596e+20,2.5179071347022794e+20,2.6029599749094136 e+20,2.6814628859286045e+20,2.7666685766872282e+20,2.8306959743671196e+20,2.871122114670577e+20,2.9195021447830746e+20,2.92938648 04461242e+20,2.996867168363391e+20,3.049057140000077e+20,3.0689390339569104e+20,3.0249010917330485e+20,2.9604434480563696e+20,2.8 518516229197057e+20
−28.167285749999998,2.000909958246191e+20,2.1476181820193456e+20,2.3054618515251564e+20,2.4008122899975173e+20,2.478550948259255 7e+20,2.5582200523750035e+20,2.6101496119610012e+20,2.6541196206064232e+20,2.7109913480239458e+20,2.7283313939208074e+20,2.793157 0111473707e+20,2.8669158939388205e+20,2.8977633996822095e+20,2.8843578402033428e+20,2.8915021882026374e+20,2.781953631836274e+2 0,2.717252559503518e+20
−30.163,1.8921482991634268e+20,2.0252301793277837e+20,2.140213421969363e+20,2.213921354576983e+20,2.2903296469276013e+20,2.376554 3413628282e+20,2.3683909896869305e+20,2.4273063855891962e+20,2.5008275008561057e+20,2.595221208098767 2e+20,2.587340912999704e+2 0,2.6667779107224473e+20,2.691720856273075e+20,2.668317738500425e+20,2.653938464397249e+20,2.608734329076831e+20,2.4749051795496 318e+20
−32.15871425,1.7070066533877278e+20,1.8728664851494827e+20,1.9551566931667147e+20,2.038375326755963e+20,2.114732668922751e+20,2.1 783920930527573e+20,2.2542965446966107e+20,2.2874877588961863e+20,2.2939471099795417e+20,2.3683230561084906e+20,2.3735143137276 294e+20,2.4666229441114636e+20,2.447754392699697 2e+20,2.4544571724391304e+20,2.4569480703152718e+20,2.4456597740311246e+20,2.2862 536322211892e+20
−34.15442855,1.5194420433143287e+20,1.6682505468875925e+20,1.76350474612383e+20,1.8479518452565384e+20,1.8850888681371825e+20,1.9 37358095615382e+20,1.9914728519745395e+20,2.0148193584318228e+20,2.0579515196098147e+20,2.1012195479446874e+20,2.15984622613858 2e+20,2.1970341992030565e+20,2.1958283781857427e+20,2.2226168526173297e+20,2.1783920930527573e+20,2.133210602258486e+20,2.0040179 194598307e+20
−36.15014285,1.261645435396833e+20,1.4102954272870698e+20,1.4842977387345972e+20,1.5841317933840115e+20,1.6100427924274122e+20,1. 6381899384278027e+20,1.652869252502423e+20,1.713890589336335e+20,1.7215274557793213e+20,1.7517352536590644e+20,1.77916343595429 66e+20,1.8326328233182727e+20,1.849972869215134 4e+20,1.8424945144551755e+20,1.856788871585301e+20,1.809246688940233e+20,1.7226200 54165901e+20

C.121 "NBSR_upper_Cycle13_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,2.76613643032278e+19,2.898703147517299e+19,3.1274298449939243e+19,3.1014735568982307e+19,3.275440129020516e+19,3.2788
424690740625e+19,3.4974244188369514e+19,3.4845170389333127e+19,3.349951942306345e+19,3.5044272385478287e+19,3.4904046157314638e
+19,3.5655278312232546e+19,3.5149909099952804e+19,3.5468347748891255e+19,3.520974726029555e+19,3.43884302969535e+19,3.2253334537
8933e+19
24.15442855,3.368916732954235e+19,3.4712473466113266e+19,3.5902669760385614e+19,3.8458500815252144e+19,4.031625774033021e+19,4.03
55376159248695e+19,4.253661014033287e+19,4.087212424591716e+19,4.134086593715457e+19,4.2296974442385056e+19,4.103329666076655e+
19,4.5716750781064856e+19,4.382474401018984e+19,4.313935081504357e+19,4.236049233822664e+19,4.023700189881664e+19,3.879692325851
508e+19
22.158714250000003,3.5823583752818147e+19,3.833299352908386e+19,4.103188137788238e+19,4.297257387997287e+19,4.502643240148166e+1
9,4.5505081072908255e+19,4.489730199112991e+19,4.5612302904213045e+19,4.819134459837997e+19,4.915702041590745e+19,4.958868169557
956e+19,4.953427822151204e+19,4.928739627519727e+19,4.954565709590078e+19,4.660905833387691e+19,4.7072761618046665e+19,4.45413866
51418616e+19
20.163,4.040338255467977e+19,4.249981278534443e+19,4.51962097362668e+19,4.755107062158106e+19,4.738757714280164e+19,4.98925712364
6874e+19,5.166750580716514e+19,5.104370572313798e+19,5.157602192153232e+19,5.333074625264276e+19,5.549222288466392e+19,5.49162593
6212172e+19,5.394475257911147e+19,5.365835593467064e+19,5.307752383900691e+19,5.2410812377931686e+19,4.9908026125563896e+19
18.167285749999998,4.332050702421744e+19,4.645020698295757e+19,4.912271395879515e+19,5.055849013912882e+19,5.482811554409556e+19
,5.313028558492879e+19,5.63718495028338e+19,5.627351564804161 5e+19,5.634354384515039e+19,5.768409979303705e+19,5.790997894135072e
+19,5.858874861259908e+19,5.863913268327556e+19,5.89425693336418e+19,5.772033103487183e+19,5.5469691581147914e+19,5.366288483989
999e+19
16.171571450000002,4.565328949653863e+19,4.871998106127548e+19,5.323405412599621e+19,5.569036248844784e+19,5.7645037985433936e+1
9,5.742085717658126e+19,5.8799908818917384e+19,6.1704635410389705e+19,6.082772613535742e+19,6.171312710769473e+19,6.162141677680
046e+19,6.55581676474102e+19,6.4065893374340415e+19,6.250964831490611e+19,6.398493919336586e+19,6.0708276259933405e+19,5.68117194
2323411e+19
14.17585715,4.9622082371646005e+19,5.11954240483211e+19,5.492418494627308e+19,5.708854875537793e+19,6.006007669898314e+19,6.06969
5399686004e+19,6.289573748570793e+19,6.437668949570434e+19,6.588255048446217e+19,6.597086413643443e+19,6.7189705756282405e+19,6.6
8426783930837e+19,6.775751724941177e+19,6.606653725940439e+19,6.561138228385502e+19,6.33808964584017e+19,6.245699979161495e+19
12.18014285,5.051880560705668e+19,5.597913680813407e+19,5.932073292029226e+19,5.96043556102801e+19,6.220055053300319e+19,6.480636
9379338625e+19,6.6154284798222975e+19,6.833608489246082e+19,6.88229422046156e+19,7.047202982125152e+19,7.051278996831565e+19,7.26
9911896778283e+19,7.246644646162514e+19,7.051109162885463e+19,6.974061162671201e+19,6.8290795840167354e+19,6.5284168881034715e+
19
10.18442855,5.494841458925009e+19,5.846080703987004e+19,6.200750594760228e+19,6.495978604398275e+19,6.452218057619711e+19,6.76397
6571344874e+19,6.8879553519982445e+19,7.165633853872573e+19,7.2471541480008155e+19,7.322333974807973e+19,7.461541199295021e+19,7.
392532005862849e+19,7.55047757573632e+19,7.470089507915414e+19,7.203280378591518e+19,6.979212792369583e+19,6.783167807254232e+1
9
8.18871425,5.539943693877767e+19,6.049371937469309e+19,6.303160464258834e+19,6.684041394046902e+19,6.9252622088249885e+19,6.66756
7501275153e+19,7.2699685080936505e+19,7.414950086748116e+19,7.458314354319113e+19,7.427857466651754e+19,7.67355057534382e+19,7.62
0109493637528e+19,7.796963242843523e+19,7.622487168882934e+19,7.580991074719044e+19,7.260627641058122e+19,7.144064942717808e+19
6.193,5.686946296490829e+19,6.122060866400327e+19,6.5192458550140445e+19,6.831060980054574e+19,7.007971340575936e+19,7.2560987358
28776e+19,7.540174316339556e+19,7.6892885210158e+19,7.576009278966763e+19,7.863085259191986e+19,7.898467331296258e+19,7.990856997
974934e+19,7.888673573737795e+19,7.968325694458932e+19,7.83313787336293e+19,7.444274748108136e+19,7.222528225816242e+19
4.19728575,5.703476800577945e+19,6.295008434846009e+19,6.499375283320286e+19,6.694118208182199e+19,7.040466235596498e+19,7.253494
615321901e+19,7.499470780590801e+19,7.496074101668792e+19,7.868067054944267e+19,7.824419730796438e+19,8.030145250839518e+19,8.005
972219177878e+19,8.145688945503229e+19,7.787226096600426e+19,7.693930648875881e+19,7.537117305309748e+19,7.322333974807973e+19
2.20157145,5.798300753817394e+19,6.022991064508363 6e+19,6.396399300668011e+19,6.793244621389529e+19,6.935508856906386e+19,6.90924
1206576175e+19,7.290631638202545e+19,7.533380958495536e+19,7.669304726691308e+19,7.73831392012348e+19,7.981346296993304e+19,7.893
825203436179e+19,7.907242085178117e+19,7.824080062904236e+19,7.646377143967739e+19,7.69540254307542e+19,7.197958914947036e+19
0.20585714999999993,5.144440061330444e+19,5.511473863379785e+19,5.793488792011213e+19,6.134685189727131e+19,6.400645149320524e+1
9,6.318388908092513e+19,6.445311477144957e+19,6.6989867813037466e+19,6.833778323192182e+19,6.874595081571671e+19,7.1555004284219
08e+19,7.097813498063104e+19,7.1971097452165325e+19,7.416365369632286e+19,7.100530841200711e+19,6.956228598330648e+19,6.806208612
608533e+19

C.122 "NBSR_lower_Cycle13_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,5.5031746445470065e+19,5.849307548962913e+19,6.280176270219898e+19,6.329654559850512e+19,6.584348867685905e+19,6.857102185123318e+19,7.076131364277606e+19,7.206224166990593e+19,7.353300364313632e+19,7.538928867401487e+19,7.49018652487064e+19,7.480449378627545e+19,7.775903833527058e+19,7.528682219320087e+19,7.578839844735105e+19,7.441330959709061e+19,7.216187758495157e+19
−12.20157145,6.239019843948209e+19,6.549136629527734e+19,6.988780104666578e+19,7.329240555282727e+19,7.417214539362789e+19,7.785414534508688e+19,7.965608351321326e+19,8.112344880752162e+19,8.195280457764576e+19,8.430217416536944e+19,8.688195180663613e+19,8.584313416965469e+19,8.585728699849641e+19,8.485639894281075e+19,8.575595274398977e+19,8.351074797654108e+19,7.993347895851074e+19
−14.197285749999999,6.343977222638322e+19,6.85993275089166e+19,7.268213557317279e+19,7.594351345145617e+19,7.833647375201231e+19,7.935377908915434e+19,8.309691926120951e+19,8.495603485785639e+19,8.417536481894772e+19,8.610411233349581e+19,8.78879348807048e+19,8.835780879824953e+19,9.089626017929845e+19,8.745825499707051e+19,8.825477620428189e+19,8.408705116697546e+19,8.002971819463436e+19
−16.192999999999998,6.424988014928263e+19,7.093850705987424e+19,7.2642507652416e+19,7.542835048161798e+19,7.775960444842426e+19,7.991423111128601e+19,8.213905580520264e+19,8.50183073047599e+19,8.496735712092976e+19,8.649303207006598e+19,8.685251392264538e+19,8.91758423053003e+19,8.80345581875049e+19,8.915433000546091e+19,8.968081523837247e+19,8.437180608327066e+19,8.437916555426834e+19
−18.188714249999997,6.690155416106521e+19,6.868028168989118e+19,7.0717722929943585e+19,7.487016291210097e+19,8.03473076738423e+19,8.157067819891963e+19,8.264516096458216e+19,8.537042968634163e+19,8.438709113841969e+19,8.679703483358588e+19,8.707329805257605e+19,8.765865905346912e+19,8.80255003770462e+19,8.713160770740388e+19,8.622016552999782e+19,8.482696105882001e+19,8.230549307238115e+19
−20.18442855,6.154782206682357e+19,6.700402064187919e+19,7.135912913304983e+19,7.38302130488122e+19,7.74867379083561e+19,7.953550141148188e+19,7.997763578449687e+19,8.359679717589867e+19,8.348300843201133e+19,8.536759912057328e+19,8.732691674541947e+19,8.654001946182045e+19,8.667758495816186e+19,8.474091185946242e+19,8.37841806297629e+19,8.174277659763481e+19,7.955644759816762e+19
−22.180142850000003,6.235453331080099e+19,6.667963780482722e+19,6.796188409788604e+19,7.1360827472510845e+19,7.3746994415222964e+19,7.586312538363527e+19,7.87265257148898e+19,7.93724608232254e+19,8.002745374201969e+19,8.349999182662138e+19,8.152878582554817e+19,8.157633933045632e+19,8.519889740078011e+19,8.299615111985655e+19,8.194431288034075e+19,7.925584151356971e+19,7.690420747323136e+19
−24.17585715,5.893351152318311e+19,6.155235097205292e+19,6.728934167132803e+19,6.934546464545149e+19,7.0123304118591816e+19,7.458144520373012e+19,7.497262939291494e+19,7.870274896243574e+19,7.871860013073845e+19,7.915111058014108e+19,8.053469112770655e+19,8.189279658335692e+19,8.147160839702766e+19,7.966344298421094e+19,7.686118287355258e+19,7.929263886855817e+19,7.52902188721229e+19
−26.171571450000002,5.441802317557821e+19,6.050390941145912e+19,6.4493308805360026e+19,6.69230664609046e+19,6.922488254372014e+19,6.965852521943009e+19,7.190033330795677e+19,7.467315553462439e+19,7.344016105934715e+19,7.480449378627545e+19,7.721217302882696e+19,7.675928250589228e+19,7.3651321292253e+19,7.634658601686804e+19,7.39966503159907e+19,7.367113525263 14e+19,7.100757286462179e+19
−28.167285749999998,5.357694886317313e+19,5.762012900667252e+19,6.094887435024244e+19,6.1764643404678545e+19,6.3104067126257885e+19,6.543532109306417e+19,6.711328048053717e+19,6.751635304594905e+19,7.16014255628199e+19,7.087793295243173e+19,7.047995540540288e+19,7.104663467222491e+19,7.162010729689095e+19,7.191844892887418e+19,7.072225183517294e+19,6.847478261510957e+19,6.647753540896761e+19
−30.163,4.801737802625769e+19,5.17426856339723e+19,5.4107227054214275e+19,5.7828458647222485e+19,5.864083102273657e+19,6.172614771022911e+19,6.09409487660911e+19,6.327276884605105e+19,6.396908802506313e+19,6.495016212037039e+19,6.6976847210503094e+19,6.533228849909653e+19,6.643903971451817e+19,6.465238660154084e+19,6.558647330509362e+19,6.379529128688694e+19,5.931280733614091e+19
−32.15871425,4.303728061343716e+19,4.802711517250078e+19,5.004445939559796e+19,5.275217860959370 4e+19,5.401489399885097e+19,5.401619605910442e+19,5.7194978028267586e+19,5.842174523226691e+19,5.870593403540843e+19,6.022934453192997e+19,6.058033468720435e+19,6.075186697276587e+19,6.183993645411645e+19,6.150649580660578e+19,6.043088081463591e+19,5.858818249944541e+19,5.3928108852393615e+19
−34.15442855,3.8404720065653645e+19,4.245554273672756e+19,4.357361621522256e+19,4.543403387212288e+19,4.856288466113251e+19,4.779336705135111e+19,5.014539737089703e+19,4.995648541151791e+19,5.236184359013937e+19,5.357825092342657e+19,5.383430390283076e+19,5.396241530950591e+19,5.393880839099794e+19,5.2686849151660384e+19,5.336816633210024e+19,5.1993586983678116e+19,4.716843474101662e+19
−36.15014285,3.2751061222598517e+19,3.5381619213749264e+19,3.717257478669447e+19,3.726162438576651e+19,3.949460110909596e+19,4.039913670602727e+19,4.131629662628536e+19,4.271284116506983e+19,4.234939652041474e+19,4.218109107982914e+19,4.281559070246063e+19,4.430554391160038e+19,4.35735596039072e+19,4.2830139810509914e+19,4.290832003703151e+19,4.153951504277679e+19,3.900836652141022e+19

C.123 "NBSR_upper_Cycle13_cladding_fluence_position_5.csv" File

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26.15014285,1.836835091676335e+20,1.9689777108886418e+20,2.1082950181611166e+20,2.1277404343989548e+20,2.1812967507466174e+20,2.2
322276309967197e+20,2.3233632375561653e+20,2.3648595642272576e+20,2.3913359831836053e+20,2.4070072644472032e+20,2.44968181091989
78e+20,2.4868579983184452e+20,2.5137558942769925e+20,2.488323588803366e+20,2.4824803849092376e+20,2.4769437097439805e+20,2.304061
5066599863e+20
24.15442855,2.2207431665433756e+20,2.3315245649623915e+20,2.4454191457840166e+20,2.5460755171274686e+20,2.6318652451206126e+20,2.
667029837735935e+20,2.7372344956704765e+20,2.8274593112094917e+20,2.785713929946191e+20,2.8789752957055975e+20,2.91960951032673
6e+20,2.912626402722113e+20,2.9619008958883404e+20,2.9949389454471116e+20,2.9410856793933536e+20,2.8748946320024846e+20,2.7792001
944576537e+20
22.158714250000003,2.4093541250276318e+20,2.6107243418511992e+20,2.7424550630840828e+20,2.872068820283193e+20,2.9620828973211084
e+20,2.989373533213523e+20,3.038255286445883e+20,3.1066016139616354e+20,3.1551481013967906e+20,3.253955721344231e+20,3.1836265361
135855e+20,3.291131908742779e+20,3.3490083643629856e+20,3.3027416843488186e+20,3.286227449080821e+20,3.260718511424977e+20,3.1352
428920656436e+20
20.163,2.6711583965529342e+20,2.86459718251693e+20,3.0102749609135047e+20,3.098200810985978e+20,3.1986847599194394e+20,3.23340871
74870073e+20,3.285058808301996e+20,3.39227681024787e+20,3.487022924537753e+20,3.5256455443756666e+20,3.573799291876954e+20,3.6204
29974756658e+20,3.6259474918763594e+20,3.699504807782941e+20,3.600831294154383e+20,3.555972730488473e+20,3.4231691587045347e+20
18.167285749999998,2.920126777556708e+20,3.054692889531661e+20,3.223762641550306e+20,3.327522616273591e+20,3.4349992518344514e+2
0,3.4749437768156276e+20,3.620995137100516e+20,3.667759926299101e+20,3.7650157445566254e+20,3.818303848266524e+20,3.924784265458
551e+20,3.909151300286062e+20,3.961893399697657e+20,3.9896438286833805e+20,3.955992721666865e+20,3.7874593949237464e+20,3.6938819
2141269e+20
16.171571450000002,3.048303681339229e+20,3.288325255069041e+20,3.454808670937826e+20,3.572755178394233e+20,3.672606911824395e+20
,3.7638087876866907e+20,3.886027539301755e+20,3.973311594848152e+20,4.004587104215906e+20,4.119832327249126e+20,4.19688598646940
8e+20,4.2192817417226433e+20,4.240901596130919e+20,4.269006448959401 5e+20,4.2279028622221764e+20,4.135292870011624e+20,3.93944974
9330536e+20
14.17585715,3.251359806171594e+20,3.4404018206808264e+20,3.667740768253547e+20,3.8083608226237284e+20,3.9011432372442713e+20,4.04
26254036644076e+20,4.1065174855887336e+20,4.2427120314358215e+20,4.325915423279105e+20,4.3899895066361994e+20,4.439369369052976
e+20,4.448603547010255e+20,4.5602566365020016e+20,4.537084980403809e+20,4.51757251100653e+20,4.429282658068522e+20,4.173637698188
9974e+20
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6110435996e+20,4.4145597000598716e+20,4.4793234730569295e+20,4.553896165377901e+20,4.59652281673671e+20,4.70484240630197e+20,4.80
56712000554127e+20,4.759816418020667e+20,4.722314543847693e+20,4.73480558954924e+20,4.7097372869411504e+20,4.388926235107924e+2
0
10.18442855,3.519946025823197e+20,3.830315942829209e+20,4.0453841622242584e+20,4.175448133493899e+20,4.372479052999368e+20,4.4842
37511741665e+20,4.6137554787130016e+20,4.725092460453098e+20,4.77067902985008e+20,4.821093426726802e+20,4.9157150137205785e+20,4.
9282347964904584e+20,4.9412431094219774e+20,4.965267298547347e+20,4.920341681721996e+20,4.8399641015980045e+20,4.61364053043967
4e+20
8.18871425,3.636082097974701e+20,3.8619554550625e+20,4.081832343891734e+20,4.319258002448909e+20,4.443842772689957e+20,4.59984673
7640419e+20,4.7349780119592305e+20,4.8506159749263176e+20,4.94606135787894e+20,5.0887696392145646e+20,5.0847560286708924e+20,5.14
4299234254342e+20,5.194742368199396e+20,5.219149718235855e+20,5.171110919007895e+20,5.05196703370433e+20,4.780947742267303e+20
6.193,3.750972897165162e+20,4.0273851784257864e+20,4.254590019679625e+20,4.457588670375326e+20,4.600201161483178e+20,4.7420090146
7774e+20,4.9038657625451076e+20,4.941779534697504e+20,5.0875914194129623e+20,5.118550821029067e+20,5.244878973415577e+20,5.317440
070953324e+20,5.346694406515076e+20,5.39982924586054e+20,5.322631901298599e+20,5.236593118713246 5e+20,4.95176087643141e+20
4.19728575,3.77402002596725e+20,4.1496614041775145e+20,4.378006149141847e+20,4.5188561000586825e+20,4.7165288140902275e+20,4.8484
798528469887e+20,4.9321621958291345e+20,5.00259675031033e+20,5.140122780323457e+20,5.228067788441484e+20,5.346713564560631e+20,5.
384972181533009e+20,5.468692840606265e+20,5.354846154898525e+20,5.360085803576865e+20,5.215279793033842e+20,5.020164678083826e
+20
2.20157145,3.724985008370454e+20,4.053334751129384e+20,4.298346995726149e+20,4.4334974280905156e+20,4.6032760277946774e+20,4.8193
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383018060886447e+20,5.395451632451331e+20,5.367308463531741e+20,5.3281685764638546e+20,5.182538693181166e+20,4.909124646049824e
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0.20585714999999993,3.4595119711214613e+20,3.7849209538877725e+20,3.9699110417622244e+20,4.098030471408081e+20,4.234723126439586
e+20,4.3588193665189944e+20,4.486718478640975e+20,4.595287122798443e+20,4.7111549823121844e+20,4.786254520885905e+20,4.8617180623
25162e+20,4.9003311031402994e+20,4.885656040245536e+20,4.909737703507568e+20,4.942124379517485e+20,4.784118398806576e+20,4.518530
413284256e+20

C.124 "NBSR_lower_Cycle13_cladding_fluence_position_5.csv" File

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86484772526e+20,4.7787924621424 2e+20,4.84473445494108e+20,4.917946926027 6806e+20,5.027186101779558e+20,5.135333268934827e+20,5.20
7702786017029e+20,5.1773851789270005e+20,5.175028739323794e+20,5.1789561386624706e+20,5.0311613962321206e+20,4.826591785800949e
+20
−12.20157145,4.068709082686888e+20,4.415594234519817e+20,4.673145419932014 3e+20,4.841151900422384e+20,5.0177603433667337e+20,5.18
4435339691064e+20,5.389598849534426e+20,5.467351777417448e+20,5.558879340054171e+20,5.682103889060848e+20,5.752873709339246 6e+2
0,5.838941228992931e+20,5.768209724805641e+20,5.794590353534216e+20,5.796582790271887e+20,5.579723293617489e+20,5.32582171588342
7e+20
−14.197285749999999,4.129277243707505e+20,4.509487815782522e+20,4.784961352810975e+20,4.924824664381754e+20,5.1282926871935405e+
20,5.2724474009685315e+20,5.478540076021289e+20,5.5440510127949716e+20,5.728926152396097e+20,5.8170818990152234e+20,5.9077473496
01992e+20,5.928687093393084e+20,6.078617957902763e+20,5.972118382665181e+20,5.925861281673791e+20,5.780547506142752e+20,5.4493432
14596199e+20
−16.192999999999998,4.18761349242102e+20,4.564222351931788e+20,4.8489779620314055e+20,4.986120831133443e+20,5.15161760765617e+20
,5.296452432048346e+20,5.423240377528164e+20,5.566552137298755 6e+20,5.689278577121015e+20,5.7835648983175895e+20,5.8593158104401
65e+20,5.911818434282328e+20,6.037907111099407e+20,6.003575893465705e+20,5.8881965641136046e+20,5.7666100280018384e+20,5.46500491
6837019e+20
−18.188714249999997,4.2128542174391016e+20,4.5368359258116004e+20,4.7446432459414864e+20,4.904881138959498e+20,5.130294702953988
e+20,5.238393974995371e+20,5.404618757249 17e+20,5.476336900782518 4e+20,5.672658972602 4696e+20,5.684833910552367e+20,5.81231154567
2147e+20,5.871883488323929 4e+20,5.885974230829279 5e+20,5.918734488727 51e+20,5.870408318816231e+20,5.7180060664300195e+20,5.504777
0194082044e+20
−20.18442855,4.0965169858092735e+20,4.447760593005857e+20,4.667436322356769e+20,4.798917988997443e+20,4.931367136938623e+20,5.131
693240279468e+20,5.289344797147619e+20,5.425424394721379 6e+20,5.481021042920599e+20,5.575671366982708e+20,5.747269981014549 5e+2
0,5.7071721916689383e+20,5.723734322050821e+20,5.829879473445643e+20,5.78782756345347e+20,5.656987691338871e+20,5.35369667216525
36e+20
−22.180142850000003,4.001684660314397e+20,4.2830492963508657e+20,4.5372382447682454e+20,4.771943460856678e+20,4.8755405921927496
e+20,4.9988609314271986e+20,5.141751214195591e+20,5.1642427596765993e+20,5.3257067676101e+20,5.4452433928475306e+20,5.49516925956
2612e+20,5.556340899018197e+20,5.622158364520752e+20,5.615271047143903e+20,5.6303580080180874e+20,5.509020526498531e+20,5.2155001
10557719e+20
−24.17585715,3.759689807892469e+20,4.111556051569573e+20,4.256151400392318e+20,4.5172564032548805e+20,4.693730739880348e+20,4.789
693390062942e+20,4.91050402532975e+20,5.067839474446251e+20,5.120466625584519e+20,5.3353624225695773e+20,5.325658872496213e+20,5.
355305947991834e+20,5.405921504346878e+20,5.469574110701773e+20,5.4202708804672134e+20,5.263031221578484e+20,4.93266988403633e+
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−26.171571450000002,3.628303931479566e+20,3.986415698007444e+20,4.065452214942619e+20,4.265146102780165e+20,4.437261984041979e+2
0,4.615039067765154e+20,4.730322606889482e+20,4.858920987674202e+20,4.908320008136534e+20,4.935888435689489e+20,4.99454079215465
4e+20,5.0840280229398204e+20,5.173668518089423e+20,5.194454997516078e+20,5.1227943281193923e+20,5.030146019817731e+20,4.770468291
34898e+20
−28.167285749999998,3.40017950403911e+20,3.72384510465996e+20,3.902206508772547 6e+20,4.023218303517678e+20,4.2285925518621396e+2
0,4.330953989259944e+20,4.445327521220432 4e+20,4.505589153512177 7e+20,4.6210163779781643e+20,4.656822765119564e+20,4.73945141559
6212e+20,4.802969915632224e+20,4.900311945094744e+20,4.879554202736421e+20,4.866996103875433e+20,4.770851452260071e+20,4.52516867
60688976e+20
−30.163,3.2433996382436904e+20,3.488775885705991e+20,3.596980526997924e+20,3.778435955467567e+20,3.883939312336312e+20,4.00162718
61777334e+20,4.047577758440251e+20,4.119564114611362e+20,4.307150117658452e+20,4.3978538843363305e+20,4.408591968869639e+20,4.471
679941288067 5e+20,4.556453764459429e+20,4.5120262568184965e+20,4.5033189251139666e+20,4.38537241765756e+20,4.2115610493641707e+2
0
−32.15871425,2.975158263412069e+20,3.160751829721488e+20,3.348970048271877e+20,3.4974544803421893e+20,3.5403589233615385e+20,3.67
629483559364e+20,3.828198978795435e+20,3.839818333424251e+20,3.913471439558606e+20,4.0229692489254686e+20,4.046878489777511e+20,
4.126001217917681e+20,4.200209907373117e+20,4.163244458475669 4e+20,4.148147918578707 3e+20,4.02471263107093e+20,3.870519101425370
4e+20
−34.15442855,2.625504773996512e+20,2.8913705511793725e+20,3.030639963337961e+20,3.1453104450045444e+20,3.2420106799409876e+20,3.2
695120543345023e+20,3.378703334972493e+20,3.412498127330667e+20,3.494542457417902e+20,3.628016560796247 7e+20,3.650498527254478e
+20,3.7412310310006876e+20,3.719735703888515 6e+20,3.7813288203462994e+20,3.719237594704098e+20,3.5894226780266653e+20,3.40540965
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−36.15014285,2.206221368013049e+20,2.4099384454170444e+20,2.5601758386555958e+20,2.6781798202486663e+20,2.7507696548547455e+20,2.
7640940755379143e+20,2.850765073626566 2e+20,2.863600964148094 6e+20,2.947168358856914e+20,2.985379080715405e+20,3.05754743831928
5e+20,3.125903344857815e+20,3.14529128695899e+20,3.1961558979062425e+20,3.1827644240636327e+20,3.03934729504249e+20,2.91176429067
21596e+20

C.125 "NBSR_upper_Cycle13_cladding_fluence_position_6.csv" File


```
,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
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541200190485588e+20,1.3954462792663487e+20,1.4362630376458371e+20,1.455029688689943e+20,1.4840939379992763e+20,1.48010284026591
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062085e+20
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C.126 "NBSR_lower_Cycle13_cladding_fluence_position_6.csv" File

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C.127 "NBSR_upper_Cycle13_cladding_fluence_position_7.csv" File

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10.18442855,2.00162892195135e+20,2.1684228404166577e+20,2.2686248686159564e+20,2.353468246956233e+20,2.4369189869384853e+20,2.5121610861925463e+20,2.5976271890018576e+20,2.6574879938707495e+20,2.70604918019242 1e+20,2.742382322394856e+20,2.8371270197927916e+20,2.8447016137888743e+20,2.8896000480063118e+20,2.896291505482672e+20,2.8908794637336025e+20,2.8317149780437225e+20,2.684497252432267e+20
8.18871425,2.07762395169979e+20,2.2538946043575055e+20,2.390299568733896e+20,2.4909035372722992e+20,2.5151105357231577e+20,2.6467261828819514e+20,2.6678931536351737e+20,2.731909229051991e+20,2.81738099299284e+20,2.8355079361733008e+20,2.8977237717614526e+20,2.922281760367586e+20,2.965232765336404e+20,3.0290110732286814e+20,3.047913591429667e+20,2.981927442238084e+20,2.7781719959697693e+20
6.193,2.149775073134822e+20,2.275786200009861e+20,2.400031053845455e+20,2.5142104158088256e+20,2.61303678904471e+20,2.6686857120503097e+20,2.7498946439440348e+20,2.8070777335960758e+20,2.8776833661215932e+20,2.87098058638216e+20,2.9994146775549e+20,3.006932660235615e+20,3.058845236427003e+20,3.102175537208779e+20,3.0733660388185965e+20,3.01732083660543e+20,2.8525366198356443e+20
4.19728575,2.142619402872454e+20,2.2990534506256305e+20,2.438673937714857e+20,2.5366454800887025e+20,2.685459644793503e+20,2.7287163508653 02e+20,2.761935870722561e+20,2.856052182519925e+20,2.9129748601212784e+20,2.96279847877563e+20,3.018096411625955e+20,3.0457397169195804e+20,3.08798874157785e+20,3.10471738526875e+20,3.100477197747774e+20,3.0447150521114414e+20,2.885003209198525e+20
2.20157145,2.0880177892011413e+20,2.2993138626763175e+20,2.4194657184108895e+20,2.5332771068243758e+20,2.639762991029393e+20,2.6899885500228498e+20,2.759954474684722e+20,2.7983482687665098e+20,2.915261957262098e+20,2.931798122480751e+20,3.0057664671390584e+20,3.0777194489703065e+20,3.079078120539111e+20,3.096859734695834e+20,3.0898682372480297e+20,2.9990523651365516e+20,2.807989175773482e+20
0.20585714999999993,1.957404162386778e+20,2.107486420555795 7e+20,2.247101246513486e+20,2.3359753505078804e+20,2.422341573231525e+20,2.461703420806086e+20,2.5344149942632494e+20,2.6096514323857736e+20,2.6640718898479127e+20,2.7281106097908764e+20,2.7273406959018882e+20,2.81751686014972e+20,2.8334869122147045e+20,2.851976167813513e+20,2.845641361623964e+20,2.7794061226447664e+20,2.6148936401887425e+20

C.128 "NBSR_lower_Cycle13_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.0522450990208375e+20,2.2354053487586977e+20,2.399153578457269e+20,2.4561894786893555e+20,2.5326034321715102e+20,2.5961496336707833e+20,2.7027713850326814e+20,2.7941250646401437e+20,2.798670953264101e+20,2.8447016137888743e+20,2.8928438763768316e+20,2.9552861572264506e+20,2.9526480699303566e+20,2.977992955820089e+20,2.988109397876142e+20,2.9359816986863606e+20,2.775998121459683e+20
−12.20157145,2.2536964647537213e+20,2.4257042853643148e+20,2.5877824812595652e+20,2.7101987896088104e+20,2.8166337236299976e+20,2.951866833778294e+20,2.9796233617026535e+20,3.0752851624095325e+20,3.1402975969768066e+20,3.182473026925099e+20,3.2847809960560434e+20,3.27696863453542e+20,3.305670571426406e+20,3.312192194956665e+20,3.2802973798789906e+20,3.2466306306303335e+20,3.060000107260487e+20
−14.197285749999999,2.3376170786535188e+20,2.5348735459177202e+20,2.6582465854966655e+20,2.795461091682801e+20,2.8901491777653703e+20,2.963358930797762e+20,3.0781836617563144e+20,3.1295640915832544e+20,3.206776264612081e+20,3.220198807485558e+20,3.3177287815995425e+20,3.353563744226749e+20,3.408001185083498e+20,3.378953919168775e+20,3.3809919265219805e+20,3.308331303248647e+20,3.1893852685313894e+20
−16.192999999999998,2.3469749290836568e+20,2.5705783025195834e+20,2.6797532242045262e+20,2.7872637732176835e+20,2.89881070901649 6e+20,2.986609198018921e+20,3.085435571254806e+20,3.1263429077388814e+20,3.18846250409091e+20,3.301509639746944e+20,3.3363935322759876e+20,3.3553073727400477e+20,3.4085956038948494e+20,3.389930853218404e+20,3.412824469152752e+20,3.375732735324402e+20,3.1614758900555396e+20
−18.188714249999997,2.348616657229295e+20,2.5102589459962207e+20,2.6721956136030536e+20,2.796021543704933e+20,2.8392782497767316e+20,2.953972774709941e+20,3.016171626903483e+20,3.080300924951034e+20,3.164164927535464 4e+20,3.2245069285849747e+20,3.2810955994 25663e+20,3.295740946711063e+20,3.3257958940393164e+20,3.3481686858722895e+20,3.374017412468787e+20,3.2732549322473564e+20,3.158826480496371e+20
−20.18442855,2.3008933183750527e+20,2.4641716741560795e+20,2.5756167095872317e+20,2.7204794044794277e+20,2.8002843757520555e+20,2.8735507400998142e+20,2.964089216765994e+20,3.0432941080957334e+20,3.087983080446314e+20,3.1554467849689714e+20,3.2005829867109496e+20,3.265680338251274e+20,3.30687639244372e+20,3.295905119525627e+20,3.359547560261023e+20,3.2280394746638644e+20,3.085695983305493e+20
−22.180142850000003,2.2450349335025962e+20,2.406597966428008e+20,2.506064047527538e+20,2.6384779141705662e+20,2.726032974516914e+20,2.7953874969728244e+20,2.8640513613812587e+20,2.979006298365155e+20,2.9888340227128374e+20,3.0397162729645493e+20,3.08331264692855e+20,3.160383291668959e+20,3.187494450598137e+20,3.1635535253295025e+20,3.2093464183297357e+20,3.131358670280383e+20,3.0116144160164525e+20
−24.17585715,2.1463387662920548e+20,2.3039446682733255e+20,2.4206488949020564e+20,2.5453070113398283e+20,2.6307278250968462e+20,2.6989331378508097e+20,2.7550292902478073e+20,2.8451148763910524e+20,2.8918701617525218e+20,2.9452602932749844e+20,3.0111162364412245e+20,3.049164701499275e+20,3.085311026360999e+20,3.09789006063551e+20,3.095591641231616e+20,3.036902690590818e+20,2.891304048598854e+20
−26.171571450000002,2.0562701635434198e+20,2.230168802087265e+20,2.349437521302114e+20,2.3798944089694716e+20,2.4930660895193124e+20,2.5594088899977072e+20,2.6199603529140742e+20,2.692779487870435e+20,2.7834425094304216e+20,2.8129936160519 1e+20,2.835105995834196e+20,2.9173565759306714e+20,2.92017581943594e+20,2.9520819567766884e+20,2.9264936422308787e+20,2.8934949065035504e+20,2.7566710183934453e+20
−28.167285749999998,1.9361296300719214e+20,2.0752915655066767e+20,2.168643624546588e+20,2.3067129615947635e+20,2.368521195712274 4e+20,2.409745555562404e+20,2.5206074944452775e+20,2.5622790836868055e+20,2.608247471764676e+20,2.6690480244686573e+20,2.6742449432193332e+20,2.731094026110709e+20,2.7240345950844646e+20,2.813650307310165e+20,2.7310091091376593e+20,2.726650037854412e+20,2.574739234199046e+20
−30.163,1.835508678138908e+20,1.9620236457207117e+20,2.0724553386067978e+20,2.142687336450894e+20,2.185434540684392e+20,2.2454085 68184017e+20,2.3241152799385282e+20,2.348169427837897e+20,2.4547968403313315e+20,2.463786717211585e+20,2.4864652101475395e+20,2.5 50639797247384e+20,2.581985482566001e+20,2.574393905175309e+20,2.591156515655428 3e+20,2.5436030107472868e+20,2.421231991450335e+20
−32.15871425,1.6488385268483044e+20,1.8066765352225787e+20,1.8701378197488012e+20,1.974025244578481e+20,2.0074032761187674e+20,2.0956093665918337e+20,2.1094961222513187e+20,2.1652752512822614e+20,2.209041459192361 8e+20,2.276663675398047e+20,2.2754861600384172e+20,2.325943825424877e+20,2.3403910331064943e+20,2.358461364971588e+20,2.3751956697940243e+20,2.3383926536740445e+20,2.222730075248063e+20
−34.15442855,1.4686220655095203e+20,1.613496082664789e+20,1.6985545840034595e+20,1.7919802377533483e+20,1.8201160614906656e+20,1.854365907287601e+20,1.9308930834004892e+20,1.9810450476839685e+20,2.0037292017514598e+20,2.0254849302469345e+20,2.0612349759010914e+20,2.086172260320182e+20,2.099860876375883e+20,2.1006024846071887e+20,2.1297629731526456e+20,2.083596445470991e+20,2.0368921102933518e+20
−36.15014285,1.2472152111098264e+20,1.3420108586915927e+20,1.4185097291467976e+20,1.475138027908243e+20,1.5224367818972342e+20,1.5729680419936715e+20,1.6159077246994163e+20,1.6711094183136176e+20,1.6646217615725783e+20,1.7281226740195577e+20,1.726831936029 1937e+20,1.77355891573298e+20,1.7852151855670113e+20,1.836742804813905e+20,1.8096769349370207e+20,1.7753025442462784e+20,1.6954239782636732e+20

C.129 "NBSR_upper_Cycle14_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,4.473211129009671e+19,4.909107219644558e+19,4.969173929206219e+19,5.385376356903113e+19,5.383570931533665e+19,5.52533
83270667395e+19,5.767275702580513e+19,5.86520446486121e+19,5.999532262751219e+19,6.197392356831595e+19,6.136537070671785e+19,6.09
52198073319105e+19,6.2398198530136e+19,6.381390104397137e+19,6.127852352084272e+19,6.063790879492515e+19,5.700796620154955e+19
24.15442855,5.386839374012837e+19,5.857453586982031e+19,6.207581596560313 5e+19,6.187836053582 97e+19,6.789281349789853e+19,6.90565
86540640944e+19,6.9617409766035276e+19,7.444526246804683e+19,7.282048339562321e+19,7.615087064178077e+19,7.568509164847917e+19,7.
63237349329014e+19,7.394731785034706e+19,7.537713173488614e+19,7.484463501097797e+19,7.646567872056826e+19,6.878587649530259e+1
9
22.158714250000003,6.200027862830673e+19,6.472730101680139e+19,6.710029401675851e+19,7.099472105075021e+19,7.414321687893993e+19
,7.773466448311638e+19,7.992721870620749e+19,8.177923235099258e+19,8.247546247909541e+19,8.30426150692907e+19,8.586509673019289e+
19,8.47690790188441e+19,8.708656037669465e+19,8.340650166530508e+19,8.60725131275219e+19,8.155521434107096e+19,8.145280314339025e
+19
20.163,6.809286292963956e+19,7.222116518102976e+19,7.575990266522465e+19,8.055050549898155e+19,8.20896825064745e+19,8.46420766825
1058e+19,8.319431230435572e+19,8.829588409398806e+19,9.100163566634843e+19,9.17362570235717e+19,8.899180294185478e+19,9.345078856
407417e+19,9.396263703232027e+19,9.507629395704873e+19,9.290438798961969e+19,9.042587098962074e+19,8.848763271943276e+19
18.167285749999998,7.353404145686961e+19,7.691682754277261e+19,8.250835442404455e+19,8.589197059057718e+19,8.846252278038643e+19
,9.155094152300662e+19,9.332814415104621e+19,9.581008523364237e+19,9.892050111294777e+19,9.908496083769337e+19,1.026564865067587e
+20,1.0280558973985613e+20,1.0265015714195775e+20,1.0127533609913172e+20,1.0254878354506411e+20,9.701712622920429e+19,9.51139588
6561822e+19
16.171571450000002,7.910481632240076e+19,8.316567452263344e+19,8.549757853142385e+19,8.838698544309002e+19,9.544318959534555e+19
,9.677941188889377e+19,1.01531000933058e+20,1.0089080124745523e+20,1.0527912625607729e+20,1.0498029722941019e+20,1.0537769833084
594e+20,1.1119760114534503e+20,1.0850087669983191e+20,1.0995248020089884e+20,1.080910243889517e+20,1.0535798391589221e+20,1.02652
85490400407e+20
14.17585715,8.222591948981266e+19,8.910884431063274e+19,9.32201299091155e+19,9.51749697918961e+19,9.954420294603663e+19,1.0500104
924515095e+20,1.0558936889140181e+20,1.0990578816548207e+20,1.1226114195205952e+20,1.1372104625942276e+20,1.1568729975086077e+2
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12.18014285,8.665066428605976e+19,9.301343983233743e+19,9.842151889446094e+19,1.0107310770573789e+20,1.0072582272231609e+20,1.065
3566080918092e+20,1.121978483040502e+20,1.1538639552261959e+20,1.1552958443123091e+20,1.1895677983081914e+20,1.1851994989947594
e+20,1.1799907430438262e+20,1.2136193845517448e+20,1.2258423218230587e+20,1.1920580401970843e+20,1.1766185404859508e+20,1.1584709
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81286192838e+20,1.1602036960350013e+20,1.1896196783475437e+20,1.2074767878924766e+20,1.2588276508430134e+20,1.2587757708036615e
+20,1.2516163253730956e+20,1.2889699537064826e+20,1.2946248779958428e+20,1.2232898238869438e+20,1.2327423670568649e+20,1.1789531
422567875e+20
8.18871425,9.414328332934368e+19,9.957678361074965e+19,1.0366337431050088e+20,1.1006142828353785e+20,1.121335170552538e+20,1.1720
219689993701e+20,1.2247528409966682e+20,1.2407111411013208e+20,1.2405762529990058e+20,1.2668690569425622e+20,1.2971358719004 76e
+20,1.3271951667009821e+20,1.322505211143568e+20,1.331822866211174e+20,1.3106661861634585e+20,1.2774940890018365e+20,1.228851364
1054703e+20
6.193,9.552318861602623e+19,1.0247044468564268e+20,1.0834627418256317e+20,1.1169564952312355e+20,1.1939153456058832e+20,1.2373078
105198343e+20,1.2567005692295843e+20,1.2419770140615077e+20,1.3054366781967843e+20,1.3039321570555784e+20,1.3510288567792571e+2
0,1.3332547552972872e+20,1.3784007655413336e+20,1.3526993940463891e+20,1.3308163934477466e+20,1.3427591785065601e+20,1.2448200040
2179934e+20
4.19728575,9.548811770942431e+19,1.0477796507593766e+20,1.0852577911872084e+20,1.1130551162719705e+20,1.1781853176743789e+20,1.22
44208087448158e+20,1.2366126179925187e+20,1.2728041334444224e+20,1.3230551395606983e+20,1.3113717546986445e+20,1.33558935706812
38e+20,1.3808391273908742e+20,1.3738872021177159e+20,1.3636253303339049e+20,1.3390964477 28314e+20,1.3346658923676595e+20,1.301514
5472217786e+20
2.20157145,9.325665345681926e+19,1.0295479673303228e+20,1.0549287201820721e+20,1.1268240787159718e+20,1.14185891412016e+20,1.1750
10259266041e+20,1.22156740658046e+20,1.2309991977346399e+20,1.2777016091592443e+20,1.2703657715948821e+20,1.2964718073967711e+2
0,1.3382144870593313e+20,1.35810529414686e+20,1.3580222860838968e+20,1.3337735556908065e+20,1.3309201535264504e+20,1.26460708722
6818e+20
0.20585714999999993,8.355290713635622e+19,9.082078184916761e+19,9.603524460442973e+19,9.847485157491473e+19,1.055188120378832e+2
0,1.0938906297353688e+20,1.0733046301205247e+20,1.1248422612127282e+20,1.140427025034047e+20,1.1791295343905841e+20,1.1398563446
011758e+20,1.201292687201727e+20,1.2304181412938985e+20,1.250308948381427e+20,1.2074767878924766e+20,1.2019048716660798e+20,1.180
8934557285496e+20

C.130 "NBSR_lower_Cycle14_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,8.906256731553083e+19,9.42833594355939e+19,1.0247501012910567e+20,1.0511103492857704e+20,1.0485059713103036e+20,1.11555557341687335e+20,1.1346994686895943e+20,1.179233294469288e+20,1.1871294364586512e+20,1.2214947745253 67e+20,1.2278345153341725e+20,1.2322858227105679e+20,1.3172030711218012e+20,1.280606891362952e+20,1.2793825224342467e+20,1.2349420807253867e+20,1.2540028071832843e+20
−12.20157145,1.0099352372537206e+20,1.0943679260974069e+20,1.1319809546275534e+20,1.1810802238702166e+20,1.2070721235855316e+20,1.2644825751323735e+20,1.2991903214588122e+20,1.3479471824417525e+20,1.3692076225681721e+20,1.3808806314223557e+20,1.3865874357510676e+20,1.4164080823705 55e+20,1.3976793881645094e+20,1.4587421944817269e+20,1.4123095592617527e+20,1.3792931022181868e+20,1.3456540847023974e+20
−14.197285749999999,1.0267049411738372e+20,1.1230887158826331e+20,1.1966649876915352e+20,1.245567112784661e+20,1.2653022797541338e+20,1.3077920319833616e+20,1.3342923560843257e+20,1.382852072917729e+20,1.4392871797247543e+20,1.434244439899747e+20,1.4816316678437965e+20,1.4343170719548398e+20,1.505911526260498e+20,1.4839870216303742e+20,1.4542805110974611e+20,1.434296319939099e+20,1.4109606782386025e+20
−16.192999999999998,1.0224787931682292e+20,1.1089877211867795e+20,1.1653916999701941e+20,1.2463868174064214e+20,1.2933590050356557e+20,1.2971877519398278e+20,1.3665306125376127e+20,1.395676818645525e+20,1.4195727647710224e+20,1.4470380576039322e+20,1.4618549968428424e+20,1.4636604222122895e+20,1.4604853638039516e+20,1.494757317799834e+20,1.461761612772009e+20,1.4627473335196955e+20,1.4166156025279626e+20
−18.188714249999997,1.0520026859626235e+20,1.1110006667136342e+20,1.2019878797290429e+20,1.2783034176157267e+20,1.2722645810351625e+20,1.3043160693467827e+20,1.3383078711301648e+20,1.4156402577881463e+20,1.416055298102962e+20,1.4312353976173355e+20,1.4903786424785315e+20,1.4328125508136339e+20,1.4600806994970067e+20,1.4953798782720573e+20,1.4776991608609207e+20,1.4190954684089847e+20,1.4111785744038807e+20
−20.18442855,1.019970912065957e+20,1.102108427968714e+20,1.1735161141327056e+20,1.2080785963489591e+20,1.2603321719842193e+20,1.3108633303129958e+20,1.3191848886250447e+20,1.3771763966126 28e+20,1.3673918211908547e+20,1.3788261818640194e+20,1.4243353523835291e+20,1.4009997106830326e+20,1.491436995281311e+20,1.4451288721557814e+20,1.4670222487622943e+20,1.4106597740103613e+20,1.3466190534343433e+20
−22.180142850000003,9.937995074144844e+19,1.0612165809515256e+20,1.1328110352571843e+20,1.1765355324229878e+20,1.247019753886515e+20,1.2938881814370455e+20,1.2574061377647708e+20,1.3336179155727506e+20,1.4007403104862732e+20,1.3703801114575258e+20,1.4123821913168454e+20,1.4394116918191989e+20,1.3821776324061538e+20,1.4290356839488137e+20,1.3859233712473629e+20,1.3657627879552043e+20,1.316850286854208e+20
−24.17585715,9.637080469895799e+19,1.0222629722045252e+20,1.0747468952145083e+20,1.158232254539628e+20,1.1666990769618625e+20,1.2110253825841485e+20,1.2653230317698744e+20,1.2742879025698877e+20,1.294687134043065e+20,1.3234182998361619e+20,1.3052602860629877e+20,1.3466916854894361e+20,1.367132420994095e+20,1.3760869157862375e+20,1.322909875450513e+20,1.323449427859773e+20,1.2833669094564746e+20
−26.171571450000002,9.443619803152466e+19,9.84069924834424e+19,1.0554163925519802e+20,1.0842928224552626e+20,1.1549845640761975e+20,1.1773863650683594e+20,1.1713786565114064e+20,1.2211731182813854e+20,1.2595747234096811e+20,1.2669728170212658e+20,1.2613178927319058e+20,1.2814784760240646e+20,1.2814784760240646e+20,1.2865212158490717e+20,1.3170474310037453e+20,1.2109735025447964e+20,1.2005559906429297e+20
−28.167285749999998,8.626529935375365e+19,9.107769180403835e+19,9.686542899413926e+19,1.0713124366094105e+20,1.0847701188173002e+20,1.0950216145932408e+20,1.0986428413400054e+20,1.1329666753752398e+20,1.161438440971577e+20,1.1932201530785674e+20,1.1747197310456704e+20,1.1580662384137018e+20,1.2338837279226072e+20,1.238023755062891e+20,1.1804680394058639e+20,1.1768260606433585e+20,1.1209616342692042e+20
−30.163,8.038552697384242e+19,9.037129318822252e+19,9.33170418226249e+19,9.681707679746327e+19,9.917917498915647e+19,1.03844124367603e+20,1.0639973510607888e+20,1.1153170859877147e+20,1.096422375655742 8e+20,1.1354984212956139e+20,1.1235763882525411e+20,1.1870775564192991e+20,1.1367539182479306e+20,1.130092521195143e+20,1.1533036508011951e+20,1.0758882560802505e+20,1.0676289538154237e+20
−32.15871425,7.326052988940625e+19,8.04793260849907e+19,8.41119664404126e+19,8.884259594867864e+19,9.119794973525611e+19,9.604624317277233e+19,9.770609315179789e+19,9.764456342512648e+19,1.0089100876761265e+20,1.0155445071084508e+20,1.0656160082885691e+20,1.0322270525624561e+20,1.0514838855691043e+20,1.0250873215468439e+20,1.0158889905697476e+20,1.0011498713898652e+20,9.724062543873239e+19
−34.15442855,6.7429317226328424e+19,7.264492134245629e+19,7.43175338111624e+19,7.869226624947424e+19,8.209621939143284e+19,8.367129738615733e+19,8.903268441286412e+19,8.766782433759363e+19,8.986255752233753e+19,9.074960243517678e+19,9.087463333001491e+19,9.238330487436893e+19,9.297069067991143e+19,9.223098507883168e+19,8.934406840905436e+19,8.875668260351186e+19,8.397790841872722e+19
−36.15014285,5.653170744029886e+19,5.734788421938338e+19,6.560739400436747 5e+19,6.550685048810344e+19,6.992070047608663e+19,6.816684386575541e+19,7.352356168892051e+19,7.358737413732338e+19,7.340911432211017e+19,7.232648166091416e+19,7.732201065011116e+19,7.632788533604955e+19,7.588265083833131e+19,7.562812736527077e+19,7.378348068607366e+19,7.239963251640038e+19,7.271246915369249e+19

C.131 "NBSR_upper_Cycle14_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.079067216866407e+20,1.1785060611441246e+20,1.2563290984362131e+20,1.2848374350819402e+20,1.3042947274409188e+20,1.3
424979834750935e+20,1.3938056308648113e+20,1.44585526765744e+20,1.4788584657274123e+20,1.4600266520386563e+20,1.5082988882346467
6e+20,1.5102550421150 49e+20,1.5423568150426422e+20,1.5156022715310683e+20,1.5145230142177434e+20,1.5158230287087937e+20,1.4747131
365012324e+20
24.15442855,1.2629947387747603e+20,1.3805602002012773e+20,1.4711748895137976e+20,1.517889561178058e+20,1.5630098754476343e+20,1.6
176043519279327e+20,1.6364361656166885e+20,1.6771352041879277e+20,1.7613663317778835e+20,1.7508190443976622e+20,1.7741211909353
61e+20,1.808829124988881e+20,1.795172840689024e+20,1.825717049084887e+20,1.8282557566287312e+20,1.7935968797258167e+20,1.68546878
7647068e+20
22.158714250000003,1.4238408713231401e+20,1.5245306729413024e+20,1.6177331269482725e+20,1.646229199306348e+20,1.7517388659715185
e+20,1.7574969490238607e+20,1.8421021373871846e+20,1.904913686593934 8e+20,1.911622251939262e+20,1.989371703505442e+20,1.982098980
92815e+20,2.0127780964881783e+20,2.0452171373261855e+20,2.046363848221593e+20,2.0349887214249008e+20,1.993713261334045e+20,1.9113
89230473885e+20
20.163,1.549739916208797e+20,1.6817711049201648e+20,1.7755070554399716e+20,1.830886446329961e+20,1.9134557629431493e+20,1.9443924
28543857e+20,2.0226324516160943e+20,2.041274168846254e+20,2.0909016088277267e+20,2.144882870925454e+20,2.1750530185479487e+20,2.2
161567786116845e+20,2.255427027671533e+20,2.2459222047416818e+20,2.239213639396355e+20,2.1819026232012672e+20,2.075491531393717
e+20
18.167285749999998,1.6891787346616225e+20,1.8061984352876515e+20,1.8992966428495843e+20,1.977039962271939e+20,2.0528639206768463
e+20,2.123788296165011e+20,2.1632793024025847e+20,2.2249134799947982e+20,2.2670474402212535e+20,2.271861173124436e+20,2.323904677
7732393e+20,2.3808416331949585e+20,2.3946082960836782e+20,2.4297209516296952e+20,2.408632509013078e+20,2.367675920401159e+20,2.30
4925692632666e+20
16.171571450000002,1.774887708913575e+20,1.9195572460497312e+20,2.0525511813417347e+20,2.1319011224464255e+20,2.1913399925490352
e+20,2.252612373655532e+20,2.3103342434869422e+20,2.3610838657885202e+20,2.4239444721458758e+20,2.4597255313688956e+20,2.49810048
74301902e+20,2.5002099449062346e+20,2.5897392447616044e+20,2.5889052732013075e+20,2.5996119963209977e+20,2.5243828558671843e+20,
2.4282860299744795e+20
14.17585715,1.9090528836762894e+20,2.0460940338932623e+20,2.15576742621609e+20,2.2451066296128625e+20,2.2978921236645763e+20,2.41
35995455119032e+20,2.4524896016545576e+20,2.4836776851521195e+20,2.551854860206366e+20,2.6240915144732328e+20,2.637864309505778
e+20,2.7003447229459402e+20,2.742871140377241e+20,2.7016815302999448e+20,2.725866705548546e+20,2.693059736080996e+20,2.5972633852
357506e+20
12.18014285,2.0015501411433028e+20,2.1485682893647063e+20,2.2545072060976765e+20,2.325413185154364e+20,2.460124120717567e+20,2.50
53364171445282e+20,2.5821169899862462e+20,2.670585428959768e+20,2.6748901939254166e+20,2.6952366471391232e+20,2.817462537873180
3e+20,2.8631531415185474e+20,2.878741051123503e+20,2.8876203953831313e+20,2.8670409207040472e+20,2.8007279173728164e+20,2.7051032
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525507939683543e+20,2.672339222093921e+20,2.7304474169863517e+20,2.813157772907531e+20,2.893728010633538e+20,2.951486673327903e
+20,2.964327382498941e+20,3.011256679197101e+20,3.0036221601340916e+20,2.983864392727653e+20,2.939749750045494e+20,2.83934815918
714e+20
8.18871425,2.1456616531913192e+20,2.3158102479233014e+20,2.468506761327313e+20,2.574310770896117e+20,2.6578612305214197e+20,2.734
7031248013944e+20,2.8094171651738485e+20,2.8335103582650637e+20,2.9001422330752313e+20,2.9339364776987207e+20,3.011207622046495
5e+20,3.060467133398425e+20,3.0785937505472256e+20,3.117600317422569e+20,3.1145219812220613e+20,3.0278012032388674e+20,2.92571327
2828443e+20
6.193,2.170999671479154e+20,2.3474950350707468e+20,2.524493234456047e+20,2.586734494287006e+20,2.6889941247245504e+20,2.810478026
0556957e+20,2.8563525940158346e+20,2.9452318866256786e+20,2.9651613540592366e+20,3.068788452569915e+20,3.1306065944769e+20,3.1277
1835473499e+20,3.1729613118810803e+20,3.178192030564411e+20,3.178541562762477e+20,3.1011539076820133e+20,3.0093618467549564e+20
4.19728575,2.184055005684091e+20,2.3756845002375365e+20,2.5412339871002365e+20,2.6360369306457165e+20,2.7649897831565777e+20,2.84
58972887929985e+20,2.9052687053135253e+20,2.9670132614946016e+20,3.016125601394715e+20,3.1136880096617654e+20,3.125565972252166
e+20,3.223888766353604e+20,3.2058725277936674e+20,3.174537272844288e+20,3.194705893887049e+20,3.15202004071628e+20,3.057419457917
048e+20
2.20157145,2.1649472455231783e+20,2.3573432580548372e+20,2.501025520035054e+20,2.6105456087622395e+20,2.6689726751336058e+20,2.77
22011842956126e+20,2.8334490368268072e+20,2.926768001566465e+20,2.9956503731606684e+20,3.037655558366784e+20,3.106887462159052e
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0,2.543239198131243e+20,2.586783551437612e+20,2.7203845689683608e+20,2.7457287194000215e+20,2.790456576464752e+20,2.793154719748
0645e+20,2.8611111376245857e+20,2.8890185241753928e+20,2.887835020417031e+20,2.9348072421219713e+20,2.862159734218782e+20,2.73009
17526444605e+20

C.132 "NBSR_lower_Cycle14_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.005020934548654 7e+20,2.259406789014419e+20,2.362935773223885e+20,2.5037236633183663e+20,2.5880774337848368e+20,2.6782383444542544e+20,2.7492976271065848e+20,2.809282258009682e+20,2.8589219622788075e+20,2.9486781514557276e+20,2.9540928344538304e+20,2.9988145593747354e+20,3.031155485911531e+20,3.022417180959894e+20,3.047785859966857e+20,2.988089439823568e+20,2.869291417488083e+20
−12.20157145,2.3128054974487036e+20,2.502901956045721e+20,2.6944517327294323e+20,2.779529096167337e+20,2.849061475007065e+20,2.9606910212102 94e+20,3.059866183303506e+20,3.144698260988382e+20,3.180693945245 3015e+20,3.264495822767457e+20,3.36272663471151e+20,3.413985224950622e+20,3.408404974069225e+20,3.435711410525113e+20,3.3775112334752966e+20,3.325363482381457e+20,3.2130900610765285e+20
−14.197285749999999,2.3581465688960054e+20,2.6347001232917116e+20,2.727160587895771e+20,2.892532242587525e+20,2.966939675768693e+20,3.069830917020285e+20,3.157588027310025e+20,3.224354809284358e+20,3.291820655654822e+20,3.35785158037007e+20,3.4281627414756 64e+20,3.484443557508033e+20,3.521788313406608e+20,3.493917719718755e+20,3.48153078919082e+20,3.3946260468928545e+20,3.2519433243562285e+20
−16.192999999999998,2.3860171625838582e+20,2.6194924066039507e+20,2.758949621488253e+20,2.8600441445989125e+20,2.970790662091239e+20,3.062343569409094e+20,3.2142245076842847e+20,3.218369836910465e+20,3.271314766701647e+20,3.35819498042431e+20,3.4410647720849585e+20,3.505194732214236e+20,3.452213009560099e+20,3.528766693080266e+20,3.4978545560548606e+20,3.455561160088937e+20,3.3011415142699015e+20
−18.188714249999997,2.355865411392841e+20,2.5337159787699156e+20,2.716852454124752e+20,2.8854005593182244e+20,2.9234566439005815e+20,3.042322119818149e+20,3.157759727337145e+20,3.200255484049316e+20,3.259853789891395e+20,3.350505272066869e+20,3.388714660244 8696e+20,3.485498286246055e+20,3.494304044779775e+20,3.5172382626879314e+20,3.507340982553235e+20,3.422643812032525e+20,3.280868646782104e+20
−20.18442855,2.329791535845921e+20,2.5336178644687038e+20,2.6721613899229757e+20,2.8158068590402385e+20,2.9140621995595933e+20,3.025654952899869e+20,3.061019026342739e+20,3.176867487498058e+20,3.246455055632 2175e+20,3.295659377689717e+20,3.335082930345208e+20,3.4102568815045896e+20,3.4296467202814856e+20,3.434515642479099e+20,3.41072905657917e+20,3.359439805620929e+20,3.207515942338958e+20
−22.180142850000003,2.2364725711062627e+20,2.4280652727967534e+20,2.605173850770917e+20,2.7531915384359112e+20,2.826930567940076 7e+20,2.9429752576978174e+20,3.006596249889561e+20,3.052667046452123e+20,3.10155862917451e+20,3.189561025217278e+20,3.2534456995935276e+20,3.2892696838233267e+20,3.323879503575636e+20,3.330201743859943e+20,3.3106156764806244e+20,3.2431436979663346e+20,3.110910148508719e+20
−24.17585715,2.1843125557247712e+20,2.4077678767336525e+20,2.517662026234206e+20,2.649325286316031e+20,2.7240147981131812e+20,2.8050633430575943e+20,2.8893496599419814e+20,2.9776647953198606e+20,3.015126061951124e+20,3.080255561523993e+20,3.153571473104185e+20,3.1755490765755305e+20,3.185501546004658e+20,3.2281506063124726e+20,3.189462910916067e+20,3.167920689656346e+20,3.005222649672603e+20
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−28.167285749999998,1.9826202131533352e+20,2.1467041176416898e+20,2.2425863185004957e+20,2.3648490020975064e+20,2.4054560585113597e+20,2.5391245296241918e+20,2.5961902600662516e+20,2.658302744876 8705e+20,2.722181287109294e+20,2.8049468323249052e+20,2.811477565499287e+20,2.8982412684892612e+20,2.9356044208193136e+20,2.9252962870482947e+20,2.920678782747535e+20,2.862815873608133e+20,2.7642416616098405e+20
−30.163,1.8496875992995883e+20,2.0346023963638812e+20,2.1277864539393743e+20,2.220246918543433e+20,2.3441959416925143e+20,2.3919408135194945e+20,2.4662010752488453e+20,2.5032146953808327e+20,2.5504199385511497e+20,2.5755249353736077e+20,2.6482337647150547e+20,2.7063542238951362e+20,2.7106651210046102e+20,2.7390814754929518e+20,2.689687056976856e+20,2.7006451979934e+20,2.546887823707541e+20
−32.15871425,1.7006519757595266e+20,1.843150733981381e+20,1.922862471571789e+20,2.050962956090876e+20,2.1229665888923655e+20,2.1640028953740188e+20,2.2418381969537588e+20,2.331116078912274e+20,2.3778920720147923e+20,2.4575302238792915e+20,2.428120462091185e+20,2.4857012926146046e+20,2.5173615511867464e+20,2.4917230578514516e+20,2.4592840170134444e+20,2.4680713791156873e+20,2.3325755291427933e+20
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−36.15014285,1.3105985712937 49e+20,1.4277654433715947e+20,1.4977025437038202e+20,1.5811364925964337e+20,1.592033312174721e+20,1.6520731323722498e+20,1.7203484217277078e+20,1.7462260686722053e+20,1.806584760348671e+20,1.8132626649748695e+20,1.85975657946140 48e+20,1.854795675106405e+20,1.892765909675203e+20,1.890963059390444e+20,1.9206548997945334e+20,1.8717387884968424e+20,1.7543021020906655e+20

C.133 "NBSR_upper_Cycle14_cladding_fluence_position_2.csv" File

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26.15014285,1.1221271308105445e+20,1.217892820936661e+20,1.2751547799811436e+20,1.3335757142086853e+20,1.3850734580569999e+20,1.4172856095734556e+20,1.4544402690134342e+20,1.510322495697132e+20,1.5395115003075124e+20,1.5165588859678792e+20,1.56550565798469 8e+20,1.5777944742114217e+20,1.5900955547257966e+20,1.6074004646019505e+20,1.62354026715122e+20,1.6074004646019505e+20,1.53138027 75946206e+20
24.15442855,1.3272596060682037e+20,1.4243007821100684e+20,1.4924534285890123e+20,1.5727354555552108e+20,1.6114660759583965e+20,1.6528703110695925e+20,1.730460326896304e+20,1.7406335535031573e+20,1.779627856090849e+20,1.8156296724915936e+20,1.84782342757657 25e+20,1.911327909035628e+20,1.9104878053315055e+20,1.915129838207568e+20,1.9339984447592786e+20,1.891625330923621e+20,1.84061202 64375373e+20
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C.134 "NBSR_lower_Cycle14_cladding_fluence_position_2.csv" File

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−10.20585715,2.1823809304196724e+20,2.40321169387115e+20,2.6105210801869364e+20,2.695328629296509e+20,2.8285985110606702e+20,2.90180404440519986e+20,2.9387195498827747e+20,3.036821586806487e+20,3.10609028346171e+20,3.146176107650378e+20,3.2119862751879004e+20,3.2763553889263806e+20,3.2994306461425284e+20,3.318851145638553e+20,3.2950522954509713e+20,3.209159356884248e+20,3.0630548980928754e+20
−12.20157145,2.4626812246928877e+20,2.688227606746336e+20,2.814181809264248e+20,2.978302538277734e+20,3.062693101607159e+20,3.1858449460588977e+20,3.29797119591201e+20,3.363180413354612e+20,3.4515568701707485e+20,3.559359958626735e+20,3.596441032340805e+20,3.653715255672938e+20,3.696989794650976e+20,3.704569124419553e+20,3.718722112369293e+20,3.588377263209996e+20,3.392228378657002e+20
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−18.188714249999997,2.508837871269009e+20,2.7411418758183908e+20,2.8902020279337553e+20,3.007951453675043e+20,3.113387534614305e+20,3.2255383130427195e+20,3.300485374880551e+20,3.4085337490895654e+20,3.524645892429389e+20,3.5711704676350525e+20,3.660233724559668e+20,3.6917406795361675e+20,3.74797243841793e+20,3.790676688020177e+20,3.798004598919005e+20,3.6542426200419493e+20,3.5559382223719884e+20
−20.18442855,2.4859220497923296e+20,2.6786675945220545e+20,2.824992810491152e+20,2.9470899262048692e+20,3.0588237188531356e+20,3.1642843283677e+20,3.2466328878031634e+20,3.332274408473033e+20,3.4500728913649264e+20,3.526534592727708e+20,3.5461635851138066e+20,3.637029692323181e+20,3.6539666735697927e+20,3.710541832505795e+20,3.689851979237849e+20,3.6105633595714155e+20,3.4631956791519465e+20
−22.180142850000003,2.3680131883115728e+20,2.5745805852244486e+20,2.7446065370799165e+20,2.8953162358843972e+20,2.9695029118878404e+20,3.065397377034297e+20,3.212084389489112e+20,3.242401708563423e+20,3.322052124715574e+20,3.427089616306164e+20,3.4512073379726826e+20,3.5251058032163 17e+20,3.5861942200080427e+20,3.643180232580368e+20,3.6229564222431756e+20,3.5766710006467147e+20,3.40360963759752e+20
−24.17585715,2.294911901765281e+20,2.4701440437287774e+20,2.600384646443037e+20,2.7551047673095327e+20,2.903447458597289e+20,2.958041935077588e+20,3.051753357022092e+20,3.1039624295541886e+20,3.1830670849058505e+20,3.2841309472973816e+20,3.335217837509374e+20,3.322469110495722e+20,3.4073502453312035e+20,3.410324335086672e+20,3.470689158906964e+20,3.417964986293508e+20,3.194644572448792e+20
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−28.167285749999998,2.069886751937018e+20,2.247614676437578e+20,2.3478874922755922e+20,2.4620864067417938e+20,2.571422531154208e+20,2.641512935082076e+20,2.7436867155060602e+20,2.7680558550694324e+20,2.8520784897693142e+20,2.9588329816311045e+20,3.011600079251341e+20,3.0439839307949166e+20,3.0978977393105614e+20,3.1742061370777e+20,3.13691043832973e+20,3.0793725328130907e+20,2.9538536808446276e+20
−30.163,1.926412982846875e+20,2.1150561233571997e+20,2.2087614131578775e+20,2.3109842507324677e+20,2.423043047003497e+20,2.4672251432677392e+20,2.5303923568163806e+20,2.59440580621297e+20,2.72227326926668e+20,2.7159878218453264e+20,2.810563876069256e+20,2.8330627117657873e+20,2.8975606005246073e+20,2.8751046898348563e+20,2.9754265628234763e+20,2.9005224259924253e+20,2.7387258111510603e+20
−32.15871425,1.7540874770567656e+20,1.956920398379784e+20,2.043604383500024e+20,2.172452989565847e+20,2.2304753344447178e+20,2.2597072640618788e+20,2.3100337684394826e+20,2.3690863134810728e+20,2.4340318487391696e+20,2.525958816830392e+20,2.5719866883861732e+20,2.6096441836198596e+20,2.6465903501697637e+20,2.633958133888801e+20,2.6751293475346193e+20,2.6638952600459182e+20,2.5565398180892092e+20
−34.15442855,1.5932720052275144e+20,1.740412796325432e+20,1.8254288383250786e+20,1.9122906556162644e+20,1.9785055446462834e+20,2.0304938600006553e+20,2.0952922238069355e+20,2.1202561813214016e+20,2.208546788123978e+20,2.2343569814863 92e+20,2.300467624071374 8e+20,2.3266028210565516e+20,2.3780085827474807e+20,2.3865138662337407e+20,2.4367361241663075e+20,2.4168863746024836e+20,2.2801763601520994e+20
−36.15014285,1.3571415429308898e+20,1.4586469196778714e+20,1.5415657684891255e+20,1.6147590371928028e+20,1.693802371106208e+20,1.7246665450981008e+20,1.7878081359543 46e+20,1.7927874367408 23e+20,1.8257354551636 4e+20,1.8979659676394052e+20,1.9534802656935602e+20,1.983773056192569e+20,2.022644715903746e+20,2.03411795700165e+20,1.9964543296241376e+20,1.9907637001538788e+20,1.9140812416133716e+20

C.135 "NBSR_upper_Cycle14_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.1063307283155147e+20,1.2094365946010069e+20,1.2674712037675287e+20,1.3356115859588209e+20,1.357748625169635e+20,1.4142072733729494e+20,1.457831344549052e+20,1.456236987154367 3e+20,1.483273609281924e+20,1.4963657363498153e+20,1.524542937228954e+20,1.5441903260465296e+20,1.5708344909692404e+20,1.5838469001673975e+20,1.5792539244419406e+20,1.5324656670517713e+20,1.4491420967480204e+20
24.15442855,1.3259105344265473e+20,1.4121775337666391e+20,1.4874802599463613e+20,1.551818712965713e+20,1.6156788587666597e+20,1.6647666700914 7e+20,1.6760743433060798e+20,1.7231508114560575e+20,1.8063578710271197e+20,1.794584154881756e+20,1.8392752190835324e+20,1.865545323232875e+20,1.844763487807543e+20,1.885125258468368e+20,1.8506564780240506e+20,1.8542805750250455e+20,1.737714653042119e+20
22.158714250000003,1.5081762453581332e+20,1.5689887156777016e+20,1.6607317194541528e+20,1.7865510464700757e+20,1.815261743862051e+20,1.857757500574223e+20,1.896199910217601e+20,1.950132115164722e+20,1.9586312665071567e+20,1.9889915105882482e+20,2.044383165765889e+20,2.066158408490986e+20,2.110849472692762e+20,2.069512691163649e+20,2.075681627852314e+20,2.054096481585814e+20,1.976537126478231e+20
20.163,1.6145996014533345e+20,1.75792619909166e+20,1.8243127881487994e+20,1.9246469254250702e+20,1.9545472587192336e+20,2.031045752944969e+20,2.0979964992340735e+20,2.1357582409127972e+20,2.1575948050761515e+20,2.1981895972023537e+20,2.2223870367386062e+20,2.246578344131033e+20,2.2945010481289583e+20,2.2854622681298616e+20,2.2970642842481056e+20,2.2589591425151422e+20,2.13561106946098e+20
18.167285749999998,1.730699480505507e+20,1.8920116559846405e+20,1.986949506694287e+20,2.0700768483956146e+20,2.1517079470034695e+20,2.2072222450576245e+20,2.258217153112231e+20,2.342049691353516e+20,2.3431412129544923e+20,2.385575648228407e+20,2.4089329840605374e+20,2.4881050929942823e+20,2.4640057677592412e+20,2.504244895543552e+20,2.4683227970125408e+20,2.446590479294224e+20,2.2814211853487188e+20
16.171571450000002,1.8520668711039638e+20,2.0269065558626147e+20,2.1317294224193058e+20,2.2137345818005288e+20,2.27262769110265e+20,2.3861765983233268e+20,2.3992012718091354e+20,2.462442071083685e+20,2.5115850717029273e+20,2.5618686510737506e+20,2.58814488736691 9e+20,2.614893298734668e+20,2.6670839748352875e+20,2.6919804787676712e+20,2.636956752219573e+20,2.5847170189683474e+20,2.4796059416518482e+20
14.17585715,1.937677731054705e+20,2.1091692652845177e+20,2.2542496560569965e+20,2.327565567637188e+20,2.3581097760330508e+20,2.483880045898368e+20,2.5684055163919584e+20,2.6077861440406693e+20,2.6532314619330085e+20,2.698885272715421e+20,2.7562269496296378e+20,2.7423192474329268e+20,2.8170210235177288e+20,2.8257347998940633e+20,2.824569692567178e+20,2.77860927459348e+20,2.5990049140822527e+20
12.18014285,2.0444260907726687e+20,2.2740871413331693e+20,2.3654376879047747e+20,2.459970817121924e+20,2.528325824347116e+20,2.6061856545021595e+20,2.668856164400918e+20,2.7488929056140883e+20,2.8088714043733603e+20,2.8325721402597304e+20,2.871658425004807 3e+20,2.915785331974618e+20,2.9504196803022297e+20,2.962585853652439e+20,2.9374011389602467e+20,2.858885169415853e+20,2.705268834437985e+20
10.18442855,2.1508249182925675e+20,2.315062126376565e+20,2.466231735967974e+20,2.574537660217668e+20,2.6392317775789117e+20,2.762487868475688e+20,2.7882735332627992e+20,2.849760539403196e+20,2.9444960293665938e+20,2.954583405959887e+20,3.0355767616098684e+20,3.036263561718347e+20,3.035239493699454e+20,3.038287169180832e+20,3.028874328408366e+20,3.006958046375279e+20,2.876404704325907e+20
8.18871425,2.1835521698903828e+20,2.4012984649975284e+20,2.538174047331207e+20,2.629095343835012e+20,2.7574963034015593e+20,2.836275955130459e+20,2.859388005209561e+20,2.9771312988070235e+20,3.0024570528072067e+20,3.1089601267721424e+20,3.1388236672033515e+20,3.16920230771592e+20,3.23287235705827e+20,3.238385154357583e+20,3.158550773890661e+20,3.120451764301524e+20,2.964082096745913e+20
6.193,2.251312359164481e+20,2.4813536026421756e+20,2.616898509765675e+20,2.7101806816423795e+20,2.7926948089611364e+20,2.9228924866686163e+20,2.9647321039914376e+20,3.0883622556615816e+20,3.092808059935222e+20,3.1468199827520776e+20,3.1923204899388477e+20,3.253917874668107e+20,3.273706302793674e+20,3.298277803103295e+20,3.260914650773242 5e+20,3.228518534942015 5e+20,3.04272070916682e+20
4.19728575,2.26209266801008e+20,2.4814517169433867e+20,2.651600311675369e+20,2.7553193923434322e+20,2.8497728036908476e+20,2.9384068105476637e+20,3.009214675303139e+20,3.110566748454479e+20,3.133525494937938e+20,3.209993328444545e+20,3.268506244829472e+20,3.268199637638186e+20,3.271455806009638e+20,3.318501613440487e+20,3.297725910158981e+20,3.2205774086877204e+20,3.05169816772766e+20
2.20157145,2.246063244049673e+20,2.4607373351001375e+20,2.640127070577465e+20,2.762776079235496e+20,2.854862483066187e+20,2.922444840169339e+20,2.992884776295273e+20,3.068788452569915e+20,3.08497731226979e+20,3.172679233265098e+20,3.243352190856408e+20,3.218621254807319e+20,3.276619071110887e+20,3.221742516014606e+20,3.292685287934247 6e+20,3.136376941816893e+20,3.0017273276919474e+20
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C.136 "NBSR_lower_Cycle14_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.1407068809801453e+20,2.387764823574186e+20,2.509144478460294e+20,2.603456850499718e+20,2.7311280849510053e+20,2.82
4913092621418e+20,2.8607186804197402e+20,2.9417488289326753e+20,3.020142155600555e+20,3.039654637253965e+20,3.0683285417829866e
+20,3.139277445846454e+20,3.2019295593137354e+20,3.106537929960987e+20,3.1336971949650575e+20,3.052446289274397e+20,2.9337157205
20995e+20
−12.20157145,2.4054928513743136e+20,2.652710229707823e+20,2.7850480256104753e+20,2.971851522973087e+20,3.046160841853044e+20,3.15
61347092233316e+20,3.2530839031078106e+20,3.2901220518151e+20,3.399439779796037e+20,3.373623454289797e+20,3.449692698447733e+20,
3.506568332431195e+20,3.532090315033801e+20,3.538504537475495e+20,3.472001437685666e+20,3.4087238455481624e+20,3.217198597439754
e+20
−14.197285749999999,2.499228801894121e+20,2.7773153922462555e+20,2.9251368513088265e+20,3.042420234119361e+20,3.161782413686811e
+20,3.254696656933972e+20,3.303097668150304e+20,3.40827006690506e+20,3.472375498459035e+20,3.519623666636132e+20,3.5797002796966
157e+20,3.6016288260173555e+20,3.585587137769297e+20,3.6490670906530497e+20,3.624759272527935e+20,3.526822803487516e+20,3.3460778
642247064e+20
−16.192999999999998,2.4837696673095054e+20,2.7348012391026062e+20,2.931790227359722e+20,3.0531453536705275e+20,3.134334937922932
e+20,3.273099220554929e+20,3.351210468606826e+20,3.385489152592546 4e+20,3.482922785839256e+20,3.54731642815304e+20,3.649821344343
612e+20,3.666212564789736e+20,3.67319707660722e+20,3.6618526105296555e+20,3.6239804902620686e+20,3.555251422263509e+20,3.35708506
23918565e+20
−18.188714249999997,2.49464195831249e+20,2.6910913179129433e+20,2.8632083308129786e+20,2.988616804192579e+20,3.085001840845093e+
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42154e+20,3.602119397523413e+20,3.60305148338492e+20,3.580656894133426e+20,3.562076498341524e+20,3.5372474479912236e+20,3.3039071
11135297e+20
−20.18442855,2.4662072073926712e+20,2.6871115565700573e+20,2.834092911928506e+20,2.9215127543078316e+20,3.0501038103329754e+20,3.
101031264805499e+20,3.222656205444636e+20,3.314104866317453e+20,3.386225009851631 5e+20,3.4287943522897114e+20,3.479899638933181
6e+20,3.4915629764896824e+20,3.5231373850482645e+20,3.576033257688841e+20,3.574205878828779e+20,3.468144319219295e+20,3.263459490
460912e+20
−22.180142850000003,2.3912601455548406e+20,2.5850849475978904e+20,2.766706783427776e+20,2.882064673077038e+20,2.982791267558154e
+20,3.085050897995698e+20,3.124394732781455e+20,3.237557314941112e+20,3.268604359130683e+20,3.330385708174714e+20,3.378400394330
0255e+20,3.400629415698225e+20,3.3909712891727315e+20,3.4488893876065645e+20,3.4014695194023474e+20,3.310358126439945e+20,3.16916
5514852966e+20
−24.17585715,2.3066672214791678e+20,2.4977938802389048e+20,2.6599093665592066e+20,2.7583486713933332e+20,2.8459340816559528e+20,
2.9295826555824662e+20,3.020142155600555e+20,3.1380878099442663e+20,3.161156935016588e+20,3.189935085990646e+20,3.21985381571628
7e+20,3.270677023743773e+20,3.290864041218011e+20,3.331673458378113e+20,3.3574591231652246e+20,3.2132740253912996e+20,3.076079571
578685e+20
−26.171571450000002,2.200716040458546e+20,2.387709634279754e+20,2.5318395427592484e+20,2.63330199449945e+20,2.6790600517268996e+
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417e+20,3.136591566850793e+20,3.161745620823856e+20,3.098958600192409e+20,3.1268782510308675e+20,3.056548693493797e+20,2.91198953
4946504e+20
−28.167285749999998,2.0736334918145268e+20,2.2531949273189745e+20,2.3618687801982114e+20,2.4799616059937397e+20,2.58575335127489
28e+20,2.6528880618787684e+20,2.7408843257777104e+20,2.7760153777552045e+20,2.8470378675445803e+20,2.8407953451300074e+20,2.95289
7066407817e+20,2.9769105416292984e+20,2.989407850746096e+20,2.991670611817782e+20,2.9171895929107055e+20,2.8944516036049725e+20,
2.7451155050174503e+20
−30.163,1.9318583265641057e+20,2.1083843508748268e+20,2.2690403869646098e+20,2.331741557582497e+20,2.3960922748895e+20,2.48700130
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0653818435207e+20,2.764174208027758e+20,2.768975676643289e+20,2.7724832629115958e+20,2.7046310914801113e+20,2.6223929106335115e
+20
−32.15871425,1.7887954111102855e+20,1.9536764942959827e+20,2.077049095925447e+20,2.151867382742938e+20,2.1929159535122422e+20,2.2
5997707839021e+20,2.324707988614407e+20,2.374684960793946e+20,2.4365398955638853e+20,2.4578306989267508e+20,2.503791116900449e+
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234e+20
−34.15442855,1.609448600639738e+20,1.737782106624202e+20,1.8914965559032807e+20,1.918263363702506e+20,1.9757951370753196e+20,2.05
09997489538302e+20,2.1202439170337502e+20,2.1530079614945205e+20,2.1924989677320942e+20,2.181896491057441e+20,2.247363258540723
7e+20,2.2568067600323176e+20,2.32477544219649e+20,2.3172513017223433e+20,2.321365970229395e+20,2.2854193431230815e+20,2.112584869
395438e+20
−36.15014285,1.3199316941964799e+20,1.4499454075891886e+20,1.5705830730723862e+20,1.5949031554851534e+20,1.6530358789528866e+20,
1.7124624847678446e+20,1.7320362878595115e+20,1.7734957122651387e+20,1.8089946928721 75e+20,1.8756572284014713e+20,1.869126495227
0897e+20,1.9109293196869566e+20,1.935237137812072e+20,1.929466790472079e+20,1.9385055704711758e+20,1.880569075605865e+20,1.773765
5265934698e+20

C.137 "NBSR_upper_Cycle14_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,2.8041803143466914e+19,2.986734236038086e+19,3.1645786712713343e+19,3.3021410537134928e+19,3.382680630720371e+19,3.55
635552081521365e+19,3.5523509182339478e+19,3.79230170613399e+19,3.745918170236318e+19,3.6759320127534866e+19,3.804406558045942e+
19,3.75992398673424e+19,3.864470906818774e+19,3.8498089509315e+19,3.774254806854925e+19,3.648355761969267e+19,3.5160915517925237e
+19
24.15442855,3.482573253641191e+19,3.719752312532012e+19,3.8858536923390255e+19,3.918875286840476e+19,4.155704813533231e+19,4.2428
42577296574e+19,4.314533470762952e+19,4.423293173657494e+19,4.447147213137762e+19,4.471258802660454e+19,4.559629127332765e+19,4.
580269923450106e+19,4.606392856147631e+19,4.732868322552906e+19,4.4720375849263194e+19,4.389854593374151e+19,4.1964958342618554
e+19
22.158714250000003,3.876428587278909e+19,4.057516926595958e+19,4.345635704246954e+19,4.550651668771925e+19,4.637519618206937e+19
,4.77568908288784e+19,5.0192394392135746e+19,5.006227030015417e+19,5.061128113687001e+19,5.122921727018683e+19,5.170433577380286
5e+19,5.223084164267834 6e+19,5.284988156188379e+19,5.099196462557009e+19,4.909891050513509e+19,5.02331731485767e+19,4.71904647036
9755e+19
20.163,4.147659440833899e+19,4.46288842633836e+19,4.7133558408994955e+19,5.014511556323951e+19,5.122431155512626e+19,5.2673705069
77114e+19,5.465634981149977e+19,5.411077297532633e+19,5.731794551761105e+19,5.624310334784056e+19,5.557966670733697e+19,5.5999718
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18.167285749999998,4.601566858956797 5e+19,4.93770645490693e+19,5.181354925533876e+19,5.303329398371078e+19,5.472355810782955e+19
,5.724889757813355e+19,5.985665306145509e+19,6.074323841577628e+19,6.074820545227511e+19,6.189724656233669 4e+19,6.06751102978726
3e+19,6.096197198603936e+19,6.219710839541393e+19,6.213517374277424e+19,6.059005746301004e+19,5.97237081833137e+19,5.709522605386
1245e+19
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6.093597169621837e+19,6.171818796262596e+19,6.350877395973334e+19,6.56071935768914e+19,6.713777667578867e+19,6.8416941877831836e
+19,6.742537422121448e+19,6.622224760261013e+19,6.822623220485223e+19,6.440161410075676e+19,6.284466278340892e+19,5.996789015045
3486e+19
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46994845e+19,6.475973130017823e+19,6.788773786567303e+19,6.87560494313936e+19,6.929506487367354e+19,6.978870245164321e+19,7.06882
8795087492e+19,7.027743431455232e+19,7.004747892108819e+19,7.17043841827951e+19,6.695871807607794e+19,6.432434908855281e+19
12.18014285,5.354097417104133e+19,5.910037576343025e+19,6.159186579981632e+19,6.351306646041134e+19,6.7839907143832486e+19,6.6568
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9
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511397265e+19,7.139225806206644e+19,7.531866975366871e+19,7.568107945376819e+19,7.742628758656534e+19,7.788926444540646e+19,7.732
633364220625e+19,7.934564860401264e+19,7.82798820071042e+19,7.618084917556358e+19,7.554004014577685e+19,7.068276902143177e+19
8.18871425,5.850390213350342e+19,6.359952968853866e+19,6.560658036250882e+19,6.856779261594431e+19,7.184664991955157e+19,7.36292
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756500531177e+19,8.197511187647719e+19,8.254724089541596e+19,7.83878077384367e+19,7.719142647804061e+19,7.351765911205762e+19
6.193,5.834569282280009e+19,6.202663479705918e+19,6.683178269888574e+19,6.983224067280576e+19,7.461163357056433e+19,7.40769106289
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8461317e+19,8.197879116277262e+19,8.245709838117803e+19,8.077995704484626e+19,7.900224854977287e+19,7.483790967773302e+19
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3185e+19,7.291732223152058e+19,7.402049490576586e+19,7.502126077812177e+19,7.316996655713984 5e+19,7.300685153137596e+19,6.8164910
766595146e+19

C.138 "NBSR_lower_Cycle14_cladding_fluence_position_4.csv" File

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3434256987e+19,7.316567405646185e+19,7.264076254498105e+19,7.719449254995349e+19,7.789171730293673e+19,7.540635941037638e+19,7.77
1756441828657e+19,7.73888815092285e+19,8.004532621452617e+19,7.912489142628706e+19,7.801987910889408e+19,7.378808665477145e+19
−12.20157145,6.149191185545725e+19,6.617748295268244e+19,7.462880357327631e+19,7.586810984045235e+19,7.949343327021226e+19,8.2020
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−14.197285749999999,6.745848779787331e+19,7.3309779436366045e+19,7.575037267899872e+19,7.831483522691074e+19,8.086703348717135e+
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2e+19,9.256777712100907e+19,9.202201632052086e+19,9.205635632594484e+19,9.116167654177373e+19,8.776508207671281e+19,8.603091180028
0195e+19
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,8.475849195896707e+19,8.57230781827513e+19,8.959246093677447e+19,9.12781872744622e+19,8.979052918234492e+19,9.309514149002016e+1
9,9.25101349690474e+19,9.313316078173956e+19,9.254570140323652e+19,9.182394807495043e+19,8.985982240757542e+19,8.577888069156526e
+19
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,8.327757922505805e+19,8.465915122899057e+19,8.679742978101572e+19,9.075327576298142e+19,8.962005558399017e+19,9.147012337620695e
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562e+19
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19
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0011e+19
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,7.280265114197981e+19,7.47348896614611e+19,7.62673124035061e+19,7.73146825689374e+19,7.977244581428208e+19,7.899243711965173e+19
,7.8580357054564e+19,7.952470720372338e+19,7.836941130695957e+19,7.67854785567786e+19,7.617042453105987e+19,7.420936493559774e+19
−28.167285749999998,5.488262591866848e+19,5.969384464288249e+19,6.136865576456047e+19,6.455553091078208e+19,6.5525636064009445e+
19,7.084895011910851e+19,7.102187657499354e+19,7.081154404177169e+19,7.428662994780168e+19,7.331223229389632e+19,7.5477492278754
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703e+19
−30.163,5.133009103611976e+19,5.552668498468282e+19,5.663556055268601e+19,6.162007366141458e+19,6.2416025929991774e+19,6.55299285
6468745e+19,6.633569226338577e+19,6.76620749728869e+19,6.840835687647584e+19,7.1006546215429276e+19,6.932817845033238e+19,6.88308
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761379607e+19,6.105395414342503e+19,5.986352106253988e+19,6.3925146525499064e+19,6.348179252690022e+19,6.334136643329145e+19,6.57
40874312291885e+19,6.367863434370551e+19,6.357438789866845e+19,6.446293553901385e+19,6.073992705811039e+19,5.749890508190776e+1
9
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26335997846e+19,5.387284579488877e+19,5.27083516823864e+19,5.628216510401033e+19,5.530519194969818e+19,5.473907243170 86e+19,5.661
8390549974024e+19,5.6753910928522215e+19,5.6015355526153675e+19,5.461023608993043e+19,5.574376287611297e+19,4.876133598752974e+
19
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32534314401095e+19,4.320806653896654e+19,4.594030454194999e+19,4.508070062046193e+19,4.570544343342529e+19,4.587971896095198e+1
9,4.633171928234508e+19,4.601211194614906e+19,4.559880545229619e+19,4.588701621210457e+19,4.427303595717763e+19,4.116587868069 02
46e+19

C.139 "NBSR_upper_Cycle14_cladding_fluence_position_5.csv" File


```
,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
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78563864016096e+20,2.4744703569294836e+20,2.499746312101742e+20,2.5196578712050113e+20,2.5558908906883967e+20,2.590557132983354
e+20,2.6279833933718336e+20,2.6565174150153932e+20,2.6737623400959735e+20,2.6645691971228123e+20,2.5631852242212776e+20,2.4314929
32330348e+20
24.15442855,2.262415884082419e+20,2.4741694527012425e+20,2.562822063945814e+20,2.707276845517318e+20,2.7733512636359318e+20,2.849
469657373078e+20,2.8490027370189108e+20,2.981099693216786e+20,3.034754029914548e+20,3.078800183324334e+20,3.116558475964666e+20,
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e+20
22.158714250000003,2.5710709902027706e+20,2.759167260877115e+20,2.8454956463587207e+20,2.977468090462151e+20,3.107147436826226e+
20,3.1819895815953154e+20,3.281370984977866e+20,3.285095971803334e+20,3.339902045374709e+20,3.404669086501654e+20,3.424954181888
2576e+20,3.479397095184169e+20,3.553305399244923e+20,3.4832673461198225e+20,3.5453677532240786e+20,3.425545614336869e+20,3.340607
613909895e+20
20.163,2.8062535845929232e+20,3.0179760251881344e+20,3.1463272425448014e+20,3.258211735411166e+20,3.3360525464547964e+20,3.453913
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17551191680025e+20,3.8373693667124616e+20,3.868487014315747e+20,3.825903878015686e+20,3.750677820955392e+20,3.5815177646445e+20
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12e+20,4.2053856138592865e+20,4.1815207957574e+20,4.2198912728620846e+20,4.221281657916718e+20,4.0352709648243196e+20,3.814791173
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97e+20
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9692175e+20,5.0291057506655633e+20,5.175926262031515e+20,5.2557800186020004e+20,5.399715999779985e+20,5.426029555739282e+20,5.515
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C.140 "NBSR_lower_Cycle14_cladding_fluence_position_5.csv" File

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765225910767775e+20,4.8510638316176225e+20,4.9721622194728894e+20,5.112798630148092e+20,5.205881796753318e+20,5.186136253775974e
+20,5.3596853614160386e+20,5.3371486723215616e+20,5.2960389291390947e+20,5.282197334640001e+20,5.1385622576902577e+20,4.92621725
6622823e+20
−12.20157145,4.170875011682386e+20,4.509267756359260 6e+20,4.7952927893143015e+20,5.002927082808582e+20,5.228127957627423e+20,5.40
899921508161 1e+20,5.436862107955964e+20,5.632242336155319e+20,5.7448635255804815e+20,5.796141756475925 5e+20,5.933208820443715e+2
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05e+20
−14.197285749999999,4.278754365510781e+20,4.7711685710156556e+20,4.947280552599705 4e+20,5.172948347772715e+20,5.339078609785454e
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−32.15871425,3.115821779405868e+20,3.368238922868731e+20,3.5570822661097436e+20,3.655094036453403e+20,3.764467535415134e+20,3.841
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C.141 "NBSR_upper_Cycle14_cladding_fluence_position_6.csv" File

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88e+20,1.601887667302637e+20,1.601384831508929e+20,1.6244600887250767e+20,1.602500881685208e+20,1.5931922873577803e+20,1.52954063
44469076e+20
24.15442855,1.3638746368515195e+20,1.4688446748600276e+20,1.5225683869170752e+20,1.5799100638312913e+20,1.6396371446937095e+20,1.
685057934010745e+20,1.7031232297212877e+20,1.756871470353637 7e+20,1.816396190469807 5e+20,1.8441625377126233e+20,1.8872469802320
635e+20,1.909525058750869e+20,1.896199910217601e+20,1.961366202653423 4e+20,1.9194652638923453e+20,1.890772962931847e+20,1.8229085
27212712e+20
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745153e+20,2.0868053367521522e+20,2.1017003141048022e+20,2.1363407945762397e+20,2.1459314675196507e+20,2.1142037553654258e+20,2.0
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e+20,2.2129803281099666e+20,2.285517457424293e+20,2.3370642584232133e+20,2.3853794196259845e+20,2.4646189821418124e+20,2.47235161
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e+20,2.8351844335294828e+20,2.8980879648936185e+20,2.964088228889738e+20,2.953209805742928e+20,2.8694999103781572e+20,2.734371989
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e+20,3.023091716780722e+20,3.059167118907375e+20,3.083149933409728e+20,3.089177830790401e+20,3.040482476670436e+20,2.908512609397
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C.142 "NBSR_lower_Cycle14_cladding_fluence_position_6.csv" File

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−14.197285749999999,2.536585822080348e+20,2.732305456565542e+20,2.9152089104550016e+20,3.07893101845764e+20,3.1934304079713016e+20,3.311670405218646e+20,3.363805892024835e+20,3.43620198203117e+20,3.556158979549714e+20,3.599850504307899e+20,3.6903732114630345e+20,3.7295882712284 51e+20,3.840187617268962e+20,3.800371607408625e+20,3.7477148883772506e+20,3.749468681511404e+20,3.492004490845133e+20
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−28.167285749999998,2.0939983414597104e+20,2.273161187615487e+20,2.395865385567949e+20,2.5085006033585947e+20,2.588684516023582e+20,2.6502328436022356e+20,2.7536453170790138e+20,2.7712936270094077e+20,2.8841312055463022e+20,2.966810900748353e+20,3.0269427031032686e+20,3.063520941023 63e+20,3.109837023339219e+20,3.1894935716351954e+20,3.106703497844281e+20,3.046430656181375e+20,2.8893251313666785e+20
−30.163,1.96854694307333e+20,2.1027489106989993e+20,2.2102453919636993e+20,2.3357703760759885e+20,2.4656185215854032e+20,2.52507578811949e+20,2.5116034681344043e+20,2.622276399900823e+20,2.6975362010737643e+20,2.7518118060751264e+20,2.8066393040208013e+20,2.8422854560796547e+20,2.878146233172409e+20,2.9043488837396686e+20,2.934243084890006e+20,2.8795259655331938e+20,2.6949729649546183e+20
−32.15871425,1.7910336436066697e+20,1.9706564005493745e+20,2.0656249119781493e+20,2.1731275253866755e+20,2.2506132947683505e+20,2.2711805051597826e+20,2.3132286153726778e+20,2.3785359471164916e+20,2.4755096695762734e+20,2.5294786673863498e+20,2.6144885772421703e+20,2.6633985563960354e+20,2.6359878734951108e+20,2.6795628875206076e+20,2.688282796040768e+20,2.6771222942779752e+20,2.5091506106041 2e+20
−34.15442855,1.6115151331090022e+20,1.7744216659828205e+20,1.854390953613908e+20,1.9052509545043486e+20,1.981614541565919e+20,2.0701504341215235e+20,2.1135292195445975e+20,2.1510702040455958e+20,2.2449901188801738e+20,2.2394589251493832e+20,2.3002775276127778e+20,2.3386156908111184e+20,2.3539337860877422e+20,2.3935964923524366e+20,2.4278751763381564e+20,2.3761444110244 65e+20,2.2371655033585677e+20
−36.15014285,1.3669468409082003e+20,1.479269319363735e+20,1.5636782791246368e+20,1.6302181717774185e+20,1.662884101936977e+20,1.7701659581677778e+20,1.759729049376419e+20,1.8003361057902723e+20,1.8555070037901876e+20,1.893507899078114e+20,1.955001037362336 2e+20,1.9572392698587208e+20,2.025017855564296e+20,2.0638159295495648e+20,2.025348991330884e+20,1.9753597548636943e+20,1.8880993482238375e+20

C.143 "NBSR_upper_Cycle14_cladding_fluence_position_7.csv" File

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C.144 "NBSR_lower_Cycle14_cladding_fluence_position_7.csv" File

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−20.18442855,2.387298780643432e+20,2.5732928350210494e+20,2.7197652224419643e+20,2.849404875061305e+20,2.936971888892447e+20,3.02 7359688883416e+20,3.0910726632325454e+20,3.2190137120121645e+20,3.279820050187907e+20,3.300620282044716e+20,3.3346843409965364e +20,3.3966619186429906e+20,3.4580385461945244e+20,3.4275740556683955e+20,3.412936628356425e+20,3.338572120182037e+20,3.184661442 300536e+20
−22.180142850000003,2.2819485497177296e+20,2.4889574609860562e+20,2.6248764288829237e+20,2.7665228191130046e+20,2.84227319179200 4e+20,2.9148593782569356e+20,3.0262559029947885e+20,3.070867249326831e+20,3.163119221040815e+20,3.206620649340404e+20,3.216039622 256695e+20,3.300638678476194e+20,3.3222360890303454e+20,3.344980210479904e+20,3.339933456111345e+20,3.2546230712080636e+20,3.1162 94170787692e+20
−24.17585715,2.220461543577333e+20,2.419897257220907e+20,2.5389589617408973e+20,2.6857992777913552e+20,2.733593206768941e+20,2.85 23299076661687e+20,2.9146018282162563e+20,2.999495227339389e+20,3.0467863205232666e+20,3.098376046528967e+20,3.134095784313729e +20,3.179479780767811e+20,3.191952561309305e+20,3.2660165944362336e+20,3.2199274014421955e+20,3.151486544203443e+20,3.0063203034 17404e+20
−26.171571450000002,2.1231566853509628e+20,2.3114441615193958e+20,2.447209825820621e+20,2.5523638281439003e+20,2.609423426442134 5e+20,2.7033801341396664e+20,2.7407064936067645e+20,2.8165427162993236e+20,2.8703829390890592e+20,2.9437785685389856e+20,2.940859 668077947e+20,3.005486331857108e+20,3.048429735068557e+20,3.081445197426181e+20,3.05165524272088e+20,3.01560436916953e+20,2.91328 9549437554e+20
−28.167285749999998,1.9995817229752497e+20,2.1608755020229063e+20,2.2526798272376147e+20,2.3765859253799164e+20,2.43000916238950 33e+20,2.538400936652758e+20,2.5899232090763756e+20,2.681543569976312e+20,2.757128374772017e+20,2.8229937316039708e+20,2.84878552 8534908e+20,2.8759509256828043e+20,2.9254802513630665e+20,2.9685708260263317e+20,2.9228189009427078e+20,2.90737816278957e+20,2.73 54267177728282e+20
−30.163,1.8922876024567975e+20,2.054280445900585e+20,2.160458516242758e+20,2.2395999644573745e+20,2.326737728220717e+20,2.3977540 85866268e+20,2.4433281787789466e+20,2.4901900218950238e+20,2.5517812744804578e+20,2.6255019075531463e+20,2.6429907817440723e+20, 2.7035150413038322e+20,2.699345183502349e+20,2.7151047931344244e+20,2.6837572738973935e+20,2.661172588187303e+20,2.5291413994759 356e+20
−32.15871425,1.7084827234249585e+20,1.8505154387160593e+20,1.9618874348786087e+20,2.0591922931049788e+20,2.140504520233897e+20,2. 1912725389669522e+20,2.2660785614967914e+20,2.3328514756149504e+20,2.3499110997380763e+20,2.44016399256488e+20,2.42070670020590 1e+20,2.4821201206203892e+20,2.531097553356337e+20,2.5003080592074464e+20,2.5096227856786997e+20,2.4216203896359318e+20,2.3664372 27348365e+20
−34.15442855,1.5679953083779367e+20,1.7078020554603047e+20,1.7700678438665663e+20,1.8528640498013064e+20,1.9255606148551016e+20, 1.9774324194767784e+20,2.0502638916947447e+20,2.0525757099170375e+20,2.1121126943208587e+20,2.159710394696021e+20,2.2096505740126 054e+20,2.2244229084887417e+20,2.2644167505200238e+20,2.2478415657591292e+20,2.2638158004251043e+20,2.260860107101112e+20,2.10130 78568999574e+20
−36.15014285,1.3337658106672824e+20,1.4329287084728426e+20,1.5314109383137493e+20,1.6209157095938163e+20,1.6159118802320366e+20, 1.6579048011505002e+20,1.7057539194225177e+20,1.7423382894867048e+20,1.7771014128346563e+20,1.8170155369962042e+20,1.87846575027 36466e+20,1.8587141150110338e+20,1.886370083664987e+20,1.8902578628504876e+20,1.9146086059823828e+20,1.861675940478852e+20,1.7464 468258499304e+20

C.145 "NBSR_upper_Cycle15_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,4.454964758624013e+19,4.877636464359097e+19,5.1293428706906096e+19,5.183896551007637e+19,5.403926961521751e+19,5.4721
76126437196e+19,5.60418192230512e+19,5.767676957382282e+19,5.610365641108647e+19,5.956653894106197e+19,6.0296653525297275e+19,5.8
51924938396787e+19,6.31363915060448e+19,6.164441371418694e+19,5.996505679214166e+19,5.763537185901397e+19,5.657469807833506e+19
24.15442855,5.467299703890789e+19,5.85518280367247e+19,5.781362066613578e+19,6.074455812386823e+19,6.671807198954446e+19,6.779628
014065619e+19,6.705848778475207e+19,7.08670775471664e+19,7.313419901656042e+19,7.313492529225881e+19,7.307661572904435e+19,7.3851
7594065805e+19,7.5859911712659e+19,7.405957800999436e+19,7.52141488631094e+19,7.042021048530745e+19,6.902887375451373e+19
22.158714250000003,6.0505613419094335e+19,6.392958832236978e+19,6.5920102504344265e+19,7.136385012487261e+19,7.0648572315618935e
+19,7.48204036809049e+19,7.785384976578308e+19,7.81139602194818e+19,8.064824739221615e+19,8.292273537226537e+19,8.42818047113158
7e+19,8.399814217425471e+19,8.580843622935455e+19,8.329956870606425e+19,8.030212514509252e+19,7.918791447007434e+19,7.64265105110
8291e+19
20.163,6.6524778900098605e+19,7.156367969560407e+19,7.478025101015047e+19,7.905614730765017e+19,8.190729819222971e+19,8.317682811
30345e+19,8.361487611284144e+19,8.883710964535751e+19,8.572512203138086e+19,9.188674130294291e+19,9.089412993057128e+19,9.6308411
5084788e+19,8.840061795061857e+19,9.210337896840877e+19,9.008350249748467e+19,8.729875396247321e+19,8.424434963601262e+19
18.167285749999998,7.141437816274708e+19,7.516081947611254e+19,8.023624156388044e+19,8.320681292401133e+19,8.754724400499751e+19
,8.80282460246813e+19,9.035876098717861e+19,9.382319982222211e+19,9.569937745853404e+19,9.674573323258734e+19,9.998201774466176e+
19,9.626161860276753e+19,9.778389246661583e+19,9.675268472855775e+19,9.904097194687658e+19,9.466785845946237e+19,8.9410037417724
64e+19
16.171571450000002,7.694507135974115e+19,8.119440671741364e+19,8.612291360676215e+19,9.130489071485257e+19,9.374818591794441e+19
,9.351411763571692e+19,9.793340150681521e+19,9.928500058153932e+19,1.0191256230468254e+20,1.06285260777412e+20,1.061431182478678
1e+20,1.0172175930334552e+20,1.0373364674158721e+20,1.0388025067899298e+20,1.0555483493216307e+20,1.0118141018373525e+20,9.711198
369192378e+19
14.17585715,8.173060569017588e+19,8.314985215852246e+19,9.24362207456188e+19,9.288101273405373e+19,9.80015676687937e+19,1.0218605
698196611e+20,1.0357479987097982e+20,1.077907265465258 5e+20,1.1107038009316195e+20,1.0798682098509411e+20,1.1284975555424413e+2
0,1.1212140478241923e+20,1.1102576601454587e+20,1.1189937192605096e+20,1.1045304574952115e+20,1.0932628088028776e+20,1.044135445
4896165e+20
12.18014285,8.559802378416069e+19,8.916227365089871e+19,9.591871271945113e+19,9.915198837506074e+19,1.0457747534945786e+20,1.0768
386026518972e+20,1.083043072189665e+20,1.153242806123672e+20,1.1262876023458788e+20,1.1385409109146137e+20,1.1605574399432812e+
20,1.1643651996763259e+20,1.1831446141635492e+20,1.1506489643436689e+20,1.1679447013327303e+20,1.148439011147106e+20,1.124171027
4533961e+20
10.18442855,9.021060074470482e+19,9.420667339097778e+19,9.891885387587404e+19,1.0182084405934165e+20,1.0742032594034139e+20,1.110
0086513345783e+20,1.1562101611199957e+20,1.1688992351077715e+20,1.1743151767444183e+20,1.2043207384554956e+20,1.21586852206007e
+20,1.2127144104555859e+20,1.1847942975356314e+20,1.215173372463029e+20,1.2005026033553308e+20,1.1761204906233006e+20,1.18723250
88088343e+20
8.18871425,9.204621069557752e+19,9.738734593528894e+19,1.0028176210075892e+20,1.0598022498408361e+20,1.0811443800067024e+20,1.129
5869690900428e+20,1.1832587432018693e+20,1.1892453300301169e+20,1.2569342251210808e+20,1.2746449767949428e+20,1.286338015539197
2e+20,1.2946175585009677e+20,1.2669672051261335e+20,1.2944411772599273e+20,1.273908325729422e+20,1.226524024092323e+20,1.19988008
13281301e+20
6.193,9.06026858681701e+19,1.0105794331500708e+20,1.0507549297121847e+20,1.0884590138263115e+20,1.124191778187636e+20,1.178880338
277224e+20,1.1928144563194012e+20,1.2272814258920841e+20,1.2382274382036977e+20,1.2455316966561866e+20,1.2446186643496257e+20,1.3
00583394594975e+20,1.2607731109554856e+20,1.2768964314599855e+20,1.2556061781297191e+20,1.2557410579022794e+20,1.21400095597846
76e+20
4.19728575,9.037930421407623e+19,9.47882127180545e+19,1.0465944074970595e+20,1.0776582566543783e+20,1.1361026996414108e+20,1.1627
88143874084e+20,1.1986869141093288e+20,1.2407590277809809e+20,1.2289311092641661e+20,1.279500648607109e+20,1.2798845371905494e+
20,1.2858607486516768e+20,1.3248098768202052e+20,1.2893987488396013e+20,1.3154097942094737e+20,1.2792516397962284e+20,1.20207965
91575726e+20
2.20157145,8.989000190069644e+19,9.77683294159358e+19,1.038003603521689e+20,1.0546560677493095e+20,1.0736014881104532e+20,1.15783
90937578378e+20,1.1730390065886567e+20,1.1969438524331663e+20,1.2085642636075809e+20,1.244110271360745e+20,1.2673095922410937e+
20,1.2334340185942517e+20,1.2473992627377891e+20,1.2783178567554274e+20,1.2412051685671413e+20,1.2152460000328691e+20,1.22599488
03692026e+20
0.20585714999999993,8.242212390872475e+19,8.857243403012599e+19,9.024349065847526e+19,9.511213167954131e+19,9.93637496179802e+19
,1.0302957432882315e+20,1.0266259759378828e+20,1.1178420535101882e+20,1.074950285836055e+20,1.1182363174607485e+20,1.13001235914
19634e+20,1.1552763780791946e+20,1.1677268186232101e+20,1.1670939212288893e+20,1.1573203254018372e+20,1.144351116501821e+20,1.087
7638642292708e+20

C.146 "NBSR_lower_Cycle15_cladding_fluence_position_0.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,8.177407847840874e+19,9.447549915305732e+19,9.826966715517483e+19,9.859638746578403e+19,1.053328020757948e+20,1.0920281401155962e+20,1.0603625196653167e+20,1.1249076785189169e+20,1.1462498086847835e+20,1.1413941368726174e+20,1.1793576051647445e+20,1.2311929392963287e+20,1.2307156724088082e+20,1.2327077428958506e+20,1.2009591195086114e+20,1.2130982990390264e+20,1.1920466791525202e+20
−12.20157145,9.459689094836147e+19,1.02746949328474e+20,1.090513336516074e+20,1.1447038789839015e+20,1.1892142039287569e+20,1.220101671845035e+20,1.2636885891162092e+20,1.261177750273166e+20,1.2946694353365677e+20,1.3573366527414456e+20,1.3789796685537924e+20,1.3650559258787352e+20,1.3713330229863429e+20,1.406049001369906e+20,1.3792909295673929e+20,1.316613336795395e+20,1.3091741985703458e+20
−14.197285749999999,1.0044673043796713e+20,1.1096247627511379e+20,1.1297114734954827e+20,1.2067278236273384e+20,1.2321993499069702e+20,1.262713304606928e+20,1.314206251623552e+20,1.3356936369290987e+20,1.3531346290578403e+20,1.3643296501803334e+20,1.4191427146753624e+20,1.4047520804799044e+20,1.4267063573058516e+20,1.4337304808461004e+20,1.4052915995701451e+20,1.3809302375723549e+20,1.3424998777598271e+20
−16.192999999999998,1.0158397442799174e+20,1.0863009374653489e+20,1.1417157732533377e+20,1.2061156769672579e+20,1.2154016305396692e+20,1.2559693159789198e+20,1.2671954632027736e+20,1.3091741985703458e+20,1.363406242506653e+20,1.3856199035106006e+20,1.418042925760641e+20,1.4322053018794585e+20,1.4278476476890531e+20,1.422110069671686e+20,1.4132910076196751e+20,1.4177731662155206e+20,1.324664621680525e+20
−18.188714249999997,9.842602393767343e+19,1.0736533649460534e+20,1.1436870930061404e+20,1.1751763322153792e+20,1.2156195132491896e+20,1.2516220371556342e+20,1.3195184395889985e+20,1.3572432744373654e+20,1.3678884011024987e+20,1.3585816967958472e+20,1.3956736342498932e+20,1.3956217574142932e+20,1.4177316647470406e+20,1.3979250889149358e+20,1.3919281267195684e+20,1.3557699723063235e+20,1.3588618317080873e+20
−20.18442855,9.970250535444862e+19,1.0633194992945204e+20,1.1310291451197243e+20,1.1956365561760449e+20,1.1993509376050094e+20,1.2407797785152209e+20,1.2807353172943903e+20,1.2905711653241627e+20,1.3371876897943804e+20,1.3617669345016909e+20,1.36115478784161e+20,1.379477686175553e+20,1.3907142087665268e+20,1.4003217987196589e+20,1.3823620382349166e+20,1.3598993684200887e+20,1.3478639425608737e+20
−22.180142850000003,9.560029270253791e+19,1.0137532579520828e+20,1.0673970185726856e+20,1.1462394333176634e+20,1.1960411954937253e+20,1.2328529980355309e+20,1.2695610469061365e+20,1.2882055816207996e+20,1.297367030787771e+20,1.3185535304468375e+20,1.3272999649290083e+20,1.396358408479814e+20,1.3757840554808284e+20,1.3844059855575592e+20,1.3511010571023176e+20,1.3364302879946197e+20,1.2535103539714767e+20
−24.17585715,9.210690659322959e+19,9.72067107937295e+19,1.0374433336972083e+20,1.1219714496239534e+20,1.1325543240863664e+20,1.1903036174763581e+20,1.2193753961466341e+20,1.2238679301095997e+20,1.257141732463481e+20,1.2678698620655744e+20,1.2933828898136862e+20,1.3179828852552365e+20,1.3017454357124165e+20,1.3168000934035553e+20,1.3056673244837814e+20,1.2731924253981411e+20,1.2512692746735537e+20
−26.171571450000002,8.686537487787028e+19,9.568526695925082e+19,9.947642610490352e+19,1.0475074398036207e+20,1.051605709816026e+20,1.1246690450751567e+20,1.146675198736704e+20,1.1789529658470641e+20,1.185250813688912e+20,1.2411947932000215e+20,1.2299686459761674e+20,1.2291904934421663e+20,1.2364636257932955e+20,1.2902910304119221e+20,1.241817315227222e+20,1.218296357966153e+20,1.130033109876203 2e+20
−28.167285749999998,8.380567911417846e+19,8.907148918859861e+19,9.750479509108747e+19,9.851172447008473e+19,1.0077075315312512e+20,1.0352873324096697e+20,1.0698974820486087e+20,1.1149680768179447e+20,1.1319110513249255e+20,1.1455857851891026e+20,1.1527759146032716e+20,1.1794924849373046e+20,1.186163845995473e+20,1.2028474363244537e+20,1.1843792828508307e+20,1.1236003822617954e+20,1.0984816184642 44e+20
−30.163,7.993151703156566e+19,8.564585422658396e+19,8.464328250177711e+19,9.064314979993816e+19,9.536062172206562e+19,9.823387213861077e+19,1.0402758089209718e+20,1.0459615101027389e+20,1.0792249370895001e+20,1.0690778280461276e+20,1.1027873958190495e+20,1.1035862990872904e+20,1.1054746159031329e+20,1.0930864275618374e+20,1.072740332639492e+20,1.0657058337321234e+20,1.0138850251145072e+20
−32.15871425,7.168517524457941e+19,7.709603295133734e+19,8.389884991091619e+19,8.70483963538673e+19,8.758750042942317e+19,9.155773841156 73e+19,9.20503608424255e+19,9.72210288003551e+19,1.0001864279059541e+20,9.86337387874161e+19,9.922845483073523e+19,1.0296825590914387e+20,1.0157536287328213e+20,9.901928742959576e+19,9.8564223827712e+19,9.629430100919558e+19,9.428230981728267e+19
−34.15442855,6.681352536704858e+19,6.877944992894861e+19,7.325766588528857e+19,7.536874183319758e+19,7.735956727618567e+19,7.905438349523978e+19,8.349244678082527e+19,8.558017815271427e+19,8.734990452237486e+19,8.687180760548468e+19,8.706022427238 41e+19,9.091228682303 13e+19,8.997798501387413e+19,8.93797413457342e+19,9.07608064630791e+19,8.974111538252425e+19,8.278993067312921e+19
−36.15014285,5.1360868593186185e+19,5.617856656169296e+19,6.0322280682083705e+19,6.064453958483131e+19,6.538224347284279e+19,6.6048757056632414e+19,6.873919350452296e+19,6.861157648894681e+19,6.829886292394961e+19,6.994211356841725e+19,7.525845168071185e+19,7.212125192463357e+19,7.43132557360787e+19,7.324500793740216e+19,7.426480277162821e+19,7.1910113203741295e+19,6.761740881150716e+19

C.147 "NBSR_upper_Cycle15_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0455763293334828e+20,1.1450826141387453e+20,1.205296547882132e+20,1.2691650136531214e+20,1.2859169960346826e+20,1.3185993042681093e+20,1.3297959074265029e+20,1.3790646403825225e+20,1.3771453978915162e+20,1.4228515752971377e+20,1.4370404798472597e+20,1.4825197819424427e+20,1.4851503091904987e+20,1.4858186715915203e+20,1.4701520116409754e+20,1.4621439263590136e+20,1.3888877281479914e+20
24.15442855,1.2403579809925887e+20,1.358811420102033e+20,1.4173391184299988e+20,1.4845310009106537e+20,1.5335667267910877e+20,1.569848381165794e+20,1.6181850858194724e+20,1.6316872326731317e+20,1.69019040394051e+20,1.721493065015865e+20,1.7410227370090103e+20,1.759890178366281e+20,1.8042289721441252e+20,1.7661261835207642e+20,1.775268645354918e+20,1.7606259901839196e+20,1.6937406959606137e+20
22.158714250000003,1.3848101043252466e+20,1.5105787392550434e+20,1.6001086421661624e+20,1.6510206881815737e+20,1.7055566073988527e+20,1.7575601076104272e+20,1.813297852796515e+20,1.8496224295272507e+20,1.8878110628626694e+20,1.9090882879227052e+20,1.944045481025663e+20,1.9745203538061754e+20,1.9828534226409272e+20,1.979799803597729e+20,1.9676527768415532e+20,1.933529503798585e+20,1.8606534750266756e+20
20.163,1.514227139517499e+20,1.6320857974076857e+20,1.7209596014480772e+20,1.7777458784742998e+20,1.843153417297182e+20,1.8995901837100266e+20,1.9876178041601343e+20,2.027051185820391e+20,2.0346852334283863e+20,2.0677231840403384e+20,2.0719173114008755e+20,2.1463998626412947e+20,2.147846959215983e+20,2.1830003688036447e+20,2.1744710834841888e+20,2.1448362625288136e+20,2.0515721146431812e+20
18.167285749999998,1.629645354879186e+20,1.7777642737697407e+20,1.859721446724334e+20,1.9288509669914355e+20,2.0036094476634695e+20,2.0548771360574063e+20,2.1237429904231875e+20,2.1573511951938087e+20,2.1875133479518254e+20,2.2221332939716988e+20,2.273394850600488e+20,2.2807836276026045e+20,2.2902878635804302e+20,2.365714706653485e+20,2.31826710794612e+20,2.280109133436436e+20,2.1930993860007277e+20
16.171571450000002,1.7462347373839473e+20,1.895052677501258e+20,1.979854989484052e+20,2.041166509188749e+20,2.1459522437855648e+20,2.1794684720789815e+20,2.254889183386889e+20,2.288239854021337e+20,2.3385203282266096e+20,2.4188219245915167e+20,2.4263823910017749e+20,2.4668581727529927e+20,2.49685476785204e+20,2.5094187546382113e+20,2.490563576811234e+20,2.4704513871291256e+20,2.3553520235550815e+20
14.17585715,1.8152477541132563e+20,1.978052250530839e+20,2.084162446399403e+20,2.1500421311386033e+20,2.2204225314956902e+20,2.3168629337274604e+20,2.3679221421063995e+20,2.4260941980558406e+20,2.4935252193772274e+20,2.5612934877816993e+20,2.5683020953447023e+20,2.61940422607967e+20,2.59766711863361e+20,2.6780729550060166e+20,2.6518167366466293e+20,2.5773893712925334e+20,2.5009201281444905e+20
12.18014285,1.9327875602157997e+20,2.1045076431570973e+20,2.206675114036151e+20,2.311632538057083e+20,2.3400716648087965e+20,2.4463596818666235e+20,2.4939728382329576e+20,2.5617104478116938e+20,2.6175830918310157e+20,2.674418422978414e+20,2.685676343788277e+20,2.7283043750901128e+20,2.7581047537044564e+20,2.767271742599198e+20,2.77534114553263e+20,2.730916507042728e+20,2.6025602672209035e+20
10.18442855,1.981081342513449e+20,2.1511887712210895e+20,2.304991836402899e+20,2.4005492644535032e+20,2.481126790250025e+20,2.5921791888270593e+20,2.6085571335346546e+20,2.671536493359331e+20,2.7121103833369264e+20,2.7891682759390796e+20,2.8519820781047874e+20,2.8761473645490527e+20,2.8745837644365722e+20,2.8895084808043315e+20,2.871971632483956e+20,2.8186681980612215e+20,2.713716905805437e+20
8.18871425,2.053172505346544e+20,2.1942582896135078e+20,2.3710125517404796e+20,2.485443552913502e+20,2.552193948303574e+20,2.6244813327621373e+20,2.6926542925255452e+20,2.7895545771433397e+20,2.8110954681046958e+20,2.866753500343873e+20,2.927071674094758e+20,2.9493299815783118e+20,3.010659896578449e+20,2.975316402271231e+20,2.960440740024647e+20,2.944013741195876e+20,2.7757519737974774e+20
6.193,2.0770128082380194e+20,2.2782757356574877e+20,2.4361257658363065e+20,2.5171509104885583e+20,2.6014381541990056e+20,2.6988902976800286e+20,2.7521998638679104e+20,2.842956119808426e+20,2.888233073653759e+20,2.9558235208689674e+20,2.9849555370822907e+20,3.020299031389509e+20,3.047977819262996e+20,3.067010818279236e+20,3.0618110814345934e+20,3.0037003431366223e+20,2.8427782986191638e+20
4.19728575,2.0656445156555103e+20,2.2827887148056684e+20,2.4452253053144318e+20,2.5372385731100797e+20,2.6283014173079444e+20,2.7130914657604443e+20,2.809354046802952e+20,2.8844559063232168e+20,2.9176594145941373e+20,2.965934801596345e+20,3.009298644715819e+20,3.0549680315305584e+20,3.088018245672804e+20,3.108559658915202e+20,3.0764414730752976e+20,2.965180594483266e+20,2.8652757449434497e+20
2.20157145,2.0593410610844107e+20,2.273161843524903e+20,2.397802233667654e+20,2.4977684008589407e+20,2.5995005164125582e+20,2.637707545043418e+20,2.6922250689652562e+20,2.824248104344977e+20,2.901158834583602e+20,2.9207069018721885e+20,2.98674601250521e+20,3.009562310617139e+20,3.046696280347277e+20,3.001805627706204e+20,3.032679065221271e+20,2.967847912322205e+20,2.8260631068284846e+20
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C.148 "NBSR_lower_Cycle15_cladding_fluence_position_1.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,1.9640166401093914e+20,2.1607359295549443e+20,2.267551278415412e+20,2.3742623872683816e+20,2.486400108276432e+20,2.5167952681100332e+20,2.627295807823839e+20,2.6868229838707632e+20,2.7506791861114585e+20,2.7995003002117485e+20,2.84553146117016e+20,2.8267743915855348e+20,2.8458196541320685e+20,2.8976514649195273e+20,2.8895820619860958e+20,2.826639492752301e+20,2.7397278535589442e+20
−12.20157145,2.1846252865675952e+20,2.4170866350549202e+20,2.55025017875198e+20,2.6550972310002657e+20,2.7624889657845508e+20,2.8386884112661253e+20,2.9536835348326705e+20,3.0462302661961056e+20,3.098754966445174e+20,3.1295302957178887e+20,3.1668604819327294e+20,3.157687361272841e+20,3.201633722081278e+20,3.242949555641658e+20,3.189302742370692e+20,3.1452460097896094e+20,3.0097891859275776e+20
−14.197285749999999,2.2688328173311322e+20,2.474958234512159e+20,2.6200357978898098e+20,2.7672104249477284e+20,2.829245492939769 4e+20,2.9293465589642897e+20,3.039681541019127e+20,3.096253206265204e+20,3.1651926418127494e+20,3.2172451961455e+20,3.2571507237220745e+20,3.2807273607122295e+20,3.3867885024596184e+20,3.331124338455294e+20,3.3153718337926916e+20,3.273356979005554e+20,3.0918 32203594228e+20
−16.192999999999998,2.2787417498086585e+20,2.5093942275776233e+20,2.642104020653806e+20,2.779909310567133e+20,2.8402949337346353 e+20,2.936956079511697e+20,3.039454665708688e+20,3.092758100131423e+20,3.1538979304120045e+20,3.214651459488326e+20,3.26114863459 7908e+20,3.33780183070036e+20,3.381858563281443e+20,3.337685327162567e+20,3.3420020898260445e+20,3.282315487885298e+20,3.08914035 8694702e+20
−18.188714249999997,2.2648349064552982e+20,2.4675142716237198e+20,2.6135545221294473e+20,2.7502928849071984e+20,2.84158260441550 2e+20,2.917383485162523e+20,3.035107244219476e+20,3.0696536090575864e+20,3.137820442196611e+20,3.2246217096173224e+20,3.270205251 720004e+20,3.300864077454926e+20,3.346343379550109e+20,3.3235638720290626e+20,3.303261597627397e+20,3.241925550862112e+20,3.12269 95093441485e+20
−20.18442855,2.22717360492252e+20,2.4292213982808018e+20,2.59907129285227e+20,2.667894224862023e+20,2.7815710189219653e+20,2.8547 90426542107e+20,3.004423891423966e+20,3.036701503157692e+20,3.0435568165920204e+20,3.146300673394891e+20,3.199585712522185e+20,3. 2544343517619605e+20,3.215184923056113e+20,3.278219468767113e+20,3.2818188149083924e+20,3.19967768899939e+20,3.069635213762145e +20
−22.180142850000003,2.17853644377664e+20,2.3915294379222893e+20,2.4987249562218702e+20,2.5830857811140814e+20,2.719909988603891e +20,2.8155716566619944e+20,2.893107826945612e+20,2.9442344847411675e+20,3.040607437556321e+20,3.046972209778891e+20,3.1312226628 984563e+20,3.14815246646928e+20,3.198126352417203e+20,3.207544743682971e+20,3.1970839523422156e+20,3.115930040621878e+20,2.976236 167043279e+20
−24.17585715,2.1210143549327796e+20,2.296303125189622e+20,2.4203671294085572e+20,2.5421623805231075e+20,2.648039569316087e+20,2.7 397830394452676e+20,2.77188282998973e+20,2.8572492643660467e+20,2.8874727347755336e+20,2.9610661800696395e+20,3.023548866917410 4e+20,3.052331372517355e+20,3.053392167887784e+20,3.078679567353947e+20,3.0886007633617676e+20,3.0091269552917034e+20,2.872633863 119831e+20
−26.171571450000002,2.0046457159733097e+20,2.219723510268934e+20,2.3208669763684414e+20,2.4505476774620142e+20,2.503446415385048 2e+20,2.581148143327634e+20,2.6714690439427142e+20,2.7099152114143065e+20,2.7576939254396086e+20,2.8095318679922144e+20,2.8727197 07831889e+20,2.8771100516771296e+20,2.891801760969304e+20,2.9346198769906958e+20,2.9236992032639166e+20,2.8826225085442687e+20,2. 7375388813401471e+20
−28.167285749999998,1.9055073370768664e+20,2.1129878743553766e+20,2.2250888047725455e+20,2.304912123455988e+20,2.337471796386475 e+20,2.424493807352477e+20,2.5087442604720426e+20,2.570190679009973e+20,2.642968599539531e+20,2.6814147670111227e+20,2.7098722890 582776e+20,2.7671613708265524e+20,2.828577130538748e+20,2.8245240337765915e+20,2.786182106312498e+20,2.7436031291318383e+20,2.625 8793700748857e+20
−30.163,1.770645294434092e+20,1.9214315311635844e+20,2.0430121704979913e+20,2.1182489288514893e+20,2.199850459427558e+20,2.271040 2527840462e+20,2.342217782610241e+20,2.3421380696633305e+20,2.4275474263956757e+20,2.4715121824995534e+20,2.5394705356235817e+2 0,2.573900396923899e+20,2.6167369082407318e+20,2.6297423821174856e+20,2.6063006439605646e+20,2.5339887375821773e+20,2.4703716741 82215e+20
−32.15871425,1.66703072698035e+20,1.811415400896391e+20,1.8939796186005357e+20,2.023415049088229e+20,2.0279464235318503e+20,2.121 2964161295408e+20,2.160110489509952e+20,2.2135059004098917e+20,2.296805929931674 4e+20,2.3031707021542444e+20,2.3154035736224783 e+20,2.3872249387891065e+20,2.388954096560556e+20,2.377604199273488e+20,2.4149159901928876e+20,2.3483372842269318e+20,2.229411699 201169e+20
−34.15442855,1.4911103849133677e+20,1.6430248664299058e+20,1.7472894009892284e+20,1.8107960926165457e+20,1.8468999258019896e+20, 1.9146743259715608e+20,1.934203997964753e+20,2.0158729779574384e+20,2.0165168132978714e+20,2.1143859168088893e+20,2.1138647167713 955e+20,2.1600062495024533e+20,2.1781501425723798e+20,2.2003777912301984e+20,2.2010829442221018e+20,2.1607297977897973e+20,2.0494 934462583538e+20
−36.15014285,1.2537865466644844e+20,1.3498529112222887e+20,1.4409280189504476e+20,1.496960088863591e+20,1.550717273907203e+20,1.6 11501461809259e+20,1.623458403845879e+20,1.6956844655122078e+20,1.7297893432597348e+20,1.738091753268752e+20,1.7771756243156304 e+20,1.811127207934483e+20,1.8026776355619383e+20,1.813346906917691e+20,1.805951998150428e+20,1.7739932382043455e+20,1.6992286257 671648e+20

C.149 "NBSR_upper_Cycle15_cladding_fluence_position_2.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0950842011302347e+20,1.1778507670842298e+20,1.2564783915640106e+20,1.2955070767245663e+20,1.3545130527339967e+20,1.4144020029245933e+20,1.3952157097796792e+20,1.4215700363814181e+20,1.459985545027275e+20,1.5264538792205858e+20,1.5105664757247495e+20,1.53630762581179e+20,1.5344006468510774e+20,1.573840160276481e+20,1.5398456543015997e+20,1.571080865960338e+20,1.4866403281212162e+20
24.15442855,1.2670495546774113e+20,1.390163135298564e+20,1.4650687783341253e+20,1.5180043068480412e+20,1.5853985375785468e+20,1.6313499855900475e+20,1.6553313190799036e+20,1.6957641784591185e+20,1.7204997190620535e+20,1.800795183661814e+20,1.7952581997340872e+20,1.835378339090806e+20,1.8688516450281937e+20,1.8526944438658898e+20,1.8806123705801097e+20,1.862909964600766e+20,1.8061298193396905e+20
22.158714250000003,1.441627040177204e+20,1.5744594685563267e+20,1.6380336096002608e+20,1.715949949322991e+20,1.7798674692151565e+20,1.7867411779449258e+20,1.850621907246209e+20,1.8878049310975223e+20,1.9058936382811265e+20,1.9945160399504916e+20,2.0219556889832466e+20,2.0506768769317215e+20,2.089300865592576e+20,2.0841808416948444e+20,2.0771354435409592e+20,2.054729973693878e+20,1.9714176806418013e+20
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C.150 "NBSR_lower_Cycle15_cladding_fluence_position_2.csv" File

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7535652004e+20
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+20,3.489188980414257e+20,3.516800318871128e+20,3.5021024778138065e+20,3.447468450354176e+20,3.38212836094791e+20,3.191694130778
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486967491908665e+20,1.7220387921139466e+20,1.7667577553309037e+20,1.8139968740232715e+20,1.8885223476197196e+20,1.9033060333890
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C.151 "NBSR_upper_Cycle15_cladding_fluence_position_3.csv" File

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3e+20,1.4986217972184236e+20,1.516612396159676e+20,1.5071449507727324e+20,1.4952861169784642e+20,1.4913127331632182e+20,1.4007159
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28048e+20
10.18442855,2.044201732936506e+20,2.2408106506094133e+20,2.375317050873663e+20,2.4793792371831348e+20,2.5694364718968943e+20,2.62
82830220125035e+20,2.7141645246611666e+20,2.7936751233221132e+20,2.8200601087495868e+20,2.8856025464057027e+20,2.90637083495853
9e+20,2.9095900116607058e+20,2.926017010489477e+20,2.950262009880653e+20,2.9630406084469686e+20,2.8703835053108874e+20,2.74330880
44047833e+20
8.18871425,2.1241292916274473e+20,2.323092807116797e+20,2.445384731208253e+20,2.5586874875942306e+20,2.6562622663781928e+20,2.766
92223198582e+20,2.8040500699508107e+20,2.8691387569860497e+20,2.932020008568374 2e+20,2.979860040245147e+20,2.9900816927451696e+
20,3.026387874180464e+20,3.0308885897983505e+20,3.033059234660383e+20,3.0153629604461865e+20,2.921209706614241e+20,2.84631019534
3827e+20
6.193,2.165843689922382e+20,2.370037601082109e+20,2.5215964402201218e+20,2.5773219218759162e+20,2.7311924364743423e+20,2.81083180
2203375e+20,2.85126466158259e+20,2.906340176132804e+20,2.9813316638804233e+20,3.0478858427857915e+20,3.040723941094114e+20,3.0915
86932988349e+20,3.118775179650078e+20,3.117585617211563e+20,3.076318837772358e+20,2.983404200500104e+20,2.856893621987522e+20
4.19728575,2.182816415849235e+20,2.3891380495149652e+20,2.5574856616254213e+20,2.6591564595275694e+20,2.7318240082844815e+20,2.84
21712538696126e+20,2.8858294217161415e+20,2.968252608821905e+20,3.019691986639957e+20,3.0764966589616207e+20,3.138593044605132e
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3301632e+20
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7842947769595e+20,2.8436367457397418e+20,2.9569578974211604e+20,2.987028073701972e+20,2.996023373172598e+20,3.102237809048661e+
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9124e+20
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8e+20,2.5274645394657863e+20,2.586127136626986e+20,2.6419568582902782e+20,2.6983568341122413e+20,2.73880195702175e+20,2.784495870
897077e+20,2.7524819250646716e+20,2.8213968335516294e+20,2.7951222198968015e+20,2.7673575873112562e+20,2.7145385623351325e+20,2.5
689459306851356e+20

C.152 "NBSR_lower_Cycle15_cladding_fluence_position_3.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,2.031741986157834e+20,2.2610700026550495e+20,2.3688664339390356e+20,2.5055434790653172e+20,2.5723981144628878e+20,2.656286793438781e+20,2.721798572269162e+20,2.798715434272934e+20,2.8270687163125898e+20,2.874565369141131e+20,2.8971670554729153e+20,2.9374895430794848e+20,2.9201918335998417e+20,2.927537688245929e+20,2.8838243345130776e+20,2.835162646306609e+20,2.7025019073516025e+20
−12.20157145,2.303740956312914e+20,2.5091857475626254e+20,2.7052918604934798e+20,2.7831100919738588e+20,2.880770715469879e+20,2.9920867799482335e+20,3.100367620678831e+20,3.1216632410343073e+20,3.175555324911153e+20,3.2859209657917256e+20,3.261577858158197e+20,3.3385008519271 17e+20,3.329652714820018e+20,3.3396536237747497e+20,3.25770871435045e+20,3.16513132416128e+20,3.0092802494203776e+20
−14.197285749999999,2.336331288069136e+20,2.6007084741465145e+20,2.787665993478068e+20,2.8908942597275507e+20,2.998929829852268e+20,3.108449287142556e+20,3.152665445617461e+20,3.230827055946071e+20,3.2872515588286205e+20,3.334637839884516e+20,3.411395275994466e+20,3.499422896444574e+20,3.470137586102577e+20,3.4454940219768 47e+20,3.3912462957214754e+20,3.3333563009687965e+20,3.156595907076677e+20
−16.192999999999998,2.4123038582402725e+20,2.6552811839546756e+20,2.7945642292684256e+20,2.9175980969426675e+20,3.01969811840510 4e+20,3.111196317928405e+20,3.191130008384494e+20,3.263356070050823e+20,3.317842935146926e+20,3.371256741342307e+20,3.448020309217404e+20,3.456071316855395e+20,3.483333144698887e+20,3.475650042969716e+20,3.428165653671469e+20,3.338979129608581e+20,3.207213628365034e+20
−18.188714249999997,2.3819025666415238e+20,2.65356428971352e+20,2.772900703004 13e+20,2.9118219741742082e+20,2.995036158983933e+20,3.095351836788598e+20,3.210947873339547e+20,3.283008377346908e+20,3.2976816913436416e+20,3.400039246942252e+20,3.4164662457710 23e+20,3.444635574856269e+20,3.433567738765962e+20,3.419483074223339e+20,3.4105184335784477e+20,3.302029112832853e+20,3.155584165827425e+20
−20.18442855,2.3517342821183606e+20,2.5936324171668947e+20,2.7024589849955733e+20,2.8096299762345666e+20,2.936606568898319e+20,3.080917661632596e+20,3.119780789134183e+20,3.187647165781007e+20,3.263049481793473e+20,3.344908546505715e+20,3.3466806266331936e+20,3.3512426599025495e+20,3.4184038835574696e+20,3.3787436265867746e+20,3.3478947161322966e+20,3.271241520029845e+20,3.144669623865793e+20
−22.180142850000003,2.286915392749589e+20,2.5085909663433684e+20,2.6557226710452584e+20,2.7393538158849786e+20,2.866594074450051 7e+20,2.939770559714163e+20,3.033917681780961e+20,3.104641460986279e+20,3.147796824090755e+20,3.154474316335821e+20,3.215650937207284e+20,3.274117317888 78e+20,3.282640471438088e+20,3.307609019116609e+20,3.242244402649755e+20,3.155289841100 37e+20,3.0191830501327575e+20
−24.17585715,2.1961652685742198e+20,2.4099799192495656e+20,2.5424321781895748e+20,2.659125800701835e+20,2.772692222989132e+20,2.863356502452444e+20,2.9238892879834736e+20,3.018091595936594e+20,3.059818257761823e+20,3.077207943718671e+20,3.1235395611692846e+20,3.175334581365862e+20,3.207195233069593e+20,3.1705824633769486e+20,3.1400278776495256e+20,3.0631662015320765e+20,2.9270778058599052e+20
−26.171571450000002,2.1068867680341272e+20,2.284456554925648e+20,2.422783044876469e+20,2.5373305495872838e+20,2.5921178711755896e+20,2.7114788115267872e+20,2.8139038165420147e+20,2.865471961428153e+20,2.9206455842207184e+20,2.9332647568932127e+20,3.0226536292059506e+20,3.039583432776775e+20,3.046702412112423e+20,3.0213046408736145e+20,3.007287425747608e+20,2.9356684088308302e+20,2.7939755798143153e+20
−28.167285749999998,2.0303562072346152e+20,2.1886231974434292e+20,2.2956960804400706e+20,2.4122425405888024e+20,2.4754181168981826e+20,2.5963303938315675e+20,2.6160869411351 51e+20,2.7172181437043648e+20,2.7177761343327404e+20,2.7917865396568415e+20,2.8035656605041987e+20,2.8670784838966628e+20,2.8760860468975832e+20,2.9093263457593855e+20,2.851975946339641e+20,2.7755864161385087e+20,2.6388173945350226e+20
−30.163,1.8518176014498713e+20,2.0598990517127863e+20,2.1611344942894983e+20,2.2307974781243877e+20,2.3267044167883702e+20,2.378707916999945e+20,2.4485057996680682e+20,2.4920903863328332e+20,2.519290896524856e+20,2.5622071207886e+20,2.635518504885945e+20,2.6838061554184467e+20,2.681396371715682e+20,2.7048381098726038e+20,2.6922679913212854e+20,2.5962384173543634e+20,2.453975334179178e+20
−32.15871425,1.7471790292165827e+20,1.877742704491321e+20,1.97904559648465e+20,2.0884117596462635e+20,2.1289365955026833e+20,2.2055530010142527e+20,2.237358466831661e+20,2.292771228464959e+20,2.3348167420778306e+20,2.4035476976103788e+20,2.4176139668575614e+20,2.46218576771099e+20,2.483162536278824e+20,2.502520518847854e+20,2.5072542415413258e+20,2.4189261645990153e+20,2.2839660137138894e+20
−34.15442855,1.547663654864005e+20,1.716906504685921e+20,1.816989175415e+20,1.8669140072417467e+20,1.928777385809672e+20,1.9757528386007194e+20,2.0328211768237028e+20,2.1273055459735857e+20,2.0942430683010456e+20,2.1705344902598253e+20,2.1748205940975675e+20,2.2241015905838806e+20,2.2860998679850402e+20,2.2546255174855688e+20,2.2164246206198563e+20,2.1688298595489635e+20,2.0636700872781814e+20
−36.15014285,1.3049132044600402e+20,1.4210488363439242e+20,1.4744933013650401e+20,1.5457505441381456e+20,1.629725067826097e+20,1.6675212681921085e+20,1.7214501426598 36e+20,1.7407468075773962e+20,1.7923946654104456e+20,1.7952888585598222e+20,1.8360283061963863e+20,1.878956793990424e+20,1.861229860950492e+20,1.8906010160045474e+20,1.841252570101617e+20,1.847911667051242e+20,1.7052868097323853e+20

C.153 "NBSR_upper_Cycle15_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,2.799199843719546e+19,2.9417817786823737e+19,3.0753745459397226e+19,3.35780978037497e+19,3.3911297921836827e+19,3.325
427928633746e+19,3.4746505652507574e+19,3.4521899095173538e+19,3.569410863832254e+19,3.5138570716005753e+19,3.665440437799176e+
19,3.5750336924720386e+19,3.55858216658268e+19,3.6215431311119155e+19,3.695179498762051e+19,3.516922954174068e+19,3.354351464832
071e+19
24.15442855,3.229778524105936e+19,3.336949515344929e+19,3.743019530438823e+19,3.9489916534911754e+19,3.96644878886464e+19,4.05005
5406643772e+19,4.160243226335081e+19,4.272515846176366e+19,4.427275466721105e+19,4.362885800912621e+19,4.46811302260002e+19,4.323
844852221771e+19,4.3974321657507316e+19,4.333024104646807e+19,4.4620548386348e+19,4.1611262005162476e+19,3.95386027501788e+19
22.158714250000003,3.6820207307566223e+19,3.988222686901581e+19,4.315701868106577e+19,4.5276585939423855e+19,4.619861946457589e+
19,4.750854845292617e+19,4.8289735332651975e+19,4.9542516269832356e+19,4.985315149217858e+19,4.943888943884832e+19,4.92469651897
4771e+19,4.993685008643492e+19,5.067983606929502e+19,5.177325243030528e+19,4.98051397710777e+19,4.759157255301634e+19,4.624227763
242242e+19
20.163,4.090984938999892e+19,4.480339762303107e+19,4.602325098137214e+19,4.826318478956553e+19,5.048233191391066e+19,5.0640592772
35433e+19,5.233265336466466e+19,5.331618849424096e+19,5.574013657449536e+19,5.498402861422071e+19,5.616574239334754e+19,5.4522613
28691012e+19,5.5730448385563116e+19,5.566238579243158e+19,5.478382648217167e+19,5.213999330374641e+19,5.1439316500400505e+19
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,5.595689447244125e+19,5.7866694045121e+19,5.664095419223883e+19,5.920623945913121e+19,5.97514146983496e+19,6.054302557882528e+19
,6.16371164340017e+19,6.033969624655127e+19,6.029990109074735e+19,5.903859700001266e+19,5.600938238209944e+19,5.4949752047049064e
+19
16.171571450000002,4.7149472285918765e+19,5.197977028045571e+19,5.39025078775956e+19,5.611061782467615e+19,5.76375499815782e+19,
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e+19
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279447629545449e+19,7.264915346147095e+19,6.971571701515362e+19,6.98782087915487e+19,6.9349037459363955e+19,6.619547064426988e+
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e+19
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903240071e+19,7.055208978120229e+19,7.237322402985665e+19,7.214941460199172e+19,7.679729258340588e+19,7.63159490193676e+19,7.7382
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6.193,5.817751322042164e+19,6.2926626644412514e+19,6.52824508138839e+19,6.90596181444263e+19,7.111682535123955e+19,7.175514210304
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5605e+19,7.819288233085952e+19,7.915311675287729e+19,7.856998588739908e+19,7.3202238677728944e+19,7.1276251245061145e+19
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2.20157145,5.620259430188091e+19,6.198724022389451e+19,6.394511283532662e+19,6.615684052384389e+19,6.922394945036547e+19,7.047482
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325e+19,7.112418346941593e+19,7.107942158384295e+19,7.066062202430391e+19,6.884684589382593e+19,6.836918138887585e+19,6.684911680
893843e+19

C.154 "NBSR_lower_Cycle15_cladding_fluence_position_4.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,5.513272391903508e+19,5.866983264407301e+19,6.21871357676862e+19,6.2173032707848135e+19,6.523400986922273e+19,6.7950
99500585151e+19,6.720230648140472e+19,7.008300974745798e+19,7.1206962298900234e+19,7.396932249761669e+19,7.544401201546643e+19,7.
501540163169223e+19,7.788874677956911e+19,7.439977241093498e+19,7.442184676546413e+19,7.259519392817748e+19,7.161533785768938e+
19
−12.20157145,6.251027979093228e+19,6.76474726310758e+19,6.9980609269503345e+19,7.199428094377302e+19,7.281593747346892e+19,7.7451
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2510397004615e+19,8.379670249868855e+19,8.155125010186288e+19,8.141144585651164e+19,8.11477799551913e+19,7.735650956481085e+19
−14.197285749999999,6.270894898169456e+19,6.9323897222261326e+19,7.403309285514533e+19,7.547283131165725e+19,7.746504180791248e+
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62e+19,8.779645290406647e+19,8.865980543676187e+19,8.616172431588042e+19,8.485811104563156e+19,8.216197391050252e+19,7.9779783150
89909e+19
−16.192999999999998,6.3579659632566346e+19,6.801783124595364e+19,7.206111718387514 6e+19,7.540906095412861e+19,7.86343694214424e+
19,8.096750605986996e+19,8.198660542729876e+19,8.638001515511307e+19,8.738378510967441e+19,8.670009329578566e+19,8.8553112723204
34e+19,8.908657629099198e+19,8.82189315226937e+19,8.73856246392185e+19,8.652595116561131e+19,8.254582240870374e+19,8.167081952222
908e+19
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416789384431e+19,8.047328578902301e+19,8.569571016470962e+19,8.360600460261735e+19,8.577235722904692e+19,8.771796631018507e+19,8.
536214214071366e+19,8.567486216320986e+19,8.635058268240755e+19,8.443624560351904e+19,8.374458249493919e+19,7.986010927432458e+
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,7.867790495398599e+19,8.104108724163378e+19,8.276840548353927e+19,8.018999823923234e+19,8.070322698203495e+19,8.394263850918678e
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,7.034974153135181e+19,7.2570666867589554e+19,7.48621075030176e+19,7.607129159000295e+19,7.591922381435773e+19,7.71002017816669e+
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7e+19
−28.167285749999998,5.290327542924303e+19,5.588834133809797e+19,6.035717177722018e+19,6.300388688526451e+19,6.688529422330564e+1
9,6.665412667726432e+19,6.8738926827239e+19,7.067779096671548e+19,7.1186114297400476e+19,7.321940762014049e+19,7.296923160214353e
+19,7.14099237252654e+19,7.297536336729052e+19,7.304219960739265e+19,7.022220081629453e+19,6.679883633473316e+19,6.5695731784790
66e+19
−30.163,4.921587714045251e+19,5.337805800457404e+19,5.693503364868819e+19,5.759064197820377e+19,5.952460070556264e+19,6.151570748
40914e+19,6.429830250779293e+19,6.569573178479066e+19,6.603911063302178e+19,6.827965761772988e+19,6.827107314652411e+19,6.6664550
67801419e+19,6.758922086217944e+19,6.576072849534869 5e+19,6.825206467456845e+19,6.420019426544116 5e+19,6.214666611771611e+19
−32.15871425,4.36774829067418e+19,4.804979936245048e+19,5.116485869242148 5e+19,5.361388569212704e+19,5.502314927585847e+19,5.6388
57073878894e+19,5.781089498228344e+19,5.836594236338847e+19,6.091056358173553e+19,5.965606575031399e+19,6.2402973900860056e+19,6.
2307318364567085e+19,5.944960921781502e+19,6.111217601976837e+19,5.960756348800134e+19,5.9657782644555145e+19,5.433301910856538
e+19
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e+19,5.323107959400081e+19,5.393929846847751e+19,5.374461492506075e+19,5.279946464530458e+19,5.1755838217287836e+19,4.90436971751
2517e+19
−36.15014285,3.338672541351232e+19,3.444304459538332e+19,3.6478422718273315e+19,3.8554944985299575e+19,3.99585060274443e+19,4.032
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23915043219746e+19,4.615851772051462e+19,4.367319067113891e+19,4.4018838272474415e+19,4.320515303746959e+19,4.080278877053258e+
19

C.155 "NBSR_upper_Cycle15_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
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177408972620 07e+20,2.3254414079756078e+20,2.4247647974154912e+20,2.4447685052228757e+20,2.477772548031637e+20,2.518392110306487
5e+20,2.594941568917942e+20,2.561626265095581e+20,2.598635199612667e+20,2.5621969102871816e+20,2.451408740179684e+20,2.3675342723
815e+20
24.15442855,2.2109596071733854e+20,2.3864381912741632e+20,2.5192947672459287e+20,2.6163770773878892e+20,2.6800195793020482e+20,2.
7394496821654818e+20,2.7676603053647967e+20,2.8792888802090156e+20,2.9547593006399894e+20,2.982295524976504e+20,3.0238799963935
146e+20,3.015019432873024e+20,3.0138158902871025e+20,3.0287875450412814e+20,3.014334658643103e+20,2.9200433222564263e+20,2.785879
4500275395e+20
22.158714250000003,2.4843920322542053e+20,2.7081679502986432e+20,2.7871867462846615e+20,2.8744020822954895e+20,3.03617480643073e
+20,3.0003486637653256e+20,3.1614988658733665e+20,3.1795831307635484e+20,3.248651949681474e+20,3.30948272710611e+20,3.3747956631
26591e+20,3.4326383348206626e+20,3.412126234024397e+20,3.439434200284271e+20,3.381000132664359e+20,3.287289816836402e+20,3.117081
919232591e+20
20.163,2.6564986220409786e+20,2.8917704468543914e+20,3.034151609842328e+20,3.186275242555957e+20,3.273864091783106e+20,3.36341388
5395936e+20,3.436902610706988e+20,3.516647682391407e+20,3.648155460637 5705e+20,3.657555543248302e+20,3.680868993166971e+20,3.7419
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30805803298e+20,4.6544415668733805e+20,4.7011825957490373e+20,4.7657070038683976e+20,4.868609894964686e+20,4.940096174421575e+2
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94148e+20,4.921845903657472e+20,4.923381457991234e+20,5.019166847243193e+20,5.128180829573168e+20,5.1579892593089656e+20,5.194469
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238e+20,4.6833369643026153e+20,4.773810165589128e+20,4.760913584258952e+20,4.6956732758083115e+20,4.561779163124545e+20,4.3672410
29624303e+20

C.156 "NBSR_lower_Cycle15_cladding_fluence_position_5.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
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02368829846e+20,4.64719956062361e+20,4.7512541174702205e+20,4.8004437329862014e+20,4.8453794479829765e+20,4.933341810426446e+20,
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9857573445601e+20,5.277503113164394e+20,5.433351502674107e+20,5.570544982102037e+20,5.6425189038135666e+20,5.7234778934510276e+
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3916e+20
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398970964e+20
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7862762057256e+20,4.93641291909397e+20,5.110843591115626e+20,5.2148670218608765e+20,5.265768572951659e+20,5.28280492576272e+20,5.
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+20
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184386699e+20
−28.167285749999998,3.4170752841406435e+20,3.762377877261792e+20,3.9200004545490276e+20,4.0664902629164895e+20,4.227277327175329
4e+20,4.353047527404125e+20,4.524178832681618e+20,4.591089575238581e+20,4.660729039348107e+20,4.75886963693631e+20,4.795733316313
715e+20,4.842837483038574e+20,4.88001242342958e+20,4.801304888457162e+20,4.7753457199228905e+20,4.6154509372363715e+20,4.40243427
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−30.163,3.13243746257021e+20,3.5323248621097465e+20,3.658644956795903e+20,3.78530743859702e+20,3.983445824487907e+20,4.0308301261
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+20,3.14863341064455e+20,3.1503972230549524e+20,3.179261494382828e+20,3.1053681297540966e+20,3.05975801589452e+20,2.891998704931
031e+20

C.157 "NBSR_upper_Cycle15_cladding_fluence_position_6.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.1071699102349407e+20,1.1858711158964853e+20,1.2637322697328932e+20,1.2982111851543862e+20,1.3467011839367386e+20,1.3974967264143573e+20,1.4245868648337342e+20,1.44122847544265e+20,1.48745585288576 5e+20,1.535890665781795e+20,1.5462288218196106e+20,1.5484239937422308e+20,1.563673693662781e+20,1.584865074010759e+20,1.561552102921 9244e+20,1.551195551588668e+20,1.460819465087265e+20
24.15442855,1.2912638952428531e+20,1.4078042236264381e+20,1.4853097350843207e+20,1.5491720690901628e+20,1.591610015672442e+20,1.6378803154715863e+20,1.651590942340243e+20,1.68981636626654 4e+20,1.743923061923534e+20,1.797637324611117e+20,1.8123351656684385e+20,1.834550550795963e+20,1.8571951594837764e+20,1.867619160233649 8e+20,1.869072388573485e+20,1.860567630314618e+20,1.7620117691071378e+20
22.158714250000003,1.4775653157036792e+20,1.5469339748115137e+20,1.6424975346272652e+20,1.7488162105108274e+20,1.781590495221459e+20,1.81029941963964e+20,1.8665583648632216e+20,1.8983331718548944e+20,1.902067416829408e+20,1.9962758565476765e+20,2.0296694495 38153e+20,2.0694094194557587e+20,2.0959047766558786e+20,2.0930902964534126e+20,2.0876514207680374e+20,2.069483000637523e+20,1.982736919103134 4e+20
20.163,1.5945226041172594e+20,1.736705974345533e+20,1.8245251147806435e+20,1.9026560662835187e+20,1.9592645221204784e+20,2.0130217071640907e+20,2.0387996478420127e+20,2.072499829089839e+20,2.1422179988110518e+20,2.192976750697788e+20,2.225358602439013e+20,2.24331854255453e+20,2.2827028700936104e+20,2.2976766405825466e+20,2.2883134352031013e+20,2.240589907064122e+20,2.157749759928363e+20
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C.158 "NBSR_lower_Cycle15_cladding_fluence_position_6.csv" File

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−20.18442855,2.414511293693187e+20,2.6367141990896068e+20,2.7434007808819885e+20,2.9023790558478523e+20,3.014283769780317e+20,3.033034707599795e+20,3.1655666794867157e+20,3.236848449320409e+20,3.3623534183488854e+20,3.447842488028142e+20,3.4936713007367035e+20,3.4911572770264397e+20,3.551530636663648e+20,3.55260369556437e+20,3.548936900006473e+20,3.4445006760230355e+20,3.292365450961206e+20
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−28.167285749999998,2.039106236099362e+20,2.1944545060982112e+20,2.3253125061000053e+20,2.453110755293454e+20,2.511276679477747e+20,2.605156013564322 5e+20,2.708425192483589e+20,2.7483797741813396e+20,2.8000215002492417e+20,2.874191331467165e+20,2.9032129759078416e+20,2.967326712284711e+20,3.0431459883271723e+20,3.0439001954402514e+20,3.0246648481741616e+20,2.9464480519592287e+20,2.754039393412006e+20
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−32.15871425,1.7672115059517807e+20,1.9286302234461445e+20,2.0008992074685025e+20,2.1056052291184076e+20,2.16149626843317e+20,2.2076194058687866e+20,2.2854560326446062e+20,2.3323517724887427e+20,2.3769787592284953e+20,2.414523557223481e+20,2.511987964234798e+20,2.5402492697972487e+20,2.560490226547443 4e+20,2.5805227032826423e+20,2.56987182722233e+20,2.5888802991779817e+20,2.393632633367705e+20
−34.15442855,1.5836571162768032e+20,1.731702453985594e+20,1.7934554607808735e+20,1.8677724543623247e+20,1.9507107097404352e+20,2.0346177840117694e+20,2.060788157659099e+20,2.0988602874567246e+20,2.1471050156331978e+20,2.2054487610067547e+20,2.2520869667147176e+20,2.275173062493114e+20,2.2866271997876807e+20,2.3201005057250682e+20,2.338655227059843e+20,2.3160903313189405e+20,2.202861156114727e+20
−36.15014285,1.2870329772914337e+20,1.4494511725047559e+20,1.5332785338291795e+20,1.5480622195985587e+20,1.6345385034664795e+20,1.692005406424017e+20,1.7322543128488228e+20,1.7660280752784127e+20,1.8354764473331578e+20,1.8848555520618227e+20,1.910467935080 7767e+20,1.8991303013240026e+20,1.9487853354842823e+20,1.9188439262715573e+20,1.9471542859551844e+20,1.9333148920184406e+20,1.805056760438968e+20

C.159 "NBSR_upper_Cycle15_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
26.15014285,1.0686746886421728e+20,1.1788011906820124e+20,1.2192769724172562e+20,1.2583792387595756e+20,1.295924036754561e+20,1.3
22971252817909e+20,1.3675675807319264e+20,1.4037756539248692e+20,1.4006300584044662e+20,1.443987769758793e+20,1.450787897306798
7e+20,1.4756460732126732e+20,1.512688066465606e+20,1.5258161756452995e+20,1.490282596618525e+20,1.439303101186497e+20,1.379101430
9734043e+20
24.15442855,1.2744996493309975e+20,1.393842194386754 6e+20,1.4308228699882173e+20,1.4860148880762238e+20,1.555769848388318e+20,1.5
906350650140715e+20,1.608411052175179e+20,1.6525720247637606e+20,1.6910979051822635e+20,1.7028647624993266e+20,1.75915436654864
3e+20,1.7487732881547985e+20,1.8106795890787526e+20,1.797649588141411e+20,1.7789722315036967e+20,1.737263964973909e+20,1.66899289
18273848e+20
22.158714250000003,1.4189149820727738e+20,1.5293971264911383e+20,1.593897164072267e+20,1.6458822689884006e+20,1.7243443358092126
e+20,1.7802966927754455e+20,1.822808220539488e+20,1.8568272535749575e+20,1.8821146530411207e+20,1.930340985922153e+20,1.959393289
1885653e+20,1.9517469780502757e+20,2.002928821732154e+20,2.0368681418207127e+20,1.9800696012641963e+20,1.930402303573623e+20,1.82
01654297611377e+20
20.163,1.5566957449255128e+20,1.6341767293228076e+20,1.755763500422361e+20,1.7822895164482152e+20,1.8453792480455374e+20,1.902662
1980486654e+20,1.958737190317838e+20,2.019411006447248e+20,2.034844659322208e+20,2.0521301052715568e+20,2.1183654323892824e+20,2.
1113690883565732e+20,2.1575780705042475e+20,2.169743492555864e+20,2.151268484168e+20,2.11125258481878e+20,2.014707942579511e+20
18.167285749999998,1.6503768528411404e+20,1.7806523351539702e+20,1.9006693743758957e+20,1.950722973270729e+20,2.0007827039307094
e+20,2.046157766018394e+20,2.136558379580385e+20,2.15055719941095e+20,2.1944545060982112e+20,2.246850439279193e+20,2.272346318760
3538e+20,2.2875224374991402e+20,2.3427328508825875e+20,2.3257417296602936e+20,2.3050654175846626e+20,2.308603446074473e+20,2.2180
066160277783e+20
16.171571450000002,1.7873727497550653e+20,1.8909873172088072e+20,1.9967357389337e+20,2.0887306114339068e+20,2.1555913785966245e+
20,2.1900518987226762e+20,2.2912444189433597e+20,2.354334150540682e+20,2.3446459616084466e+20,2.411249194634991e+20,2.4235311202
244008e+20,2.4882028472229645e+20,2.4929610969837047e+20,2.513864284369775e+20,2.4903367015007958e+20,2.4555266707613653e+20,2.363
7586735715972e+20
14.17585715,1.8421662031085178e+20,2.0071106855623977e+20,2.1288752778512135e+20,2.187243550285358e+20,2.242472358964246e+20,2.35
0397557316318e+20,2.42621683335878e+20,2.4515471551809726e+20,2.4719046154689603e+20,2.5482941456700916e+20,2.5925348312055847e
+20,2.6197660002233418e+20,2.675932968969719e+20,2.7107184726485618e+20,2.6526016025854434e+20,2.5943191748633566e+20,2.48724629
18667154e+20
12.18014285,1.964396809548504e+20,2.1413472881601797e+20,2.2308771910712984e+20,2.3089039025666746e+20,2.3705465375893088e+20,2.4
358191775789577e+20,2.5453631619298335e+20,2.5641386268098993e+20,2.6394121757542803e+20,2.6687526719826002e+20,2.6988719023845
874e+20,2.7429470302611115e+20,2.759410819680764e+20,2.7836803461325285e+20,2.7715823734975283e+20,2.7191251226650765e+20,2.58711
435081565e+20
10.18442855,2.0205085924085583e+20,2.193914910765277e+20,2.2779630156349915e+20,2.384361404465465e+20,2.485069515239536e+20,2.580
9396633126368e+20,2.6033941872808935e+20,2.6930896478509813e+20,2.7696079451201995e+20,2.7989668366439606e+20,2.839914764295522
2e+20,2.8365852158207102e+20,2.901643244030214e+20,2.911815842409061e+20,2.8847195722245367e+20,2.829030881159625e+20,2.682923181
237281e+20
8.18871425,2.0997309981075967e+20,2.2626581298281185e+20,2.3776532533946637e+20,2.4790358583349035e+20,2.5519548094628417e+20,2.6
31171083396733e+20,2.7333201589803457e+20,2.822727426588525e+20,2.881291915507373e+20,2.87731239992698e+20,2.9633839872951996e+
20,2.957491360988948e+20,3.006484164513353e+20,2.988095000837547e+20,2.98011757438132e+20,2.96004830705524e+20,2.759545718513998
2e+20
6.193,2.1217133761595353e+20,2.2927589649346652e+20,2.4365365941011546e+20,2.537845617859631e+20,2.625744471241652e+20,2.67500094
0667377e+20,2.761415906883828e+20,2.8679307992520943e+20,2.8756629551024413e+20,2.9528802735984155e+20,3.0019834488954664e+20,3.0
32666801690976e+20,3.050050355882677e+20,3.060382380155345e+20,3.023800269288437e+20,2.986782803096092e+20,2.8223227300888245e+
20
4.19728575,2.1190583218508913e+20,2.3257478614254407e+20,2.475289349830096e+20,2.562924537310797e+20,2.604381401469558e+20,2.7024
467214652798e+20,2.8245608243674733e+20,2.9049666607398794e+20,2.9316153120686735e+20,3.018109991232035e+20,3.002633416001047e+
20,3.0523681631082375e+20,3.096056989780501e+20,3.1000119783003056e+20,3.0436671883646665e+20,2.9685040111929315e+20,2.809022931
4850148e+20
2.20157145,2.0956840331105868e+20,2.311632538057083e+20,2.4158848090861116e+20,2.498357050313051e+20,2.5693199683591012e+20,2.689
116264035735e+20,2.7305792599596437e+20,2.8649262343300714e+20,2.9223134243406984e+20,2.9554924055510306e+20,3.038479715050317e
+20,3.0371555253778569e+20,3.029251408504106e+20,3.020299031389509e+20,3.018281680656151e+20,2.95573767615691e+20,2.8004323285140
898e+20
0.20585714999999993,1.9260119597283823e+20,2.075541184602743e+20,2.1939394378258645e+20,2.291912781344381e+20,2.353156851632461e
+20,2.4521664634608176e+20,2.5003130833949393e+20,2.5963487891270087e+20,2.618000051861011e+20,2.688901652255591e+20,2.670763890
950811e+20,2.7480609223936963e+20,2.7660821801606837e+20,2.7507588990583695e+20,2.6974432011053403e+20,2.6626515656613508e+20,2.5
353193306190732e+20

C.160 "NBSR_lower_Cycle15_cladding_fluence_position_7.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
−10.20585715,1.962998767094992e+20,2.1853794936806746e+20,2.3231541247682665e+20,2.420354865878263e+20,2.4883500095931728e+20,2.5467121502621706e+20,2.661725669124156e+20,2.7045253898501072e+20,2.7709691969828297e+20,2.790480473680534e+20,2.829214834114034e+20,2.873277698460264e+20,2.8881533607068484e+20,2.8609221916890907e+20,2.8691939428723723e+20,2.823628796065132e+20,2.6589541112777196e+20
−12.20157145,2.2293135909588176e+20,2.4318457937637117e+20,2.5479752938824486e+20,2.6638411280998656e+20,2.774666651366461e+20,2.8588496550694103e+20,2.980644906183961e+20,3.060321062503876e+20,3.0780479955438076e+20,3.163034260481011e+20,3.205527392949613e+20,3.1820856547926914e+20,3.2245787872612935e+20,3.2729338872104124e+20,3.209617280302651e+20,3.122264154018712e+20,2.95348118658282e+20
−14.197285749999999,2.3223324682385706e+20,2.5176659787609054e+20,2.665631603522785e+20,2.7636601329276246e+20,2.848511499031594 3e+20,2.9464971060804053e+20,3.074890136493111e+20,3.145068188600347e+20,3.151635309072767e+20,3.2060301976916656e+20,3.308491993297775e+20,3.3146421537402e+20,3.379111532495594e+20,3.330284286630157e+20,3.271603294173517e+20,3.233708985565153e+20,3.047395301574033e+20
−16.192999999999998,2.3141097711764645e+20,2.5485026256850895e+20,2.698970010626939e+20,2.7901186995368618e+20,2.876337449268609e+20,2.969748759517769e+20,3.049608868792094e+20,3.1284695003474605e+20,3.1613357615352965e+20,3.25095764092362e+20,3.3222945966436365e+20,3.3284386253209154e+20,3.3733722003180166e+20,3.3653641150360545e+20,3.304635113020322e+20,3.256065401291058e+20,3.118603490225963e+20
−18.188714249999997,2.3115405615798783e+20,2.502164876469329e+20,2.661750196184744e+20,2.7805654094378605e+20,2.8333844344139838e+20,2.958656396366875e+20,3.0541954291220 39e+20,3.104770228054366e+20,3.180877697058736e+20,3.243004741527981e+20,3.262742893536124e+20,3.3131521348094833e+20,3.353676970665903e+20,3.3313328184702914e+20,3.294566754648973e+20,3.205288254108881e+20,3.109350656619163e+20
−20.18442855,2.2713958951625717e+20,2.5031520906579932e+20,2.5892604686170948e+20,2.7515069744063015e+20,2.8132109270804054e+20,2.898062293184375e+20,2.982055212167767e+20,3.021421144411407e+20,3.0807030498524516e+20,3.133755081904161e+20,3.189548012976572e+20,3.202486037436709e+20,3.2647909030952174e+20,3.2753436709131774e+20,3.2587695097208786e+20,3.170950369285768e+20,3.0445624260761256e+20
−22.180142850000003,2.2303314639732172e+20,2.394074120458288e+20,2.511503554788186e+20,2.643299714857468e+20,2.7121900962838377e+20,2.8072017972363605e+20,2.8524787510816935e+20,2.9825212263189375e+20,3.0503692076703207e+20,3.063386945077368e+20,3.0941745378803764e+20,3.147183647576057e+20,3.190197980082152e+20,3.222818970664109e+20,3.2038227622387515e+20,3.1051320021980375e+20,2.9465890825576094e+20
−24.17585715,2.1504039052822757e+20,2.3177091173177446e+20,2.4544168212697616e+20,2.5546466543623678e+20,2.642931808948649e+20,2.710902425602971e+20,2.7958273728887043e+20,2.852815998164777e+20,2.8971609237077688e+20,2.977787503625466e+20,3.0105372612755096e+20,3.0576966670209666e+20,3.07737350137764e+20,3.124465457706479e+20,3.068151326596575e+20,3.00577901152145e+20,2.8171659156002102e+20
−26.171571450000002,2.055238910201078e+20,2.232177125282459e+20,2.3420215661255374e+20,2.4555389342916595e+20,2.4981424385329067e+20,2.6028484601828124e+20,2.6600087748830006e+20,2.7115462609434044e+20,2.7746850466619025e+20,2.8096177127042725e+20,2.8674586533357758e+20,2.880384414265619e+20,2.9224728502345197e+20,2.9628014696062367e+20,2.923208662052158e+20,2.8535518099824155e+20,2.7257780878495547e+20
−28.167285749999998,1.965009986063203e+20,2.132511414583375e+20,2.2072698952554086e+20,2.3161025948492346e+20,2.372447384784874e+20,2.450615126878631e+20,2.5331118951661588e+20,2.6000033211546114e+20,2.6509582895260516e+20,2.688846466369268e+20,2.7032806415252695e+20,2.742303194920678e+20,2.8063617454112234e+20,2.8055155618209396e+20,2.7766226844483496e+20,2.7369440321822133e+20,2.5795232255636836e+20
−30.163,1.8123719562593206e+20,1.956364197205955e+20,2.06704868987417e+20,2.138919109161974e+20,2.2081406059062804e+20,2.2817953688518568e+20,2.352378117458794e+20,2.376058994456448e+20,2.452785771740663e+20,2.479863646629746e+20,2.5250854145887558e+20,2.5859002613165474e+20,2.6164057929227947e+20,2.6307970457227667e+20,2.589039725071803e+20,2.5501888611005098e+20,2.3929029533152136e+20
−32.15871425,1.7022209271588933e+20,1.8226426628805195e+20,1.9199476439980147e+20,2.012089678861749e+20,2.0405778597346384e+20,2.1227803032951112e+20,2.1716259444559885e+20,2.1898924728288553e+20,2.3023980997457245e+20,2.3239144636464926e+20,2.326281324993229e+20,2.4020392833842207e+20,2.381454947785794e+20,2.3891748401058474e+20,2.4036580693830245e+20,2.3310641018078763e+20,2.2665027465752776e+20
−34.15442855,1.5213461188531477e+20,1.6848680317929275e+20,1.7416543088191498e+20,1.8159896976960415e+20,1.8618123786394557e+20,1.934786515653717e+20,1.958485787946811e+20,2.00895634687164e+20,2.0713041348861772e+20,2.0723710620217524e+20,2.117795178230612 6e+20,2.1567686775048454e+20,2.154052305544731e+20,2.231846009964522e+20,2.209201401276709e+20,2.1473318909436366e+20,2.046795469593681e+20
−36.15014285,1.28663441255688e+20,1.378028372072682e+20,1.4807047794589355e+20,1.5238969331542934e+20,1.5478046854623858e+20,1.584889601071347e+20,1.666626030480649e+20,1.6970027950188095e+20,1.7364729672699478e+20,1.7296544444265015e+20,1.7944058843786563e+20,1.8139846104929776e+20,1.8269655573091434e+20,1.8439260197057023e+20,1.798569352913459e+20,1.7885009945421103e+20,1.6835558340514726e+20

C.161 "NBSR_upper_Cycle6_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.5658099364044464e+19,4.954504385063135e+19,5.161385310893763e+19,5.294761486020878e+19,5.4149651797000815e+19,5.58140697152
2977e+19,5.7364044500179485e+19,5.908760417184806e+19,6.050316588079867e+19,6.1814269427829965e+19,6.270946051398403e+19,6.355933
519165839e+19,6.540885911740612e+19,6.604444094820191e+19,6.546531260932575e+19,6.564581016852455e+19,6.588276121964298e+19
bottom,9.761730442567247e+19,6.50048518146888e+19,6.961368417201824e+19,7.3293914192873824e+19,7.577709976261735e+19,7.8771439058
51749e+19,8.206839979409562e+19,8.427469442728098e+19,8.666148661966515e+19,8.870034096389163e+19,9.055216911379934e+19,9.1905132
73307036e+19,9.404498890297618e+19,9.582500206656437e+19,9.67401630954383e+19,9.736883225375413e+19,9.747559463983342e+19

C.162 "NBSR_lower_Cycle6_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,1.0458412617338626e+20,6.943625891169941e+19,7.345751410823274e+19,7.763123213668506e+19,8.068932163434476e+19,8.336529395880
701e+19,8.650825571302616e+19,8.884589112333066e+19,9.120272840163503e+19,9.41459907286555e+19,9.619867041784188e+19,9.9220276366
30186e+19,1.0078177227205149e+20,1.0222421659620191e+20,1.0400269361035012e+20,1.0444740887322717e+20,1.04774224666585e+20
bottom,8.330922450424739e+19,5.9892162441042715e+19,6.308735327622153e+19,6.551408535404543e+19,6.8307573110666895e+19,6.98809741
7457646e+19,7.257538029231859e+19,7.445255490798613e+19,7.557624822333868e+19,7.713505586756834e+19,7.91892717061947e+19,8.078072
252602416e+19,8.209528240929543e+19,8.250313008561273e+19,8.371707218056467e+19,8.31118293012087e+19,8.43449732641605e+19

C.163 "NBSR_upper_Cycle7_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.418362980232155e+19,4.754229107971283e+19,4.992440037900344e+19,5.1712519198180614e+19,5.311873210711668e+19,5.535753063727
255e+19,5.677911199330081e+19,5.800781933832199e+19,5.927225832283254e+19,6.003683856566936e+19,6.168395188277563e+19,6.233980036
243515e+19,6.246851110683231e+19,6.326997562269041e+19,6.400189791545631e+19,6.448792505474706e+19,6.438572488158395e+19
bottom,9.224871945973793e+19,6.232212664827912e+19,6.609738567647743e+19,7.002018179676084e+19,7.235464891006562e+19,7.5224322193
3562e+19,7.814010081792336e+19,8.02363570012991e+19,8.30852828810152e+19,8.49375649668023e+19,8.676295227017804e+19,8.80562070929
8646e+19,9.027272137485864e+19,9.06807536451565e+19,9.289649950467405e+19,9.417822799216332e+19,9.370680087761017e+19

C.164 "NBSR_lower_Cycle7_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,9.984726391329766e+19,6.59314064478817e+19,7.023726111193812e+19,7.324332936317195e+19,7.65629139350866e+19,7.957013481985234
e+19,8.208479697531314e+19,8.470089088158247e+19,8.729393211721346e+19,8.962839923051826e+19,9.210848238002163e+19,9.375367464124
137e+19,9.563592519885826e+19,9.715471198274467e+19,9.907807313633318e+19,9.93055261532977e+19,1.0067370215568058e+20
bottom,8.088836336918557e+19,5.8177256467513475e+19,6.1504141051796914e+19,6.359424985633577e+19,6.626029121565472e+19,6.81417733
50917e+19,7.034983498688847e+19,7.19012797208458e+19,7.444245244754066e+19,7.549403843982426e+19,7.773091591409361e+19,7.83245221
8302974e+19,7.947946098200837e+19,8.055717333434873e+19,8.022367803244803e+19,8.144854326569617e+19,8.062133660095865e+19

C.165 "NBSR_upper_Cycle8_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.3687664882582675e+19,4.572776952595301e+19,4.847826291800535e+19,5.069710957893143e+19,5.235163418541547e+19,5.463429383069
9426e+19,5.55211405470467e+19,5.700805996569918e+19,5.841117679043587e+19,5.939835597014103e+19,6.111015849631979e+19,6.052892215
686722e+19,6.203813767848812e+19,6.26013064862249e+19,6.344356099663533e+19,6.390370643203528e+19,6.380606487765634e+19
bottom,9.037379364236929e+19,6.077840786077246e+19,6.4549909002551e+19,6.8023872494371684e+19,7.020043344277146e+19,7.29417008611
8168e+19,7.562645919103404e+19,7.773574741496013e+19,8.024713432345459e+19,8.215421995839123e+19,8.417624584828576e+19,8.51503548
986646e+19,8.759139375813802e+19,8.91059910977168e+19,8.992864041413776e+19,9.094042219022187e+19,9.18303442311165e+19

C.166 "NBSR_lower_Cycle8_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,9.70445570011791e+19,6.426851680646682e+19,6.795352444535064e+19,7.171618402905549e+19,7.402690601083419e+19,7.76361837827390
8e+19,8.014949276907566e+19,8.212192905064387e+19,8.398096273952788e+19,8.713432364728544e+19,8.924707161132732e+19,9.14836013884
0075e+19,9.280368445035851e+19,9.452817269029518e+19,9.501292072207406e+19,9.618577262132658e+19,9.711644271247385e+19
bottom,7.833966427294953e+19,5.671436647142552e+19,6.015181048424621e+19,6.318062074783535e+19,6.493855314222464e+19,6.6816423193

96128e+19,6.82291504079085e+19,7.074130614753982e+19,7.210675024657125e+19,7.390004887325527e+19,7.522551375317092e+19,7.58390410
0037085e+19,7.770460975391801e+19,7.862528504028633e+19,7.93706668274547e+19,7.967973694446518e+19,7.955287980688625e+19

C.167 "NBSR_upper_Cycle9_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.72860089566128e+19,5.068529034110235e+19,5.271551750864943e+19,5.401034491403523e+19,5.6331949171542704e+19,5.7906285779797
82e+19,5.990131852355755e+19,6.1055368233793946e+19,6.298983382964057e+19,6.402479543148023e+19,6.501555938484275e+19,6.549068410
597199e+19,6.64884050965948e+19,6.698481016699082e+19,6.7122313962253025e+19,6.842573535484272e+19,6.732856965514635e+19
bottom,9.493982282878101e+19,6.435341313265747 6e+19,6.8393405593456665e+19,7.069291102672558e+19,7.379083879498428e+19,7.67074743
5698953e+19,7.908473491257932e+19,8.19804993628037e+19,8.419079102414653e+19,8.676284862302447e+19,8.857126907321405e+19,9.035513
527424968e+19,9.116951787118954e+19,9.324476116218557e+19,9.347352491680335e+19,9.482482709524328e+19,9.549188419725935e+19

C.168 "NBSR_lower_Cycle9_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,1.0074772122866577e+20,6.7971890982980256e+19,7.168940430489069e+19,7.516628598509226e+19,7.855272618091007e+19,8.16428784369
3668e+19,8.532601581003155e+19,8.717495077132519e+19,8.966311468559375e+19,9.25069059751089e+19,9.41540868558541e+19,9.5806997061
40192e+19,9.800214693576647e+19,9.830539191281795e+19,9.896958435243273e+19,1.0124248934911816e+20,1.0118110372623326e+20
bottom,8.373776512725587e+19,6.096533598689608e+19,6.386560204946534e+19,6.646958017224343e+19,6.882064952873568e+19,7.1561722209
29007e+19,7.338855834634513e+19,7.514500563582548e+19,7.765485913684671e+19,7.951484351025963e+19,8.067053017043963e+19,8.2174887
16860595e+19,8.276746304818831e+19,8.37868736255638e+19,8.36399573681259e+19,8.46434076835513e+19,8.338541165189646e+19

C.169 "NBSR_upper_Cycle10_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.657882579742906e+19,4.9554063990173426e+19,5.135124718419659e+19,5.3750218786689606e+19,5.502211556570578e+19,5.72873109550
3113e+19,5.881285070702575e+19,6.044802967959816e+19,6.196825111119162e+19,6.2939867338326835e+19,6.329701300834328e+19,6.4264538
2197853e+19,6.520997194648402e+19,6.5857988832287056e+19,6.590462641118954e+19,6.625031723726502e+19,6.6440549466998874e+19
bottom,9.137897203118234e+19,6.200057013516791e+19,6.602531137413897e+19,6.8504466884218724e+19,7.105030595009765e+19,7.409197611
799254e+19,7.703423460454266e+19,7.914478960066501e+19,8.088428947341404e+19,8.334830822542895e+19,8.491762184534083e+19,8.740782
309779222e+19,8.903686554682483e+19,8.966606376043912e+19,9.058613318984001e+19,9.131187937381386e+19,9.141988218811436e+19

C.170 "NBSR_lower_Cycle10_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,9.673533887829031e+19,6.534661187063694e+19,6.9384035258256925e+19,7.264702937515399e+19,7.602661743930725e+19,7.862727611547
509e+19,8.086587990279463e+19,8.236564625592209e+19,8.551122822242429e+19,8.841625846616624e+19,9.053785920466023e+19,9.257559412
144362e+19,9.342775269033737e+19,9.505229502210746e+19,9.546221479456619e+19,9.702784650035418e+19,9.746967619521988e+19
bottom,8.222041519881345e+19,5.878503180031198e+19,6.235853400832299e+19,6.451695388805 5796e+19,6.697729072594683e+19,7.031924144
572266e+19,7.143486142525854e+19,7.327540938562964e+19,7.567274458184539e+19,7.689145815684994e+19,7.859168427894424e+19,7.981080
69555181e+19,8.019863524323356e+19,8.209768472801744e+19,8.228668965304331e+19,8.233005441939123e+19,8.182113206715703e+19

C.171 "NBSR_upper_Cycle11_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.568400365093851e+19,4.8128466809112175e+19,4.95848467447794e+19,5.23339224423946e+19,5.422283780711501e+19,5.59357926850871
95e+19,5.7998213273950044e+19,5.92475327298624e+19,6.011096674623156e+19,6.1118852045907984e+19,6.23755372896377e+19,6.2871500336
0071e+19,6.447683286975887e+19,6.483571042063838e+19,6.488522488318846e+19,6.568072996746412e+19,6.5487582642475385e+19
bottom,8.8499940625668e+19,5.967147473979531e+19,6.388102247742072e+19,6.6872350752140435e+19,6.9233904168475255e+19,7.2178991664
11927e+19,7.4134199119196e+19,7.610782104381612e+19,7.811008769883713e+19,7.989547282368009e+19,8.246531435107269e+19,8.392987849
542587e+19,8.511290586098193e+19,8.63564963708141e+19,8.68761936223728e+19,8.825932489030066e+19,8.844469721703773e+19

C.172 "NBSR_lower_Cycle11_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,9.368954735343755e+19,6.241809517480472e+19,6.6113674606951604e+19,7.000199215365294e+19,7.279689941990952e+19,7.585902109976
283e+19,7.724624447203366e+19,7.999736622182035e+19,8.285242742191877e+19,8.514687032702868e+19,8.688642388323027e+19,8.844060511
269475e+19,8.994363503787198e+19,9.13345413040515e+19,9.21701490108884e+19,9.375420260205666e+19,9.429190511272447e+19
bottom,7.968759392305662e+19,5.7767418589005865e+19,6.0406007469360546e+19,6.361912779946988e+19,6.5860782558555324e+19,6.7687907
14769671e+19,7.022337499860825e+19,7.215239298588989e+19,7.361204660503151e+19,7.543262382722412e+19,7.646833543643282e+19,7.7857
60486087516e+19,7.813013901011775e+19,7.919326771842441e+19,7.97383360169096e+19,7.979317021510556e+19,8.048514505950378e+19

C.173 "NBSR_upper_Cycle12_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.371942214187586e+19,4.754292372000193e+19,4.957501548450699e+19,5.139546513735833e+19,5.3134860364469944e+19,5.515917417903
029e+19,5.651826859042369e+19,5.775987502107313e+19,5.9688806607364e+19,6.079040993900847e+19,6.192066887571244e+19,6.23767023619
4504e+19,6.2815951850928505e+19,6.334116815509048e+19,6.3967497808533815e+19,6.396422288224131e+19,6.354585104837262e+19
bottom,8.602494512019015e+19,5.687482619052116e+19,6.084403685704839e+19,6.389995245375072e+19,6.657270167422887e+19,6.8864331347
415695e+19,7.134672547714147e+19,7.3031265688853455e+19,7.625379316068745e+19,7.794774878549043e+19,7.957088412921777e+19,8.13720
9359010059e+19,8.165824027490902e+19,8.33051188342571e+19,8.433917681111848e+19,8.508463190845202e+19,8.516650506576486e+19

C.174 "NBSR_lower_Cycle12_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,8.97063716387627e+19,5.999460284992752e+19,6.347748696201639e+19,6.641878013848071e+19,6.957335288974502e+19,7.21957501184757
7e+19,7.431708362445185e+19,7.672415444944981e+19,7.898794724915025e+19,8.084810538329832e+19,8.255106705540571e+19,8.51857452577
3339e+19,8.604254784901241e+19,8.718672522245957e+19,8.725918296668145e+19,8.88217321739973e+19,8.97612266541623e+19
bottom,7.765750844281635e+19,5.690757545344632e+19,5.954266302156056e+19,6.261577198129858e+19,6.453324132556565e+19,6.6877269819
43269e+19,6.838496401134893e+19,6.957826527918379e+19,7.172416073235372 4e+19,7.2484762363790156e+19,7.43416455716457e+19,7.610028
099072583e+19,7.651824345880797e+19,7.744791316009543e+19,7.811354192904895e+19,7.811395129483552e+19,7.804067481904051e+19

C.175 "NBSR_upper_Cycle13_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.3241804100094394e+19,4.618046986101414e+19,4.896612758349171e+19,5.093820938878686e+19,5.253054050162073e+19,5.309135894604
304e+19,5.495365306462879e+19,5.640752060243362e+19,5.791464336140347e+19,5.889699736412424e+19,6.0214859260184584e+19,6.09071771
3532975e+19,6.178916552893183e+19,6.212917963329307e+19,6.336060420884711e+19,6.313857090214 3754e+19,6.364039894773714e+19
bottom,8.262957219299084e+19,5.570414202750804e+19,5.9295592211766985e+19,6.1274228505820905e+19,6.419261462566358e+19,6.69196916
0485963e+19,6.914371157182004e+19,7.062748396458675e+19,7.2370568318872355e+19,7.404933675221251e+19,7.530165376068515e+19,7.6878
00830716951e+19,7.814138601542258e+19,7.908932895215978e+19,8.02474251847251e+19,8.175413828814751e+19,8.156405811414317e+19

C.176 "NBSR_lower_Cycle13_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,8.701452517260847e+19,5.835911963035762e+19,6.119721326290534e+19,6.4378188588646285e+19,6.686848466143173e+19,6.927152410261
6056e+19,7.258113127024787e+19,7.412225543965383e+19,7.660681633477537e+19,7.839086624380328e+19,7.989061520291951e+19,8.23579705
6504929e+19,8.325757414719057e+19,8.346445019863926e+19,8.502851507870094e+19,8.555123555721291e+19,8.674169457802464e+19
bottom,7.570639344153924e+19,5.463821829311295e+19,5.712605643261387e+19,6.013702470617418e+19,6.2498688937068765e+19,6.464897090
5493e+19,6.659483475575305e+19,6.901917628540338e+19,7.002774824315924e+19,7.1585258634462134e+19,7.306165722737525e+19,7.3513507
296183e+19,7.42103313823498e+19,7.560602783242056e+19,7.557612297745867e+19,7.626229601939248e+19,7.639748235004214e+19

C.177 "NBSR_upper_Cycle14_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.7163941802090635e+19,4.937221321935143e+19,5.145024451883934e+19,5.338716671022621e+19,5.568129906520218e+19,5.748643067449
469e+19,5.904306825694261e+19,6.06542857774296e+19,6.159146325010063e+19,6.315431318321966e+19,6.4072409864543715e+19,6.440033323
2111845e+19,6.504331152657222e+19,6.646727104825847e+19,6.698910850463212e+19,6.690790420657398e+19,6.723893374947766e+19
bottom,8.5855573753474 19e+19,5.7578728455894114e+19,6.02970756138405e+19,6.399741791715669e+19,6.621700206407927e+19,6.9131925746

83308e+19,7.114872101827714e+19,7.292989070355249e+19,7.519296130517289e+19,7.736018421072467e+19,7.873755110237756e+19,7.9781669
75445849e+19,8.192803691132867e+19,8.228613455358507e+19,8.398033132946483e+19,8.419421368828464e+19,8.565944096800042e+19

C.178 "NBSR_lower_Cycle14_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,9.056320434417828e+19,6.02145400977814e+19,6.285700782967346e+19,6.723538383480846e+19,7.026612348364519e+19,7.22669441390778
1e+19,7.380982580218254e+19,7.60742276218039e+19,7.859599825494286e+19,8.021564682276923e+19,8.179136519656415e+19,8.422261300563
83e+19,8.586932967281738e+19,8.636099285450275e+19,8.761988134407081e+19,8.832143323057314e+19,8.991356995971314e+19
bottom,8.001773907996087e+19,5.755787270721251e+19,6.045593429528758e+19,6.3863408638394065e+19,6.582207405712984e+19,6.824888447
286747e+19,7.00393726841495e+19,7.138567782244681e+19,7.3664279300744995e+19,7.432988830122158e+19,7.60977458064874e+19,7.7321135
14936338e+19,7.803244930120602e+19,7.945019647222117e+19,8.041444204424492e+19,8.06194496163917e+19,8.140220580095217e+19

C.179 "NBSR_upper_Cycle15_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,6.563031919480439e+19,4.772343677995826e+19,5.008842137423261e+19,5.251375079117512e+19,5.4335631981435994e+19,5.497191489104
0014e+19,5.691093603118615e+19,5.891385168945151e+19,6.01393840439329e+19,6.199853703642999e+19,6.1992768781322e+19,6.34525810355
739e+19,6.414210937694395e+19,6.382751761759301e+19,6.473845821272344e+19,6.510673911157718e+19,6.496342016193491e+19
bottom,8.28711900006438e+19,5.524701628849782e+19,5.779303535077665e+19,6.141638698245476e+19,6.34201900645829e+19,6.564008085729
482e+19,6.800950257088302e+19,7.033410937940145e+19,7.217373904691768e+19,7.435946402091306e+19,7.553041980783428e+19,7.677547548
729652e+19,7.833645406190387e+19,7.932548795695782e+19,8.0443198312113e+19,8.067392851643243e+19,8.182136757099032e+

C.180 "NBSR_lower_Cycle15_cladding_fluence_top_bottom_edge.csv" File

,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17
top,8.764641780619362e+19,5.772337257754944e+19,6.088570751252003e+19,6.331414291298224e+19,6.661713453020141e+19,6.8020595369167
6e+19,7.0252022672095494e+19,7.378840676522245e+19,7.580019666211547e+19,7.689084058945624e+19,7.850151490037855e+19,7.9538469684
02192e+19,8.078663134700387e+19,8.241327928745597e+19,8.333753124052792e+19,8.435052557987658e+19,8.654201880897995e+19
bottom,7.854189268613446e+19,5.641619722769349e+19,5.984475932149422e+19,6.241207655647946e+19,6.379024581535678 5e+19,6.522875989
690227e+19,6.817722568094599 6e+19,6.932865814288632e+19,7.180989526318316e+19,7.319605133682539e+19,7.45311805383585e+19,7.573275
244854518e+19,7.619820626456651e+19,7.709406065402991e+19,7.78443775299995e+19,7.8246380539833e+19,7.937784596486106e+19

APPENDIX D: Fast neutron fluence in NBSR-DDE Outside and Side Plates

Appendices D.1-D.10 report the fast neutron fluence (E>0.1 MeV) for each cell position in the top and bottom outside plates (see Figure 2.2) during irradiation cycles 6-15. Fast neutron fluence in the left and right side plates reported in D.11-D.20 for irradiation cycles 6-15.

The text under D.1-D.20 is generated in the comma-separated csv file format, which could be parsed using any spreadsheet software, text editor, or Python. Once the text is imported (copied) to spreadsheet software, the first line represents the plate position (outside bottop&top, side left&right) as a column title. The fast neutron fluence data listed under the plate number corresponds to axial coordinate (z-coordinate) given in the first column of the spreadsheet first line in the .csv file contains information about the position where the fluence is reported (bottom outside plate, top outside plate, or left/right side plate). The first column of the file contains the number of axial nodes (see Figure 2.2), where the fluence is calculated. The text could be copied and pasted into Excel or a text file for further processing with Python routines.

D.1 "NBSR_Cycle6_outside_plates_fluence.csv" File

```
,bottom outside plate,top outside plate
1,6.5153354977775256e+19,4.6669373124446085e+19
2,1.1436105924408779e+20,8.401485772724098e+19
3,1.397930845540422e+20,1.0242145761463457e+20
4,1.5794983297796755e+20,1.156760943760622e+20
5,1.77506364141813e+20,1.2823681133441416e+20
6,1.962550772406156e+20,1.4017974968437293e+20
7,2.1408274173594744e+20,1.5219299986542825e+20
8,2.3128280796290525e+20,1.6108200685618836e+20
9,2.4551228933407008e+20,1.723889188484792e+20
10,2.596035949216577e+20,1.813217895166596e+20
11,2.7508849969553713e+20,1.882526561717891e+20
12,2.842374559929486e+20,1.9386477726346933e+20
13,2.907078395063924e+20,1.9552940553651343e+20
14,2.9280650546720276e+20,1.9372250005806075e+20
15,2.733257073669975e+20,1.7865510132792184e+20
16,2.1265780503351555e+20,1.3395535409041555e+20
17,1.7708776761535018e+20,1.0617588318536571e+20
18,2.261387995004142e+20,1.4295725426115527e+20
19,2.956630396445874e+20,1.912209930563719e+20
20,3.2448827607960506e+20,2.121626329439223e+20
21,3.291684817960603e+20,2.1852953197576503e+20
22,3.302877909301425e+20,2.2038779599334074e+20
23,3.247289186624152e+20,2.182724072892832e+20
24,3.155749996854472e+20,2.1179718843529642e+20
25,3.037439090519376e+20,2.0437929480764873e+20
26,2.8890533138455583e+20,1.9715231682417246e+20
27,2.7095766054941983e+20,1.8662276114037198e+20
28,2.519331098134016e+20,1.7355081374743806e+20
29,2.32726456641961e+20,1.593961467266737e+20
30,2.1025039775265125e+20,1.4469771894209723e+20
31,1.8411413601718518e+20,1.277519449943092e+20
32,1.4961286166111987e+20,1.0510468141023982e+20
```

D.2 "NBSR_Cycle7_outside_plates_fluence.csv" File

```
,bottom outside plate,top outside plate
1,6.4035635186237915e+19,4.479908785759571e+19
2,1.1123246275894002e+20,8.087980171961124e+19
3,1.3701301580135101e+20,9.9265353848243e+19
4,1.5498740815209205e+20,1.126072144083542 2e+20
5,1.727054278331091e+20,1.2426418744287771e+20
6,1.888214111702519e+20,1.3593846808677828e+20
7,2.078755425481573e+20,1.4550092544572134e+20
8,2.233946005355726e+20,1.5552778781130074e+20
9,2.3806408643350807e+20,1.6532093917457098e+20
10,2.4854546855692806e+20,1.725460241250352e+20
11,2.6137966233444732e+20,1.793822866685495e+20
12,2.7171467352111484e+20,1.8500875809510274e+20
```

13,2.7825528759766496e+20,1.8759767312998167e+20
14,2.7967498209425513e+20,1.8513249240959943e+20
15,2.6151282045314004e+20,1.707239896905391e+20
16,2.0214247793793525e+20,1.2854397010752134e+20
17,1.7140107241117037e+20,1.0198601447239903e+20
18,2.1846770701635033e+20,1.3540802227906273e+20
19,2.8358668958359362e+20,1.826567439912536e+20
20,3.1117845514791394e+20,2.0350728960334807e+20
21,3.168927357857369e+20,2.085373442436667e+20
22,3.175308907606461e+20,2.1074405723104428e+20
23,3.128441839140588e+20,2.0806298447631827e+20
24,3.039833943999176 4e+20,2.0414373604790426e+20
25,2.90993465740407e+20,1.9683226082006496e+20
26,2.782712685945343e+20,1.8686465705596656e+20
27,2.6217853241540313e+20,1.7834677559778846e+20
28,2.4322834045232936e+20,1.6714328853366637e+20
29,2.2273042694405905e+20,1.543921449265448e+20
30,2.01559170141436e+20,1.3883843649034848e+20
31,1.774997646944995e+20,1.2317921102294824e+20
32,1.4587218233750625e+20,1.0150659159578203e+20

D.3 "NBSR_Cycle8_outside_plates_fluence.csv" File

,bottom outside plate,top outside plate
1,6.3010198409269084e+19,4.428796495600645e+19
2,1.0892641682704289e+20,7.939367661703494e+19
3,1.3369246404755561e+20,9.626219322826559e+19
4,1.5275927873003946e+20,1.102280875664975e+20
5,1.6906655195509965e+20,1.213778215695237e+20
6,1.8569510539143155e+20,1.3218723367742705e+20
7,2.023588792268557e+20,1.4282684800807736e+20
8,2.1730326172519134e+20,1.5208546321637625e+20
9,2.3178425557729883e+20,1.622644281014427e+20
10,2.462821955259351e+20,1.686817392940827e+20
11,2.5703932125244277e+20,1.76232458190957e+20
12,2.644992790898354e+20,1.8052517316212117e+20
13,2.6979300289877793e+20,1.8223161469138377e+20
14,2.6968625712198543e+20,1.8022595913012602e+20
15,2.5339278876552482e+20,1.6616091120078012e+20
16,1.993968835483185e+20,1.2371315163292423e+20
17,1.670156953351785e+20,9.95412593928781e+19
18,2.1111245327977352e+20,1.321593286154611e+20
19,2.716394395416875e+20,1.771819285836504e+20
20,2.9973102544592654e+20,1.9681132031687075e+20
21,3.063728087057657e+20,2.0354360235570885e+20
22,3.088480227890859e+20,2.0261201490304523e+20
23,3.019192359381991e+20,2.024157469331571e+20
24,2.929271486908161e+20,1.988358013865895e+20
25,2.818192580237152e+20,1.9217135214187158e+20
26,2.702652975216766e+20,1.829094347373962e+20
27,2.5421063916417293e+20,1.732259459987107e+20
28,2.37320746296681e+20,1.634969303246562e+20
29,2.172922368529421e+20,1.5044624571958198e+20
30,1.9574418646727295e+20,1.3602369197456155e+20
31,1.7291385895822836e+20,1.2148324611323924e+20
32,1.4154217929429587e+20,9.907146249941359e+19

D.4 "NBSR_Cycle9_outside_plates_fluence.csv" File

,bottom outside plate,top outside plate
1,6.711710454726585e+19,4.786246217264248e+19
2,1.1727611347892026e+20,8.580487815587144e+19
3,1.432544906370023e+20,1.04305207384778e+20
4,1.6285887253076276e+20,1.1793961781860716e+20
5,1.8129829616254147e+20,1.3135217107638845e+20
6,1.9794011238770246e+20,1.4181952145099273e+20
7,2.144548230637346e+20,1.5263275730243214e+20
8,2.295451830481612e+20,1.6306935309511865e+20
9,2.4363466386521483e+20,1.71637725066473e+20
10,2.5747761074379676e+20,1.7936649390708228e+20
11,2.6911866692507304e+20,1.867642367646173e+20
12,2.787870646325133e+20,1.92745313681437e+20
13,2.8608049533972444e+20,1.9384111235611846e+20
14,2.829277654044795e+20,1.917618874823579e+20
15,2.6258730506695052e+20,1.7390031141111982e+20

16,2.0770905329613226e+20,1.3309274404554293e+20
17,1.7665347549609407e+20,1.0520849491612662e+20
18,2.2225410413711065e+20,1.402431126108445e+20
19,2.8486557923844833e+20,1.870057597546198e+20
20,3.1439974723084163e+20,2.0912578660331504e+20
21,3.204641908141416e+20,2.1406906704200086e+20
22,3.224741168548036e+20,2.1794663733028205e+20
23,3.177594191232293e+20,2.1661996650073326e+20
24,3.056290414451053e+20,2.108641107581572e+20
25,2.9495642242625385e+20,2.047974583380061e+20
26,2.8167522860153097e+20,1.955475009131108e+20
27,2.6736788094868626e+20,1.8626723565549404e+20
28,2.4957303382130924e+20,1.724066142186021e+20
29,2.285270793416825e+20,1.6042632963278317e+20
30,2.0612169009255953e+20,1.4629955940835457e+20
31,1.815594991431226e+20,1.290446489969437e+20
32,1.4993347193789025e+20,1.0495575624585825e+20

D.5 "NBSR_Cycle10_outside_plates_fluence.csv" File

,bottom outside plate,top outside plate
1,6.6215376451147964e+19,4.625288302537712e+19
2,1.1450425806631816e+20,8.358850678726895e+19
3,1.4129664609634448e+20,1.0352792985517955e+20
4,1.587098419825496e+20,1.1641666074264935e+20
5,1.7562949610702632e+20,1.2855266459162927e+20
6,1.9256491878904906e+20,1.3872184951989507e+20
7,2.0831252475328014e+20,1.4946904854164357e+20
8,2.2499566133193494e+20,1.584794040313193e+20
9,2.392098127152206e+20,1.6770400637101815e+20
10,2.500098508646541e+20,1.767271499200105e+20
11,2.6180981744312376e+20,1.8178074853421364e+20
12,2.702099244707243e+20,1.854946083117464e+20
13,2.7509147177558475e+20,1.875008845876803e+20
14,2.7121587732623973e+20,1.8388130849932257e+20
15,2.521788480747413e+20,1.6917765106478055e+20
16,2.015680668279644e+20,1.2693133480257095e+20
17,1.7255532452919398e+20,1.0236827030155579e+20
18,2.1295241794084074e+20,1.3530869379466795e+20
19,2.7241998535267405e+20,1.7957013586540557e+20
20,3.000222371078957e+20,2.0039072377372508e+20
21,3.0895961431111677e+20,2.085566258601452e+20
22,3.088775245978148e+20,2.0810745090262145e+20
23,3.0511121495257855e+20,2.064965041056283e+20
24,2.985094428034696e+20,2.03172647708899e+20
25,2.882686879160017e+20,1.969712013651306e+20
26,2.7438354191744152e+20,1.905739801403449e+20
27,2.591482906505975e+20,1.801475587691856e+20
28,2.4051964214936083e+20,1.6968449762754688e+20
29,2.2105251435572557e+20,1.5708235470307662e+20
30,2.0172250083528498e+20,1.4226205091727394e+20
31,1.7649119916290962e+20,1.2490509032533546e+20
32,1.4581859649873505e+20,1.0277791547631239e+20

D.6 "NBSR_Cycle11_outside_plates_fluence.csv" File

,bottom outside plate,top outside plate
1,6.512699801783219e+19,4.584957120655619e+19
2,1.1270824783255387e+20,8.179755313735033e+19
3,1.351751519919041e+20,9.998135599589555e+19
4,1.5571728147539513e+20,1.131277455394349e+20
5,1.7315814407477084e+20,1.246835597575505e+20
6,1.878415285798271e+20,1.34187778116689e+20
7,2.039299686479789e+20,1.4443422553292884e+20
8,2.1838670172142063e+20,1.5342575408544897e+20
9,2.3114365773852416e+20,1.6115707500815178e+20
10,2.43814060405941e+20,1.6882902256617737e+20
11,2.5263794511585713e+20,1.7524715976789118e+20
12,2.5951499319706033e+20,1.800403168937432e+20
13,2.630128166322686e+20,1.8077870211673185e+20
14,2.6179595882885153e+20,1.7868879428742126e+20
15,2.4358199587920455e+20,1.626091136057385e+20
16,1.943460250600023e+20,1.2388204342068971e+20
17,1.6875435916278265e+20,9.887127356795055e+19
18,2.0755040326746083e+20,1.3019654976331404e+20

19,2.606950355460403e+20,1.72202590933951e+20
20,2.8593150668389712e+20,1.9300214728075552e+20
21,2.952875381822198e+20,1.993923843200174e+20
22,2.9688932313658825e+20,2.0284202027302404e+20
23,2.941204466716079e+20,2.01143664031976e+20
24,2.8808136162168993e+20,1.9766511551209374e+20
25,2.7634993759715315e+20,1.9122344966762195e+20
26,2.6267284195269355e+20,1.8438944847821288e+20
27,2.4941782997077135e+20,1.737550385716981e+20
28,2.327303600807268e+20,1.6299486244972374e+20
29,2.1708750912660685e+20,1.5127659663443172e+20
30,1.9431294717384147e+20,1.3836967729840038e+20
31,1.7258593346176382e+20,1.236371527771496e+20
32,1.3986364705609453e+20,1.0018881290937863e+20

D.7 "NBSR_Cycle12_outside_plates_fluence.csv" File

,bottom outside plate,top outside plate
1,6.378975742669525e+19,4.453256178726385e+19
2,1.0896712488558472e+20,8.035579364231635e+19
3,1.3411899561096638e+20,9.809286154196972e+19
4,1.5156915858141908e+20,1.1099757138112253e+20
5,1.6775395687114333e+20,1.215242341392513e+20
6,1.8201259028209274e+20,1.3198225452192545e+20
7,1.9894304729310816e+20,1.4206182962582823e+20
8,2.1220728373219243e+20,1.4899343555356882e+20
9,2.2581446307274526e+20,1.5883466742029474e+20
10,2.3644482375357448e+20,1.649611313455873e+20
11,2.4635818874503954e+20,1.7156150593106972e+20
12,2.516639172873055e+20,1.744089241265901e+20
13,2.5637347553471696e+20,1.7584668016789497e+20
14,2.5058517058101726e+20,1.7212310846704425e+20
15,2.3190413910786556e+20,1.5503722042330923e+20
16,1.8950882352749137e+20,1.1843562074222612e+20
17,1.6505230142171775e+20,9.497002229272014e+19
18,2.0051001648842064e+20,1.2298888450923631e+20
19,2.5001258420764833e+20,1.6532324866367727e+20
20,2.754693093065947e+20,1.8578912505013394e+20
21,2.833592348731453e+20,1.9392405427465522e+20
22,2.8530654527979317e+20,1.9472093878381547e+20
23,2.8272718754799242e+20,1.9399379149717684e+20
24,2.77004598734608e+20,1.8944250337040305e+20
25,2.6709610158062027e+20,1.836186048862792e+20
26,2.5551326348857465e+20,1.7632196484891535e+20
27,2.404024776092183e+20,1.6924575984105272e+20
28,2.257517840924871e+20,1.592247132571682e+20
29,2.0816741065339005e+20,1.4680646998130082e+20
30,1.8844153507039933e+20,1.3457184258986502e+20
31,1.672952850916554e+20,1.1897737222239227e+20
32,1.3733087445305942e+20,9.720335232344444e+19

D.8 "NBSR_Cycle13_outside_plates_fluence.csv" File

,bottom outside plate,top outside plate
1,6.34093309089699e+19,4.429092694097388e+19
2,1.0622896923014572e+20,7.826560778770075e+19
3,1.307251325426029e+20,9.617851257475047e+19
4,1.4862227650132568e+20,1.0856563658351275e+20
5,1.6393688307791174e+20,1.1981353970106832e+20
6,1.7916760508243568e+20,1.2952178092118529e+20
7,1.927385668437103e+20,1.3775392690263572e+20
8,2.071303907530125e+20,1.4705558175113e+20
9,2.178478426817676e+20,1.5559285235605966e+20
10,2.2907803030385636e+20,1.605061802016721e+20
11,2.3774097649983344e+20,1.656757516580087e+20
12,2.4417415173864738e+20,1.6844365009538066e+20
13,2.4464268236998037e+20,1.7029001083250308e+20
14,2.414608305107022e+20,1.6541322679244213e+20
15,2.230844875665015e+20,1.5082329516084442e+20
16,1.830306934688556e+20,1.1477841931981554e+20
17,1.6045708875142024e+20,9.348512978967165e+19
18,1.9517965553039262e+20,1.2013243288886986e+20
19,2.4014743622082432e+20,1.5922815526975422e+20
20,2.6335868916429613e+20,1.7942246522357062e+20
21,2.7179695017840627e+20,1.859630954282165e+20

22,2.748763659111829e+20,1.8919898112123676e+20
23,2.7124572873263576e+20,1.8811653004534134e+20
24,2.6750481610593003e+20,1.8522501825775183e+20
25,2.5829539837757158e+20,1.807453402404241e+20
26,2.4812529097198158e+20,1.7288607614317645e+20
27,2.3296032001483453e+20,1.642875341666803e+20
28,2.1991834006919276e+20,1.5483140259733576e+20
29,2.0172999852424053e+20,1.45367254077646e+20
30,1.839309399497837e+20,1.3318279528009564e+20
31,1.629502078435188e+20,1.1637717536540567e+20
32,1.333031710324504e+20,9.380747502903624e+19

D.9 "NBSR_Cycle14_outside_plates_fluence.csv" File

,bottom outside plate,top outside plate
1,6.766353830373523e+19,4.684805427169915e+19
2,1.1411604762180567e+20,8.305247062028152e+19
3,1.3874149445588635e+20,1.0113208499339543e+20
4,1.5711524807717143e+20,1.1386812436009486e+20
5,1.739614643952124e+20,1.2584834525636346e+20
6,1.895835254304059e+20,1.3587612431806315e+20
7,2.033348655114016e+20,1.4628042239000568e+20
8,2.1707292198966175e+20,1.535861675574295e+20
9,2.2828786753087635e+20,1.6106635397647224e+20
10,2.3918094074531188e+20,1.6867059492705868e+20
11,2.4898336650428695e+20,1.7347664807365234e+20
12,2.5526636019804566e+20,1.7638455107288085e+20
13,2.5611571442411366e+20,1.7736525191057588e+20
14,2.5249476357303475e+20,1.7291913484215088e+20
15,2.3242386716663113e+20,1.5690406336576976e+20
16,1.902645956472994e+20,1.19397015651118e+20
17,1.707404977176709e+20,9.692077689177416e+19
18,2.0496120933593727e+20,1.2431551034375845e+20
19,2.503985388917974e+20,1.645383165714984e+20
20,2.7334129133604245e+20,1.863041831891767e+20
21,2.834846299714467e+20,1.949532593337217e+20
22,2.853913228741087e+20,1.9572799678270017e+20
23,2.8291531199984183e+20,1.9630610846936583e+20
24,2.77739969429598e+20,1.9259676636370335e+20
25,2.6892176074265428e+20,1.8675153945403926e+20
26,2.5788640848436154e+20,1.8007539340767098e+20
27,2.4475721106374817e+20,1.722787507627084e+20
28,2.2787512772107698e+20,1.6196800007645476e+20
29,2.1180886817500794e+20,1.51326259335235e+20
30,1.9322355424949276e+20,1.3827290635134157e+20
31,1.7057889545504168e+20,1.2260718204761517e+20
32,1.3965942313346205e+20,9.983676505962144e+19

D.10 "NBSR_Cycle15_outside_plates_fluence.csv" File

,bottom outside plate,top outside plate
1,6.556771173502612e+19,4.572245350569436e+19
2,1.1100265468892465e+20,8.074038791400561e+19
3,1.3627812471072706e+20,9.888470755492533e+19
4,1.5370082989760007e+20,1.1110886732179882e+20
5,1.6761560683101464e+20,1.2307477430109284e+20
6,1.8459383946020123e+20,1.328309025807564e+20
7,1.9738588241607115e+20,1.4189440000931973e+20
8,2.101235907542328e+20,1.497970502542867e+20
9,2.2135822070783895e+20,1.5669621321252078e+20
10,2.3365058158379198e+20,1.629560536421621e+20
11,2.3798680155757917e+20,1.6711001593542414e+20
12,2.4451502298565247e+20,1.7022634273984876e+20
13,2.4589121259499785e+20,1.7031330834717494e+20
14,2.4092087366757584e+20,1.6495150518997782e+20
15,2.216509708667811e+20,1.4863348903045077e+20
16,1.8545556716683975e+20,1.1416996153196036e+20
17,1.655123827374142e+20,9.376843609878864e+19
18,1.969338551061874e+20,1.2067419200936968e+20
19,2.3729507709932195e+20,1.5781400698076876e+20
20,2.611078093056312e+20,1.775526378322699e+20
21,2.7100080123105745e+20,1.868002792941019e+20
22,2.727605997582574e+20,1.884494917471137e+20
23,2.7118718517479716e+20,1.8647646467622244e+20
24,2.6591514677175422e+20,1.847124916646712e+20

25,2.5766909459415338e+20,1.8081444827521063e+20
26,2.4648832451186622e+20,1.7419933562642045e+20
27,2.335375590932756e+20,1.6624848057523166e+20
28,2.1854291117736854e+20,1.5674415995487132e+20
29,2.039710380730205e+20,1.4611361759551155e+20
30,1.868840134638557e+20,1.3375861591771087e+20
31,1.677285962613873e+20,1.1974784286490211e+20
32,1.3541713685126991e+20,9.676178039033353e+19

D.11 "NBSR_Cycle6_side_plates_fluence.csv" File

,left,right
1,5.9133605747842154e+19,6.0982467802581656e+19
2,9.7016622775342e+19,9.923470342435085e+19
3,1.1770107722215127e+20,1.2244108790180042e+20
4,1.3472983938349936e+20,1.3924606292522133e+20
5,1.5183907158251287e+20,1.5521514241082622e+20
6,1.6573115579040547e+20,1.702052883345689e+20
7,1.7968122022184672e+20,1.8368572238055434e+20
8,1.9199617562140606e+20,1.9778065426632376e+20
9,2.0428597523351565e+20,2.1139351218104402e+20
10,2.1495172941928196e+20,2.2231882291603046e+20
11,2.233866512190269e+20,2.316602229932036e+20
12,2.325803279519975e+20,2.399069006441973e+20
13,2.3537725479217857e+20,2.4666928443059002e+20
14,2.344938154919812e+20,2.4320904250364117e+20
15,2.169965495425953e+20,2.2583715707434634e+20
16,1.7545624125584782e+20,1.8325538333385333e+20
17,1.493377788786201e+20,1.548119952427451e+20
18,1.858899232897887e+20,1.9483142615796092e+20
19,2.3408557999830047e+20,2.4537289947683363e+20
20,2.564937749582843e+20,2.7089063395469486e+20
21,2.6409683198407852e+20,2.7773370993288272e+20
22,2.6392523704166443e+20,2.7876737644632074e+20
23,2.6104264404245568e+20,2.753527893529836e+20
24,2.5607658499073688e+20,2.6831861107599566e+20
25,2.4720115780287142e+20,2.5969332450610784e+20
26,2.3424371879617266e+20,2.4669242637095757e+20
27,2.2104840398965463e+20,2.3162058753117 13e+20
28,2.0629019926648786e+20,2.1619184565257503e+20
29,1.9058606391214617e+20,1.966979134379052e+20
30,1.7234565254831735e+20,1.7869162516060537e+20
31,1.5152681859921678e+20,1.5537070248204812e+20
32,1.2403865324804776e+20,1.2618194433700135e+20

D.12 "NBSR_Cycle7_side_plates_fluence.csv" File

,left,right
1,5.728088746011739e+19,5.909721023587959e+19
2,9.417444152058428e+19,9.649554861706412e+19
3,1.1518812132460152e+20,1.1775658654174934e+20
4,1.315675011735868e+20,1.3431158218951582e+20
5,1.463491884815016e+20,1.5015383400554915e+20
6,1.5954882109117363e+20,1.6535894944766332e+20
7,1.729344476438366e+20,1.7811360319145132e+20
8,1.859018668303377e+20,1.915459813994799e+20
9,1.973365816188224e+20,2.0302712226117494e+20
10,2.071731633320843e+20,2.1472655721937515e+20
11,2.171311820763656e+20,2.2364240288896955e+20
12,2.2309344415926423e+20,2.3027672477222896e+20
13,2.268735800921264e+20,2.3541318695318055e+20
14,2.2516123678881835e+20,2.337038282940415e+20
15,2.0799221288990022e+20,2.1726064481849852e+20
16,1.6967283333647335e+20,1.7728001755493063e+20
17,1.4448244928876515e+20,1.5060992864261272e+20
18,1.7978187826994033e+20,1.8726359992268626e+20
19,2.2455693197159694e+20,2.3536743429925313e+20
20,2.4658796136883228e+20,2.5927165680464327e+20
21,2.5395914666583717e+20,2.661437154549639e+20
22,2.542203996411341e+20,2.6697806453405293e+20
23,2.5149442644656374e+20,2.638641727384144e+20
24,2.458768963757733e+20,2.5726883192627213e+20
25,2.370098101392624e+20,2.4782133938577434e+20
26,2.2637061235954863e+20,2.3627013226152922e+20
27,2.1450702879136865e+20,2.2325005012206997e+20

28,1.9894907026135715e+20,2.0788648350388925e+20
29,1.8397940250130946e+20,1.9156214343553902e+20
30,1.666252313606213e+20,1.7075589963622062e+20
31,1.4684850672466757e+20,1.5024962491225706e+20
32,1.1970989548930577e+20,1.225141416261997e+20

D.13 "NBSR_Cycle8_side_plates_fluence.csv" File

,left,right
1,5.656156955895501e+19,5.84429234265586e+19
2,9.358087217615399e+19,9.474109865753005e+19
3,1.1401474394190232e+20,1.1609035459293063e+20
4,1.2880776137359216e+20,1.3222900498884541e+20
5,1.440155170742365e+20,1.483447482069086e+20
6,1.5675125837614957e+20,1.607339846487807e+20
7,1.6989115094426347e+20,1.7608508581592688e+20
8,1.8230057593523873e+20,1.874555552639949e+20
9,1.947488651698757e+20,1.9833262020811938e+20
10,2.039776483780018e+20,2.0882410258789745e+20
11,2.109648312575965e+20,2.1747761873410592e+20
12,2.1710981629240826e+20,2.2598526860848208e+20
13,2.1980203070009652e+20,2.2903546065176753e+20
14,2.1754942216724398e+20,2.2762899803496717e+20
15,2.0119742586472707e+20,2.110323951380307e+20
16,1.6575966224172968e+20,1.716140070875864e+20
17,1.4104895068362128e+20,1.4676602942169981e+20
18,1.7412689763365418e+20,1.828389175297466e+20
19,2.168362164029721e+20,2.2864764618864443e+20
20,2.3885388984870486e+20,2.518163176335916e+20
21,2.4451366737697918e+20,2.5902812062712064e+20
22,2.4570668615021347e+20,2.6009364074132183e+20
23,2.4397876312967286e+20,2.579803936975786e+20
24,2.384681559933545e+20,2.507803469205807e+20
25,2.3048250078670324e+20,2.423478915103388e+20
26,2.2035468092865767e+20,2.3150093730406102e+20
27,2.0819205065046757e+20,2.1726102839547825e+20
28,1.9440072703145945e+20,2.0294933938068164e+20
29,1.7905100783760654e+20,1.870602591529654e+20
30,1.6339121460835638e+20,1.6771962206252427e+20
31,1.4394119639890683e+20,1.4694631889301458e+20
32,1.1762534206887238e+20,1.1990810036607563e+20

D.14 "NBSR_Cycle9_side_plates_fluence.csv" File

,left,right
1,6.098960084937031e+19,6.2480835965368844e+19
2,9.825948203811113e+19,1.0302149025627293e+20
3,1.2005525314917877e+20,1.2556388667502587e+20
4,1.3720491006834703e+20,1.4231419440827066e+20
5,1.5269932432318018e+20,1.5818804417728535e+20
6,1.6658542378256528e+20,1.7282288278557055e+20
7,1.8051327390162508e+20,1.873053099782662e+20
8,1.9317210121028714e+20,2.0083005675154516e+20
9,2.04186709151261e+20,2.1341015629636148e+20
10,2.135296015068293e+20,2.2341785490957147e+20
11,2.2172239024539923e+20,2.3248627356034043e+20
12,2.295654792917514e+20,2.390366365230127e+20
13,2.3203450565446258e+20,2.4422967211996597e+20
14,2.2969391056355852e+20,2.41953684390288e+20
15,2.108502261546255e+20,2.2339013955323665e+20
16,1.7310455884012048e+20,1.8233827011988052e+20
17,1.4894543526222538e+20,1.5677237575459416e+20
18,1.8329843636920192e+20,1.9354863751789173e+20
19,2.2766520217714242e+20,2.4211564567623575e+20
20,2.5034808753745546e+20,2.661770616252778e+20
21,2.5755331279439544e+20,2.7504943717985206e+20
22,2.5983890039290485e+20,2.750123195506406e+20
23,2.5658529865957592e+20,2.7234489724302164e+20
24,2.5183010284554253e+20,2.6648652607137972e+20
25,2.4323257161989502e+20,2.577454587437468e+20
26,2.3201634976966276e+20,2.4587008133392574e+20
27,2.1881549057836794e+20,2.3144090163870315e+20
28,2.0582570507266913e+20,2.15707539195998e+20
29,1.897551468837788e+20,1.984596493658548e+20
30,1.7196795778394348e+20,1.8028040495496495e+20

31,1.5278498317909446e+20,1.5741452628870427e+20
32,1.244252950718758e+20,1.2865339895627537e+20

D.15 "NBSR_Cycle10_side_plates_fluence.csv" File

,left,right
1,5.949217248750963e+19,6.176402163996942e+19
2,9.731341078447258e+19,1.0149031554650192e+20
3,1.1784608610887947e+20,1.2317374715246753e+20
4,1.3435823755612293e+20,1.3974463010712823e+20
5,1.497442701722951e+20,1.55753415029195e+20
6,1.6373317770556663e+20,1.7001583311292396e+20
7,1.764983459439638e+20,1.8253644299552578e+20
8,1.8770537239220237e+20,1.947627259778595e+20
9,1.9935798924391095e+20,2.0769897524982058e+20
10,2.0966401050771607e+20,2.1718698355163093e+20
11,2.1656831552584997e+20,2.2623107073135644e+20
12,2.2286508144585664e+20,2.3369608897182823e+20
13,2.2526576630776024e+20,2.3697856690908253e+20
14,2.225982105205554e+20,2.3351530336262957e+20
15,2.0489988564737288e+20,2.1508630541533564e+20
16,1.6747685887124357e+20,1.7674833248016017e+20
17,1.442724374726409e+20,1.5189116877072364e+20
18,1.7609314079856738e+20,1.8724253901885453e+20
19,2.1959510647029236e+20,2.3223833568702097e+20
20,2.4281675115946184e+20,2.5741675060121466e+20
21,2.4943149941169085e+20,2.6548688071635032e+20
22,2.516987854448734e+20,2.6708522108476994e+20
23,2.490892997209442e+20,2.6559710022112546e+20
24,2.424338264493579e+20,2.585038386316303e+20
25,2.361466703032429e+20,2.4896514876226666e+20
26,2.2574303524806315e+20,2.3829875256290587e+20
27,2.138580869462825e+20,2.266583310677214e+20
28,1.998584035612038e+20,2.1079016454772444e+20
29,1.8370374077530915e+20,1.94038685792878e+20
30,1.6705957590772623e+20,1.746407317460713e+20
31,1.4729651471700425e+20,1.5302109297932776e+20
32,1.206922094087828e+20,1.2461301563272834e+20

D.16 "NBSR_Cycle11_side_plates_fluence.csv" File

,left,right
1,5.822435905768964e+19,6.0620879088175325e+19
2,9.510712620914734e+19,9.918211693138536e+19
3,1.164080353058161e+20,1.2109259563117081e+20
4,1.321760307090714e+20,1.3750306909160296e+20
5,1.4655608131714297e+20,1.5139226298509994e+20
6,1.5981011479912713e+20,1.6581831847993452e+20
7,1.7211141657503023e+20,1.788235261676968e+20
8,1.8363956537434626e+20,1.8968491004695137e+20
9,1.9342733295605147e+20,2.0136806847306986e+20
10,2.015779879916482e+20,2.1205111613257292e+20
11,2.1080048181172955e+20,2.197609361140717e+20
12,2.1656907040884104e+20,2.2549027042550658e+20
13,2.1812748749642934e+20,2.2857385808554387e+20
14,2.1443925148019337e+20,2.256329410521481e+20
15,1.9688118670637017e+20,2.079187636409047e+20
16,1.6314551566209715e+20,1.7096439079800904e+20
17,1.4138889779785818e+20,1.482679022036031e+20
18,1.71702945720235e+20,1.818633536262478e+20
19,2.1206666546859062e+20,2.2323094422054443e+20
20,2.3298308589467655e+20,2.4740943232230166e+20
21,2.4125827487299587e+20,2.5553523535097958e+20
22,2.4304288517052364e+20,2.573763673099455e+20
23,2.4076333150055095e+20,2.5607592803572305e+20
24,2.3666713449236952e+20,2.5036037396726245e+20
25,2.2874013064650965e+20,2.4191414414182085e+20
26,2.1910761451497387e+20,2.3045177966085656e+20
27,2.0672200994596372e+20,2.1896343330466847e+20
28,1.9417430663667548e+20,2.0289895443328077e+20
29,1.7981272606677425e+20,1.8857396780784094e+20
30,1.6316936127690244e+20,1.6989050639396703e+20
31,1.437365081781137e+20,1.489258858385136e+20
32,1.1801403644477756e+20,1.2092133407089556e+20

D.17 "NBSR_Cycle12_side_plates_fluence.csv" File

```
,left,right
1,5.728012236768284e+19,6.026209077828886e+19
2,9.364617660880275e+19,9.663884059627907e+19
3,1.135416334950611e+20,1.1719121957188349e+20
4,1.2939290451496047e+20,1.3376211769731539e+20
5,1.4351101075960734e+20,1.4746290454849046e+20
6,1.556046190635737e+20,1.6172160760491885e+20
7,1.677820226824619e+20,1.739835879452511e+20
8,1.7832893502256818e+20,1.8576753332179567e+20
9,1.8875039697454262e+20,1.9628943946992547e+20
10,1.9709645326960964e+20,2.0426783378879765e+20
11,2.0458861610573385e+20,2.135852632614466e+20
12,2.0959941071258377e+20,2.1835733843915968e+20
13,2.1206763796125014e+20,2.2110179196119977e+20
14,2.0761463626421643e+20,2.1759262163507115e+20
15,1.914437345825684e+20,1.9977181716774948e+20
16,1.59515861282019e+20,1.662296507143788e+20
17,1.3777523545869289e+20,1.4423128145384995e+20
18,1.6555210969512778e+20,1.761770854696409e+20
19,2.027045967332724e+20,2.1489650409261577e+20
20,2.245051981640233e+20,2.3816032727447942e+20
21,2.3361105338347477e+20,2.463272653898994e+20
22,2.3403982455682174e+20,2.486807456204719e+20
23,2.3233832415873506e+20,2.4697236160647574e+20
24,2.2871902087853326e+20,2.4161996323017618e+20
25,2.2135324194427263e+20,2.3362475910697796e+20
26,2.1127933880358424e+20,2.247066785089139e+20
27,2.0080405888752848e+20,2.11188570948523e+20
28,1.8914208167292394e+20,1.974447991442202e+20
29,1.7459806825391017e+20,1.831588865914582e+20
30,1.5858717759143176e+20,1.6605176394485206e+20
31,1.3997876170060713e+20,1.4613177824023459e+20
32,1.1466447229487192e+20,1.1925259419761515e+20
```

D.18 "NBSR_Cycle13_side_plates_fluence.csv" File

```
,left,right
1,5.6300264386386665e+19,5.899517828775728e+19
2,9.118068296292468e+19,9.569949075715205e+19
3,1.115514797456438e+20,1.1534065264267642e+20
4,1.2657639989955192e+20,1.3100454580819553e+20
5,1.392050823276227e+20,1.4573818671261173e+20
6,1.5335913891624328e+20,1.580488638941398e+20
7,1.644718271490266e+20,1.7029272394607077e+20
8,1.7387931037074047e+20,1.806699383066309e+20
9,1.8443295794151293e+20,1.9171343125754348e+20
10,1.926807313482738e+20,1.9901028245771944e+20
11,1.9903781539230663e+20,2.06593686878002e+20
12,2.0329473866537193e+20,2.1287736304364352e+20
13,2.0400559807576293e+20,2.1390299914255525e+20
14,2.0093237528262535e+20,2.0927589836841465e+20
15,1.8493299689741166e+20,1.929984697167389e+20
16,1.543933967897859e+20,1.6024396175644475e+20
17,1.337952434056824e+20,1.414038454435044e+20
18,1.605609102605929e+20,1.691519129549698e+20
19,1.9675592941544653e+20,2.0662649471333918e+20
20,2.182124809548054e+20,2.299661410969733e+20
21,2.265727359634355e+20,2.3739126397533084e+20
22,2.2860847317972297e+20,2.399770244819113e+20
23,2.2658458784580338e+20,2.3914451658899807e+20
24,2.207252940045155e+20,2.3329586286991947e+20
25,2.152592545783268e+20,2.2691786224686976e+20
26,2.0462618438051303e+20,2.1654572295083213e+20
27,1.940995216727963e+20,2.0546044884785453e+20
28,1.8307550068990296e+20,1.9307290569834896e+20
29,1.6922007539902336e+20,1.7747544836802288e+20
30,1.5446377181782285e+20,1.6131138944511905e+20
31,1.3634531733549592e+20,1.4172183989007607e+20
32,1.122807749455238e+20,1.1547072109787972e+20
```

D.19 "NBSR_Cycle14_side_plates_fluence.csv" File

```
,left,right
1,6.037155502235625e+19,6.2969453855747105e+19
2,9.795186294517925e+19,1.0090056151754347e+20
3,1.1835069421511187e+20,1.2275254757855789e+20
4,1.3448508408579106e+20,1.3913971319043442e+20
5,1.4763668865458346e+20,1.5329493405113503e+20
6,1.602996020941933e+20,1.6695373917736203e+20
7,1.729839899566469e+20,1.807660050881483e+20
8,1.8485714954599488e+20,1.912430342407148e+20
9,1.9210902911451077e+20,2.01964698357872e+20
10,2.0217071634125583e+20,2.1082817624526275e+20
11,2.0901379483131986e+20,2.171027491363078e+20
12,2.1323554265760285e+20,2.222102256977219e+20
13,2.1537930528687905e+20,2.249575472600312e+20
14,2.0953169271726745e+20,2.1977632455199058e+20
15,1.92704147281763e+20,2.0181039796232674e+20
16,1.6144691593057626e+20,1.6874989358244082e+20
17,1.4108550865427844e+20,1.4826471589156825e+20
18,1.680728481962184e+20,1.7746059472832863e+20
19,2.0358101925141217e+20,2.166266676165683e+20
20,2.254016676947656e+20,2.3875724648584135e+20
21,2.3509962599841595e+20,2.4776410037784648e+20
22,2.3750095962522183e+20,2.510065959197475e+20
23,2.362199087626669e+20,2.492567642411206e+20
24,2.3015572216873212e+20,2.4393392404382895e+20
25,2.2413049433865545e+20,2.3661597806494378e+20
26,2.158849270297289e+20,2.274025988391214e+20
27,2.054242684642285e+20,2.1458808479045868e+20
28,1.9333178134284055e+20,2.0155041305618724e+20
29,1.8016806456373107e+20,1.871323668941495e+20
30,1.638668940726629e+20,1.6903780764030403e+20
31,1.4539972408875606e+20,1.4857238565697903e+20
32,1.189945255666007e+20,1.2104657205665353e+20
```

D.20 "NBSR_Cycle15_side_plates_fluence.csv" File

```
,left,right
1,5.8514840090477126e+19,6.1054666722929385e+19
2,9.58091037217004e+19,9.818969216796094e+19
3,1.1628352208719716e+20,1.1890879760078309e+20
4,1.3131938558546967e+20,1.3719987378263928e+20
5,1.4387967541256015e+20,1.5002053438733563e+20
6,1.562755899623245e+20,1.6309155701233225e+20
7,1.6790649892067964e+20,1.7432706351938457e+20
8,1.7876640404597037e+20,1.840588236452871e+20
9,1.8765476002895243e+20,1.9486356431829523e+20
10,1.961240491740484e+20,2.0256046511002747e+20
11,2.0341197121649462e+20,2.0815787075921455e+20
12,2.05652601752867e+20,2.1299206988796078e+20
13,2.0624009069318527e+20,2.1448306633919778e+20
14,2.0170721188774416e+20,2.1072877937720377e+20
15,1.8449550584204424e+20,1.9199816255238026e+20
16,1.5451302223816553e+20,1.6280244576345748e+20
17,1.3654603396282637e+20,1.4335490058891198e+20
18,1.621403367377767e+20,1.7147492782360435e+20
19,1.961370155123163e+20,2.0718113570650626e+20
20,2.1744060250844588e+20,2.2901888889895947e+20
21,2.2732356780453646e+20,2.3814487596751058e+20
22,2.2755406732794836e+20,2.4043084647147933e+20
23,2.2665804640089804e+20,2.3866214085148875e+20
24,2.229828259714021e+20,2.3579451565840007e+20
25,2.1638126109138413e+20,2.286479935057202e+20
26,2.084541560086898e+20,2.194848904863753e+20
27,1.9795473753649847e+20,2.0844304639772262e+20
28,1.865383577253414e+20,1.9654301428535555e+20
29,1.738619292655997e+20,1.8158874881208e+20
30,1.5837733786576005e+20,1.6353513106679326e+20
31,1.406820113122656e+20,1.4441727832319518e+20
32,1.1477980313329099e+20,1.1747794264765617e+20
```

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